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Nevanlinna Theory in Several Complex Variables and Diophantine Approximation

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Junjiro Noguchi • Jörg Winkelmann

Nevanlinna Theory in Several Complex Variables and Diophantine Approximation

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Preface

Based on theorems of Picard and Borel, R. Nevanlinna published his paper [25] and evolved a theory that carries his name. The development of the theory into complex analysis in several variables was quite fast and a number of papers such as A. Bloch [26b], H. Cartan [33], and H. Weyl–J. Weyl [38] were published. Following H. Weyl–J. Weyl [38], the paper of A.L. Ahlfors [41] was published, and this work was succeeded by W. Stoll [53b], [54], being developed into the theory of meromorphic mappings from parabolic spaces into complex projective spaces.

S.-S. Chern studied the value distribution of holomorphic curves from the viewpoint of differential geometry, and published a number of papers; one of them was a joint paper, Bott–Chern [65], where they introduced a "connection" to the study of the value distribution theory and found a relation with characteristic classes. S.-S. Chern introduced with Osserman the Nevanlinna theoretic method into the study of Gauss' maps of minimal surfaces in Euclidean spaces. This approach was advanced much, especially by H. Fujimoto, and finally reached the solution of Gauss' map conjecture (H. Fujimoto [88a]).

From the late 1960s to the early 1970s the theory of Kobayashi hyperbolic manifolds and the value distribution theory of equidimensional holomorphic mappings were initiated (Kobayashi [67], [70]; Carlson–Griffiths [72]; Griffiths–King [73]), and the value distribution theory in several complex variables advanced considerably and was enlarged, so that it has broadened the border with more related subjects.

T. Ochiai [77] revived Bloch's paper (Bloch [26b]) and proved a lemma on holomorphic differentials, which is a counterpart to the lemma on logarithmic derivatives in Nevanlinna theory, and obtained the so-called the Bloch–Ochiai Theorem. In the study of the extension of the Bloch–Ochiai Theorem to the non-compact case, the relation with P. Deligne's logarithmic forms (Deligne [71], [74]) was found and the initial step to deal with varieties other than complex projective space began for the value distribution theory of holomorphic curves (Noguchi [77b]). As a result, the Bloch–Ochiai Theorem and the classical theorem of Borel were unified.

On the other hand, S. Lang formulated a conjecture of a higher dimensional version of Mordell's conjecture on rational points such that a Kobayashi hyperbolic

algebraic manifold defined over a number field would have at most finitely many rational points (Lang's conjecture), and proposed a number of related conjectures to enhance the study of problems over function fields and of holomorphic curves (Mordell [22]; Lang [74], [86]). In 1987 P. Vojta [87], completing the viewpoint of Osgood [81], formulated an analogue between Nevanlinna theory and Diophantine approximation theory. He proposed the well-known Vojta conjecture, which has attracted the attention of many researchers.

As for monographs describing these developments of the study of Nevanlinna theory and Diophantine approximation theory, we may list, in order of year, e.g., Kobayashi [70], Wu [70], Stoll [70], [73], Griffiths [76], Stoll [77a], Lang [83], Shiffman [83], Ochiai–Noguchi [84], Shabat [85], Vojta [87], Lang [87], Noguchi–Ochiai [90], Lang [91], Fujimoto [93], Kobayashi [98], Ru [01] and Bombieri–Gubler [06].

On Kobayashi's hyperbolicity, his book [98] provides in great detail the results obtained up to the present. Fujimoto [93] describes the theory of Gauss' maps of minimal surfaces in a self-contained manner.

The core of Nevanlinna theory consists of two Main Theorems. The First Main Theorem is considered to be a non-compact version of Poincaré's duality, and we now have a satisfactory theory for it. In the present book the second chapter is devoted to it. On the other hand, there are not many cases where the Second Main Theorem is established for meromorphic mappings $f : \mathbf{C}^n \to M$ and divisors D on M; the following is essentially the full list:

(i) $n = 1$, $M = \mathbf{P}^m(\mathbf{C})$, D is a union of hyperplanes in general position (Cartan [33]; H. Weyl–J. Weyl [38]; Ahlfors [41]), Nochka's solution of Cartan's conjecture (Nochka [83]; Chen [90]).

(ii) $n \geqq 1$, $M = \mathbf{P}^m(\mathbf{C})$, D is a union of hyperplanes in general position (Stoll [53b], [54]; Chen [90]).

(iii) $n = 1$, $M = \mathbf{P}^m(\mathbf{C})$, D is a union of hypersurfaces in general position (Eremenko–Sodin [92]; Corvaja–Zannier [04b]; Ru [04]).

(iv) $n \geqq \dim M$ and f is differentiably non-degenerate (of maximal rank at some point) (Griffiths' theory, Carlson–Griffiths [72]; Griffiths–King [73]; Sakai [74a]; Shiffman [75]; Noguchi [76a]).

(v) M is an abelian or semi-abelian variety and D is an arbitrary divisor on it. When M is a semi-abelian variety, a compactification of M is taken so that the closure $\bar{D}$ is in general position (in a sense) (Noguchi–Winkelmann–Yamanoi [00], [02], [08]).

(vi) $M = \mathbf{P}^1(\mathbf{C})$, $n = 1$ and D is a moving divisor for which the counting functions are truncated to level one (K. Yamanoi [04d], [05], [06]).

There is no book among the list above mentioned that deals with all of the above cases. Because of the applications to the degeneracy problem and Kobayashi hyperbolicity, we deal here with the cases of (i), (ii), (iv), and (v), and describe the theory of the all Second Main Theorems in these cases from a uniformized viewpoint in a self-contained way (Chaps. 3, 4 and 6). For the other cases, we limit ourselves only to introducing some important results. The Kobayashi hyperbolicity

will be discussed in Chap. 7, and we will describe some new results which were not mentioned in Kobayashi [98]. These are the first purpose of this book.

The second purpose is to describe and discuss the analogy between Nevanlinna theory and Diophantine approximation theory. The case of function fields is dealt with in Chap. 8, and the case of number fields in Chap. 9.

The readers we have in mind are graduate students and researchers who are interested in these subjects. For an easier understanding of the idea of the development of the theory we start with Nevanlinna theory for meromorphic functions on the one-dimensional complex plane. As prerequisites the basic theory of analytic functions in one variable, the very elementary part of analytic functions in several complex variables, and the terminologies of the theory of complex manifolds are assumed. Other materials which we use in the text without giving the proofs will be explained at least so that the readers are able to understand the contents.

We will briefly mention the content of each chapter in the sequel.

In Chap. 1 we prove the First and the Second Main Theorems in a self-contained manner for meromorphic functions on the Gaussian plane, which is the most basic part of Nevanlinna theory. This provides the prototype of the theory extended to the higher dimensional case.

In Chap. 2 plurisubharmonic functions are described and Jensen's formula is proved. Then we show the First Main Theorem for meromorphic mappings from the m-dimensional complex space $\mathbf{C}^m$ into a compact complex manifold with respect to holomorphic line bundles and coherent ideal sheaves of the structure sheaf. We will introduce a various fundamental properties of order functions. The content of this chapter serves the basic for the research of this subject, and we intended that the readers naturally master the fundamental method and knowledge of this subject.

In Chap. 3 we first extend the lemma on logarithmic derivatives to the case of several complex variables. We will give a simplified proof for it. Then we prove the Second Main Theorem by using the lemma on logarithmic derivatives, which is different to the metric method due to Griffiths et al. We also give some applications.

In Chap. 4 we deal with the theory of holomorphic curves from the complex plane, which we call "*entire curves*". We describe, following to Chen [90], the notion of the Nochka weights, which plays an important role in the proof of Cartan's conjecture, now Nochka's theorem. Lemma on logarithmic jet differentials is proved in a most generalized and uniformized form due to Noguchi [77b], [81b], [85b], [86], and Vitter [77]. Using this we prove an inequality of the Second Main Theorem type for entire curves in an algebraic variety, and then deduce the Logarithmic Bloch–Ochiai Theorem (Noguchi [77b], [81b]).

Chapter 5 is devoted to the semi-tori and their compactifications. The contents are the preparation for the next chapter.

In Chap. 6 we then establish the late result of the Second Main Theorem for entire curves in semi-abelian varieties (Noguchi–Winkelmann–Yamanoi [00], [02], [08]). From this Lang's conjecture for entire curves is derived, and more applications are given (Yamanoi [04b]; Noguchi–Winkelmann–Yamanoi [07], [13]; Winkelmann [11]; Corvaja–Noguchi [12]; Lu–Winkelmann [12]).

In Chap. 7 we describe the applications of the results of Chaps. 4 and 6 to the algebraic degeneracy problem of entire curves and the Kobayashi hyperbolicity problem. In particular, being related to the Kobayashi conjecture, the existence of an algebraic curve of degree 5 in $\mathbf{P}^2(\mathbf{C})$ whose complement is hyperbolic and hyperbolically embedded into $\mathbf{P}^2(\mathbf{C})$ is proved (Zaidenberg [89]). We construct explicit examples of hyperbolic hypersurfaces of $\mathbf{P}^n(\mathbf{C})$ for arbitrary n, and those whose complement are hyperbolic and hyperbolically embedded into $\mathbf{P}^n(\mathbf{C})$.

In Chap. 8 we describe Nevanlinna theory over function fields. This is motivated by the conjectures of S. Lang and P. Vojta. Nevanlinna theory over function fields is understood as an approximation theory of rational functions by rational functions. We prove an analogue of the so-called abc-conjecture of Masser and Oesterlé over function fields (cf. Oesterlé [88], Sect. 3; Granville–Tucker [02]).

In Chap. 9 we describe Diophantine approximation theory over number fields in an analogous way to Nevanlinna theory, based on Vojta's idea. As application we prove finiteness theorems of integral points. Readers will find analogies not only in conjectures or results but also even in proofs.

We would like to draw attention to Nevanlinna theory of p-adic analytic functions, started by Ha (Ha Hui Khoai [83]; Ha Hui Khoai–Tu [95]), which is not mentioned in this book.

There is a large undeveloped area in the relation between Nevanlinna theory of entire curves and Diophantine approximation theory, which is expected to be explored more. We hope that the present book will serve for readers an introduction to this subject.

In writing this book the authors received a valuable comment on Nochka's weights from Professor N. Toda, and helpful advice on Chap. 9 from Professor N. Hirata. Professors Y. Aihara and K. Yamanoi have given a number of invaluable comments all through the book. The authors express their sincere gratitude to all of them. Taking this opportunity, the authors thank Professors Mitsuru Ozawa, Wilhelm Stoll, Shoshichi Kobayashi, Hirotaka Fujimoto, Takushiro Ochiai, and Alan Huckleberry for their teachings, various suggestions and comments all through their research lives.

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Conventions

(i) The set of natural numbers (positive integers) is denoted by $\mathbf{N}$, the set of integers by $\mathbf{Z}$, the set of rational numbers by $\mathbf{Q}$, the set of real numbers by $\mathbf{R}$, the set of complex numbers by $\mathbf{C}$, and the imaginary unit by i, as usual. The set of non-negative integers (resp. numbers) is denoted by $\mathbf{Z}^+$ (resp. $\mathbf{R}^+$).

(ii) For a complex number $z = x + iy \in \mathbf{C}$ we set $\Re z = x$ and $\Im z = y$.

(iii) In general, we set $k^* = k \setminus \{0\}$ for a field k.

(iv) For a function $\phi : X \to [-\infty, \infty]$ we set the positive part by $\phi^+ = \max\{\phi(x), 0\}$; e.g., $\log^+ |z| = \max\{\log |z|, 0\}$ for $z \in \mathbf{C}$.

(v) Theorems, equations etc. are numbered consecutively. Here an equations is numbered as (1.1.1) with parentheses; the first 1 stands for the chapter number and the second 1 for the section number.

(vi) *Monotone increasing*, and *monotone decreasing* are used in the sense including the case of equality: e.g., a sequence of functions $\{\varphi_\nu(x)\}_{\nu=1}^\infty$ is said to be monotone increasing if for every point x of the defining domain $\varphi_\nu(x) \leqq \varphi_{\nu+1}(x)$ for all $\nu = 1, 2, \ldots$.

(vii) A map $f : X \to Y$ between locally compact topological spaces is said to be *proper* if for every compact subset $K \subset Y$, the inverse image $f^{-1}K$ is compact, too.

(viii) Manifolds are assumed to be connected, unless anything else is specified.

(ix) The symbol $\Subset$ stands for the relative compactness; e.g., $\Delta(a; r) \Subset U$ means that the closure $\overline{\Delta(a; r)}$ is compact in U.

(x) The symbols $O(1)$, $o(1)$, etc. follow after Landau's.

(xi) For a set S, $|S|$ denotes its cardinality.

(xii) A map $f : X \to Y$ is said to be *injective* or an *injection* if $f(x_1) \neq f(x_2)$ for every distinct $x_1, x_2 \in X$, and to be *surjective* or a *surjection* if $f(X) = Y$. If f is proper and the inverse image $f^{-1}\{y\}$ is always finite for all $y \in Y$, f is called a *finite map*. The restriction of f to a subset $E \subset X$ is denoted by $f|_E$.

(xiii) A function f defined on an open subset U of a differentiable manifold is said to be of C^k-class if f is k-times continuously differentiable. $C^k(U)$ denotes the set of all functions of C^k-class on U. $C_0^k(U)$ stands for the set of all $f \in C^k(U)$ with compact support.

(xiv) In general, for a differential form α we write $\alpha^k = \alpha \wedge \cdots \wedge \alpha$ (k-times).

(xv) $\mathbf{P}^n_k$ denotes the n-dimensional projective space defined over a field k. Thus, a subspace of $\mathbf{P}^n_k$ means that it is defined by equations with coefficients in k.

(xvi) Complex spaces are assumed to be connected and to have reduced structures (may be reducible) unless otherwise mentioned. Subschemes, however, are assumed to be possibly non-connected or to have non-reduced structures in general.

Chapter 1
Nevanlinna Theory of Meromorphic Functions

The value distribution theory of meromorphic functions on $\mathbf{C}$ established by R. Nevanlinna in 1925 is described. It not only deepened complex function theory in one variable but also led to the covering theory by L. Ahlfors and the value distribution theory in several complex variables. The aim of this chapter is to introduce the most fundamental part of the theory in a self-contained manner. We try to give the proofs by making use of those by H.L. Selberg and H. Cartan so that they are straightforward and compatible with those in the case of several complex variables presented in the later chapters.

1.1 The First Main Theorem

We begin with Jensen's formula, which will be generalized to several complex variables and is fundamental throughout the present book.

We introduce some notation. We write the complex coordinate $z = x + iy$ $(x, y \in \mathbf{R})$ on $\mathbf{C}$. For $a \in \mathbf{C}$ and $r > 0$ we set

$$\Delta(a; r) = \{z \in \mathbf{C}; |z - a| < r\}, \qquad \Delta(r) = \Delta(0; r).$$

For a differentiable function $\varphi(z) = \varphi(x, y)$ we define the following partial differential operators:

$$\begin{aligned}
&\frac{\partial\varphi}{\partial z} = \frac{1}{2}\left(\frac{\partial\varphi}{\partial x} + \frac{1}{i}\frac{\partial\varphi}{\partial y}\right), \qquad \frac{\partial\varphi}{\partial\bar{z}} = \frac{1}{2}\left(\frac{\partial\varphi}{\partial x} - \frac{1}{i}\frac{\partial\varphi}{\partial y}\right),\\
&dz = dx + i\,dy, \qquad d\bar{z} = dx - i\,dy,\\
&\partial\varphi = \frac{\partial\varphi}{\partial z}dz, \qquad \bar{\partial}\varphi = \frac{\partial\varphi}{\partial\bar{z}}d\bar{z},\\
&d^c\varphi = \frac{i}{4\pi}(\bar{\partial}\varphi - \partial\varphi) = \frac{1}{4\pi}\left(\frac{\partial\varphi}{\partial x}dy - \frac{\partial\varphi}{\partial y}dx\right).
\end{aligned} \tag{1.1.1}$$

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With the above notation we have

$$d\varphi = \partial\varphi + \bar{\partial}\varphi, \qquad dd^c\varphi = \frac{i}{2\pi}\partial\bar{\partial}\varphi,$$

$$\partial\bar{\partial}\varphi = \frac{\partial^2\varphi}{\partial z\partial\bar{z}}dz \wedge d\bar{z}.$$

If α, β are 1-forms on a domain $U \subset \mathbf{C}$, then

$$d\alpha \wedge d^c\beta = d\beta \wedge d^c\alpha.$$

In polar coordinates, $z = re^{i\theta}$ ($\log z = \log r + i\theta$), we have by the last formula of (1.1.1)

$$d^c\varphi = \frac{1}{4\pi}\left(r\frac{\partial\varphi}{\partial r}d\theta - \frac{1}{r}\frac{\partial\varphi}{\partial\theta}dr\right). \tag{1.1.2}$$

Let $D \Subset \mathbf{C}$ be a bounded domain such that the boundary ∂D is of C^1-class. Let $\eta = Pdz + Qd\bar{z}$ be a 1-form of C^1-class in a neighborhood of the closure $\bar{D}$. Then one has Stokes' theorem,

$$\int_{\partial D}\eta = \int_D d\eta = \int_D\left(-\frac{\partial P}{\partial\bar{z}} + \frac{\partial Q}{\partial z}\right)dz \wedge d\bar{z}. \tag{1.1.3}$$

Let $\varphi(z)$ be a function on $\mathbf{C}$ such that $-\infty \leqq \varphi \leqq \infty$ and that the singular point set $\{\varphi = -\infty, \infty\}$ of φ is a discrete set denoted by $Z = \{a_\nu\}_{\nu=1}^{\infty}$. Assume that φ is of C^2-class on $\mathbf{C} \setminus Z$, and that in a neighborhood of every point a_ν there are a C^2-function ψ_ν and a real number λ_ν satisfying the following:

$$\varphi(z) = \lambda_\nu \log|z - a_\nu| + \psi_\nu(z). \tag{1.1.4}$$

(For example, if f is meromorphic and $\alpha \in \mathbf{R}$, then $\phi = \log(|f|^\alpha)$ is such a function.) Since $\partial\bar{\partial}\log|z - a_\nu| \equiv 0$, there exists a continuous $(1,1)$ form $\partial\bar{\partial}\varphi(z)$ on $\mathbf{C}$ which is defined locally in a neighborhood of a_ν by $\partial\bar{\partial}\psi_\nu(z)$. We simply use a notation such as

$$\frac{1}{2\pi}\int_{|z|=r}\varphi(z)d\theta = \frac{1}{2\pi}\int_0^{2\pi}\varphi\big(re^{i\theta}\big)d\theta.$$

If $\varphi(0) \neq \pm\infty$, we understand $\frac{1}{2\pi}\int_{|z|=0}\varphi(z)d\theta = \varphi(0)$.

Lemma 1.1.5 (Jensen's formula) *Let $\varphi(z)$ be as above. Then for $0 \leqq s < r$ if $\varphi(0) \neq \pm\infty$, and for $0 < s < r$ in general,*

$$\frac{1}{2\pi}\int_{|z|=r}\varphi(z)d\theta - \frac{1}{2\pi}\int_{|z|=s}\varphi(z)d\theta$$
$$= 2\int_s^r\frac{dt}{t}\int_{\Delta(t)}\frac{i}{2\pi}\partial\bar{\partial}\varphi + \int_s^r\frac{dt}{t}\left(\sum_{|a_\nu|<t}\lambda_\nu\right).$$

Proof We first assume that φ is of C^2-class. It follows from (1.1.2) and a repeated use of Stokes' theorem that

$$\begin{aligned}
&\frac{1}{2\pi}\int_{|z|=r}\varphi(z)d\theta-\frac{1}{2\pi}\int_{|z|=s}\varphi(z)d\theta\\
&\quad=2\int_{|z|=r}\varphi(z)d^c\log|z|-2\int_{|z|=s}\varphi(z)d^c\log|z|\\
&\quad=2\int_{\Delta(r)\setminus\Delta(s)}d\varphi\wedge d^c\log|z|=2\int_{\Delta(r)\setminus\Delta(s)}d\log|z|\wedge d^c\varphi \qquad (1.1.6)\\
&\quad=2\int_s^r\frac{dt}{t}\int_{|z|=t}d^c\varphi=2\int_s^r\frac{dt}{t}\int_{\Delta(t)}dd^c\varphi. \qquad (1.1.7)
\end{aligned}$$

If φ has a logarithmic singularity $\lambda_\nu\log|z-a_\nu|$ such as (1.1.4), we take a circle of a small radius $\varepsilon>0$ with center a_ν and apply Stokes' theorem. The contribution from the arc integration over the circle is estimated as $O(\varepsilon|\log\varepsilon|)$ in (1.1.6), so that it converges to 0 as $\varepsilon\to 0$. In (1.1.7) the contribution is $\lambda_\nu+O(\varepsilon)$, from which the integral $\int_s^r\frac{dt}{t}(\sum_{|a_\nu|<t}\lambda_\nu)$ is deduced. □

A *meromorphic function* $f(z)$ on a domain U of $\mathbf{C}$ is so defined as at each point $a\in U$ there are connected neighborhood V and two holomorphic functions $g(z),h(z)$ on V such that $h(z)\not\equiv 0$ and $f(z)=\frac{g(z)}{h(z)}$ $(z\in V)$ holds. The set of meromorphic functions on U naturally forms a field. If $f\not\equiv 0$, there holds a unique expression

$$f(z)=(z-a)^m g(z)$$

in a neighborhood of every point $a\in U$, where $m\in\mathbf{Z}$, $g(z)$ is holomorphic and $g(a)\neq 0$. When $m>0$, a is called a zero of *order* m of $f(z)$, and when $m<0$, a is called a *pole* of *order* $|m|$. The sets of zeros and poles of $f(z)$ are discrete.

A *divisor* on U with coefficients in $\mathbf{Z}$ (or $\mathbf{R}$) is a formal sum $\sum_{\nu=1}^\infty\lambda_\nu\{z_\nu\}$ with a discrete subset $\{z_\nu\}_{\nu=1}^\infty$ of U and with coefficients $\lambda_\nu\in\mathbf{Z}$ (or $\mathbf{R}$), $\nu=1,2,\ldots$. Here, $\{z_\nu\}$ denotes the one-point set. (A divisor on a complex manifold will be defined in Sect. 2.2.) Let $\{a_\nu\}_{\nu=1}^\infty$ be the union of zeros and poles of $f(z)$. Then in a neighborhood of every point a_ν there is a holomorphic function $g(z)$ such that

$$f(z)=(z-a_\nu)^{\lambda_\nu}g(z),\quad \lambda_\nu\in\mathbf{Z},\ g(a_\nu)\neq 0.$$

The zero divisor of $f(z)$ is defined by

$$(f)_0=\sum_{\lambda_\nu>0}\lambda_\nu\{a_\nu\},$$

and the polar divisor of $f(z)$ is defined by

$$(f)_\infty=\sum_{\lambda_\nu<0}-\lambda_\nu\{a_\nu\}.$$

Let $E = \sum \mu_\nu \{z_\nu\}$ be a divisor on $\mathbf{C}$, where z_ν are distinct. Let $k \in \mathbf{N} \cup \{\infty\}$. Then the *truncated counting functions* to level k are defined by

$$\begin{aligned} n_k(t, E) &= \sum_{|z_\nu|<t} \min\{k, \mu_\nu\}, \quad t > 0, \\ N_k(r, E) &= \int_1^r \frac{n_k(t, E)}{t} dt, \quad r > 1. \end{aligned} \tag{1.1.8}$$

In particular, we write $n(t, E) = n_\infty(t, E)$, $N(r, E) = N_\infty(r, E)$, which are simply called *counting functions*. For convenience we write

$$(f - \infty)_0 = (f)_\infty.$$

Remark In the definition of $N_k(r, E)$ the initial value of integration was set to 1, but if we take an arbitrarily fixed number $r_0 > 0$, and carry out all the arguments, the conclusions are the same; this will be seen in what follows. It is also possible to set $r_0 = 0$ as follows; for instance if $z_1 = 0$ and $n_k(0, E) = \min\{k, \mu_1\}$, we define

$$N_k(r, E) = \int_0^r \frac{n_k(t, E) - n_k(0, E)}{t} dt + n_k(0, E) \log r, \quad r > 0. \tag{1.1.9}$$

The difference of (1.1.9) from (1.1.8) is a constant, $\int_0^1 \frac{n_k(t,E)-n_k(0,E)}{t} dt$. In some literature definition (1.1.9) is used, but in the present book we would take (1.1.8) to keep a compatibility with the case of several complex variables in what follows. It is also convenient to get an estimate to prove the lemma on logarithmic derivatives.

In what follows in this chapter, a *meromorphic function* $f(z)$ *is defined on* $\mathbf{C}$*, and not a constant*.

The *proximity function* of $f(z)$ (to ∞) is defined by

$$m(r, f) = \frac{1}{2\pi} \int_{|z|=r} \log^+ \left| f(z) \right| d\theta. \tag{1.1.10}$$

(Cf. Convention (iv) for "$\log^+$".) The following properties hold:

$$\begin{aligned} &\log s = \log^+ s - \log^+ \frac{1}{s}, \qquad |\log s| = \log^+ s + \log^+ \frac{1}{s}, \\ &\log^+ \sum_{j=1}^N s_j \leqq \sum_{j=1}^N \log^+ s_j + \log N, \qquad \log^+ \prod_{j=1}^N s_j \leqq \sum_{j=1}^N \log^+ s_j. \end{aligned} \tag{1.1.11}$$

The proximity function $m(r, f)$ measures the average approximation of $f(z)$ to ∞ over the circle $|z| = r$. In comparison to Diophantine approximation, $m(r, f)$ is also called an *approximation function* of f to ∞. Therefore the approximation function of f to $a \in \mathbf{C}$ is set to be $m(r, 1/(f - a))$.

Let $r \geqq 1$. Nevanlinna's *order function* (also called characteristic function) is defined as follows:

$$T(r, f) = N\big(r, (f)_\infty\big) + m(r, f). \tag{1.1.12}$$

For N meromorphic functions $f_j(z)$, $1 \leqq j \leqq N$, we have by (1.1.11)

$$T\left(r, \sum_{j=1}^{N} f_j\right) \leqq \sum_{j=1}^{N} T(r, f_j) + \log N,$$

$$T\left(r, \prod_{j=1}^{N} f_j\right) \leqq \sum_{j=1}^{N} T(r, f_j).$$

These properties and (1.1.11) are basic and will be used without specific mention.

Let f be a holomorphic function in a neighborhood of the closed disk $\bar{\Delta}(r) = \{|z| \leqq r\}$. Note that Jensen's formula (Lemma 1.1.5) holds for φ defined in a neighborhood of $\bar{\Delta}(r)$. Setting $\varphi(z) = \log|f(z)|$, we have that $\lambda_\nu \geqq 0$ and $\partial\bar{\partial}\varphi(z) \equiv 0$ where $f(z) \neq 0$. Therefore, we get

$$\log\big|f(0)\big| \leqq \frac{1}{2\pi}\int_{|z|=r} \log\big|f(z)\big| d\theta. \tag{1.1.13}$$

Let $w = \psi(z) = r^2(z+\zeta)/(\bar{\zeta}z + r^2)$ $(|\zeta| < r)$ be a linear fractional transformation of $\Delta(r)$, which is holomorphic in a neighborhood of $\bar{\Delta}(r)$. Applying (1.1.13) for $f \circ \psi(z)$, we have

$$\begin{aligned}
\log\big|f\circ\psi(0)\big| &\leqq \frac{1}{2\pi}\int_{|z|=r} \log\big|f\circ\psi(z)\big| d\theta = \frac{1}{2\pi}\int_{|z|=r} \log\big|f\circ\psi(z)\big| \frac{dz}{iz} \\
&= \frac{1}{2\pi}\int_{|w|=r} \big(\log\big|f(w)\big|\big) \frac{\psi^{-1\prime}(w)}{i\psi^{-1}(w)} dw \\
&= \frac{1}{2\pi}\int_0^{2\pi} \big(\log\big|f\big(re^{i\theta}\big)\big|\big) \frac{r^2 - |\zeta|^2}{|re^{i\theta} - \zeta|^2} d\theta.
\end{aligned}$$

Since $\psi(0) = \zeta$, we have

$$\log\big|f(\zeta)\big| \leqq \frac{1}{2\pi}\int_0^{2\pi} \big(\log\big|f\big(re^{i\theta}\big)\big|\big) \frac{r^2 - |\zeta|^2}{|re^{i\theta} - \zeta|^2} d\theta. \tag{1.1.14}$$

The right-hand side is called the Poisson integral.

If $u(z)$ is harmonic, i.e., $\partial\bar{\partial}u = 0$, in a neighborhood of $\bar{\Delta}(r)$, it follows from Lemma 1.1.5 with $s = 0$ that

$$u(0) = \frac{1}{2\pi}\int_{|z|=r} u(z) d\theta.$$

Therefore we have in the same way as above that

$$u(\zeta) = \frac{1}{2\pi}\int_0^{2\pi} u\big(re^{i\theta}\big)\frac{r^2-|\zeta|^2}{|re^{i\theta}-\zeta|^2}d\theta.$$

Since $\frac{r^2-|\zeta|^2}{|re^{i\theta}-\zeta|^2} = \Re\frac{re^{i\theta}+\zeta}{re^{i\theta}-\zeta}$,

$$u(\zeta) = \frac{1}{2\pi}\int_0^{2\pi} u\big(re^{i\theta}\big)\Re\frac{re^{i\theta}+\zeta}{re^{i\theta}-\zeta}d\theta.$$

Thus we see that

$$\Im f(\zeta) = \frac{1}{2\pi}\int_0^{2\pi} \Im f\big(re^{i\theta}\big)\Re\frac{re^{i\theta}+\zeta}{re^{i\theta}-\zeta}d\theta.$$

The function $\frac{1}{2\pi}\int_0^{2\pi}\Im f(re^{i\theta})\frac{re^{i\theta}+\zeta}{re^{i\theta}-\zeta}d\theta$ is holomorphic in $|\zeta|<r$, and its real part is $\Im f(\zeta)$. Therefore we get

$$f(\zeta) = \frac{i}{2\pi}\int_0^{2\pi}\frac{re^{i\theta}+\zeta}{re^{i\theta}-\zeta}\Im f\big(re^{i\theta}\big)d\theta + \Re f(0); \tag{1.1.15}$$

this is called the complex Poisson integral.

A holomorphic function on $\mathbf{C}$ is called an *entire function*.

Theorem 1.1.16 *Let $f(z)$ be an entire function. Then*

$$T(r,f) \leqq \max_{|z|=r}\log^+\big|f(z)\big| \leqq \frac{R+r}{R-r}T(R,f), \quad 1<r<R.$$

Proof Since $N(r,(f)_\infty)=0$, the first inequality is immediate.

To show the second inequality, we use (1.1.14). It follows that

$$\begin{aligned}\log\big|f(z)\big| &\leqq \frac{1}{2\pi}\int_{|w|=R}\big(\log\big|f(w)\big|\big)\frac{R^2-|z|^2}{|w-z|^2}d\theta\\ &\leqq \frac{1}{2\pi}\int_{|w|=R}\big(\log^+\big|f(w)\big|\big)\frac{R^2-r^2}{(R-r)^2}d\theta\\ &= \frac{R+r}{R-r}T(R,f).\end{aligned}$$

□

Theorem 1.1.17 (Nevanlinna's First Main Theorem) *Let $f(z)$ be a meromorphic function and $a\in\mathbf{C}$. Then*

$$T\left(r,\frac{1}{f-a}\right) = T(r,f)+O(1),$$

$$\big|O(1)\big| \leqq \log^+|a| + \log 2 + \left|\frac{1}{2\pi}\int_{|z|=1}\log\big|f(z)-a\big|d\theta\right|.$$

Proof We apply Jensen's formula (Lemma 1.1.5) to $\varphi(z) = \log|f(z) - a|$. Outside those points z such that $\varphi(z) = \pm\infty$, $\partial\bar{\partial}\varphi(z) \equiv 0$, and hence

$$\begin{aligned} &N\big(r, (f-a)_0\big) - N\big(r, (f-a)_\infty\big) \\ &\quad - \frac{1}{2\pi}\int_{|z|=r} \log\big|f(z)-a\big|d\theta + \frac{1}{2\pi}\int_{|z|=1} \log\big|f(z)-a\big|d\theta \\ &= 0. \end{aligned}$$

Noting that $(f-a)_\infty = (f)_\infty$, we get

$$\begin{aligned} &N\big(r, (f-a)_0\big) + \frac{1}{2\pi}\int_{|z|=r} \log^+ \frac{1}{|f(z)-a|}d\theta \\ &\quad = N\big(r, (f)_\infty\big) + \frac{1}{2\pi}\int_{|z|=r} \log^+\big|f(z)-a\big|d\theta \\ &\qquad - \frac{1}{2\pi}\int_{|z|=1} \log\big|f(z)-a\big|d\theta. \end{aligned}$$

Since $|\log^+|f(z)-a| - \log^+|f(z)|| \leqq \log^+|a| + \log 2$, the required formulae follow. □

It will be seen later that Nevanlinna's First Main Theorem is an open manifold version of Poincaré's duality theorem over a compact manifold (cf. Theorem 2.3.29 and the discussion after it).

Let $\hat{\mathbf{C}} = \mathbf{C} \cup \{\infty\}$ denote the Riemann sphere.

Theorem 1.1.18 (Nevanlinna's inequality) *Let $f(z)$ be a meromorphic function. Then there is a constant C such that*

$$N\big(r, (f-a)_0\big) \leqq T(r, f) + C, \quad {}^\forall a \in \hat{\mathbf{C}}.$$

Proof For $a = \infty$, this follows from the definition. Suppose that $a \neq \infty$. We show first that the integral

$$I_1(w) = \frac{1}{2\pi}\int_{|z|=1} \log\big|f(z)-w\big|d\theta$$

is bounded on $|w| \leqq 1$. We write $f(z) = \frac{h(z)}{g(z)}$ with entire functions $g(z), h(z)$. Then

$$I_1(w) = \frac{1}{2\pi}\int_{|z|=1} \log\frac{|h(z)-wg(z)|}{M}d\theta - \frac{1}{2\pi}\int_{|z|=1} \log\big|g(z)\big|d\theta + \log M,$$

where $M = \max\{|h(z) - wg(z)|;\, |z| = 1, |w| \leqq 1\}$. Thus it suffices to show the lower-boundedness of

$$I_2(w) = \frac{1}{2\pi}\int_{|z|=1} \log\big|h(z)-wg(z)\big|d\theta, \quad |w| \leqq 1$$

with the assumption, $|h(z) - wg(z)| \leqq 1$ for $|z| = 1$ and $|w| \leqq 1$.

For every $|w'| \leqq 1$ there exists a point $|\zeta| < \frac{1}{2}$ such that $|h(\zeta) - w'g(\zeta)| = 2\delta > 0$. Then there is a neighborhood $U(w')$ of w' such that $|h(\zeta) - wg(\zeta)| > \delta$ for $w \in U(w')$. It follows from (1.1.14) that for $w \in U(w')$

$$\begin{aligned}\log\delta < \log\left|h(\zeta) - wg(\zeta)\right| &\leqq \frac{1}{2\pi}\int_0^{2\pi}\left(\log\left|h\left(e^{i\theta}\right) - wg\left(e^{i\theta}\right)\right|\right)\frac{1-|\zeta|^2}{|e^{i\theta}-\zeta|^2}d\theta \\ &\leqq \frac{1}{2\pi}\int_0^{2\pi}\left(\log\left|h\left(e^{i\theta}\right) - wg\left(e^{i\theta}\right)\right|\right)\frac{1-|\zeta|^2}{(1+|\zeta|)^2}d\theta \\ &\leqq \frac{1}{2\pi}\int_0^{2\pi}\left(\log\left|h\left(e^{i\theta}\right) - wg\left(e^{i\theta}\right)\right|\right)\frac{1-|\zeta|}{1+|\zeta|}d\theta.\end{aligned}$$

Since $\frac{1-|\zeta|}{1+|\zeta|} = \frac{2}{1+|\zeta|} - 1 > \frac{1}{3}$ for $|\zeta| < \frac{1}{2}$, it follows that

$$\log\delta < \frac{1}{3}I_2(w), \quad w \in U\left(w'\right).$$

Taking a finite covering of $\bar{\Delta}(1)$ by such $U(w')$'s, we see that $\inf_{|w|\leqq 1} I_2(w) > -\infty$.

By Theorem 1.1.17 and the boundedness of $I_1(a)$ $(|a| \leqq 1)$ there is a constant $C_1 > 0$ such that

$$N\left(r, (f-a)_0\right) \leqq T(r,f) + C_1, \quad r \geqq 1,\ |a| \leqq 1.$$

For $|a| > 1$, we consider $1/f(z)$ and $1/a$. Since $(f-a)_0 = (1/f - 1/a)_0$ and $|1/a| < 1$, the fact shown above implies that there is a constant $C_2 > 0$ satisfying

$$N\left(r, (f-a)_0\right) = N\left(r, \left(\frac{1}{f} - \frac{1}{a}\right)_0\right) \leqq T\left(r, \frac{1}{f}\right) + C_2.$$

Theorem 1.1.17 with $a = 0$ implies the existence of a constant $C_3 > 0$ satisfying

$$T\left(r, \frac{1}{f}\right) \leqq T(r,f) + C_3.$$

Hence we may set $C = C_1 + C_2 + C_3$. □

The geometric meaning of the order function $T(r,f)$ is not clear by Nevanlinna's definition (1.1.12). We give a geometric understanding of $T(r,f)$ due to Shimizu [29] and Ahlfors [30]. Let w be the complex coordinate of the finite part $\mathbf{C}$ of the Riemann sphere $\hat{\mathbf{C}}$. We define a surface element on $\hat{\mathbf{C}}$ by

$$\Omega = dd^c\log\left(1+|w|^2\right) = \frac{1}{(1+|w|^2)^2}\frac{i}{2\pi}dw \wedge d\bar{w}.$$

This is called the *Fubini–Study metric form* on $\hat{\mathbf{C}}$ and

$$\int_{\hat{\mathbf{C}}} \Omega = 1.$$

For a meromorphic function $f(z)$ we define *Shimizu's order function* $T_f(r, \Omega)$ by

$$T_f(r, \Omega) = \int_1^r \frac{dt}{t} \int_{\Delta(t)} f^* \Omega .$$

We apply Lemma 1.1.5 for $\varphi(z) = \log \sqrt{1 + |f(z)|^2}$:

$$T_f(r, \Omega) = N\big(r, (f)_\infty\big) + \frac{1}{2\pi} \int_{|z|=r} \log \sqrt{1 + \big|f(z)\big|^2} d\theta$$
$$- \frac{1}{2\pi} \int_{|z|=1} \log \sqrt{1 + \big|f(z)\big|^2} d\theta .$$

It is immediate that

$$m(r, f) \leqq \frac{1}{2\pi} \int_{|z|=r} \log \sqrt{1 + \big|f(z)\big|^2} d\theta \leqq m(r, f) + \frac{1}{2} \log 2.$$

Therefore we have the following.

Theorem 1.1.19 (Shimizu–Ahlfors) *For a meromorphic function* $f(z)$,

$$T(r, f) = T_f(r, \Omega) + O(1) = \int_1^r \frac{dt}{t} \int_{\Delta(t)} f^* \Omega + O(1) \tag{1.1.20}$$
$$= \int_1^r \frac{dt}{t} \int_{\Delta(t)} \frac{|f'(z)|^2}{(1 + |f(z)|^2)^2} \frac{i}{2\pi} dz \wedge d\bar{z} + O(1)$$
$$= \int_1^r \frac{dt}{t} \int_{\Delta(t)} dd^c \log\big(1 + \big|f(z)\big|^2\big) + O(1).$$

The term $\int_{\Delta(t)} f^* \Omega$ is the surface area of the image $f(\Delta(t))$ with counting multiplicities with respect to the Fubini–Study metric form on the Riemann sphere. Therefore $T(r, f)$ is measuring its growth. We deduce from (1.1.20) and Fubini's theorem that

$$T(r, f) = \int_1^r \frac{dt}{t} \int_{w \in \hat{\mathbf{C}}} n\big(t, (f - w)_0\big) \Omega(w) \tag{1.1.21}$$
$$= \int_{w \in \hat{\mathbf{C}}} N\big(r, (f - w)_0\big) \Omega(w).$$

We prepare a simple lemma.

Lemma 1.1.22 *Let* $a(t)$, $t \in [1, \infty)$, *be a monotone increasing function, and set* $b(r) = \int_1^r \frac{a(t)}{t} dt$, $r \geqq 1$. *Then the following conditions are mutually equivalent*:

(i) $b(r) = O(\log r)$; *i.e.*, $\overline{\lim}_{r \to \infty} \frac{b(r)}{\log r} < \infty$.

(ii) $\varliminf_{r\to\infty}\frac{b(r)}{\log r}<\infty$.
(iii) $a(t)=O(1)$.

Proof The implications from (i) to (ii), and from (iii) to (i) are clear. Suppose (ii) to hold. Then there are a monotone increasing sequence $1<r_\nu\nearrow\infty$ $(\nu\to\infty)$ and a constant $C>0$ such that

$$b(r_\nu)\leqq C\log r_\nu,\quad \nu\to\infty.$$

For every $1\leqq r<r_\nu$ we have

$$\begin{aligned}a(r)&\leqq\frac{1}{\log r_\nu-\log r}\int_r^{r_\nu}\frac{a(t)}{t}dt\\&\leqq\frac{1}{\log r_\nu-\log r}\int_1^{r_\nu}\frac{a(t)}{t}dt\\&=\frac{b(r_\nu)}{\log r_\nu-\log r}\leqq C\left(1-\frac{\log r}{\log r_\nu}\right)^{-1}.\end{aligned}$$

Letting $r_\nu\to\infty$, we have that $a(r)\leqq C$. □

From this lemma, the First Main Theorem 1.1.17 and Theorem 1.1.19 we obtain the following.

Corollary 1.1.23 *The equivalence of three conditions of Lemma 1.1.22 holds for $T(r,f)$ and $N(r,(f)_\infty)$.*

When $f(z)$ is a rational function $P(z)/Q(z)$ with co-prime polynomials P and Q, we may compute directly $N(r,(f)_\infty)$ and $m(r,f)$;

$$T\left(r,\frac{P}{Q}\right)=\max\{\deg P,\deg Q\}\log r+O(1).\tag{1.1.24}$$

For $f(z)=e^{z^p}$ $(p\in\mathbf{N})$ a direct computation yields that

$$\begin{aligned}T(r,f)=m(r,f)&=\int_{|z|=r}\log^+\left|e^{z^p}\right|\frac{d\theta}{2\pi}=\int_{|z|=r}\max\left\{0,\Re z^p\right\}\frac{d\theta}{2\pi}\\&=\int_0^{2\pi}r^p\cos^+p\theta\frac{d\theta}{2\pi}=\frac{r^p}{\pi}.\end{aligned}\tag{1.1.25}$$

Theorem 1.1.26 *A meromorphic function $f(z)$ is rational if and only if $T(r,f)=O(\log r)$, or equivalently,*

$$\varliminf_{r\to\infty}\frac{T(r,f)}{\log r}<\infty.$$

Proof The necessity follows from (1.1.24). We show the sufficiency. Assume that there are a number $r_0 > 1$ and a positive constant C such that

$$T(r, f) \leqq C \log r, \quad r \geqq r_0.$$

Then $N(r, (f)_\infty) \leqq C \log r$ $(r \geqq r_0)$, and hence $f(z)$ has at most finitely many poles. Take a polynomial $Q(z)$ so that $g(z) = f(z)Q(z)$ is entire. There are also a constant $C_1 > 0$ and a number $r_1 \geqq r_0$ such that $T(r, g) \leqq C_1 \log r$ $(r \geqq r_1)$. Theorem 1.1.16 implies that for a sufficiently large r

$$\begin{aligned} \max_{|z|=r} \log |g(z)| &\leqq \frac{2r + r}{2r - r} T(2r, g) \\ &\leqq 3C_1(\log r + \log 2) < 4C_1 \log r. \end{aligned}$$

Therefore $\max_{|z|=r} |g(z)|^2 < r^{8C_1}$. Letting $g(z) = \sum_{j=0}^{\infty} a_j z^j$ be a power series expansion, we have

$$\frac{1}{2\pi} \int_0^{2\pi} \left|g\left(re^{i\theta}\right)\right|^2 d\theta = \sum_{j=0}^{\infty} |a_j|^2 r^{2j} \leqq r^{8C_1}.$$

Thus $a_j = 0$ for all $j > 4C_1$, and hence $g(z)$ is a polynomial. □

The following is clear.

Corollary 1.1.27 *It is a necessary and sufficient condition for $f(z)$ to be transcendental that*

$$\overline{\lim_{r\to\infty}} \frac{T(r, f)}{\log r} = \infty.$$

An *order* ρ_f of a meromorphic function $f(z)$ to measure the transcendency is defined as

$$\rho_f = \overline{\lim_{r\to\infty}} \frac{\log T(r, f)}{\log r} \leqq \infty. \tag{1.1.28}$$

If $\rho_f < \infty$ (resp. $=\infty$), $f(z)$ is said to have a finite (resp. infinite) order. For example, if $f(z) = e^{z^p}$ with $p \in \mathbf{N}$, then $\rho_f = p$.

Historical Notes J.L.W.V. Jensen [1899] proved his formula (Lemma 1.1.5) for meromorphic functions. As the title of the paper had suggested, R. Nevanlinna [25] used the formula systematically in establishing the value distribution theory for meromorphic functions.

Theorem 1.1.19 is sometimes called the Ahlfors–Shimizu Theorem (e.g., Hayman [64]). However, historical fact shows that it should be rightly called the Shimizu–Ahlfors Theorem or Shimizu's Theorem. This theorem was first found and

proved by Shimizu [29]. According to the comment given in Ahlfors' Collected Papers, Vol. 1, p. 56, Ahlfors met R. Nevanlinna in Paris in 1929, and told him this result. Nevanlinna became very much interested in it and encouraged Ahlfors. But in a short time Ahlfors was informed by Nevanlinna that the theorem had been already obtained by Shimizu. Since the result was quite interesting by itself, Ahlfors gave a talk on this result at the 7th Scandinavian Congress of Mathematicians, August 1929, the proceedings of which were published in 1930. In Hayman [64], the publication year of Ahlfors [30] is stated as 1929, but this should be 1930 as in Ahlfors' Collected Papers. As mentioned in the above comment of Ahlfors' Collected Papers, Shimizu gave a talk in 1927 on his result.

In the same comment of Ahlfors he remarks that H. Cartan's work (Cartan [29a]) on order functions done in the similar time should be paid more attentions. Looking into the papers in Cartan's Œuvres around 1929 on the value distribution theory, we find that he seems to have been trying to find a natural definition of order function from a potential theoretic viewpoint, rather than geometric one.

Writing a meromorphic function $f(z) = f_2(z)/f_1(z)$ with two entire functions $f_1(z)$, $f_2(z)$ without common zero, Cartan ([29a]) set

$$U(z) = \max\{|f_1(z)|, |f_2(z)|\}$$

and then defined the order function $T(r)$ by

$$T(r) = \frac{1}{2\pi}\int_{|z|=r} \log U(z)d\theta - \log|f_1(0)|.$$

Here, for the sake of simplicity, $f_1(0) \neq 0$ is assumed. In a later paper of the same year, Cartan dealt with a system of entire functions $\{f_j(z), 1 \leqq j \leqq p\}$ without common zero, and defined $U(z)$ in the same way (cf. Theorem 2.5.12) with the second term "$-\log U(0)$", which is more natural from the viewpoint of Jensen's formula. In his papers a formula in terms of metric forms as in Shimizu [29] and Ahlfors [30] (1.1.20) did not appear.

Reading the comments of Ahlfors' Collected Papers, we may feel the elevation of the time when R. Nevanlinna started the theory that bears his name. (In the same sense the introduction of H. Weyl–J. Weyl [38] is interesting, too.) Ahlfors advanced the research of Nevanlinna theory from geometric viewpoint, and completed the covering theory (Ahlfors [35]). Because of this work Ahlfors was awarded the first Fields medal at the International Congress of Mathematicians in Oslo in 1936 with J. Douglas.

1.2 The Second Main Theorem

The Second Main Theorem is in a sense an inequality opposite to Nevanlinna's inequality (Theorem 1.1.18); that is, it claims an estimate of order function $T(r, f)$ by counting function $N(r, (f-a)_0)$ from above. It is impossible by one counting function, and so requires several counting functions.

Here we introduce a notation. For non-negatively valued functions $\varphi(r)$, $\psi(r)$ $(r \geqq r_0 \geqq 0)$ we write

$$\varphi(r) \leqq \psi(r)\|_E$$

if there is a Borel subset $E \subset [r_0, \infty)$ of finite Lebesgue measure and the inequality $\varphi(r) \leqq \psi(r)$ holds for $r \notin E$. Therefore,

$$\varphi(r) \leqq O\big(\psi(r)\big)\big\|_E$$

means that there is a positive constant C such that

$$\varphi(r) \leqq C\psi(r)\|_E.$$

Lemma 1.2.1 (Borel's Lemma) *Let $\phi(r) \geqq 0$ $(r \geqq r_0 \geqq 0)$ be a monotone increasing function. For every $\delta > 0$*

$$\frac{d}{dr}\phi(r) \leqq \phi(r)^{1+\delta}\|_{E(\delta)},$$

where $E(\delta)$ depends on the choice of δ.

Proof Since $\phi(r)$ is monotone increasing, the derivative $\frac{d}{dr}\phi(r)$ exists almost everywhere. We may assume that $\phi(r) \not\equiv 0$. Take $r_1 \geqq r_0$ so that $\phi(r_1) > 0$. Set $E(\delta) = \{r \geqq r_1;\ \frac{d}{dr}\phi(r) > \phi(r)^{1+\delta}\}$. On $E(\delta)$ we have

$$\frac{d\phi(r)}{\phi(r)^{1+\delta}} > dr.$$

Therefore,

$$\int_{E(\delta)} dr \leqq \int_{E(\delta)} \frac{d\phi(r)}{\phi(r)^{1+\delta}} \leqq \int_{r_1}^{\infty} \frac{d\phi(r)}{\phi(r)^{1+\delta}} \leqq \frac{1}{\delta\phi(r_1)^{\delta}}. \qquad \square$$

Lemma 1.2.2 (Nevanlinna's Lemma on Logarithmic Derivatives) *Let $f(z)$ be a meromorphic function. Then for $\delta > 0$*

$$m\left(r, \frac{f'}{f}\right) \leqq \left(1 + \frac{(1+\delta)^2}{2}\right)\log^+ T(r, f) + \frac{\delta}{2}\log r + O(1)\|_{E(\delta)}.$$

Proof For $w \in \mathbf{C}$, we define an surface element,

$$\Phi = \frac{1}{(1 + (\log|w|)^2)|w|^2}\frac{i}{4\pi^2}dw \wedge d\bar{w}.$$

This is a $(1, 1)$-form on $\hat{\mathbf{C}}$ with singularities at $w = 0, \infty$. By computation

$$\int_{\hat{\mathbf{C}}} \Phi = \int_{\mathbf{C}} \frac{1}{(1 + (\log r)^2)r^2}\frac{1}{2\pi^2} r dr d\theta = 1.$$

On the other hand, setting $\mu(r) = \int_1^r \frac{dt}{t} \int_{\Delta(t)} f^*\Phi$, we have

$$\begin{aligned}\mu(r) &= \int_1^r \frac{dt}{t} \int_{\Delta(t)} \frac{|f'|^2}{(1+(\log|f|)^2)|f|^2} \frac{i}{4\pi^2} dz \wedge d\bar{z} \\ &= \int_{w\in\mathbf{C}} \int_1^r \frac{dt}{t} n\big(t, (f-w)_0\big)\Phi(w) \\ &= \int_{w\in\mathbf{C}} N\big(r, (f-w)_0\big)\Phi(w).\end{aligned}$$

By Theorem 1.1.18 there is a constant $C > 0$ such that

$$\mu(r) \leqq T(r, f) + C.$$

By making use of this, Lemma 1.2.1 and the concavity of the logarithm function, we carry the following computations:

$$\begin{aligned} m\left(r, \frac{f'}{f}\right) &= \frac{1}{4\pi} \int_{|z|=r} \log^+\left(\frac{|f'|^2}{(1+(\log|f|)^2)|f|^2}\big(1+\big(\log|f|\big)^2\big)\right) d\theta \\ &\leqq \frac{1}{4\pi} \int_{|z|=r} \log^+ \frac{|f'|^2}{(1+(\log|f|)^2)|f|^2} d\theta \\ &\quad + \frac{1}{4\pi} \int_{|z|=r} \log^+\left(1+\left(\log^+|f| + \log^+\frac{1}{|f|}\right)^2\right) d\theta \\ &\leqq \frac{1}{4\pi} \int_{|z|=r} \log\left(1+\frac{|f'|^2}{(1+(\log|f|)^2)|f|^2}\right) d\theta \\ &\quad + \frac{1}{2\pi} \int_{|z|=r} \log^+\left(\log^+|f| + \log^+\frac{1}{|f|}\right) d\theta + \frac{1}{2}\log 2 \\ &\leqq \frac{1}{2}\log\left(1+\frac{1}{2\pi}\int_{|z|=r} \frac{|f'|^2}{(1+(\log|f|)^2)|f|^2} d\theta\right) \\ &\quad + \frac{1}{2\pi} \int_{|z|=r} \log\left(1+\log^+|f| + \log^+\frac{1}{|f|}\right) d\theta + \frac{1}{2}\log 2 \\ &\leqq \frac{1}{2}\log\left(1+\frac{1}{r}\frac{d}{dr}\int_{\Delta(r)} \frac{|f'|^2}{(1+(\log|f|)^2)|f|^2} \frac{1}{2\pi} r\,dr\,d\theta\right) \\ &\quad + \log\left(1+m(r,f)+m\left(r,\frac{1}{f}\right)\right) + \frac{1}{2}\log 2 \\ &\leqq \frac{1}{2}\log\left(1+\frac{2\pi}{r}\frac{d}{dr}\int_{\Delta(r)} f^*\Phi\right) + \log^+ T(r,f) + O(1) \\ &\leqq \frac{1}{2}\log\left(1+\frac{2\pi}{r}\left(\int_{\Delta(r)} f^*\Phi\right)^{1+\delta}\right) \end{aligned}$$

(continued)

$$
\begin{aligned}
&+\log^+ T(r,f)+O(1)\|_{E_1(\delta)}\\
&=\frac{1}{2}\log\left(1+2\pi r^{\delta}\left(\frac{d}{dr}\int_1^r\frac{dt}{t}\int_{\Delta(t)}f^*\Phi\right)^{1+\delta}\right)\\
&\quad+\log^+ T(r,f)+O(1)\|_{E_1(\delta)}\\
&\leqq\frac{1}{2}\log\left(1+2\pi r^{\delta}\mu(r)^{(1+\delta)^2}\right)+\log^+ T(r,f)+O(1)\|_{E_2(\delta)}\\
&\leqq\left(1+\frac{(1+\delta)^2}{2}\right)\log^+ T(r,f)+\frac{\delta}{2}\log^+ r+O(1)\|_{E_2(\delta)}. \qquad \square
\end{aligned}
$$

Remark 1.2.3 If $\rho_f<\infty$, then in fact

$$m\left(r,\frac{f'}{f}\right)=O(\log r).$$

The proof of this gets a bit longer than the above (cf. R. Nevanlinna [29]; Hayman [64]). For example, if $f(z)=e^{P(z)}$ with a polynomial $P(z)$, then a direct computation yields

$$m\left(r,\frac{f'}{f}\right)=m\big(r,P'\big)=O(\log r).$$

A term such as the right-hand side in the estimate of $m(r,f'/f)$ of Lemma 1.2.2 is denoted by the notation $S(r,f)$ in the sense of a small term compared with $T(r,f)$; i.e., for every $\delta>0$

$$S(r,f)=O\big(\log T(r,f)\big)+\delta\log r\|_{E(\delta)}. \tag{1.2.4}$$

Here the constant in the estimate $O(*)$ is independent of δ. If $f(z)$ is a rational function, $m(r,f'/f)=O(1)$; in this case we understand that $S(r,f)=O(1)$.

Theorem 1.2.5 (Nevanlinna's Second Main Theorem) *Let $f(z)$ be a meromorphic function, and let $a_1,\dots,a_q\in\hat{\mathbf{C}}$ be distinct points. Then*

$$(q-2)T(r,f)\leqq\sum_{i=1}^{q}N_1\big(r,(f-a_i)_0\big)+S(r,f).$$

Proof By the First Main Theorem 1.1.17 we may assume that $a_1,\dots,a_{q-1}\in\mathbf{C}$, $a_q=\infty$. For $w,a\in\hat{\mathbf{C}}$ we set

$$\|w,a\|=\begin{cases}\dfrac{|w-a|}{\sqrt{1+|w|^2}\sqrt{1+|a|^2}}, & a\in\mathbf{C},\\[2ex] \dfrac{1}{\sqrt{1+|w|^2}}, & a=\infty.\end{cases}$$

This defines a distance function on the Riemann sphere. We put

$$\Psi(w) = \sum_{1 \leqq i_1 < \cdots < i_{q-2} \leqq q} \prod_{j=1}^{q-2} \|w, a_{i_j}\|.$$

Since $\Psi(w)$ is positively valued and continuous on the Riemann sphere, there is a constant $C > 0$ such that

$$C^{-1} \leqq \Psi(w) \leqq C. \tag{1.2.6}$$

We transform

$$\begin{aligned}
&\Psi\big(f(z)\big) \\
&= \prod_{i=1}^{q} \|f(z), a_i\| \times \sum_{1 \leqq i_1 < i_2 \leqq q} \frac{1}{\|f(z), a_{i_1}\| \cdot \|f(z), a_{i_2}\|} \\
&= \frac{1}{(\sqrt{1+|f(z)|^2})^{q-2}} \times \frac{1}{|f'(z)|} \prod_{i=1}^{q-1} \frac{|f(z)-a_i|}{\sqrt{1+|a_i|^2}} \\
&\quad \times \left\{ \sum_{1 \leqq i_1 < i_2 \leqq q-1} \frac{\sqrt{(1+|a_{i_1}|^2)(1+|a_{i_2}|^2)}}{|a_{i_1}-a_{i_2}|} \left| \frac{(f(z)-a_{i_1})'}{f(z)-a_{i_1}} - \frac{(f(z)-a_{i_2})'}{f(z)-a_{i_2}} \right| \right. \\
&\qquad \left. + \sum_{i=1}^{q-1} \sqrt{1+|a_i|^2} \left| \frac{(f(z)-a_i)'}{f(z)-a_i} \right| \right\}.
\end{aligned}$$

This with (1.2.6) implies that

$$\begin{aligned}
&\left(\sqrt{1+|f(z)|^2}\right)^{q-2} \\
&\leqq \frac{C}{|f'(z)|} \prod_{i=1}^{q-1} \frac{|f(z)-a_i|}{\sqrt{1+|a_i|^2}} \\
&\quad \times \left\{ \sum_{1 \leqq i_1 < i_2 \leqq q-1} \frac{\sqrt{(1+|a_{i_1}|^2)(1+|a_{i_2}|^2)}}{|a_{i_1}-a_{i_2}|} \left| \frac{(f(z)-a_{i_1})'}{f(z)-a_{i_1}} - \frac{(f(z)-a_{i_2})'}{f(z)-a_{i_2}} \right| \right. \\
&\qquad \left. + \sum_{i=1}^{q-1} \sqrt{1+|a_i|^2} \left| \frac{(f(z)-a_i)'}{f(z)-a_i} \right| \right\}.
\end{aligned} \tag{1.2.7}$$

We take the logarithms of the both sides of (1.2.7) and compute the circle means. It follows from Jensen's formula (Lemma 1.1.5) that

$$\int_{|z|=r}\log\left(\sqrt{1+\left|f(z)\right|^2}\right)^{q-2}\frac{d\theta}{2\pi}-\int_{|z|=1}\log\left(\sqrt{1+\left|f(z)\right|^2}\right)^{q-2}\frac{d\theta}{2\pi}$$
$$=(q-2)\int_1^r\frac{dt}{t}\int_{\Delta(t)}dd^c\log\bigl(1+\left|f(z)\right|^2\bigr)-(q-2)N\bigl(r,(f)_\infty\bigr). \tag{1.2.8}$$

Next we calculate the expansion of $\frac{C}{|f'(z)|}\prod_{i=1}^{q-1}\frac{|f(z)-a_i|}{\sqrt{1+|a_i|^2}}$ in a neighborhood of b such that $f(b)=\infty(=a_q)$, a_i $(1\leqq i\leqq q-1)$. If b is a pole of f of order m, there is a non-vanishing holomorphic function $g(z)$ in a neighborhood of b such that

$$\frac{C}{|f'(z)|}\prod_{i=1}^{q-1}\frac{|f(z)-a_i|}{\sqrt{1+|a_i|^2}}=|z-b|^{1-(q-2)m}\cdot\left|g(z)\right|.$$

If $f(b)=a_i$ $(1\leqq i\leqq q-1)$, it follows that

$$\frac{C}{|f'(z)|}\prod_{i=1}^{q-1}\frac{|f(z)-a_i|}{\sqrt{1+|a_i|^2}}=|z-b|\cdot\left|g(z)\right|\quad\bigl(g(b)\neq 0\bigr).$$

Since there is a possibility of the existence of zeros of $f'(z)$ other than these,

$$\int_{|z|=r}\log\left|\frac{C}{|f'(z)|}\prod_{i=1}^{q-1}\frac{|f(z)-a_i|}{\sqrt{1+|a_i|^2}}\right|\frac{d\theta}{2\pi}$$
$$-\int_{|z|=1}\log\left|\frac{C}{|f'(z)|}\prod_{i=1}^{q-1}\frac{|f(z)-a_i|}{\sqrt{1+|a_i|^2}}\right|\frac{d\theta}{2\pi}$$
$$\leqq\sum_{i=1}^{q}N_1\bigl(r,(f-a_i)_0\bigr)-(q-2)N\bigl(r,(f)_\infty\bigr). \tag{1.2.9}$$

The remaining term of the right-hand side of (1.2.7) satisfies by (1.1.11)

(1.2.10)
$$\int_{|z|=r}\log\left|\sum_{1\leqq i_1<i_2\leqq q-1}\frac{\sqrt{(1+|a_{i_1}|^2)(1+|a_{i_2}|^2)}}{|a_{i_1}-a_{i_2}|}\left|\frac{(f(z)-a_{i_1})'}{f(z)-a_{i_1}}-\frac{(f(z)-a_{i_2})'}{f(z)-a_{i_2}}\right|\right.$$
$$\left.+\sum_{i=1}^{q-1}\sqrt{1+|a_i|^2}\left|\frac{(f(z)-a_i)'}{f(z)-a_i}\right|\right|\frac{d\theta}{2\pi}$$
$$=O\left(m\left(r,\frac{f'}{f}\right)+\sum_{i=1}^{q-1}m\left(r,\frac{(f-a_i)'}{f-a_i}\right)\right)+O(1).$$

It follows from (1.2.7)–(1.2.10), the Shimizu–Ahlfors Theorem 1.1.19 and Lemma 1.2.2 that

$$(q-2)T(r,f) \leqq \sum_{i=1}^{q} N_1\big(r,(f-a_i)_0\big) + S(r,f). \qquad \square$$

The *defects* $\delta(f,a)$ and $\delta_k(f,a)$ $(k \geqq 1)$ of a meromorphic function $f(z)$ for $a \in \hat{\mathbf{C}}$ are defined by

$$\delta(f,a) = 1 - \varlimsup_{r\to\infty} \frac{N(r,(f-a)_0)}{T(r,f)},$$
$$\delta_k(f,a) = 1 - \varlimsup_{r\to\infty} \frac{N_k(r,(f-a)_0)}{T(r,f)}.$$

These satisfy

$$0 \leqq \delta(f,a) \leqq \delta_k(f,a) \leqq \delta_1(f,a) \leqq 1.$$

Theorem 1.2.11 (Defect Relation) *For a meromorphic function $f(z)$, the number of points a with $\delta(f,a) > 0$ is at most countable and*

$$\sum_{a\in\hat{\mathbf{C}}} \delta(f,a) \leqq \sum_{a\in\hat{\mathbf{C}}} \delta_1(f,a) \leqq 2.$$

This immediately follows from Theorem 1.2.5. The next theorem follows immediately, too.

Corollary 1.2.12 (The Little Picard Theorem) *The number of points $a \in \hat{\mathbf{C}}$ omitted (i.e., not to be taken as a value) by a meromorphic function is at most two.*

Assume that for $a \in \hat{\mathbf{C}}$, every multiplicity of the roots of $f(z) = a$ is not less than ν_a. Then

$$\begin{aligned}\delta_1(f,a) &= 1 - \varlimsup_{r\to\infty} \frac{N_1(r,(f-a)_0)}{T(r,f)} \\ &\geqq 1 - \varlimsup_{r\to\infty} \frac{1}{\nu_a} \frac{N(r,(f-a)_0)}{T(r,f)} \\ &\geqq 1 - \frac{1}{\nu_a}.\end{aligned}$$

If $f(z)$ omits a, we put $\nu_a = \infty$, and then the above holds. Hence we have the following.

Theorem 1.2.13 (Ramification Theorem) *Let $f(z)$ be a non-constant meromorphic function. Then the number of points $a \in \hat{\mathbf{C}}$ with $\nu_a \geqq 2$ is at most four, and*

$$\sum_{a\in\hat{\mathbf{C}}} \left(1 - \frac{1}{\nu_a}\right) \leqq 2.$$

If $\nu_a \geqq 2$, then $1 - 1/\nu_a \geqq 1/2$ so that there are at most four such points. The estimate of the above ramification theorem makes sense when the number of points with $\nu_a \geqq 2$ is three or four.In the case of three points, the possible combinations of ν_a are $(2, 2, 2)$, $(2, 2, 3)$, $(2, 3, 3)$, and $(3, 3, 3)$; in the case of four points, $(2, 2, 2, 2)$ is the only one. R. Nevanlinna [29] gave examples to show that all these cases exist.

Nevanlinna's inverse problem to ask the inverse of the defect relation was solved by D. Drasin in 1977.

Theorem 1.2.14 (Drasin [77]) *Let $0 < \delta_j \leqq 1$ and let $a_j \in \hat{\mathbf{C}}$ $(j = 1, 2, \ldots)$ be arbitrarily given so that $\sum \delta_j \leqq 2$. Then there exists a meromorphic function $f(z)$ such that $\delta(f, a_j) = \delta_j$ for all $j \geqq 1$ (or $\delta_1(f, a_j) = \delta_j$ for all $j \geqq 1$) and $\delta(f, a) = \delta_1(f, a) = 0$ for all $a \notin \{a_j\}_{j \geqq 1}$.*

A point a such that $\delta(f, a) > 0$ is called Nevanlinna's exceptional value, and it is a problem to measure the number of them. As for this problem there is a deep result. For a meromorphic function $f(z)$ the lower order λ_f is defined by

$$\lambda_f = \varliminf_{r \to \infty} \frac{\log T(r, f)}{\log r} \leqq \infty.$$

Theorem 1.2.15 (i) (Hayman [64]; Weitsman [72]) *Let $f(z)$ be a meromorphic function with $\lambda_f < \infty$. If $\alpha \geqq 1/3$, then $\sum \delta(f, a)^\alpha < \infty$.*

(ii) (Hayman [64]) *If $0 \leqq \alpha < 1/3$, there is a meromorphic function $f(z)$ with $\lambda_f < \infty$ such that $\sum \delta(f, a)^\alpha = \infty$.*

1.3 Examples of Functions of Finite Order

Many naturally occurring entire or meromorphic functions are of finite order. Here we present a number of examples.

(a) **Rational functions**. Every rational function is a function of finite order $\rho = 0$ (Theorem 1.1.26).

(b) **Exponential function**. For $f(z) = e^{z^p}$ $(p \in \mathbf{N})$ one has $T(r, f) = \frac{r^p}{\pi}$ by (1.1.25). Therefore, $\rho_f = p$.

(c) **Derived functions**. Let f be an entire function. For $0 < r < R$ and $k \in \mathbf{N}$ the Cauchy formula yields

$$\sup_{|z| \leqq r} \left| f^{(k)}(z) \right| \leqq \frac{k! R}{(R - r)^{k+1}} \sup_{|z| \leqq R} \left| f(z) \right|.$$

Using Theorem 1.1.16 this implies that if an entire function f is of finite order, so are all of its derived functions.

This remains true for a meromorphic function f of finite order. By Remark 1.2.3 we have

$$m\big(r, f'/f\big) = O(\log r),$$

which implies

$$m\big(r, f'\big) \leqq m\big(r, f \cdot f'/f\big) \leqq m(r, f) + O(\log r).$$

Note that $N(r, f') \leq 2N(r, f)$. Thus we have

$$T\big(r, f'\big) \leq 2T(r, f) + O(\log r),$$

and hence $\rho_{f'} \leqq \rho_f$.

(d) **Theta functions**. Let $\tau \in \mathbf{H}^+ = \{z \in \mathbf{C}; \Im z > 0\}$. Then $\mathbf{Z}[\tau] := \mathbf{Z} + \tau\mathbf{Z}$ is a lattice in $\mathbf{C}$ and a "theta function" is an entire function θ such that there exists $a, b \in \mathbf{C}$ satisfying $\theta(z+1) = \theta(z)$ and $\theta(z+\tau) = e^{az+b}\theta(z)$. Then

$$\theta(z + n + \tau m) = e^{maz + \frac{m(m-1)}{2}a\tau + mb}\theta(z)$$

for all $m, n \in \mathbf{Z}$.

They correspond to holomorphic sections of a certain line bundle L over the quotient curve $E = \mathbf{C}/\mathbf{Z}[\tau]$. More precisely, we can define an equivalence relation $\sim$ on $\mathbf{C} \times \mathbf{C}$ via setting

$$(z, t) \sim \big(z + n + \tau m, e^{maz + \frac{m(m-1)}{2}a\tau + mb}t\big)$$

for all $n, m \in \mathbf{Z}$. Let $L = (\mathbf{C} \times \mathbf{C})/\sim \to \mathbf{C}/\mathbf{Z}[\tau] = E$, where the projection map is induced by $(z, t) \mapsto z$. Then L is a line bundle over E whose sections correspond to the theta functions for the parameters (a, b).

Let F denote a fundamental region, i.e., a relatively compact subset of $\mathbf{C}$ such that for every $z \in \mathbf{C}$ there exist uniquely integers $m, n \in \mathbf{Z}$ such that $z - n - \tau m \in F$. Let $C_1 = \max_{z \in F} |z|$. Then $z - n - \tau m \in F$ for some $m, n \in \mathbf{Z}$ with

$$m < \frac{|z| + C_1}{|\tau|}.$$

Then for any theta function θ we obtain

$$\log\big|\theta(z)\big| \leqq MAC_1 + \frac{M(M-1)}{2}A\tau + MB\log\|\theta\|_F,$$

where $A = |a|$, $B = |b|$, $M = \frac{|z|+C_1}{|\tau|}$ and $\|\theta\|_F$ is the supremum norm over F. It follows that $\log|\theta(z)| = O(|z|^2)$ and consequently $T_\theta(r) = O(r^2)$ and $\rho_\theta \leqq 2$.

Assume that $\theta(z)$ is non-trivial, in the sense that $\theta \not\equiv 0$ and it has zeros. Then, $N(r, (\theta)_0) \sim r^2$, that is, there is a positive constant C with

$$\frac{r^2}{C} \leqq N\big(r, (\theta)_0\big) \leqq Cr^2,$$

$$r \gg 1 \quad \text{(i.e., all sufficiently large } r\text{)}.$$

Since $T(r, \theta) > N(r, (\theta)_0) + O(1)$, it follows that $\rho_\theta = 2$.

(e) **Elliptic functions**. Let $\tau \in \mathbf{H}^+$. An elliptic function is a meromorphic function on $\mathbf{C}$ invariant under translations by elements of $\mathbf{Z}[\tau]$. In other words, an elliptic function is a pull-back of a meromorphic function on the quotient elliptic curve $E = \mathbf{C}/\mathbf{Z}[\tau]$. For any $w \in \hat{\mathbf{C}}$ a meromorphic function on E assumes the value w only finitely many times, since E is compact. Thus there is a finite set S_w such that

$$f^{-1}(w) = \big\{s + \omega : s \in S_w, \omega \in \mathbf{Z}[\tau]\big\}.$$

There is a positive constant C independent of w such that

$$\frac{r^2}{C} \leqq N_f\big(r, \{w\}\big) \leqq Cr^2, \quad r \gg 1.$$

It follows from (1.1.21) that $T_f(r) \sim r^2$ and $\rho_f = 2$.

(f) **Gamma function**. For $\Re z > 1$ the Gamma function is defined via

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt.$$

It extends to a meromorphic function on the whole complex plane via the functional equation

$$\Gamma(z+1) = z\Gamma(z). \tag{1.3.1}$$

There are simple poles at 0 and at every negative integer. Thus, we have

$$\begin{aligned} r - 1 = \int_1^r \frac{t}{t} dt < \int_1^r \frac{[t]+1}{t} dt = N\big(r, (\Gamma)_\infty\big) \\ \leqq \int_1^r (t+1)\frac{dt}{t} = r + \log r - 1. \end{aligned} \tag{1.3.2}$$

We are going to compute $m(r, \Gamma)$ at $r = n + \frac{1}{2}$, $n = 1, 2, \ldots$. Observe that

$$\begin{aligned} |\Gamma(z)| &= \left| \int_0^\infty t^{z-1} e^{-t} dt \right| \leqq \int_0^\infty \left| t^{z-1} e^{-t} \right| dt \\ &= \int_0^\infty t^{\Re z - 1} e^{-t} dt = \Gamma(\Re z), \end{aligned}$$

whenever $\Re z > 1$. From the convexity of the real Gamma function we obtain $\max\{\Gamma(z) : 1 \leqq z \leqq 2\} \leqq 1 = \max\{\Gamma(1), \Gamma(2)\}$. Then

$$\big|\Gamma(z)\big| \leqq 1 \quad \text{for all } z \text{ with } 1 \leqq \Re z \leqq 2. \tag{1.3.3}$$

Now we can employ (1.3.1) in order to deduce that

$$\big|\Gamma(z)\big| = \big|(z-1)(z-2)\cdots(z-n)\Gamma(z-n)\big| \leqq |z-1|^n$$

for all $z \in \mathbf{C}$ and $n \in \mathbf{N}$ with $n+1 \leqq \Re z \leqq n+2$. It follows that

$$\left|\Gamma(z)\right| \leqq |z-1|^{\Re z - 1}, \quad \Re z \geqq 2. \tag{1.3.4}$$

Next let us consider the case where $\Re z \leqq 2$ and $|\Im z| \geqq 1$. Then there exists a natural number $n \in \mathbf{Z}^+$ such that $1 \leqq \Re z + n \leqq 2$. Using (1.3.1), we deduce

$$\Gamma(z+n) = (z+n-1)\cdots z\Gamma(z).$$

Now $|\Gamma(z+n)| \leqq 1$ and

$$|z+n-k| \geqq |\Im z + n - k| \geqq |\Im z|, \quad {}^\forall k \in \mathbf{N}.$$

This implies

$$\left|\Gamma(z)\right| \leqq 1, \quad \Re z \leqq 2 \text{ and } |\Im z| \geqq 1. \tag{1.3.5}$$

For a fourth estimate we consider the case where

$$-(n+2/3) \leqq \Re z \leqq -(n+1/3)$$

for some natural number $n \in \mathbf{N}$. Let $w = z+n+2$. Then $1+1/3 \leqq \Re w \leqq 2-1/3$ and

$$\Gamma(w) = (w-1)(w-2)\cdots\big(w-(n+2)\big)\Gamma(z).$$

A small calculation shows that

$$\left|(w-1)(w-2)(w-3)(w-4)(w-5)\right| \geqq \frac{280}{243} > 1$$

for $1+1/3 \leqq \Re w \leqq 2-1/3$. It follows that $|\Gamma(z)| \leqq 1$ for

$$-(n+2/3) \leqq \Re z \leqq -(n+1/3)$$

with $n \in \mathbf{N}$ and $n \geqq 3$. Using

$$\sqrt{\left(n+\frac{1}{2}\right)^2 - 1} > n + \frac{1}{3}, \quad {}^\forall n \geq 3,$$

we see that $|\Gamma(z)| \leqq 1$ for every natural number $n \geqq 3$ and $z \in \mathbf{C}$ with $|z| = n + \frac{1}{2}$ and $\Re z \leqq 2$. Combined with (1.3.4), this implies

$$m(r, \Gamma) \leqq \frac{1}{2}(r+1)\log(r+1)$$

for $r = n + \frac{1}{2}$ with $n \in \mathbf{N}$ and $n \geqq 3$. It follows from this and (1.3.2) that

$$r - 1 < T(r, \Gamma) \leqq r\left(1 + \frac{1}{2}\log(r+1)\right) + O(\log r).$$

Therefore, $\rho_\Gamma = 1$.

(g) **Zeta function**. The Riemann zeta function is defined as

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \left(1 - p^{-s}\right)^{-1}$$

for $\Re s > 1$, where p runs over all primes >1. It extends to a meromorphic function on $\mathbf{C}$, fulfilling the functional equation

$$\zeta(1-s) = \frac{2}{(2\pi)^s}\left(\cos\frac{\pi s}{2}\right)\Gamma(s)\zeta(s). \tag{1.3.6}$$

However, these two equations do not suffice to control the growth behavior of the zeta function on the "critical strip" where $0 < \Re s < 1$. For this we need a third relation. The following inequality holds (see Lang [99], Chap. XVI, Proof of Theorem 1.2):

$$\left|\zeta(s) - \frac{1}{s-1}\right| \leqq \sum_{n\in\mathbf{N}} \frac{|s|}{n^{\Re s+1}}, \quad \Re s > 0.$$

It follows that

$$\left|\zeta(s)\right| < \frac{1}{|s|-1} + |s|\zeta(1.25), \quad \Re s > 0.25.$$

In combination with (1.3.1) we obtain

$$\begin{aligned}\left|\zeta(1-s)\right| &= \left|\frac{2}{(2\pi)^s}\left(\cos\frac{\pi s}{2}\right)\Gamma(s)\zeta(s)\right| \\ &\leqq \frac{2}{(2\pi)^{\Re s}} e^{\frac{\pi}{2}|\Im s|}\left|\Gamma(s)\right|\left(\frac{1}{|s|-1} + |s|\zeta(1.25)\right)\end{aligned}$$

for $\Re s > 0.25$. For $|s| = n + \frac{1}{2}$ with $n \geqq 3$ it follows that

$$\left|\zeta(s)\right| \leqq 2e^{|s|(\log(2\pi)+\frac{\pi}{2})}\left(|s|+1\right)^{|s|-1}\left(0.4 + |s|\zeta(1.25)\right),$$

which implies

$$\log^+\left|\zeta(s)\right| = O(r\log r), \quad r = |s|.$$

Since ζ has only one pole at 1, it follows that ζ is a function of finite order $\rho_\zeta \leqq 1$.

On the other hand, $\zeta(s)$ has the trivial zeros at negative even $s \in \mathbf{Z}$ by (1.3.6). Therefore we have

$$N\left(r, (\zeta)_0\right) \geqq \frac{r}{2} + O(\log r).$$

In all we get

$$\frac{r}{2} + O(\log r) \leqq T(r,\zeta) \leqq O(r\log r),$$

and hence $\rho_\zeta = 1$.

Notes For more results of the value distribution theory in one variable, confer R. Nevanlinna [53], Hayman [64], Ozawa [76], etc.

The Second Main Theorem is very powerful for the investigation of meromorphic functions. For example, it implies a unicity theorem such that a meromorphic function is uniquely determined by the inverse image sets of five points of $\hat{\mathbf{C}}$, and a finiteness theorem such that there are at most finitely many meromorphic functions if the inverse images of three points are given. For these results, see Cartan [28], R. Nevanlinna [29], etc. There are numerous papers dealing with unicity theorems for meromorphic functions in one variable.

F. Nevanlinna (elder brother of R. Nevanlinna) gave another geometric proof of the Second Main Theorem 1.2.5 by analysis of singularities at given a_i of the metric induced from the Poincaré metric of the universal cover of $\hat{\mathbf{C}} \setminus \{a_i\}$ (F. Nevanlinna [27]). This method is close to Griffiths' metric method (cf. Carlson–Griffiths [72]; Griffiths–King [73]). It was one of the main motivations of the research then by Borel, R. Nevanlinna and others to give a proof of Picard's Theorem, elementary in the sense that it does not rely on the uniformization theorem. F. Nevanlinna's method is not elementary since it uses the uniformization theorem. It is unknown how much Griffiths et al. had in mind F. Nevanlinna's metric method to establish the Second Main Theorem in several complex variables, but it is interesting so see that the both are close in spite of the long difference of time.

On the other hand, the proof of the Second Main Theorem of Griffiths et al. which we will give in Chap. 3 will be close to R. Nevanlinna's; it will be very elementary and will not rely on the original metric method due to them.

Chapter 2
The First Main Theorem

The value distribution theory with domains in several complex variables was pioneered by Wilhelm Stoll [53a], [53b], [54]. While his presentation may not be familiar or easy to us in modern terminologies, the works which he has contributed, beginning with the integrations over singular analytic subvarieties and the extension of Stokes' theorem, were fundamental. In the 1960s there were many works on the First Main Theorem; these were summarized by W. Stoll (see Stoll [70], in particular its preface and the listed references). The relation to characteristic classes was made explicit first by Bott–Chern [65].[1] In the present chapter we follow Carlson–Griffiths [72], Griffiths–King [73], Noguchi [03b] and Noguchi–Winkelmann–Yamanoi [08] which may be most comprehensive.

2.1 Plurisubharmonic Functions

2.1.1 One Variable

We first investigate subharmonic functions. Let U be an open subset of $\mathbf{C}$. Set

$$d(a;\partial U) = \inf\{|a-w|;\, w\in\partial U\}.$$

Definition 2.1.1 A function $\varphi : U \to [-\infty,\infty)$ is said to be *subharmonic* if φ is upper semicontinuous and has the submean property; that is,

(i) (upper semicontinuity) $\overline{\lim}_{z\to a}\varphi(z) \leqq \varphi(a)$, $^\forall a \in U$:
(ii) (submean property) On an arbitrary disk $\Delta(a;r) \Subset U$

$$\varphi(a) \leqq \frac{1}{2\pi}\int_0^{2\pi} \varphi\big(a+re^{i\theta}\big)d\theta.$$

[1] Readers may find a number of interesting papers on the theory of holomorphic mappings in Chern, *Selected Papers* (Chern [78]).

J. Noguchi, J. Winkelmann, *Nevanlinna Theory in Several Complex Variables and Diophantine Approximation*, Grundlehren der mathematischen Wissenschaften 350,
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Remark 2.1.2 (i) If $\varphi : U \to [-\infty, \infty)$ is upper semicontinuous, φ is bounded from above on every compact subset $K \Subset U$.

(ii) The upper semicontinuity of $\varphi : U \to [-\infty, \infty)$ is equivalent to that for every $c \in \mathbf{R}$ the sublevel set $\{z \in U; \varphi(z) < c\}$ is open.

(iii) The function $\varphi : U \to [-\infty, \infty)$ is upper semicontinuous if and only if there is a monotone decreasing sequence of continuous functions $\psi_\nu : U \to \mathbf{R}$, $\nu = 1, 2, \ldots$, such that $\lim_{\nu\to\infty} \psi_\nu(z) = \varphi(z)$.

(iv) It follows from the above Definition 2.1.1 (ii) that

$$\varphi(a) \leqq \frac{1}{\pi r^2} \int_0^r t dt \int_0^{2\pi} \varphi\big(a + te^{i\theta}\big) d\theta \tag{2.1.3}$$
$$= \frac{1}{r^2} \int_{|\zeta|<r} \varphi(a+\zeta) \frac{i}{2\pi} d\zeta \wedge d\bar{\zeta} < \infty.$$

Theorem 2.1.4 (i) *Let φ be a subharmonic function on U. Let $a \in U$ be a point such that $\varphi(a) > -\infty$. Then φ is locally integrable on the connected component of U containing a.*

(ii) *Let φ be a subharmonic function on U. If φ takes the maximum value at $a \in U$, then φ is constant on the connected component of U containing a.*

(iii) *Assume that $\varphi \in C^2(U)$. Then φ is subharmonic if and only if $dd^c\varphi = (i/2\pi)\partial\bar{\partial}\varphi \geqq 0$.*

(iv) *Let $\varphi : U \to [-\infty, \infty)$ be subharmonic, and let λ be a monotone increasing convex function defined on $\mathbf{R}$. Then $\lambda \circ \varphi$ is subharmonic. Here we put $\lambda(-\infty) = \lim_{t\to-\infty} \lambda(t)$.*

(v) *Let $\varphi_\nu : U \to [-\infty, \infty)$, $\nu = 1, 2, \ldots$, be a monotone decreasing sequence of subharmonic functions. Then the limit function $\varphi(z) = \lim_{\nu\to\infty} \varphi_\nu(z)$ is subharmonic, too.*

(vi) *Let $\varphi_\nu : U \to [-\infty, \infty)$, $1 \leqq \nu \leqq l$, be finitely many subharmonic functions. Then $\varphi(z) = \max_{1\leqq\nu\leqq l} \varphi_\nu(z)$ is subharmonic.*

Proof (i) Without loss of generality we may assume that U is connected. Notice that if $\varphi(a) > -\infty$, then φ is integrable on every relatively compact disk $\Delta(a; r) \Subset U$ by (2.1.3). Suppose that there is a point $a \in U$ with $\varphi(a) > -\infty$. Denote by U_0 the set of all points $z \in U$ with a neighborhood W such that the restriction $\varphi|_W$ of φ to W is integrable. Clearly, U_0 is non-empty and open.

We show that U_0 is closed in U. Let $a \in U$ be an accumulation point of U_0. Take a sequence of points $z_\nu \in U_0$, $\nu = 1, 2, \ldots$, convergent to a. One may assume that $\varphi(z_\nu) > -\infty$, $\nu = 1, 2, \ldots$. There are some $r > 0$ and a sufficiently large ν such that $a \in \Delta(z_\nu; r) \Subset U$. By the remark at the beginning, $\varphi|_{\Delta(z_\nu;r)}$ is integrable. Therefore $a \in U_0$. Since U is connected, $U_0 = U$.

(ii) Assume that U is connected and $\varphi(a)$ is the maximum. It follows from (2.1.3) that for every $\Delta(a; r) \Subset U$

$$\int_{\Delta(a;r)} \big\{\varphi(\zeta) - \varphi(a)\big\} \frac{i}{2\pi} d\zeta \wedge d\bar{\zeta} = 0. \tag{2.1.5}$$

By assumption $\varphi(\zeta) - \varphi(a) \leqq 0$. Suppose that $\varphi(b) - \varphi(a) = \delta_0 < 0$ at a point $b \in \Delta(a; r)$. The upper semicontinuity of φ implies that $\varphi(\zeta) - \varphi(a) < \frac{\delta_0}{2}$ in a neighborhood of b. Then (2.1.5) does not hold. Hence $\varphi|_{\Delta(a;r)} \equiv \varphi(a)$. Denote by U_1 the set of all points $z \in U$ with a neighborhood W such that $\varphi|_W \equiv \varphi(a)$. By a similar argument to (i) U_1 is open and closed in U. Therefore $U_1 = U$.

(iii) About every point $a \in U$ we expand φ to a Taylor series up to degree two:

$$\varphi\big(a + \varepsilon e^{i\theta}\big) = \varphi(a) + \frac{\partial\varphi}{\partial z}(a)\varepsilon e^{i\theta} + \frac{\partial\varphi}{\partial\bar{z}}(a)\varepsilon e^{-i\theta} + \varepsilon^2\left(\frac{\partial^2\varphi}{\partial z^2}(a)e^{2i\theta} + 2\frac{\partial^2\varphi}{\partial z\partial\bar{z}}(a) + \frac{\partial^2\varphi}{\partial z^2}(a)e^{-2i\theta}\right)\big(1 + o(1)\big).$$

Taking the integration in θ, we have

$$\frac{1}{2\pi}\int_0^{2\pi}\varphi\big(a + \varepsilon e^{i\theta}\big)d\theta = \varphi(a) + \varepsilon^2\big(1 + o(1)\big)2\frac{\partial^2\varphi}{\partial z\partial\bar{z}}(a).$$

The submean property implies that $\frac{\partial^2\varphi}{\partial z\partial\bar{z}}(a) \geqq 0$.

Conversely, assume that $\frac{\partial^2\varphi}{\partial z\partial\bar{z}} \geqq 0$. It follows from Jensen's formula, Lemma 1.1.5 that about every point $a \in U$

$$\text{(2.1.6)}\quad \frac{1}{2\pi}\int_{|\zeta|=s}\varphi(a+\zeta)d\theta \leqq \frac{1}{2\pi}\int_{|\zeta|=r}\varphi(a+\zeta)d\theta, \quad 0 < s < r < d\ (a; \partial U).$$

Let $s \searrow 0$. Then

$$\varphi(a) \leqq \frac{1}{2\pi}\int_{|\zeta|=r}\varphi(a+\zeta)d\theta.$$

(iv) Note that λ is continuous. The remaining part is immediate, for λ is monotone increasing and convex.

(v) The upper semicontinuity of φ follows immediately from the assumption. Since upper semicontinuous functions are bounded from above on every relatively compact subset, φ_ν are uniformly bounded from above on every relatively compact subset. Take an arbitrary disk $\Delta(a; r) \Subset U$. By Fatou's lemma in integration theory we have

$$\varphi(a) = \lim_{\nu\to\infty}\varphi_\nu(a) \leqq \varlimsup_{\nu\to\infty}\frac{1}{2\pi}\int_0^{2\pi}\varphi_\nu\big(a + re^{i\theta}\big)d\theta \leqq \frac{1}{2\pi}\int_0^{2\pi}\varlimsup_{\nu\to\infty}\varphi_\nu\big(a + re^{i\theta}\big)d\theta = \frac{1}{2\pi}\int_0^{2\pi}\varphi\big(a + re^{i\theta}\big)d\theta.$$

(vi) Both the upper-semicontinuity and the submean property are immediate by definition. □

Example 2.1.7 Let $f : U \to \mathbf{C}$ be a holomorphic function. Then $\log|f|$ and $|f|^c$ with $c > 0$ are subharmonic. Because a direct computation of partial derivatives of

$\log(|f|^2+C)$ with $C>0$ implies the subharmonicity $\log(|f|^2+C)$. Setting $C=1/\nu$, $\nu=1,2,\ldots$, and taking the limit, we see by Theorem 2.1.4 (v) that $\log|f|^2=2\log|f|$ is subharmonic, and so is $\log|f|$. Since the exponential function e^{ct}, $t\in\mathbf{R}$ with $c>0$ is monotone increasing and convex, Theorem 2.1.4 (iv) implies that $|f|^c$ is subharmonic.

Let $\chi\in C_0^\infty(\mathbf{C})$ be a function such that $\operatorname{Supp}\chi\subset\Delta(1)$, $\chi(z)=\chi(|z|)\geqq 0$ and

$$\int\chi(z)\frac{i}{2}dz\wedge d\bar z=1.$$

Set $\chi_\varepsilon(z)=\chi(\varepsilon^{-1}z)\varepsilon^{-2}$, $\varepsilon>0$. Then

$$\int\chi_\varepsilon(z)\frac{i}{2}dz\wedge d\bar z=1.$$

Consider a subharmonic function φ on U such that $\varphi\not\equiv-\infty$ on every connected component of U. Put

$$U_\varepsilon=\{z\in U;\,d(z;\partial U)>\varepsilon\}.$$

The *smoothing* $\varphi_\varepsilon(z)$ $(z\in U_\varepsilon)$ of φ is defined by

$$\begin{aligned}\varphi_\varepsilon(z)=\varphi*\chi_\varepsilon(z)&=\int_{\mathbf{C}}\varphi(w)\chi_\varepsilon(w-z)\frac{i}{2}dw\wedge d\bar w\\&=\int_{\mathbf{C}}\varphi(z+w)\chi_\varepsilon(w)\frac{i}{2}dw\wedge d\bar w\\&=\int_0^1\chi(t)tdt\int_0^{2\pi}\varphi\big(z+\varepsilon te^{i\theta}\big)d\theta\\&\geqq\varphi(z)\int_0^1 2\pi\chi(t)tdt=\varphi(z).\end{aligned}\tag{2.1.8}$$

Note that $\varphi_\varepsilon(z)$ is C^∞ on U_ε, and subharmonic. Therefore Theorem 2.1.4 implies

$$\frac{\partial^2}{\partial z\partial\bar z}\varphi_\varepsilon(z)\geqq 0.$$

Taking $\varepsilon_1>\varepsilon_2>0$, and $\delta>0$, we consider the double smoothing $(\varphi_\delta)_{\varepsilon_i}=(\varphi_{\varepsilon_i})_\delta$, $i=1,2$. Note that φ_δ is C^∞ and subharmonic. Combining (2.1.6) applied to φ_δ with (2.1.8) applied to $\varphi=\varphi_\delta$, we deduce that $(\varphi_\delta)_{\varepsilon_1}\geqq(\varphi_\delta)_{\varepsilon_2}$. Hence $(\varphi_{\varepsilon_1})_\delta\geqq(\varphi_{\varepsilon_2})_\delta$. Letting $\delta\to 0$, we see that $\varphi_{\varepsilon_1}\geqq\varphi_{\varepsilon_2}$. Thus as $\varepsilon\searrow 0$, $\varphi_\varepsilon(z)$ monotonically decreases, and it follows from (2.1.8) that

$$\varphi(z)\leqq\lim_{\varepsilon\to 0}\varphi_\varepsilon(z).$$

Here we show the equality by making use of the upper semicontinuity.

Suppose that $\varphi(z) = -\infty$. For every $K < 0$ there exists a disk neighborhood $\Delta(z;r) \subset U$ such that $\varphi|_{\Delta(z;r)} < K$. By definition (2.1.8) $\varphi_\varepsilon(z) < K$ for $\varepsilon < r$, and so $\lim_{\varepsilon\to 0}\varphi_\varepsilon(z) = -\infty$.

Suppose that $\varphi(z) > -\infty$. For every $\varepsilon' > 0$ there is a disk $\Delta(z;r) \subset U$ such that $\varphi|_{\Delta(z;r)} < \varphi(z) + \varepsilon'$. By the same reasoning as above, $\varphi_\varepsilon(z) \leqq \varphi(z) + \varepsilon'$ for $\varepsilon < r$. Thus $\lim_{\varepsilon\to 0}\varphi_\varepsilon(z) = \varphi(z)$.

Now we have the convergence, $\varphi_\varepsilon(z) \searrow \varphi(z)$ $(\varepsilon \searrow 0)$. For $\eta \in C_0^\infty(U)$

$$\int \eta(z) dd^c \varphi_\varepsilon(z) = \int \varphi_\varepsilon(z) dd^c \eta(z). \tag{2.1.9}$$

If $\eta \geqq 0$, this integral is non-negative. We set

$$dd^c[\varphi] = \frac{i}{2\pi}\frac{\partial^2}{\partial z \partial \bar{z}}[\varphi] dz \wedge d\bar{z}$$

in the sense of the Schwartz distribution. As $\varepsilon \searrow 0$, (2.1.9) implies

$$\int \eta(z) dd^c\big[\varphi(z)\big] = \int \varphi(z) dd^c \eta(z) \geqq 0, \quad \eta \geqq 0.$$

We see that $dd^c[\varphi]$ is a positive Radon measure. We may also regard $dd^c[\varphi]$ as a differential form with coefficients in Radon measures.

We apply (2.1.6) to the C^∞ subharmonic function φ_ε; for $\Delta(a;r) \Subset U, 0 < s < r$ and sufficiently small $\varepsilon > 0$ we obtain

$$\frac{1}{2\pi}\int_{|\zeta|=s} \varphi_\varepsilon(a+\zeta) d\theta \leqq \frac{1}{2\pi}\int_{|\zeta|=r} \varphi_\varepsilon(a+\zeta) d\theta.$$

As $\varepsilon \searrow 0$, Lebesgue's monotone convergence theorem implies that

$$\frac{1}{2\pi}\int_{|\zeta|=s} \varphi(a+\zeta) d\theta \leqq \frac{1}{2\pi}\int_{|\zeta|=r} \varphi(a+\zeta) d\theta. \tag{2.1.10}$$

By Theorem 2.1.4 (i) φ is locally integrable on U. Fubini's theorem and (2.1.3) imply that for almost all $s \in (0, r)$ with respect to the Lebesgue measure

$$\frac{1}{2\pi}\int_{|\zeta|=s} \varphi(a+\zeta) d\theta > -\infty. \tag{2.1.11}$$

This with (2.1.10) implies (2.1.11) for all $s \in (0, r]$.

Summarizing the above we have the next theorem.

Theorem 2.1.12 *Let $\varphi : U \to [-\infty, \infty)$ be a subharmonic function on U such that $\varphi \not\equiv -\infty$ on every connected component of U.*

(i) *$dd^c[\varphi]$ is a positive Radon measure.*
(ii) *The smoothing $\varphi_\varepsilon(z)$ is subharmonic; as $\varepsilon \searrow 0$ it is monotone decreasing and converges to $\varphi(z)$.*

(iii) *For $\Delta(a;r) \Subset U$ and any $s \in (0,r)$*

$$-\infty < \frac{1}{2\pi}\int_{|\zeta|=s} \varphi(a+\zeta)d\theta \leqq \frac{1}{2\pi}\int_{|\zeta|=r} \varphi(a+\zeta)d\theta < \infty.$$

Theorem 2.1.13 (i) *The subharmonicity is a local property*; *i.e., if $\varphi : U \to [-\infty,\infty)$ is subharmonic in a neighborhood of every point $a \in U$, then φ is subharmonic in U.*

(ii) *If an upper semicontinuous function $\varphi : U \to [-\infty,\infty)$ satisfies*

$$\varphi(a) \leqq \frac{1}{r^2}\int_{\Delta(a;r)} \varphi(z)\frac{i}{2\pi}dz\wedge d\bar{z}$$

for every disk $\Delta(a;r) \subset U$, then φ is subharmonic.

Proof (i) Take the smoothing $\varphi_\varepsilon(z)$. Let φ be subharmonic in $\Delta(a;r) \subset U$. Then φ_ε, $0 < \varepsilon < r/2$ is subharmonic in $\Delta(a;r/2)$. Therefore $dd^c\varphi_\varepsilon(z) \geqq 0$, and hence by Theorem 2.1.4 (iii) $\varphi_\varepsilon(z)$ is subharmonic in U_ε.

For Definition 2.1.1 (ii), it suffices to show that φ is subharmonic in an arbitrarily fixed U_δ ($\delta > 0$). As $\delta > \varepsilon \searrow 0$, $\varphi_\varepsilon(z) \searrow \varphi(z)$ in U_δ. We infer from Theorem 2.1.4 (v) that φ is subharmonic in U_δ.

(ii) We first assume that φ is of C^2-class. By the same computation as in the proof of Theorem 2.1.4 (iii) we get

$$\int_0^\varepsilon tdt\frac{1}{2\pi}\int_0^{2\pi}\varphi\big(a+te^{i\theta}\big)d\theta = \int_0^\varepsilon\left(t\varphi(a)+t^3\big(1+o(1)\big)2\frac{\partial^2\varphi}{\partial z\partial\bar{z}}(a)\right)dt.$$

It follows that

$$\frac{1}{\varepsilon^2}\int_{\Delta(\varepsilon)}\varphi(a+\zeta)\frac{i}{2\pi}d\zeta\wedge d\bar{\zeta} = \varphi(a)+\varepsilon^2\big(1+o(1)\big)\frac{\partial^2\varphi}{\partial z\partial\bar{z}}(a).$$

This combined with the assumption implies that $\frac{\partial^2\varphi}{\partial z\partial\bar{z}}(a) \geqq 0$. Thus $\varphi(z)$ is subharmonic.

For the general case we may assume that U is connected, and $\varphi \not\equiv -\infty$. By the proof of Theorem 2.1.4 (i) φ is locally integrable in U.

We take the smoothing $\varphi_\varepsilon(z)$ $z \in U_\varepsilon$. Since $\varphi(z)$ is upper semicontinuous, by Remark 2.1.2 (iii) there is a monotone decreasing sequence of continuous functions $\psi_\nu(z)$, $\nu = 1,2,\ldots$, such that

$$\lim_{\nu\to\infty}\psi_\nu(z) = \varphi(z), \quad z \in U.$$

We are going to show that for every compact subset $K \subset U$

$$\lim_{\varepsilon\to 0}\int_K \big|\varphi_\varepsilon(z)-\varphi(z)\big|\frac{i}{2\pi}dz\wedge d\bar{z} = 0. \tag{2.1.14}$$

Take $W \Subset U$ be an open subset such that $W \ni K$. Set

$$d(K, \partial W) = \inf\{d(z, \partial W); z \in K\},$$

and

$$\delta_0 = \min\{d(K, \partial W), d(\bar{W}, \partial U)\} > 0.$$

Take any $\varepsilon' > 0$. By Lebesgue's monotone convergence theorem there is a number ν_0 such that for $\|w\| < \delta_0$

$$0 \leqq \int_K \big(\psi_{\nu_0}(z+w) - \varphi(z+w)\big)\frac{i}{2\pi}dz \wedge d\bar{z} \leqq \int_{\bar{W}} \big(\psi_{\nu_0}(z) - \varphi(z)\big)\frac{i}{2\pi}dz \wedge d\bar{z} < \varepsilon'. \tag{2.1.15}$$

Let $0 < \varepsilon < \delta_0$. Then $(\psi_{\nu_0})_\varepsilon(z) \geqq \varphi_\varepsilon(z)$ $(z \in \bar{W})$, and

$$\begin{aligned} 0 &\leqq \int_K \big((\psi_{\nu_0})_\varepsilon(z) - \varphi_\varepsilon(z)\big)\frac{i}{2\pi}dz \wedge d\bar{z} \\ &= \int_{z\in K}\left(\int_{w\in\mathbf{C}^m} \big((\psi_{\nu_0})(z+w) - \varphi(z+w)\big)\chi_\varepsilon(w)\frac{i}{2\pi}dw \wedge d\bar{w}\right)\frac{i}{2\pi}dz \wedge d\bar{z} \\ &= \int_{w\in\mathbf{C}^m}\left(\int_{z\in K} \big((\psi_{\nu_0})(z+w) - \varphi(z+w)\big)\frac{i}{2\pi}dz \wedge d\bar{z}\right)\chi_\varepsilon(w)\frac{i}{2\pi}dw \wedge d\bar{w} \\ &\leqq \varepsilon'. \end{aligned} \tag{2.1.16}$$

Since ψ_{ν_0} is uniformly continuous on $\bar{W}$, $(\psi_{\nu_0})_\varepsilon$ uniformly approximates ψ_{ν_0} on K as $\varepsilon \to 0$. Thus there is some $0 < \varepsilon_0 < \delta_0$ such that for every $0 < \varepsilon < \varepsilon_0$

$$\int_K \big|(\psi_{\nu_0})_\varepsilon(z) - \psi_{\nu_0}(z)\big|\frac{i}{2\pi}dz \wedge d\bar{z} < \varepsilon'. \tag{2.1.17}$$

It follows from (2.1.15)–(2.1.17) that for every $0 < \varepsilon < \varepsilon_0$

$$\begin{aligned} \int_K |\varphi_\varepsilon - \varphi|\frac{i}{2\pi}dz \wedge d\bar{z} &\leqq \int_K \big((\psi_{\nu_0})_\varepsilon - \varphi_\varepsilon\big)\frac{i}{2\pi}dz \wedge d\bar{z} \\ &\quad + \int_K \big|(\psi_{\nu_0})_\varepsilon - \psi_{\nu_0}\big|\frac{i}{2\pi}dz \wedge d\bar{z} + \int_K (\psi_{\nu_0} - \varphi)\frac{i}{2\pi}dz \wedge d\bar{z} \\ &< 3\varepsilon'. \end{aligned}$$

Thus (2.1.14) is deduced.

We see by the assumption and Fubini's theorem that for $\Delta(z; r) \subset U_\delta$

$$\varphi_\delta(z) \leqq \frac{1}{r^2}\int_{\Delta(r)} \varphi_\delta(z+\zeta)\frac{i}{2\pi}d\zeta \wedge d\bar{\zeta}.$$

Therefore φ_δ is subharmonic in U_δ. Take $\delta > \varepsilon_1 > \varepsilon_2 > 0$, arbitrarily. For $z \in U_{\delta+\varepsilon_1}$

$$(\varphi_{\varepsilon_1})_\delta(z) = (\varphi_\delta)_{\varepsilon_1}(z) \geqq (\varphi_\delta)_{\varepsilon_2}(z) = (\varphi_{\varepsilon_2})_\delta(z).$$

As $\delta \to 0$, $\varphi_{\varepsilon_1}(z) \geqq \varphi_{\varepsilon_2}(z)$. Put $\psi(z) = \lim_{\varepsilon\to 0}\varphi_\varepsilon(z)$. Since ψ is the limit of a monotone decreasing sequence of subharmonic functions, it is subharmonic. It follows from the upper semicontinuity that for every $z \in U$ and $\varepsilon' > 0$ there is a neighborhood $\Delta(z; r) \subset U$ satisfying

$$\varphi(\zeta) < \varphi(z) + \varepsilon', \quad \zeta \in \Delta(z; r).$$

Hence, for $0 < \varepsilon < r$, $\varphi_\varepsilon(z) \leqq \varphi(z) + \varepsilon'$. We have

$$\varphi(z) - \psi(z) \geqq 0, \quad z \in U.$$

This combined with (2.1.14) implies that for any compact subset $K \subset U$

$$\begin{aligned} 0 &\leqq \int_K \big(\varphi(z) - \psi(z)\big)\frac{i}{2\pi}dz \wedge d\bar{z} \\ &= \lim_{\varepsilon\to 0}\int_K \big(\varphi(z) - \varphi_\varepsilon(z)\big)\frac{i}{2\pi}dz \wedge d\bar{z} \\ &\leqq \lim_{\varepsilon\to 0}\int_K \big|\varphi(z) - \varphi_\varepsilon(z)\big|\frac{i}{2\pi}dz \wedge d\bar{z} = 0. \end{aligned}$$

Therefore $\psi(z) = \varphi(z)$ for almost all $z \in U$ with respect to the Lebesgue measure. For every $\Delta(a; r) \subset U$,

$$\begin{aligned} \varphi(a) &\leqq \frac{1}{r^2}\int_{\Delta(a;r)} \varphi(z)\frac{i}{2\pi}dz \wedge d\bar{z} \\ &= \frac{1}{r^2}\int_{\Delta(a;r)} \psi(z)\frac{i}{2\pi}dz \wedge d\bar{z} \\ &\to \psi(a) \quad (r \to 0). \end{aligned}$$

Hence, $\varphi(a) \leqq \psi(a)$, and $\psi = \varphi$, by which φ is subharmonic. □

Proposition 2.1.18 *A function ϕ on U is subharmonic if and only if there exists a decreasing sequence of C^2 subharmonic functions ϕ_n with $\lim \phi_n = \phi$.*

Proof If there is such a sequence, subharmonicity of ϕ follows from Theorem 2.1.4 (v). The converse is obtained by making use of the smoothing (Theorem 2.1.12 (ii)). □

Proposition 2.1.18 together with Theorem 2.1.4 (iii) implies the following.

Theorem 2.1.19 *Let U, V be open in* **C**.

(i) *If ϕ is subharmonic on U and $f : V \to U$ is holomorphic, then $\phi \circ f$ is subharmonic.*

(ii) *If ϕ is a function on U and $f : V \to U$ is biholomorphic, then ϕ is subharmonic if and only if $\phi \circ f$ is subharmonic.*

Proof (i) Due to Proposition 2.1.18 there is a decreasing sequence of C^2 subharmonic functions ϕ_n $(n = 1, 2, \dots)$ with $\lim \phi_n = \phi$. Subharmonicity of the ϕ_n is equivalent to $dd^c\varphi_n \geqq 0$ (Theorem 2.1.4 (iii)). Hence the functions $\phi_n \circ f$ $(n = 1, 2, \dots)$ form a decreasing sequence of subharmonic functions converging to $\phi \circ f$. Now the subharmonicity of $\phi \circ f$ follows from Theorem 2.1.4 (v).

Statement (ii) is a direct consequence of (i). □

2.1.2 Several Variables

We deal with the case of several complex variables. The notion of plurisubharmonic functions was first introduced by K. Oka [42] VI. We let $U \subset \mathbf{C}^m$ be an open set. Let $z = (z_1, \dots, z_m)$ be the standard coordinate system of $\mathbf{C}^m$. As usual we set

$$\|z\| = \sqrt{\sum |z_j|^2},$$

$$d(z; \partial U) = \inf\{\|z - w\|; w \in \partial U\}, \quad z \in U,$$

$$U_\varepsilon = \{z \in U; d(z; \partial U) > \varepsilon\}, \quad \varepsilon > 0.$$

We write $z_j = x_j + iy_j$ $(1 \leqq j \leqq m)$. As in (1.1.1), we define the following differential operators:

$$\frac{\partial\varphi}{\partial z_j} = \frac{1}{2}\left(\frac{\partial\varphi}{\partial x_j} + \frac{1}{i}\frac{\partial\varphi}{\partial y_j}\right), \qquad \frac{\partial\varphi}{\partial \bar{z}_j} = \frac{1}{2}\left(\frac{\partial\varphi}{\partial x_j} - \frac{1}{i}\frac{\partial\varphi}{\partial y_j}\right),$$

$$dz_j = dx_j + i dy_j, \qquad d\bar{z}_j = dx_j - i dy_j,$$

$$\partial\varphi = \sum_{j=1}^m \frac{\partial\varphi}{\partial z_j} dz_j, \qquad \bar{\partial}\varphi = \sum_{j=1}^m \frac{\partial\varphi}{\partial \bar{z}_j} d\bar{z}_j, \tag{2.1.20}$$

$$d^c\varphi = \frac{i}{4\pi}(\bar{\partial}\varphi - \partial\varphi) = \frac{1}{4\pi}\sum_{j=1}^m \left(\frac{\partial\varphi}{\partial x_j} dy_j - \frac{\partial\varphi}{\partial y_j} dx_j\right).$$

With this notation we have

$$d\varphi = \partial\varphi + \bar{\partial}\varphi, \qquad dd^c\varphi = \frac{i}{2\pi}\partial\bar{\partial}\varphi,$$

$$\partial\bar{\partial}\varphi = \sum_{j,k=1}^m \frac{\partial^2\varphi}{\partial z_j \partial \bar{z}_k} dz_j \wedge d\bar{z}_k.$$

We further introduce the following notation:

$$
\begin{aligned}
B(a;r) &= \left\{z \in \mathbf{C}^m;\, \|z-a\| < r\right\}, \quad a \in \mathbf{C}^m,\ r > 0,\\
B(r) &= B(0;r),\\
\alpha &= dd^c\|z\|^2, \qquad \beta = dd^c \log\|z\|^2,\\
\gamma &= d^c \log\|z\|^2 \wedge \beta^{m-1}.
\end{aligned}
\tag{2.1.21}
$$

Let ϕ be a differential form on $\mathbf{C}^m$ and let $\iota : \{\|z\| = r\} \hookrightarrow \mathbf{C}^m$ be the inclusion of the sphere $\{\|z\| = r\}$ into $\mathbf{C}^m$. Associated to the map ι there is the "pull-back" of differential forms. In this way $\iota^*\phi$ is the differential form induced from ϕ on the sphere $\{\|z\| = r\}$. In the present case, $\iota^*(d\|z\|^2) = 0$, and so as differential forms induced over $\{\|z\| = r\}$, $d\|z\|^2 = \partial\|z\|^2 + \bar\partial\|z\|^2 = 0$. Therefore as differential forms induced over $\{\|z\| = r\}$,

$$
d\|z\|^2 \wedge d^c\|z\|^2 = 0, \qquad \partial\|z\|^2 \wedge \bar\partial\|z\|^2 = 0.
$$

Hence we have, as induced forms on $\{\|z\| = r\}$,

$$
\beta = \frac{1}{r^2}\alpha. \tag{2.1.22}
$$

It follows that

$$
\int_{B(r)} \alpha^m = r^{2m}, \qquad \int_{\|z\|=r} \gamma = 1.
$$

Definition 2.1.23 A function $\varphi : U \to [-\infty, \infty)$ is said to be *plurisubharmonic* if the following conditions are satisfied:

(i) φ is upper semicontinuous.
(ii) For every point $z \in U$ and every vector $v \in \mathbf{C}^m$ the function

$$
\zeta \in \mathbf{C} \to \varphi(z + \zeta v) \in [-\infty, \infty)
$$

is subharmonic where it is defined.

We have the following examples by Example 2.1.7.

Example 2.1.24 If $f : U \to \mathbf{C}$ is a holomorphic function, $\log|f|$ and $|f|^c$ $(c > 0)$ are both plurisubharmonic.

Let φ be a plurisubharmonic function on U, and let $B(a;r) \Subset U$. By making use of the invariance of α with respect to the rotation $z \mapsto e^{i\theta}z$ $(\theta \in [0, 2\pi])$ we have by Definition 2.1.1 (ii) that

$$
\int_{z\in B(r)} \varphi(a+z)\alpha^m(z) = \int_{z\in B(r)} \varphi\big(a + e^{i\theta}z\big)\alpha^m(z)
$$

(continued)

$$= \frac{1}{2\pi}\int_0^{2\pi} d\theta \int_{z\in B(r)} \varphi\big(a + e^{i\theta}z\big)\alpha^m(z)$$

$$= \int_{z\in B(r)} \left(\frac{1}{2\pi}\int_0^{2\pi} \varphi\big(a + e^{i\theta}z\big)d\theta\right)\alpha^m(z)$$

$$\geqq \int_{z\in B(r)} \varphi(a)\alpha^m(z) = r^{2m}\varphi(a).$$

Thus as in (2.1.3) the following is obtained:

$$\varphi(a) \leqq \frac{1}{r^{2m}}\int_0^r 2mt^{2m-1}dt \int_{\|z\|=t} \varphi(a+z)\gamma(z) \tag{2.1.25}$$

$$= \frac{1}{r^{2m}}\int_{B(a;r)} \varphi(z)\alpha^m \quad \big(B(a;r) \Subset U\big).$$

Identifying $\mathbf{C}^m \cong \mathbf{R}^{2m}$, we see that φ is a *subharmonic function*[2] on $U \subset \mathbf{R}^{2m}$. When φ is of C^2-class, we have by definition

$$dd^c\varphi = \sum_{1\leqq j,k\leqq m} \frac{\partial^2\varphi}{\partial z_j \partial \bar{z}_k}\frac{i}{2\pi}dz_j \wedge d\bar{z}_k.$$

We write $dd^c\varphi \geqq 0$ if the hermitian matrix $(\frac{\partial^2\varphi}{\partial z_j \partial \bar{z}_k})$ is semi-positive definite.

The next theorem follows from the above and the same arguments as used in the proof of Theorem 2.1.4:

Theorem 2.1.26 (i) *A plurisubharmonic function is subharmonic with identification* $\mathbf{C}^m \cong \mathbf{R}^{2m}$.

(ii) *If φ is a plurisubharmonic function on U and $\varphi(a) > -\infty$ at a point $a \in U$, then φ is locally integrable in the connected component U' of U containing a.*

(iii) *Let φ be a plurisubharmonic function on U. If φ admits the maximum value at $a \in U$, then it is constant on the connected component of U containing a.*

(iv) *Let φ be of C^2-class. Then φ is plurisubharmonic if and only if $dd^c\varphi \geqq 0$.*

(v) *Let $\varphi : U \to [-\infty, \infty)$ be plurisubharmonic and let λ be a monotone increasing convex function defined on $\mathbf{R}$. Then $\lambda \circ \varphi$ is plurisubharmonic. Here, $\lambda(-\infty) = \lim_{t\to-\infty}\lambda(t)$.*

(vi) *Let $\varphi_\nu : U \to [-\infty, \infty)$, $\nu = 1, 2, \ldots$, be monotone decreasing plurisubharmonic functions. Then the limit function $\varphi(z) = \lim_{\nu\to\infty}\varphi_\nu(z)$ is plurisubharmonic.*

(vii) *For finitely many plurisubharmonic functions $\varphi_\nu : U \to [-\infty, \infty)$, $1 \leqq \nu \leqq l$, $\varphi(z) = \max_{1\leqq\nu\leqq l}\varphi_\nu(z)$ is plurisubharmonic, too.*

[2] In general, a function $\psi : W \to [-\infty, \infty)$ defined on an open subset W of $\mathbf{R}^n$ is said to be subharmonic if ψ is upper semicontinuous and satisfies the submean property in the sense of (2.1.25).

Here we explain the notion of currents, limited to what we will need. Cf. Noguchi–Ochiai [90] (Ochiai–Noguchi [84]) for more detailed treatment. In general a differential form with coefficients in distributions in the sense of Schwartz is called a *current*. We introduce only $(1,1)$ currents that will be needed. We consider only the case where the domain is an open subset U of $\mathbf{C}^m$. A complex-valued measure of the form $\mu = \mu' + i\mu''$ with real-valued Radon measures μ' and μ'' on U is called a *complex Radon measure* on U. Its complex conjugate is defined by $\bar{\mu} = \mu' - i\mu''$. We consider a $(1,1)$ current $T = \sum T_{j\bar{k}} \frac{i}{2} dz_j \wedge d\bar{z}_k$ with coefficients of complex-valued Radon measures $T_{j\bar{k}}$, $1 \leqq j, k \leqq m$. The complex conjugate of T is defined by

$$\bar{T} = \sum \bar{T}_{j\bar{k}} \frac{-i}{2} d\bar{z}_j \wedge dz_k = \sum \bar{T}_{j\bar{k}} \frac{i}{2} dz_k \wedge d\bar{z}_j .$$

If $T = \bar{T}$, i.e., $\bar{T}_{j\bar{k}} = T_{k\bar{j}}$ (hermitian), T is called a *real current*. If for every vector $(\xi_j) \in \mathbf{C}^m$

$$\sum_{j,k} T_{j\bar{k}} \xi_j \bar{\xi}_k$$

is a positive Radon measure, T is called a $(1,1)$ *positive current*, and we write $T \geqq 0$.

For two real $(1,1)$ currents T, S on U we write $T \geqq S$ ($S \leqq T$) if $T - S \geqq 0$.

Take $\chi(z) = \chi(\|z\|) \in C_0^\infty(\mathbf{C}^m)$ so that $\chi(z) \geqq 0$, $\operatorname{Supp} \chi \subset B(1)$, and

$$\int \chi(z) \alpha^m = 1.$$

Set $\chi_\varepsilon(z) = \chi(\varepsilon^{-1} z)\varepsilon^{-2m}$, $\varepsilon > 0$. Let φ be a plurisubharmonic function on U. The *smoothing* $\varphi_\varepsilon(z)$ of φ is defined by

$$\begin{aligned} \varphi_\varepsilon(z) = \varphi * \chi_\varepsilon(z) &= \int_{\mathbf{C}^m} \varphi(w) \chi_\varepsilon(w - z) \alpha^m(w) \\ &= \int_{\mathbf{C}^m} \varphi(z + w) \chi_\varepsilon(w) \alpha^m(w), \quad z \in U_\varepsilon. \end{aligned}$$

Then $\varphi_\varepsilon(z) \in C^\infty(U_\varepsilon)$ and it is plurisubharmonic. Since $\chi(w) = \chi(\|w\|)$, one gets

$$\begin{aligned} \varphi_\varepsilon(z) &= \int_{\mathbf{C}^m} \varphi(z + \varepsilon w) \chi(w) \alpha^m(w) \\ &= \int_{\mathbf{C}^m} \alpha^m(w) \frac{1}{2\pi} \int_0^{2\pi} d\theta \, \varphi\big(z + \varepsilon e^{i\theta} w\big) \chi(w) \\ &\geqq \varphi(z) \int_{\mathbf{C}^m} \chi(w) \alpha^m = \varphi(z). \end{aligned}$$

Therefore by Theorem 2.1.12 (iii) φ_ε is monotone decreasing as $\varepsilon \searrow 0$. Since φ is upper semicontinuous, in the same way as in the proof of Theorem 2.1.12 (ii), one verifies that $\varphi_\varepsilon(z) \searrow \varphi(z)$.

Since $\sum \frac{\partial^2 \varphi_\varepsilon}{\partial z_j \partial \bar{z}_k} \xi_j \bar{\xi}_k \geqq 0$ for every vector $(\xi_1, \ldots, \xi_m) \in \mathbf{C}^m$,

$$\sum \frac{\partial^2 [\varphi]}{\partial z_j \partial \bar{z}_k} \xi_j \bar{\xi}_k,$$

where the notation $\frac{\partial^2 [\varphi]}{\partial z_j \partial \bar{z}_k}$ is used in the sense of Schwartz distributions, defines a positive Radon measure, so that

$$dd^c[\varphi] = \sum \frac{\partial^2 [\varphi]}{\partial z_j \partial \bar{z}_k} \frac{i}{2\pi} dz_j \wedge d\bar{z}_k \geqq 0.$$

Hence we have the following.

Theorem 2.1.27 *Let $\varphi : U \to [-\infty, \infty)$ be a plurisubharmonic function such that $\varphi \not\equiv -\infty$ on each connected component of U.*

(i) *$dd^c[\varphi] \geqq 0$ and the coefficients $\frac{\partial^2 [\varphi]}{\partial z_j \partial \bar{z}_k}$ are complex Radon measures which are absolutely continuous with respect to the trace $\sum_{j=1}^m \frac{\partial^2 [\varphi]}{\partial z_j \partial \bar{z}_j}$.*

(ii) *The smoothing $\varphi_\varepsilon(z)$ converges monotone decreasingly to $\varphi(z)$ as $\varepsilon \searrow 0$.*

(iii) *For every $B(a; R) \subset U$ with $0 < s < r < R$,*

$$-\infty < \int_{\|z\|=s} \varphi(a+z)\gamma(z) \leqq \int_{\|z\|=r} \varphi(a+z)\gamma(z) < \infty. \tag{2.1.28}$$

Proof The absolute continuity of (i) follows from the positivity $dd^c[\varphi] \geqq 0$. (ii) was already shown. Only (iii) remains.

First note that by Theorem 2.1.26 (ii) φ is locally integrable. We infer from (2.1.25) and Fubini's theorem that there is a subset $E \subset (0, R)$ of Lebesgue measure zero with finite $\int_{\|z\|=t} \varphi(a+z)\gamma(z)$ for $t \in (0, R) \setminus E$. On the other hand, for every $t \in (0, R)$ and $\vartheta \in [0, 2\pi]$ the $\mathbf{C}^*$-invariance $\gamma(te^{i\vartheta} z) = \gamma(z)$ implies

$$\begin{aligned} \int_{\|z\|=t} \varphi(a+z)\gamma(z) &= \int_{\|z\|=1} \varphi\big(a + te^{i\vartheta} z\big)\gamma(z) \\ &= \int_{\|z\|=1} \int_0^{2\pi} \varphi\big(a + te^{i\theta} z\big) \frac{d\theta}{2\pi} \gamma(z). \end{aligned}$$

It follows from this and Theorem 2.1.12 (iii) that for every $0 < s < r < R$

$$\int_{\|z\|=s} \varphi(a+z)\gamma(z) \leqq \int_{\|z\|=r} \varphi(a+z)\gamma(z) < \infty.$$

Applying this to $0 < t < s$, $t \notin E$, we see that

$$-\infty < \int_{\|z\|=t} \varphi(a+z)\gamma(z) \leqq \int_{\|z\|=s} \varphi(a+z)\gamma(z).$$

Thus (2.1.28) is shown. □

As in Theorem 2.1.13 (i) the following holds.

Theorem 2.1.29 *The plurisubharmonicity is a local property.*

Let $\varphi \not\equiv -\infty$ be a plurisubharmonic function on $\mathbf{C}^m$. In the sense of currents

$$dd^c[\varphi] = \sum \frac{\partial^2[\varphi]}{\partial z_j \partial \bar{z}_k} \frac{i}{2\pi} dz_j \wedge d\bar{z}_k \geqq 0.$$

Then

$$dd^c[\varphi] \wedge \alpha^{m-1} = \left((m-1)! \sum_{j=1}^{m} \frac{\partial^2[\varphi]}{\partial z_j \partial \bar{z}_j} \right) \bigwedge_{j=1}^{m} \frac{i}{2\pi} dz_j \wedge d\bar{z}_j$$

is a volume form with a positive Radon measure as coefficient. Therefore, for a Borel measurable subset $E \subset \mathbf{C}^m$ and a Borel measurable function ψ the integral

$$\int_E \psi dd^c[\varphi] \wedge \alpha^{m-1}$$

is defined; in particular, we set

$$n\big(t, dd^c[\varphi]\big) = \frac{1}{t^{2m-2}} \int_{B(t)} dd^c[\varphi] \wedge \alpha^{m-1}, \quad t > 0. \tag{2.1.30}$$

Lemma 2.1.31 *The function $n(t, dd^c[\varphi])$ is left-continuous in $t > 0$ and monotone increasing.*

Proof By the inner regularity of Radon measure, $\int_{B(t)} dd^c[\varphi] \wedge \alpha^{m-1}$ is left-continuous in $t > 0$, and so is $n(t, dd^c[\varphi])$.

Let φ_ε, $\varepsilon > 0$ be the smoothing of φ. By making use of (2.1.22) one gets for $t > s > 0$

$$\begin{aligned}
&\frac{1}{t^{2m-2}} \int_{B(t)} dd^c \varphi_\varepsilon \wedge \alpha^{m-1} - \frac{1}{s^{2m-2}} \int_{B(s)} dd^c \varphi_\varepsilon \wedge \alpha^{m-1} \\
&= \frac{1}{t^{2m-2}} \int_{\|z\|=t} d^c \varphi_\varepsilon \wedge \alpha^{m-1} - \frac{1}{s^{2m-2}} \int_{\|z\|=s} d^c \varphi_\varepsilon \wedge \alpha^{m-1} \\
&= \int_{\|z\|=t} d^c \varphi_\varepsilon \wedge \beta^{m-1} - \int_{\|z\|=s} d^c \varphi_\varepsilon \wedge \beta^{m-1} \\
&= \int_{s<\|z\|<t} dd^c \varphi_\varepsilon \wedge \beta^{m-1}.
\end{aligned}$$

Since $\beta \geqq 0$ (semi-positive), the integral is non-negative. Now we let $\varepsilon \to 0$. Then there is a subset $E \subset (0, \infty)$ of Lebesgue measure zero such that for $0 < s < t$

outside E

$$n\big(t, dd^c[\varphi]\big) - n\big(s, dd^c[\varphi]\big) = \int_{s<\|z\|<t} dd^c[\varphi] \wedge \beta^{m-1} \geqq 0. \tag{2.1.32}$$

For every $0 < s < t$ we take sequences $s_\nu \nearrow s$ and $s < t_\nu \nearrow t$ with $s_\nu \notin E, t_\nu \notin E$, $\nu = 1, 2, \ldots$. Then $n(s_\nu, dd^c[\varphi]) \leqq n(t_\nu, dd^c[\varphi])$. As $\nu \to \infty$, the left-continuity implies

$$n\big(s, dd^c[\varphi]\big) \leqq n\big(t, dd^c[\varphi]\big). \qquad \square$$

It follows from Lemma 2.1.31 that at every point $a \in \mathbf{C}^m$ the following limit exists:

$$\mathscr{L}\big(a; dd^c[\varphi]\big) = \lim_{t\to 0} \frac{1}{t^{2m-2}} \int_{B(a;t)} dd^c[\varphi] \wedge \alpha^{m-1}.$$

The limit $\mathscr{L}(a; dd^c[\varphi])$ is called the *Lelong number* of the current $dd^c[\varphi]$ at a, and plays an important role in various aspects of complex analysis. For example, for a given $\delta > 0$ the set $\{a \in \mathbf{C}^m; \mathscr{L}(a; dd^c[\varphi]) \geqq \delta\}$ forms an analytic subset (Y.-T. Siu's Theorem; cf. Siu [74]; Ohsawa [98]; Hörmander [89]).

Lemma 2.1.33 (Jensen's formula) *Let $\varphi \not\equiv -\infty$ be a plurisubharmonic function on $\mathbf{C}^m$. Then for every $0 < s < r$*

$$\begin{aligned} \int_{\|z\|=r} \varphi\gamma - \int_{\|z\|=s} \varphi\gamma &= 2\int_s^r \frac{dt}{t^{2m-1}} \int_{B(t)} dd^c[\varphi] \wedge \alpha^{m-1} \\ &= 2\int_s^r \frac{dt}{t} \int_{B(t)\setminus\{0\}} dd^c[\varphi] \wedge \beta^{m-1} \\ &\quad + 2\mathscr{L}\big(0; dd^c[\varphi]\big) \log \frac{r}{s}. \end{aligned} \tag{2.1.34}$$

Proof Take the smoothing φ_ε of φ. Since $d\gamma = 0$,

$$\begin{aligned} \int_{\|z\|=r} \varphi_\varepsilon\gamma - \int_{\|z\|=s} \varphi_\varepsilon\gamma &= \int_{\{s<\|z\|<r\}} d\varphi_\varepsilon \wedge \gamma \\ &= \int_{\{s<\|z\|<r\}} d\log\|z\|^2 \wedge d^c\varphi_\varepsilon \wedge \big(dd^c \log\|z\|^2\big)^{m-1} \\ &= 2\int_s^r \frac{dt}{t} \int_{\|z\|=t} d^c\varphi_\varepsilon \wedge \big(dd^c \log\|z\|^2\big)^{m-1} \\ &= 2\int_s^r \frac{dt}{t} \int_{\|z\|=t} d^c\varphi_\varepsilon \wedge \frac{\alpha^{m-1}}{t^{2(m-1)}} \end{aligned} \tag{2.1.35}$$

(continued)

$$= 2\int_s^r \frac{dt}{t^{2m-1}} \int_{\|z\|=t} d^c\varphi_\varepsilon \wedge \alpha^{m-1}$$

$$= 2\int_s^r \frac{dt}{t^{2m-1}} \int_{B(t)} dd^c\varphi_\varepsilon \wedge \alpha^{m-1}.$$

As $\varepsilon \searrow 0$, $\varphi_\varepsilon \searrow \varphi$. By the monotone convergence theorem of Lebesgue the first integral of (2.1.35) converges to the integral of φ.

On the other hand, for almost all t with respect to the Lebesgue measure

$$\int_{B(t)} dd^c\varphi_\varepsilon \wedge \alpha^{m-1} \to \int_{B(t)} dd^c\varphi \wedge \alpha^{m-1} \quad (\varepsilon \to 0).$$

It follows from the definition that for $0 < \varepsilon < 1$ and $t \leqq r$

$$0 \leqq \int_{B(t)} dd^c\varphi_\varepsilon \wedge \alpha^{m-1} \leqq \int_{B(r+1)} dd^c[\varphi] \wedge \alpha^{m-1} < \infty.$$

By Lebesgue's bounded convergence theorem we have

$$2\int_s^r \frac{dt}{t^{2m-1}} \int_{B(t)} dd^c\varphi_\varepsilon \wedge \alpha^{m-1} \to 2\int_s^r \frac{dt}{t^{2m-1}} \int_{B(t)} dd^c[\varphi] \wedge \alpha^{m-1} \tag{2.1.36}$$

as $\varepsilon \to 0$. Therefore the first equality of (2.1.34) is inferred.

Letting $s \to 0$ ($s \notin E$) in (2.1.32), we have

$$\frac{1}{t^{2m-2}} \int_{B(t)} dd^c[\varphi] \wedge \alpha^{m-1} = \int_{B(t)\setminus\{0\}} dd^c[\varphi] \wedge \beta^{m-1} + \mathscr{L}\big(0; dd^c[\varphi]\big).$$

This implies the second equality of (2.1.34). □

Corollary 2.1.37 *Jensen's formula* (2.1.34) *holds for Borel measurable functions* φ *on* $\mathbf{C}^m$ *if for every* $a \in \mathbf{C}^m$ *there exist plurisubharmonic functions* φ_1, φ_2 *in a neighborhood* U *of* a *such that* φ *is written as* $\varphi = \varphi_1 - \varphi_2$ *on* U.

Proof Let $\{U_j\}_{j=1}^\infty$ be a locally finite open covering of $\mathbf{C}^m$ such that there are plurisubharmonic functions $\varphi_{j1}, \varphi_{j2}$ on U_j satisfying $\varphi = \varphi_{j1} - \varphi_{j2}$. Let $\{\eta_j\}$ be a partition of unity subordinated to $\{U_j\}$. It suffices to show the first equality of (2.1.34). Take a number j_0 so that $U_j \cap \overline{B(r)} = \emptyset$, $j \geqq j_0$. Put $\delta = \min\{d(\operatorname{Supp}\eta_j, \partial U_j); 1 \leqq j \leqq j_0\}$. For $0 < \varepsilon < \delta$ and $z \in \overline{B(r)} \cap \operatorname{Supp}\eta_j$

$$\varphi_\varepsilon(z) = \varphi_{j1\varepsilon}(z) - \varphi_{j2\varepsilon}(z).$$

Thus

$$\varphi_\varepsilon(z) = \sum_{j=1}^{j_0} \eta_j(z)\big(\varphi_{j1\varepsilon}(z) - \varphi_{j2\varepsilon}(z)\big), \quad z \in \overline{B(r)}.$$

By the same computation as in (2.1.35) we have

$$\sum_{j=1}^{j_0} \int_{\|z\|=r} \eta_j \varphi_{j1\varepsilon} \gamma - \sum_{j=1}^{j_0} \int_{\|z\|=s} \eta_j \varphi_{j2\varepsilon} \gamma$$
$$= 2 \int_s^r \frac{dt}{t^{2m-1}} \int_{B(t)} dd^c \varphi_\varepsilon \wedge \alpha^{m-1}.$$

The monotone convergence theorem of Lebesgue implies that

$$\int_{\{\|z\|=r\}} \eta_j \varphi_{ji\varepsilon} \gamma \to \int_{\{\|z\|=r\}} \eta_j \varphi_{ji} \gamma \quad (\varepsilon \to 0,\ i = 1, 2).$$

Since the coefficients of $dd^c[\varphi]$ are complex Radon measures, the convergence of (2.1.36) holds. Thus the required formula follows. □

Supplement (Currents) In general, on a differentiable manifold M satisfying the second countability axiom a differential form with coefficients in Schwartz' distributions is called a *current*. In the space of currents, the exterior differential operator d is defined in the sense of derivations of Schwartz' distributions. If M is a complex manifold, the operators $\partial, \bar{\partial}, d^c$ are defined similarly (cf., e.g., Lelong [68]; Noguchi–Ochiai [90] (Ochiai–Noguchi [84])). Let $(z_1, \ldots, z_m)$ be a holomorphic local coordinate system of M ($m = \dim M$). If a current T on M is written as

$$T = \sum_{|I|=p, |J|=q} T_{I\bar{J}} dz^I \wedge d\bar{z}^J,$$
$$dz^I = \bigwedge_{i \in I} dz_i, \quad d\bar{z}^J = \bigwedge_{j \in J} d\bar{z}_j$$

with multi-index sets $I, J \subset \{1, \ldots, m\}$, T is said to be of type (p, q), or called a (p, q) current. The complex conjugate $\bar{T}$ is defined by

$$\bar{T}_{I\bar{J}}(\phi) = \overline{T_{I\bar{J}}(\bar{\phi})},$$
$$\bar{T} = \sum_{|I|=p, |J|=q} \bar{T}_{I\bar{J}} d\bar{z}^I \wedge dz^J,$$

where ϕ is a test function. A current T of type (p, p) is called a *positive current* if T is real, i.e., $\bar{T} = T$, and for every C^∞ $(1, 0)$ form η_j, $1 \leqq j \leqq m - p$

$$T \wedge i\eta_1 \wedge \bar{\eta}_1 \wedge \cdots \wedge i\eta_{m-p} \wedge \bar{\eta}_{m-p}$$

is a positive Radon measure. In this case, we write $T \geqq 0$. For two real (p, p) currents T, T', we write $T \geqq T'$ for $T - T' \geqq 0$.

2.2 Poincaré–Lelong Formula

Let $U \subset \mathbf{C}^m$ be an open subset.

Definition 2.2.1 A closed subset $A \subset U$ is said to be *analytic*, if for every point $a \in A$ there are a neighborhood $W \subset U$ of a and holomorphic functions $g_1, \ldots, g_l$ on W $(l < \infty)$ satisfying

$$A \cap W = \{g_1 = \cdots = g_l = 0\}.$$

When the closedness of A is not assumed, A is called a locally closed analytic subset. In particular, if the above g_j, $1 \leqq j \leqq l$, can be taken so that their differentials at $a \in A$

$$dg_1(a), \ldots, dg_l(a)$$

are linearly independent, a is called a *regular* or *non-singular point* of A; in this case, if W is chosen sufficiently small, $A \cap W$ is a closed complex submanifold of W. A point of A which is not non-singular is called a *singular point*; the subset of all singular points of A is denoted by $S(A)$. Set $R(A) = A \setminus S(A)$. If $S(A) = \emptyset$, A is said to be non-singular.

We describe elementary and useful properties of analytic subsets and of plurisubharmonic functions without proofs, for which cf. Oka [Iw] VII, [50], [51], Grauert–Remmert [84], Gunning–Rossi [65], Hervé [63], Narasimhan [66], Noguchi–Ochiai [90] (Ochiai–Noguchi [84]), Nishino [96], Ohsawa [98], Noguchi [13].

Theorem 2.2.2 *An analytic subset $A \subset U$ satisfies the following*:

(i) *$S(A)$ is an analytic subset and nowhere dense in A.*

(ii) *Let $R(A) = \bigcup_\lambda A'_\lambda$ be the decomposition into the connected components. Then the closure $A_\lambda = \bar{A}'_\lambda$ is analytic, and $A = \bigcup_\lambda A_\lambda$ is a locally finite covering.*

(iii) *If $A_\lambda \subset U$, $\lambda \in \Lambda$ are analytic subsets of U, then so is the intersection $\bigcap_{\lambda \in \Lambda} A_\lambda$.*

An analytic subset B is said to be *irreducible* if there are no analytic subsets $B_i \subsetneqq B$, $i = 1, 2$, with $B = B_1 \cup B_2$. The above A_λ are known to be irreducible. Each A_λ is called an *irreducible component* of A. Since A'_λ is a locally closed submanifold, its (complex) dimension is denoted by $\dim A'_\lambda$, and one sets

$$\dim A_\lambda = \dim A'_\lambda, \qquad \dim_a A = \max_{a \in A_\lambda} \dim A_\lambda,$$
$$\operatorname{codim}_a A = m - \dim_a A, \qquad \dim A = \max_a \dim_a A,$$
$$\operatorname{codim} A = m - \dim A.$$

In particular, the following holds:

$$\dim S(A) < \dim A.$$

If $\dim_a A = \dim A$ at all points $a \in A$, A is said to be of pure dimension $\dim A$.

Let $a \in A$. Then there is a linear subspace $L \cong \mathbf{C}^l$ passing through a such that a is isolated in $L \cap A$. Moreover, there is a direct decomposition $\mathbf{C}^m \cong \mathbf{C}^{m-l} \times L \ni a = (a_1, a_2)$ by linear subspaces, and there are neighborhoods $U_1 \subset \mathbf{C}^{m-l}$ of a_1, $U_2 \subset L$ of a_2 such that the projection $p : A \cap (U_1 \times U_2) \to U_1$ satisfies the following.

Theorem 2.2.3 *Let the notation be as above.*

(i) $p : A \cap (U_1 \times U_2) \to U_1$ *is surjective, proper, finite, and* $p^{-1}a_1 = \{a\}$.
(ii) *There is a proper analytic subset* $S \subset U_1$ *such that the restriction of* p

$$p|_{A\cap(U_1\times U_2)\setminus p^{-1}S} : A \cap (U_1 \times U_2) \setminus p^{-1}S \to U_1 \setminus S$$

is a finite unramified covering.
(iii) *The collection of such* l*-dimensional linear subspaces* L *of* $\mathbf{C}^m$ *satisfying the above* (i) *and* (ii) *forms an open dense subset in the Grassmann space of* l*-dimensional linear subspaces of* $\mathbf{C}^m$.

If there is one holomorphic function $\phi \not\equiv 0$ in a neighborhood W of every point $a \in A$ satisfying

$$A \cap W = \{\phi = 0\},$$

A is called a *complex hypersurface* or simply *hypersurface*. A hypersurface is an analytic subset of pure dimension $m - 1$.

Let $U_i \subset \mathbf{C}^{n_i}$, $i = 1, 2$, be open sets, and let $X_i \subset U_i$, $i = 1, 2$, be analytic subsets. A map $\varphi : X_1 \to X_2$ is said to be holomorphic if for every point $a \in X_1$ there are a neighborhood W of a in U_1 and holomorphic functions φ_j, $1 \leqq j \leqq n_2$, on W satisfying

$$\varphi(x) = \big(\varphi_1(x), \ldots, \varphi_{n_2}(x)\big), \quad x \in W \cap X_1.$$

In particular, when $X_2 = \mathbf{C}$, φ is called a holomorphic function. If the inverse $\varphi^{-1} : X_2 \to X_1$ exists and is holomorphic, φ is called a biholomorphic map or a *biholomorphism*, and X_1 is said to be biholomorphic to X_2.

Let X be a Hausdorff topological space. Let X carry an open covering $\{X_\lambda\}_{\lambda\in\Lambda}$ such that there exist analytic subsets $Z_\lambda \subset U_\lambda$ with open subsets $U_\lambda \subset \mathbf{C}^{n_\lambda}$, homeomorphisms $\varphi_\lambda : X_\lambda \to Z_\lambda$, $\lambda \in \Lambda$, and the restrictions

$$\varphi_\mu \circ \varphi_\lambda^{-1}|_{Z_\lambda\cap\varphi_\lambda(X_\lambda\cap X_\mu)} : Z_\lambda \cap \varphi_\lambda(X_\lambda \cap X_\mu) \to Z_\mu \cap \varphi_\mu(X_\lambda \cap X_\mu)$$

are biholomorphisms for all $\lambda, \mu \in \Lambda$. Then X is called a *complex space*.

If we can take $Z_\lambda = U_\lambda$, then X is called a *complex manifold*.

Holomorphic functions on a complex space, holomorphic maps between complex spaces, and analytic subsets are defined using the coordinate charts ϕ_λ; e.g., a function $f : X \to \mathbf{C}$ is *holomorphic*, if all the functions $f|_{X_\lambda} \circ \phi_\lambda^{-1} : Z_\lambda \to \mathbf{C}$ are holomorphic.

Similarly, a function $f : X \to \mathbf{C}$ on a complex manifold X is called *plurisubharmonic* if all the functions $f|_{X_\lambda} \circ \phi_\lambda^{-1} : Z_\lambda \to \mathbf{C}$ are plurisubharmonic. (Note that given a biholomorphic map $\alpha : Z \to Z'$ a function f on Z' is plurisubharmonic if and only if $f \circ \alpha$ is plurisubharmonic; this is proven as in Theorem 2.1.19.)

Theorem 2.2.4 (Remmert) *Let $\phi : X \to Y$ be a proper holomorphic map between complex spaces, and let $A \subset X$ be an analytic subset. Then $\phi(A)$ is an analytic subset of Y and $\dim \phi(A) \leqq \dim A$.*

Theorem 2.2.5 (Remmert) *Let X be a complex space, and let $E \subset X$ be an analytic subset. Let $A \subset X \setminus E$ be an analytic subset such that $\min\{\dim_a A; a \in A\} > \dim E$. Then the closure $\bar{A}$ of A in X is an analytic subset of X.*

For the analytic continuation of holomorphic functions we have the following.

Theorem 2.2.6 *Let $U \subset \mathbf{C}^m$ be a domain, and let $E \subsetneqq U$ be a proper analytic subset. Let $f : U \setminus E \to \mathbf{C}$ be a holomorphic function.*

(i) (Riemann extension) *If there is a neighborhood $V \subset U$ of every point $x \in E$ such that $f|_{V \setminus E}$ is bounded, then f is uniquely extended to a holomorphic function on U.*

(ii) (Hartogs extension) *If $\operatorname{codim} E \geqq 2$, f is necessarily extended uniquely to a holomorphic function on U.*

We know similar theorems for plurisubharmonic functions (cf. Grauert–Remmert [56]; Noguchi–Ochiai [90] (Ochiai–Noguchi [84])).

Theorem 2.2.7 *Let $U \subset \mathbf{C}^m$ be a domain, and let $E \subset U$ be a proper analytic subset. Let $\psi : U \setminus E \to [-\infty, \infty)$ be a plurisubharmonic function.*

(i) (Riemann type) *If there is a neighborhood $V \subset U$ of every point $x \in E$ such that $\psi|_{V \setminus E}$ is bounded from above, then ψ extends uniquely to a plurisubharmonic function on U.*

(ii) (Hartogs type) *If $\operatorname{codim} E \geqq 2$, then ψ necessarily extends uniquely to a plurisubharmonic function on U.*

Since the above two extension theorems are local in nature, the analogous statements hold for arbitrary complex manifolds.

Take a homogeneous coordinate system $[w_0, \ldots, w_n]$ of the n-dimensional complex projective space $\mathbf{P}^n(\mathbf{C})$. A subset $X \subset \mathbf{P}^n(\mathbf{C})$ is said to be *algebraic* if there are finitely many homogeneous polynomials $P_\alpha(w_0, \ldots, w_n)$ satisfying

$$X = \bigcap_\alpha \{P_\alpha(w_0, \ldots, w_n) = 0\}.$$

If a complex space Z is biholomorphic to an algebraic subset of $\mathbf{P}^n(\mathbf{C})$, Z is called a (complex) *projective algebraic variety*. Then an algebraic subset of Z is naturally defined.

Theorem 2.2.8 (Chow) *Every analytic subset of $\mathbf{P}^n(\mathbf{C})$ is algebraic.*

The *Zariski topology* is defined on $\mathbf{P}^n(\mathbf{C})$ by taking the algebraic subsets as closed subsets. Similarly, the Zariski topology on a complex space Z (resp. its subset Y) is defined so that analytic subsets X of Z (resp. $X \cap Y$) are closed subsets.

Let U be an open subset of $\mathbf{C}^m$. Let $A \subset U$ be an analytic subset of pure dimension l, and let $\iota_{R(A)} : R(A) \to \mathbf{C}^m$ be the inclusion map. For a compact subset $K \subset U$ the integral

$$\int_{K \cap R(A)} \alpha^l = \int_{K \cap R(A)} \iota_{R(A)}^* \alpha^l$$

is considered as a measure of $K \cap A$. We show that this is finite.

Lemma 2.2.9 *Let the notation be as above.*

(i) *If $l = \dim A < m$, the Lebesgue measure of A in $\mathbf{C}^m$ is zero.*
(ii) $\int_{K \cap R(A)} \alpha^l < \infty$.

Proof (i) We use induction on l. When $l = 0$, A is a discrete subset, and so the measure is zero. Suppose that the statement holds for $\dim A < l$. We decompose $A = R(A) \cup S(A)$. The induction hypothesis implies that the measure of $S(A)$ is zero. Since $R(A)$ is a locally closed submanifold of dimension l ($<m$), its measure is zero, and hence so is the measure of A.

(ii) It suffices to show the claim in a neighborhood W of every point $a \in A$. By making use of a translation and a unitary transformation of coordinates combined with Theorem 2.2.3, we may assume that $a = 0$ and for arbitrary $1 \leqq i_1 < \cdots < i_l \leqq m$ there is a neighborhood W of 0 with projection $W = U_1 \times U_2 \subset \mathbf{C}^l \times \mathbf{C}^{m-l}$ satisfying the following properties:

$$p : (z_1, \ldots, z_m) \in A \cap W \to (z_{i_1}, \ldots, z_{i_l}) \in U_1$$

is a proper finite map, and there is an analytic subset $S \subset U_1$ of $\dim S < l$ such that

$$p|_{A \cap W \setminus p^{-1}S} : (z_1, \ldots, z_m) \in A \cap W \setminus p^{-1}S \to (z_{i_1}, \ldots, z_{i_l}) \in U_1 \setminus S$$

is a unramified covering. Denote by k its covering number. Note that

$$\alpha^l = n \cdots (n-l+1) \sum_{i_1 < \cdots < i_l} \left(\frac{i}{2\pi}\right)^l dz_{i_1} \wedge d\bar{z}_{i_1} \wedge \cdots \wedge dz_{i_l} \wedge d\bar{z}_{i_l}.$$

It is sufficient to show that for every $i_1 < \cdots < i_l$

$$\int_{R(A) \cap W} \left(\frac{i}{2\pi}\right)^l dz_{i_1} \wedge d\bar{z}_{i_1} \wedge \cdots \wedge dz_{i_l} \wedge d\bar{z}_{i_l} < \infty.$$

By (i) the Lebesgue measure of $R(A) \cap W \cap p^{-1}S$ is zero in $R(A) \cap W$. Therefore

$$\begin{aligned} &\int_{R(A) \cap W} \left(\frac{i}{2\pi}\right)^l dz_{i_1} \wedge d\bar{z}_{i_1} \wedge \cdots \wedge dz_{i_l} \wedge d\bar{z}_{i_l} \\ &= \int_{R(A) \cap W \setminus p^{-1}S} \left(\frac{i}{2\pi}\right)^l dz_{i_1} \wedge d\bar{z}_{i_1} \wedge \cdots \wedge dz_{i_l} \wedge d\bar{z}_{i_l} \end{aligned}$$

(continued)

$$= k \int_{U_1 \setminus S} \left(\frac{i}{2\pi}\right)^l dz_{i_1} \wedge d\bar{z}_{i_1} \wedge \cdots \wedge dz_{i_l} \wedge d\bar{z}_{i_l}$$

$$= k \int_{U_1} \left(\frac{i}{2\pi}\right)^l dz_{i_1} \wedge d\bar{z}_{i_1} \wedge \cdots \wedge dz_{i_l} \wedge d\bar{z}_{i_l} < \infty. \qquad \square$$

From now on we write

$$\int_{K \cap A} \alpha^l = \int_{K \cap R(A)} \alpha^l.$$

The following is immediate.

Corollary 2.2.10 *Let $B \subset A$ be an analytic subset of* $\dim B < \dim A$. *Then*

$$\int_B \alpha^l = 0.$$

The following is deduced from the above.

Theorem 2.2.11 *Let $A \subset U$ be an analytic subset of pure dimension l. Let η be a $2l$ form with coefficients which are bounded Borel functions on U with compact supports. Then*

$$\int_A \eta = \int_{R(A)} \iota^*_{R(A)} \eta$$

is defined. Let $B \subset A$ be an analytic subset of $\dim B < \dim A$ *and let χ_B be the characteristic function of the set B. Then*

$$\int_A \chi_B \eta = 0.$$

Lemma 2.2.12 *Let $A \subset U$ be an analytic subset of dimension at most $m-2$. Let φ be a plurisubharmonic function on U such that $\varphi \not\equiv -\infty$ on every connected component of U, and set $dd^c[\varphi] = \sum T_{j\bar{k}} \frac{i}{2\pi} dz_j \wedge d\bar{z}_k$. Then A is of measure zero with respect to Radon measures $T_{j\bar{k}}$.*

Proof One may assume that $0 \in A$, and it suffices to show the lemma in a neighborhood of $0 \in A$. Suppose that $\dim_0 A = l$. It follows from Theorem 2.2.3 (iii) that the coordinate system $(z_1, \ldots, z_m)$ is chosen to satisfy the property: For arbitrary l coordinates $z_{\nu_1}, \ldots, z_{\nu_l}$ and the others $z_{\nu_{l+1}}, \ldots, z_{\nu_m}$ there are neighborhoods U_1 and U_2 of the origins of the $(z_{\nu_1}, \ldots, z_{\nu_l})$-space and the $(z_{\nu_{l+1}}, \ldots, z_{\nu_m})$-space, respectively such that the projection

$$p : (z_j) \in \left\{(z_j) \in A; (z_{\nu_1}, \ldots, z_{\nu_l}) \in U_1, (z_{\nu_{l+1}}, \ldots, z_{\nu_m}) \in U_2\right\}$$
$$\to (z_{\nu_1}, \ldots, z_{\nu_l}) \in U_1$$

satisfies Theorem 2.2.3 (i), (ii). By Theorem 2.1.27 (i) $T_{j\bar{k}}$ is absolutely continuous with respect to $\sum_j T_{j\bar{j}}$. Therefore it is sufficient to prove that $T_{j\bar{j}}(A \cap (U_1 \times U_2)) = 0$, $1 \leqq j \leqq m$. For instance, let $j = m$. Then there are neighborhoods V_1 of $0 \in \mathbf{C}^{m-1}$ and V_2 of $0 \in \mathbf{C}$ such that

$$q : (z_1, \ldots, z_{m-1}, z_m) \in (V_1 \times V_2) \cap A \to (z_1, \ldots, z_{m-1}) \in V_1$$

is proper and finite. By φ being plurisubharmonic and the definition we have

$$\begin{aligned} (2.2.13) \qquad & T_{m\bar{m}}\big((V_1 \times V_2) \cap A\big) \\ &= \int_{(V_1 \times V_2) \cap A} \frac{\partial^2[\varphi]}{\partial z_m \partial \bar{z}_m} \bigwedge_{j=1}^{m} \frac{i}{2} dz_j \wedge d\bar{z}_j \\ &= \int_{q(A)} \bigwedge_{j=1}^{m-1} \frac{i}{2} dz_j \wedge d\bar{z}_j \int_{V_2} \frac{\partial^2[\varphi(\cdot, z_m)]}{\partial z_m \partial \bar{z}_m} \frac{i}{2} dz_m \wedge d\bar{z}_m. \end{aligned}$$

It follows from Fubini's theorem that for $z' = (z_1, \ldots, z_{m-1})$

$$z' \in V_1 \to \int_{V_2} \frac{\partial^2[\varphi(z', z_m)]}{\partial z_m \partial \bar{z}_m} \geqq 0$$

is an integrable function on V_1. By Theorem 2.2.4 $q(A)$ is an analytic subset of V_1 of dimension at most l ($\leqq m-1$). Lemma 2.2.9 (i) implies that $q(A)$ is of Lebesgue measure zero in V_1. Thus (2.2.13) implies that $T_{m\bar{m}}((V_1 \times V_2) \cap A) = 0$. □

Let M be a complex manifold of dimension m. It is clear that Theorem 2.2.11 and Lemma 2.2.12 hold generally on M.

Let $\{A_\lambda\}$ be a locally finite family of hypersurfaces of M. The formal sum

$$\sum_\lambda k_\lambda A_\lambda$$

with integral coefficients $k_\lambda \in \mathbf{Z}$ is called a *divisor*, and the $\mathbf{Z}$-module generated by them is called the *divisor group*. For a given divisor D on M there are uniquely distinct irreducible hypersurfaces D_λ of M and $k_\lambda \in \mathbf{Z} \setminus \{0\}$ such that $D = \sum k_\lambda D_\lambda$ (the *irreducible decomposition*). Each D_λ is called an irreducible component of D. The hypersurface $\operatorname{Supp} D = \bigcup D_\lambda$ is called the *support* of D. If $D = \sum_\lambda k_\lambda A_\lambda$ with $k_\lambda \geqq 0$, D is called an *effective divisor*, and written as $D \geqq 0$. If there is no confusion, the notation D may be used for $\operatorname{Supp} D$. For two divisors D, D' we write $D \geqq D'$ if $D - D' \geqq 0$.

Let $D = \sum k_\lambda D_\lambda$ be the irreducible decomposition of a divisor D. If all $k_\lambda = 1$, then D is called a *reduced divisor*.

Let f be a holomorphic function on M. Then a hypersurface $A = \{f = 0\}$ is defined. Let $A = \bigcup_\lambda A_\lambda$ be the irreducible decomposition. The zero degree m_λ of f at a point x of every $R(A_\lambda) \setminus S(A)$ is naturally defined, independently from the choice of x because $R(A_\lambda) \setminus S(A)$ is connected. Thus a divisor

$$(f) = \sum m_\lambda A_\lambda$$

is determined by f.

A *meromorphic function* f on M is a function locally expressed by the ratio $f = g/h$ of two holomorphic functions g, h with $h \not\equiv 0$. Then the divisor (f) locally expressed by $(g) - (h)$ is defined globally on M. Let $(f) = \sum_\lambda m_\lambda D_\lambda$ be the irreducible decomposition and set

$$(f)_0 = \sum_{m_\lambda > 0} m_\lambda D_\lambda, \qquad (f)_\infty = \sum_{m_\lambda < 0} -m_\lambda D_\lambda.$$

Then $(f)_0$ is called the *zero divisor* of f and $(f)_\infty$ is called the *polar divisor* of f.

Every divisor is locally expressed by the divisor of a meromorphic function; for some special M there exists a global expression.

Theorem 2.2.14 *If M is biholomorphic to $\mathbf{C}^m$ or $B(r)$, then for a divisor D on M, there exists a meromorphic function (f) on M such that $(f) = D$.*

As for Hartogs extension (cf. Theorem 2.2.7 (ii)) we have the following.

Theorem 2.2.15 *Let $E \subset M$ be an analytic subset which has a codimension of at least two everywhere. Then every meromorphic function on $M \setminus E$ extends meromorphically over M.*

Proof This is a local property, so that M is assumed to be an open ball of $\mathbf{C}^m$. Let f be a meromorphic function on $M \setminus E$. Then the support of the polar divisor $(f)_\infty$ of f extends uniquely to an analytic subset of M by Theorem 2.2.5. Therefore, $(f)_\infty$ is an effective divisor on M. By Theorem 2.2.14 there is a holomorphic function g on M such that $(g) = (f)_\infty$, so that gf is holomorphic in $M \setminus E$. Then Theorem 2.2.7 (ii) implies that gf extends holomorphically to a holomorphic function h on M. Thus, $f = h/g$ is meromorphic on M. □

Let $D = \sum k_\lambda A_\lambda$ be a divisor on M. For a $2m-2$ form η on M whose coefficients are locally bounded Borel-measurable functions and whose support is compact, a current by integration

$$D(\eta) = \int_D \eta = \sum_\lambda k_\lambda \int_{A_\lambda} \eta$$

is defined.

Theorem 2.2.16 (Poincaré–Lelong formula) *Let $f \not\equiv 0$ be a meromorphic function on M and let η be a $2m-2$ form of C^2-class on M with compact support. Then,*

$$\int_{(f)} \eta = \int_M \log|f|^2 dd^c\eta = \int_M dd^c\big[\log|f|^2\big] \wedge \eta.$$

Here dd^c is taken in the sense of currents. That is, as currents,

$$dd^c\big[\log|f|^2\big] = (f).$$

Proof The given formula trivially holds outside $\mathrm{Supp}(f)$. It suffices to show it locally in a neighborhood of every point of $\mathrm{Supp}(f)$. Hence one may assume that f is holomorphic. Note that $\log|f|^2$ $(\not\equiv -\infty)$ is a plurisubharmonic function. The dimension of the set S of singular points of $\mathrm{Supp}(f)$ is at most $m-2$. It follows from Theorem 2.2.11 and Lemma 2.2.12 that S is a measure-zero set with respect to the currents of both sides of the formula. Therefore it is sufficient to show

$$\int_{(f)\setminus S} \eta = \int_{M\setminus S} \log|f|^2 dd^c\eta \quad (S \cap \mathrm{Supp}\,\eta = \emptyset).$$

Take an arbitrary point $a \in \mathrm{Supp}(f) \setminus S$. Choosing a sufficiently small neighborhood W of a, we have $W \cap S = \emptyset$ and a coordinate system $(w_1, \ldots, w_m)$ in W such that

(i) $\mathrm{Supp}(f) \cap W = \{w_1 = 0\}$,
(ii) $f(w) = (w_1)^k h(w)$ and $h(w) \neq 0$, $^\forall w \in W$,
(iii) $\mathrm{Supp}\,\eta \subset W$.

It is immediate that $\int_M \log|h|^2 dd^c\eta = \int_M dd^c \log|h|^2 \wedge \eta = 0$. We get

(2.2.17)
$$\begin{aligned}\int_M \log|f|^2 dd^c\eta &= \int_M \log|w_1|^{2k} dd^c\eta \\ &= \lim_{\varepsilon\to 0}\left(-2k\log\varepsilon \int_{|w_1|=\varepsilon} d^c\eta - k\int_{|w_1|\geqq\varepsilon} d\log|w_1|^2 \wedge d^c\eta\right).\end{aligned}$$

Since $\int_{|w_1|=\varepsilon} d^c\eta = O(\varepsilon)$, the first term of the right-hand side of (2.2.17) converges to 0. We calculate the second term:

$$\begin{aligned}-k\int_{|w_1|\geqq\varepsilon} d\log|w_1|^2 \wedge d^c\eta &= k\int_{\{|w_1|\geqq\varepsilon\}} d^c\log|w_1|^2 \wedge d\eta \\ &= -k\int_{\{|w_1|\geqq\varepsilon\}} d\big(d^c\log|w_1|^2 \wedge \eta\big) \\ &= k\int_{\{|w_1|=\varepsilon\}} d^c\log|w_1|^2 \wedge \eta \\ &= k\int_{\{|w_1|=\varepsilon\}} \frac{1}{2\pi} d(\arg w_1) \wedge \eta\end{aligned}$$

(continued)

$$\to k \int_{\{w_1=0\}} \eta \quad (\varepsilon \to 0).$$

Hence, this combined with (2.2.17) implies the required identity. □

It follows from Theorem 2.2.16 that the integration over D defines naturally a current with coefficients in Radon measures (Lelong [68]; Noguchi–Ochiai [90] (Ochiai–Noguchi [84])).

Let $M = \mathbf{C}^m$. Let D be an effective divisor on $\mathbf{C}^m$ with the irreducible decomposition $D = \sum_\lambda k_\lambda D_\lambda$. For $1 \leqq k \leqq \infty$ the *truncated counting functions* to level k are defined by

$$n_k(t, D) = \frac{1}{t^{2m-2}} \int_{B(t)\cap(\sum_\lambda \min\{k,k_\lambda\}D_\lambda)} \alpha^{m-1},$$

$$N_k(r, D) = \int_1^r \frac{n_k(t, D)}{t} dt, \quad r > 1, \tag{2.2.18}$$

$$n(t, D) = n_\infty(t, D), \qquad N(r, D) = N_\infty(r, D).$$

In particular, $n(t, D)$ and $N(r, D)$ are simply called the *counting functions*.

Theorem 2.2.19 *Let D be an effective divisor on $\mathbf{C}^m$. Then $n_k(t, D)$ is increasing in $t > 0$.*

Proof If we reconsider D to be $\sum_\lambda \min\{k, k_\lambda\}D_\lambda$ in (2.2.18), we may take $k = \infty$. By Theorem 2.2.14 there is a holomorphic function f on $\mathbf{C}^m$ such that $(f) = D$ and $dd^c[\log|f|^2] = D$. Therefore we see by Lemma 2.1.31 that $n(t, D)$ is increasing in $t > 0$. □

2.3 The First Main Theorem

2.3.1 Meromorphic Mappings, Divisors and Line Bundles

Let M and N be complex spaces (cf. Convention (xvi)). A *meromorphic mapping* $f : M \to N$ from M into N is a correspondence such that for a point $x \in M$ a subset $f(x) \subset N$ is assigned and the graph $\Gamma(f) = \{(x, f(x)); x \in M\} \subset M \times N$ forms an irreducible analytic subset and satisfies the following:

(i) The first projection $p : \Gamma(f) \to M$ is proper.
(ii) There is a nowhere dense analytic subset $S \subset M$ such that the restriction $p|_{\Gamma(f)\setminus p^{-1}S} : \Gamma(f) \setminus p^{-1}S \to M \setminus S$ is biholomorphic.

Therefore the restriction $f|_{M\setminus S} : M \setminus S \to N$ is a holomorphic mapping. The set $I(f)$ of points $x \in M$ with $f(x)$ containing more than one point is called the *indeterminacy locus* of f.

Theorem 2.3.1 *Let $f : M \to N$ be a meromorphic mapping. Assume that M is non-singular. Then $I(f)$ is an analytic subset of codimension $\geqq 2$ and f is holomorphic on $M \setminus I(f)$.*

Proof Note that for every $x \in M$, $p^{-1}x$ is connected. Set

$$Z = \left\{ z \in \Gamma(f); \dim_z p^{-1} p(x) \geqq 1 \right\}.$$

Then Z is an analytic subset, $\dim Z < \dim \Gamma(f) = \dim M$, and $p(Z) = I(f)$. By definition $p(Z) < \dim Z$, and hence $\operatorname{codim} I(f) \geqq 2$. Let $x \notin I(f)$. Then $p^{-1}(x) = \{(x, y)\} \in \Gamma(f)$ is a point set. By Theorem 2.2.3 there are a holomorphic local coordinate neighborhood $(V, (y_1, \ldots, y_n))$, $|y_j| < 1$ of $y \in N$, a neighborhood U of $x \in M$, and a proper analytic subset $S \subset U$ such that $f(U) \subset V$, $f|(U \setminus S)$ is represented by holomorphic functions $f_j(x) \in \Delta(1)$. It follows from Theorem 2.2.6 (i) that $f_j(x)$ are holomorphic functions on U. Thus $f|U$ is a holomorphic mapping. □

Remark 2.3.2 The above theorem holds if M is a normal complex space. For a holomorphic function which is defined outside an analytic subset Z of codimension $\geqq 2$ in a normal complex space extends holomorphically over Z; the same extension holds for plurisubharmonic functions in a normal complex space (Grauert–Remmert [56]).

Corollary 2.3.3 *If M is a Riemann surface, then a meromorphic mapping $f : M \to N$ is necessarily holomorphic.*

Remark 2.3.4 There is a one-to-one correspondence between non-constant meromorphic functions and non-constant meromorphic mappings to $\mathbf{P}^1(\mathbf{C})$.

Definition 2.3.5 A meromorphic mapping $f : M \to N$ is said to be *analytically degenerate*, if the image $f(M)$ is contained in a proper analytic subset of N; otherwise, f is said to be *analytically non-degenerate*. When N is contained in an projective algebraic manifold, we similarly define f to be *algebraically (non-)degenerate* by using algebraic subsets in place of analytic subsets.

Remark 2.3.6 If N is projective algebraic, by Theorem 2.2.8 the analytic degeneracy is the same as the algebraic degeneracy. If N is an open subset of a projective algebraic variety, the two notions are different.

Let $f : M \to N$ be a meromorphic mapping. Let $A \subset N$ be a hypersurface which is *Cartier*; that is, it is locally defined as zero locus of a single holomorphic function. Assume that $f(M) \not\subset A$. Unless $(f|_{M \setminus I(f)})^{-1} A$ is empty, it is a hypersurface of $M \setminus I(f)$. By Theorem 2.2.5 the closure of $(f|_{M \setminus I(f)})^{-1} A$ is a hypersurface, and is denoted by $f^{-1}A$, which is called the pull-back of A by f. Therefore the pull-back $(f|_{M \setminus I(f)})^* A$ as divisor, extends uniquely to a divisor f^*A on M.

On singular N we only deal with Cartier divisors defined as follows. Let $N = \bigcup_\alpha U_\alpha$ be an open covering and let ϕ_α be a meromorphic function on each U_α. We assume that ϕ_α/ϕ_β restricted to $U_\alpha \cap U_\beta$ is a holomorphic function without zero for every α and β. Let $N = \bigcup V_\lambda$ and $\{\psi_\lambda\}$ be another such family. We say that $\{\phi_\alpha\}$ and $\{\psi_\lambda\}$ are equivalent if there is a refinement $\{W_\nu\}$ of $\{U_\alpha\}$ and $\{V_\lambda\}$ such that $\phi_{\alpha(\nu)}/\psi_{\lambda(\nu)}$ restricted to W_ν is holomorphic and zero free for every ν. The equivalence class $D = [\{\phi_\alpha\}]$ is called a *Cartier divisor* on N. The restriction $D|_U$ to an open subset $U \subset N$ is naturally defined.

Let M be non-singular. Let $f : M \to N$ be a meromorphic mapping, and let D be a Cartier divisor on N. Assume that D has a representation $\phi_\alpha = \phi_{1\alpha}/\phi_{2\alpha}$ such that $\phi_{2\alpha} \circ f|_{M\setminus I(f)}$ does not vanish identically on any open subset where they are defined. Then the pull-back $(f|_{M\setminus I(f)})^*D$ by $f|_{M\setminus I(f)}$ is defined as a divisor on $M \setminus I(f)$. Then the support $\mathrm{Supp}(f|_{M\setminus I(f)})^*D$ is a hypersurface of $M \setminus I(f)$. Since $\mathrm{codim}\, I(f) \geqq 2$, it follows from Theorem 2.2.5 that the closure of $\mathrm{Supp}(f|_{M\setminus I(f)})^*D$ in M is a hypersurface of M, and hence the divisor $(f|_{M\setminus I(f)})^*D$ has a unique extension as divisor over M, which is denoted by f^*D.

For a holomorphic function ψ on N $(f|_{M\setminus I(f)})^*\psi$ is a holomorphic function on $M \setminus I(f)$. Since $\mathrm{codim}\, I(f) \geqq 2$, Theorem 2.2.6 (ii) implies that it uniquely extends to a holomorphic function on M denoted by $f^*\psi$. Let ϕ be a meromorphic function on N with local representations $\phi|_{U_\alpha} = \phi_{1\alpha}/\phi_{2\alpha}$, with an open covering $N = \bigcup_\alpha U_\alpha$. Assume that $\phi_{2\alpha} \circ f|_{N\setminus I(f)}$ does not vanish identically on any open subset where they are defined. Then $(f|_{M\setminus I(f)})^*\phi$ is a meromorphic function on $M \setminus I(f)$. Then we have a divisor $f^*(\psi)$ on M as above and so there is a holomorphic function g in a neighborhood U of an arbitrary point $x \in M$ such that $(g) + f^*(\psi)|_U$ is effective. Thus $g \cdot (f|_{U\setminus I(f)})^*\psi$ is holomorphic and by Theorem 2.2.6 (ii) extends to a holomorphic function h on U. Hence the pull-back meromorphic function $f^*\psi$ is defined locally by $\frac{h}{g}$ on U. For a plurisubharmonic function on N its pull-back by f is defined as well by Theorem 2.2.7.

Example 2.3.7 Let $M = \mathbf{C}^m$, $N = \mathbf{P}^n(\mathbf{C})$, and let $f : \mathbf{C}^m \to \mathbf{P}^n(\mathbf{C})$ be a meromorphic mapping. Let $[w_0, \ldots, w_n]$ be a homogeneous coordinate system of $\mathbf{P}^n(\mathbf{C})$. The hyperplane $\{w_j = 0\}$ is itself an effective divisor. There is an index j with $f(\mathbf{C}^m) \not\subset \{w_j = 0\}$. Changing the indices, we may assume without loss of generality that $f(\mathbf{C}^m) \not\subset \{w_0 = 0\}$. For the pull-back $f^*\{w_0 = 0\}$ as divisor, Theorem 2.2.14 implies the existence of an entire function f_0 on $\mathbf{C}^m$ such that $(f_0) = f^*\{w_0 = 0\}$. As w_j/w_0 is a meromorphic function on $\mathbf{P}^n(\mathbf{C})$, the pull-back $f^*(w_j/w_0)$ is a meromorphic function on $\mathbf{C}^m$, and $f_j = f_0 \cdot f^*(w_j/w_0)$ is holomorphic. By the construction we have

$$\mathrm{codim}\{f_0 = \cdots = f_n = 0\} \geqq 2, \qquad I(f) = \{f_0 = \cdots = f_n = 0\}.$$

We represent $f = [f_0, \ldots, f_n]$, which is called the *reduced representation*.

Let L be a complex space and let $\pi : L \to N$ be a surjective holomorphic mapping. If the following three conditions are satisfied, the triple (L, π, N) or simply L is called a *holomorphic line bundle* (or simply a *line bundle*) over N:

Condition 2.3.8 (i) There is an open covering $\{V_\lambda\}_{\lambda\in\Lambda}$ of N such that the restriction $L|_{V_\lambda} = \pi^{-1}(V_\lambda)$ admits a biholomorphic mapping $\phi_\lambda : L|_{V\lambda} \to V_\lambda \times \mathbf{C}$.

(ii) Whenever $V_\lambda \cap V_\mu \neq \emptyset$, there is a holomorphic function $\phi_{\lambda\mu}$ without zero on $V_\lambda \cap V_\mu$ such that

$$\phi_\lambda \circ \phi_\mu^{-1}|_{(V_\lambda\cap V_\mu)\times\mathbf{C}} : (x, \xi_\mu) \in (V_\lambda \cap V_\mu) \times \mathbf{C}$$
$$\longrightarrow \big(x, \phi_{\lambda\mu}(x)\xi_\mu\big) \in (V_\mu \cap V_\lambda) \times \mathbf{C}.$$

(iii) If $\lambda = \mu$, $\phi_{\lambda\lambda} = 1$.

In this case $\{V_\lambda\}$ is called a *local trivialization covering* of L and $\{\phi_{\lambda\mu}\}$ is called the *transition function system*.

At every $x \in N$ the inverse image $L_x = \pi^{-1}(x)$ is a one-dimensional complex vector space. For $y_i \in L_x$, $c_i \in \mathbf{C}$, $i = 1, 2$ the natural operation as vector space

$$c_1 y_1 + c_2 y_2 \in L_x$$

is defined by Condition 2.3.8 (ii).

A mapping $\sigma : W \to L$ from a subset $W \subset N$ into L satisfying $\pi \circ \sigma = \mathrm{id}_W$, is called a *section* of L over W. In particular, when W is open, we denote the set of holomorphic sections of L over W by $H^0(W, L)$, which naturally forms a complex vector space. We denote by $\mathscr{O}(L)$ the sheaf of germs of holomorphic sections of L.

The transition function system $\{\phi_{\lambda\mu}\}$ in Condition 2.3.8 (ii) satisfies the so-called cocycle condition:

$$\begin{aligned} \phi_{\lambda\lambda} &= 1, \\ \phi_{\lambda\mu}\phi_{\mu\lambda} &= 1 \quad (\text{on } V_\lambda \cap V_\mu), \\ \phi_{\lambda\mu}\phi_{\mu\nu}\phi_{\nu\lambda} &= 1 \quad (\text{on } V_\lambda \cap V_\mu \cap V_\nu). \end{aligned} \tag{2.3.9}$$

On the other hand, suppose that we are given a system $\{\phi_{\lambda\mu}\}$ of holomorphic functions satisfying the cocycle condition (2.3.9). Then we may construct as follows a holomorphic line bundle over N whose transition function system is $\{\phi_{\lambda\mu}\}$. First we consider the disjoint union $\sqcup_{\lambda\in\Lambda} V_\lambda \times \mathbf{C}$ of topological spaces. We introduce an equivalence relation $\sim$ for two elements (x_λ, ξ_λ), (x_μ, ξ_μ) by

$$x_\lambda = x_\mu \in N, \qquad \xi_\lambda = \phi_{\lambda\mu}(x_\mu)\xi_\mu.$$

The quotient space $L = (\sqcup_{\lambda\in\Lambda} V_\lambda \times \mathbf{C})/\sim$ constitutes a complex space, as is easily checked, and the mapping $\pi : L \to N$ projecting an equivalence class $[(x_\lambda, \xi_\lambda)]$ to $x_\lambda \in N$ is a holomorphic surjection. It is easily checked that this is what was desired.

Let $W \subset N$ be an open subset and let $S \subset W$ be an analytic subset which is nowhere dense in W. Let $\sigma : W \setminus S \to L$ be a holomorphic section. On an arbitrary $V_\lambda \cap W$ we represent $\phi_\lambda(\sigma(x)) = (x, \sigma_\lambda(x)) \in V_\lambda \times \mathbf{C}$. If $\sigma_\lambda(x)$ is a meromorphic function on $V_\lambda \cap W$, then σ is called a meromorphic section of L on W. Since the transition functions have no zero, a meromorphic section $\sigma : W \to L$ (we write it in this way) defines a Cartier divisor (σ) on W. In particular, when $W = N$, a Cartier

divisor (σ) on N is obtained. It is equivalent for σ to be a holomorphic section that $(\sigma) \geqq 0$.

Conversely, let a Cartier divisor D on N be given. We take an open covering $\{V_\lambda\}$ of N such that there exist meromorphic functions σ_λ on V_λ defining $D|_{V_\lambda}$. Then $\sigma_\lambda/\sigma_\mu$ has no zero on $V_\lambda \cap V_\mu$. Put $\phi_{\lambda\mu} = \sigma_\lambda/\sigma_\mu$. Then these are holomorphic functions on $V_\lambda \cap V_\mu$ without zero, which satisfy the cocycle condition (2.3.9). Thus a holomorphic line bundle $L(D)$ is obtained as above. By the construction $L(D)|_{V_\lambda} = V_\lambda \times \mathbf{C}$ and $\{\phi_{\lambda\mu}\}$ is the transition function system. Setting locally $x \in V_\lambda \to (x, \sigma_\lambda(x))$, we obtain a meromorphic section σ on N satisfying $(\sigma) = D$.

In the present book we deal only with holomorphic line bundles, which we call simply line bundles.

Let $\pi_i : L_i \to N$, $i = 1, 2$ be two line bundles over N. If there is a biholomorphic mapping $\psi : L_1 \to L_2$ such that $\pi_1 = \pi_2 \circ \psi$ and $\psi|_{L_{1x}} : L_{1x} \to L_{2x}$ $(x \in N)$ is a linear isomorphism, $\psi : L_1 \to L_2$ is called an *isomorphism*, and L_1 is said to be isomorphic to L_2. We identify isomorphic line bundles with each other. The line bundle $\mathbf{1}_N = N \times \mathbf{C}$ is called a trivial line bundle. A line bundle $L \to N$ is trivial if and only if there exists a holomorphic section on N without zero.

Take a local trivialization covering of both L_1 and L_2. Let $\{\phi_{i\lambda\mu}\}$ be the transition function system of L_i. The product $\{\phi_{1\lambda\mu} \cdot \phi_{2\lambda\mu}\}$ yields a line bundle $L_3 \to N$. This is called the tensor product of L_1 and L_2 and is denoted by $L_1 \otimes L_2$.

The line bundle given by $\{\phi_{1\lambda\mu}^{-1}\}$ is denoted by L_1^{-1}. Then $L_1 \otimes L_1^{-1} = \mathbf{1}_N$. We set

$$L_1^k = L_1 \otimes \cdots \otimes L_1 \quad (k\text{-times}, k \geqq 0),$$
$$L_1^k = L_1^{-1} \otimes \cdots \otimes L_1^{-1} \quad \big(|k|\text{-times}, k < 0\big).$$

For two given divisors D_i, $i = 1, 2$ on N,

$$L(D_1 + D_2) = L(D_1) \otimes L(D_2).$$

Assume for a moment that N is non-singular. Let $\{V_\lambda(x_{\lambda 1}, \ldots, x_{\lambda n})\}_{\lambda \in \Lambda}$ be a covering of holomorphic local coordinate neighborhood system of N. For $V_\lambda \cap V_\mu \neq \emptyset$ we consider the following transition of holomorphic forms:

$$dx_{\lambda 1} \wedge \cdots \wedge dx_{\lambda n} = \kappa_{\mu\lambda}(x) dx_{\mu 1} \wedge \cdots \wedge dx_{\mu n},$$
$$\kappa_{\mu\lambda} = \frac{\partial(x_{\lambda 1}, \ldots, x_{\lambda n})}{\partial(x_{\mu 1}, \ldots, x_{\mu n})} \quad \text{(Jacobian)}.$$

Since $\{\kappa_{\lambda\mu}\}$ satisfies the cocycle condition (2.3.9), a line bundle K_N over N is obtained from it, and K_N is called the *canonical bundle* over N. Meromorphic sections of K_N are identified with meromorphic n-forms.

2.3.2 *Differentiable Functions on Complex Spaces*

Here we have to be precise on the definition of C^∞ functions on a singular complex space N. There have been more than one such definitions, but here we follow

Fujiki [78a]. Since it is a local notion, we restrict ourselves for a moment to the case where N is an analytic subset of an open set $\Omega \subset \mathbf{C}^n$. Let $\mathscr{I}\langle N\rangle$ be the ideal sheaf of N in the structure sheaf $\mathscr{O}_\Omega$ of holomorphic functions over Ω. The quotient $\mathscr{O}_N = \mathscr{O}_\Omega/\mathscr{I}\langle N\rangle$ is called the structure sheaf (of holomorphic functions) over N. Since $\mathscr{I}\langle N\rangle$ is coherent[3] and we are concerned here only with a local property, we may assume that there are finitely many holomorphic functions,

2.3.10 $\tau_1, \ldots, \tau_l \in \Gamma(\Omega, \mathscr{I}\langle N\rangle)$ over Ω, generating $\mathscr{I}\langle N\rangle_x$ at every point $x \in \Omega$.

Let $\mathscr{E}_\Omega$ denote the sheaf of germs of complex-valued C^∞ functions over Ω. We denote by $\mathscr{E}\mathscr{I}\langle N\rangle$ the sheaf of ideals of $\mathscr{E}_\Omega$ generated by τ_j and $\bar{\tau}_j$, $1 \leqq j \leqq l$, over $\mathscr{E}_\Omega$; that is,

$$\mathscr{E}\mathscr{I}\langle N\rangle = \sum_{j=1}^{l}(\mathscr{E}_\Omega \tau_j + \mathscr{E}_\Omega \bar{\tau}_j) = \sum_{j=1}^{l}(\mathscr{E}_\Omega \Re\tau_j + \mathscr{E}_\Omega \Im\tau_j). \tag{2.3.11}$$

Then we have that $\mathscr{I}\langle N\rangle \subset \mathscr{E}\mathscr{I}\langle N\rangle \subset \mathscr{E}_\Omega$. We define the sheaf of germs of C^∞ functions over N by the quotient

$$\mathscr{E}_N = \mathscr{E}_\Omega/\mathscr{E}\mathscr{I}\langle N\rangle. \tag{2.3.12}$$

Because of the definition, $\mathscr{E}_N$ is well-defined for a complex space N. It is clear that $\mathscr{E}_N$ is a sheaf of rings with unit 1. A section ϕ of $\mathscr{E}_N$ over an open subset $U \subset N$ is called a differentiable or C^∞ function on U, and it uniquely defines a continuous function

$$\phi : U \to \mathbf{C}.$$

If $\phi(x_0) \neq 0$, then in a neighborhood of x_0, ϕ is invertible; this is seen as follows. Since it is a local problem, we let $N \subset \Omega \subset \mathbf{C}^n$ and τ_j be as in 2.3.10. Let ϕ be represented by $\tilde{\phi}, \tilde{\phi}' \in \Gamma(\Omega, \mathscr{E}_\Omega)$ satisfying

$$\tilde{\phi} = \tilde{\phi}' + \sum_{j=1}^{l}(a_j \tau_j + b_j \bar{\tau}_j).$$

Taking $a'_j = a_j/\tilde{\phi}'$ and $b'_j = b_j/\tilde{\phi}'$, one gets

$$\tilde{\phi} = \tilde{\phi}'\left(1 + \sum_{j=1}^{l}(a'_j \tau_j + b'_j \bar{\tau}_j)\right).$$

[3] K. Oka, in [Iw] VII, [50] and [51], proved three fundamental coherence theorems for (i) $\mathscr{O}_\Omega$, (ii) $\mathscr{I}\langle N\rangle$, and (iii) the normalization of $\mathscr{O}_N$. Cf. H. Cartan [50] for another proof of the coherence of $\mathscr{I}\langle N\rangle$, and Grauert–Remmert [84]. K. Oka called $\mathscr{I}\langle N\rangle$ the *geometric ideal sheaf* (*l'idéal géométrique de domaines indéterminés*). It is interesting to see the comments in Oka [Sp] and Cartan [79].

We may assume that

$$\sum_{j=1}^{l}|a'_j\tau_j| < \frac{1}{2}, \qquad \sum_{j=1}^{l}|b'_j\tau_j| < \frac{1}{2}.$$

Then $|\sum_{j=1}^{l}(a'_j\tau_j + b'_j\bar{\tau}_j)| < 1$, and we have

$$\begin{aligned}\frac{1}{\tilde{\phi}} &= \frac{1}{\tilde{\phi}'}\cdot\frac{1}{1+\sum_{j=1}^{l}(a'_j\tau_j + b'_j\bar{\tau}_j)}\\ &= \frac{1}{\tilde{\phi}'} + \sum_{k=1}^{\infty}(-1)^k\left(\sum_{j=1}^{l}\big(a'_j\tau_j + b'_j\bar{\tau}_j\big)\right)^k\\ &= \frac{1}{\tilde{\phi}'} + \sum_{k=1}^{\infty}(-1)^k\left(\sum_{j=1}^{l}\big(a'_j\tau_j + b'_j\bar{\tau}_j\big)\right)^k.\end{aligned}$$

The second term above is written as an absolute convergent power series in $a'_j\tau_j$, $b'_j\bar{\tau}_j$ without constant term. Therefore there are C^∞ functions a''_j, b''_j in Ω such that

$$\frac{1}{\tilde{\phi}} = \frac{1}{\tilde{\phi}'} + \sum_{j=1}^{l}\big(a''_j\tau_j + b''_j\bar{\tau}_j\big). \tag{2.3.13}$$

Thus $\frac{1}{\tilde{\phi}}$ and $\frac{1}{\tilde{\phi}'}$ define the same section $\frac{1}{\phi} \in \Gamma(N, \mathscr{O}_N)$ such that $\phi\cdot\frac{1}{\phi} = 1$.

We then define the *tangent space* over N. This is again a local object, so that we let $N \subset \Omega \subset \mathbf{C}^n$ and τ_j be as above (cf. 2.3.10). Let $(x_1, \ldots, x_n)$ be the natural coordinate system of $\mathbf{C}^n$. Let $X = \sum_{i=1}^{n} X^i\frac{\partial}{\partial x_i}$ be a holomorphic tangent vector of $\mathbf{T}(\Omega)$ at $x \in N$. We define

$$\mathbf{T}(N)_x = \left\{(x, X);\, X(\tau_j) = \sum_{i=1}^{n} X^i\frac{\partial\tau_j}{\partial x_i}(x) = 0,\, 1 \leqq j \leqq l_\lambda\right\}. \tag{2.3.14}$$

This yields an analytic subset

$$\mathbf{T}(N) = \bigcup_{x\in N}\mathbf{T}(N)_x \subset N\times\mathbf{C}^n$$

with the natural projection $\pi : \mathbf{T}(N) \to N$, the fibers of which are vector spaces. We call $\mathbf{T}(N)$ the holomorphic tangent space over N. Because of the coherence of $\mathscr{I}\langle N\rangle$ it is easy to see that the definition of $\mathbf{T}(N)$ is compatible with the change of coordinates (i.e., the embedding into Ω), so that for a general complex space N the holomorphic tangent space $\mathbf{T}(N)$ is defined with the natural projection $\pi : \mathbf{T}(N) \to N$, the fibers of which are vector spaces.

In terms of (2.3.14) we set

$$\bar{\mathbf{T}}(N) = \left\{ Y = \sum_{j=1}^{n} Y^j \frac{\partial}{\partial \bar{x}_j};\ Y^j \in \mathbf{C},\ \bar{Y} = \sum_{j=1}^{n} \bar{Y}^j \frac{\partial}{\partial x_j} \in \mathbf{T}(N) \right\}, \tag{2.3.15}$$

$$T(N) = \mathbf{T}(N) \oplus \bar{\mathbf{T}}(N).$$

We call $\bar{\mathbf{T}}(N)$ the anti-holomorphic tangent space over N, and $T(N)$ the tangent space over N.

Let $\phi \in \Gamma(U, \mathscr{O}_N)$ be a C^∞ function over an open subset U of a complex space N. We want to define the *differential* $d\phi(X)$ for a tangent vector $X \in \mathbf{T}(N)|_U$. There is locally a C^∞ function $\tilde{\phi} \in \Gamma(\Omega, \mathscr{O}_\Omega)$ which represents ϕ on U. If we take another $\tilde{\phi}' \in \Gamma(\Omega, \mathscr{O}_\Omega)$, then there are C^∞ functions a_j, b_j on Ω such that

$$\tilde{\phi} = \tilde{\phi}' + \sum_{j=1}^{l} (a_j \tau_j + b_j \bar{\tau}_j). \tag{2.3.16}$$

For the differentials we get

$$\begin{aligned} \partial\tilde{\phi} &= \partial\tilde{\phi}' + \sum_{j=1}^{l} (\tau_j \partial a_j + a_j \partial \tau_j + \bar{\tau}_j \partial b_j), \\ \bar{\partial}\tilde{\phi} &= \bar{\partial}\tilde{\phi}' + \sum_{j=1}^{l} (\tau_j \bar{\partial} a_j + b_j \bar{\partial}\bar{\tau}_j + \bar{\tau}_j \bar{\partial} b_j). \end{aligned} \tag{2.3.17}$$

It follows from (2.3.14) that for $X \in \mathbf{T}(N)_x$ $(x \in N)$,

$$\partial\tilde{\phi}(X) = \partial\tilde{\phi}'(X), \qquad \bar{\partial}\tilde{\phi}(\bar{X}) = \bar{\partial}\tilde{\phi}'(\bar{X}).$$

Thus $\partial\phi : \mathbf{T}(N) \to \mathbf{C}$ and $\bar{\partial}\phi : \bar{\mathbf{T}}(N) \to \mathbf{C}$ are well-defined, and we have the differential of ϕ:

$$d\phi = \partial\phi \oplus \bar{\partial}\phi : T(N) = \mathbf{T}(N) \oplus \bar{\mathbf{T}}(N) \to \mathbf{C}.$$

From the second equation of (2.3.17) it follows that

$$\partial\bar{\partial}\tilde{\phi} = \partial\bar{\partial}\tilde{\phi}' + \sum_{j=1}^{l} (\partial\tau_j \wedge \bar{\partial} a_j + \tau_j \partial\bar{\partial} a_j + \partial b_j \wedge \bar{\partial}\bar{\tau}_j + \bar{\tau}_j \partial\bar{\partial} b_j).$$

Therefore we can define the *Levi-form* of $\partial\bar{\partial}\phi$ along with the real form $\frac{i}{2}\partial\bar{\partial}\phi$ by

$$\partial\bar{\partial}\phi(X, \bar{Y}) = \partial\bar{\partial}\tilde{\phi}(X, \bar{Y}) = \partial\bar{\partial}\tilde{\phi}'(X, \bar{Y}), \quad X, Y \in \mathbf{T}(N). \tag{2.3.18}$$

2.3.3 Metrics and Curvature Forms of Line Bundles

Let $L \to N$ be a line bundle over a complex space N. Let $N = \bigcup_\lambda V_\lambda$ be an open covering such that there is a transition function system $\{\phi_{\lambda\mu}\}$ of L. A *hermitian metric* in L is a family $h = \{h_\lambda\}$ of positive real-valued C^∞ functions $h_\lambda \in \Gamma(V_\lambda, \mathscr{E}_N)$ satisfying

$$h_\lambda(x) = \left|\phi_{\lambda\mu}(x)\right|^2 h_\mu(x), \quad x \in V_\lambda \cap V_\mu. \tag{2.3.19}$$

The line bundle L endowed with h is called a *hermitian line bundle* and is denoted by the pair (L, h); we sometimes write simply L for a hermitian line bundle when there is no confusion. We take a local trivialization

$$L|_{V_\lambda} \cong V_\lambda \times \mathbf{C}.$$

For $v = (x, \xi_\lambda) \in V_\lambda \times \mathbf{C} \subset L$ we set the norm of v with respect to h by

$$\|v\| = \frac{|\xi_\lambda|}{\sqrt{h_\lambda(x)}}.$$

Then this is independent of the choice of V_λ. The norm function $\|v\|$ is also called a hermitian metric in L.

If N is paracompact, we may construct such a hermitian metric in every line bundle L over N. In fact, let $\{V_\lambda\}$ and $\{\phi_{\lambda\mu}\}$ be as above. We may assume that $\{V_\lambda\}$ is locally finite. Taking a partition $\{c_\lambda\}$ of unity subordinated to $\{V_\lambda\}$, we set C^∞ functions on V_λ by $c_\nu(x)\log|\phi_{\lambda\nu}(x)|^2$ extending as 0 on $V_\lambda \setminus V_\nu$, and set

$$h_\lambda(x) = \exp\left\{\sum_\nu c_\nu(x)\log\left|\phi_{\lambda\nu}(x)\right|^2\right\}, \quad x \in V_\lambda.$$

It follows from the cocycle condition (2.3.9) that

$$h_\lambda = |\phi_{\lambda\mu}|^2 h_\mu \quad (\text{on } V_\lambda \cap V_\mu \neq \emptyset). \tag{2.3.20}$$

Going back to (2.3.19), we want to define the Levi-form $\partial\bar\partial \log h_\lambda$. As in (2.3.16), let h_λ and h'_λ be C^∞ functions such that

$$\tilde h_\lambda = \tilde h'_\lambda + \sum_{j=1}^{l}(a_{\lambda j}\tau_{\lambda j} + b_{\lambda j}\bar\tau_{\lambda j}),$$

where $a_{\lambda j}$ and $b_{\lambda j}$ are C^∞ functions on Ω_λ into which V_λ is embedded, and $\tau_{\lambda j}$ are generators of $\mathscr{I}\langle N\rangle|_{V_\lambda}$. Since h'_λ is positive valued, we have

$$\tilde h_\lambda = \tilde h'_\lambda \cdot \left(1 + \sum_{j=1}^{l}\left(a'_{\lambda j}\tau_{\lambda j} + b'_{\lambda j}\bar\tau_{\lambda j}\right)\right),$$

where $a'_{\lambda j} = a_{\lambda j}/h'_\lambda$ and $b'_{\lambda j} = b_{\lambda j}/h'_\lambda$. Since $\tau_{\lambda j}$ vanish on V_λ, we may assume that $\sum_{j=1}^{l} |a'_{\lambda j}\tau_{\lambda j}| < \frac{1}{2}$ and $\sum_{j=1}^{l} |b'_{\lambda j}\bar{\tau}_{\lambda j}| < \frac{1}{2}$. By making use of the power expansion $\log(1+t) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} t^k$ $(|t| < 1)$, we deduce in the same way as in (2.3.13) that

$$\log \tilde{h}_\lambda = \log \tilde{h}'_\lambda + \sum_{j=1}^{l} \big(a''_{\lambda j}\tau_{\lambda j} + b''_{\lambda j}\bar{\tau}_{\lambda j}\big),$$

where $a''_{\lambda j}$ and $b''_{\lambda j}$ are C^∞ functions. Hence the Levi-form $\Theta_\lambda = \partial\bar{\partial}\log h_\lambda$ of $\log h_\lambda$ is well-defined, and (2.3.19) implies

$$\Theta_\lambda(x) = \Theta_\mu(x), \quad x \in V_\lambda \cap V_\mu.$$

The *curvature form* $\Theta_{(L,h)}$ of the hermitian line bundle (L, h) is globally defined by

$$\Theta_{(V,h)}|_{V_\lambda}(X, \bar{Y}) = \Theta_\lambda(X, \bar{Y}), \quad X, Y \in \mathbf{T}(N)|_{V_\lambda},$$

and the real form

$$\omega_{(L,h)} = \frac{i}{2\pi}\Theta_{(L,h)} = dd^c \log h_\lambda \text{ in } V_\lambda$$

is called the *Chern form* of (L, h); here we recall

$$d^c = \frac{i}{4\pi}(\bar{\partial} - \partial).$$

When it is not necessary to specify the hermitian metric h, we write ω_L for $\omega_{(L,h)}$. Let $\hat{h}$ be another hermitian metric in L. After taking a refinement $\{V_\lambda\}$ of open coverings of N, we have $h = \{h_\lambda\}$ and $\hat{h} = \{\hat{h}_\lambda\}$, so that $h_\lambda(x)/\hat{h}_\lambda(x)$ is globally defined, independently from the choice $V_\lambda \ni x$. Thus there is a C^∞ function b on N such that

$$\omega_{(L,h)} = \omega_{(L,\hat{h})} + dd^c b. \tag{2.3.21}$$

If N is non-singular, the Chern form $\omega_{(L,h)}$ defines a cohomology class $c_1(L) = [\omega_L] \in H^2(N, \mathbf{R})$, which is independent of the choice of the hermitian metric by (2.3.21) and is called the *Chern class* of L.

If $\omega_{(L,h)}$ is positive (resp. semi-)definite at all points of N, we write $\omega_{(L,h)} > 0$ (resp. $\omega_{(L,h)} \geqq 0$). We write $L > 0$ (resp. $L \geqq 0$) if there exists a hermitian metric h in L with Chern form $\omega_{(L,h)} > 0$ (resp. $\omega_{(L,h)} \geqq 0$); in this case we say that L is positive (resp. semi-positive). If N is non-singular and $L > 0$, ω_L defines a Kähler metric on N.

Let σ_i $(1 \leqq i \leqq p)$ and σ'_j $(1 \leqq j \leqq q)$ be holomorphic sections of L over N (may be singular) and let $\{V_\lambda\}$ be as above. On each V_λ they are given by holomorphic functions $\sigma_{i\lambda}$ $(1 \leqq i \leqq p)$ and $\sigma'_{j\lambda}$ $(1 \leqq j \leqq q)$. Assume that $\sum_j |\sigma'_{j\lambda}|^2 \not\equiv 0$. For $x \in N$ we take $V_\lambda \ni x$ and set

$$\frac{\sum_i |\sigma_i(x)|^2}{\sum_j |\sigma'_j(x)|^2} = \frac{\sum_i |\sigma_{i\lambda}(x)|^2}{\sum_j |\sigma'_{j\lambda}(x)|^2}. \tag{2.3.22}$$

This is independent of the choice of $V_\lambda \ni x$, and defines a function on N outside where the denominator vanishes.

Assume that $\dim H^0(N, L) \geqq 2$ and take linearly independent elements $\sigma_0, \ldots, \sigma_n$ of $H^0(N, L)$. Then we get a meromorphic mapping

$$\Phi : x \in N \to \left[\sigma_0(x), \ldots, \sigma_n(x)\right] \in \mathbf{P}^n(\mathbf{C}).$$

Note that this is not necessarily reduced representation. If $\dim H^0(N, L) < \infty$ and $\{\sigma_i\}$ is a base of $H^0(N, L)$, we write for the above Φ

$$\Phi_L : N \to \mathbf{P}^n(\mathbf{C}). \tag{2.3.23}$$

Let $S(N)$ denote the set of singular points of N. Then $S(N)$ is a nowhere dense analytic subset of N. Let N be compact. If the meromorphic mapping $\Phi_{L^l} : N \to \mathbf{P}^{n_l}(\mathbf{C})$ $(n_l = \dim H^0(N, K_N^l) - 1)$ for some l has the differential $d\Phi_{K_N^l x}$ of rank equal to $\dim N$ at some point $x \in N \setminus I(\Phi_{K_N^l}) \cup S(N)$, L is said to be *big*. If N is non-singular and K_N is big, N is said to be of *general type*.

If Φ_L gives rise to a holomorphic embedding, L is said to be *very ample*. If there is a number $k \in \mathbf{N}$ with very ample L^k, L is said to be *ample*.

Going back to Example 2.3.7, we denote by $H \to \mathbf{P}^n(\mathbf{C})$ the line bundle determined by the transition function system $\{\phi_{jk} = \frac{w_k}{w_j}\}$ associated to the affine open covering, $\mathbf{P}^n(\mathbf{C}) = \bigcup_{j=0}^n U_j$ with $U_j = \{w_j \neq 0\}$. We call H the *hyperplane bundle* over $\mathbf{P}^n(\mathbf{C})$. Then the functions

$$\rho_j = 1 + \sum_{i \neq j} \frac{|w_i|^2}{|w_j|^2} \tag{2.3.24}$$

on U_i define a hermitian metric in H with the curvature form $\omega_H > 0$. That is, $H > 0$. The Kähler metric h_H is called the *Fubini–Study metric*. The associated Kähler form ω_H is called the Fubini–Study metric form on $\mathbf{P}^n(\mathbf{C})$.

Let $f : \mathbf{C}^m \to \mathbf{P}^n(\mathbf{C})$ be a meromorphic mapping with reduced representation $f = [f_0, \ldots, f_n]$. The pull-back of ω_H by f is written as

$$f^*\omega_H = \frac{i}{2\pi} f^* \partial\bar{\partial} \log \rho_j = \frac{i}{2\pi} \partial\bar{\partial} \log \left(\sum_{i=0}^n |f_i|^2 \right). \tag{2.3.25}$$

The function $\log(\sum_{i=0}^n |f_i|^2)$ is a plurisubharmonic function on $\mathbf{C}^m$.

Let N be a complex projective algebraic variety with a holomorphic embedding $\Psi : N \hookrightarrow \mathbf{P}^n(\mathbf{C})$. The pull-back $\Psi^* H$ is a positive line bundle and a meromorphic mapping into $\mathbf{P}^n(\mathbf{C})$ given by $n+1$ linearly independent sections of $H^0(N, \Psi^* H)$ coincides with Φ. The next theorem gives the converse for it; the non-singular case is due to Kodaira [54], [74] (cf. Nakano [81] for a further generalization).

Theorem 2.3.26 (Grauert [62]) *Let N be a compact complex space and let $L \to N$ be a positive line bundle over N.*

(i) *Let $\mathscr{I} \to N$ be a coherent sheaf over a compact complex space N. Then there is a number l_0 such that $H^q(N, \mathscr{I} \otimes \mathscr{O}(L^l)) = 0$, $l \geqq l_0$, $q \geqq 1$.*

(ii) *Let $E \to N$ be a line bundle over N. Then there is a number l_0 such that for every $l \geqq l_0$ the meromorphic mapping $\Phi_{L^l \otimes E} : N \to \mathbf{P}^{n_l - 1}(\mathbf{C})$ with $n_l = \dim H^0(N, L^l \otimes E)$ is a holomorphic embedding, and hence N is projective algebraic.*

A divisor D on N defines a line bundle $L(D)$. If $L(D)$ satisfies the above property (ii), D is said to be *ample*; hence, D is ample iff $L(D)$ is positive. If the holomorphic sections of $L(D)$ over N give a holomorphic embedding into a projective space, D is said to be *very ample*.

Let N be a compact complex space. Take a vector subspace $E \subset H^0(N, L)$, $\dim E = l + 1 \geqq 2$. For the divisor (σ) given by $\sigma \in E \setminus \{0\}$ and for $c \in \mathbf{C}^*$ $(c\sigma) = (\sigma)$ holds clearly. Conversely, if $\sigma, \tau \in E \setminus \{0\}$ satisfy $(\sigma) = (\tau)$, then $\sigma = c\tau$ for some $c \in \mathbf{C}^*$. Therefore we have the following isomorphism:

$$\big\{(\sigma); \sigma \in E \setminus \{0\}\big\} \cong \big(E \setminus \{0\}\big)/\mathbf{C}^* = P(E) \cong \mathbf{P}^l(\mathbf{C}).$$

The space $P(E)$ is called a linear system of D, and in particular, when $E = H^0(N, L)$, it is called the *complete linear system* of L denoted by $|L|$. The analytic subset $B(E) = \{x \in N; \sigma(x) = 0, \ {}^\forall\sigma \in E\}$ is called the *base locus* of E.

In what follows we assume N to be *projective algebraic* unless otherwise mentioned. For any line bundle L over N there are by Theorem 2.3.26 very ample line bundles L_i, $i = 1, 2$, such that $L = L_1 \otimes L_2^{-1}$. By making use of the Chern forms $\omega_{L_i} > 0$ of L_i we obtain

$$\omega_L = \omega_{L_1} - \omega_{L_2}.$$

Noting (2.3.25), we see the following.

Lemma 2.3.27 *Let N, L be as above. Let $f : \mathbf{C}^m \to N$ be a meromorphic mapping. Then there are plurisubharmonic functions ξ_i, $i = 1, 2$ on $\mathbf{C}^m$ such that*

$$f^*\omega_L = dd^c\xi_1 - dd^c\xi_2.$$

Let $L \to N$ be a line bundle with a hermitian metric $\|\cdot\|$. For an element $D \in |L|$ we take and fix a holomorphic section $\sigma \in H^0(N, L) \setminus \{0\}$ such that $(\sigma) = D$ and $\|\sigma\| \leqq 1$. The next follows from the Poincaré–Lelong Theorem 2.2.16.

Lemma 2.3.28 *Let $\|\sigma\|$ and ω_L be as above. Let $U \subset \mathbf{C}^m$ be an open subset and let $f : \mathbf{C}^m \to N$ be a meromorphic mapping such that the pull-back f^*D is defined. Then we have a current equation on U:*

$$dd^c\left[\log \frac{1}{\|\sigma \circ f(z)\|^2}\right] = f^*\omega_L - f^*D.$$

Theorem 2.3.29 (Poincaré duality) *Let N be a compact complex manifold of dimension n. Let $L \to N$ be a hermitian line bundle and let $D \in |L|$. Then*

$$\int_D \eta = \int_N \omega_L \wedge \eta$$

for all d-closed $(n-1, n-1)$ forms η on N.

Proof Let $U \subset N$ be a holomorphic local coordinate neighborhood of N and let $\iota : U \to N$ be the inclusion. Suppose that $\operatorname{Supp} \eta \subset U$. By Lemma 2.3.28 we have

$$\int_U \omega_L \wedge \eta - \int_D \eta = \int_U \left(\log \frac{1}{\|\sigma\|^2} \right) dd^c \iota^* \eta = 0.$$

By making use of the partition of unity, we get the required formula. □

Let $f : \mathbf{C}^m \to N$ be a meromorphic mapping such that the pull-back f^*D is defined. We apply Lemma 2.3.28 for $U = B(r)$ and $\eta = \alpha^{m-1} = (dd^c \|z\|^2)^{m-1}$. Then the equality such as in Theorem 2.3.29 does not hold, and a boundary integral appears:

$$\begin{aligned} &\int_{B(r)} f^* \omega_L \wedge \alpha^{m-1} - \int_{f^*D \cap B(r)} \alpha^{m-1} \\ &\quad = \int_{\partial B(r)} d^c \log \frac{1}{\|\sigma \circ f(z)\|} \wedge \alpha^{m-1}. \end{aligned}$$

The first term of the above equation should be the order function, the second should be the counting function, and the third might be some remainder. Since the boundary integral contains a differentiation, it is inconvenient to handle this formula directly, and we need some more modifications as follows.

The counting function $N(r, f^*D)$ is already defined by (2.2.18). By Lemma 2.3.27 and the result in Sect. 2.1, in particular, by Corollary 2.1.37, we may define the following two quantities:

$$\begin{aligned} m_f(r, D) &= \int_{\|z\|=r} \log \frac{1}{\|\sigma \circ f(z)\|} \gamma(z), \\ T_f(r, \omega_L) &= \int_1^r \frac{dt}{t^{2m-1}} \int_{B(t)} f^* \omega_L \wedge \alpha^{n-1}, \quad r \geqq 1. \end{aligned} \tag{2.3.30}$$

We call $m_f(r, D)$ the *proximity function* or the *approximate function* of f for D. We call $T_f(r, \omega_L)$[4] the *order function* of f with respect to ω_L. If $\omega_L \geqq 0$, it follows from

[4] In S. Lang [87] the notation $T_{f,D}$ is used, but this is not proper and mises an essential point: the order function is not dependent on each D, but determined solely by the complete linear system or by its cohomology class, and this is where the First Main Theorem 2.3.31 makes sense.

Lemma 2.1.31 that $T_f(r, \omega_L)$ is a monotone increasing convex function in $\log r$. Let ω'_L be the Chern form of another hermitian metric in L. By (2.3.21) we see that

$$T_f(r, \omega_L) - T_f\big(r, \omega'_L\big) = \frac{1}{2}\int_{\|z\|=r} b \circ f(z)\gamma(z) - \frac{1}{2}\int_{\|z\|=1} b \circ f(z)\gamma(z)$$
$$= O(1) \quad (r \to \infty).$$

Thus the order function of f with respect to L is defined by

$$T_f(r, L) = T_f(r, \omega_L)$$

modulo up to addition with a bounded term in $r \geqq 1$. In the same sense the proximity function $m_f(r, D)$ is determined modulo up to addition with a bounded term in $r \geqq 1$.

We obtain the following important formula from Corollary 2.1.37.

Theorem 2.3.31 (The First Main Theorem) *Let $L \to N$ be a line bundle and let $f : \mathbf{C}^m \to N$ be a meromorphic mapping. For $D \in |L|$ with* Supp $D \not\supset f(\mathbf{C}^m)$,

$$T_f(r, \omega_L) = N\big(r, f^*D\big) + m_f(r, D) - m_f(1, D),$$
$$T_f(r, L) = N\big(r, f^*D\big) + m_f(r, D) + O(1).$$

Corollary 2.3.32 *With the conditions in Theorem 2.3.31 we suppose one of the following*:

(i) $f^*D \neq 0$.
(ii) $f^*\omega_L \geqq 0$ *and* $f^*\omega_L(z_0) > 0$ *at some point* $z_0 \in \mathbf{C}^m$.

Then there is a constant $C > 0$ such that

$$C \log r \leqq T_f(r, \omega_L) + O(1).$$

Proof In the case of (i) there is a $t_0 > 0$ with $n(t_0, f^*D) > 0$. It follows from Theorem 2.2.19 that $n(t, f^*D)$ is a monotone increasing function in t. Therefore

$$N\big(r, f^*D\big) \geqq \int_{t_0}^{r} \frac{n(t_0, f^*D)}{t} dt = n\big(t_0, f^*D\big)(\log r - \log t_0).$$

It follows from Theorem 2.3.31 that

$$T_f(r, \omega_L) \geqq N\big(r, f^*D\big) + O(1).$$

Thus the claim is deduced.

To deal with the case of (ii) we may assume that (i) does not hold; that is, $f^{-1}D = \emptyset$. Then $dd^c \log 1/\|\sigma \circ f\|^2 = f^*\omega_L \geqq 0$, and so $\log 1/\|\sigma \circ f(z)\|^2$ is a plurisubharmonic function. By Lemma 2.1.31

$$\frac{1}{t^{2m-2}}\int_{B(t)} f^*\omega_L \wedge \alpha^{m-1}$$

is a monotone increasing function in t. By assumption we have for $t_0 = \|z_0\| + 1$

$$C_0 = \frac{1}{t_0^{2m-2}} \int_{B(t_0)} f^*\omega_L \wedge \alpha^{m-1} > 0.$$

By definition

$$T_f(r, \omega_L) \geqq \int_{t_0}^{r} \frac{dt}{t^{2m-1}} \int_{B(t)} f^*\omega_L \wedge \alpha^{m-1} \geqq C_0(\log r - \log t_0). \qquad \square$$

Remark 2.3.33 In the above First Main Theorem N is assumed to be projective algebraic, but in fact suffices to be a compact complex manifold. In that case we consider the integral over $\Gamma(f)$ (cf. Theorem 2.2.11). To show Jensen's formula on $\Gamma(f)$ (Lemma 2.1.33) one needs Stokes' Theorem on $\Gamma(f)$ with singularities in general, or a desingularization.

Example 2.3.34 Let $[w_0, \ldots, w_n]$ be a homogeneous coordinate system of $\mathbf{P}^n(\mathbf{C})$ and let $H \to \mathbf{P}^n(\mathbf{C})$ be the hyperplane bundle. Let $f : \mathbf{C}^m \to \mathbf{P}^n(\mathbf{C})$ be a meromorphic mapping with reduced representation $f = [f_0, \ldots, f_n]$. Then w_j, $0 \leqq j \leqq n$ form a base of $H^0(\mathbf{P}^n(\mathbf{C}), H)$. We take a holomorphic section $\sigma = \sum_{j=0}^{n} c_j w_j$, $(c_j) \in \mathbf{C}^{n+1} \setminus \{0\}$ and the hyperplane $D = (\sigma)$ defined by it. The coefficients (c_j) may be normalized so as

$$\sum_{j=0}^{n} |c_j|^2 = 1.$$

The length of σ with respect to the hermitian metric in H given by (2.3.24) is

$$\|\sigma\| = \frac{|\sum_j c_j w_j|}{\sqrt{(\sum_j |w_j|^2)}} \leqq 1.$$

Then the quantities appearing in the First Main Theorem 2.3.31 are as follows:

$$f^*D = \left(\sum_j c_j f_j\right) = \left(\sum_j c_j f_j\right)_0,$$

$$m_f(r, D) = \int_{\|z\|=1} \frac{\sqrt{\sum_j |f_j(z)|^2}}{|\sum_j c_j f_j(z)|} \gamma(z),$$

$$T_f(r, H) = \int_1^r \frac{dt}{t^{2m-1}} \int_{B(t)} dd^c \log\left(\sum_j |f_j|^2\right) \wedge \alpha^{m-1}.$$

Here, Jensen's formula (Lemma 2.1.33) applied to the above expression of $T_f(r, H)$ yields

$$T_f(r,H)=\int_{\|z\|=r}\log\left(\sum_{j=0}^{n}\left|f_j(z)\right|^2\right)^{1/2}\gamma(z)$$
$$-\int_{\|z\|=1}\log\left(\sum_{j=0}^{n}\left|f_j(z)\right|^2\right)^{1/2}\gamma(z). \tag{2.3.35}$$

We take a vector subspace $E\subset H^0(N,L)$ of $\dim E=l+1\geqq 2$ and introduce a homogeneous coordinate system $[u_0,\dots,u_l]$ of $P(E)$. The volume element $\Omega=\omega_0^l$ defined by the Fubini–Study metric form $\omega_0=dd^c\log\sum|u_j|^2$ satisfies

$$\int_{|L|}\Omega=1.$$

The unitary transformation group of the homogeneous coordinate system $u_0,\dots,u_l$ naturally acts on $|L|$ by

$$\big(U,\big[(w_j)\big]\big)\in \mathrm{U}(l+1)\times|L|\to\big[U(w_j)\big]\in|L|.$$

This action is transitive and leaves ω_0 and Ω invariant. For an arbitrary $D\in P(E)$ we take $\sigma=\sum c_ju_j$ so that $(\sigma)=D$, $\sum|c_j|^2=1$. While the vector (c_j) is not uniquely determined, $\frac{(\sum|u_j|^2)^{1/2}}{|\sum c_ju_j|}$ depends only on D. In this sense the following holds:

$$\int_{D=(\sigma)\in P(E)}\log\frac{(\sum|u_j|^2)^{1/2}}{|\sum c_ju_j|}\Omega(D)=C(l)>0. \tag{2.3.36}$$

Here $C(l)$ is a constant dependent only on l, and a computation yields (H. Weyl–J. Weyl [38])

$$C(l)=\frac{1}{2}\left(1+\frac{1}{2}+\cdots+\frac{1}{l}\right).$$

Theorem 2.3.37 *Let $L\to N$ be a line bundle and let $E\subset H^0(N,L)$ be a vector subspace. Suppose that $B(E)=\emptyset$. Then for an arbitrary meromorphic mapping $f:\mathbf{C}^m\to N$,*

$$T_f(r,L)=\int_{D\in P(E)}N\big(r,f^*D\big)\Omega(D)+O(1).$$

Proof We take bases $\sigma_0,\dots,\sigma_l$ of E. By assumption $l\geqq 1$ and a hermitian metric in L may be assumed to satisfy that for a section $\sigma\in E$

$$\big\|\sigma(x)\big\|^2=\frac{|\sigma(x)|^2}{\sum|\sigma_j(x)|^2}. \tag{2.3.38}$$

It follows from (2.3.36) and Fubini's theorem that

$$\int_{D\in P(E)} m_f(r,D)\Omega(D) = \int_{\|z\|=r}\int_{D=(\sigma)\in P(E)} \log\frac{1}{\|\sigma(f(z))\|}\Omega(D)\gamma(z)$$
$$= C(l).$$

By the First Main Theorem 2.3.31

$$T_f(r,\omega_L) = \int_{D\in P(E)} N\big(r, f^*D\big)\Omega(D) = T_f(r,L) + O(1). \qquad \square$$

The properties of the order function $T_f(r,L)$ will be summarized in the next section, but here we assume that

$$T_f(r,L)\to\infty \quad (r\to\infty)$$

(cf. Corollary 2.3.32). In this case we define *Nevanlinna's defect*

$$\delta(f,D) = 1 - \varlimsup_{r\to\infty}\frac{N(r,f^*D)}{T_f(r,L)}$$

of $D\in|L|$. This satisfies

$$0 \leqq \delta(f,D) \leqq 1.$$

With $k\in\mathbf{N}$ we define the *k-defect* $\delta_k(f,D)$ by

$$\delta_k(f,D) = 1 - \varlimsup_{r\to\infty}\frac{N_k(r,f^*D)}{T_f(r,L)}.$$

In particular, when $f(\mathbf{C}^m)\cap D=\emptyset$, $\delta(f,D)=1$. A divisor D with $\delta(f,D)>0$ is called *Nevanlinna's exceptional divisor*. Theorem 2.3.37 implies the following.

Theorem 2.3.39 (Casorati–Weierstrass) *Let the assumption be the same as in Theorem 2.3.37 and moreover $T_f(r,\omega_L)\to\infty$ $(r\to\infty)$. Then almost all $D\in P(E)$ with respect to the measure Ω satisfy $\delta(f,D)=0$, i.e., they are not Nevanlinna's exceptional divisors.*

2.4 The First Main Theorem for Coherent Ideal Sheaves

2.4.1 Proximity Functions for Coherent Ideal Sheaves

In Theorems 2.3.37 and 2.3.39 the assumption $B(E)=\emptyset$ is essential. To deal with the case of $B(E)\neq\emptyset$ it is necessary to define the proximity function $m_f(r,D)$

not only for divisors but also for cycles of higher codimension. W. Stoll [70] and Bott–Chern [65] dealt with such a case. The First Main Theorem for coherent ideal sheaves was dealt with by Noguchi [03b] and Noguchi–Winkelmann–Yamanoi [08].

Because of a technical reason we assume in what follows that N is a *projective algebraic variety*, possibly singular with reduced structure (cf. Hartshorne [77]). Let $\mathscr{I} \subset \mathscr{O}_N$ be a coherent ideal sheaf of the structure sheaf $\mathscr{O}_N$. Then by definition there are a finite open covering $N = \bigcup U_\lambda$ of N and holomorphic functions $\zeta_{\lambda 1}, \dots, \zeta_{\lambda l_\lambda}$ on U_λ such that their germs $\underline{\zeta_{\lambda 1}}_x, \dots, \underline{\zeta_{\lambda l_\lambda}}_x$ generate the stalk $\mathscr{I}_x$ of $\mathscr{I}$ at every point $x \in U_\lambda$.

Lemma 2.4.1 *Let $\mathscr{I}$ be a coherent ideal sheaf of $\mathscr{O}_N$. Then there is a very ample line bundle over N such that $\Gamma(N, \mathscr{I}\otimes\mathscr{O}(L))$ generates the stalk $\mathscr{I}_x\otimes\mathscr{O}_x(L)$ at all $x \in N$; that is, there are bases ϕ_j $(1 \leqq j \leqq l)$ of $\Gamma(N, \mathscr{I}\otimes\mathscr{O}(L))$ such that the germs $\underline{\phi_j}_x$ $(1 \leqq j \leqq l)$ generate $\mathscr{I}_x\otimes\mathscr{O}_x(L)$ as $\mathscr{O}_{N,x}$ module at all $x \in N$.*

Proof In the algebraic sense of coherence the proof is immediate, since U_λ of the aforementioned open covering $\{U_\lambda\}$ are affine varieties and ζ_λ are regular rational functions on U_λ.

If the coherence is taken in the analytic sense, we then use Theorem 2.3.26. For an every fixed point $x \in N$ we denote by the same $\mathscr{I}_x$ the coherent sheaf over N extending $\mathscr{I}_x$ to be a zero sheaf on $N \setminus \{x\}$. Then there is a natural morphism $\kappa_x : \mathscr{I} \to \mathscr{I}_x$ whose kernel is denoted by $\mathscr{K}_x$:

$$0 \to \mathscr{K}_x \to \mathscr{I} \to \mathscr{I}_x \to 0.$$

We fix a positive line bundle L_0 over N. Then we have

$$0 \to \mathscr{K}_x\otimes\mathscr{O}\big(L^\nu\big) \to \mathscr{I}\otimes\mathscr{O}\big(L^\nu\big) \to \mathscr{I}_x\otimes\mathscr{O}\big(L^\nu\big) \to 0.$$

There is a number $\nu_0 \in \mathbf{N}$ such that for $\nu \geqq \nu_0$ $H^1(N, \mathscr{K}_x\otimes\mathscr{O}(L^\nu)) = 0$. Therefore,

$$H^0\big(N, \mathscr{I}\otimes\mathscr{O}\big(L^\nu\big)\big) \to H^0\big(N, \mathscr{I}_x\otimes\mathscr{O}\big(L^\nu\big)\big) \to 0.$$

By the coherence of $\mathscr{I}$, $H^0(N, \mathscr{I}\otimes\mathscr{O}(L^\nu))$ generates $\mathscr{I}_y$ in a neighborhood $U(x)$ of x. Since N is compact we can cover N by a finite number of such $U(x)$'s. If ν is large enough, then L^ν is very ample. □

We extend the bases ϕ_j $(1 \leqq j \leqq l)$ of $\Gamma(N, \mathscr{I}\otimes\mathscr{O}(L))$ in Lemma 2.4.1 to bases ϕ_j $(1 \leqq j \leqq l')$ of $\Gamma(N, \mathscr{O}(L))$. Then in the sense of (2.3.22) we set

$$d_{\mathscr{I}}(x) = \sqrt{\frac{\sum_{j=1}^{l} |\phi_i(x)|^2}{\sum_{j=1}^{l'} |\phi_j(x)|^2}}, \quad x \in N. \tag{2.4.2}$$

A different choice of L yields another $d'_{\mathscr{I}}(x)$, for which there is a constant $C > 0$ satisfying

$$C^{-1} d_{\mathscr{I}}(x) \leqq d'_{\mathscr{I}}(x) \leqq C d_{\mathscr{I}}(x), \quad {}^{\forall}x \in N. \tag{2.4.3}$$

Let $Y = (\mathrm{Supp}\, \mathscr{O}_N/\mathscr{I}, \mathscr{O}_N/\mathscr{I})$ be the subspace (subscheme), possibly non-reduced of N defined by $\mathscr{I}$ and set $d_Y(x) = d_{\mathscr{I}}(x)$. We call

$$\phi_{\mathscr{I}}(x) = -\log d_{\mathscr{I}}(x), \quad x \in N, \qquad \phi_Y(x) = -\log d_Y(x)$$

the *proximity (approximation) potential* of the coherent ideal sheaf $\mathscr{I}$ (resp. the subspace Y).

Let $f : \mathbf{C}^m \to N$ be a meromorphic mapping such that $f(\mathbf{C}^m) \not\subset \mathrm{Supp}\, \mathscr{O}_N/\mathscr{I}$. Then as in (2.3.25) and Lemma 2.3.27 we see that $f^*\phi_{\mathscr{I}}$ is written as a difference of two plurisubharmonic functions ξ_1 and ξ_2:

$$f^*\phi_{\mathscr{I}}(z) = \phi_{\mathscr{I}} \circ f(z) = \xi_1(z) - \xi_2(z), \quad z \in \mathbf{C}^m. \tag{2.4.4}$$

The *proximity function* (or approximation function) for $\mathscr{I}$ (or, Y) is defined by

$$m_f(r, \mathscr{I}) = m_f(r, Y) = \int_{\|z\|=r} \phi_{\mathscr{I}} \circ f(z) \gamma(z). \tag{2.4.5}$$

It follows from (2.4.4) that the integral is finite, and then from (2.4.3) that $m_f(r, \mathscr{I})$ is well-defined up to addition of $O(1)$-term.

Moreover, because of (2.4.4) the current $dd^c[\phi_{\mathscr{I}} \circ f(z)]$ is of degree 0; i.e., its coefficients are Radon measures, and the differential form $dd^c\phi_{\mathscr{I}} \circ f(z)$ has coefficients that are locally integrable. We set

$$\begin{aligned} \omega_{\mathscr{I},f} = \omega_{Y,f} &= -2dd^c\phi_{\mathscr{I}} \circ f(z) = -\frac{i}{\pi}\partial\bar{\partial}\phi \circ f(z) \\ &= 2dd^c \log \frac{1}{d_{\mathscr{I}} \circ f(z)}, \quad z \in \mathbf{C}^m. \end{aligned} \tag{2.4.6}$$

The order function of f with respect to $\mathscr{I}$ or Y is defined by

$$T(r, \omega_{\mathscr{I},f}) = T(r, \omega_{Y,f}) = \int_1^r \frac{dt}{t} \int_{B(t)} \omega_{\mathscr{I},f}. \tag{2.4.7}$$

We have to be careful to consider $f^*\mathscr{I}$, or f^*Y. Let $\Gamma(f) \subset \mathbf{C}^m \times N$ be the graph of f, let $p : \Gamma(f) \to \mathbf{C}^m$ and $q : \Gamma(f) \to N$ be respectively the natural projections. Then we have an analytic cycle (a locally finite formal sum of analytic subsets with integral coefficients) $p_*(q^*Y)$, whose supports are of codimension one or more in general. We denote by f^*Y or $f^*\mathscr{I}$ the sum of only those components of $p_*(q^*Y)$ whose supports are of codimension one. Then we have the current equation by Theorem 2.2.11 and Lemma 2.2.12:

$$\begin{aligned} 2dd^c\big[f^*\phi_{\mathscr{I},f}\big] &= \omega_{\mathscr{I},f} - f^*\mathscr{I}, \\ 2dd^c\big[f^*\phi_{Y,f}\big] &= \omega_{Y,f} - f^*Y. \end{aligned} \tag{2.4.8}$$

The counting function for f^*Y or for $f^*\mathscr{I}$ is defined by

$$N\big(r, f^*\mathscr{I}\big) = N\big(r, f^*Y\big) = \int_1^r \frac{dt}{t^{2m-1}} \int_{f^*Y\cap B(t)} \alpha^{m-1}, \quad r \geqq 1.$$

We have also the truncated counting function $N_k(r, f^*\mathscr{I})$ or $N_k(r, f^*Y)$ as in (2.2.18). Then by Jensen's formula (Corollary 2.1.37) and (2.4.8) we have the following.

Theorem 2.4.9 (The First Main Theorem) *Let $f : \mathbf{C}^m \to M$ and let $\mathscr{I}$ be as above. Then we have*

$$T(r, \omega_{\mathscr{I},f}) = N\big(r, f^*\mathscr{I}\big) + m_f(r, \mathscr{I}) - m_f(1, \mathscr{I}).$$

Let $\mathscr{I}_i$ $(i = 1, 2)$ be two coherent ideal sheaves of $\mathscr{O}_M$ and let Y_i be the subspace defined by $\mathscr{I}_i$, which is possibly non-reduced. We write $Y_1 \supset Y_2$ if $\mathscr{I}_1 \subset \mathscr{I}_2$.

Theorem 2.4.10 *The proximity function for coherent ideal sheaves has the following properties.*

(i) *If $\mathscr{I} \subset \mathscr{J}$, $m_f(r, \mathscr{J}) \leqq m_f(r, \mathscr{I}) + O(1)$; if $Y = \mathscr{O}_N/\mathscr{J} \subset Z = \mathscr{O}_N/\mathscr{I}$, $m_f(r, Y) \leqq m_f(r, Z) + O(1)$.*

(ii) *$m_f(r, \mathscr{I}_1\otimes\mathscr{I}_2) = m_f(r, \mathscr{I}_1) + m_f(r, \mathscr{I}_2) + O(1)$. In particular, $m_f(r, \mathscr{I}^k) = k m_f(r, \mathscr{I}) + O(1)$, $k \in \mathbf{N}$.*

(iii) *$m_f(r, \mathscr{I}_1 + \mathscr{I}_2) \leqq \min\{m_f(r, \mathscr{I}_1), m_f(r, \mathscr{I}_2)\} + O(1)$.*

Remark 2.4.11 Let D be an effective Cartier divisor on N. Let $\mathscr{I}$ be the ideal sheaf determined by D, and let $L(D)$ be the line bundle determined by D. Then we have

$$\begin{aligned} T(r, \omega_{\mathscr{I},f}) &= T_f\big(r, L(D)\big) + O(1), \\ m_f(r, \mathscr{I}) &= m_f(r, D) + O(1), \\ N\big(r, f^*\mathscr{I}\big) &= N\big(r, f^*D\big). \end{aligned}$$

These follow from

$$\big|-\log\|\sigma(x)\| - \phi_{\mathscr{I}}(x)\big| \leqq C, \quad x \in N,$$

where C is a positive constant.

Let $E \subset H^0(N, L)$ be a vector subspace and let $\mathscr{I}_0$ be the coherent ideal sheaf generated by $\{\underline{\sigma}_x; \sigma \in E\}$ at every $x \in N$. Then $\mathscr{I}_0$ decomposes into the common divisor part $\mathscr{I}_1$ and the remaining part $\mathscr{I}_2$, that is,

$$\mathscr{I}_0 = \mathscr{I}_1\otimes\mathscr{I}_2, \qquad \operatorname{codim}\operatorname{Supp}\mathscr{O}_N/\mathscr{I}_1 = 1, \qquad \operatorname{codim}\operatorname{Supp}\mathscr{O}_N/\mathscr{I}_2 \geqq 2.$$

It may happen that there is no $\mathscr{I}_1$-factor, that is, $\mathscr{I}_0 = \mathscr{I}_2$. Let D_1 be the effective divisor given by $\mathscr{I}_1$, if exists. Note that $D - D_1$ is an effective divisor. The next theorem is due to R. Kobayashi in the case of $D_1 = \emptyset$.

Theorem 2.4.12 *Let the notation be as above. For a meromorphic mapping $f : \mathbf{C}^m \to N$ with $f(\mathbf{C}^m) \not\subset B(E)$ we have the following.*

$$\int_{D\in P(E)} m_f(r, D)\Omega(D) = m_f(r, D_1) + m_f(r, \mathscr{I}_2) + O(1),$$

$$T_f(r, L) = \int_{D\in P(E)} N\big(r, f^*D\big)\Omega(D) + m_f(r, D_1) + m_f(r, \mathscr{I}_2) + O(1).$$

Proof Take $\tau_1 \in H^0(N, L(D_1))$ so that $(\tau_1) = D_1$. Let $\sigma_0, \ldots, \sigma_l$ be the bases of E. Every σ_j is written as

$$\sigma_j = \tau_1 \otimes \tau_{2j}, \quad \tau_{2j} \in H^0\big(N, L(D - D_1)\big),\ 0 \leqq j \leqq l.$$

For an arbitrary $\sigma = \sum c_j\sigma_j = \tau_1 \otimes (\sum c_j\tau_{2j})$,

$$-\log\big\|\sigma(x)\big\| = -\log\big\|\tau_1(x)\big\| + \phi_{\mathscr{I}_2} + \log\frac{(\sum|\tau_{2j}(x)|^2)^{1/2}}{|\sum c_j\tau_{2j}(x)|} + b(x),$$

where $b(x)$ is a C^∞ function on N. Therefore

$$m_f\big(r, (\sigma)\big) = m_f(r, D_1) + m_f(r, \mathscr{I}_2) + \int_{\|z\|=r} \log\frac{(\sum|\tau_{2j}(f(z))|^2)^{1/2}}{|\sum c_j\tau_{2j}(f(z))|}\gamma(z) + O(1).$$

We integrate this with respect to $\Omega([c_j]), [c_j] \in \mathbf{P}^l(\mathbf{C})$; by (2.3.36)

$$\int_{D\in P(E)} m_f(r, D)\Omega(D) = m_f(r, D_1) + m_f(r, \mathscr{I}_2) + O(1).$$

The claimed second formula follows from this and the First Main Theorem 2.3.31. □

Example 2.4.13 In general, even if $D_1 = 0$, $m_f(r, \mathscr{I}_2)$ is not bounded. Consider the following holomorphic map:

$$f : z \in \mathbf{C} \to \big[1, e^z, e^{cz}\big] = [w_0, w_1, w_2] \in \mathbf{P}^2(\mathbf{C}), \quad c > 1.$$

Let $H \to \mathbf{P}^2(\mathbf{C})$ be the hyperplane bundle, and let $E \subset H^0(\mathbf{P}^2(\mathbf{C}), H)$ be the subspace generated by holomorphic sections w_1, w_2. Then $B(E) = \{[1, 0, 0]\}$ and $\mathscr{I}_2$ is the maximal ideal of $\mathscr{O}_{\mathbf{P}^2(\mathbf{C}),[1,0,0]}$. Its proximity potential is

$$\phi_{\mathscr{I}_2} = \frac{1}{2}\log\frac{|w_0|^2 + |w_1|^2 + |w_2|^2}{|w_1|^2 + |w_2|^2}.$$

Following the definition, we calculate $m_f(r, \mathscr{I}_2)$:

$$\begin{aligned} m_f(r,\mathscr{I}_2) &= \frac{1}{4\pi}\int_{|z|=r} \log\frac{1+|e^z|^2+|e^{cz}|^2}{|e^z|^2+|e^{cz}|^2}d\theta \\ &= \frac{1}{4\pi}\int_{|z|=r} \log\left(1+\frac{1}{e^{2r\cos\theta}+e^{2cr\cos\theta}}\right)d\theta \\ &= \frac{1}{4\pi}\int_{\cos\theta<0} \log\left(1+\frac{1}{e^{2r\cos\theta}+e^{2cr\cos\theta}}\right)d\theta + O(1) \\ &= \frac{1}{4\pi}\int_{-\pi/2}^{\pi/2} \log\left(1+\frac{1}{e^{-2r\cos\theta}+e^{-2cr\cos\theta}}\right)d\theta + O(1) \\ &= \frac{1}{4\pi}\int_{-\pi/2}^{\pi/2} \log\left(1+\frac{e^{2r\cos\theta}}{1+e^{(1-c)2r\cos\theta}}\right)d\theta + O(1) \\ &= \frac{1}{4\pi}\int_{-\pi/2}^{\pi/2} 2r\cos\theta d\theta + O(1) \\ &= \frac{r}{\pi} + O(1). \end{aligned}$$

The order function $T_f(r, H)$ is calculated by (2.3.35) as

$$\begin{aligned} T_f(r,H) &= \frac{1}{4\pi}\int_{|z|=r} \log\bigl(1+|e^z|^2+|e^{cz}|^2\bigr)d\theta + O(1) \\ &= \frac{1}{4\pi}\int_{|z|=r} \log\bigl(1+e^{2r\cos\theta}+e^{2cr\cos\theta}\bigr)d\theta + O(1) \\ &= \frac{1}{4\pi}\int_{\cos\theta>0} \log\bigl(1+e^{2r\cos\theta}+e^{2cr\cos\theta}\bigr)d\theta + O(1) \\ &= \frac{1}{4\pi}\int_{-\pi/2}^{\pi/2} 2cr\cos\theta d\theta + O(1) \\ &= \frac{cr}{\pi} + O(1). \end{aligned}$$

2.4.2 The Case of $m = 1$

In the preceding subsection the assumption for N to be projective algebraic was used to deduce (2.4.4), which was used to apply Jensen's formula, Corollary 2.1.37. For Jensen's formula it is sufficient to know that (2.4.4) holds locally in $\mathbf{C}^m$.

In the case of $m = 1$ we give here another simpler way to define the proximity potential function $\phi_{\mathscr{I}}$ without the projective algebraic assumption for N such that (2.4.4) holds in every disk $\Delta(r)$ (cf. Noguchi–Winkelmann–Yamanoi [08]).

Let N be a compact complex space in general, and let $\mathscr{I}$ be a coherent ideal sheaf of $\mathscr{O}_N$. Take an open covering $\{U_j\}$ of N such that

(i) there is a partition of unity $\{c_j\}$ subordinate to $\{U_j\}$,
(ii) there are finitely many sections $\sigma_{jk} \in \Gamma(U_j, \mathscr{I}), k = 1, 2, \ldots$, generating every fiber $\mathscr{I}_x$ over $x \in U_j$.

We set

$$\rho_{\mathscr{I}}(x) = C\left(\sum_j c_j(x) \sum_k \left|\sigma_{jk}(x)\right|^2\right)^{1/2}. \tag{2.4.14}$$

Here a positive constant C is chosen so that

$$\rho_{\mathscr{I}}(x) \leqq 1, \quad x \in N.$$

Using the compactness of N, one easily verifies that, up to addition by a bounded function on N, $\log \rho_{\mathscr{I}}$ is independent of the choices of the open covering, the partition of unity, the local generators of the ideal sheaf $\mathscr{I}$, and the constant C.

Let $f : \mathbf{C} \to N$ be a holomorphic mapping, which we will call an *entire curve*, such that $f(\mathbf{C}) \not\subset \operatorname{Supp} Y$. Then the function $\rho_{\mathscr{I}} \circ f(z)$ is smooth over $\mathbf{C} \setminus f^{-1}(\operatorname{Supp} Y)$. For $z_0 \in f^{-1}(\operatorname{Supp} Y)$ there is an open neighborhood U of z_0 and a positive integer ν such that $f^*\mathscr{I} = ((z - z_0)^{\nu_0})$, and then

$$\log \rho_{\mathscr{I}} \circ f(z) = \nu_0 \log|z - z_0| + \psi_0(z), \quad z \in U$$

for some C^∞ function $\psi_0(z)$ defined on U. Setting $\bar{\Delta}(r) \cap \operatorname{Supp} f^*Y = \{z_j\}_{j=1}^h$, we have a finite sum $\sum_{j=1}^h \nu_j \log|z - z_j|$ with $\nu_j \in \mathbf{N}$ such that

$$\psi(z) = \log \rho_{\mathscr{I}} \circ f(z) - \sum_{j=1}^h \nu_j \log|z - z_j|$$

is a C^∞ function in a neighborhood of $\bar{\Delta}(r)$. It is easy to see that a C^∞ function in a neighborhood of $\bar{\Delta}(r)$ is written as a difference of two subharmonic functions in a (possibly smaller) neighborhood of $\bar{\Delta}(r)$. Therefore, at least in a neighborhood of $\bar{\Delta}(r)$, (2.4.4) holds for $\log \rho_{\mathscr{I}} \circ f(z)$.

We then define the proximity function of f for $\mathscr{I}$ or for Y by

$$m_f(r, Y) = m_f(r, \mathscr{I}) = \int_{|z|=r} \log \frac{1}{\rho_{\mathscr{I}}(f(re^{i\theta}))} \frac{d\theta}{2\pi} \quad (\geqq 0). \tag{2.4.15}$$

We define the counting function $N(r, f^*\mathscr{I})$ and $N_l(r, f^*\mathscr{I})$ by using the divisor $\sum \nu_j \{z_j\}$. Moreover we define

$$\begin{aligned} \omega_{\mathscr{I}, f} = \omega_{Y, f} &= -2dd^c \psi(z) = -\frac{i}{\pi} \partial\bar{\partial} \psi(z) \\ &= dd^c \log \frac{1}{\rho_{\mathscr{I}} \circ f(z)} \quad (z \in U), \end{aligned} \tag{2.4.16}$$

which is well-defined on $\mathbf{C}$ as a C^∞ (1,1)-form. The order function of f for $\mathscr{I}$ or Y is defined by

$$T(r, \omega_{\mathscr{I},f}) = T(r, \omega_{Y,f}) = \int_1^r \frac{dt}{t} \int_{\Delta(t)} \omega_{\mathscr{I},f}. \tag{2.4.17}$$

If N is projective algebraic, the difference of the proximity potential functions (cf. (2.4.2), (2.4.14))

$$\log \rho_{\mathscr{I}}(x) - \log d_{\mathscr{I}}(x) \quad (x \in N)$$

is a bounded function.

Therefore we see the following.

Proposition 2.4.18 (i) *The proximity function* $m_f(r, \mathscr{I})$ $(=m_f(r, Y))$ *in the present subsection differs from that defined in the previous subsection only by a bounded function.*

(ii) *There is no change in the counting function for* f^*I $(=f^*Y)$.

(iii) *Thus the difference of the order functions* $T(r, \omega_{\mathscr{I},f})$ $(=T(r, \omega_{Y,f}))$ *is bounded, too.*

2.5 Order Functions

Let N be an n-dimensional compact complex space and let $f : \mathbf{C}^m \to N$ be a meromorphic mapping. We will define several order functions of f and will give their comparison. Then we will give a characterization of rational f in terms of the order function, when N is projective algebraic.

2.5.1 Metrics

Let

$$h = \sum_{j,k} h_{j\bar{k}} dx_j \otimes d\bar{x}_k$$

be a hermitian metric on the holomorphic tangent space $\mathbf{T}(N)$ of N; that is, the coefficients of h are C^∞ functions, the quadratic form $h(X, \bar{Y})$ in $X, Y \in \mathbf{T}(N)_x$ $(x \in N)$ is hermitian and positive definite. Let $\omega = \sum_{j,k} \frac{i}{2} h_{j\bar{k}} dx_j \wedge d\bar{x}_k$ be the associated $(1,1)$-form. If $d\omega = 0$, we call ω a Kähler form, h is called a Kähler metric, and N is called a Kähler manifold if N is non-singular. As in (2.3.30), we define the *order function* of f with respect to ω by

$$T_f(r, \omega) = \int_1^r \frac{dt}{t^{2m-1}} \int_{B(t)} f^*\omega \wedge \alpha^{m-1}, \quad r > 1. \tag{2.5.1}$$

Lemma 2.5.2 *Let ω and ω' be two hermitian metrics on N. Then there is a constant $C > 0$ such that for an arbitrary meromorphic mapping $f : \mathbf{C}^m \to N$*

$$C^{-1}T_f(r,\omega) \leqq T_f(r,\omega) \leqq CT_f\big(r,\omega'\big).$$

Proof Since ω and ω' are both positive definite and N is compact, there is a constant $C > 0$ such that

$$C\omega - \omega' \geqq 0, \qquad C\omega' - \omega \geqq 0.$$

Therefore

$$Cf^*\omega \wedge \alpha^{m-1} - f^*\omega' \wedge \alpha^{m-1} \geqq 0,$$
$$f^*\omega' \wedge \alpha^{m-1} - Cf^*\omega \wedge \alpha^{m-1} \geqq 0.$$

Thus the required inequalities follow. □

We define the symbol $S_f(r,\omega)$ with respect to $T_f(r,\omega)$ as (1.2.4). That is,

$$S_f(r,\omega) = O\big(\log T_f(r,\omega)\big) + \delta \log r\|_{E(\delta)}. \tag{2.5.3}$$

The above definition is independent of the choice of ω by Lemma 2.5.2. When it is not necessary to specify ω, we simply write

$$S_f(r) = S_f(r,\omega).$$

Theorem 2.5.4 *Let ω be a hermitian metric form on N. A meromorphic mapping $f : \mathbf{C}^m \to N$ is constant if and only if $\underline{\lim}_{r\to\infty} T_f(r,\omega)/\log r = 0$. This condition is equivalent to $T_f(r,\omega) = S_f(r,\omega)$, provided that $d\omega = 0$, or $m = 1$.*

Proof The "only if" part is trivial. Assume that ω is a Kähler form on N. Suppose that $d\omega = 0$. Then $f^*\omega$ is a d-closed positive semi-defined $(1,1)$-form. By Poincaré's Lemma for d and for $\bar\partial$ on $\mathbf{C}^m$ there exists a plurisubharmonic function φ on $\mathbf{C}^m$ such that $dd^c\varphi = f^*\omega$ in the sense of currents. (For the present argument, the existence on every ball $B(R)$ $(R > 0)$ is sufficient.) It follows from Lemma 2.1.31 that the function

$$n\big(t, f^*\omega\big) = \frac{1}{t^{2m-2}} \int_{B(t)} f^*\omega \wedge \alpha^{m-1}$$

is monotone increasing in $t > 0$; this monotonicity is trivial without the assumption $d\omega = 0$ if $m = 1$. The constancy of f is equivalent to $f^*\omega \wedge \alpha^{m-1} \equiv 0$ and so to $n(t, f^*\omega) \equiv 0$. Assume that f is not constant. Then there is a $t_0 > 0$ with $n(t_0, f^*\omega) > 0$. For $r > t_0$

$$T_f(r,\omega) = \int_1^r \frac{n(t, f^*\omega)}{t} dt \geqq \int_{t_0}^r \frac{n(t, f^*\omega)}{t} dt$$

(continued)

$$\geqq n\big(t_0, f^*\omega\big)\int_{t_0}^{r}\frac{dt}{t} = n\big(t_0, f^*\omega\big)(\log r - \log t_0).$$

Hence $\underline{\lim}_{r\to\infty} T_f(r,\omega)/\log r \geqq n(t_0, f^*\omega) > 0$.

Assume that $T_f(r,\omega) = S_f(r,\omega)$. Then for every $\delta > 0$ we have

$$\overline{\lim_{r\to\infty}}\frac{\log r}{T_f(r,\omega)} \geqq \frac{1}{\delta}.$$

It follows that $\underline{\lim}_{r\to\infty} T_f(r,\omega)/\log r = 0$, and that f is constant. □

Let f be a meromorphic function on $\mathbf{C}^m$. Following after (1.1.10) and (1.1.12), we set

$$\begin{aligned} m(r, f) &= \int_{\|z\|=r} \log^+ \big|f(z)\big|\gamma(z), \\ T(r, f) &= m(r, f) + N\big(r, (f)_\infty\big). \end{aligned} \tag{2.5.5}$$

We call $T(r, f)$ *Nevanlinna's order function*. This is convenient in calculating estimates. There are co-prime holomorphic functions f_0 and f_1 on $\mathbf{C}^m$ (i.e., $\operatorname{codim}\{f_0 = f_1 = 0\} \geqq 2$) such that $f = f_1/f_0$. Let $[w_0, w_1]$ be the natural homogeneous coordinate system of $\mathbf{P}^1(\mathbf{C})$, and identify f with a meromorphic mapping

$$f : z \in \mathbf{C}^m \to \big[f_0(z), f_1(z)\big] \in \mathbf{P}^1(\mathbf{C}).$$

By taking the Fubini–Study metric form ω on $\mathbf{P}^1(\mathbf{C})$, we compare $T_f(r,\omega)$ and $T(r, f)$. The following is the Shimizu–Ahlfors theorem and the First Main Theorem.

Theorem 2.5.6 *For a meromorphic function f on $\mathbf{C}^m$*

$$T(r, f) - T_f(r,\omega) = O(1).$$

In particular, for any $a \in \mathbf{C}$

$$T\left(r, \frac{1}{f-a}\right) = T(r, f) + O(1).$$

Proof Noting that for every $s \geqq 0$

$$0 \leqq \log(1+s) - \log^+ s \leqq \log 2,$$

we get

$$\begin{aligned} T(r, f) &= \frac{1}{2}\int_{\|z\|=r} \log\left(1 + \left|\frac{f_1(z)}{f_0(z)}\right|^2\right)\gamma(z) + \frac{1}{2}\int_{\|z\|=r}\log|f_0|^2\gamma + O(1) \\ &= \frac{1}{2}\int_{\|z\|=r}\log\big(|f_0|^2 + |f_1|^2\big)\gamma + O(1). \end{aligned}$$

By making use of Lemma 2.1.33 (Jensen's formula), we have

$$\begin{aligned} T(r, f) &= \int_1^r \frac{dt}{t^{2m-1}} \int_{B(t)} f^*\omega \wedge \alpha^{m-1} + O(1) \\ &= T_f(r, \omega) + O(1). \end{aligned}$$

For the latter half, we notice that $T_{1/(f-a)}(r, \omega) = T_{(f-a)}(r, \omega)$. It follows from what was shown that

$$\begin{aligned} T\left(r, \frac{1}{f-a}\right) &= T(r, f-a) + O(1) \\ &= m(r, f-a) + N\big(r, (f-a)_\infty\big) + O(1) \\ &= m(r, f) + N\big(r, (f)_\infty\big) + O(1) \\ &= T(r, f) + O(1). \end{aligned}$$

□

Theorem 2.5.7 *Let $f : \mathbf{C}^m \to N$ be a meromorphic mapping and let $L \to N$ be a hermitian line bundle. Assume that $L \geqq 0$ or that N is projective algebraic. Let $\sigma_0, \sigma_1 \in H^0(N, L)$ be linearly independent sections such that $f(\mathbf{C}^m) \not\subset \{\sigma_0 = 0\}$. Then the meromorphic function $g(z) = \sigma_1 \circ f(z)/\sigma_0 \circ f(z)$ satisfies*

$$T(r, g) \leqq T_f(r, L) + O(1).$$

Proof Let $\|\cdot\|$ be a hermitian metric in L. Then

$$\begin{aligned} \log^+ \left|\frac{\sigma_1}{\sigma_0}\right| &= \log^+ \frac{\|\sigma_1\|}{\|\sigma_0\|} \\ &\leqq \log^+ \frac{1}{\|\sigma_0\|} + \log^+ \|\sigma_1\|. \end{aligned}$$

Since $\|\sigma_1\|$ is bounded on N, there is a constant C such that

$$\log^+ \left|\frac{\sigma_1}{\sigma_0}\right| \leqq \log^+ \frac{1}{\|\sigma_0\|} + C.$$

By definition $(g)_\infty \leqq f^*(\sigma_0)$. Thus

$$N\big(r, (g)_\infty\big) \leqq N\big(r, f^*(\sigma_0)\big).$$

It follows that

$$\begin{aligned} T(r, g) &\leqq m_f\big(r, (\sigma_0)\big) + N\big(r, f^*(\sigma_0)\big) + C \\ &= T_f(r, L) + O(1). \end{aligned}$$

□

Corollary 2.5.8 (Noguchi [76b]) *If σ_0 and σ_1 have no common zero,*

$$T(r, g) = T_f(r, L) + O(1).$$

Proof By the assumption there is a constant $C > 0$ such that

$$\left|\log^+\left|\frac{\sigma_1}{\sigma_0}\right| - \log^+\frac{1}{\|\sigma_0\|}\right| \leqq C.$$

Furthermore, $N(r, (g)_\infty) = N(r, f^*(\sigma_0))$, so that the claim holds. □

If $\dim N = 1$, then the assumption of Corollary 2.5.8 is always satisfied. Let $N = \mathbf{P}^1(\mathbf{C})$ and $L \to \mathbf{P}^1(\mathbf{C})$ be the hyperplane bundle. Then the above corollary is the Shimizu–Ahlfors Theorem 1.1.19. For the composition of a rational function with f we have

Corollary 2.5.9 *Let $Q \circ f$ be the composition of a meromorphic function $f(z)$ on $\mathbf{C}^m$ with a rational function Q in one variable. Assume that f and Q are non-constant. Denoting the degree of Q by d, we have*

$$T(r, Q \circ f) = dT(r, f) + O(1).$$

Proof There are sections $\sigma_0, \sigma_1 \in H^0(\mathbf{P}^1(\mathbf{C}), L^d)$ without common zero such that $Q = \sigma_1/\sigma_0$. Therefore

$$T(r, Q \circ f) = dT_f(r, L) + O(1) = dT(r, f) + O(1).$$ □

2.5.2 Cartan's Order Function

Here we set $N = \mathbf{P}^n(\mathbf{C})$. Let ω be the Fubini–Study metric form on $\mathbf{P}^n(\mathbf{C})$. Let $w = [w_0, \ldots, w_n]$ be a homogeneous coordinate system of $\mathbf{P}^n(\mathbf{C})$. Take a meromorphic mapping $f : \mathbf{C}^m \to \mathbf{P}^n(\mathbf{C})$. Then there are holomorphic functions $f_0, \ldots, f_n$ on $\mathbf{C}^m$ such that $\operatorname{codim}\{f_0 = \cdots = f_n = 0\} \geqq 2$ and

$$f(z) = \big[f_0(z), \ldots, f_n(z)\big].$$

Let $f(z) = [g_0(z), \ldots, g_n(z)]$ be another such representation. Then there is a holomorphic function $h(z)$ on $\mathbf{C}^m$ without zero such that

$$f_j(z) = h(z)g_j(z), \quad 0 \leqq j \leqq n,\ z \in \mathbf{C}^m. \tag{2.5.10}$$

H. Cartan [33] defined the order function $T_f(r)$ of $f : \mathbf{C}^m \to \mathbf{P}^n(\mathbf{C})$ by

$$T_f(r) = \int_{\|z\|=r} \log \max_{0 \leqq j \leqq n} \big|f_j(z)\big| \gamma(z) - \log \max_{0 \leqq j \leqq n} \big|f_j(0)\big|, \tag{2.5.11}$$

which is called Cartan's order function; in fact, he dealt with the case of $m = 1$. It follows from (2.5.10) that $T_f(r)$ is independent of the representation of f.

Theorem 2.5.12 *Let the notation be as above. Then we have*

$$T_f(r) = \int_{\|z\|=r} \log\Bigg(\sum_{0\leqq j\leqq n} \big|f_j(z)\big|^2\Bigg)^{1/2} \gamma(z) + O(1)$$

$$= T_f(r,\omega) + O(1).$$

This is immediate by (2.3.35).

A linear form $F_j = \sum c_{jk} w_k$ in the homogeneous coordinates $[w_0, \ldots, w_n]$ is identified with a holomorphic section of the hyperplane bundle on $\mathbf{P}^n(\mathbf{C})$. For two linearly independent linear forms F_1 and F_2 we consider the compositions $F_j(f(z)) = \sum c_{jk} f_k(z)$ with $f(z) = [f_0(z), \ldots, f_n(z)]$. If $F_1(f(z)) \not\equiv 0$, $g(z) = F_2(f(z))/F_1(f(z))$ is defined independently of the representation of f. The next theorem is due to Toda [70a], Lemma 1 in the case of $m = 1$ and the case of $m \geqq 2$ is similarly proved.

Theorem 2.5.13 *Let $f(z) = [f_0(z), \ldots, f_n(z)]$ and $g(z)$ be as above. The following hold*:

(i) $T(r, g) \leqq T_f(r, \omega) + O(1)$.

(ii) *If $f_k \not\equiv 0$,*

$$\frac{1}{n}\sum_{j=0}^{n} T\left(r, \frac{f_j}{f_k}\right) + O(1) \leqq T_f(r,\omega) \leqq \sum_{j=0}^{n} T\left(r, \frac{f_j}{f_k}\right) + O(1).$$

Proof (i) is a special case of Theorem 2.5.7. The first inequality in (ii) is clear by (i). The latter is deduced as follows. We may assume $f_0 \not\equiv 0$ without loss of generality. Then

$$T_f(r,\omega) = \int_{\|z\|=r} \log\Bigg(\max_{1\leqq j\leqq n}\left\{1, \frac{|f_j(z)|}{|f_0(z)|}\right\} \cdot \big|f_0(z)\big|\Bigg)\gamma(z) + O(1)$$

$$= \int_{\|z\|=r} \log^+ \max_{1\leqq j\leqq n}\left\{\frac{|f_j(z)|}{|f_0(z)|}\right\}\gamma(z)$$

$$+ \int_{\|z\|=r} \log\big|f_0(z)\big|\gamma(z) + O(1)$$

$$\leqq \sum_{j=1}^{n} \int_{\|z\|=r} \log^+\left|\frac{f_j}{f_0}\right|\gamma + N\big(r, (f_0)_0\big) + O(1)$$

$$\leqq \sum_{j=1}^{n}\left\{\int_{\|z\|=r} \log^+\left|\frac{f_j}{f_0}\right|\gamma + N\left(r, \left(\frac{f_j}{f_0}\right)_\infty\right)\right\} + O(1)$$

$$= \sum_{j=1}^{n} T\left(r, \frac{f_j}{f_0}\right) + O(1).$$

□

2.5.3 A Family of Rational Functions

First note that w_j/w_k, $0 \leqq j(\neq k) \leqq n$, form a transcendental base of the field extension of the rational function field of $\mathbf{P}^n(\mathbf{C})$ over $\mathbf{C}$. It follows from Theorem 2.5.13 (ii) that up to a positive constant multiple, $T_f(r,\omega)$ and $\sum_{\substack{j=0\\ j\neq k}}^{n} T(r, f_j/f_k)$ are equivalent. We generalize this to a general projective algebraic manifold.

Lemma 2.5.14 *For an arbitrary point $a \in \mathbf{C}$*

$$\frac{1}{2\pi}\int_0^{2\pi} \log\left|e^{i\theta} - a\right| d\theta = \log^+ |a|.$$

The proof is left to the reader.

Lemma 2.5.15 *Let $g(z)$ and $A_1(z), \ldots, A_l(z)$ be meromorphic functions on $\mathbf{C}^m$ satisfying*

$$\big(g(z)\big)^l + A_1(z)\big(g(z)\big)^{l-1} + \cdots + A_l(z) = 0.$$

Then

$$T(r,g) \leqq \sum_{j=1}^{l} T(r, A_j) + \log(l+1).$$

Proof Introducing a variable t, we set

$$B(z,t) = t^l + A_1(z)t^{l-1} + \cdots + A_l(z).$$

For $z \in \mathbf{C}^m \setminus \bigcup_j \mathrm{Supp}(A_j)_\infty$ we let $t_1(z) = g(z), t_2(z), \ldots, t_l(z)$ be roots of $B(z,t) = 0$. Then

$$B(z,t) = \prod_{j=1}^{l} \big(t - t_j(z)\big).$$

Therefore we have

$$\begin{aligned}
\frac{1}{2\pi}\int_0^{2\pi} \log\left|B\big(z, e^{i\theta}\big)\right| d\theta &= \sum_{j=1}^{l} \frac{1}{2\pi}\int_0^{2\pi} \log\left|e^{i\theta} - t_j(z)\right| d\theta \\
&\qquad \text{(by Lemma 2.5.14)} \\
&= \log^+\left|g(z)\right| + \sum_{j=2}^{l} \log^+\left|t_j(z)\right| \\
&\geqq \log^+\left|g(z)\right|.
\end{aligned}$$

On the other hand,

$$\begin{aligned}&\frac{1}{2\pi}\int_0^{2\pi}\log\left|B\left(z,e^{i\theta}\right)\right|d\theta\\&=\frac{1}{2\pi}\int_0^{2\pi}\log\left|e^{il\theta}+A_1(z)e^{i(l-1)\theta}+\cdots+A_l(z)\right|d\theta\\&\leqq\sum_{j=1}^{l}\log^+\left|A_j(z)\right|+\log(l+1).\end{aligned}$$

It follows that

$$m(r,g)\leqq\sum_{j=1}^{l}m(r,A_j)+\log(l+1). \tag{2.5.16}$$

By making use of co-prime holomorphic functions g_0, g_1 on $\mathbf{C}^m$, we represent $g=g_1/g_0$. Let (A_0) be the minimum common divisor of the polar divisors of $A_1,\ldots,A_l$, where A_0 is a holomorphic function on $\mathbf{C}^m$. Then

$$A_0(g_1)^l=-g_0(z)\left\{A_0A_1(g_1)^{l-1}+\cdots+A_0A_l(g_0)^{l-1}\right\}.$$

Therefore $(g_0)\leqq(A_0)$ as divisors, and

$$N\left(r,(g)_\infty\right)\leqq N\left(r,(A_0)_0\right)\leqq\sum_{j=1}^{l}N\left(r,(A_j)_\infty\right). \tag{2.5.17}$$

It follows from (2.5.16) and (2.5.17) that

$$\begin{aligned}T(r,g)&\leqq\sum_{j=1}^{l}\left\{m(r,A_j)+N\left(r,(A_j)_\infty\right)\right\}+\log(l+1)\\&=\sum_{j=1}^{l}T(r,A_j)+\log(l+1).\end{aligned}$$

□

In general, let N be an n-dimensional complex projective algebraic variety and let $\{\phi_j\}_{j=1}^s$ be a finite subset of the rational function field $\mathbf{C}(N)$ of N. Assume that a meromorphic mapping $f:\mathbf{C}^m\to N$ satisfies $f(\mathbf{C}^m)\not\subset\bigcup_{j=1}^s\mathrm{Supp}(\phi_j)_\infty$. We define the *order function* of f with respect to $\{\phi_j\}$ by

$$T_f\left(r,\{\phi_j\}\right)=\sum_{j=1}^{l}T(r,\phi_j\circ f).$$

In what follows, it is the essential case when $\{\phi_j\}_{j=1}^s$ contains a transcendental base of $\mathbf{C}(N)$ over $\mathbf{C}$.

Let $N \hookrightarrow \mathbf{P}^l(\mathbf{C})$ be an embedding and let L be the restriction of the hyperplane bundle over $\mathbf{P}^l(\mathbf{C})$ to N.

Theorem 2.5.18 (i) *Assume that $\{\phi_j\}_{j=1}^s$ contains a transcendental base of $\mathbf{C}(N)$ over $\mathbf{C}$ and that f is algebraically non-degenerate. Then there is a constant $C > 0$ independent of f such that*

$$C^{-1}T_f(r, L) + O(1) \leqq T_f\big(r, \{\phi_j\}\big) \leqq CT_f(r, L) + O(1).$$

(ii) *Let $f : \mathbf{C}^m \to N$ be an algebraically non-degenerate meromorphic mapping and let $\Phi : N \to M$ be a birational mapping onto another complex projective algebraic variety M. Let $H \to M$ be a positive line bundle. Then there is a constant $C_1 = C_1(L, H) > 0$ such that*

$$C_1^{-1}T_f(r, L) + O(1) \leqq T_{\Phi\circ f}(r, H) \leqq C_1^{-1}T_f(r, L) + O(1).$$

Proof (i) Let $[w_0, \ldots, w_l]$ be a homogeneous coordinate system of $\mathbf{P}^l(\mathbf{C})$ and let $f(z) = [f_0, \ldots, f_l]$ be a reduced representation of f as a meromorphic mapping into $\mathbf{P}^l(\mathbf{C})$. We may assume $f_0 \not\equiv 0$ without loss of generality. The restrictions ψ_k of w_k/w_0 to N give a transcendental base of $\mathbf{C}(N)$ and then ϕ_j are represented as rational functions in ψ_k,

$$\phi_j = Q_j(\psi_1, \ldots, \psi_l).$$

Substituting $f(z)$, we have

$$\phi_j \circ f(z) = Q_j\big(\psi_1 \circ f(z), \ldots, \psi_l \circ f(z)\big).$$

Therefore there is a constant $C > 0$ determined by Q_j, $1 \leqq j \leqq n$, such that

$$\begin{aligned} T\big(r, Q_j(\psi_k \circ f)\big) &\leqq C\sum_{k=1}^{l} T(r, \psi_k \circ f) + O(1) \\ &\leqq lCT_f(r, L) + O(1). \end{aligned}$$

We deduce that

$$T\big(r, \{\phi_j\}\big) \leqq lnCT_f(r, L) + O(1).$$

Conversely, the rational functions ψ_k are algebraic over the field generated by $\{\phi_j\}$ over $\mathbf{C}$. Thus there are algebraic relations

$$(\psi_k)^{d_k} + A_{k1}(\phi_j)(\psi_k)^{d_k-1} + \cdots + A_{kd_k}(\phi_j) = 0, \quad 1 \leqq k \leqq l.$$

Substituting f, we obtain algebraic relations of meromorphic functions on $\mathbf{C}^m$:

$$(\psi_k \circ f)^{d_k} + A_{k1}(\phi_j \circ f)(\psi_k \circ f)^{d_k-1} + \cdots + A_{kd_k}(\phi_j \circ f) = 0, \quad 1 \leqq k \leqq l.$$

Lemma 2.5.15 implies that

$$T(r, \psi_k \circ f) \leqq \sum_{h=1}^{d_k} T\big(r, A_{kh}(\phi_j \circ f)\big) + \log(d_k + 1)$$
$$\leqq C' \sum_{j=1}^{n} T(r, \phi_j \circ f) + O(1).$$

Here $C' > 0$ is a constant depending only on $\{\psi_k\}$ and $\{\phi_j\}$. It follows from Theorem 2.5.13 that

$$T_f(r, L) \leqq \sum_{k=1}^{l} T(r, \psi_k \circ f) \leqq lC' \sum_{j=1}^{n} T(r, \phi_j \circ f) + O(1).$$

(ii) Since $\Phi^* : \mathbf{C}(M) \to \mathbf{C}(N)$ is a field isomorphism over $\mathbf{C}$, the claim is clear by (i). □

In general let N be a compact complex space and let ω be a hermitian metric form on it. For a meromorphic mapping $f : \mathbf{C}^m \to N$ we define the *order* ρ_f by

$$\rho_f = \varlimsup_{r\to\infty} \frac{\log T_f(r, \omega)}{\log r}. \tag{2.5.19}$$

By Lemma 2.5.2 ρ_f is independent of the choice of ω. If N is projective algebraic and f is algebraically non-degenerate, $T_f(r, \omega)$ in (2.5.19) may be replaced with $T_f(r, \{\phi_j\})$ by Theorem 2.5.18 to define the same ρ_f.

The next three propositions are easily deduced from Theorem 2.5.18.

Proposition 2.5.20 *Let $\eta : V \to W$ be a rational mapping between quasi-projective algebraic varieties V and W. Then for an algebraically non-degenerate meromorphic mapping $f : \mathbf{C}^m \to V$*

$$T_{\eta\circ f}(r) = O\big(T_f(r)\big).$$

Moreover, if η is generically finite, then

$$T_f(r) = O\big(T_{\eta\circ f}(r)\big).$$

Let V be a quasi-projective algebraic variety, and let $f : \mathbf{C}^m \to V$ be a meromorphic mapping. Taking a projective compactification $\bar{V} \supset V$, and regarding $f : \mathbf{C}^m \to \bar{V}$, we may define the *order* ρ_f of f, which is independent of the choice of the compactification $\bar{V}$.

Corollary 2.5.21 *The above order ρ_f of f is independent of the choice of the compactification $\bar{V}$ of V.*

This is immediate from Proposition 2.5.20.

Proposition 2.5.22 *Let $f : \mathbf{C}^m \to N$ be a meromorphic mapping into a complex projective variety N and let H be a line bundle on N. Assume that H is big, and that f is algebraically non-degenerate. Then*

$$T_f(r, L) = O\big(T_f(r, H)\big)$$

for every line bundle L on M.

If $f : \mathbf{C}^m \to N$ is algebraically degenerate, we may consider the Zariski closure X of $f(\mathbf{C})$ and a desingularization $\tau : \tilde{X} \to X$. Then f lifts to a map to $\tilde{X}$ and $\tau^*(H|_X)$ is big on $\tilde{X}$ for every ample line bundle H on N. As a consequence we obtain

Proposition 2.5.23 *Let $f : \mathbf{C}^m \to N$ be a meromorphic mapping into a complex projective variety N. Let $h(r)$ be a non-negative valued function in $r > 1$. Then $h(r) = S_f(r, H)$ holds for every ample line bundle if and only if it holds for at least one ample line bundle.*

Similarly, the statement "$h(r) \leqq \varepsilon T_f(r, H)\|_\varepsilon$, $^\forall\varepsilon > 0$", respectively "$h(r) = O(T_f(r, H))$" holds for every ample line bundle H if and only if it holds for at least one ample line bundle.

If f is algebraically non-degenerate, the same statements as above hold for big line bundles.

2.5.4 Characterization of Rationality

Let $g \not\equiv 0$ be a holomorphic function on $\mathbf{C}^m$. Then $\log|g(z)|$ is a plurisubharmonic function and hence a subharmonic function on $\mathbf{C}^m \cong \mathbf{R}^{2m}$ (Theorem 2.1.26 (i)). Taking the Poisson integral over the sphere $\{\|z\| = R\}$ of $\mathbf{C}^m$ we have that for $\|z\| < R$

$$\begin{aligned}
\log\big|g(z)\big| &\leqq \int_{\|\zeta\|=R} \log\big|g(\zeta)\big| \frac{(R^2 - \|z\|^2)R^{2m-2}}{\|\zeta - z\|^{2m}} \gamma(\zeta) \\
&\leqq \int_{\|\zeta\|=R} \log^+\big|g(\zeta)\big| \frac{(R^2 - \|z\|^2)R^{2m-2}}{\|\zeta - z\|^{2m}} \gamma(\zeta) \\
&\leqq \frac{(R^2 - \|z\|^2)R^{2m-2}}{(R - \|z\|)^{2m}} \int_{\|\zeta\|=R} \log^+\big|g(\zeta)\big| \gamma(\zeta).
\end{aligned}$$

Therefore we obtain

Lemma 2.5.24[5] *Let g be a holomorphic function on $\mathbf{C}^m$. Then for $0 < r < R$,*

$$T(r,g) \leqq \log^+ \max_{\|z\|=r} |g(z)| \leqq \frac{1-(r/R)^2}{(1-r/R)^{2m}} T(R,g).$$

Lemma 2.5.25 *A holomorphic function g on $\mathbf{C}^m$ is polynomial if and only if $T(r,g) = O(\log r)$.*

Proof If $g(z)$ is a polynomial function, an easy computation yields that $T(r,g) = O(\log r)$. Conversely we assume that $T(r,g) \leqq d\log r + C$. Putting $R = \tau r$, $\tau > 1$, we get by Lemma 2.5.24

$$\log^+ \max_{\|z\|=r} |g(z)| \leqq \frac{(\tau+1)\tau^{2m-2}}{(\tau-1)^{2m-1}} (d\log r + d\log\tau + C).$$

Set $d(\tau) = \frac{(\tau+1)\tau^{2m-2}}{(\tau-1)^{2m-1}}$ and $C(\tau) = d(\tau)(d\log\tau + C)$. We expand $g(z)$ to a Taylor series with multi-indices α,

$$g(z) = \sum_{|\alpha|=0}^{\infty} a_\alpha z^\alpha.$$

From this we obtain

$$\begin{aligned}
&\left(\frac{1}{2\pi}\right)^m \int_0^{2\pi} \cdots \int_0^{2\pi} \left|g\left(e^{i\theta_1} z_1, \ldots, e^{i\theta_m} z_m\right)\right|^2 d\theta_1 \cdots d\theta_m \\
&= \sum_{|\alpha|=0}^{\infty} |a_\alpha|^2 |z_1|^{2\alpha_1} \cdots |z_m|^{2\alpha_m} \leqq \max_{\|z\|=r} |g(z)|^2 \\
&\leqq r^{2dd(\tau)} \cdot e^{2d(\tau)C(\tau)}.
\end{aligned}$$

Hence $\sum_{|\alpha|>dd(\tau)}^{\infty} |a_\alpha|^2 |z_1|^{2\alpha_1} \cdots |z_m|^{2\alpha_m} = 0$ and so

$$g(z) = \sum_{|\alpha|=0}^{dd(\tau)} a_\alpha z^\alpha.$$

Since $dd(\tau) \to d$ as $\tau \to \infty$, $g(z)$ is a polynomial of degree at most d. □

Theorem 2.5.26 (Stoll [64a], [64b]) *Let E be an effective divisor on $\mathbf{C}^m$. Then E is a divisor determined by a polynomial of degree at most d if and only if*

$$N(r,E) \leqq d\log r + O(1).$$

[5] An estimate of this type in several complex variables is found in Kneser [38] without an explicit formula; cf. Noguchi [75].

This is shown by the Weierstrass–Stoll canonical product which generalizes Weierstrass' canonical product. Cf. Noguchi–Ochiai [90] (Ochiai–Noguchi [84]) for a proof simplified by Lelong. Here we omit the proof.

Theorem 2.5.27 *A meromorphic function $g(z)$ on $\mathbf{C}^m$ is a rational function if and only if*

$$T(r, g) = O(\log r).$$

Proof If g is a rational function, we write $g(z) = \frac{P(z)}{Q(z)}$ with co-prime polynomials $P(z)$, $Q(z)$. By the Shimizu–Ahlfors Theorem 1.1.19 and Theorem 2.5.12 we have

$$\begin{aligned} T(r, g) &= \int_{\|z\|=r} \log \sqrt{|P(z)|^2 + |Q(z)|^2}\gamma + O(1) \\ &= O(\log r). \end{aligned}$$

For the converse, we first note that

$$N\big(r, (g)_\infty\big) \leqq T(r, g) = O(\log r).$$

By Stoll's Theorem 2.5.26 there is a polynomial g_0 such that $(g_0) = (g)_\infty$. If we set $g_1 = gg_0$, g_1 is a holomorphic function and satisfies

$$T(r, g_1) \leqq T(r, g) + T(r, g_0) = O(\log r).$$

By Lemma 2.5.25 g_1 is a polynomial. Thus g is a rational function. □

Theorem 2.5.28 *Let N be a projective algebraic variety and let ω be a hermitian metric form on it. A meromorphic mapping $f : \mathbf{C}^m \to N$ is a rational mapping if and only if*

$$T_f(r, \omega) = O(\log r).$$

Proof Taking an embedding $N \hookrightarrow \mathbf{P}^l(\mathbf{C})$, we may assume $N = \mathbf{P}^l(\mathbf{C})$ with ω the Fubini–Study metric form. The "only if" part is immediate from Theorems 2.5.27 and 2.5.13.

Assume that $T_f(r, \omega) = O(\log r)$. Let $w = [w_0, \ldots, w_l]$ be a homogeneous coordinate system of $\mathbf{P}^l(\mathbf{C})$ such that $f(\mathbf{C}^m) \not\subset \{w_0 = 0\}$. It follows from Theorem 2.5.13 that

$$T\left(r, f^* \frac{w_j}{w_0}\right) = O(\log r).$$

Therefore $f^*(w_j/w_0)$ are rational, and hence f is rational. □

Remark By Lemma 1.1.22 we see that $T_f(r, \omega) = O(\log r)$ if and only if $\varliminf_{r\to\infty} \frac{T_f(r,\omega)}{\log r} < \infty$.

2.6 Nevanlinna's Inequality

We generalize Theorem 1.1.18 to meromorphic functions on $\mathbf{C}^m$ and moreover to the case of meromorphic mappings. This plays an essential role in the proof of the lemma on logarithmic derivatives in the next chapter.

Let $f_1, \dots, f_n$ be entire functions on $\mathbf{C}^m$ which are linearly independent over $\mathbf{C}$. For a vector $w = (w_j) \in \mathbf{C}^n$ we set

$$I(w) = \int_{\|z\|=1} \log \left| \sum_{j=1}^{n} w_j f_j(z) \right| \gamma(z).$$

Lemma 2.6.1 *The function $I(w)$ is bounded on $\{\|w\| = 1\}$.*

Proof[6] Set $M = \sup\{|\sum_{j=1}^n w_j f_j(z)|;\ \|w\| = 1, \|z\| = 1\}$ $(<\infty)$. Setting $\Gamma = \{w \in \mathbf{C}^n;\ \|w\| = 1\}$, we see that $I(w) \leqq \log M$ on Γ. Now we show the boundedness of $I(w)$ from below. The function $\log|\sum_{j=1}^n w_j f_j(\zeta)|$ in $\zeta \in \mathbf{C}^m \cong \mathbf{R}^{2m}$ is subharmonic by Theorem 2.1.26 (i). By taking the Poisson integral we have

$$\log \left| \sum_{j=1}^{n} w_j f_j(\zeta) \right| \leqq \int_{\|z\|=1} \left(\log \left| \sum_{j=1}^{n} w_j f_j(z) \right| \right) \frac{1 - \|\zeta\|^2}{\|z - \zeta\|^{2m}} \gamma(z), \quad \|\zeta\| < 1.$$

For an arbitrary $a = (a_j) \in \Gamma$ we take $\zeta_0 \in B(1) (\subset \mathbf{C}^m)$ so that

$$\sum_{j=1}^{n} a_j f_j(\zeta_0) \neq 0.$$

There is a neighborhood W of a in Γ such that for every $w \in W$

$$\left| \sum_{j=1}^{n} w_j f_j(\zeta_0) \right| \geqq \frac{1}{2} \left| \sum_{j=1}^{n} a_j f_j(\zeta_0) \right| > 0.$$

Thus for $w \in W$

$$\begin{aligned} \log \frac{1}{2} \frac{|\sum_{j=1}^n a_j f_j(\zeta_0)|}{M} &\leqq \int_{\|z\|=1} \left(\log \frac{|\sum_{j=1}^n w_j f_j(z)|}{M} \right) \frac{1 - \|\zeta_0\|^2}{\|z - \zeta_0\|^{2m}} \gamma(z) \\ &\leqq \frac{1 - \|\zeta_0\|^2}{(1 + \|\zeta_0\|)^{2m}} \int_{\|z\|=1} \log \frac{|\sum_{j=1}^n w_j f_j(z)|}{M} \gamma(z) \\ &\leqq \frac{1 - \|\zeta_0\|^2}{2^{2m}} \big(I(w) - \log M \big). \end{aligned}$$

[6] A discussion on the proof of this lemma with Professors Phong and Demailly at Hayama Symposium on Complex Analysis in Several Variables 2002 was very helpful.

Hence $I(w)$ $(w \in W)$ is bounded from below. Since Γ is compact, $I(w)$ is bounded from below on Γ. □

Remark 2.6.2 In fact, it is shown that $I(w)$ is continuous in $w \in \mathbf{C}^n \setminus \{0\}$. Since the convergence theorem of Lebesgue integrals cannot be used, some more contrivance is necessitated to the proof.

Theorem 2.6.3 (Nevanlinna's inequality) *Let f be a non-constant meromorphic function on $\mathbf{C}^m$. Then there is a constant C such that for every $a \in \mathbf{C} \cup \{\infty\}$*

$$N\big(r, (f-a)_0\big) < T(r, f) + C, \quad r \geqq 1.$$

Proof We write $f = f_2/f_1$ with co-prime entire functions f_1, f_2. We regard f to be a meromorphic mapping $f : z \in \mathbf{C}^m \to [f_1(z), f_2(z)] \in \mathbf{P}^1(\mathbf{C})$. Let $a = [a_2, -a_1] \in \mathbf{P}^1(\mathbf{C}) \cong \mathbf{C} \cup \{\infty\}$. We may assume that $|a_1|^2 + |a_2|^2 = 1$. By Example 2.3.34 we have

$$(f-a)_0 = (a_1 f_1 + a_2 f_2)_0 = f^* a,$$

$$m_f(r, a) = \int_{\|z\|=r} \log \frac{\sqrt{|f_1(z)|^2 + |f_2(z)|^2}}{|a_1 f_1(z) + a_2 f_2(z)|} \gamma(z) \geqq 0.$$

Let ω be the Fubini–Study metric form on $\mathbf{P}^1(\mathbf{C})$. The First Main Theorem 2.3.31 implies that

$$N\big(r, f^* a\big) \leqq T_f(r, \omega) + m_f(1, a),$$

$$\begin{aligned} m_f(1, a) &= \int_{\|z\|=1} \log \frac{\sqrt{|f_1(z)|^2 + |f_2(z)|^2}}{|a_1 f_1(z) + a_2 f_2(z)|} \gamma(z) \\ &= \int_{\|z\|=1} \log \sqrt{\big|f_1(z)\big|^2 + \big|f_2(z)\big|^2} \gamma(z) \\ &\quad - \int_{\|z\|=1} \log \big|a_1 f_1(z) + a_2 f_2(z)\big| \gamma(z). \end{aligned}$$

By Lemma 2.6.1 there is a constant C such that

$$m_f(1, a) < C, \quad {}^\forall a \in \mathbf{P}^1(\mathbf{C}).$$

By Theorem 2.5.6 $T_f(r, \omega) = T(r, f) + O(1)$. Thus the required formula is obtained. □

The above Nevanlinna inequality will suffice for the application in the next chapter, but we extend it to a meromorphic mapping $f : \mathbf{C}^m \to N$ into a projective algebraic variety N. Let $L \to N$ be a hermitian line bundle and take an arbitrary linear subspace $E \subset H^0(N, L)$.

Theorem 2.6.4 *Let the notation be as above. Assume that* $f(\mathbf{C}^m) \not\subset \mathrm{Supp}(\sigma)$ *for every* $\sigma \in E \setminus \{0\}$. *Then there is a constant* C *such that for all* $\sigma \in E \setminus \{0\}$

$$N\big(r, f^*(\sigma)\big) < T_f(r, L) + C, \quad r \geqq 1.$$

Proof Let $\|\cdot\|$ be the hermitian metric in L. The pull-back f^*L is a line bundle on $\mathbf{C}^m$. On $\mathbf{C}^m$ every line bundle is globally trivial. We fix an isomorphism, $f^*L \cong \mathbf{C}^m \times \mathbf{C}$. Take bases $\sigma_1, \ldots, \sigma_n$ of E. Because of the isomorphism $f^*L \cong \mathbf{C}^m \times \mathbf{C}$, there are entire functions $f_j(z) = (f^*\sigma_j)(z)$, $1 \leqq j \leqq n$, on $\mathbf{C}^m$ and a C^∞ positive-valued function $h(z)$ such that

$$f^*\omega_L = dd^c \log h(z),$$

$$\sum_{j=1}^{n} \big\|\sigma_j\big(f(z)\big)\big\|^2 = \frac{\sum_j |f_j(z)|^2}{h(z)} \leqq 1, \quad 1 \leqq j \leqq n.$$

Write $\sigma = \sum w_j \sigma_j$ with $\|(w_j)\| = 1$. Then by the First Main Theorem 2.3.31 we have

$$N\big(r, f^*(\sigma)\big) = T_f(r, L) + m_f\big(1, (\sigma)\big) - m_f\big(r, (\sigma)\big),$$

$$m_f\big(1, (\sigma)\big) = \int_{\|z\|=1} \log \frac{\sqrt{h(z)}}{|\sum w_j f_j(z)|} \gamma(z).$$

Notice that $m_f(r, (\sigma)) \geqq 0$. By the choice, the functions f_j, $1 \leqq j \leqq n$, are linearly independent over $\mathbf{C}$. By Lemma 2.6.1 $m_f(1, (\sigma))$ is bounded in $\sigma = \sum w_j \sigma_j$, $\|(w_j)\| = 1$. Therefore there is a constant C such that

$$N\big(r, f^*(\sigma)\big) < T_f(r, L) + C, \quad r \geqq 1. \qquad \square$$

2.7 Ramified Covers over $\mathbf{C}^m$

Let X be an irreducible normal complex space. We call $X \xrightarrow{\pi} \mathbf{C}^m$ a finite ramified cover over $\mathbf{C}^m$ if π is a proper finite surjective holomorphic mapping. For example, if X is a normal affine algebraic variety, then there exists such a $\pi : X \to \mathbf{C}^m$ due to the "Noether Normalization Lemma".

In this section we summarize known facts on meromorphic mappings $f : X \to N$ from such X into a compact complex space N.

The case where $m = 1$ and $N = \mathbf{P}^1(\mathbf{C})$ is classical and was studied by Rémoundos [27], A. Valiron [29], [31], H.L. Selberg [30], [34], and Ullrich [32], etc. It is an essential case when $X \xrightarrow{\pi} \mathbf{C}^m$ is not algebraic but transcendental.

Let p be the covering number of $\pi : X \to \mathbf{C}^m$. We denote by $S(X)$ the set of all singular points of X. Since X is assumed to be normal, $\operatorname{codim} S(X) \geqq 2$. Let $R(X) = X \setminus S(X)$ be the set of regular (non-singular) points of X. The zero divisor

of $\det d\pi|_{R(X)}$ naturally extends to a divisor on X by Theorem 2.2.5. It is called the *ramification divisor* of $\pi : X \to \mathbf{C}^m$ and is denoted by Ξ. Set

$$X(r) = \{x \in X;\ \|\pi(x)\| < r\}, \qquad \partial X(r) = \{x \in X;\ \|\pi(x)\| = r\}.$$

Let ω be a hermitian metric form on N. We define the order function $T_f(r, \omega)$ of f with respect to ω by

$$T_f(r, \omega) = \frac{1}{p}\int_1^r \frac{dt}{t^{2m-1}} \int_{X(t)} f^*\omega \wedge \pi^*\alpha^{m-1}. \tag{2.7.1}$$

For a line bundle L over N we define $T_f(r, L)$ as done previously.

Let E be a Weil divisor on X and let $E = \sum_\lambda k_\lambda E_\lambda$ be the irreducible decomposition. As in (2.2.18) the counting functions are similarly defined:

$$\begin{aligned} n_k(t, E) &= \frac{1}{p}\int_{X(t)\cap(\sum_\lambda \min\{k,k_\lambda\}E_\lambda)} \alpha^{m-1}, \\ N_k(r, E) &= \int_1^r \frac{n_k(t, E)}{t^{2m-1}}dt, \\ n(t, E) &= n_\infty(t, E), \qquad N(r, E) = N_\infty(r, E). \end{aligned} \tag{2.7.2}$$

As in (2.3.30) we define a proximity function for an effective Cartier divisor D on N by

$$m_f(r, D) = \frac{1}{p}\int_{\partial X(r)} \log \frac{1}{\|\sigma \circ f\|}\pi^*\gamma. \tag{2.7.3}$$

For the proofs of the following results, cf. Noguchi [76a], [76b].

Theorem 2.7.4 (The First Main Theorem) *Assume that $f(X) \not\subset \operatorname{Supp} D$. Then*

$$T_f\big(r, L(D)\big) = N\big(r, f^*D\big) + m_f(r, D) + O(1).$$

We say that a meromorphic mapping $f : X \to N$ *separates the fiber* of π if there is a point $z \in \mathbf{C}^m$ satisfying that $\pi^{-1}(z) \cap (\Xi \cup I(f)) = \emptyset$ and f takes distinct values on $\pi^{-1}(z)$.

Lemma 2.7.5 (Noguchi [76a]) *For every meromorphic mapping $f : X \to N$ there exist a finite ramified cover $\pi' : X' \to \mathbf{C}^m$, a proper finite holomorphic mapping $\eta : X \to X'$ and a meromorphic mapping $f' : X' \to N$ satisfying the following*:

(i) $\pi = \pi' \circ \eta$, $f = f' \circ \eta$.
(ii) *f' separates the fiber of π'.*
(iii) $T_{f'}(r, \omega) = T_f(r, \omega)$, $N(r, f'^*D) = N(r, f^*D)$, $m_{f'}(r, D) = m_f(r, D)$.

Lemma 2.7.6 (Characterization of algebraicity I; Noguchi [76a]) *The complex space X is affine algebraic and $\pi : X \to \mathbf{C}^m$ is rational if and only if*

$$N(r, \Xi) = O(\log r).$$

In this case we say that $X \overset{\pi}{\to} \mathbf{C}^m$ is algebraic.

Theorem 2.7.7 (Characterization of algebraicity II; Noguchi [76a]) *Let L be the hyperplane bundle on $\mathbf{P}^n(\mathbf{C})$. If a meromorphic mapping $f : X \to \mathbf{P}^n(\mathbf{C})$ separates the fiber of π, then the following holds*:

(i) $N(r, \Xi) \leqq (2p-2)T_f(r, L) + O(1)$.
(ii) *It is necessary and sufficient for X to be algebraic and for $f : X \to \mathbf{P}^n(\mathbf{C})$ to be rational that*

$$T_f(r, L) = O(\log r).$$

H.L. Selberg [30] proved the above (i) in the case of $m = n = 1$.

Chapter 3
Differentiably Non-degenerate Meromorphic Maps

At the beginning of the 1970s, P.A. Griffiths et al. extended Nevanlinna theory to the higher dimensional case, and established the theory for differentiably non-degenerate holomorphic mappings $f : W \to V$ from an affine algebraic variety W into a projective algebraic variety V with $\operatorname{rank} df = \dim V$. This theory was very different to the Nevanlinna–Weyl–Ahlfors theory extended by W. Stoll, and was very fresh. The theory has been generalized in various ways, including the case of meromorphic mappings, applications have been developed, and a new phase was brought into the value distribution theory.

3.1 Lemma on Logarithmic Derivatives

As we have seen in Chap. 1, the lemma on logarithmic derivatives played an essential role in the analytic part of the proof of the Second Main Theorem for meromorphic functions. In this section we will extend it to the case of several variables which is due to Vitter [77]. The proof of Vitter [77] relied on the negative curvature method based on the idea of Griffiths–King [73]. A different proof was given by Biancofiore–Stoll [81]. Here we will give the third proof by making use of Nevanlinna's inequality, which seems to be simplest. This is based on the same idea as the proof of Lemma 1.2.2 in one variable.

Let $z = (z_1, \ldots, z_m)$ be the standard complex coordinate system on $\mathbf{C}^m$. We keep the notation introduced in (2.1.21). Let g be a meromorphic function on $\mathbf{C}^m$. Then we take partial derivatives $\partial g/\partial z_j$, $1 \leqq j \leqq m$, and set

$$\|dg\| = \left(\sum_{j=1}^{m} \left| \frac{\partial g}{\partial z_j} \right|^2 \right)^{1/2}.$$

Using the notation $m(r, \bullet)$ defined by (2.5.5) for real-valued functions as well, we set

$$m\left(r, \frac{\|dg\|}{|g|}\right) = \int_{\|z\|=r} \log^+ \frac{\|dg\|}{|g|} \gamma.$$

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Lemma 3.1.1 *For a meromorphic function $g \not\equiv 0$ on $\mathbf{C}^m$ we have*

$$m\left(r, \frac{\frac{\partial g}{\partial z_j}}{g}\right) \leqq m\left(r, \frac{\|dg\|}{|g|}\right) = S(r, g), \quad 1 \leqq j \leqq m.$$

Here, $S(r, g)$ stands for a term defined by (1.2.4).

Proof We take the following singular metric on $\mathbf{P}^1(\mathbf{C})$,

$$\Phi = \frac{1}{|w|^2(1 + (\log|w|)^2)} \frac{i}{4\pi^2} dw \wedge d\bar{w}.$$

By calculation we see that

$$\begin{aligned} &\int_{\mathbf{P}^1(\mathbf{C})} \Phi = 1, \\ &2m\pi g^*\Phi \wedge \alpha^{m-1} = \frac{\|dg\|^2}{|g|^2(1 + (\log|g|)^2)} \alpha^m. \end{aligned} \tag{3.1.2}$$

Put

$$\mu(r) = \int_1^r \frac{dt}{t^{2m-1}} \int_{\|z\|<t} g^*\Phi \wedge \alpha^{m-1}.$$

It follows from Fubini's theorem that

$$\begin{aligned} \mu(r) &= \int_{w \in \mathbf{P}^1(\mathbf{C})} \int_1^r \frac{dt}{t^{2m-1}} \int_{\{\|z\|<t\} \cap (g-w)_0} \alpha^{m-1} \Phi(w) \\ &= \int_{w \in \mathbf{P}^1(\mathbf{C})} N\big(r, (g-w)_0\big) \Phi(w). \end{aligned}$$

By Nevanlinna's inequality (Theorem 2.6.3) we get

$$\mu(r) \leqq \int_{w \in \mathbf{P}^1(\mathbf{C})} \big(T(r, g) + C\big) \Phi(w) = T(r, g) + C. \tag{3.1.3}$$

The following calculations are based on the concavity of "log", (3.1.2) and Lemma 1.2.1:

(3.1.4)

$$\begin{aligned} m\left(r, \frac{\|dg\|}{|g|}\right) &= \frac{1}{2} \int_{\|z\|=r} \log^+ \left(\frac{\|dg\|^2}{(1 + (\log|g|)^2)|g|^2} \big(1 + \big(\log|g|\big)^2\big) \right) \gamma \\ &\leqq \frac{1}{2} \int_{\|z\|=r} \log^+ \frac{\|dg\|^2}{(1 + (\log|g|)^2)|g|^2} \gamma \\ &\quad + \frac{1}{2} \int_{\|z\|=r} \log^+ \left(1 + \left(\log^+|g| + \log^+ \frac{1}{|g|} \right)^2 \right) \gamma \end{aligned}$$

(continued)

$$
\begin{aligned}
&\leqq \frac{1}{2}\int_{\|z\|=r} \log\left(1+\frac{\|dg\|^2}{(1+(\log|g|)^2)|g|^2}\right)\gamma \\
&\quad + \int_{\|z\|=r} \log^+\left(\log^+|g| + \log^+\frac{1}{|g|}\right)\gamma + \frac{1}{2}\log 2 \\
&\leqq \frac{1}{2}\log\left(1+\int_{\|z\|=r}\frac{\|dg\|^2}{(1+(\log|g|)^2)|g|^2}\gamma\right) \\
&\quad + \int_{\|z\|=r} \log\left(1+\log^+|g| + \log^+\frac{1}{|g|}\right)\gamma + \frac{1}{2}\log 2 \\
&\leqq \frac{1}{2}\log\left(1+\frac{1}{2mr^{2m-1}}\frac{d}{dr}\int_{B(r)}\frac{\|dg\|^2}{(1+(\log|g|)^2)|g|^2}\alpha^m\right) \\
&\quad + \log\left(1+m(r,g)+m\left(r,\frac{1}{g}\right)\right) + \frac{1}{2}\log 2 \\
&\leqq \frac{1}{2}\log\left(1+\frac{\pi}{r^{2m-1}}\frac{d}{dr}\int_{B(r)} g^*\Phi\wedge\alpha^{m-1}\right) \\
&\quad + \log^+ T(r,g) + O(1) \\
&\leqq \frac{1}{2}\log\left(1+\frac{\pi}{r^{2m-1}}\left(\int_{B(r)} g^*\Phi\wedge\alpha^{m-1}\right)^{1+\delta}\right) \\
&\quad + \log^+ T(r,g) + O(1)\|_{E_1(\delta)} \\
&\leqq \frac{1}{2}\log\left(1+\pi r^{\delta(2m-1)}\left(\frac{d}{dr}\int_1^r \frac{dt}{t^{2m-1}}\int_{B(t)} g^*\Phi\wedge\alpha^{m-1}\right)^{1+\delta}\right) \\
&\quad + \log^+ T(r,g) + O(1)\|_{E_1(\delta)} \\
&\leqq \frac{1}{2}\log\left(1+\pi r^{\delta(2m-1)}\mu(r)^{(1+\delta)^2}\right) + \log^+ T(r,g) + O(1)\|_{E_2(\delta)} \\
&\leqq \frac{(1+\delta)^2}{2}\log^+\mu(r) + \log^+ T(r,g) + \frac{\delta(2m-1)}{2}\log^+ r \\
&\quad + O(1)\|_{E_2(\delta)}.
\end{aligned}
$$

Combining this with (3.1.3), we obtain

$$
m\left(r,\frac{\|dg\|}{|g|}\right) = S(r,g). \qquad \square
$$

3.2 The Second Main Theorem

Let V be an n-dimensional complex projective algebraic manifold. We consider a meromorphic mapping $f : \mathbf{C}^m \to V$. We say that f is *differentiably non-*

degenerate if $m \geqq n$ and at some point $z \in \mathbf{C}^m \setminus I(f)$ the differential df has rank n.

The Second Main Theorem for a differentiably non-degenerate holomorphic mapping $f : \mathbf{C}^n \to V$ and a divisor on V was first established by Carlson–Griffiths [72], and then the domain was extended to complex affine algebraic varieties by Griffiths–King [73]. More generalizations were done by Sakai [74a] in terms of Kodaira dimension, and to the case of singular divisors by Shiffman [75]. The case of meromorphic mappings from a finite covering space into V is due to Noguchi [76a]. Stoll [77a] extended the domain to a general parabolic manifold.

Readers may now have the following question. If $f : \mathbf{C}^n \to V$ is differentiably non-degenerate ($\dim V = n$), then outside a subset of measure zero almost all points $x \in f(\mathbf{C}^n)$ have the discrete inverse images $f^{-1}x$. Then it should be natural to investigate the distribution of $f^{-1}x$. What is the reason to think of the inverse of a divisor? There seems to be a stability problem in the background, although it has not been made clear. At least the pull-backs of divisors by a transcendental mapping carry a sort of stability. In a sense, it is what the Second Main Theorem should imply. On the other hand, the inverse of a point does not carry a stability for the following reasons.

(i) By Fatou [22] and Bieberbach [33] there exists an injective holomorphic mapping $f : \mathbf{C}^2 \to \mathbf{C}^2$ with the Jacobian $J(f) \equiv 1$ such that $\mathbf{C}^2 \setminus f(\mathbf{C}^2)$ contains a non-empty open set. Furthermore, there is another injective holomorphic mapping $g : \mathbf{C}^2 \to \mathbf{C}^2$ such that $f(\mathbf{C}^2) \cap g(\mathbf{C}^2) = \emptyset$. For such an f the inverse $f^{-1}x$ is empty over a non-empty open set, and a point set over another non-empty open set; hence, there is no stability for $f^{-1}x$.

(ii) Buzzard–Lu [00] constructed a differentiably non-degenerate holomorphic mapping $f : \mathbf{C}^n \to N$ ($n \geqq 2$) into a complex torus N of dimension n such that $f(\mathbf{C}^n)$ omits (does not intersect) a non-empty open subset $U \subset N$. Let $\pi : \mathbf{C}^n \to N$ be the universal covering and set $\tilde{U} = \pi^{-1}U$. Then the lift $\tilde{f} : \mathbf{C}^n \to \mathbf{C}^n$ of f is differentiably non-degenerate and omits a non-empty open set $\tilde{U}$ that is invariant by a lattice.

(iii) Cornalba–Shiffman [72] constructed a holomorphic mapping $f : z \in \mathbf{C}^2 \to (f_1(z), f_2(z)) \in \mathbf{C}^2$ such that the zeros of each holomorphic function f_i ($i = 1, 2$) is of order zero, but the common zeros $f^{-1}0$ is of infinite order. It is called the transcendental Bezout problem to get a bound of the growth of the intersection $\bigcap_\nu A_\nu$ in terms of the growths of analytic subsets A_ν of $\mathbf{C}^m$. Their example shows that it does not hold in general. This appears to be a slightly weaker reason than those given in (i) and (ii), but it still represents the difficulty in dealing with the pointwise inverse image.

In general we assume that a non-negative $2n$-form Ω is given on an n-dimensional complex manifold N. Let $x = (x_1, \ldots, x_n)$ be a holomorphic local coordinate system of N and write $\Omega(x) = A(x) \bigwedge_{j=1}^n \frac{i}{2\pi} dx_j \wedge d\bar{x}_j$. Then the *Ricci form* $\mathrm{Ric}\,\Omega$ is defined by

$$\mathrm{Ric}\,\Omega = dd^c \log A(x) = \frac{i}{2\pi} \partial\bar{\partial} \log A(x).$$

This is defined where $A(x)$ is positive and differentiable at least twice. For the Fubini–Study metric form ω_0 on $\mathbf{P}^n(\mathbf{C})$ we have[1]

$$\operatorname{Ric}\omega_0^n = -(n+1)\omega_0. \tag{3.2.1}$$

Let a reduced divisor $D = \sum_{j=1}^q D_j$ be given on V. We say that D is of *normal crossing type* if at every point $a \in V$ there is a holomorphic local coordinate neighborhood $U(x_1, \ldots, x_n)$ and a number $0 \leqq k \leqq n$ satisfying

$$U \cap D = \{x_1 \cdots x_k = 0\}.$$

If $k = 0$, $U \cap D = \emptyset$. Moreover, if every D_j is non-singular, then D is said to be of *simple normal crossing type*, or *s.n.c.* for short. By K_V we denote the canonical bundle (the line bundle of holomorphic n-forms) over V (cf. the end of Sect. 2.3.1).

Theorem 3.2.2 (The Second Main Theorem, Carlson–Griffiths [72]; Griffiths–King [73]; Shiffman [75]; Noguchi [76a]) *Let $L \to V$ be a line bundle on a complex projective manifold V, and let $f : \mathbf{C}^m \to V$ $(m \geqq n)$ be a differentiably non-degenerate meromorphic mapping. Let $D = \sum_{j=1}^q D_j \in |L|$ be a divisor of simple normal crossing type. We fix a hermitian metric form ω on V. Then we have*

$$T_f(r, L) + T_f(r, K_V) \leqq \sum_{j=1}^q N_1\big(r, f^* D_j\big) + S_f(r, \omega).$$

Proof Equip a hermitian metric $\|\cdot\|$ in each $L(D_j)$, and take sections $\sigma_j \in H^0(V, L(D_j))$ with $(\sigma_j) = D_j$ and $\|\sigma_j\| < 1$. Note that $L = \otimes_{j=1}^q L(D_j)$ and $\omega_L = \sum_{j=1}^q \omega_{L(D_j)}$. We define a singular volume form Φ on V as follows:

$$\Phi = \frac{1}{\prod_{j=1}^q \|\sigma_j\|^2}\omega^n.$$

By the condition of simple normal crossing type for D there is an affine open covering $\{U_\lambda\}$ of V and rational holomorphic functions $x_{\lambda 1}, \ldots, x_{\lambda n}$ on U_λ such that

$$U_\lambda \cap D = \{x_{\lambda 1} \cdots x_{\lambda k_\lambda} = 0\}, \quad {}^{\exists} k_\lambda \leqq n,$$
$$dx_{\lambda 1} \wedge \cdots \wedge dx_{\lambda n}(a) \neq 0, \quad {}^{\forall} a \in U_\lambda.$$

Furthermore, we may assume the trivialization, $L(D_j)|_{U_\lambda} \cong U_\lambda \times \mathbf{C}$, $1 \leqq j \leqq q$. On every U_λ we get

$$\Phi|_{U_\lambda} = \frac{\phi_\lambda}{|x_{\lambda 1}|^2 \cdots |x_{\lambda k_\lambda}|^2} \bigwedge_{k=1}^n \frac{i}{2\pi} dx_{\lambda k} \wedge d\bar{x}_{\lambda k},$$

[1] A Kähler metric is called *Kähler–Einstein* if $\operatorname{Ric}\omega = c\omega$ for a constant c. Thus the Fubini–Study metric is a Kähler–Einstein metric.

where ϕ_λ is a positively valued C^∞ function. Let $\{c_\lambda\}$ be a partition of unity subordinate to $\{U_\lambda\}$. We set

$$\Phi_\lambda = c_\lambda \phi_\lambda \bigwedge_{k=1}^{k_\lambda} \left(\frac{i}{2\pi} \frac{dx_{\lambda k} \wedge d\bar{x}_{\lambda k}}{|x_{\lambda k}|^2} \right) \wedge \bigwedge_{k=k_\lambda+1}^{n} \frac{i}{2\pi} dx_{\lambda k} \wedge d\bar{x}_{\lambda k},$$

and $\Phi = \sum_\lambda c_\lambda \Phi|_{U_\lambda} = \sum_\lambda \Phi_\lambda$, where Φ_λ is extended on V as 0 outside U_λ. If we write $x_{\lambda k} \circ f(z) = f_{\lambda k}(z)$, we have

$$f^*\Phi_\lambda = c_\lambda \circ f \cdot \phi_\lambda \circ f \cdot \bigwedge_{k=1}^{k_\lambda} \left(\frac{i}{2\pi} \frac{df_{\lambda k} \wedge d\bar{f}_{\lambda k}}{|f_{\lambda k}|^2} \right) \wedge \bigwedge_{k=k_\lambda+1}^{n} \frac{i}{2\pi} df_{\lambda k} \wedge d\bar{f}_{\lambda k}. \tag{3.2.3}$$

Since $f_{\lambda k}$ is a pull-back of a rational function $x_{\lambda k}$ on V by f, it follows from Theorem 2.5.18 that

$$T(r, f_{\lambda k}) = O\big(T_f(r, \omega)\big) + O(1). \tag{3.2.4}$$

Since f is differentiably non-degenerate, $f^*\Phi \wedge \alpha^{m-n} \not\equiv 0$. Setting

$$f^*\Phi \wedge \alpha^{m-n} = \xi \alpha^m, \qquad f^*\Phi_\lambda \wedge \alpha^{m-n} = \xi_\lambda \alpha^m,$$

we have $\xi = \sum_\lambda \xi_\lambda$ and a current equation

$$\begin{aligned} dd^c[\log \xi] &= dd^c \left[\log \left(\sum_\lambda \xi_\lambda \right) \right] \geqq f^* \operatorname{Ric} \Phi - \operatorname{Supp} f^*D \\ &= f^*\omega_L + f^* \operatorname{Ric} \omega^n - \operatorname{Supp} f^*D. \end{aligned} \tag{3.2.5}$$

This applied with Jensen's formula Lemma 2.1.33 yields

$$\begin{aligned} &T_f(r, \omega_L) + T_f\big(r, \operatorname{Ric} \omega^n\big) \\ &\quad \leqq N_1\big(r, f^*D\big) + \int_{\|z\|=r} \log \left(\sum_\lambda \xi_\lambda \right) \gamma - \int_{\|z\|=1} \log \left(\sum_\lambda \xi_\lambda \right) \gamma. \end{aligned} \tag{3.2.6}$$

We put

$$f^*\omega \wedge \alpha^{m-1} = \zeta \alpha^m.$$

Notice that c_λ is multiplied in (3.2.3). We see that ξ_λ is bounded from above by a polynomial $P_\lambda(\zeta, |(\partial f_{\lambda k}/\partial z_j)/f_{\lambda k}|)$ in ζ and the absolute values of logarithmic partial derivatives $|(\partial f_{\lambda k}/\partial z_j)/f_{\lambda k}|$, $1 \leqq j \leqq m$, $1 \leqq k \leqq n$. Hence,

$$\xi_\lambda \leqq P_\lambda\big(\zeta, \big|(\partial f_{\lambda k}/\partial z_j)/f_{\lambda k}\big|\big).$$

Therefore we obtain

$$\log^+ \xi_\lambda \leqq O\left(\log^+ \zeta + \sum_k \log^+ \frac{\|df_{\lambda k}\|}{|f_{\lambda k}|}\right) + O(1).$$

The following estimate is deduced:

$$\begin{aligned}\int_{\|z\|=r} \log\left(\sum_\lambda \xi_\lambda\right)\gamma &\leqq \sum_\lambda \int_{\|z\|=r} \log^+ \xi_\lambda \gamma + O(1)\\ &\leqq O\left(\sum_{\lambda,k} \int_{\|z\|=r} \log^+ \frac{\|df_{\lambda k}\|}{|f_{\lambda k}|}\gamma\right)\\ &\quad + O\left(\int_{\|z\|=r} \log^+ \zeta\gamma\right) + O(1).\end{aligned}$$

Here, applying Lemma 3.1.1 and (3.2.4) to the first term, we see that

$$\int_{\|z\|=r} \log\left(\sum_\lambda \xi_\lambda\right)\gamma \leqq S_f(r,\omega) + O\left(\int_{\|z\|=r} \log^+ \zeta\gamma\right).$$

Hence the proof is completed if we prove

$$\int_{\|z\|=r} \log^+ \zeta\gamma = S_f(r,\omega). \tag{3.2.7}$$

We use a similar calculation to (3.1.4). We take an arbitrary $0 < \delta < 1$ and apply Lemma 1.2.1 (Borel's Lemma) as follows:

$$\begin{aligned}&\int_{\|z\|=r} \log^+ \zeta\gamma \qquad (3.2.8)\\ &\leqq \int_{\|z\|=r} \log(1+\zeta)\gamma\\ &\leqq \log\left(1 + \int_{\|z\|=r} \zeta\gamma\right)\\ &\leqq \log\left(1 + \frac{1}{2mr^{2m-1}}\frac{d}{dr}\int_{B(r)} \zeta\alpha^m\right)\\ &\leqq \log\left(1 + \frac{1}{2mr^{2m-1}}\left(\int_{B(r)} f^*\omega\wedge\alpha^{m-1}\right)^{1+\delta}\right)\Bigg\|_{E_1(\delta)}\\ &\leqq \log\left(1 + \frac{r^{(2m-1)\delta}}{2m}\left(\frac{d}{dr}\int_1^r \frac{dt}{t^{2m-1}}\int_{B(t)} f^*\omega\wedge\alpha^{m-1}\right)^{1+\delta}\right)\Bigg\|_{E_1(\delta)}\\ &\leqq \log\left(1 + \frac{r^{(2m-1)\delta}}{2m}\left(\int_1^r \frac{dt}{t^{2m-1}}\int_{B(t)} f^*\omega\wedge\alpha^{m-1}\right)^{(1+\delta)^2}\right)\Bigg\|_{E_2(\delta)}\\ &\leqq \log\left(1 + \frac{r^{(2m-1)\delta}}{2m}\left(T_f(r,\omega)\right)^{(1+\delta)^2}\right)\Bigg\|_{E_2(\delta)}\end{aligned}$$

(continued)

$$\leqq 4\log^+ T_f(r,\omega) + (2m-1)\delta\log r + O(1)\|_{E_2(\delta)}$$
$$= S_f(r,\omega).$$

Therefore (3.2.7) is deduced and the proof is finished. □

Corollary 3.2.9 *Let D be a reduced divisor of simple normal crossing type on V. Assume one of the following*:

(i) $c_1(L(D)\otimes K_V) > 0$.
(ii) *With the notation in* (2.3.23), $\sup\{\dim \Phi_{(L(D)\otimes K_V)^\nu}(V);\ \nu\in\mathbf{N}\} = \dim V$.

Then any meromorphic mapping $f:\mathbf{C}^m\to V\setminus D$ $(m\geqq \dim V)$ is differentiably degenerate.

In particular, if $\sup\{\dim \Phi_{K_V^\nu}(V);\ \nu\in\mathbf{N}\} = \dim V$ *(in this case V is said to be of general type in the classification theory of algebraic varieties), any meromorphic mapping $f:\mathbf{C}^m\to V$ $(m\geqq\dim V)$ is differentiably degenerate.*

Proof By Theorem 2.3.26 condition (i) implies condition (ii). It is also directly deduced from Corollary 2.3.32 that for a non-constant meromorphic mapping $f:\mathbf{C}^m\to V$ there is a constant $C>0$ satisfying

$$T_f\big(r, L(D)\otimes K_V\big) \geqq C\log r + O(1).$$

Therefore the assertion follows from Theorem 3.2.2.

In the case of (ii) there is a number $\nu\in\mathbf{N}$ such that

$$\dim \Phi_{(L(D)\otimes K_V)^\nu}(V) = \dim V.$$

Take bases σ_j, $0\leqq j\leqq N$ of $H^0(V,(L(D)\otimes K_V)^\nu)$. Set $\phi_j = \sigma_j/\sigma_0$. Then the family $\{\phi_j\}_{j=1}^N$ of rational functions on V contains a transcendental base of the rational function field $\mathbf{C}(V)$ of V over $\mathbf{C}$. Let $f:\mathbf{C}^m\to V\setminus D$ be a differentiably non-degenerate meromorphic mapping. Then it is, of course, algebraically non-degenerate, and hence $f^*\phi_j$ are defined. Take a line bundle $L_0>0$ on V. One deduces from Theorems 2.5.7 and 3.2.2 that

$$T_f\big(r,\{\phi_j\}\big) \leqq T_f\big(r,\big(L(D)\otimes K_V\big)^\nu\big) + O(1) \leqq S_f(r,\omega_{L_0}).$$

By Theorem 2.5.18 there is a constant $C_1>0$ such that

$$C_1^{-1}T_f\big(r,\{\phi_j\}\big) + O(1) \leqq T_f(r,L_0) \leqq C_1 T_f\big(r,\{\phi_j\}\big) + O(1).$$

By Corollary 2.3.32 there is a constant $C_2>0$ satisfying

$$\log r \leqq C_2 T_f(r,L_0) + O(1).$$

Therefore, for a given $\delta>0$ we have

$$T_f(r,L_0) \leqq O\big(\log T_f(r,L_0)\big) + \delta\log r\|_{E(\delta)}.$$

This implies a contradiction, $1\leqq C_2\delta$. □

The above proof of the Second Main Theorem 3.2.2 is different to the original one due to Griffiths et al., and the calculations are considerably simplified. It is remarked that a similar approach was taken in M. Ru [01].

The above method makes it clear how to construct a metric from the given divisor and to decompose the metric into an invariant differential part (logarithmic differential) and the adjunction bundle $L\otimes K_V$. In the present book we give the proofs to all the second main theorems, based on this idea. The Second Main Theorem for meromorphic functions in one variable given in Chap. 1 was proved, being based on this idea.

The method used by Griffiths et al. is called the *negative curvature method*; under the assumption of $L(D)\otimes K_V > 0$ a singular metric Ψ with singularities on D is constructed so that $\operatorname{Ric}\Psi \geqq \Psi$. The negative curvature method is interesting by itself and quite effective; it is worthwhile to learn the method. For instance, cf. Carlson–Griffiths [72] and Noguchi–Ochiai [90]; it should not be difficult to read that part with the knowledge learned so far.

In general we set for a line bundle L on V

$$\begin{aligned}\underline{\left[\frac{c_1(L)}{\omega}\right]} &= \sup\bigl\{s\in\mathbf{R};\ {}^{\exists}\omega_L \in c_1(L), \omega_L > s\omega\bigr\},\\ \overline{\left[\frac{c_1(L)}{\omega}\right]} &= \inf\bigl\{s\in\mathbf{R};\ {}^{\exists}\omega_L \in c_1(L), \omega_L < s\omega\bigr\}.\end{aligned} \tag{3.2.10}$$

By definition it is clear that

$$\underline{\left[\frac{c_1(L)}{\omega}\right]} \leqq \varliminf_{r\to\infty}\frac{T_f(r,L)}{T_f(r,\omega)} \leqq \varlimsup_{r\to\infty}\frac{T_f(r,L)}{T_f(r,\omega)} \leqq \overline{\left[\frac{c_1(L)}{\omega}\right]}. \tag{3.2.11}$$

It follows from Theorem 3.2.2 that

$$\sum_{i=1}^{q}\bigl(T_f\bigl(r,L(D_i)\bigr) - N_1\bigl(r,f^*D_i\bigr)\bigr) \leqq T_f\bigl(r,K_V^{-1}\bigr) + S_f(r,\omega).$$

Thus

$$\sum\left(1-\frac{N_1(r,f^*D_i)}{T_f(r,L(D_i))}\right)\cdot\frac{T_f(r,L(D_i))}{T_f(r,\omega)} \leqq \frac{T_f(r,K_V^{-1})}{T_f(r,\omega)} + \frac{S_f(r,\omega)}{T_f(r,\omega)}.$$

This implies that

$$\begin{aligned}\sum\delta_1(f,D_i)\varliminf_{r\to\infty}\frac{T_f(r,L(D_i))}{T_f(r,\omega)} &\leqq \varliminf_{r\to\infty}\frac{T_f(r,K_V^{-1})}{T_f(r,\omega)} + \varliminf_{r\to\infty}\frac{S_f(r,\omega)}{T_f(r,\omega)},\\ &\leqq \varlimsup_{r\to\infty}\frac{T_f(r,K_V^{-1})}{T_f(r,\omega)}.\end{aligned} \tag{3.2.12}$$

Let $f^*D_i = \sum_\lambda \mu_{i\lambda} A_{i\lambda}$ be the decomposition into irreducible components. Put $\min_\lambda\{\mu_{i\lambda}\} = \mu_i$. Then we say that f is *completely μ_i-ramified* over D_i. In the case of $f^{-1}D_i = \emptyset$, we set $\mu_i = \infty$. Then the following is obtained from (3.2.11) and (3.2.12).

Theorem 3.2.13 *Assume the same condition as in Theorem 3.2.2.*

(i) (Defect Relation) *We have*

$$\sum_{i=1}^{q} \delta_1(f, D_i) \underline{\left[\frac{c_1(L(D_i))}{\omega}\right]} \leqq \overline{\left[\frac{c_1(K_V^{-1})}{\omega}\right]}.$$

In particular, if $c_1(L(D_1)) = \cdots = c_1(L(D_q)) = [\omega] > 0$,

$$\sum \delta_1(f, D_i) \leqq \overline{\left[\frac{c_1(K_V^{-1})}{\omega}\right]}.$$

(ii) (Ramification Theorem) *If f is completely μ_i-ramified over each D_i, then*

$$\sum \left(1 - \frac{1}{\mu_i}\right) \underline{\left[\frac{c_1(L(D_i))}{\omega}\right]} \leqq \overline{\left[\frac{c_1(K_V^{-1})}{\omega}\right]}.$$

The proof of (ii) is similar to that of Theorem 1.2.13.

Example 3.2.14 (i) Let $V = \mathbf{P}^n(\mathbf{C})$ and let D_i, $1 \leqq i \leqq q$, be hypersurfaces of degree d_i such that $\sum D_i$ is of simple normal crossing type. Let ω_0 be the Fubini–Study metric form. By (3.2.1), $c_1(K_{\mathbf{P}^n(\mathbf{C})}) = -(n+1)[\omega_0]$ and $c_1(L(D_i)) = d_i[\omega_0]$. If $f : \mathbf{C}^n \to \mathbf{P}^n(\mathbf{C})$ is a differentiably non-degenerate meromorphic mapping, we have

$$\sum \delta_1(f, D_i) d_i \leqq n + 1.$$

If all D_i are hyperplanes, then $d_i = 1$ and

$$\sum \delta_1(f, D_i) \leqq n + 1.$$

In particular, for

$$f : (z_1, \ldots, z_n) \in \mathbf{C}^n \to \left[1, e^{z_1}, \ldots, e^{z_n}\right] = [x_0, \ldots, x_n] \in \mathbf{P}^n(\mathbf{C}),$$
$$D_i = \{x_i = 0\}, \quad 0 \leqq i \leqq n,$$

we have that $\delta_1(f, D_i) = 1$, $0 \leqq i \leqq n$, $\sum_i \delta_1(f, D_i) = n + 1$.

(ii) (F. Sakai) In Theorem 3.2.2 the condition for D to be of simple normal crossing type is necessary. In fact, with the homogeneous coordinate system $[w_0, w_1, w_2]$ of $\mathbf{P}^2(\mathbf{C})$ we set

$$D = \left\{F = (w_0)^{k-1} w_2 - (w_1)^k = 0\right\}, \quad k > 1.$$

Then D has a singularity at $[0, 0, 1]$. The differentiably non-degenerate holomorphic mapping

$$f : (z_1, z_2) \in \mathbf{C}^2 \to \left[1, z_1, (z_1)^k + e^{z_2}\right] \in \mathbf{P}^2(\mathbf{C})$$

satisfies $F \circ f(z) = e^{z_2}$. Therefore $f(\mathbf{C}^2) \cap D = \emptyset$, $^\forall k > 1$. On the other hand,

$$c_1\big(L(D)\big) + c_1(K_{\mathbf{P}^2(\mathbf{C})}) = (k-3)[\omega_0] > 0, \quad k \geqq 4,$$

and so Theorem 3.2.2 does not hold for $k \geqq 4$.

(iii) Let $V \subset \mathbf{P}^{n+1}(\mathbf{C})$ be a non-singular hypersurface of degree d. By a simple calculation of the normal bundle of V (cf., e.g., Noguchi–Ochiai [90], p. 252 (Ochiai–Noguchi [84])) we have

$$c_1(K_V) = (n+1-d)[\omega_0|V].$$

Let $D_i, 1 \leqq i \leqq q$, be the cuts of hypersurfaces of degree d_i and V, and assume that $\sum D_i$ is of simple normal crossing type (in V). Then for a differentiably non-degenerate meromorphic mapping $f : \mathbf{C}^n \to V$

$$\sum \delta_1(f, D_i) d_i \leqq n + 1 - d.$$

That is, $d \leqq n+1$ is a necessary condition for the existence of such an f. If $n = 3$, then $d \leqq 4$. If $d \leqq 3$, V is a rational surface, and hence there is certainly such an f. When $d = 4$, V is a K3-surface. The hypersurface of Fermat type of degree 4 is known to be a Kummer surface; that is, V is the once blowing-up of the quotient $A/\{\pm 1\}$ of an abelian variety A (which is a product of elliptic curves in this case) by the involution -1 with center at 16 fixed points of the involution. (Pjateckiĭ-Šapiro–Shafarevič [71].) Therefore the construction implies the existence of such a differentiably non-degenerate meromorphic mapping $f : \mathbf{C}^2 \to V$. Furthermore, Buzzard–Lu [00] prove that there exists a differentiably non-degenerate holomorphic mapping $g : \mathbf{C}^2 \to A$ omitting those 16 points, and so the composition of it and the quotient mapping gives an example of a differentiably non-degenerate holomorphic mapping $h : \mathbf{C}^2 \to V$.

By the way, it is remarked that this gives a counter-example to Green [78], Theorem 2.

(iv) Let V be an n-dimensional abelian variety A. Then $K_A = \mathbf{1}_A$. It is first noticed that any meromorphic mapping $f : \mathbf{C}^m \to A$ is holomorphic: For $\operatorname{codim} I(f) \geqq 2$ implies the simply connectedness of $\mathbf{C}^m \setminus I(f)$, and so $f|(\mathbf{C}^n \setminus I(f))$ has a lifting $F : \mathbf{C}^m \setminus I(f) \to \mathbf{C}^n$. By the extension Theorem 2.2.6 (ii) F extends holomorphically over $\mathbf{C}^m$, so that f is holomorphic.

For a non-singular reduced irreducible divisor D on A, set

$$\mathrm{St}(D) = \{x \in A; x + D = D\}$$

and denote by $\mathrm{St}(D)^0$ its identity component. Then $\mathrm{St}(D)^0$ is an abelian subvariety of A. We denote by D_0 the image of D by the quotient map $\lambda : A \to A/\mathrm{St}(D)^0$.

Then we see that $\lambda^{-1}(D_0) = D$ and $L(D_0) > 0$ (Weil [58]). Therefore we may suppose $L(D) > 0$ from the beginning. Theorem 3.2.2 implies that for a differentiably non-degenerate holomorphic mapping $f : \mathbf{C}^n \to A$

$$\delta_1(f, D) = 0. \tag{3.2.15}$$

In fact, a similar result holds for singular D on the abelian variety A (cf. Theorem 6.3.14). This is a special nature of the case of abelian varieties.

3.3 Applications and Generalizations

3.3.1 Applications

The applications that we are going to describe are not direct ones of the theorems obtained up to this point, but are those of the methods of proofs used and developed there.

A compact complex manifold which is bimeromorphic to $\mathbf{P}^n(\mathbf{C})$ is called a *rational variety*. Two-dimensional compact complex manifolds are called (complex) *surfaces*. There is a fine classification theory of surfaces (cf. Kodaira [75]; Barth–Peters–Van de Ven [84]), from which we know the following fact, where $b_1(N) = \dim H_1(N, \mathbf{R})$ denotes the first Betti number of N.

Theorem 3.3.1 (Kodaira [68], Theorem 54) *If a complex surface N satisfies $b_1(N) = 0$ and $H^0(N, K_N^l) = \{0\}$ for all $l > 0$, then N is rational.*

Let Ω_N^k denote the sheaf of holomorphic k-forms over a complex manifold N. We denote by $S^l\Omega_N^k$ its l-th symmetric tensor power.

Theorem 3.3.2 *Let N be an n-dimensional compact complex manifold. Assume that there exists a differentiably non-degenerate meromorphic mapping $f : \mathbf{C}^m \to N$ $(m \geqq n)$ with $\rho_f < 2$. Then for arbitrary $l_k \geqq 0$ with $\sum_{k=1}^n l_k > 0$*

$$H^0\big(N, S^{l_1}\Omega_N^1 \otimes \cdots \otimes S^{l_n}\Omega_N^n\big) = \{0\}.$$

Proof Assume the existence of an element

$$\tau \in H^0\big(N, S^{l_1}\Omega_N^1 \otimes \cdots \otimes S^{l_n}\Omega_N^1\big) \setminus \{0\}.$$

We take a hermitian metric h on N, and denote by ω its associated form. Let $\|\tau\|_h$ denote the norm of τ with respect to h. Then there is a constant $c_1 > 0$ such that

$$\|\tau\|_h \leqq c_1. \tag{3.3.3}$$

We define a function ζ on $\mathbf{C}^m$ by

$$f^*\omega \wedge \alpha^{m-1} = \zeta \alpha^m.$$

Since f is differentiably non-degenerate, $f^*\tau \not\equiv 0$. We denote by ξ_λ the coefficient functions of $f^*\tau$ with respect to the standard coordinate system $(z_1, \ldots, z_m)$ on $\mathbf{C}^m$, and set

$$\|f^*\tau\|^2_{\mathbf{C}^m} = \sum_\lambda |\xi_\lambda|^2 \not\equiv 0. \tag{3.3.4}$$

By (3.3.3) there are positive constants c_2 and c_3 such that

$$\zeta \geqq c_2 \|f^*\tau\|^{2c_3}_{\mathbf{C}^m}. \tag{3.3.5}$$

By (3.3.4) $\|f^*\tau\|^{2c_3}_{\mathbf{C}^m}$ is plurisubharmonic. Since $f^*\tau \not\equiv 0$ is holomorphic, it follows that

$$\int_{\|z\|=1} \|f^*\tau\|^{2c_3}_{\mathbf{C}^m} \gamma = c_4 > 0.$$

By applying Lemma 2.1.33 and noting that being restricted to $\{\|z\| = r\}$ $\gamma = (1/r^{2m-1}) d^c \|z\|^2 \wedge \alpha^{m-1}$, we see

$$\int_{\|z\|=r} \|f^*\tau\|^{2c_3}_{\mathbf{C}^m} d^c \|z\|^2 \wedge \alpha^{m-1} \geqq c_4 r^{2m-1}, \quad r > 1.$$

Therefore

$$\int_{\|z\| \leqq r} \|f^*\tau\|^{2c_3}_{\mathbf{C}^m} \alpha^m \geqq \frac{c_4}{2m}\left(r^{2m} - 1\right), \quad r > 1.$$

We deduce from this that

$$\begin{aligned} T_f(r, \omega) &= \int_1^r \frac{dt}{t^{2m-1}} \int_{B(t)} \zeta \alpha^m \\ &\geqq c_2 \int_1^r \frac{dt}{t^{2m-1}} \int_{B(t)} \|f^*\tau\|^{2c_3}_{\mathbf{C}^m} \alpha^m \\ &\geqq \frac{c_2 c_4}{2m} \int_1^r \left(t - \frac{1}{t^{2m-1}}\right) dt \\ &= \frac{c_2 c_4}{4m} r^2 + C_m(r), \end{aligned}$$

where $C_1(r) = O(\log r)$ and $C_m(r) = O(1)$ for $m \geqq 2$. Thus,

$$\rho_f = \varlimsup_{r \to \infty} \frac{\log T_f(r, \omega)}{\log r} \geqq 2.$$

This is a contradiction. □

Remark 3.3.6 Let N be a complex torus and let $f : \mathbf{C}^n \to N$ be the universal covering mapping. Then $\rho_f = 2$ and for any $l_k \geqq 0$ with $\sum_{k=1}^n l_k > 0$

$$H^0\big(N, S^{l_1}\Omega_N^1 \otimes \cdots \otimes S^{l_n}\Omega_N^1\big) \neq \{0\}.$$

Theorem 3.3.7 *Let N be a compact Kähler surface. Assume that there is a differentiably non-degenerate meromorphic mapping $f : \mathbf{C}^2 \to N$ with $\rho_f < 2$. Then N is rational.*

Proof By Theorem 3.3.2 $\dim H^0(N, \Omega_N^1) = 0$. Due to the Kähler assumption we have $b_1(N) = 2\dim H^0(N, \Omega_N^1) = 0$. Moreover $H^0(N, K_N^l) = \{0\}$, $^\forall l > 0$. It follows from Theorem 3.3.1 that N is rational. □

In the present text, a *compactification* N of a non-compact complex manifold X means that there is a holomorphic open embedding $f : X \to N$ with open dense image in X and that $N \setminus f(X)$ is a thin analytic subset of N (a "thin" set means that it is nowhere dense).

Theorem 3.3.8 (Kodaira [71]) *Let N be a smooth compactification of $\mathbf{C}^n$. Then $b_1(N) = 0$ and $H^0(N, K_N^l) = \{0\}$ for all $l > 0$. In particular, any compactification of $\mathbf{C}^2$ is a rational surface.*

Proof Suppose the existence of $\tau \in H^0(N, K_N^l) \setminus \{0\}$. Take a holomorphic local coordinate neighborhood $U(x_1, \ldots, x_n)$ of N, and write

$$\tau = \tau_U(x)(dx_1 \wedge \cdots \wedge dx_n)^l.$$

Set

$$|\tau|^{2/l} = \big|\tau_U(x)\big|^{2/l} \bigwedge_{j=1}^n \frac{i}{2} dx_j \wedge d\bar{x}_j.$$

Then this is a continuous (n, n)-form defined on N. Since N is compact,

$$0 < \int_N |\tau|^{2/l} = C < \infty.$$

Let $(z_1, \ldots, z_n)$ be the standard holomorphic coordinate system on $\mathbf{C}^n$. By making use of the holomorphic open embedding $f : \mathbf{C}^n \to N$ we set

$$f^*\tau(z) = \xi(z)(dz_1 \wedge \cdots \wedge dz_n)^l.$$

Then $\xi(z)$ is holomorphic and $\xi \not\equiv 0$. By a translation we may assume $\xi(0) \neq 0$. Since $f^*|\tau|^{2/l} = |\xi|^{2/l}\frac{\pi^n}{n!}\alpha^n$, and $|\xi|^{2/l}$ is a plurisubharmonic function (2.1.25),

$$\int_{B(r)} \big|\xi(z)\big|^{2/l}\alpha^n \geqq r^{2n}\big|\xi(0)\big|^{2/l}.$$

On the other hand,

$$\int_{B(r)} \left|\xi(z)\right|^{2/l} \alpha^n = \frac{n!}{\pi^n} \int_{B(r)} f^* |\tau|^{2/l}$$
$$\leqq \frac{n!}{\pi^n} \int_N |\tau|^{2/l} = \frac{n!}{\pi^n} C.$$

Therefore, $|\xi(0)|^{2/l} \leqq Cn!/\pi^n r^{2n}$. Letting $r \to \infty$, we have a contradiction, $|\xi(0)| = 0$.

Since $N \setminus f(\mathbf{C}^n)$ is a thin analytic subset, $b_1(N) \leqq b_1(\mathbf{C}^n) = 0$. In the case of $n = 2$, Theorem 3.3.1 implies that N is a rational surface. □

3.3.2 *Non-Kähler Counter-Example*

We here give a non-Kähler compact surface S for which Theorem 3.3.7 does not hold.

Let $\lambda \in \mathbf{C}$ with $|\lambda| > 1$. Then we consider a Hopf surface S defined as the quotient of $\mathbf{C}^2 \setminus \{(0,0)\}$ under the $\mathbf{Z}$-action given by

$$n : (x, y) \mapsto \left(\lambda^n x, \lambda^n y\right).$$

Such a surface S is known to be diffeomorphic to $S^1 \times S^3$. As a consequence $b_1(S) = 1$ and S is *not Kähler*.

Now

$$\omega = \frac{i}{2\pi} \cdot \frac{dx \wedge d\bar{x} + dy \wedge d\bar{y}}{|x|^2 + |y|^2} = \frac{dd^c \|(x, y)\|^2}{\|(x, y)\|^2}$$

is a positive $(1,1)$-form on $\mathbf{C}^2 \setminus \{(0,0)\}$ which is invariant under the above given $\mathbf{Z}$-action. Therefore it induces a positive $(1,1)$-form on the quotient surface S which by abuse of notation is again denoted by ω.

Let α and γ be as in (2.1.21). We take the holomorphic map

$$f : \mathbf{C}^2 \to S \tag{3.3.9}$$

induced by $(z, w) \mapsto (z, 1 + zw)$.

Proposition 3.3.10 *Let $f : \mathbf{C}^2 \to S$ be as above. Then the order $\rho_f \leqq 1$.*[2]

Proof By definition $\rho_f \leqq 1$ means that

$$\overline{\lim_{r \to \infty}} \frac{1}{\log r} \log \int_1^r \frac{dt}{t^3} \int_{B(t)} f^* \omega \wedge \alpha \leqq 1.$$

[2] *Remark* There was a gap in the proof to claim $\rho_f = 1$ in Noguchi–Winkelmann [12].

Note that

$$f^*\omega \wedge \alpha = \frac{1+|z|^2+|w|^2}{2(|z|^2+|1+zw|^2)}\alpha^2.$$

We define

$$I_r = \int_{S(r)} \frac{r^2}{|z|^2+|1+zw|^2} dV, \quad r = \|(z,w)\|.$$

Here dV is the Euclidean volume element on $S(r)$, and therefore a constant multiple of $r^3\gamma$. It is sufficient to show

$$I_r = O\big(r^{2+\varepsilon}\big), \quad {}^\forall\varepsilon > 0. \tag{3.3.11}$$

Indeed, assume that this holds. Because of $\lim_{r\to\infty} \frac{1+r^2}{r^2} = 1$, (3.3.11) is equivalent to the assertion

$$I_r' = O\big(r^{2+\varepsilon}\big)$$

with

$$I_r' = \int_{S(r)} \frac{1+r^2}{|z|^2+|1+zw|^2} dV.$$

From this we first obtain

$$\int_{B(r)} \frac{1+r^2}{|z|^2+|1+zw|^2}\alpha^2 = O\left(\int^r I_r' dr\right) = O\big(r^{3+\varepsilon}\big), \quad {}^\forall\varepsilon > 0,$$

implying

$$T_f(r) = \frac{1}{2}\int_1^r \frac{dt}{t^3}\int_{B(r)} \frac{1+r^2}{|z|^2+|1+zw|^2}\alpha^2 = O\big(r^{1+\varepsilon}\big), \quad {}^\forall\varepsilon > 0,$$

and

$$\rho_f = \varlimsup_{r\to\infty} \frac{\log T_f(r)}{\log r} \leqq 1.$$

Hence it suffices to show (3.3.11). We define

$$\eta = \frac{r^2}{|z|^2+|1+zw|^2}.$$

Thus we have to show

$$I_r = \int_{S(r)} \eta dV = O\big(r^{2+\varepsilon}\big).$$

We set

$$\eta = \frac{r^2}{\phi(z,w)}, \quad \phi(z,w) = |z|^2+|1+zw|^2.$$

(a) Geometric Estimates

For $(z, w) \in S(r)$ we let $\theta \in [0, 2\pi)$ such that $e^{i\theta}|zw| = zw$. Let $K > 0$, $-\infty < \lambda < 1$ and $\mu \geq 0$. We set

$$\Omega_{K,\lambda,\mu} = \big\{(z, w) \in S(r);\, |z| \leq Kr^{\lambda}, |\sin\theta| \leq r^{-\mu}\big\}.$$

We need some volume estimates.

First we note that $(\sin\theta)/\theta \geq 2/\pi$ for all $\theta \in [0, \pi/2]$, because "sin" is concave on $[0, \pi/2]$. It follows that for every $C \in (0, 1)$ (open interval) we have the following bound for the Lebesgue measure:

$$\mathrm{vol}\big(\big\{\theta \in [0, 2\pi];\, |\sin\theta| \leq C\big\}\big) \leq 4(C\pi/2) = 2C\pi. \tag{3.3.12}$$

Second we define a map $\zeta : \mathbf{C}^2 \to \mathbf{C} \times \mathbf{R}^2$ as follows:

$$\zeta : (z, w) \mapsto \big(z, r\arg(zw), r\big),$$

where $r = \|(z, w)\| = \sqrt{|z|^2 + |w|^2}$.

An explicit calculation shows that the Jacobian of this map (where defined) is constant with value "-1". Furthermore, the gradient $\mathrm{grad}(r)$ is of length one and normal on the level set $S(r)$. Correspondingly, the map ζ is volume preserving and $S(r)$ has the same volume as its image

$$\zeta\big(S(r)\big) = \big\{z \in \mathbf{C};\, |z| \leq r\big\} \times [0, 2\pi r) \times \{r\}, \tag{3.3.13}$$

namely $2\pi^2 r^3$.

Similarly, the Euclidean volume of $\Omega_{K,\lambda,\mu}$ agrees with the Euclidean volume of

$$\zeta(\Omega_{K,\lambda,\mu}) = \big\{z \in \mathbf{C};\, |z| \leq Kr^{\lambda}\big\} \times \big\{\theta r;\, \theta \in [0, 2\pi), |\sin\theta| \leq r^{-\mu}\big\} \times \{r\}.$$

By using (3.3.12), it follows that for $r \geq 1$ the volume of $\Omega_{K,\lambda,\mu}$ is bounded by

$$\pi\big(Kr^{\lambda}\big)^2 \cdot 2r^{-\mu}\pi r = 2K^2\pi^2 r^{2\lambda+1-\mu}.$$

In particular,

$$\mathrm{vol}(\Omega_{K,\lambda,\mu}) = O\big(r^{2\lambda+1-\mu}\big). \tag{3.3.14}$$

(b) Arithmetic Estimates

Besides the Landau O-symbols we also use the notation "$\gtrsim$": If f, g are functions of a real parameter r, then $f(r) \gtrsim g(r)$ indicates that

$$\varliminf_{r\to+\infty} \frac{f(r)}{g(r)} \geq 1.$$

Similarly, $f \sim g$ indicates

$$\lim_{r\to+\infty} \frac{f(r)}{g(r)} = 1.$$

In the sequel, we will work with domains $\Omega \subset S(r)$ (i.e. for each $r > 0$ some subset $\Omega = \Omega_r \subset S(r)$ is chosen). In this context, given functions f, g on $\mathbf{C}^2$ we say "$f(z,w) \gtrsim g(z,w)$ holds on Ω" if for every sequence $(z_n, w_n) \in \mathbf{C}^2$ with $\lim \|(z_n, w_n)\| = +\infty$ and $(z_n, w_n) \in \Omega_r$ $(r = \|(z_n, w_n)\|)$ we have

$$\varliminf_{r\to+\infty} \frac{f(z_n, w_n)}{g(z_n, w_n)} \geq 1.$$

We develop some estimates for $\phi(z,w) = |z|^2 + |1+zw|^2$. Fix $\mu > 0$, $-\infty < \lambda < 1$.

(i) For all z, w: $\phi \geq |z|^2$.
(ii) If $(z,w) \in S(r)$ and $|z| \leq \frac{1}{2r}$, then

$$|w| \leq r \quad \Longrightarrow \quad |zw| \leq \frac{1}{2} \quad \Longrightarrow \quad |1+zw| \geq \frac{1}{2}$$

and therefore $\phi \geq \frac{1}{4}$.
(iii) For $|z| \leq r^\lambda$ we have $|w| \sim r$, i.e., for fixed λ, μ and any choice of $(z_r, w_r) \in S(r)$ with $|z_r| \leq r^\lambda$ we have $\lim_{r\to\infty} |w_r|/r = 1$.
(iv) For $|z| \geq \frac{3}{2r}$ and $|z| \leq r^\lambda$ we have that $\phi \gtrsim \frac{1}{9}|zw|^2$, because $|w| \sim r$ and $|zw| \gtrsim \frac{3}{2}$ (equivalently, $1 \lesssim \frac{2}{3}|zw|$), implying $|1+zw| \geq |zw| - 1 \gtrsim \frac{1}{3}|zw|$.
(v) For all z, w, $\phi \geq |\Im(1+zw)|^2 = (|zw| \sin\theta)^2$.

(c) Putting Things Together

We are going to prove first the claim

$$"I(r) = O\big(r^{2+\varepsilon}\big), \quad {}^\forall\varepsilon > 0"$$

by dividing $S(r)$ into regions A, B, C, D_{-2}, D_{-1}, D_0, D_1, E, F, each of which is investigated separately.

- Region A consists of those points with $|z| \leq \frac{1}{2r}$, i.e., $A = \Omega_{\frac{1}{2},-1,0}$. The volume $\mathrm{vol}(A)$ is thus of order $O(r^{-1})$ Due to (b)(ii) the integrand η is bounded by $\eta|_A = O(r^2)$. It follows that

$$\int_A \eta dV \leq \mathrm{vol}(A) \cdot \sup_{(z,w)\in A} \eta(z,w) = O(r).$$

Hence the contribution of A to the integral $I_r = \int_{S(r)} \eta dV$ is bounded by $O(r)$.

- Region B consists of those points with $\frac{1}{2r} \le |z| \le \frac{3}{2r}$ and $|\sin\theta| < \frac{1}{r}$. Thus $B \subset \Omega_{3/2,-1,1}$. Due to (3.3.14) this implies $\mathrm{vol}(B) = O(r^{-2})$. For the integrand $\eta|_B$ we have the bound $\eta|_B = O(r^4)$ (using (b)(i) and $|z| \ge \frac{1}{2r}$). Hence

$$\int_B \eta dV \le \mathrm{vol}(B) \cdot \sup_{(z,w)\in B} \eta(z,w) = O\big(r^2\big);$$

  i.e., the contribution of B to the integral I_r is bounded by $O(r^2)$.
- Region C consists of those points with $\frac{1}{2r} \le |z| \le \frac{3}{2r}$ and $|\sin\theta| > \frac{1}{r}$. Since $|w| \sim r$, $\frac{1}{2} \lesssim |zw| \lesssim \frac{3}{2}$. We take the volume-compatible parameter $\psi = r\theta$ due to (3.3.13). Then $\frac{1}{r} < |\sin\frac{\psi}{r}| < \frac{\psi}{r}$, and so $\psi > 1$. Therefore,

$$\begin{aligned} J_r &:= \int_{1<\psi<2\pi r, |\sin\frac{\psi}{r}|>\frac{1}{r}} \eta d\psi \\ &\leqq \int_{1<\psi<2\pi r, |\sin\frac{\psi}{r}|>\frac{1}{r}} \frac{2r^2}{(\sin\frac{\psi}{r})^2} d\psi = O\big(r^4\big). \end{aligned}$$

  Here in fact we have that there is a constant $c > 1$ such that

$$J_r \le cr^4.$$

  Therefore it follows that

$$\int_C \eta dV = \int_{\frac{1}{2r}\le|z|\le\frac{3}{2r}} J_r \frac{i}{2} dz \wedge d\bar{z} \le c' r^2, \tag{3.3.15}$$

  where c' is a positive constant. Thus the contribution of C to the integral I_r is bounded by $O(r^2)$.
- For $\gamma \in \{-2,-1,0,1\}$ let D_γ denote the set of those points where $|z| \ge \frac{3}{2r}$, $|z| \le r^{1-\varepsilon}$ and $r^{\frac{\gamma}{2}} \le |z| \le r^{\frac{\gamma+1}{2}}$. For each γ the integrand η is bounded on D_γ by $O(r^{-\gamma})$ (due to (b)(iv)), and the volume $\mathrm{vol}(D_\gamma)$ is bounded by $O(r^{2+\gamma})$, because $D_\gamma \subset \Omega_{1,\frac{\gamma+1}{2},0}$. Thus the contribution of D_γ to the integral I_r is bounded by $O(r^2)$.
- Let E denote the region where $|z| \ge r^{1-\varepsilon}$, $|w| \ge r^{\frac{1}{2}}$. For the integrand we have that $\eta|_E = O(r^{2\varepsilon-1})$ (using (b)(iv)). The volume of E is bounded by the total volume of $S(r)$, i.e., $\mathrm{vol}(E) = O(r^3)$. Together this shows that the contribution of E to I_r is bounded by $O(r^{2+2\varepsilon})$.
- Let F denote the region where $|w| \le r^{\frac{1}{2}}$. In analogy to (b)(iii) we have $|z| \sim r$. With (b)(i) it follows that $\sup_{(z,w)\in F} \eta(z,w) = O(1)$. On the other hand, the volume of F agrees with the volume of $\{(z,w) \in S(r) : |z| \le r^{\frac{1}{2}}\}$ which according to (3.3.14) is bounded by $O(r^2)$. Together this yields that the contribution of F to I_r is bounded by $O(r^2)$.

Thus we have a collection of nine regions $(A, B, C, D_{-2}, D_{-1}, D_0, D_1, E, F)$ covering the sphere $S(r)$. For each such region Ω we have verified

$$\int_{\Omega} \eta dV = O\big(r^{2+\varepsilon}\big), \quad \varepsilon > 0.$$

This establishes (3.3.11):

$$I_r = O\big(r^{2+\varepsilon}\big), \quad \varepsilon > 0.$$

As a consequence, we see that $\rho_f \leqq 1$. □

3.3.3 *Generalizations*

We introduce various generalizations of the Second Main Theorem for differentiably non-degenerate meromorphic mappings. F. Sakai [74a] dealt with a generalization of the results of Griffiths et al. in terms of Kodaira dimensions; Theorem 3.2.2 is of more general form. Sakai [74b] and [76] dealt with the case of domains to be open balls in $\mathbf{C}^n$.

B. Shiffman [75] extended the Second Main Theorem due to Griffiths et al. for differentiably non-degenerate meromorphic mappings, and dealt with singular divisors D. His Second Main Theorem is as follows. We use the same notation as in Theorem 3.2.2, and by $\mathrm{Sing}(D)$ the set of singular points of D.

$$T_f(r, L) + T_f(r, K_V) \leqq \sum_{j=1}^{q} N_1\big(r, f^* D_j\big) + m_f\big(r, \mathrm{Sing}(D)\big) + S_f(r, \omega). \tag{3.3.16}$$

The case of differentiably non-degenerate meromorphic mappings f defined on a finite analytic covering space X over $\mathbf{C}^m$ into a complex projective algebraic manifold was dealt with by Noguchi [76a]. Here X is an irreducible normal complex space. Let $\pi : X \to \mathbf{C}^m$ be a finite analytic covering space, let p be the covering number, and let R be the ramification divisor of π. Let g a meromorphic function on X. We define the partial derivatives $\partial g/\partial z_j$ of g with respect to z_j by the following relation:

$$dg = \sum_{j=1}^{m} \frac{\partial g}{\partial z_j} \pi^* dz_j.$$

The meromorphic functions $\partial g/\partial z_j$ are first defined over the set $X \setminus S(X)$ of non-singular points, and then extended meromorphically over X since X is a normal complex space. As in Lemma 3.1.1 the following is proved.

Lemma 3.3.17 *Let the notation be as above. Then*

$$m\left(r, \frac{\|dg\|}{|g|}\right) = S(r, g).$$

In the case of $m = 1$ this is due to Valiron [29], [31]. By making use of Lemma 3.3.17 and Theorem 2.7.7 (i) we have the following Second Main Theorem.

Theorem 3.3.18 (Noguchi [76a]) *Let $D_j \in |L|$, $1 \leqq j \leqq q$, with $L > 0$ and let $\sum_j D_j$ be of simple normal crossing type. For a differentiably non-degenerate meromorphic mapping $f : X \to V$ we have*

$$qT_f(r, L) + T_f(r, K_V) \leqq \sum_{j=1}^{q} N_1\big(r, f^*D_j\big) + \mu(2p-2)T_f(r, L) + S_f(r). \tag{3.3.19}$$

Here μ depends only on L and if L is very ample, $\mu = 1$.

Being specialized to the case of $\dim X = \dim V = 1$, Theorem 3.3.18 gives rise to the following. Let g be the genus of Y and let $f : X \to Y$ be a holomorphic mapping. Let ω be a metric form on Y with $\int_Y \omega = 1$. For arbitrary distinct q points $a_i \in Y$, $1 \leqq i \leqq q$, we have

$$\big(q + 2g - 2 - 2(p-1)(2g+1)\big)T_f(r, \omega) \leqq \sum_{i=1}^{p} N_1\big(r, f^*a_i\big) + S_f(r). \tag{3.3.20}$$

If $g = 0$ and $p = 1$, $q + 2g - 2 - 2(p-1)(2g+1) = q - 2$ so that Nevanlinna's Second Main Theorem 1.2.5 follows. If $Y = \mathbf{P}^1(\mathbf{C})$, $g = 0$ so that it is the same as the result obtained by H.L. Selberg [30], [34].

Furthermore, W. Stoll [77a] generalized the domain to a parabolic spaces M, and proved

$$qT_f(r, L) + T_f(r, K_V) \leqq \sum_{j=1}^{q} N\big(r, f^*D_j\big) + N(r, \Xi) + S_f(r). \tag{3.3.21}$$

Here Ξ is some divisor on M depending on f. Therefore in this estimate it is indispensable to get an effective bound of the term $N(r, \Xi)$ to make sense of (3.3.21).

Chapter 4
Entire Curves in Algebraic Varieties

In Chap. 3 the Second Main Theorem for differentiably non-degenerate meromorphic mappings $f : \mathbf{C}^m \to V$ into a projective algebraic manifold V of $\dim V \leqq m$ was established in a satisfactory form. It is the next problem to deal with the case of $m < \dim V$. In this case the freeness of the images inside V is very large, and causes the difficulties. It is still an open problem to establish the general Second Main Theorem in the case of $m < \dim V$. We take the typical case of $m = 1$ in this chapter. In the first half we will show the Cartan–Nochka Second Main Theorem for $V = \mathbf{P}^n(\mathbf{C})$. The readers who first learn the theory of holomorphic curves in $\mathbf{P}^n(\mathbf{C})$ should skip Sect. 4.1 and are recommended to read Sect. 4.2, assuming that holomorphic curves are linearly non-degenerate and the given hyperplanes are in general position; in this case, $N = n$, $\omega(j) = 1$, and $\tilde{\omega} = 1$.

4.1 Nochka Weights

In general, a holomorphic mapping from a Riemann surface into a complex manifold is called a *holomorphic curve*. When the domain is the complex plane $\mathbf{C}$, it is called an *entire curve*.

H. Cartan [33] established the Second Main Theorem for linearly non-degenerate entire curves (see the beginning of Sect. 4.2 for definition) $f : \mathbf{C} \to \mathbf{P}^n(\mathbf{C})$ and for hyperplanes of $\mathbf{P}^n(\mathbf{C})$ in general position (see Definition 4.1.1). In the paper he posed a conjecture of a form of the Second Main Theorem for linearly degenerate f (Cartan's Conjecture). This conjecture was affirmatively solved by Nochka [83] after a half century. The paper of Nochka [83] is, however, rather difficult to follow. W. Chen [90] considerably improved the presentation, and H. Fujimoto [93] is based on it.

In this section we describe the Nochka weights following W. Chen [90] and H. Fujimoto [93], which play an important role in the proof of Cartan's Conjecture. The construction of the Nochka weights is rather involved and technical, and not so transparent. As mentioned in the introductory part, the readers who first learn

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the theory of entire curves should skip this section and move to Sect. 4.2, limiting the case where f is linearly non-degenerate, and hyperplanes are in general position (Definition 4.1.1). For they will have a better understanding of the meaning of the Nochka weights.

We take a homogeneous coordinate system $[w_0, \dots, w_n]$ of $\mathbf{P}^n(\mathbf{C})$. Let H_j, $1 \leqq j \leqq q$, be hyperplanes of $\mathbf{P}^n(\mathbf{C})$ defined by the following equations:

$$H_j : \quad \sum_{k=0}^{n} h_{jk} w_k = 0, \quad 1 \leqq j \leqq q.$$

Let $Q = \{1, \dots, q\}$ be the index set. For a subset $R \subset Q$, $|R|$ stands for the cardinality.

Definition 4.1.1 Suppose that $N \geqq n$ and $q \geqq N+1$. The family $\{H_j\}_{j\in Q}$ is said to be in *N-subgeneral position* if for an arbitrary $R \subset Q$ with $|R| = N+1$

$$\bigcap_{j\in R} H_j = \emptyset.$$

If it is in n-subgeneral position, it is simply said to be in *general position*.

$\{H_j\}_{j\in Q}$ being in N-subgeneral position is equivalent to

$$\operatorname{rank}(h_{jk})_{j\in R, 0\leqq k\leqq n} = n+1$$

for every $(N+1, n+1)$ matrix $(h_{jk})_{j\in R, 0\leqq k\leqq n}$.

In general we set for $R \subset S \subset Q$

$$\begin{aligned} V(R) &= \text{the vector subspace of } \mathbf{C}^{n+1} \text{ spanned by vectors} \\ &\qquad (h_{jk})_{0\leqq k\leqq n}, \quad j \in R, \\ \operatorname{rk}(R) &= \dim V(R), \quad \operatorname{rk}(\emptyset) = 0, \\ \operatorname{rk}_R(S) &= \operatorname{rk}(S) - \operatorname{rk}(R), \\ \operatorname{sl}_R(S) &= \frac{\operatorname{rk}(S) - \operatorname{rk}(R)}{|S| - |R|}, \quad \text{provided that } |S| > |R|. \end{aligned} \tag{4.1.2}$$

The index $\operatorname{sl}_R(S)$ represents the slant or slope of the increase of the rank $\operatorname{rk}(S)$ as S increases, and of course, $0 \leqq \operatorname{sl}_R(S) \leqq 1$.

Lemma 4.1.3 *Let $\{H_j\}_{j\in Q}$ be hyperplanes of $\mathbf{P}^n(\mathbf{C})$ in N-subgeneral position.*

(i) *For $R \subset S_i \subset Q$, $i = 1, 2$,*

$$\operatorname{rk}_R(S_1 \cup S_2) + \operatorname{rk}_R(S_1 \cap S_2) \leqq \operatorname{rk}_R(S_1) + \operatorname{rk}_R(S_2).$$

(ii) *For $R \subset S \subset Q$ with $|S| \leqq N+1$,*

$$|R| - \mathrm{rk}(R) \leqq |S| - \mathrm{rk}(S) \leqq N - n.$$

Proof (i) By the elementary theory of vector space we get

$$\dim\big(V(S_1) + V(S_2)\big) + \dim\big(V(S_1) \cap V(S_2)\big) = \dim V(S_1) + \dim V(S_2).$$

Therefore

$$\mathrm{rk}(S_1 \cup S_2) + \mathrm{rk}(S_1 \cap S_2) \leqq \mathrm{rk}(S_1) + \mathrm{rk}(S_2).$$

Deducting $2\,\mathrm{rk}(R)$ from the both sides, we obtain the required inequality.

(ii) It follows from (i) that

$$\begin{aligned}\mathrm{rk}(S) &= \mathrm{rk}\big(R \cup (S \setminus R)\big) \leqq \mathrm{rk}(R) + \mathrm{rk}(S \setminus R)\\ &\leqq \mathrm{rk}(R) + |S \setminus R| = \mathrm{rk}(R) + |S| - |R|.\end{aligned}$$

The first inequality is immediate.

We take a subset $R' \subset Q$, $R' \supset R$ with $|R'| = N+1$. The definition of N-subgeneral position implies $\mathrm{rk}(R') = n+1$. Thus

$$|R| - \mathrm{rk}(R) \leqq \big|R'\big| - \mathrm{rk}\big(R'\big) = N - n. \qquad \square$$

Lemma 4.1.4 *Let $\{H_j\}_{j\in Q}$ be in N-subgeneral position. Assume that $N > n$, $q > 2N - n + 1$. Then there are subsets N_i, $0 \leqq i \leqq s$ of Q satisfying the following conditions*:

(i) $N_0 = \emptyset \subset N_1 \subset \cdots \subset N_s$, $\mathrm{rk}(N_s) < n+1$.

(ii) $0 < \mathrm{sl}_{N_0}(N_1) < \mathrm{sl}_{N_1}(N_2) < \cdots < \mathrm{sl}_{N_{s-1}}(N_s) < \frac{n+1-\mathrm{rk}(N_s)}{2N-n+1-|N_s|} < 1$.

(iii) *Let $1 \leqq i \leqq s$ be an arbitrary index and let $R \subset Q$ be a subset such that $R \supset N_{i-1}$. If $\mathrm{rk}(N_{i-1}) < \mathrm{rk}(R) < n+1$, then*

$$\mathrm{sl}_{N_{i-1}}(N_i) \leqq \mathrm{sl}_{N_{i-1}}(R).$$

Furthermore, the equality holds only when $R \subset N_i$.

(iv) *For an arbitrary subset $R \subset Q$ with $R \supset N_s$ and $\mathrm{rk}(N_s) < \mathrm{rk}(R) < n+1$,*

$$\mathrm{sl}_{N_s}(R) \geqq \frac{n+1-\mathrm{rk}(N_s)}{2N-n+1-|N_s|}.$$

Proof We first put $N_0 = \emptyset$, and then inductively define N_i. Assume that $\{N_i\}_{i=0}^{s}$ is chosen so that it satisfies (i)–(iii). If it satisfies (iv), then it is done. Otherwise it suffices to find $N_{s+1} \subset Q$ with $N_{s+1} \supset N_s$ such that $\{N_i\}_{i=0}^{s+1}$ satisfies (i)–(iii).

By the condition of N-subgeneral position and $\mathrm{rk}(N_s) < n+1$ it is seen that $|N_s| \leqq N$. It follows from Lemma 4.1.3 (ii) that

$$|N_s| - \mathrm{rk}(N_s) \leqq N - n < 2N - 2n.$$

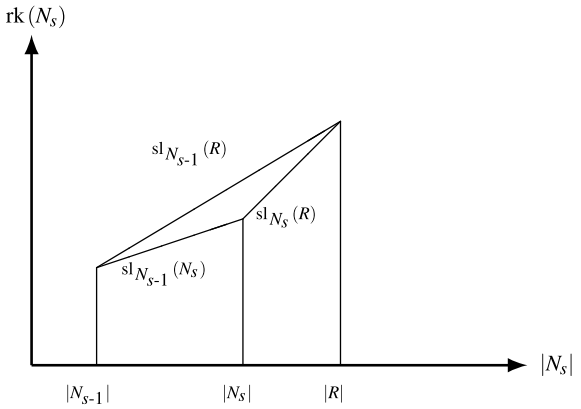


Fig. 4.1 Nochka weights

Therefore

$$\frac{n+1-\mathrm{rk}(N_s)}{2N-n+1-|N_s|}<1.$$

Assume that (iv) does not hold. Set

$$\mathscr{R}=\big\{R;\, N_s\subset R\subset Q, \mathrm{rk}(N_s)<\mathrm{rk}(R)<n+1\big\}.$$

Then $\mathscr{R}\neq\emptyset$. If $R\in\mathscr{R}$, $|R|\leqq N$. We take $\varepsilon_0=\min\{\mathrm{sl}_{N_s}(R);\, R\in\mathscr{R}\}$. By the assumption

$$\varepsilon_0<\frac{n+1-\mathrm{rk}(N_s)}{2N-n+1-|N_s|}<1. \tag{4.1.5}$$

Then (iii) implies that $\mathrm{sl}_{N_{s-1}}(N_s)<\mathrm{sl}_{N_{s-1}}(R)$ for every $R\in\mathscr{R}$. It is immediate from Fig. 4.1 that

$$\mathrm{sl}_{N_{s-1}}(N_s)<\mathrm{sl}_{N_s}(R). \tag{4.1.6}$$

Set $\mathscr{R}'=\{R\in\mathscr{R};\, \mathrm{sl}_{N_s}(R)=\varepsilon_0\}$. We are going to show the next claim.

Claim 4.1.7 *If $R_1, R_2\in\mathscr{R}'$, then $R_1\cup R_2\in\mathscr{R}'$.*

Since $R_1, R_2\in\mathscr{R}'$,

$$\varepsilon_0=\frac{\mathrm{rk}(R_1)-\mathrm{rk}(N_s)}{|R_1|-|N_s|}=\frac{\mathrm{rk}(R_2)-\mathrm{rk}(N_s)}{|R_2|-|N_s|}.$$

By making use of Lemma 4.1.3 (ii) we have

$$\begin{aligned}
&\mathrm{rk}(R_1)+\mathrm{rk}(R_2)-2\,\mathrm{rk}(N_s)\\
&\quad=\mathrm{rk}(R_1)-\mathrm{rk}(N_s)+\mathrm{rk}(R_2)-\mathrm{rk}(N_s)\\
&\quad=\varepsilon_0\big(|R_1|-|N_s|+|R_2|-|N_s|\big)\\
&\quad\leqq\varepsilon_0\big(\mathrm{rk}(R_1)+N-n+\mathrm{rk}(R_2)+N-n-2|N_s|\big)\\
&\quad=\varepsilon_0\big(\mathrm{rk}(R_1)+\mathrm{rk}(R_2)+2N-2n-2|N_s|\big)
\end{aligned}$$

(continued)

$$= \varepsilon_0\big(\mathrm{rk}(R_1) + \mathrm{rk}(R_2) - 2\,\mathrm{rk}(N_s) + 2N - 2n - 2|N_s| + 2\,\mathrm{rk}(N_s)\big).$$

Hence,

$$\begin{aligned}
&\mathrm{rk}(R_1) + \mathrm{rk}(R_2) - 2\,\mathrm{rk}(N_s)\\
&\leqq \frac{\varepsilon_0}{1-\varepsilon_0}\big(2N - 2n - 2|N_s| + 2\,\mathrm{rk}(N_s)\big)\\
&= \left(\frac{1}{1-\varepsilon_0} - 1\right)\big(2N - 2n - 2|N_s| + 2\,\mathrm{rk}(N_s)\big)\\
&\quad \text{(using (4.1.5), we continue the computation)}\\
&< \frac{(n+1-\mathrm{rk}(N_s))(2N - 2n - 2|N_s| + 2\,\mathrm{rk}(N_s))}{2N - 2n - |N_s| + \mathrm{rk}(N_s)}\\
&\quad \big(\text{as } |N_s| \geqq \mathrm{rk}(N_s)\big)\\
&\leqq n + 1 - \mathrm{rk}(N_s).
\end{aligned}$$

Thus it follows that

$$\mathrm{rk}(R_1) + \mathrm{rk}(R_2) - \mathrm{rk}(N_s) < n + 1.$$

This with Lemma 4.1.3 (i) implies that

$$\begin{aligned}
\mathrm{rk}(R_1 \cup R_2) &\leqq \mathrm{rk}(R_1) + \mathrm{rk}(R_2) - \mathrm{rk}(R_1 \cap R_2)\\
&\leqq \mathrm{rk}(R_1) + \mathrm{rk}(R_2) - \mathrm{rk}(N_s) < n + 1.
\end{aligned}$$

One infers that $\mathrm{rk}(N_s) < \mathrm{rk}(R_1) \leqq \mathrm{rk}(R_1 \cup R_2) < n + 1$ and that $R_1 \cup R_2 \in \mathscr{R}$. The following, if it is shown, implies Claim 4.1.7:

$$\varepsilon_0 = \mathrm{sl}_{N_s}(R_1 \cup R_2). \tag{4.1.8}$$

By definition $\varepsilon_0 \leqq \mathrm{sl}_{N_s}(R_1 \cup R_2)$. To show the converse, we prove the following inequality:

$$\mathrm{rk}_{N_s}(R_1 \cap R_2) = \mathrm{rk}(R_1 \cap R_2) - \mathrm{rk}(N_s) \geqq \varepsilon_0\big(|R_1 \cap R_2| - |N_s|\big). \tag{4.1.9}$$

If $\mathrm{rk}(R_1 \cap R_2) > \mathrm{rk}(N_s)$, then $R_1 \cap R_2 \in \mathscr{R}$ and hence (4.1.9) is deduced from the definition of ε_0. If $\mathrm{rk}(R_1 \cap R_2) = \mathrm{rk}(N_s)$, it remains to show that $R_1 \cap R_2 = N_s$. Suppose that $|R_1 \cap R_2| > |N_s|$. Since $\mathrm{rk}(R_1 \cap R_2) - \mathrm{rk}(N_{s-1}) = \mathrm{rk}(N_s) - \mathrm{rk}(N_{s-1}) > 0$ and $\mathrm{rk}(R_1 \cap R_2) \leqq \mathrm{rk}(R_1) < n + 1$, it follows from (iii) that

$$\begin{aligned}
\mathrm{sl}_{N_{s-1}}(N_s) \leqq \mathrm{sl}_{N_{s-1}}(R_1 \cap R_2) &= \frac{\mathrm{rk}(R_1 \cap R_2) - \mathrm{rk}(N_{s-1})}{|R_1 \cap R_2| - |N_{s-1}|}\\
&< \frac{\mathrm{rk}(N_s) - \mathrm{rk}(N_{s-1})}{|N_s| - |N_{s-1}|} = \mathrm{sl}_{N_{s-1}}(N_s).
\end{aligned}$$

This is a contradiction.

By making use of Lemma 4.1.3 (i), $R_1, R_2 \in \mathscr{R}'$ and (4.1.9) we have

$$\begin{aligned} \mathrm{sl}_{N_s}(R_1 \cup R_2) &= \frac{\mathrm{rk}_{N_s}(R_1 \cup R_2)}{|R_1 \cup R_2| - |N_s|} \\ &\leqq \frac{\mathrm{rk}_{N_s}(R_1) + \mathrm{rk}_{N_s}(R_2) - \mathrm{rk}_{N_s}(R_1 \cap R_2)}{(|R_1| - |N_s|) + (|R_2| - |N_2|) - (|R_1 \cap R_2| - |N_s|)} \\ &\leqq \varepsilon_0. \end{aligned}$$

Therefore (4.1.8) follows and Claim 4.1.7 is proved. Set

$$N_{s+1} = \bigcup_{R \in \mathscr{R}'} R.$$

By Claim 4.1.7, $N_{s+1} \in \mathscr{R}'$. By the construction $\{N_i\}_{i=0}^{s+1}$ satisfies (i)–(iii). This finishes the proof. □

Theorem 4.1.10 *Let $\{H_j\}_{j \in Q}$ be a family of hyperplanes of $\mathbf{P}^n(\mathbf{C})$ in N-subgeneral position, and let $q > 2N - n + 1$. Then there are rational constants $\omega(j)$, $j \in Q$ satisfying the following conditions*:

(i) $0 < \omega(j) \leqq 1, {}^\forall j \in Q$.

(ii) *Set $\tilde{\omega} = \max_{j \in Q} \omega(j)$. Then*

$$\sum_{j=1}^{q} \omega(j) = \tilde{\omega}(q - 2N + n - 1) + n + 1.$$

(iii) $\frac{n+1}{2N-n+1} \leqq \tilde{\omega} \leqq \frac{n}{N}$.[1]

(iv) *If $R \subset Q$ and $0 < |R| \leqq N + 1$, $\sum_{j \in R} \omega(j) \leqq \mathrm{rk}(R)$.*

The constants $\omega(j)$ appearing in the above theorem are called *Nochka's weights*, and $\tilde{\omega}$ is called *Nochka's constant*.

Proof If $N = n$, it suffices to set $\omega(j) = 1$.

Assume that $N > n$. We take $\{N_i\}_{i=0}^{s}$ as in Lemma 4.1.4 (i) of which implies $|N_s| \leqq N$. We take $N_{s+1} \subset Q$ so that $N_{s+1} \supset N_s$ and

$$|N_{s+1}| = 2N - n + 1 > N + 1. \tag{4.1.11}$$

It follows that $\mathrm{rk}(N_{s+1}) = n + 1$ and $\mathrm{sl}_{N_s}(N_{s+1}) = \frac{n+1-\mathrm{rk}(N_s)}{2N-n+1-|N_s|}$. We define constants $\omega(j)$ as follows:

$$\omega(j) = \begin{cases} \mathrm{sl}_{N_i}(N_{i+1}), & j \in N_{i+1} \setminus N_i,\ 0 \leqq i \leqq s, \\ \mathrm{sl}_{N_s}(N_{s+1}) = \frac{n+1-\mathrm{rk}(N_s)}{2N-n+1-|N_s|}, & j \notin N_{s+1}. \end{cases} \tag{4.1.12}$$

In the sequel we show that these satisfy (i)–(iv).

[1] As for the last estimate, it was $\frac{n+1}{N+1}$ in Nochka [83] and Chen [90]. This improvement is due to N. Toda.

(i) By Lemma 4.1.3 (ii), $-\operatorname{rk}(N_s) \leqq -|N_s| + N - n$. Therefore,

$$\begin{aligned}\operatorname{sl}_{N_s}(N_{s+1}) &\leqq \frac{n+1-|N_s|+N-n}{2N-n+1-|N_s|}\\ &\leqq \frac{N+1-|N_s|}{2N-n+1-|N_s|}\\ &< \frac{N+1-|N_s|}{N+1-|N_s|} = 1.\end{aligned}$$

We show that $\omega(j) \leqq \operatorname{sl}_{N_s}(N_{s+1})$, $j \in N_{s+1}$. If $j \notin N_s$, it is trivial. If $j \in N_s$, there is a unique i such that $j \in N_{i+1} \setminus N_i$, $0 \leqq i \leqq s-1$. From Lemma 4.1.4 (ii) we infer that $\omega(j) < \operatorname{sl}_{N_s}(N_{s+1})$.

(ii) By the above arguments, $\tilde{\omega} = \operatorname{sl}_{N_s}(N_{s+1})$. Decomposing $Q = (Q \setminus N_{s+1}) \cup (N_{s+1} \setminus N_s) \cup \cdots \cup (N_1 \setminus N_0)$, we have

$$\begin{aligned}\sum_{j=1}^{q}\omega(j) &= \sum_{j\in Q\setminus N_{s+1}}\omega(j) + \sum_{i=1}^{s+1}\sum_{j\in N_i\setminus N_{i-1}}\omega(j)\\ &= \tilde{\omega}\big(q-|N_{s+1}|\big) + \sum_{i=1}^{s+1}\sum_{j\in N_i\setminus N_{i-1}}\frac{\operatorname{rk}(N_i)-\operatorname{rk}(N_{i-1})}{|N_i|-|N_{i-1}|}\\ &\quad \big(\text{here we use (4.1.11)}\big)\\ &= \tilde{\omega}(q-2N+n-1) + \sum_{i=1}^{s+1}\big(\operatorname{rk}(N_i)-\operatorname{rk}(N_{i-1})\big)\\ &= \tilde{\omega}(q-2N+n-1) + \operatorname{rk}(N_{s+1})\\ &= \tilde{\omega}(q-2N+n-1) + n + 1.\end{aligned}$$

(iii) It follows from the already shown (i) and (ii) that

$$\begin{aligned}n+1 &= \sum_{j=1}^{q}\omega(j) - \tilde{\omega}(q-2N+n-1)\\ &\leqq q\tilde{\omega} - \tilde{\omega}(q-2N+n-1)\\ &= \tilde{\omega}(2N-n+1).\end{aligned}$$

Hence $\tilde{\omega} \geqq (n+1)/(2N-n+1)$. If $s=0$, then

$$\tilde{\omega} = \frac{n+1}{2N-n+1} \leqq \frac{n}{N}.$$

If $s > 0$, $\mathrm{rk}(N_s) \geqq 1$, and Lemma 4.1.3 (ii) yields that

$$\tilde{\omega} = \frac{n+1-\mathrm{rk}(N_s)}{(N+1)+(N-n-|N_s|)} \leqq \frac{n+1-\mathrm{rk}(N_s)}{N+1-\mathrm{rk}(N_s)}$$
$$= 1 - \frac{N-n}{N+1-\mathrm{rk}(N_s)} \leqq 1 - \frac{N-n}{N} = \frac{n}{N}.$$

(iv) Take an arbitrary $R \subset Q$ with $0 < |R| \leqq N+1$.

(a) Case of $\mathrm{rk}(R \cup N_s) = n+1$. It follows from Lemma 4.1.3 (ii) that

$$\begin{aligned} |R| &\leqq \mathrm{rk}(R) + N - n, \\ |N_s| &\leqq \mathrm{rk}(N_s) + N - n. \end{aligned} \tag{4.1.13}$$

The case assumption, $\mathrm{rk}(R \cup N_s) = n+1$ with Lemma 4.1.3 (i) implies that

$$\begin{aligned} n+1-\mathrm{rk}(N_s) &= \mathrm{rk}(R \cup N_s) - \mathrm{rk}(N_s) \\ &\leqq \mathrm{rk}(R) - \mathrm{rk}(R \cap N_s) \\ &\leqq \mathrm{rk}(R). \end{aligned}$$

From (i), (4.1.12) and (4.1.13) we deduce that

$$\begin{aligned} \sum_{j \in R} \omega(j) &\leqq \tilde{\omega}|R| \leqq \tilde{\omega}\big(\mathrm{rk}(R) + N - n\big) \\ &= \tilde{\omega}\,\mathrm{rk}(R)\left(1 + \frac{N-n}{\mathrm{rk}(R)}\right) \\ &\leqq \tilde{\omega}\,\mathrm{rk}(R)\left(1 + \frac{N-n}{n+1-\mathrm{rk}(N_s)}\right) \\ &= \mathrm{rk}(R)\frac{N+1-\mathrm{rk}(N_s)}{2N-n+1-|N_s|} \\ &\leqq \mathrm{rk}(R)\frac{N+1-\mathrm{rk}(N_s)}{2N-n+1-(\mathrm{rk}(N_s)+N-n)} \\ &= \mathrm{rk}(R). \end{aligned}$$

(b) Case of $\mathrm{rk}(R \cup N_s) < n+1$. Firstly, we see that $|R \cup N_s| \leqq N$. Set

$$R_i = \begin{cases} R \cap N_i, & 0 \leqq i \leqq s, \\ R, & i = s+1. \end{cases}$$

We are going to show the following:

Claim 4.1.14 *Suppose that* $1 \leqq i \leqq s+1$ *and* $|R_i| > |R_{i-1}|$. *Then*

$$\mathrm{rk}(R_i \cup N_{i-1}) > \mathrm{rk}(N_{i-1}).$$

Let $i = 1$. Since $\mathrm{rk}(R_1 \cup N_0) = \mathrm{rk}(R_1) > 0 = \mathrm{rk}(N_0)$, Claim 4.1.14 holds.
Let $i > 1$. Assume that $\mathrm{rk}(R_i \cup N_{i-1}) = \mathrm{rk}(N_{i-1})$. Then

$$\mathrm{rk}_{N_{i-2}}(R_i \cup N_{i-1}) = \mathrm{rk}(N_{i-1}) - \mathrm{rk}(N_{i-2}) > 0.$$

Since $\mathrm{rk}(R_i \cup N_{i-1}) \leqq \mathrm{rk}(R \cup N_s) < n+1$, it follows from Lemma 4.1.4 (iii) that

$$\mathrm{sl}_{N_{i-2}}(N_{i-1}) \leqq \mathrm{sl}_{N_{i-2}}(N_{i-1} \cup R_i).$$

On the other hand,

$$\begin{aligned}\mathrm{sl}_{N_{i-2}}(N_{i-1} \cup R_i) &= \frac{\mathrm{rk}(N_{i-1} \cup R_i) - \mathrm{rk}(N_{i-2})}{|N_{i-1} \cup R_i| - |N_{i-2}|} \\ &\leqq \frac{\mathrm{rk}(N_{i-1}) - \mathrm{rk}(N_{i-2})}{|N_{i-1}| - |N_{i-2}|} = \mathrm{sl}_{N_{i-2}}(N_{i-1}).\end{aligned}$$

Therefore $\mathrm{sl}_{N_{i-2}}(N_{i-1}) = \mathrm{sl}_{N_{i-2}}(N_{i-1} \cup R_i)$ and again by Lemma 4.1.4 (iii), $|R_i| = |R_{i-1}|$; this is a contradiction. Thus Claim 4.1.14 is proved.

Using Claim 4.1.14, we show the following inequality:

$$\big(|R_i| - |R_{i-1}|\big)\,\mathrm{sl}_{N_{i-1}}(N_i) \leqq \mathrm{rk}(R_i) - \mathrm{rk}(R_{i-1}), \quad 1 \leqq i \leqq s+1. \tag{4.1.15}$$

We may assume that $|R_i| - |R_{i-1}| > 0$. By Claim 4.1.14, $\mathrm{rk}(N_{i-1} \cup R_i) > \mathrm{rk}(N_{i-1})$. We apply Lemma 4.1.4 (iii) for the case of $1 \leqq i \leqq s$, and in the case of $i = s+1$ we use the assumption of $\mathrm{rk}(R \cup N_s) < n+1$. Thus, by Lemma 4.1.4 (iv) and (4.1.12) we see that

$$\mathrm{sl}_{N_{i-1}}(N_i) \leqq \mathrm{sl}_{N_{i-1}}(N_{i-1} \cup R_i). \tag{4.1.16}$$

By definition and Lemma 4.1.3 (i),

$$\begin{aligned}|R_i \cup N_{i-1}| &= |N_{i-1}| + |R_i| - |R_i \cap N_{i-1}| \\ &= |N_{i-1}| + |R_i| - |R_{i-1}|, \\ \mathrm{rk}(R_i \cup N_{i-1}) &\leqq \mathrm{rk}(N_{i-1}) + \mathrm{rk}(R_i) - \mathrm{rk}(R_i \cap N_{i-1}) \\ &= \mathrm{rk}(N_{i-1}) + \mathrm{rk}(R_i) - \mathrm{rk}(R_{i-1}).\end{aligned}$$

These with (4.1.16) implies that

$$\begin{aligned}\mathrm{sl}_{N_{i-1}}(N_i) &\leqq \mathrm{sl}_{N_{i-1}}(N_{i-1} \cup R_i) \\ &= \frac{\mathrm{rk}(R_i \cup N_{i-1}) - \mathrm{rk}(N_{i-1})}{|R_i \cup N_{i-1}| - |N_{i-1}|} \\ &\leqq \frac{\mathrm{rk}(R_i) - \mathrm{rk}(R_{i-1})}{|R_i| - |R_{i-1}|}.\end{aligned}$$

Thus (4.1.15) is deduced.

Now it follows from (4.1.12) and (4.1.15) that

$$\begin{aligned}\sum_{j\in R}\omega(j) &= \sum_{i=1}^{s+1}\sum_{j\in R_i\setminus R_{i-1}}\omega(j) = \sum_{i=1}^{s+1}\sum_{j\in R_i\setminus R_{i-1}}\mathrm{sl}_{N_{i-1}}(N_i)\\ &= \sum_{i=1}^{s+1}\big(|R_i|-|R_{i-1}|\big)\,\mathrm{sl}_{N_{i-1}}(N_i)\\ &\leqq \sum_{i=1}^{s+1}\big(\mathrm{rk}(R_i)-\mathrm{rk}(R_{i-1})\big)\\ &= \mathrm{rk}(R)-\mathrm{rk}(R_s)+\sum_{i=1}^{s}\big(\mathrm{rk}(R_i)-\mathrm{rk}(R_{i-1})\big)\\ &= \mathrm{rk}(R).\end{aligned}$$

This finishes the proof of the present theorem. □

Lemma 4.1.17 *Let $q > 2N-n+1$ and let $\{H_j\}_{j\in Q}$ be a family of hyperplanes of $\mathbf{P}^n(\mathbf{C})$ in N-subgeneral position. Let $\{\omega(j)\}_{j\in Q}$ be Nochka's weights for $\{H_j\}_{j\in Q}$. Let $E_j \geqq 1$, $j\in Q$ be arbitrary constants. Then for every subset $R\subset Q$ with $0<|R|\leqq N+1$ there are distinct $j_1,\dots,j_{\mathrm{rk}(R)}\in R$ such that $\mathrm{rk}(\{j_l\}_{l=1}^{\mathrm{rk}(R)})=\mathrm{rk}(R)$ and*

$$\prod_{j\in R}E_j^{\omega(j)} \leqq \prod_{l=1}^{\mathrm{rk}(R)}E_{j_l}.$$

Proof Changing the order of indices, we may assume that $E_1\geqq E_2\geqq\cdots\geqq E_q$. We set $j_1=\min R$, $R_1=\{j_1\}$ and $S_1=\{j\in R;\, H_j\in V(R_1)\}$ (cf. (4.1.2) for notation $V(\cdot)$). Assuming that $R_l=\{j_1,\dots,j_l\}$ and S_l are determined, we inductively set

$$\begin{aligned}j_{l+1} &= \min\big\{j\in R;\, H_j\notin V(R_l)\big\},\\ R_{l+1} &= R_l\cup\{j_{l+1}\},\\ S_{l+1} &= \left\{j\in R\setminus\bigcup_{k=1}^{l}S_k;\, H_j\in V(R_{l+1})\right\}.\end{aligned}$$

In this way we choose $j_1<j_2<\cdots<j_{\mathrm{rk}(R)}$ and mutually disjoint $S_1,S_2,\dots,S_{\mathrm{rk}(R)}$. Noting that $j_l=\min S_l$, we have

$$\begin{aligned}E_j &\leqq E_{j_l},\quad j\in S_l,\\ R &= S_1\cup\cdots\cup S_{\mathrm{rk}(R)}.\end{aligned}$$

Set $T_l = S_1 \cup \cdots \cup S_l$, $1 \leqq l \leqq \mathrm{rk}(R)$. Theorem 4.1.10 (iv) implies that

$$\sum_{j \in T_l} \omega(j) \leqq \mathrm{rk}(T_l) = l.$$

From this it follows that

$$\begin{aligned}
\prod_{j \in R} E_j^{\omega(j)} &= \prod_{l=1}^{\mathrm{rk}(R)} \prod_{j \in S_l} E_j^{\omega(j)} \leqq \prod_{l=1}^{\mathrm{rk}(R)} E_{j_l}^{\sum_{j \in S_l} \omega(j)} \\
&\leqq E_{j_1} E_{j_1}^{-1+\sum_{j \in T_1} \omega(j)} \prod_{l=2}^{\mathrm{rk}(R)} E_{j_l}^{\sum_{j \in S_l} \omega(j)} \\
&\leqq E_{j_1} E_{j_2}^{-1+\sum_{j \in T_1} \omega(j)} \prod_{l=2}^{\mathrm{rk}(R)} E_{j_l}^{\sum_{j \in S_l} \omega(j)} \\
&= E_{j_1} E_{j_2} E_{j_2}^{-2+\sum_{j \in T_2} \omega(j)} \prod_{l=3}^{\mathrm{rk}(R)} E_{j_l}^{\sum_{j \in S_l} \omega(j)} \\
&\vdots \\
&= E_{j_1} E_{j_2} \cdots E_{j_{\mathrm{rk}(R)}} \cdot E_{j_{\mathrm{rk}(R)}}^{-\mathrm{rk}(R)+\sum_{j \in R} \omega(j)}.
\end{aligned}$$

By Theorem 4.1.10 (iv), $-\mathrm{rk}(R) + \sum_{j \in R} \omega(j) \leqq 0$. We finally obtain

$$\prod_{j \in R} E_j^{\omega(j)} \leqq \prod_{l=1}^{\mathrm{rk}(R)} E_{j_l}.$$

□

4.2 The Cartan–Nochka Theorem

We consider an entire curve $f : \mathbf{C} \to \mathbf{P}^N(\mathbf{C})$ into N-dimensional complex projective space $\mathbf{P}^N(\mathbf{C})$ and a family of hyperplanes $\{H_j\}_{j=1}^q$ in general position. Let $\mathbf{P}^n(\mathbf{C}) \subset \mathbf{P}^N(\mathbf{C})$ be the smallest linear space containing the image $f(\mathbf{C})$. When $n = N$, f is said to be *linearly non-degenerate*. H. Cartan [33] established the Second Main Theorem for linearly non-degenerate $f : \mathbf{C} \to \mathbf{P}^N(\mathbf{C})$ and $\{H_j\}_{j=1}^q$, and posed a conjecture of the Second Main Theorem for the linearly degenerate case ($n < N$).

For Cartan's Second Main Theorem readers may assume that $N = n$ and all Nochka's weights and constant $\omega(j) = \tilde{\omega} = 1$.

We describe the solution of Cartan's Conjecture due to Nochka [83]. In Chen [90] and Fujimoto [93] the proofs based on the Weyl–Ahlfors approach are found. Here we give a more direct proof based on Cartan's method.

Let $Q = \{1, \dots, q\}$ denote the index set. We regard f as a linearly non-degenerate entire curve $f : \mathbf{C} \to \mathbf{P}^n(\mathbf{C})$. Thus cut loci $H_j \cap \mathbf{P}^n(\mathbf{C})$, $j \in Q$, are in N-subgeneral position. In the sequel, we denote by H_j, $j \in Q$, hyperplanes of $\mathbf{P}^n(\mathbf{C})$ in N-subgeneral position. Let $w = [w_0, \dots, w_n]$ be a homogeneous coordinate system of $\mathbf{P}^n(\mathbf{C})$ and let $f(z) = [f_0(z), \dots, f_n(z)]$ be the reduced representation of f. We may assume that H_j are defined as

$$H_j : \hat{H}_j(w) = \sum_{k=0}^{n} h_{jk} w_k = 0, \quad 1 \leqq j \leqq q,$$

$$\|\hat{H}_j\| = \left(\sum_k |h_{jk}|^2 \right)^{1/2} = 1, \tag{4.2.1}$$

$$\frac{|\hat{H}_j(w)|}{\|w\|} \leqq 1.$$

The Wronskian $W(f_0, \dots, f_n)$ and the logarithmic Wronskian $\Delta(f_0, \dots, f_n)$ are defined as

$$W(f_0, \dots, f_n) = \begin{vmatrix} f_0 & \cdots & f_n \\ \frac{d}{dz} f_0 & \cdots & \frac{d}{dz} f_n \\ \vdots & \vdots & \vdots \\ \frac{d^n}{dz^n} f_0 & \cdots & \frac{d^n}{dz^n} f_n \end{vmatrix},$$

$$\Delta(f_0, \dots, f_n) = \begin{vmatrix} 1 & \cdots & 1 \\ \frac{\frac{d}{dz} f_0}{f_0} & \cdots & \frac{\frac{d}{dz} f_n}{f_n} \\ \vdots & \vdots & \vdots \\ \frac{\frac{d^n}{dz^n} f_0}{f_0} & \cdots & \frac{\frac{d^n}{dz^n} f_n}{f_n} \end{vmatrix}.$$

Let g be a meromorphic function on $\mathbf{C}$. Then the Wronskian and the logarithmic Wronskian satisfy the following functional equations:

$$W(gf_0, \dots, gf_n) = g^{n+1} W(f_0, \dots, f_n),$$

$$\Delta(gf_0, \dots, gf_n) = \Delta(f_0, \dots, f_n), \tag{4.2.2}$$

$$\Delta\left(1, \frac{f_1}{f_0}, \dots, \frac{f_n}{f_0}\right) = \Delta(f_0, \dots, f_n).$$

The proofs are easy, and the last one follows by setting $g = 1/f_0$ in the second.

For $R \subset Q$ with $|R| = n + 1$ we denote by $W((\hat{H}_j \circ f, j \in R))$ (resp. $\Delta((\hat{H}_j \circ f, j \in R))$) the Wronskian (resp. logarithmic Wronskian) of $\hat{H}_j \circ f$, $j \in R$ with increasing order of indices.

Lemma 4.2.3 *Let $q > 2N - n + 1$ and let $\omega(j), \tilde{\omega}$ be Nochka's weights and constant of $\{H_j\}_{j\in Q}$. Then there is a positive constant C determined by $\{\hat{H}_j\}_{j\in Q}$ such that for every $z \in \mathbf{C}$*

$$\begin{aligned}\big\|f(z)\big\|^{\tilde{\omega}(q-2N+n-1)} \leqq C\frac{\prod_{j\in Q}|\hat{H}_j(f(z))|^{\omega(j)}}{|W(f_0,\dots,f_n)(z)|}\\ \cdot\left\{\sum_{R\subset Q,|R|=n+1}\big|\Delta\big((\hat{H}_j\circ f, j\in R)\big)(z)\big|\right\}.\end{aligned}$$

Proof By the definition of N-subgeneral position we see that for every point $w \in \mathbf{P}^n(\mathbf{C})$ there is a subset $S \subset Q$ with $|S| = q - N - 1$ such that $\prod_{j\in S}\hat{H}_j(w) \neq 0$. Therefore there is a constant $C_1 > 0$ such that

$$C_1^{-1} < \sum_{|S|=q-N-1}\prod_{j\in S}\left(\frac{|\hat{H}_j(w)|}{\|w\|}\right)^{\omega(j)} < C_1, \quad {}^{\forall}w \in \mathbf{P}^n(\mathbf{C}). \tag{4.2.4}$$

By setting $R = Q \setminus S$ the middle term is modified to

$$\prod_{j\in S}\left(\frac{|\hat{H}_j(w)|}{\|w\|}\right)^{\omega(j)} = \prod_{j\in R}\left(\frac{\|w\|}{|\hat{H}_j(w)|}\right)^{\omega(j)}\cdot\frac{\prod_{j\in Q}|\hat{H}_j(w)|^{\omega(j)}}{\|w\|^{\sum_{j\in Q}\omega(j)}}.$$

Here we use Theorem 4.1.10 (ii) and apply Lemma 4.1.17 for R, noting $\mathrm{rk}(R) = n + 1$, so that there are indices $\{j_1,\dots,j_{n+1}\} = R^\circ$ satisfying

$$\begin{aligned}\prod_{j\in S}\left(\frac{|\hat{H}_j(w)|}{\|w\|}\right)^{\omega(j)} &\leqq \left(\prod_{j\in R^\circ}\frac{\|w\|}{|\hat{H}_j(w)|}\right)\cdot\frac{\prod_{j\in Q}|\hat{H}_j(w)|^{\omega(j)}}{\|w\|^{\tilde{\omega}(q-2N+n-1)+n+1}}\\ &= \frac{1}{\prod_{j\in R^\circ}|\hat{H}_j(w)|}\cdot\frac{\prod_{j\in Q}|\hat{H}_j(w)|^{\omega(j)}}{\|w\|^{\tilde{\omega}(q-2N+n-1)}}.\end{aligned} \tag{4.2.5}$$

By the property of Wronskian there is a constant $c(R^\circ) > 0$ such that

$$c\big(R^\circ\big)\frac{|W((\hat{H}_j\circ f, j\in R^\circ))|}{|W(f_0,\dots,f_n)|} = 1.$$

It follows from this and (4.2.5) that

$$\begin{aligned}\prod_{j\in S}\left(\frac{|\hat{H}_j\circ f|}{\|w\|}\right)^{\omega(j)} \leqq c\big(R^\circ\big)\frac{1}{\|f\|^{\tilde{\omega}(q-2N+n-1)}}\\ \cdot\frac{\prod_{j\in Q}|\hat{H}_j\circ f|^{\omega(j)}}{|W(f_0,\dots,f_n)|}\cdot\frac{|W((\hat{H}_j\circ f, j\in R^\circ))|}{\prod_{j\in R^\circ}|\hat{H}_j\circ f|}\end{aligned}$$

(continued)

$$= c\big(R^\circ\big)\frac{1}{\|f\|^{\tilde{\omega}(q-2N+n-1)}} \cdot \frac{\prod_{j\in Q}|\hat{H}_j\circ f|^{\omega(j)}}{|W(f_0,\ldots,f_n)|}\cdot\big|\Delta\big(\big(\hat{H}_j\circ f, j\in R^\circ\big)\big)\big|.$$

Thus, setting $C = C_1 \max_{R^\circ}\{c(R^\circ)\}$, we obtain the required inequality. □

We denote by $\operatorname{ord}_a \hat{H}_j\circ f$ the order of zero of $\hat{H}_j\circ f(z)$. The next lemma is due to Fujimoto [93], Lemma 3.2.13.

Lemma 4.2.6 *Let the notation be as above. As divisors on* $\mathbf{C}$ *with rational coefficients the following inequality holds*:

$$\sum_{j\in Q}\omega(j)(\hat{H}_j\circ f) - \big(W(f_0,\ldots,f_n)\big) \leqq \sum_{j\in Q}\omega(j)\sum_{a\in\mathbf{C}}\min\{\operatorname{ord}_a \hat{H}_j\circ f, n\}\cdot\{a\}.$$

Proof For the sake of simplicity we write $W = W(f_0,\ldots,f_n)$. Note that

$$\min\{\operatorname{ord}_a \hat{H}_j\circ f, n\} + (\operatorname{ord}_a \hat{H}_j\circ f - n)^+ = \operatorname{ord}_a \hat{H}_j\circ f.$$

Hence the claimed inequality is equivalent to

$$\sum_{j\in Q}\omega(j)\sum_{a\in\mathbf{C}}(\operatorname{ord}_a \hat{H}_j\circ f - n)^+\cdot\{a\} \leqq (W). \tag{4.2.7}$$

Take an arbitrary point $a\in\mathbf{C}$ and set

$$S = \{j\in Q;\ \operatorname{ord}_a \hat{H}_j\circ f \geqq n+1\}.$$

We may assume that $S\neq\emptyset$. By the assumption of N-subgeneral position, $|S|\leqq N$. Let $m_1 > m_2 > \cdots > m_t \geqq n+1$ be the orders $\operatorname{ord}_a \hat{H}_j\circ f$, $j\in S$, in order from the large to the small. We take a sequence of subsets of S

$$S_0 = \emptyset \neq S_1\subset S_2\subset\cdots\subset S_t = S$$

such that $\operatorname{ord}_a \hat{H}_j = m_l$, $^\forall j\in S_l\setminus S_{l-1}$. For every S_l we take a subset $T_l\subset S_l$ such that $|T_l| = \operatorname{rk}(T_l) = \operatorname{rk}(S_l)$ and $T_l\supset T_{l-1}$. It follows that $|T_l\setminus T_{l-1}| = \operatorname{rk}(S_l) - \operatorname{rk}(S_{l-1})$. Put $m_l^* = m_l - n$. By making use of Theorem 4.1.10 (iv) one gets

$$\sum_{j\in Q}\omega(j)(\operatorname{ord}_a \hat{H}_j\circ f - n)^+ = \sum_{j\in S}\omega(j)(\operatorname{ord}_a \hat{H}_j\circ f - n) = \sum_{l=1}^{t}\sum_{j\in S_l\setminus S_{l-1}}\omega(j)m_l^* \tag{4.2.8}$$

(continued)

$$
\begin{aligned}
&= (m_1^* - m_2^*) \sum_{j \in S_1} \omega(j) + (m_2^* - m_3^*) \sum_{j \in S_2} \omega(j) + \cdots + m_t^* \sum_{j \in S_t} \omega(j) \\
&\leqq (m_1^* - m_2^*) \operatorname{rk}(S_1) + (m_2^* - m_3^*) \operatorname{rk}(S_2) + \cdots + m_t^* \operatorname{rk}(S_t) \\
&= \operatorname{rk}(S_1) m_1^* + \big(\operatorname{rk}(S_2) - \operatorname{rk}(S_1)\big) m_2^* + \cdots + \big(\operatorname{rk}(S_t) - \operatorname{rk}(S_{t-1})\big) m_t^* \\
&= |T_1| m_1^* + |T_2 \setminus T_1| m_2^* + \cdots + |T_t \setminus T_{t-1}| m_t^*.
\end{aligned}
$$

Now put $T_t = \{j_0, \ldots, j_k\}$. The order of zero of W at a does not change by a linear transformation of $f_0, \ldots, f_n$. Thus we may assume that $f_0 = \hat{H}_{j_0} \circ f, \ldots, f_k = \hat{H}_{j_k} \circ f$. By an easy computation of the Wronskian we see

$$
\operatorname{ord}_a W \geqq |T_1| m_1^* + |T_2 \setminus T_1| m_2^* + \cdots + |T_t \setminus T_{t-1}| m_t^*.
$$

Therefore (4.2.7) follows from (4.2.8). □

Lemma 4.2.9 *Let $g \not\equiv 0$ be a meromorphic function on* $\mathbf{C}$. *For $k \geqq 1$ the following hold*:

(i) $m(r, (\frac{d^k g}{dz^k})/g) = S(r, g)$. (*Cf.* (1.2.4) *for the notation* $S(r, g)$.)

(ii) $T(r, \frac{d^k g}{dz^k}) \leqq (k+1) T(r, g) + S(r, g)$.

Proof We use an induction on k. When $k = 1$, (i) was already proved by Lemma 1.2.2 and (ii) is deduced as

$$
\begin{aligned}
T(r, g') &= N\big(r, (g')_\infty\big) + m(r, g') \\
&\leqq 2N\big(r, (g)_\infty\big) + m(r, g) + m\big(r, g'/g\big) \\
&\leqq 2T(r, g) + S(r, g).
\end{aligned}
$$

Assume (i) and (ii) for the case $k - 1$. Then one calculates that

$$
\begin{aligned}
m\left(r, \left(\frac{d^k g}{dz^k}\right) \Big/ g\right) &\leqq m\left(r, \left(\frac{d^{k-1} g}{dz^{k-1}}\right) \Big/ g\right) \\
&\quad + m\left(r, \left(\frac{d^k g}{dz^k}\right) \Big/ \left(\frac{d^{k-1} g}{dz^{k-1}}\right)\right) \\
&= S(r, g) + S\left(r, \frac{d^{k-1} g}{dz^{k-1}}\right) \\
&= S(r, g).
\end{aligned}
$$

Thus (i) is inferred. By making use of $N(r, (\frac{d^k g}{dz^k})_\infty) \leqq (k+1) N(r, (g)_\infty)$, (ii) is deduced as in the case of $k = 1$. □

Lemma 4.2.10 *For a subset $R \subset Q$ with $|R| = n + 1$ we have*

$$
m\big(r, \Delta(\hat{H}_j \circ f, j \in R)\big) = S_f(r).
$$

Proof Let R consist of $j_0 < \cdots < j_n$. Without loss of generality we may assume that $\hat{H}_{j_0} \circ f \not\equiv 0$. It follows from (4.2.2) that

$$\Delta(\hat{H}_{j_0} \circ f, \ldots, \hat{H}_{j_n} \circ f) = \Delta\left(1, \frac{\hat{H}_{j_0} \circ f}{\hat{H}_{j_0} \circ f}, \ldots, \frac{\hat{H}_{j_n} \circ f}{\hat{H}_{j_0} \circ f}\right).$$

The required inequality follows from Theorem 2.5.13, Lemma 4.2.9 and (1.1.11). □

Theorem 4.2.11 (The Cartan–Nochka Second Main Theorem, Cartan [33], Nochka [83]) *Let $L \to \mathbf{P}^N(\mathbf{C})$ be the hyperplane bundle. Let $f : \mathbf{C} \to \mathbf{P}^N(\mathbf{C})$ be an entire curve such that the smallest linear subspace containing the image $f(\mathbf{C})$ is $\mathbf{P}^n(\mathbf{C}) \subset \mathbf{P}^N(\mathbf{C})$. Let $H_j \not\supset \mathbf{P}^n(\mathbf{C})$, $1 \leqq j \leqq q$, be hyperplanes of $\mathbf{P}^N(\mathbf{C})$ in general position. Then we have the following estimate*:

$$(q - 2N + n - 1)T_f(r, L) \leqq \sum_{j=1}^{q} N_n\big(r, f^* H_j\big) + S_f(r).$$

Proof We consider f as a linearly non-degenerate entire curve $f : \mathbf{C} \to \mathbf{P}^n(\mathbf{C})$ and rewrite H_j for $H_j \cap \mathbf{P}^n(\mathbf{C})$. Put $Q = \{1, \ldots, q\}$. Then $\{H_j\}_{j \in Q}$ is in N-subgeneral position. We may assume that $q - 2N + n - 1 > 0$.

It follows from Lemmas 4.2.3, 4.2.6 and 1.1.5 (Jensen's formula) that

$$\begin{aligned}
(4.2.12)\quad & \tilde{\omega}(q - 2N + n - 1)T_f(r, L) \\
& \leqq \sum_{j=1}^{q} \omega(j) N_n\big(r, f^* H_j\big) \\
& \quad + \frac{1}{2\pi} \int_{|z|=r} \log\Bigg(\sum_{R \subset Q, |R| = n+1} \big|\Delta\big((\hat{H}_j \circ f, j \in R)\big)\big| \Bigg) d\theta + O(1) \\
& \leqq \tilde{\omega} \sum_{j=1}^{q} N_n\big(r, f^* H_j\big) \\
& \quad + \frac{1}{2\pi} \int_{|z|=r} \log\Bigg(\sum_{R \subset Q, |R| = n+1} \big|\Delta\big((\hat{H}_j \circ f, j \in R)\big)\big| \Bigg) d\theta + O(1).
\end{aligned}$$

Therefore

$$\begin{aligned}
(4.2.13)\quad & (q - 2N + n - 1)T_f(r, L) \\
& \leqq \sum_{j=1}^{q} N_n\big(r, f^* H_j\big) \\
& \quad + \frac{1}{2\pi\tilde{\omega}} \int_{|z|=r} \log\Bigg(\sum_{R \subset Q, |R| = n+1} \big|\Delta\big((\hat{H}_j \circ f, j \in R)\big)\big| \Bigg) d\theta + O(1).
\end{aligned}$$

Using Lemma 4.2.10, we compute

$$\begin{aligned}
&\frac{1}{2\pi\tilde{\omega}}\int_{|z|=r}\log\Bigg(\sum_{R\subset Q,|R|=n+1}\left|\Delta\big((\hat{H}_j\circ f, j\in R)\big)\right|\Bigg)d\theta\\
&\leqq \frac{1}{\tilde{\omega}}\Bigg(\sum_{R\subset Q,|R|=n+1}\frac{1}{2\pi}\int_{|z|=r}\log^+\left|\Delta\big((\hat{H}_j\circ f, j\in R)\big)\right|d\theta\Bigg)+O(1)\\
&= S_f(r).
\end{aligned}$$

This with (4.2.13) implies the required estimate. □

We assume the condition given in Theorem 4.2.11. For a hyperplane $H\subset \mathbf{P}^N(\mathbf{C})$ with $H\not\supset f(\mathbf{C})$ we define *k-defect* ($k\in\mathbf{N}\cup\{\infty\}$) of f by

$$\delta_k(f,H)=1-\varlimsup_{r\to\infty}\frac{N_k(r,f^*H)}{T_f(r,L)},$$

$$\delta(f,H)=\delta_\infty(f,H).$$

Corollary 4.2.14 *Under the same condition as in Theorem 4.2.11 the following hold*:

(i) (Defect Relation) $\sum_{j=1}^q\delta_n(f,H_j)\leqq 2N-n+1$.

(ii) (Ramification Theorem) *If f is completely μ_j-ramified over all H_j* $(1\leqq j\leqq q)$,

$$\sum_{j=1}^q\left(1-\frac{n}{\mu_j}\right)\leqq 2N-n+1.$$

Proof (i) This is deduced from Theorem 4.2.11 and

$$\varlimsup_{r\to\infty}S_f(r)/T_f(r,L)=0.$$

(ii) If $\mu_j<\infty$,

$$\begin{aligned}
\varlimsup_{r\to\infty}\frac{N_n(r,f^*H_j)}{T_f(r,L)}&\leqq n\varlimsup_{r\to\infty}\frac{N_1(r,f^*H_j)}{T_f(r,L)}\\
&\leqq n\varlimsup_{r\to\infty}\frac{N_1(r,f^*H_j)}{N(r,f^*H_j)}\leqq\frac{n}{\mu_j}.
\end{aligned}$$

If $\mu_j=\infty$, the above equation is trivial. The assertion follows from this and Theorem 4.2.11. □

The above result in the case of $n=N$ was first proved by H. Cartan. It is fundamental and very useful for applications, so that we formulate it as follows.

Corollary 4.2.15 (Cartan [33]) *Let $L \to \mathbf{P}^n(\mathbf{C})$ be the hyperplane bundle, let $f : \mathbf{C} \to \mathbf{P}^n(\mathbf{C})$ be a linearly non-degenerate entire curve, and let $\{H_j\}_{j=1}^q$ be a family of hyperplanes of $\mathbf{P}^n(\mathbf{C})$ in general position. Then the following hold*:

(i) $(q-n-1)T_f(r,L) \leqq \sum_{j=1}^q N_n(r, f^*H_j) + S_f(r)$.
(ii) $\sum_{j=1}^q \delta_n(f, H_j) \leqq n+1$.
(iii) *If f is completely μ_j-ramified over H_j $(1 \leqq j \leqq q)$,*

$$\sum_{j=1}^{q} \left(1 - \frac{n}{\mu_j}\right) \leqq n+1.$$

Theorem 4.2.16 (Generalized Borel's Theorem) *Let $F_j \not\equiv 0$, $1 \leqq j \leqq n$, be entire functions on $\mathbf{C}$, and let $d \in \mathbf{N}$. Assume that*

$$F_1^d + \cdots + F_n^d = 0. \tag{4.2.17}$$

If $d > n(n-2)$, there is a partition of indices, $\{1, \ldots, n\} = \cup I_\alpha$ such that

(i) *the cardinality $|I_\alpha| \geqq 2$ for every I_α,*
(ii) *$F_i/F_j = c_{ij} \in \mathbf{C}$ for all $i, j \in I_\alpha$,*
(iii) *$\sum_{i \in I_\alpha} F_i^d = 0$.*

Proof We use an induction on n. If $n = 2$, the assertion is trivial. Assume that $n \geqq 3$, and that the case of entire functions of the number less than n is proved. Let $[w_1, \ldots, w_{n-1}]$ be a homogeneous coordinate system of $\mathbf{P}^{n-2}(\mathbf{C})$, and define an entire curve by

$$f : z \in \mathbf{C} \to \left[F_1^d(z), \ldots, F_{n-1}^d(z)\right] \in \mathbf{P}^{n-2}(\mathbf{C}).$$

Suppose that this is linearly non-degenerate. We take the following n hyperplanes in general position:

$$H_j = \{w_j = 0\}, \quad 1 \leqq j \leqq n-1,$$
$$H_n = \{w_0 + \cdots + w_{n-1} = 0\}.$$

It is clear that $\mathrm{ord}_z f^*H_j \geqq d$ for all $z \in f^{-1}H_j$. By Corollary 4.2.15 (iii)

$$\sum_{j=1}^{n} \left(1 - \frac{n-2}{d}\right) \leqq n-1.$$

This yields a contradiction, $d \leqq n(n-2)$. Hence f is linearly degenerate. There is a non-trivial linear relation,

$$c_1 F_1^d + \cdots + c_{n-1} F_{n-1}^d = 0, \quad c_l' \neq 0.$$

Changing indices, we may assume that $c_1 = 1$. It follows from (4.2.17) that

$$(1 - c_2)F_2^d + \cdots + (1 - c_{n-1})F_{n-1}^d + F_n^d = 0.$$

The induction hypothesis implies that the assertion of the present theorem for F_j with indices $1 - c_j \neq 0$, $2 \leqq j \leqq n$ holds. Combining this with (4.2.17), we see that there are entire functions F_{i_l} of number less than or equal to $(n-1)/2$ satisfying a linear relation

$$F_1^d + \sum_l c_l' F_{i_l}^d = 0.$$

By making use of the induction hypothesis again, the assertion is deduced. □

Corollary 4.2.18 (Borel's Theorem) *Let G_j, $1 \leqq j \leqq n$, be entire functions that are nowhere vanishing. Assume that*

$$G_1 + \cdots + G_n = 0.$$

Then the conclusion in Theorem 4.2.16 holds.

The proof is immediate by setting $F_j = G_j^{1/d}$ and $d > n(n-2)$.

Historical Supplement H. Cartan's Theorem as given in Corollary 4.2.15 was announced in Cartan [29b] without proof, and the details were published in Cartan [33]. In Ahlfors [82], Vol. 1, p. 363, the author comments that he feels sorry for publishing the paper Ahlfors [41] without noticing the reprint of Cartan [33] in his file box.

Corollary 4.2.18 is due to E. Borel [1897]. The aim was to prove Picard's Theorem in elementary way without the use of the modular function. The relation is given by the identity,

$$f(z) + \big(1 - f(z)\big) + (-1) = 0,$$

where $f(z)$ is assumed to be an entire function without taking values 0 and 1.

4.3 Entire Curves Omitting Hyperplanes

In this section we consider entire curves in $\mathbf{P}^n(\mathbf{C})$ omitting hyperplanes.

We begin with an easy consequence of Borel's Theorem (Corollary 4.2.18).

Lemma 4.3.1 *Let H_j, $1 \leqq j \leqq n+2$, be distinct hyperplanes of $\mathbf{P}^n(\mathbf{C})$. If an entire curve $f : \mathbf{C} \to \mathbf{P}^n(\mathbf{C})$ omits all H_j, then f is linearly degenerate.*

Proof We denote by F_j $(1 \leqq j \leqq n+2)$ the linear forms defining H_j. We may assume that F_j, $1 \leqq j \leqq h$, form a maximal linearly independent system of $\{F_j\}_{j=1}^{n+2}$. Then $2 \leqq h \leqq n+1$, and for F_{n+2} there is a linear relation

$$\sum_{j=1}^{h} c_j F_j + c_{n+2} F_{n+2} = 0,$$

where $c_{n+2} \neq 0$ and at least two $c_j \neq 0$ $(1 \leqq j \leqq h)$. Re-ordering the indices, we may suppose that

$$\sum_{j=1}^{h'} c_j F_j + c_{n+2} F_{n+2} = 0,$$

where $h' \geqq 2$ and $c_j \neq 0$, $1 \leqq j \leqq h'$. It follows from Corollary 4.2.18 (Borel's Theorem) that $F_j \circ f$, $1 \leqq j \leqq h'$, must be linearly dependent. Thus, f is linearly degenerate. □

Theorem 4.3.2 (Fujimoto [72b]; Green [72]) *Let $\{H_i\}_{i=1}^{l}$ be a family of hyperplanes of $\mathbf{P}^n(\mathbf{C})$ in general position, and assume $l > n$. Let $f : \mathbf{C} \to \mathbf{P}^n(\mathbf{C})$ be an entire curve such that either $f(\mathbf{C}) \cap H_i = \emptyset$ or $f(\mathbf{C}) \subset H_i$. Let W denote the smallest linear subspace containing the image $f(\mathbf{C})$. Then we have*

$$\dim W \leqq \frac{n}{l-n}.$$

Proof The case of $l = n+1$ is trivial. Assume that $l \geqq n+2$.

Put $k = \dim W$. Note that $f : \mathbf{C} \to W$ $(\cong \mathbf{P}^k(\mathbf{C}))$ is linearly non-degenerate. Let the distinct hyperplane cuts of $H_i \cap W \neq W$ be exactly $H_1 \cap W, \ldots, H_{l'} \cap W$. For every $1 \leqq j \leqq l'$ the number of hyperplanes H_i $(1 \leqq i \leqq l)$ with $H_i \cap W = H_j \cap W$ is denoted by s_j. Changing the indices, we may assume that

$$s_1 \geqq \cdots \geqq s_{l'}. \tag{4.3.3}$$

Moreover, let s_0 be the number of H_i such that $H_i \supset f(\mathbf{C})$. Then the intersection of those s_0 hyperplanes H_i contains W.

If $l' \leqq k$, the ampleness of H_i implies that $W \cap \bigcap_{i=1}^{l'} H_i \neq \emptyset$. Therefore, $W \cap \bigcap_{i=1}^{l} H_i \neq \emptyset$. Since $\{H_i\}$ is in general position, $l \leqq n$; this contradicts the assumption for l.

We see that $l' > k$. From Lemma 4.3.1 we get

$$l' = k+1. \tag{4.3.4}$$

It follows from (4.3.3) that

$$\frac{1}{l'} \sum_{j=1}^{l'} s_j \leqq \frac{1}{k} \sum_{j=1}^{k} s_j.$$

Hence $\sum_{j=1}^{l'} s_j \leqq \frac{l'}{k} \sum_{j=1}^{k} s_j$. As $s_0 \leqq \frac{l'}{k} s_0$ and $\bigcap_{j=1}^{s_0} H_j \cap \bigcap_{j=1}^{k} H_j \cap W \neq \emptyset$, one sees that

$$l = \sum_{j=0}^{l'} s_j \leqq \frac{l'}{k} \sum_{j=0}^{k} s_j \leqq \frac{l'n}{k} = \frac{k+1}{k} n.$$

Thus the claimed inequality is obtained. □

4.4 Generalizations and Applications

4.4.1 Derived Curves

For a given entire curve $f = [f_0, \ldots, f_N] : \mathbf{C} \to \mathbf{P}^N(\mathbf{C})$ we set $\tilde{f} = (f_0, \ldots, f_N) : \mathbf{C} \to \mathbf{C}^{N+1} \setminus \{0\}$. Let $\mathrm{Gr}(k+1, N+1)$ denote the Grassmann manifold of $k+1$-dimensional vector subspaces of $\mathbf{C}^{N+1}$. Taking the derivatives $\tilde{f}^{(k)} = (\frac{d^k}{dz^k} f_0, \ldots, \frac{d^k}{dz^k} f_N)$ of $\tilde{f}$ ($\tilde{f}^{(0)} = \tilde{f}$ for convention), we consider the following entire curve:

$$f^{(k)} = \left[\tilde{f} \wedge \tilde{f}^{(1)} \wedge \cdots \wedge \tilde{f}^{(k)}\right] : \mathbf{C} \to \mathrm{Gr}(k+1, N+1) \hookrightarrow \mathbf{P}^{\binom{N+1}{k+1}-1}(\mathbf{C}).$$

This is called the k-th *derived curve* of f. It is assumed that the above definition makes sense. If f is linearly non-degenerate, $f^{(k)}$, $0 \leqq k \leqq N-1$, are non-constant, but $f^{(k)}$ is not necessarily linearly non-degenerate as an entire curve in $\mathbf{P}^{\binom{N+1}{k+1}-1}(\mathbf{C})$ with embedding $\mathrm{Gr}(k+1, N+1) \subset \mathbf{P}^{\binom{N+1}{k+1}-1}(\mathbf{C})$.

Example 4.4.1 (H. Fujimoto) Let a_i, $1 \leqq i \leqq 4$, be distinct non-zero constants such that $a_1 + a_2 = a_3 + a_4$. The entire curve defined by

$$f : z \in \mathbf{C} \to \left[e^{a_1 z}, e^{a_2 z}, e^{a_3 z}, e^{a_4 z}\right] \in \mathbf{P}^3(\mathbf{C})$$

is linearly non-degenerate. The first derived curve is given by

$$f^{(1)} = \left[\ldots, (a_j - a_i) e^{(a_i + a_j) z}, \ldots\right]_{i<j} \in \mathbf{P}^{\binom{4}{2}}(\mathbf{C}),$$

which is actually linearly degenerate.

A hyperplane of $\mathbf{P}^{\binom{N+1}{k+1}-1}(\mathbf{C})$ is said to be *decomposable* if the defining linear form is written as $v_0 \wedge \cdots \wedge v_k$ with elements $v_j \in (\mathbf{C}^{N+1})^\vee$, $0 \leqq j \leqq k$, of the dual space of $\mathbf{C}^{N+1}$.

H. Weyl and his son Joachim [38], [43] initiated a value distribution theory to deal with all entire curves $f^{(k)}$, $0 \leqq k \leqq N-1$. The part of the Second Main Theorem was proved by L. Ahlfors [41], and the theory was completed (the Weyl–Ahlfors theory). In this theory, the estimate of H. Cartan's Second Main Theorem is split into

several steps of estimates of the k-th derived curves ($0 \leqq k \leqq N-1$) with the aid of the lemma on logarithmic derivatives. By summing up all the estimates, it yields the same Second Main Theorem as H. Cartan's.

Assume that f is linearly non-degenerate. Let E_j be decomposable hyperplanes of $\mathbf{P}^{\binom{N+1}{k+1}-1}(\mathbf{C})$ in general position. Then the defect relation for $f^{(k)}$ is written as

$$\sum_j \delta\big(f^{(k)}, E_j\big) \leqq \binom{N+1}{k+1}. \tag{4.4.2}$$

H. Fujimoto [82a] and [93] proved the Second Main Theorem with truncated counting functions for $f^{(k)}$, provided that f is linearly non-degenerate; the defect relation is

$$\sum_j \delta_{(k+1)(N-k)}\big(f^{(k)}, E_j\big) \leqq \binom{N+1}{k+1}. \tag{4.4.3}$$

W. Chen [90] dealt with the linearly degenerate case by using Nochka's weights. Let $\mathbf{P}^n(\mathbf{C})$ be the smallest linear subspace containing the image $f(\mathbf{C})$. Then

$$\sum_j \delta\big(f^{(k)}, E_j\big) \leqq 2\binom{N+1}{k+1} - \binom{n+1}{k+1}, \quad 0 \leqq k < n. \tag{4.4.4}$$

4.4.2 Generalization to Higher Dimensional Domains

W. Stoll [53b], [54] generalized the domain space of the Weyl–Ahlfors theory to general dimensional $\mathbf{C}^m$ and moreover to so-called parabolic manifolds. Thus he obtained the Second Main Theorem and defect relation (4.4.2) for linearly non-degenerate meromorphic mapping $f : \mathbf{C}^m \to \mathbf{P}^n(\mathbf{C})$ and its derived mappings $f^{(k)}$.

A.L. Vitter [77] proved Lemma 3.1.1, and gave another proof of Stoll's Second Main Theorem when the domain is $\mathbf{C}^m$ by Cartan's method.

4.4.3 Finite Ramified Covering Spaces

Let $\pi : X \to \mathbf{C}^m$ be a finite ramified covering. By definition X is a normal complex space and π is a finite proper holomorphic mapping. Let R denote the ramification divisor and let p be the sheet number (covering number or degree). We follow after the notation in Sect. 2.6. For a meromorphic mapping $f : X \to \mathbf{P}^n(\mathbf{C})$ we take a representation $f(z) = [f_0(z), \ldots, f_n(z)]$, which may not be able to be a reduced representation. The Wronskian $W(f_0, \ldots, f_n)$ is defined as in Noguchi [97] outside R, and is extended meromorphically over X. Let $\{H_j\}_{j=1}^q$ be in N-subgeneral

position. By taking account of the order of partial differentiations we have

$$\big(W(f_0,\dots,f_n)\big)+\frac{n(n+1)}{2}R\geqq 0. \tag{4.4.5}$$

Where (4.2.12) is deduced from Lemmas 4.2.3 and 4.2.6, we compute the divisor $(W(f_0,\dots,f_n))$ and get

$$\begin{aligned}\tilde{\omega}(q-2N+n-1)T_f(r,L)\leqq\tilde{\omega}\sum_{j=1}^{q}N_n\big(r,f^*H_j\big)\\ +\frac{n(n+1)}{2}N(r,R)+S_f(r).\end{aligned} \tag{4.4.6}$$

By Theorem 2.7.7 (i)

$$\frac{n(n+1)}{2}N(r,R)\leqq n(n+1)(p-1)T_f(r,L)+O(1). \tag{4.4.7}$$

Similarly to Theorem 4.2.11 we obtain the next one from (4.4.6), (4.4.7) and Theorem 4.1.10 (iii).

Theorem 4.4.8 *Let $f:X\to\mathbf{P}^n(\mathbf{C})$ be a linearly non-degenerate meromorphic mapping. Let $\{H_j\}_{j=1}^q$ be a family of hyperplanes of $\mathbf{P}^n(\mathbf{C})$ in N-subgeneral position. Then*

$$\begin{aligned}&\big(q-2N+n-1-(p-1)n(2N-n+1)\big)T_f(r,L)\\ &\leqq\sum_{j=1}^{q}N_n\big(r,f^*H_j\big)+S_f(r).\end{aligned}$$

4.4.4 The Eremenko–Sodin Second Main Theorem

Let L be the hyperplane bundle over $\mathbf{P}^n(\mathbf{C})$ and take hypersurfaces $D_j\in|L^{d_j}|$ of degree d_j $(1\leqq j\leqq q)$. To obtain the following theorem Eremenko–Sodin [92] employed a potential theoretic method, which is a totally different method to what is presented in this text.

Theorem 4.4.9 *Assume that the intersections of arbitrary $n+1$ divisors among D_j, $1\leqq j\leqq q$, are empty. Then for an entire curve $f:\mathbf{C}\to\mathbf{P}^n(\mathbf{C})$ we have*

$$(q-2n)T_f(r,L)\leqq\sum_{j=1}^{q}\frac{1}{d_j}N\big(r,f^*D_j\big)+o\big(T_f(r,L)\big)\big\|_E.$$

It is an interesting problem if the counting functions on the right hand side can be replaced by truncated ones.

4.4.5 The Second Main Theorem of Corvaja–Zannier, Evertse–Ferretti and Ru

It is noted that there is no non-degeneracy condition for f in Theorem 4.4.9. For algebraically non-degenerate entire curves into a projective algebraic variety of dimension n, Min Ru [04], [09] obtained the following Second Main Theorem without truncations:

Theorem 4.4.10 *Let $V \subset \mathbf{P}^n(\mathbf{C})$ be an embedded projective algebraic manifold of dimension n. Let L be the line bundle of hyperplane sections over V. Let D_j $(1 \leqq j \leqq q)$ be distinct hypersurface sections of degree d_j with V such that the intersections of arbitrary distinct $n+1$ D_j's are empty. Let $f : \mathbf{C} \to V$ be an algebraically non-degenerate entire curve. Then we have*

$$(q-n-1)T_f(r,L) \leqq \sum_{j=1}^{q} \frac{1}{d_j} N\big(r, f^* D_j\big) + o\big(T_f(r,L)\big)\big\|_E.$$

The proof is very different to Eremenko–Sodin [92], and is based on H. Cartan's method combined with some new method of Diophantine approximation due to Corvaja–Zannier [04b], and Evertse–Ferretti [02], [08].

This Second Main Theorem is less geometric, because the dimension n is the only geometric information in the statement.

4.4.6 Krutin's Theorem

As an extension of Theorem 1.2.15 (i) in the case of $\alpha > 1/3$ there is a result due to Krutin'.

Theorem 4.4.11 (Krutin' [79]) *Let $f : \mathbf{C} \to \mathbf{P}^n(\mathbf{C})$ be a linearly non-degenerate entire curve. Let $\{H_j\}$ be an infinite family of hyperplanes of $\mathbf{P}^n(\mathbf{C})$ in general position. Assume that the lower order of f is finite. Then for $\alpha > 1/3$*

$$\sum_j \delta(f, H_j)^\alpha < \infty.$$

It is not known if all of Theorem 1.2.15 holds for $f : \mathbf{C} \to \mathbf{P}^n(\mathbf{C})$.

4.4.7 Moving Targets

By a "moving target" or "moving hyperplane" we mean a hyperplane defined by (4.2.1) with the coefficients h_{jk}, not necessarily constants in $\mathbf{C}$, but meromorphic

functions $a_{jk}(z)$ on $\mathbf{C}$, which have orders small with respect to a given entire curve $f : \mathbf{C} \to \mathbf{P}^n(\mathbf{C})$; i.e., $T(r, a_{jk}) = o(T_f(r, L))\|_E$. The term "moving target" is due to W. Stoll, but such a study can go back at least to E. Borel [1897].

When $n = 1$, R. Nevanlinna [29] made a conjecture that the defect relation as in Theorem 1.2.11 would hold for these moving targets. This conjecture was proved by Steinmetz [85]. In the general case of $n \geqq 1$ Corollary 4.2.14 (i) was extended to moving hyperplanes by Ru–Stoll [91]. M. Shirosaki [91] gave another simple proof for it by Cartan's method when f is linearly non-degenerate over "small functions".

4.4.8 Yamanoi's Second Main Theorem

K. Yamanoi [04d], [05], [06] proved the Second Main Theorem with counting functions truncated to level one for moving targets, of which proof is very innovative. It is considered to be the "*abc Theorem*" for the field of meromorphic functions.

Theorem 4.4.12 (Yamanoi [04d], [05], [06]) *Let $p : X \to S$ be a surjective morphism between smooth projective algebraic varieties with relative canonical bundle $K_{X/S}$. Assume that*

(i) $\dim X/S = 1$;
(ii) *$D \subset X$ is a reduced divisor*;
(iii) *$f : \mathbf{C} \to X$ is algebraically nondegenerate*;
(iv) $g = p \circ f : \mathbf{C} \to S$.

Then for every $\varepsilon > 0$, there exists $C(\varepsilon) > 0$ such that

$$T_f\big(r; L(D)\big) + T_f(r; K_{X/S}) \leq N_1\big(r; f^*D\big) + \varepsilon T_f(r) + C(\varepsilon)T_g(r)\|_{E(\varepsilon)}. \tag{4.4.13}$$

His method consists of

(i) Ahlfors' covering theory;
(ii) Mumford's theory of the compactification of curve moduli;
(iii) the tree theory for point configurations.

4.4.9 Applications

The Second Main Theorem obtained in the present section has been applied to meromorphic mappings $f : \mathbf{C}^m \to \mathbf{P}^n(\mathbf{C})$ for the unicity and finiteness problems, and a number of interesting results were obtained. See H. Fujimoto [75], [88b], [00] and the references in [93], and also Y. Aihara [91], [98], [03].

As another interesting application there is an upper-bound for the level structure of abelian varieties defined over function fields (Nadel [89a]; Noguchi [91]).

The result is related to the value distribution of meromorphic mappings into the compactifications of the quotients of symmetric bounded domains by discrete subgroups (Aihara–Noguchi [91]).

For applications to the Kobayashi hyperbolicity problem, see Chap. 7.

4.5 Logarithmic Forms

Let M be an m-dimensional complex manifold and let D be a reduced divisor on it. By $\mathscr{O}_M$ we denote the sheaf of germs of holomorphic functions over M and let $\mathscr{O}_M(U) = \Gamma(U, \mathscr{O}_M)$ the set of all sections of $\mathscr{O}_M$ over a subset $U \subset M$.

Let Ω^k_M denote the sheaf of germs of holomorphic k-forms over M. In a neighborhood of an arbitrary point $x_0 \in M$ the divisor D is assumed to be defined by a local equation

$$\sigma_1 \cdots \sigma_s = 0,$$

where every σ_j is irreducible; it is called a *local reduced defining equation*. In this case we define locally a sheaf of germs of meromorphic differential forms by

$$\Omega^1_M(\log D) = \sum_{j=1}^{s} \mathscr{O}_M \frac{d\sigma_j}{\sigma_j} + \Omega^1_M, \tag{4.5.1}$$

which is called the sheaf of germs of *logarithmic* 1*-forms* with poles at most on D. If D is of normal crossing type, $\Omega^1_M(\log D)$ is locally free and gives rise to the sheaf $\mathscr{O}(W)$ of germs of holomorphic sections of a holomorphic vector bundle W over M. The sheaf of germs of logarithmic k-forms is defined by

$$\Omega^k_M(\log D) = \bigwedge^k \Omega^1_M(\log D).$$

In particular, $\Omega^m_M(\log D)$ is called the logarithmic canonical sheaf along D. If D is of normal crossing type, $\Omega^m_M(\log D) = \mathscr{O}(K_M \otimes L(D)^{-1})$.

Theorem 4.5.2 *Let M and N be complex manifolds and let D (resp. E) be a reduced divisor on M (resp. N). Let $f : M \to N$ be a meromorphic mapping such that $f^{-1}E \subset D$. If f is either holomorphic or D is of normal crossing type, then $f^*\phi \in H^0(M, \Omega^k_M(\log D))$ for every $\phi \in H^0(N, \Omega^k_N(\log E))$.*

Proof For the sake of simplicity we let $k = 1$; the case of $k \geqq 2$ is similar. First we assume that f is holomorphic. Let $x_0 \in M$ and $y_0 = f(x_0) \in N$. If $y_0 \notin E$, then $f^*\phi$ is a holomorphic differential form in a neighborhood of x_0. In the case of $y_0 \in E$ we take a local reduced defining equation of E about y_0

$$\psi_1 \cdots \psi_s = 0.$$

Let the equation of D about x_0 be

$$\sigma_1 \cdots \sigma_t = 0.$$

Because $f^{-1}E \subset D$, we may write $f^*\psi_j = a_j(x)(\sigma_1(x))^{\nu_{j1}} \cdots (\sigma_t(x))^{\nu_{jt}}$, $a_j \in \mathscr{O}^*_{M,x_0}$, where $\mathscr{O}^*_{M,x_0}$ denotes the set of all invertible elements of $\mathscr{O}_{M,x_0}$. This leads to

$$f^*\frac{d\psi_j}{\psi_j} = \sum_{k=1}^{t} \nu_{jk}\frac{d\sigma_k}{\sigma_k} + \frac{da_j}{a_j} \in \Omega^1_M(\log D).$$

Therefore $f^*\phi \in H^0(M, \Omega^1_M(\log D))$.

Assume next that f is meromorphic and D is of normal crossing type. Let $I(f)$ be the set of indeterminant points of f and let $x_0 \in I(f)$. In a small neighborhood U of x_0, there is a holomorphic local coordinate system $(x_1, \ldots, x_m)$ such that

$$D \cap U = \{x_1 \cdots x_l = 0\}.$$

By the preceding argument, $(f|_{M\setminus I(f)})^*\phi \in H^0(M \setminus I(f), \Omega^1_M(\log D))$. Hence,

$$\big(f|_{U\setminus I(f)}\big)^*\phi = \sum_{j=1}^{l} a_j\frac{dx_j}{x_j} + \sum_{j=l+1}^{m} a_j dx_j, \quad a_j \in \mathscr{O}_M\big(U \setminus I(f)\big).$$

Since $\operatorname{codim} I(f) \geqq 2$, Theorem 2.2.6 implies that a_j are uniquely holomorphically extended over U. Thus $(f|_{M\setminus I(f)})^*\phi$ uniquely extends to an element of $H^0(M, \Omega^1_M(\log D))$. □

Let D be a reduced divisor on a complex manifold M. According to the Hironaka resolution of singularities (Hironaka [64a], [64b], see also Bierstone–Milman [97]), there are a complex manifold $\tilde{M}$, a reduced divisor $\tilde{D}$ of normal crossing type and a proper holomorphic mapping $f : \tilde{M} \to M$ such that

(i) $f^{-1}D = \tilde{D}$,
(ii) $f|_{\tilde{M}\setminus\tilde{D}} : \tilde{M} \setminus \tilde{D} \to M \setminus D$ is biholomorphic.

Corollary 4.5.3 *Let M, $\tilde{M}$, D, $\tilde{D}$ and f be as above. Then*

$$f^*H^0\big(M, \Omega^k_M(\log D)\big) = H^0\big(\tilde{M}, \Omega^k_{\tilde{M}}(\log \tilde{D})\big).$$

Proof Due to Theorem 4.5.2 we have

$$f^*H^0\big(M, \Omega^k_M(\log D)\big) \subset H^0\big(\tilde{M}, \Omega^k_{\tilde{M}}(\log \tilde{D})\big).$$

On the other hand, since f is a proper map which restricts to a biholomorphism between $\tilde{M} \setminus \tilde{D}$ and $M \setminus D$, there is an "inverse" f^{-1} which is a meromorphic map.

Since $\tilde{D}$ is of normal crossing type, we may apply Theorem 4.5.2 for f^{-1} as well. This yields

$$f^* H^0\big(M, \Omega_M^k(\log D)\big) \supset H^0\big(\tilde{M}, \Omega_{\tilde{M}}^k(\log \tilde{D})\big). \qquad \square$$

In the compact Kähler case the following facts are known (A. Fujiki [78a]; Deligne [71]; Noguchi [95]).

Theorem 4.5.4 *Let M and $\tilde{M}$ be as above. Assume in addition that M is compact and Kähler. Then we have the following properties*:

(i) *$\tilde{M}$ is also compact and Kähler. If M is projective algebraic, so is $\tilde{M}$.*
(ii) *Every element of $H^0(M, \Omega_M^k(\log D))$ is d-closed.*
(iii) (Homology groups) *There are natural isomorphisms induced by integration of 1-forms over paths as follows*:

$$H^0\big(M, \Omega_M^1\big) \oplus \overline{H^0\big(M, \Omega_M^1\big)} \cong \big(H_1(M, \mathbf{Z}) \otimes \mathbf{C}\big)^*,$$
$$H^0\big(M, \Omega_M^1(\log D)\big)/H^0\big(M, \Omega_M^1\big) \cong \big(\big(H_1(M \setminus D, \mathbf{Z})/H_1(M, \mathbf{Z})\big) \otimes \mathbf{C}\big)^*.$$

If M is compact, $\bar{q}(M \setminus D) = \dim H^0(M, \Omega_M^1(\log D))$ is called the *logarithmic irregularity* of $M \setminus D$. When $D = \emptyset$, it is called the *irregularity* of M and denoted by $q(M)$.

Let M be Kähler. Set $q = q(M)$, $\bar{q} = \bar{q}(M \setminus D)$. Let b_1 denote the first Betti number, i.e., $b_1(M) = \dim_{\mathbf{R}} H^1(M, \mathbf{R})$. Then Theorem 4.5.4 (iii) implies

$$2q = b_1(M), \qquad \bar{q} - q = b_1(M \setminus D) - b_1(M). \tag{4.5.5}$$

Theorem 4.5.4 moreover implies that the natural pairing given by path integrals embeds the free parts of $H_1(M, \mathbf{Z})$ (resp. $H_1(M \setminus D, \mathbf{Z})$) as discrete additive subgroups into the additive groups $H^0(M, \Omega_M^1)^*$ (resp. $H^0(M, \Omega_M^1(\log D))^*$). Let W denote the space of linear forms on $H^0(M, \Omega_M^1(\log D))$ which vanish on $H^0(M, \Omega_M^1)$. Then we obtain an exact sequence of vector spaces,

$$0 \to W \to H^0\big(M, \Omega_M^1(\log D)\big)^* \to H^0\big(M, \Omega_M^1\big)^* \to 0,$$

inducing an exact sequence of commutative complex Lie groups

$$\begin{aligned} 0 &\to W/\big(H_1(M \setminus D, \mathbf{Z})/H_1(M, \mathbf{Z})\big) \\ &\to H^0\big(M, \Omega_M^1(\log D)\big)^*/H_1(M \setminus D, \mathbf{Z}) \\ &\to H^0\big(M, \Omega_M^1\big)^*/H_1(M, \mathbf{Z}) \to 0. \end{aligned} \tag{4.5.6}$$

Let $t = \bar{q} - q$. Then $\dim W = t$ and $W/(H_1(M \setminus D, \mathbf{Z})/H_1(M, \mathbf{Z})) \cong (\mathbf{C}^*)^t$. The second quotient group, $H^0(M, \Omega_M^1(\log D))^*/H_1(M \setminus D, \mathbf{Z})$ is called the *quasi-Albanese variety* of $M \setminus D$ and is denoted by $A_{M \setminus D}$. The third quotient is a compact complex torus, that is, the Albanese variety A_M of M.

The above construction may be made more explicit as follows: Let

$$\gamma_j \in H_1(M \setminus D, \mathbf{Z}), \quad 1 \leqq j \leqq t,$$

be a set of generators of the free part of $H_1(M \setminus D, \mathbf{Z})/H_1(M, \mathbf{Z})$. Moreover we take

$$\gamma_j \in H_1(M, \mathbf{Z}), \quad t+1 \leqq j \leqq t+2q,$$

so that they form a basis of $H_1(M, \mathbf{Z}) \otimes \mathbf{C}$.

We take a basis ω_j, $(1 \leqq j \leqq \bar{q})$ of $H^0(M, \Omega^1_M)$ such that

$$\int_{\gamma_j} \omega_k = \delta_{jk} \text{ (Kronecker's symbol)}, \quad 1 \leqq j, k \leqq t, \tag{4.5.7}$$

$$\int_{\gamma_h} \omega_i = 0, \quad 1 \leqq h \leqq t,\ t+1 \leqq i \leqq \bar{q}. \tag{4.5.8}$$

We put the vectors

$$\eta_j = \left(\int_{\gamma_j} \omega_1, \ldots, \int_{\gamma_j} \omega_{\bar{q}} \right) \in \mathbf{C}^{\bar{q}}, \quad 1 \leqq j \leqq t+2q,$$

and set $\Gamma = \sum_j \mathbf{Z} \cdot \eta_j$. Then Γ is the image of $H_1(M \setminus D, \mathbf{Z})$ under the natural map to $\mathbf{C}^q \cong H^0(M, \Omega^1_M)^*$. Equation (4.5.6) translates into an exact sequence of the form

$$0 \to \left(\mathbf{C}^*\right)^t \to \mathbf{C}^{\bar{q}}/\Gamma \to \mathbf{C}^q/\Gamma_0 \to 0, \tag{4.5.9}$$

where Γ_0 is a co-compact lattice in $\mathbf{C}^q$. We obtain $A_{M\setminus D} \cong \mathbf{C}^{\bar{q}}/\Gamma$ and $A_M \cong \mathbf{C}^q/\Gamma_0$. We fix a point $x_0 \in M \setminus D$. Then a holomorphic mapping

$$\alpha_{M\setminus D} : x \in M \setminus D \to \left(\int_{x_0}^{x} \omega_1, \ldots, \int_{x_0}^{x} \omega_{\bar{q}} \right) \in \mathbf{C}^{\bar{q}}/\Gamma$$

is defined. We call $\alpha_{M\setminus D}$ the *quasi-Albanese map* of $M \setminus D$.

These extend the ordinary Albanese variety and Albanese map in the case of $D = 0$.

Remark 4.5.10 Strictly speaking the quasi-Albanese variety $A_{M\setminus D}$ depends not only on the complex manifold $M \setminus D$, but on the pair (M, D). However, suppose that there are reduced divisors D_1, D_2 on complex manifolds M_1, M_2, respectively such that there exists a biholomorphic map $\lambda : M_1 \setminus D_1 \to M_2 \setminus D_2$, and that λ extends meromorphically from M_1 to M_2. Then $A_{M_1\setminus D_1}$ and $A_{M_2\setminus D_2}$ are isomorphic. (This can be seen by applying Corollary 4.5.3 to a desingularization of the graph of the meromorphic map.)

In particular, if everything is algebraic, the quasi-Albanese variety depends only on the open variety $M \setminus D$.

Proposition 4.5.11 *The image of the quasi-Albanese map α generates the quasi-Albanese variety $A_{M\setminus D}$ as a group in the following sense*:

There is a natural number N such that the map $\zeta_N : (M \setminus D)^{2N} \to A_{M\setminus D}$ given by

(4.5.12)
$$(M \setminus D)^{2N} \ni (x_1, y_1, \ldots, x_N, y_N) \to (x_1 - y_1) + \cdots + (x_N - y_N) \in A_{M\setminus D}$$

is surjective.

Proof Let Z_k denote the image of ζ_k. From Proposition 4.5.13 below it can be deduced that the sets Z_k are constructible subsets of $\overline{A_{M\setminus D}}$. By definition $H = \bigcup_k Z_k$ is a connected subgroup of $A_{M\setminus D}$ and therefore a Lie subgroup. Due to the construction of the quasi-Albanese variety every invariant holomorphic 1-form on $A_{M\setminus D}$ pulls back to a non-zero holomorphic 1-form on $M \setminus D$. This implies $H = A_{M\setminus D}$. Observe that there must be a large number N such that

$$\dim Z_N = \sup_k \dim Z_k = \dim A_{M\setminus D}.$$

It follows that for a large number N there is a nowhere dense analytic subset S in $A_{M\setminus D}$ such that $A_{M\setminus D} \setminus S \subset Z_N$. Then every element of $A_{M\setminus D}$ can be written as a sum of two elements of Z_N. Therefore $Z_{2N} = A_{M\setminus D}$. □

The group variety $A_{M\setminus D}$ is a special case of a semi-toric variety, which we will discuss in detail in Chap. 5. Here, referring partially to the next chapter, we describe some necessary properties that we need.

Note that $A_{M\setminus D}$ can be considered as a $(\mathbf{C}^*)^t$-principal bundle over A_M. The compactification $(\mathbf{C})^t \hookrightarrow (\mathbf{P}^1(\mathbf{C}))^t$ yields the associated compactification $\bar{A}_{M\setminus D}$ of $A_{M\setminus D}$, which is a $(\mathbf{P}^1(\mathbf{C}))^t$-fiber bundle over A_M. By the construction $\bar{A}_{M\setminus D} \setminus A_{M\setminus D}$ is a divisor of normal crossing type.

Proposition 4.5.13 *Let the notation be as above. Then the quasi-Albanese mapping $\alpha : M \setminus D \to A_{M\setminus D}$ extends to a meromorphic mapping $\bar{\alpha} : M \to \bar{A}_{M\setminus D}$ which is holomorphic on $M \setminus \operatorname{Sing} D$.*

Proof Let $p \in D \setminus \operatorname{Sing} D$. There is a neighborhood U of p with suitably chosen local coordinates $(z_i)_{1\leqq i\leqq n}$ such that $W \cong \Delta^n$ and $W \cap D = \{z_1 = 0\}$. $H_1(W \setminus (D \cap W), \mathbf{Z})$ is generated by a loop around $z_1 = 0$, e.g. $\gamma(t) = (\delta e^{2\pi i t}, 0, \ldots)$ with small fixed $\delta > 0$ and $0 \leqq t \leqq 1$. Let $\{\omega_k\}$ be a basis of $H^0(M, \Omega^1_M(\log D))$ as in (4.5.7). By construction we have $\int_\gamma \omega_k \in \mathbf{Z}$ for all $\gamma \in H_1(M \setminus D, \mathbf{Z})$. Hence for every index k there is an integer m_k and a holomorphic one-form $\eta_k \in H^0(W, \Omega^1_M)$ such that

$$\omega_k|_W = \frac{m_k}{2\pi i}\frac{dz_1}{z_1} + \eta_k. \tag{4.5.14}$$

Now fix a point $q \in W$ and let $\{q_j\}_{j=1}^{\infty}$ be a sequence in $W \setminus D$ converging to $p \in W \cap D$. Then (4.5.14) implies

$$\lim_{j\to\infty} \exp\left(2\pi i \int_q^{q_j} \omega_k\right) \begin{cases} \in \mathbf{C}^* & \text{if } m_k = 0, \\ = \infty & \text{if } m_k < 0, \\ = 0 & \text{if } m_k > 0. \end{cases}$$

Thus for every k the map $M \setminus D \to \mathbf{C}/\mathbf{Z} \cong \mathbf{C}^*$ defined by

$$p \mapsto \int^p \omega_k$$

extends to a holomorphic map from $M \setminus \operatorname{Sing} D$ to $\mathbf{C}^* \cup \{0, \infty\} = \mathbf{P}^1(\mathbf{C})$. It follows that the quasi-Albanese map $\alpha : M \setminus D \to A_{M\setminus D}$ extends to a holomorphic map from $M \setminus \operatorname{Sing} D$ to $\bar{A}_{M\setminus D}$.

Now, let $p \in \operatorname{Sing} D$. Observe that $\operatorname{Sing} D$ has codimension at least two in M. Then there is a small open neighborhood U of p in M such that $U \setminus \operatorname{Sing} D$ is simply connected. The mapping $\alpha : U \setminus \operatorname{Sing} D \to \bar{A}_{M\setminus D}$ is lifted to a mapping $\tilde{\alpha}$ from $U \setminus \operatorname{Sing} D$ to the universal covering of $\bar{A}_{M\setminus D}$. Recall that the toric variety $\overline{(\mathbf{C}^*)^\rho}$ is simply connected (see Proposition 5.3.10 in the next chapter). Therefore the universal covering of $\bar{A}_{M\setminus D}$ is biholomorphic to a direct product of the algebraic variety $\overline{(\mathbf{C}^*)^\rho}$ and $\mathbf{C}^{q(M)}$. Hence, $\tilde{\alpha}$ are described as a tuple of meromorphic functions on $U \setminus \operatorname{Sing} D$. It follows from Theorem 2.2.15 that α extends through $\operatorname{Sing} D$ to a meromorphic mapping from U to $\bar{A}_{M\setminus D}$. □

Theorem 4.5.15 *Let D and Θ be reduced effective divisors on compact Kähler manifolds X and Y, respectively. Let $X' = X \setminus D$ and $Y' = Y \setminus \Theta$. Then each holomorphic map $f : X' \to Y'$ which extends to a meromorphic map from X to Y induces a holomorphic map $\phi : A_{X\setminus D} \to A_{Y\setminus\Theta}$ such that we have the following commutative diagram*:

$$\begin{array}{ccc} X' & \longrightarrow & Y' \\ \downarrow & & \downarrow \\ A_{X\setminus D} & \longrightarrow & A_{Y\setminus\Theta}. \end{array}$$

Proof This follows from the definition of the quasi-Albanese variety together with Theorem 4.5.2. □

In the language of Sect. 5.3.2 this implies the following universality property.

Theorem 4.5.16 *We let*

(i) $A \hookrightarrow \bar{A}$ *be a semi-toric variety*,
(ii) X *be a compact Kähler manifold*,
(iii) D *be a reduced effective divisor on* X,
(iv) $f : X' = X \setminus D \to A$ *be a holomorphic mapping which extends to a meromorphic mapping* $f : X \to \bar{A}$.

Then there is an element $a \in A$ and a morphism of semi-tori $\phi : A_{X\setminus D} \to A$ which extends meromorphically to the respective compactifications such that $f(x) = a + \phi(\pi(x))$.

4.6 Logarithmic Jet Bundles

4.6.1 Jet Bundles in General

Let M be an m-dimensional complex manifold. Take a point $x \in M$ and its holomorphic local coordinate neighborhood $U(x_1, \ldots, x_m)$. We consider a holomorphic mapping $f : V \to M$ with $f(0) = x$ defined in a neighborhood V of the origin $0 \in \mathbf{C}$. Let $g : W \to M$ with $g(0) = x$ be such another one defined in W. If there is a neighborhood $0 \in V' \subset V \cap W$ such that $f|_{V'} = g|_{V'}$, we define the equivalence relation $f \sim g$. The set of the equivalence classes is denoted by $\mathrm{Hol}((\mathbf{C}, 0), (M, x))$. For $f \in \mathrm{Hol}((\mathbf{C}, 0), (M, x))$ we write $f = (f_1, \ldots, f_m)$ and $f^{(k)}(z) = (\ldots, \frac{d^k}{dz^k} f_i(z), \ldots)$ for the k-th derivative. For $f, g \in \mathrm{Hol}((\mathbf{C}, 0), (M, x))$ we write $f \overset{k}{\sim} g$ if and only if

$$f^{(j)}(0) = g^{(j)}(0), \quad 1 \leqq j \leqq k.$$

This is independent to the choice of the local coordinate system, and defines an equivalence relation. The equivalence class of f is denoted by $j_k(f)$. We set

$$J_k(M)_x = \big\{ j_k(f);\, f \in \mathrm{Hol}\big((\mathbf{C}, 0), (M, x)\big) \big\} \cong \mathbf{C}^{mk},$$

$$\pi_k : J_k(M) = \bigcup_{x \in M} J_k(M)_x \to M.$$

Here π_k is the natural projection and $J_k(M)$ becomes a complex manifold in a natural way, and is a fiber bundle over M. This is called the *k-jet bundle* over M. For $k > l$ there is a natural morphism

$$p_{lk} : j_k(f) \in J_k(M) \to j_l(f) \in J_l(M). \tag{4.6.1}$$

A holomorphic mapping $f : Z \to M$ from an open subset Z of $\mathbf{C}$ induces naturally

$$J_k(f) : w \in Z \to j_k\big(f(w + z)\big) \in J_k(M), \quad \pi_k \circ J_k(f) = f,$$

which is called the *k-jet lift* of f.

In the case of $k = 1$ $J_1(M)$ is the holomorphic tangent bundle $\mathbf{T}(M)$ over M. For $k \geqq 2$ there is no such natural linear structure in the jet bundle. However, the $\mathbf{C}^*$-action on $(\Delta, 0)$ induces a natural $\mathbf{C}^*$-action on $\mathrm{Spec}\, \mathbf{C}\{t\}/(t^{k+1})$ via

$$\lambda \cdot j(t) = j(\lambda t), \quad \lambda \in \mathbf{C}^*,\ j(t) \in \mathrm{Spec}\, \mathbf{C}\{t\}/\big(t^{k+1}\big), \tag{4.6.2}$$

and hence on $J_k(M)_x$. This $\mathbf{C}^*$-action can be used to define the notion of polynomials of weighted degree: A function ψ on $J_k(M)_x$ is called a *polynomial of weighted degree* d if $\psi(\lambda \cdot j) = \lambda^d \psi(j)$. The points of $J_k(M)_x$ are separated by polynomials of weighted degree $\leqq k$.

Readers should compute the transformations in the case of $k = 2, 3$ by themselves.

A holomorphic mapping $\Phi : M \to N$ between complex manifolds naturally induces a bundle morphism

$$\Phi_* = d^k\Phi : j_k(f) \in J_k(M) \to j_k(\Phi \circ f) \in J_k(N), \tag{4.6.3}$$

which is equivariant in the sense that

$$d^k\Phi : J_k(M) \to J_k(N), \quad \text{i.e., } \lambda d^k\Phi(j) = d^k\Phi(\lambda j)$$

for $j \in J_k(M)_x$ and $\lambda \in \mathbf{C}^*$. As in (4.6.1) we define $p_{lk} : J_k(N) \to J_l(N)$ (there will be no confusion in using the same notation). Then, we have

$$d^k\Phi \circ p_{lk} = p_{lk} \circ d^k\Phi.$$

A holomorphic differential 1-form $\omega \in H^0(M, \Omega^1_M)$ is a holomorphic functional $\omega : \mathbf{T}(M) \to \mathbf{C}$ which is linear on the fibers. For $f \in \mathrm{Hol}((\mathbf{C}, 0), (M, x))$ we set $f^*\omega = A(z)dz$ and define the following functional,

$$d^{k-1}\omega : j_k(f) \in J_k(M) \to A^{(k-1)}(0) \in \mathbf{C}.$$

If there is a holomorphic function ψ with $\omega = d\psi$, then we write $d^{k-1}d\psi = d^k\psi$. If $\omega_i, 1 \leqq i \leqq m$, generate Ω^1_M at every point, then $(d^l\omega_i)_{0 \leqq l \leqq k-1, 1 \leqq i \leqq m}$ gives a trivialization of $J_k(M)$:

$$j_k(f) \in J_k(M) \to \big(\pi_k\big(j_k(f)\big), \ldots, d^l\omega_i\big(p_{lk}\big(j_k(f)\big)\big), \ldots\big) \in M \times \mathbf{C}^{mk}.$$

In this case the projection to the second factor $\mathbf{C}^{mk}$ is called the *jet projection*, and the natural coordinates of $\mathbf{C}^{qk}$ are called *jet coordinates*. If ω is a meromorphic differential 1-form, $d^j\omega$ is a meromorphic function. In general a holomorphic (resp. meromorphic) functional on $J_k(M)$ which is a polynomial on every fiber is called holomorphic (resp. meromorphic) *k-jet differential*.

For example, we suppose $\omega = \sum \phi_i dx_i$ with a holomorphic local coordinate system $(x_1, \ldots, x_m)$ about x. For $f = (f_1, \ldots, f_m) \in \mathrm{Hol}((\mathbf{C}, 0), (M, x))$ we have $f^*\omega = (\sum \phi_i(f(z))\frac{df_i}{dz}(z))dz$. Thus

$$\omega\big(j_1(f)\big) = \sum \phi_i(x)\frac{df_i}{dz}(0),$$

$$d\omega\big(j_2(f)\big) = \sum \phi_i(x)\frac{d^2f_i}{dz^2}(0) + \sum_{ij} \frac{\partial\omega_i}{\partial x_j}(x)\frac{df_i}{dz}(0)\frac{df_j}{dz}(0).$$

4.6.2 Jet Spaces

We want to define the space of jets over a singular complex space X, which is possibly non-reduced. Let $X \subset M$ be an analytic subspace with defining ideal sheaf $\mathscr{I}\langle X\rangle$. For $x \in X$ we define the k-jet space of X by

$$J_k(X)_x = \left\{ j \in J_k(M)_x ; \left(j^* \mathscr{I}\langle X\rangle\right)_0 \subset \left(t^{k+1}\right)\right\}, \qquad J_k(X) = \bigcup_{x\in X} J_k(X)_x.$$

By virtue of the coherence of $\mathscr{I}\langle X\rangle$ due to K. Oka [50], [51], and Cartan [50], $J_k(X)$ is a complex subspace of $J_k(M)$. In the case where X is non-singular, $J_k(X)$ coincides with the one defined as above with $X = M$.

Because $J_k(X)$ is independent of the used embedding $X \hookrightarrow M$, the k-jet space $J_k(X)$ over X is defined for a general complex space X such that the natural projection $J_k(X) \to X$ is holomorphic and the $\mathbf{C}^*$-action induced from (4.6.2) is well-defined on $J_k(X)$.

When $k = 1$, $J_1(X)$ is called the *Zariski tangent space* of X and often denoted by $\mathbf{T}(X)$.

A holomorphic mapping $\phi : X \to Y$ between complex spaces induces $\mathbf{C}^*$-equivariant mappings

$$d^k\phi : J_k(X) \to J_k(Y), \quad \text{i.e., } \lambda d^k\phi(j) = d^k\phi(\lambda j)$$

for $j \in J_k(X)_x$, $\lambda \in \mathbf{C}^*$.

Remark 4.6.4 If we consider only the germs of $\mathrm{Hol}((\mathbf{C},0),(X,x))$, then $J_k(X)$ does not carry a structure of complex space. To see this, we take a simple example of $X = \{u^2 - v^3 = 0\} \subset \mathbf{C}^2$ and $x = (u,v)$. Then, the desingularization of X at 0 is given by the parameterization $u = t^3, v = t^2$. Hence, every element of $\mathrm{Hol}((\mathbf{C},0),(X,0))$ is represented by $u = (t(\zeta))^3, v = (t(\zeta))^2$ with $t(0) = 0$. It follows that $J_1(X)_0 = \{0\}$. Since $x \in X \setminus \{0\}$ is a non-singular point, $J_1(X)_x \cong \mathbf{C}$. If $J_1(X)$ were a complex space with natural holomorphic projection $J_1(X) \to X$, the semicontinuity of dimension would imply that $\dim J_1(X)_x \leqq \dim J_1(X)_0$ for x near 0, which is clearly a contradiction. Therefore, $J_1(X)$ defined in this way is not a complex space.

4.6.3 Logarithmic Jet Bundles and Logarithmic Jet Spaces

The notion of logarithmic jet bundles was first introduced by Noguchi [86]. Combined with the lemma on logarithmic forms of the next section, it is indispensable to develop the value distribution theory for higher dimensional algebraic varieties.

Let $D \subset M$ be a reduced divisor. We consider the sheaf $\Omega^1_M(\log D)$ of germs of logarithmic 1-forms along D defined by (4.5.1). In the sequel we denote by U an open subset of M. We denote by $\mathscr{J}_k(M)$ the set of local holomorphic sections $\alpha : U \to J_k(M)$ of the k-jet bundle $J_k(M) \to M$. When $k \geqq 2$, this sheaf does not

carry an additive operation or an algebraic operation. In general, $\Gamma(\cdot, *)$ stands for the sections of sheaves or fiber bundles. We denote by $\mathscr{J}_k(M; \log D)$ the sheaf of germs of local holomorphic sections α of $J_k(M)$ such that for any $\omega \in \Omega^1_M(\log D)_x$, $(d^{j-1}\omega)(p_{jk}(\alpha))$ $(1 \leqq j \leqq k)$ are all holomorphic. This is called the *logarithmic k-jet sheaf* and α is called a *logarithmic k-jet field*.

A local meromorphic k-jet differential α on U is called a *logarithmic k-jet differential*, if $\alpha(\beta)$ is holomorphic for any logarithmic k-jet section $\beta \in \Gamma(U, \mathscr{J}_k(M; \log D))$.

In particular, we consider the case where D is of normal crossing type. Let $U(x_1, \ldots, x_m)$ be a holomorphic local coordinate system about $x_0 \in D$ as the origin such that

$$D \cap U = \{x_1 \cdots x_s = 0\}.$$

For $f \in \mathrm{Hol}((\mathbf{C}, 0), (M, x))$ we see that

$$\begin{aligned} f^* \frac{dx_j}{x_j} &= \frac{f_j^{(1)}}{f_j} dz, \quad 1 \leqq j \leqq s, \\ \left(\frac{f_j^{(1)}}{f_j}\right)' &= \frac{f_j^{(2)}}{f_j} - \left(\frac{f_j^{(1)}}{f_j}\right)^2, \quad 1 \leqq j \leqq s. \end{aligned} \tag{4.6.5}$$

Using dx_i, d^2x_i, we have a local trivialization of $J_2(M)|_U$,

$$J_2(M)|_U \cong U \times \mathbf{C}^m \times \mathbf{C}^m \ni \left(x, \left(Z_i^1\right)_{1 \leqq i \leqq m}, \left(Z_i^2\right)_{1 \leqq i \leqq m}\right).$$

By the above computation (4.6.5) we infer that $\mathscr{J}_2(M; \log D)|_U$ is trivialized by the sections of the following type:

$$\left(x, \begin{pmatrix} x_1 \tilde{Z}_1^1 \\ \vdots \\ x_s \tilde{Z}_s^1 \\ Z_{s+1}^1 \\ \vdots \\ Z_m^1 \end{pmatrix}, \begin{pmatrix} x_1 \tilde{Z}_1^2 \\ \vdots \\ x_s \tilde{Z}_s^2 \\ Z_{s+1}^2 \\ \vdots \\ Z_m^2 \end{pmatrix} \right).$$

Therefore we see that the fiber bundle $J_2(M; \log D)$ over U is given by

$$\left(\begin{pmatrix} \tilde{Z}_1^1 \\ \vdots \\ \tilde{Z}_s^1 \\ Z_{s+1}^1 \\ \vdots \\ Z_m^1 \end{pmatrix}, \begin{pmatrix} \tilde{Z}_1^2 \\ \vdots \\ \tilde{Z}_s^2 \\ Z_{s+1}^2 \\ \vdots \\ Z_m^2, \end{pmatrix} \right)$$

and obtain a fiber mapping

$$\lambda : J_2(M; \log D) \to J_2(M).$$

The induced mapping

$$\lambda_* : \Gamma\big(U, J_2(M; \log D)\big) \to \Gamma\big(U, \mathscr{J}_2(M; \log D)\big)$$

is isomorphic. In general, the case of $k \geqq 2$ is similar as above, and we obtain $J_k(M; \log D)$, which is called the *logarithmic k-jet bundle*.

If $X \subset M$ is a complex subspace, we restrict ourselves to take those $\alpha : U \to J_k(X)(\subset J_k(M))$ in the above construction, and then we have the *logarithmic k-jet space* $J_k(X; \log D)$.

Let $\phi : M \to N$ be a meromorphic mapping into another complex manifold N and let E be a divisor on N such that $\phi^{-1}E = D$. Then for a logarithmic k-jet differential ξ over N along E the pull-back $\phi^*\xi$ is a logarithmic k-jet differential over M along D.

4.7 Lemma on Logarithmic Forms

In the present section we prove a lemma on logarithmic forms due to Noguchi [77b] in a slightly less general form, restricting the compact manifold to be projective algebraic.

Let M be a complex projective algebraic manifold and let D be a reduced divisor on M. Let $f : \mathbf{C} \to M$ be an entire curve such that $f(\mathbf{C}) \not\subset D$. Let ω be a logarithmic k-jet differential along D over M. Then the function

$$\xi(z) = \omega\big(J_k(f)(z)\big)$$

is meromorphic on $\mathbf{C}$.

Lemma 4.7.1 (Lemma on logarithmic jet differentials) *Let the notation be as above. We have* $m(r, \xi) = S_f(r)$.

Proof By virtue of the Hironaka desingularization (Hironaka [64a], [64b], see also Bierstone–Milman [97]) there exists a blow-up $\phi : (\tilde{M}, \tilde{D}) \to (M, D)$ with center at the singular locus of D such that $\tilde{D}$ is of simple normal crossing type. There is an entire curve $\tilde{f} : \mathbf{C} \to \tilde{M}$ with $\phi \circ \tilde{f} = f$. Since $\tilde{\xi}(z) = \phi^*\omega(J_k(\tilde{f})(z)) = \xi(z)$, it suffices to prove the assertion for $\tilde{\xi}$ and $T_{\tilde{f}(r)}$ by Theorem 2.5.18. Thus we may assume that D is of normal crossing type.

We take an affine covering $\{U_\alpha\}$ of M and rational holomorphic functions $(x_{\alpha 1}, \ldots, x_{\alpha m})$ on U_α so that

$$dx_{\alpha 1} \wedge \cdots \wedge dx_{\alpha m}(x) \neq 0, \quad {}^\forall x \in U_\alpha,$$
$$D \cap U_\alpha = \{x_{\alpha 1} \cdots x_{\alpha s(\alpha)} = 0\}.$$

On every U_α one gets

$$\omega|_{U_\alpha} = P_\alpha\left(\frac{d^i x_{\alpha j}}{x_{\alpha j}}, d^h x_{\alpha l}\right), \quad 1 \leqq i,\ h \leqq k,\ 1 \leqq j \leqq s(\alpha),\ s(\alpha)+1 \leqq l \leqq m.$$

Here P_α is a polynomial in the variables described above with coefficients which are rational holomorphic functions on U_α. Let $f(z) = (f_{\alpha 1}(z), \ldots, f_{\alpha m}(z)) \in U_\alpha$. Then

$$\xi(z) = P_\alpha\left(\frac{f_{\alpha j}^{(i)}}{f_{\alpha j}}, f_{\alpha l}^{(h)}\right).$$

Let $\{c_\alpha\}$ be the partition of unity subordinated to $\{U_\alpha\}$. Since $\xi(z) = \sum_\alpha f^*(c_\alpha P_\alpha)$, there is a constant $C > 0$ such that

$$|\xi| \leqq C\left(\sum c_\alpha \circ f\left(\left|\frac{f_{\alpha j}^{(i)}}{f_{\alpha j}}\right| + \left|f_{\alpha l}^{(h)}\right|\right)\right)^C.$$

The sum is a finite sum, of course. Therefore

$$m(r,\xi) \leqq C\left(\sum m\left(r, \frac{f_{\alpha j}^{(i)}}{f_{\alpha j}}\right) + \sum m\left(r, c_\alpha \circ f \cdot f_{\alpha l}^{(h)}\right)\right) + O(1).$$

It follows from Lemma 4.2.9 that $m(r, f_{\alpha j}^{(i)}/f_{\alpha j}) = S_f(r)$ and

$$\begin{aligned} m\left(r, c_\alpha \circ f \cdot f_{\alpha l}^{(h)}\right) &\leqq m\left(r, \frac{f_{\alpha l}^{(h)}}{f_{\alpha l}^{(1)}}\right) + m\left(r, c_\alpha \circ f \cdot f_{\alpha l}^{(1)}\right) \\ &\leqq S_f(r) + m\left(r, c_\alpha \circ f \cdot f_{\alpha l}^{(1)}\right). \end{aligned}$$

Hence there remains to show that

$$m\left(r, c_\alpha \circ f \cdot f_{\alpha l}^{(1)}\right) = S_f(r).$$

Take a hermitian metric H on M. There is a constant $C_1 > 0$ such that $c_\alpha^2 dx_{\alpha l} \cdot d\bar{x}_{\alpha l} \leqq C_1 H$. Setting $f^*H = B(z)dz \cdot d\bar{z}$, one obtains

(4.7.2)

$$\begin{aligned} m\left(r, c_\alpha \circ f \cdot f_{\alpha l}^{(1)}\right) &= \frac{1}{2\pi}\int_{|z|=r} \log^+ c_\alpha \circ f \cdot \left|f_{\alpha l}^{(1)}\right| d\theta \\ &\leqq \frac{1}{4\pi}\int_{|z|=r} \log^+ c_\alpha^2 \circ f \cdot \left|f_{\alpha l}^{(1)}\right|^2 d\theta \leqq \frac{1}{4\pi}\int_{|z|=r} \log^+ B d\theta + O(1) \\ &\leqq \frac{1}{4\pi}\int_{|z|=r} \log(1+B) d\theta + O(1) \leqq \frac{1}{2}\log\left(1 + \frac{1}{2\pi}\int_{|z|=r} B d\theta\right) + O(1) \\ &\leqq \frac{1}{2}\log\left(1 + \frac{1}{2\pi r}\frac{d}{dr}\int_{|z|\leqq r} B r dr d\theta\right) + O(1). \end{aligned}$$

The rest of the proof is the same as that of (3.2.8), and hence one sees the last term of (4.7.2) to be $S_f(r)$. □

Corollary 4.7.3 *If the above ξ is holomorphic, then $T(r,\xi) = S_f(r)$.*

This is immediate by definition (1.1.12).

Remark 4.7.4 (i) (Noguchi [77b]) Let M be just a compact complex manifold without algebraicity, and let D be a reduced divisor on M. Let ω be a logarithmic 1-form (logarithmic 1-jet differential) with integral residues on D, and set $f^*\omega = \xi(z)dz$ for an entire curve $f : \mathbf{C} \to M$ such that $f(\mathbf{C}) \not\subset D$. Then we have

$$m\big(r, \xi^{(\nu)}\big) = S_f(r), \quad \nu \in \mathbf{Z}^+.$$

(ii) (Noguchi [81a]) Let Ω_H denote the associated metric form with H. When the domain of f is a punctured disk $\Delta^*(1) = \{0 < |z| < 1\}$, fixing an arbitrary $r_0 > 1$, we set

$$T_f(r) = \int_{r_0}^{r} \frac{dt}{t} \int_{\{1/t<|z|<1/r_0\}} f^*\Omega_H, \quad r \geqq r_0,$$

$$m(r,\xi) = \frac{1}{2\pi} \int_{|z|=1/r} \log^+ |\xi| d\theta,$$

$$S_f(r) = O\big(\log T_f(r)\big) + O(\log 1/r)\|_E.$$

Then Lemma 4.7.1 holds for compact Kähler M. Note here that $T_f(r) = O(\log 1/r)$ $(r \to 0)$ if and only if f is extended holomorphically over the origin.

4.8 Inequality of the Second Main Theorem Type

Let M be an m-dimensional complex projective algebraic manifold and let D be a reduced divisor on it. Let

$$\alpha : M \setminus D \to A_{M\setminus D}$$

be the quasi-Albanese map (cf. Sect. 4.5), and let X denote the Zariski closure of the image. For a subset $B \subset A_{M\setminus D}$ we set the *stabilizer* of B by

$$\mathrm{St}(B) = \{a \in A_{M\setminus D}; a + B = B\},$$

and denote by $\mathrm{St}(B)^0$ its identity component.

Remark 4.8.1 It is not necessary in what follows but known that X is of logarithmic general type if and only if $\mathrm{St}(X)^0 = \{\mathrm{id}\}$ (cf. Proposition 5.6.21).

Theorem 4.8.2 (Noguchi [77b], [81a]) *Let M and X be as above. Assume that* $\dim X = m$ *and* $\mathrm{St}(X)^0 = \{0\}$. *For an algebraically non-degenerate entire curve $f : \mathbf{C} \to M$, there is a constant $\kappa > 0$ such that*

$$\kappa T_f(r) \leqq N_1\big(r, f^*D\big) + S_f(r).$$

We give a proof by the *jet projection method* due to Noguchi–Ochiai [90], Chap. 6 (Ochiai–Noguchi [84]); we need some preparation.

Let $\alpha : M \setminus D \to A_{M\setminus D}$ be the quasi-Albanese map. Then α is extended to a rational mapping $\bar{\alpha} : M \to \bar{A}_{M\setminus D}$. We consider an entire curve $g = \bar{\alpha} \circ f : \mathbf{C} \to \bar{A}_{M\setminus D}$. We denote by X the Zariski closure of $g(\mathbf{C})$ in $\bar{A}_{M\setminus D}$, so that $X = \bar{X} \cap A_{M\setminus D}$.

Since $A_{M\setminus D} = \mathbf{C}^q/\Gamma$, the natural coordinate system $(x_1, \ldots, x_q)$ of $\mathbf{C}^q$ induces the basis $dx_i, 1 \leqq i \leqq q$, of global holomorphic 1-forms over $A_{M\setminus D}$. We fix the associated global trivialization

$$J_k(A_{M\setminus D}) \cong A_{M\setminus D} \times \mathbf{C}^{qk}.$$

Let

$$p_2 : J_k(A_{M\setminus D}) \cong A_{M\setminus D} \times \mathbf{C}^{qk} \to \mathbf{C}^{qk}$$

be the natural projection. Since $J_k(X) \subset J_k(A_{M\setminus D})$, we set the *jet projection* of $J_k(X)$ by restricting p_2,

$$I_k = p_2|_{J_k(X)} : J_k(X) \to \mathbf{C}^{qk}.$$

Lemma 4.8.3 *Assume that* $\dim X = m$. *Take a point $z_0 \in \mathbf{C}$ so that $g(z_0)$ is a non-singular point of X. Then there is a number $k_0 \geqq 1$ such that for every $k \geqq k_0$ the differential dI_k of I_k is injective at $J_k(g)(z_0)$.*

Proof In a neighborhood $U \subset A_{M\setminus D}$ of $g(z_0)$, X may be written as follows:

$$x_h = F_h(x_1, \ldots, x_m), \quad m+1 \leqq h \leqq q.$$

Consider $(x_1, \ldots, x_m)$ to be a holomorphic local coordinate system of $U \cap X$ and let

$$J_k(X \cap U) \cong \big((x_i), \big(Z_i^j\big)\big), \quad 1 \leqq i \leqq m,\ 1 \leqq j \leqq k$$

be the induced local trivialization of $J_k(X)|_U$. In term of these coordinates I_k is written as

$$I_k\big((x_i), \big(Z_i^j\big)\big) = \big(Z_i^j, I_h^j\big), \quad 1 \leqq i \leqq m < h \leqq q,\ 1 \leqq j \leqq k. \tag{4.8.4}$$

Here I_h^j are polynomials in Z_i^j with partial derivatives of F_h as coefficients. For instance one has

$$I_h^1 = \sum_i \frac{\partial F_h}{\partial x_i} Z_i^1, \tag{4.8.5}$$

(continued)

$$I_h^2 = \sum_i \frac{\partial F_h}{\partial x_i} Z_i^2 + \sum_{1 \leqq i_1, i_2 \leqq m} \frac{\partial^2 F_h}{\partial x_{i_1} \partial x_{i_2}} Z_{i_1}^1 Z_{i_2}^1.$$

We fix a decomposition, $\mathbf{T}(J_k(X)) \subset \mathbf{T}(J_k(A_{M\setminus D})) \cong \mathbf{T}(A_{M\setminus D}) \oplus \mathbf{T}(\mathbf{C}^{qk})$. By (4.8.4) we have for the kernel, $\operatorname{Ker} dI_{kJ_k(g)(z_0)} \subset \mathbf{T}(A_{M\setminus D})_{g(z_0)}$.

Suppose that $\bigcap_{k=1}^{\infty} \operatorname{Ker} dI_{kJ_k(g)(z_0)} \neq \{0\}$. Take $v \in \bigcap_{k=1}^{\infty} \operatorname{Ker} dI_{kJ_k(g)(z_0)}$, $v \neq 0$. Then

$$vI_k \circ J_k(g)(z_0) = 0, \quad {}^{\forall}k \geqq 1. \tag{4.8.6}$$

It is inferred from (4.8.5) and (4.8.6) that

$$\frac{d^k}{dz^k} vF_h\big(g(z_0)\big) = 0, \quad {}^{\forall}k \geqq 1.$$

We regard $v = \sum_{i=1}^m v_i \frac{\partial}{\partial x_i}$ as a constant vector field on $X \cap U$. Then one sees that $vF_h(g(z))$ is constant in z. We extend v to a constant vector field $\tilde{v}$ on $A_{M\setminus D}$ by

$$\tilde{v} = \sum_{i=1}^m v_i \frac{\partial}{\partial x_i} + \sum_{h=m+1}^q \frac{\partial F_h}{\partial x_i}\big(g(z_0)\big) \frac{\partial}{\partial x_h}.$$

Then in a neighborhood of z_0

$$\tilde{v}(x_h - F_h)\big(g(z)\big) \equiv 0, \quad m+1 \leqq h \leqq q.$$

Since the image of g is Zariski dense in X, $\tilde{v}$ is tangent to X at every point of X. For $\exp(t\tilde{v}) = t\tilde{v}$ $(t \in \mathbf{C})$ we have

$$X + t\tilde{v} = X.$$

This implies that $\operatorname{St}(X)^0 \neq \{0\}$, which is a contradiction. □

Proof of Theorem 4.8.2 Take the transcendental bases $\phi_1, \ldots, \phi_m$ of the rational function field $\mathbf{C}(X)$ of X. By Theorem 2.5.18 there is a constant $C > 0$ such that

$$C^{-1} T_f(r) + O(1) \leqq T_g\big(r; \{\phi_l\}\big) \leqq C T_f(r) + O(1). \tag{4.8.7}$$

We regard each ϕ_l as a rational function on the projective compactification of the Zariski closure $X_k \subset J_k(X)$ of the image $J_k(g)|_{\mathbf{C}\setminus f^{-1}D}$. Let $\omega_1, \ldots, \omega_q$ be the bases of $H^0(M, \Omega_M^1(\log D)) \cong H^0(\bar{A}_{M\setminus D}, \Omega^1_{\bar{A}_{M\setminus D}}(\log(\bar{A}_{M\setminus D} \setminus A_{M\setminus D})))$. The components of I_k are given by

$$d^j \omega_i, \quad 1 \leqq i \leqq q, \ 0 \leqq j \leqq k-1.$$

Put

$$\zeta_i^j(z) = d^j \omega_i\big(J_k(g)(z)\big).$$

We infer from Lemma 4.7.1 that

$$\begin{aligned} &m\big(r,\zeta_i^j\big) = S_f(r),\\ &T\big(r,\zeta_i^j\big) \leqq (j+1)N_1\big(r,f^*D\big) + S_f(r). \end{aligned} \tag{4.8.8}$$

Since the differential of $I_k|_{X_k}$ is generically injective for a large k by Lemma 4.8.3, there are algebraic relations over X_k as follows:

$$\begin{aligned} &P_0\big(d^j\omega_i\big)\phi_l^{d_l} + \cdots + P_{d_l}\big(d^j\omega_i\big) \equiv 0,\\ &P_0\big(d^j\omega_i\big) \not\equiv 0. \end{aligned}$$

Therefore

$$\begin{aligned} &P_0\big(\zeta_i^j\big)f^*\phi_l^{d_l} + \cdots + P_{d_l}\big(\zeta_i^j\big) \equiv 0,\\ &P_0\big(\zeta_i^j\big) \not\equiv 0. \end{aligned}$$

It follows from Lemma 2.5.15 and (4.8.8) that

$$T\big(r,g^*\phi_l\big) = O\big(T\big(r,\zeta_i^j\big)\big) = O\big(N_1\big(r,f^*D\big)\big) + S_f(r).$$

This with (4.8.7) implies that

$$T_f(r) \leqq O\big(N_1\big(r,f^*D\big)\big) + S_f(r). \qquad \square$$

Remark 4.8.9 (i) The constant $\kappa > 0$ in Theorem 4.8.2 is independent from f as far as D is fixed (Noguchi [77b], Supplement).

(ii) Let $Z \to \mathbf{C}^m$ be a finite ramified covering space. Theorem 4.8.2 was generalized to meromorphic mappings $f : Z \to M$ (Noguchi [85b]).

Example 4.8.10 The constant $\kappa > 0$ in Theorem 4.8.2 cannot be set $\kappa = 1$ with $T_f(r) = T_f(r, c_1(D))$. This is seen by the following example.

Let $M = (\mathbf{P}^1(\mathbf{C}))^2$ and let (x_1, x_2) be an affine coordinate system. Let $m < n$ be natural numbers and set

$$D = \big\{x_1 + x_2^m + x_2^n = 0\big\} + \sum_{i=1,2}\{x_i = 0\} + \{x_i = \infty\}.$$

Then $\bar{q}(M \setminus D) = 3$ and (M, D) satisfies the condition of Theorem 4.8.2. Take a real irrational number c such that $cm < 1 < cn$, and define an entire curve by

$$f : z \in \mathbf{C} \to \big(e^z, e^{cz}\big) \in M.$$

Note that f is algebraically non-degenerate. Take the sum $\Omega_1 + \Omega_2$ on M of the Fubini–Study metric form on $\mathbf{P}^1(\mathbf{C})$. Then $c_1(D) = \Omega_1 + n\Omega_2$. By computation

$$T_f\big(r, c_1(\bar{D})\big) = \frac{1+nc}{\pi}r + O(1). \tag{4.8.11}$$

We next compute $N(r, f^*D)$:

$$f^*D = \left(e^z + e^{mcz} + e^{ncz}\right)_0. \tag{4.8.12}$$

Let $[w_0, w_1, w_2]$ be the homogeneous coordinate system of $\mathbf{P}^2(\mathbf{C})$. We consider an entire curve g given by

$$g : z \in \mathbf{C} \to \left[e^z, e^{mcz}, e^{ncz}\right] \in \mathbf{P}^2(\mathbf{C}).$$

By computing the Wronskian one sees that g is linearly non-degenerate. We compute the order function $T_g(r)$ of g with respect to the Fubini–Study metric of $\mathbf{P}^2(\mathbf{C})$:

$$\begin{aligned} T_g(r) &= \frac{1}{4\pi}\int_{\{|z|=r\}} \log\left(\left|e^z\right|^2 + \left|e^{mcz}\right|^2 + \left|e^{ncz}\right|^2\right)d\theta + O(1) \\ &= \frac{1}{4\pi}\int_{\{|z|=r\}} \log\left(1 + \left|e^{(mc-1)z}\right|^2 + \left|e^{(nc-1)z}\right|^2\right)d\theta + O(1). \end{aligned} \tag{4.8.13}$$

If $\Re z \geqq 0$ (resp. $\leqq 0$), $|e^{(mc-1)z}| \leqq 1$ (resp. $\geqq 1$) and $|e^{(nc-1)z}| \geqq 1$ (resp. $\leqq 1$). Therefore, if $z = re^{i\theta}$ and $\Re z \geqq 0$,

$$\begin{aligned} \log\left(1 + \left|e^{(mc-1)z}\right|^2 + \left|e^{(nc-1)z}\right|^2\right) &= 2\log^+\left|e^{(nc-1)z}\right| + O(1) \\ &= 2(nc-1)r\cos\theta + O(1). \end{aligned}$$

If $z = re^{i\theta}$ and $\Re z \leqq 0$,

$$\begin{aligned} \log\left(1 + \left|e^{(mc-1)z}\right|^2 + \left|e^{(nc-1)z}\right|^2\right) &= 2\log^+\left|e^{(mc-1)z}\right| + O(1) \\ &= 2(mc-1)r\cos\theta + O(1). \end{aligned}$$

These with (4.8.13) yield

$$T_g(r) = \frac{(n-m)c}{\pi}r + O(1). \tag{4.8.14}$$

We define four lines H_j of $\mathbf{P}^2(\mathbf{C})$, $1 \leqq j \leqq 4$, in general position defined by

$$H_j = \{w_{j-1} = 0\}, \quad 1 \leqq j \leqq 3, \qquad H_4 = \{w_0 + w_1 + w_2 = 0\}.$$

By Cartan's Second Main Theorem (Corollary 4.2.15) we have

$$T_g(r) \leqq \sum_{j=1}^{4} N_2\left(r, g^*H_j\right) + O(\log r)\|_E. \tag{4.8.15}$$

Since $N_2(r, g^*H_j) = 0$, $1 \leqq j \leqq 3$,

$$N_2\left(r, g^*H_4\right) = \frac{(n-m)c}{\pi}r + O(\log r)\|_E.$$

By definition $N(r, g^*H_4) = N(r, f^*D)$, the First Main Theorem implies

$$N_2\big(r, g^*D\big) \leqq N\big(r, g^*D\big) \leqq T_g(r).$$

From these, (4.8.14) and (4.8.15) we deduce that

$$N\big(r, f^*D\big) = \frac{(n-m)c}{\pi} r + O(\log r)\|_E. \tag{4.8.16}$$

Therefore we have to take $\kappa = \frac{(n-m)c}{1+nc}$. If $m = n-1$ and $n \to \infty$, then $\kappa \to 0$.

Note that $\rho_g = 1$ by (4.8.14), so that by Remark 1.2.3 (4.8.16) holds without the exceptional set E; that is, the symbol "$\|_E$" can be dropped.

Theorem 4.8.17 (The Logarithmic Bloch–Ochiai Theorem; Noguchi [77b], [81a]; Noguchi–Winkelmann [02]) *Let M be a compact Kähler manifold and D be a reduced divisor on M. If $\bar{q}(M \setminus D) > \dim M$, then for any entire curve $f : \mathbf{C} \to M \setminus D$ does not have a Zariski dense image in M.*

Here, the Zariski topology on M means that analytic subsets of M are closed subsets.

Proof We need some facts from Chap. 5 on semi-tori.

Let $\alpha : M \setminus D \to A$ be the quasi-Albanese map with $A = A_{M\setminus D}$. Then it extends to a meromorphic map $\bar{\alpha} : M \to \bar{A}$ (Proposition 4.5.13). We set $\bar{X} = \bar{\alpha}(M)$ and $X = \bar{X} \cap A$, where $\bar{\alpha}(M)$ coincides with the closure of $\alpha(M \setminus D)$. By assumption, $\dim X < \bar{q}(M \setminus D) = \dim A$.

Let $\beta : A \to A' := A/\operatorname{St}(X)^0$ be the quotient map and set $X' = \beta(X)$. Then A' is a semi-torus and β extends to a meromorphic map $\bar{\beta} : \bar{A} \to \bar{A}'$ between compactifications, such that $\bar{\beta}(\bar{X}) = \bar{X}'$ and $0 < \dim X' < \dim A'$ (see Proposition 5.3.13 (vii), Proposition 5.3.24). Note that X' generates A'. It follows from Proposition 5.6.21 that X' is of logarithmic general type; in particular, $\bar{X}'$ is Moishezon. Therefore the quotient A'/T' of A' by the maximal affine subgroup T' is Moishezon, and hence abelian (cf. Noguchi–Winkelmann [02], Lemma 2.11). Thus, A' is semi-abelian. Therefore the case is reduced to algebraic one.

If f has a Zariski dense image in M, then $g = \beta \circ \alpha \circ f : \mathbf{C} \to X'$ is algebraically non-degenerate. On the other hand, Theorem 4.8.2 implies that for every $\delta > 0$

$$T_g(r) = S_g(r) = O\big(\log T_g(r)\big) + \delta \log r\|_{E(\delta)}.$$

By Theorem 2.5.4 g is to be constant; this is a contradiction. □

Corollary 4.8.18 (The Bloch–Ochiai Theorem; Bloch [26b]; Ochiai [77]; Kawamata [80]) *If $q(M) > \dim M$, every an entire curve $f : \mathbf{C} \to M$ is algebraically degenerate.*

This means in the case of $\dim M = 1$ that if the genus of M is bigger than or equal to two, any entire curve $f : \mathbf{C} \to M$ is a constant. This is a consequence of Liouville's Theorem and the uniformization theorem by which the universal cover-

ing surface of M with genus $\geqq 2$ is the unit disk of $\mathbf{C}$. It is a crucial point that the proof given above is completely independent from the uniformization theorem.

On the other hand, we consider the case where $M = \mathbf{P}^n(\mathbf{C})$ and D is the sum of hyperplanes $H_1, H_2, \ldots, H_p$. Let $[w_0, \ldots, w_n]$ be a homogeneous coordinate system of $\mathbf{P}^n(\mathbf{C})$ and let H_j be defined as zeros of linear forms $F_j = \sum_k c_{jk} w_k$, $c_{jk} \in \mathbf{C}$. Then

$$\omega_j = d \log \frac{F_j}{F_p}, \quad 1 \leqq j \leqq p-1,$$

form a basis of $H^0(\mathbf{P}^n(\mathbf{C}), \Omega^1_{\mathbf{P}^n(\mathbf{C})}(\log \sum H_j))$, and so $\bar{q}(\mathbf{P}^n(\mathbf{C}) \setminus \sum_{j=1}^p H_j) = p-1$. Therefore, if $p \geqq n+2$, then $f : \mathbf{C} \to \mathbf{P}^n(\mathbf{C}) \setminus \sum_{j=1}^p H_j$ is always algebraically degenerate. In fact, one can see a bit more detail. If $\{H_j\}$ is in general position, then $f(\mathbf{C})$ is contained in an element of the linear system of $H^0(\mathbf{P}^n(\mathbf{C}), \Omega^n_{\mathbf{P}^n(\mathbf{C})}(\log \sum H_j)) = H^0(\mathbf{P}^n(\mathbf{C}), O(p-n-1))$, where $O(1)$ stands for the hyperplane bundle (Noguchi [77b], [81a]). In particular, applying this with $p = n+2$, one sees that f is linearly degenerate. If $p > n+2$, one gets the linear degeneracy of the number of the combination $\binom{p}{n+2}$.

Summarizing the above one may say that the Logarithmic Bloch–Ochiai Theorem 4.8.17 uniformizes Borel's Theorem (Corollary 4.2.18) in the non-compact case and the Bloch-Ochiai Theorem (Corollary 4.8.18) in the compact case.

Corollary 4.8.19 *Theorem 4.8.17 and Corollary 4.8.18 hold for meromorphic mappings from* $\mathbf{C}^m$.

The proof is immediate from the next lemma.

Lemma 4.8.20 *There is an analytically non-degenerate entire curve* $f : \mathbf{C} \to \mathbf{C}^m$.

Proof Take real numbers $\theta_1 = 1, \theta_2, \ldots, \theta_m$ that are linearly independent over $\mathbf{Q}$. Set $f(z) = (e^{i\theta_1 z}, \ldots, e^{i\theta_m z})$ for $z \in \mathbf{C}$. Then $\{f(z); z \in \mathbf{R}\}$ is dense in $(\partial\Delta(1))^m$. If $\phi(f(z)) \equiv 0$ for an arbitrary holomorphic function $\phi(w)$ on $\mathbf{C}^m$, then $\phi|_{(\partial\Delta(1))^m} \equiv 0$. Thus $\phi \equiv 0$. It follows that $f(z)$ is analytically non-degenerate. □

Remark 4.8.21 The results of M. Green [75], Part 5 and T. Nishino [84] are special cases of the Logarithmic Bloch–Ochiai Theorem 4.8.17.

Remark 4.8.22 Let $f : \Delta^*(1) \to M \setminus D$ be a holomorphic mapping. We then use Remark 4.7.4 (ii) and apply the same arguments as in this section to infer that if f is algebraically non-degenerate, then it holomorphically extends to $\tilde{f} : \Delta(1) \to M$. This is a generalization of the Big Picard Theorem (Noguchi [81a]).

Historical Supplement Corollary 4.8.18 was claimed by Bloch [26b]. The proof was, however, a sketchy one in dimension two and he was unable to prove

Lemma 4.7.1 for holomorphic forms, as he wrote himself. T. Ochiai [77] completed the missing part and made considerable improvement; he completed the proof in dimension two and made clear what should be done to complete the general case. M. Green gave a talk on the remaining part at the Taniguchi International Symposium, Katata, 1978. Later he published a joint paper with P. Griffiths in 1980 (Green–Griffiths [80]). This seemed to give a complete proof, but unfortunately a gap was found (Demailly [97]). After all a precise proof was first given by Kawamata [80]. P.-M. Wong[2] also published a paper to claim the proof, but there was a gap in a crucial step.

Ochiai–Noguchi [84], Lemma 6.3.10 (Lemma 4.8.3 in the present book) and its proof are originally due to the present first author by the *jet projection method*. It is a point to consider the jets to infinite order. Afterwards M. McQuillan [96] gave another proof based on P. Vojta's arithmetic method.

4.9 Entire Curves Omitting Hypersurfaces

Let M be a complex projective algebraic manifold, let D_i, $1 \leqq i \leqq l$, be distinct divisors on M. Since the result of this section relies only on the Logarithmic Bloch–Ochiai Theorem 4.8.17, it will suffice in fact to assume that M is compact Kähler (Remark 4.8.21 (i)).

4.9.1 (Notation) We denote by $r(\{D_i\})$ the rank of the subgroup $\sum_i \mathbf{Z} \cdot c_1(D_i)$ generated by $c_1(D_i) \in H^2(M, \mathbf{Z})$, $1 \leqq i \leqq l$, in $H^2(M, \mathbf{Z})$.

Let $\mathrm{NS}(M)$ denote the *Néron–Severi group* of M (cf. Hartshorne [77]); i.e.,

$$\mathrm{NS}(M) = \mathrm{Pic}(M)/\mathrm{Pic}^0(M).$$

We know that

$$r\left(\{D_i\}_{i=1}^l\right) \leqq \mathrm{rank}_{\mathbf{Z}} \mathrm{NS}(M). \tag{4.9.2}$$

Definition 4.9.3 We say that D_i, $1 \leqq i \leqq l$, are (or the family $\{D_i\}_{i=1}^l$ is) *independent in supports* if for every $1 \leqq j \leqq l$

$$\mathrm{Supp}\, D_j \not\subset \bigcup_{i \neq j} \mathrm{Supp}\, D_i.$$

This notion makes sense even for divisors on a singular space.

Suppose that $l > r(\{D_i\})$ and set $r_0 = r(\{D_i\})$. Then there is a maximal linearly independent system, to say, $\{c_1(D_i)\}_{i=1}^{r_0}$, and for $r_0 + 1 \leqq j \leqq l$ we have linear

[2] Wong, P.-M., Holomorphic mappings into Abelian varieties, Amer. J. Math. **102** (1980), 493–501.

relations over $\mathbf{Z}$,

$$a_j c_1(D_j) + \sum_{i=1}^{r_0} a_{ji} c_1(D_i) = 0, \quad a_j \neq 0. \tag{4.9.4}$$

We refer to the following lemma derived from the theory of harmonic integral due to Hodge–Kodaira (cf. Kodaira [75], Vol. I, pp. 325–338, and Noguchi [95]).

Lemma 4.9.5 *Let E_i, $1 \leqq i \leqq l$, be distinct irreducible reduced divisors on M and let $a_i \in \mathbf{Z}$ such that $\sum a_i c_1(E_i) = 0$ modulo torsion. Then there is a multi-valued meromorphic function ϑ on M satisfying the following properties*:

(i) *$|\vartheta|$ is one-valued.*
(ii) *The divisor (ϑ) defined by ϑ is equal to $\sum_{i=1}^{l} a_i E_i$.*

Conversely, if there is such ϑ, then $\sum a_i c_1(E_i) = 0$ modulo torsion.

The above ϑ is called a multiplicative meromorphic function. By Lemma 4.9.5 and (4.9.4) there are multiplicative meromorphic functions ϑ_j $(1 \leqq j \leqq l - r(\{D_i\}))$ on M such

$$(\vartheta_j) = a_j D_j + \sum_{i=1}^{r_0} a_{ji} D_i,$$

and then

$$d \log \vartheta_j \in H^0\left(M, \Omega^1_M\left(\log \operatorname{Supp} D_j \cup \bigcup_{i=1}^{r_0} \operatorname{Supp} D_i\right)\right).$$

It is immediate that $\{D_j\}_{j=r_0+1}^{l}$ being independent in supports implies the linear independence of $d \log \vartheta_j$, $r_0 + 1 \leqq j \leqq l$. Therefore, we obtain

Lemma 4.9.6 *Let the notation be as above. Then we have*

$$d \log \vartheta_j \in H^0\left(M, \Omega^1_M\left(\log \bigcup_{i=1}^{l} \operatorname{Supp} D_i\right)\right), \quad r_0 + 1 \leqq j \leqq l.$$

The polar locus of $d \log \vartheta_j$ is exactly $\operatorname{Supp} D_j \cup \bigcup_{i, a_i \neq 0} \operatorname{Supp} D_i$; if $\{D_j\}_{j=1}^{l}$ is independent in supports, then $d \log \vartheta_j$, $r_0 + 1 \leqq j \leqq l$, are linearly independent.

Combining this with the Logarithmic Bloch–Ochiai Theorem 4.8.17, we conclude the following.

Theorem 4.9.7 [3] *Let* $\dim M = m$ *and let* $\{D_i\}_{i=1}^{l}$ *be a family of divisors on* M *which are independent in supports. If* $l > m + r(\{D_i\}) - q(M)$ *(in particular, if* $l > m + \operatorname{rank}_{\mathbf{Z}} \mathrm{NS}(M) - q(M)$*), then every entire curve in* $M \setminus \bigcup_{i=1}^{l} \operatorname{Supp} D_i$ *is algebraically degenerate.*

4.10 The Fundamental Conjecture of Entire Curves

Let M be a compact Kähler manifold and let ω_0 be the Kähler form. Let $L \to M$ be a line bundle and let $D \in |L|$ be a divisor of simple normal crossing type. Let $f : \mathbf{C} \to M$ be an entire curve such that $f(\mathbf{C}) \not\subset D$. To measure the multiplicities of f over D we define the *multiplicity counting function* by

$$N^k\big(r, f^*D\big) = N\big(r, f^*D\big) - N_k\big(r, f^*D\big). \tag{4.10.1}$$

We set

$$T_f(r) = T_f(r, \omega_0).$$

Conjecture 4.10.2 (The fundamental conjecture for entire curves) *Let* M *and* D *be as above. Then there is a number* $k_0 \in \mathbf{N}$ *such that for an analytically non-degenerate entire curve* $f : \mathbf{C} \to M$

$$m_f(r, D) + N^{k_0}\big(r, f^*D\big) + T_f(r, K_M) \leqq \varepsilon T_f(r)\|_{\varepsilon}, \tag{4.10.3}$$

where $\varepsilon > 0$ *is arbitrarily small. Equivalently, we have*

$$T_f(r, L) + T_f(r, K_M) \leqq N_{k_0}\big(r, f^*D\big) + \varepsilon T_f(r)\|_{\varepsilon}.$$

Moreover,

$$m_f(r, D) + N^1\big(r, f^*D\big) + T_f(r, K_M) \leqq \varepsilon T_f(r)\|_{\varepsilon}. \tag{4.10.4}$$

Of course it is ideal to have $k_0 = 1$ as in (4.10.4), but it will be very hard. The case of abelian and semi-abelian varieties will be dealt with in Chap. 6.

Griffiths [72] conjectured a formula of (4.10.3) without term $N^{k_0}(r, f^*D)$.

Conjecture 4.10.5 (Griffiths Conjecture) *Under the same assumptions as in* 4.10.2 *and of* M *to be projective algebraic, we have that*

$$m_f(r, D) + T_f(r, K_M) \leqq \varepsilon T_f(r)\|_{\varepsilon}.$$

See (i) (Example) of Remark 6.3.6 for an optimal case of the above estimate with $D = \emptyset$ and M which is a blowing-up of the abelian surface A of the example.

[3] There is a related similar theorem claimed in Noguchi–Winkelmann [02]. Unfortunately, a gap in the proof of its Theorem 1.2 was lately found by A. Levin with a counter-example due to W. Cherry, both of whom the authors thank.

Chapter 5
Semi-abelian Varieties

In the forthcoming two chapters semi-abelian varieties play an important role. For that purpose we here give a notion of semi-abelian varieties from the viewpoint of complex geometry.

5.1 Semi-tori

5.1.1 Definition

A *complex Lie group* G is a complex manifold G which is simultaneously a group in such a way that the two structures are compatible, i.e. one requires that the map $(x, y) \mapsto x \cdot (y^{-1})$ is a holomorphic map from $G \times G$ to G. More fancily, one may say: A complex Lie group is a group object in the category of complex analytic spaces.

Definition 5.1.1 A *torus* is a compact quotient group $T = \mathbf{C}^g/\Gamma$ where Γ is a discrete subgroup of the additive group $(\mathbf{C}^g, +)$.

Proposition 5.1.2 *Every connected compact complex Lie group G is a torus.*

Proof Consider the map $\zeta : (g, h) \mapsto ghg^{-1}h^{-1}$. Let Ω be a holomorphic local coordinate neighborhood of e in G. Then $\zeta^{-1}(\Omega)$ is an open subset of $G \times G$ containing $(G \times \{e\}) \cup (\{e\} \times G)$. Since G is compact, $G \times \{p\} \subset \zeta^{-1}(\Omega)$ for every p in some neighborhood of e. Now every holomorphic map from G to Ω is constant, since G is compact and connected and Ω is biholomorphic to an open subset of $\mathbf{C}^g$. Thus the restriction of ζ to $G \times \{p\}$ is constant with value $e = \zeta(e, p)$ for every p in some neighborhood of e. By the identity principle it follows that $\zeta \equiv e$, i.e., G is commutative. Since G is commutative, the Campbell–Hausdorff formula (see e.g. Varadarajan [74], Sect. 2.15) implies that the exponential map from the Lie algebra of G to G is a group homomorphism. This implies that $G \cong \mathbf{C}^g/\Gamma$ for some discrete subgroup $\Gamma \subset (\mathbf{C}^g, +)$. □

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Corollary 5.1.3 *Every compact connected complex Lie group is commutative.*

Remark 5.1.4 A quotient $\mathbf{C}^n/\Gamma$ by a discrete subgroup Γ is compact (i.e. a torus) if and only if Γ generates $\mathbf{C}^n$ as a real vector space.

Definition 5.1.5 A connected commutative complex Lie group G is called a *semi-torus* if G admits a short exact sequence

$$1 \to \left(\mathbf{C}^*\right)^k \to G \to A \to 1,$$

where A is a compact complex torus and $k \in \mathbf{Z}^+$.

Definition 5.1.6 A complex Lie group G is called *reductive* if there exists a compact subgroup $K \subset G$ such that G is the smallest complex Lie subgroup of G containing K.

Definition 5.1.7 Let G be a complex Lie group. A "G-module" is a complex vector space V on which the group G acts via a holomorphic Lie group homomorphism from G to $GL(V)$.

Proposition 5.1.8 *Let G be a connected commutative complex Lie group. Then the following conditions are equivalent*:

(i) *G is a semi-torus.*
(ii) *Every linear representation of G is* completely reducible, *i.e., every short exact sequence of G-modules splits.*
(iii) *G admits no non-zero holomorphic Lie group homomorphism to the additive group $(\mathbf{C}, +)$.*
(iv) *G is isomorphic to a quotient $\mathbf{C}^n/\Gamma$ where Γ is a discrete subgroup of $\mathbf{C}^n$ generating $\mathbf{C}^n$ as a complex vector space.*
(v) *G is reductive.*

Proof (i) $\Longrightarrow$ (iii): It suffices to show that neither $\mathbf{C}^*$ nor a compact complex torus admits a non-trivial holomorphic group homomorphism to $(\mathbf{C}, +)$. If $\phi : \mathbf{C}^* \to \mathbf{C}$ is a group homomorphism, then $\phi(e^{2\pi i x}) = 0$ for all $x \in \mathbf{Q}$, because $(\mathbf{C}, +)$ has no elements of finite order. If ϕ is a *holomorphic* group homomorphism, then $\phi \equiv 0$ due to the identity principle, since $\{e^{2\pi i x}; x \in \mathbf{Q}\}$ is not discrete. For a compact complex torus every holomorphic map to $\mathbf{C}$ is constant by the maximum principle.

(iii) $\Longrightarrow$ (iv): As a connected commutative complex Lie group, G can be written as a quotient of some $(\mathbf{C}^n, +)$ by a discrete subgroup Γ. Let $\langle\Gamma\rangle_{\mathbf{C}}$ be the complex vector subspace of $\mathbf{C}^n$ generated by Γ. Then there is a surjective holomorphic Lie group homomorphism from G to the complex quotient vector space $\mathbf{C}^g/\langle\Gamma\rangle_{\mathbf{C}}$. Since every homomorphism of complex Lie groups from G to $(\mathbf{C}, +)$ is constant, it follows that $\langle\Gamma\rangle_{\mathbf{C}} = \mathbf{C}^n$, i.e. Γ generates $\mathbf{C}^n$ as a complex vector space.

(iv) $\Longrightarrow$ (i): Let V denote the real vector subspace of $\mathbf{C}^n$ generated by Γ and let $k = \dim_{\mathbf{R}} W$ with $W = V/(V \cap iV)$. Choose $v_1, \ldots, v_k \in \Gamma$ such that their im-

ages in W are $\mathbf{R}$-linearly independent. Let U denote the complex vector subspace generated by $v_1, \ldots, v_k$. Then the short exact sequence of vector spaces

$$0 \to U \to \mathbf{C}^n \to \mathbf{C}^n/U \to 0$$

induces a short exact sequence of complex Lie groups

$$1 \to \left(\mathbf{C}^*\right)^k \to G \to A \to 1,$$

where A is a compact complex torus.

(iv) $\Longrightarrow$ (v): If Γ generates $\mathbf{C}^n$ as a complex vector space, then $G = \mathbf{C}^n/\Gamma$ is the smallest complex Lie group containing $K = \langle\Gamma\rangle_{\mathbf{R}}/\Gamma$ which is compact, because it is homeomorphic to some $(S^1)^n$ (with $n = \operatorname{rank}(\Gamma)$).

(v) $\Longrightarrow$ (ii): Let M be a G-module with submodule N. Then we may equip the complex vector space M with some hermitian scalar product. Using the Haar measure on the compact group K, we may assume that this inner product on M is K-invariant. Let N' denote the orthogonal complement of N in M. Since the inner product is hermitian, this is a complex vector subspace which is K-invariant. Thus the group of all $g \in G$ stabilizing N' is a complex Lie subgroup of G containing K, hence all of G. Therefore we have seen that the sequence $0 \to N \to M \to M/N \to 0$ splits. (This line of argument is known as "Weyl's unitary trick".)

(ii) $\Longrightarrow$ (iii): If there is a surjective holomorphic Lie group homomorphism $\rho : G \to (\mathbf{C}, +)$, then the linear representation of G given by

$$g \mapsto \begin{pmatrix} 1 & \rho(g) \\ 0 & 1 \end{pmatrix}$$

is not completely reducible. □

Corollary 5.1.9 *A semi-torus A contains only countably many sub-semi-tori.*

Proof We may write A as $\mathbf{C}^g/\Gamma$. For each sub-semi-torus $S \subset A$ there is a vector subspace $V \subset \mathbf{C}^g$ such that $V/(V \cap \Gamma)$ is a semi-torus. By the above proposition V is generated as a complex vector space by $V \cap \Gamma$. Hence S is determined by $V \cap \Gamma$ and the assertion follows from the fact that $\Gamma \cong \mathbf{Z}^g$ contains only countably many subgroups. □

Corollary 5.1.10 *Let*

$$1 \to A \to B \to C \to 1$$

be a short exact sequence of connected commutative complex Lie groups. If both A and C are semi-tori, so is B.

Proof We use the equivalence (i) $\Longleftrightarrow$ (iii) of Proposition 5.1.8. Every holomorphic Lie group homomorphism from A to $(\mathbf{C}, +)$ is zero, because A is semi-torus. It follows that every holomorphic Lie group homomorphism from B to $(\mathbf{C}, +)$ is factored through $B/A = C$ and is therefore zero. Hence B must be a semi-torus. □

Corollary 5.1.11 *Let G be a complex semi-torus and let $\phi : G \to H$ be a surjective holomorphic group homomorphism onto some complex Lie group H. Then H is a semi-torus, too.*

Proof Due to Proposition 5.1.8 (iii) every holomorphic Lie group homomorphism from G to $(\mathbf{C}, +)$ is zero. It follows that every Lie group homomorphism $f : H \to (\mathbf{C}, +)$ must be zero by considering the composite homomorphism $f \circ \phi$. Now G being a semi-torus forces H to be connected and commutative. Hence Proposition 5.1.8 implies that H is a semi-torus. □

For a general commutative complex connected Lie group one has the following structure theorem (Morimoto [65]):

Proposition 5.1.12 *Let G be a connected commutative complex Lie group. Then there exists a complex vector space V and a semi-torus S such that $G \cong (V, +) \times S$.*

Proof If G is not a semi-torus, there exists a non-trivial holomorphic Lie group homomorphism $\rho : G \to (\mathbf{C}, +)$. Let $G_1 = \operatorname{Ker} \rho$. Then $G \cong (\mathbf{C}, +) \times G_1$.

Hence the statement can be proved by induction on $\dim G$. □

Remark There are many related notions to be found in the literature. Semi-tori on which all holomorphic functions are constant are often called "Cousin groups" or "(H,C)-groups" or "toroidal groups". The name "quasi-torus" means sometimes "semi-torus" and sometimes "semi-torus without non-constant holomorphic functions".

In Abe–Kopfermann [01] a semi-torus A is called a "quasi-abelian variety" if there exists a semi-abelian variety (see Sect. 5.1.4 below) B such that A and B are isomorphic as complex Lie groups.

5.1.2 Characteristic Subgroups of Complex Semi-tori

We give some properties on special subgroups of a semi-torus.

Theorem 5.1.13 *Let S be a complex semi-torus. Then we have the following facts*:

(i) *There exists a unique sequence of real Lie subgroups $\{e\} \subset T \subset L \subset K \subset S$ such that*

- *K is a maximal compact subgroup,*
- *L is a maximal connected complex Lie subgroup with compact closure,*
- *T is a maximal connected compact complex Lie subgroup.*

(ii) *There is a non-constant holomorphic function on S if and only if K is larger than the closure of L.*

(iii) *Every holomorphic function is constant along each L-orbit.*

(iv) *Every connected compact complex analytic subset $Z \subset S$ is contained in a T-orbit.*
(v) *There exists a compact complex analytic subset $Z \subset S$ with* $\dim Z > 0$ *if and only if $T \neq \{e\}$.*

Proof If C_1 and C_2 are two compact subgroups in a commutative Lie subgroup, so is $C_1 + C_2 = \{x + y : x \in C_1, y \in C_2\}$. Thus it is clear that there exists such maximal subgroups K, L and T. Since the universal covering of L is some complex vector space, it is clear that every bounded holomorphic function on L is constant due to Liouville's Theorem. However, by assumption every continuous function on S restricts to a bounded function on L. Thus every holomorphic function on S is constant along the orbits of L and its closure $\bar{L}$. If $\bar{L} = K$, it follows that every holomorphic function on S is constant, for by the definition of a semi-torus, S is the smallest complex subspace of S containing K. On the other hand, if $\bar{L} \neq K$, then there is a proper complex Lie subgroup H of S containing $\bar{L}$ as a maximal compact subgroup. Then $S/H \cong (\mathbf{C}^*)^d$ for some $d > 0$, and hence there is a non-constant holomorphic function on S.

If Z is a connected compact complex subspace of S, then

$$W_n = \left\{(a_1 - b_1) + (a_2 - b_2) + \cdots + (a_n - b_n);\, a_i, b_i \in Z\right\}$$

defines an ascending sequence of compact connected analytic subsets of S. This sequence stabilizes eventually, i.e., there is a number N such that $W_n = W_N$ for all $n \geqq N$. However, $W_N = W_{2N}$ implies that W_N is a subgroup. Therefore

$$Z - a \subset W_1 \subset W_N \subset T$$

for every $a \in Z$. Thus Z is contained in a T-orbit. □

Using this theorem, it is easy to provide examples of semi-tori which admit neither non-constant holomorphic functions nor compact complex analytic subspaces of positive dimension. Let $(a, b) \in \mathbf{C}^2 \setminus \mathbf{R}^2$ and let $S = \mathbf{C}^2/\Gamma$ be a semi-torus with

$$\Gamma = \left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} a \\ b \end{pmatrix} \right\rangle_{\mathbf{Z}}.$$

Let $\pi : \mathbf{C}^2 \to S$ be the natural projection. Then K equals the image (under π) of the $\mathbf{R}$-vector subspace of $\mathbf{C}^2$ generated by Γ. Then L is the π-image of the complex line generated by the vector $(\Im a, \Im b)$. For dimension reasons there are only two possibilities:

(i) If L is compact, then $L = T$ and $S \cong (\mathbf{C}^*) \times T$.
(ii) If L is non-compact, then $\dim_{\mathbf{R}} \bar{L} > \dim_{\mathbf{R}} L$ and $\dim_{\mathbf{C}} L > \dim_{\mathbf{C}} T$.
For dimension reasons it follows that L is dense in K and that $T = \{e\}$. Hence in this case S is a semi-torus without either non-constant holomorphic functions or positive-dimensional compact analytic subsets.

So, when is L compact? The subgroup L is isomorphic to the complex line generated by $(\Im a, \Im b)$ divided by its intersection with Γ. Hence L is compact if and

only if this intersection has $\mathbf{Z}$-rank two. If $[\Im a, \Im b] \notin \mathbf{P}^1(\mathbf{Q})$, then the intersection of this complex line with $\mathbf{Z}^2$ is trivial. Therefore it follows that the intersection with Γ has $\mathbf{Z}$-rank at most one. This implies

Proposition 5.1.14 *Let $(a, b) \in \mathbf{C}^2 \setminus \mathbf{R}^2$ with $[\Im a, \Im b] \notin \mathbf{P}^1(\mathbf{Q})$. Let $S = \mathbf{C}^2/\Gamma$ with*

$$\Gamma = \left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} a \\ b \end{pmatrix} \right\rangle_{\mathbf{Z}}.$$

Then S is a semi-torus, every holomorphic function on S is constant and every compact complex analytic subset of S is finite.

5.1.3 Holomorphic Functions

Definition 5.1.15 A complex manifold X is said to be *holomorphically separable* iff for every $p, q \in X$ with $p \neq q$ there exists a holomorphic function f on X with $f(p) \neq f(q)$.

Corollary 5.1.16 *A semi-torus G is holomorphically separable iff it is isomorphic (as a complex Lie group) to $(\mathbf{C}^*)^g$.*

Proof In the notation of Theorem 5.1.13, G being holomorphically separable implies that $L = \{e\}$. It follows that K is totally real. Hence $\dim_{\mathbf{R}} K \leqq \dim_{\mathbf{C}} G$. Now $G \cong \mathbf{C}^g/\Gamma$ where Γ is a discrete subgroup. We have $\operatorname{rank}_{\mathbf{Z}} \Gamma \leqq g$, because $\operatorname{rank}_{\mathbf{Z}} \Gamma = \dim_{\mathbf{R}} K$. Since Γ generates $\mathbf{C}^g$ as a complex vector space, it follows that $\operatorname{rank}_{\mathbf{Z}} \Gamma = g$ and that moreover $\Gamma = \mathbf{Z}^g$ after some change of coordinates. Thus $G \cong (\mathbf{C}/\mathbf{Z})^g \cong (\mathbf{C}^*)^g$.

The other direction is obvious. □

Corollary 5.1.17 *Let G be a connected commutative complex Lie group which is holomorphically separable. Then $b_1(G) \leqq \dim_{\mathbf{C}} G$.*

(Here $b_1(G)$ denotes the first Betti number. This number is equal to the dimension $\dim H^1(G, \mathbf{R})$ of the first de Rham cohomology group, but also equal to the $\mathbf{Z}$-rank of the abelianization of the fundamental group $\pi_1(G)$.)

Proof There is a semi-torus A such that $G \cong \mathbf{C}^d \times A$ for some $d \in \mathbf{Z}^+$ (Proposition 5.1.12). Without loss of generality we may assume that $A = G$, i.e., G is a semi-torus.

We use Theorem 5.1.13. The assumption that G is holomorphically separable implies that $L = \{e\}$, which in turn implies that K is totally real in G. Hence $\dim_{\mathbf{R}} K \leqq \dim_{\mathbf{C}} G$. This yields the assertion, because $b_1(G) = b_1(K) = \dim_{\mathbf{R}} K$. □

Corollary 5.1.18 *Every semi-torus G is isomorphic to a direct product of some $(\mathbf{C}^*)^k$ and a semi-torus H on which every holomorphic function is constant.*

Proof Let H denote the set of all $g \in G$ such that $f(e) = f(g)$ for every holomorphic function f on G. We claim that H is a closed complex Lie subgroup of G. It is a closed analytic subset because $H = \bigcap_f f^{-1}(f(e))$. Let $h, \tilde{h} \in H$. In order to verify that H is a subgroup, we have to show that $h^{-1}\tilde{h} \in H$. To every holomorphic function f on G we define another holomorphic function $f_{\tilde{h}} : x \mapsto f(x^{-1}\tilde{h})$. Then

$$f\left(h^{-1}\tilde{h}\right) = f_{\tilde{h}}(h) = f_{\tilde{h}}(e) = f(\tilde{h}) = f(e).$$

Hence H is a closed complex Lie subgroup of G. The quotient G/H is a holomorphically separable semi-torus and therefore isomorphic to $(\mathbf{C}^*)^k$. The fiber is commutative and connected. Hence $H \cong (\mathbf{C}^*)^m \times H_0$, where H_0 is semi-torus. Defining J as the set of all $g \in H_0$ for which $f(g) = f(e)$ for all holomorphic functions f on H_0, we obtain a fibration $G/J \to G/H$, where $G/H \cong (\mathbf{C}^*)^k$ and $H/J \cong \mathbf{C}^m \times (\mathbf{C}^*)^l$. But now, $G/J \cong \mathbf{C}^n/\Gamma$, where $n = m + k + l$ and $\Gamma = \langle \gamma_1, \ldots, \gamma_s \rangle$ (with $s = k + l$). The structures of G/H and J/H easily imply that the γ_i are $\mathbf{C}$-linearly independent. But then G/J is isomorphic to $(\mathbf{C}^*)^{k+l} \times \mathbf{C}^m$ and therefore holomorphically separable. As a consequence, we have $J = H$, i.e. every holomorphic function on H is constant. Finally, the short exact sequence $1 \to H \to G \to G/H \to 1$ is easily seen to split. □

5.1.4 Semi-abelian Varieties

We are going to introduce *semi-abelian varieties*—the algebraic analogue of the complex analytic concept of a "*semi-torus*".

We begin with recalling the definition of an abelian variety. Recall that an algebraic group is a group which is also an algebraic variety in such a way that both structures are compactible.

Definition 5.1.19 An *abelian variety* is an algebraic group which is projective, regarded as a variety.

An abelian variety is necessarily commutative (this follows from Proposition 5.1.2). One-dimensional abelian varieties are called *elliptic curves*.

Let k be an algebraically closed field (e.g., $\mathbf{C}$). Let G_a denote the additive group $(k, +)$ of all elements of k, and let G_m denote the multiplicative group $(k^*, \times)$ of non-zero elements of k.

An algebraic group G which is defined over some field k is called a "k-group".

Definition 5.1.20 A commutative algebraic k-group A is called a *semi-abelian variety* if there is a short exact sequence of algebraic groups defined over k

$$1 \to H \to A \to A_0 \to 1,$$

where A_0 is an abelian variety and $H \cong G_m^l$.

Proposition 5.1.21 *A complex semi-abelian variety is a semi-torus* (*as a complex Lie group*).

Proof This follows from the definition combined with Corollary 5.1.10. □

Definition 5.1.22 An algebraic k-group G is called *unipotent* if there exists a sequence of normal algebraic subgroups $G_j, 0 \leqq j \leqq n$, such that $G_0 = \{e\}$, $G_n = G$, and that all the quotients G_{j+1}/G_j (for $0 \leqq j < n$) are isomorphic to the additive group G_a.

Proposition 5.1.23 *An algebraic k-group G is semi-abelian if and only if G is connected and commutative and every unipotent algebraic subgroup of G is trivial.*

Proof If G satisfies the condition, by the Theorem of Chevalley (Chevalley [60]) there is a linear normal subgroup H of G such that G/H is an abelian variety. Now H is commutative, linear and contains no unipotent subgroup. It follows that $H \cong G_m^l$ for some $l \in \mathbf{Z}^+$.

Conversely, the condition $H \cong G_m^l$ implies that $\mathrm{Aut}(H) \cong GL_l(\mathbf{Z})$ is discrete. Since G is connected, it follows that the natural action of G on H given by conjugation is trivial. Therefore H is central and G is commutative. Furthermore, G contains no unipotent subgroup, because every morphism of algebraic groups from a unipotent group to G_m or an abelian variety is constant. □

Remark 5.1.24 If k is not algebraically closed, a k-group A is called a "semi-abelian variety" if it fulfills the above definition after extending the scalars to an algebraic closure $\bar{k}$. For example,

$$S^1 = \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix}; a, b \in \mathbf{R}, a^2 + b^2 = 1 \right\} = \left\{ \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}; t \in \mathbf{R} \right\}$$

is a semi-abelian variety defined over the field of real numbers $\mathbf{R}$, because $S^1 \cong G_m$ after extending the scalars to $\mathbf{C}$:

$$\begin{aligned} &\left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix}; a, b \in \mathbf{C}, a^2 + b^2 = 1 \right\} \\ &\quad = \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix}; a, b \in \mathbf{C}, (a + ib)(a - ib) = 1 \right\} \\ &\quad \cong (\mathbf{C}^*, \times). \end{aligned}$$

Remark 5.1.25 By definition a semi-abelian variety contains no algebraic subgroup isomorphic to $(\mathbf{C}, +)$. However, it may contain a complex Lie subgroup isomorphic to $(\mathbf{C}, +)$. For example, $z \mapsto (e^z, e^{iz})$ embeds $(\mathbf{C}, +)$ as a closed complex Lie subgroup into $\mathbf{C}^* \times \mathbf{C}^*$.

Proposition 5.1.26 *Let A be a semi-abelian variety and let B be a connected algebraic subgroup of A. Then both B and the quotient algebraic group A/B are semi-abelian varieties.*

Proof Since A is commutative and has no unipotent subgroup, so is B. Thus B is a semi-abelian variety. Moreover, every algebraic subgroup of A is a semi-abelian variety. The quotient A/B is an algebraic group, because A is an algebraic group and B is a normal algebraic subgroup (see Chevalley [56]).

We will now show that A/B is a semi-abelian variety. Let U be a unipotent algebraic subgroup of A/B and let $\pi : A \to A/B$ denote the natural projection. Then $S = \pi^{-1}(U)$ is an algebraic subgroup of A and therefore a semi-abelian variety. However, as a semi-abelian variety S admits a short exact sequence of the form

$$1 \to G_m^d \to S \to A \to 1,$$

where A is an abelian variety. Every morphism of algebraic groups from either G_m or A to a unipotent group is trivial. Hence every morphism from S (or any semi-abelian variety) to a unipotent algebraic group must be constant. It follows that $U = \{e\}$, i.e., A/B cannot contain any non-trivial unipotent algebraic subgroup. Therefore A/B is a semi-abelian variety. □

Remark 5.1.27 There do exist complex algebraic groups which are not semi-abelian varieties despite being semi-tori as complex Lie groups. See Example 5.1.44.

5.1.5 Presentations

Every semi-torus A admits a short exact sequence of commutative complex Lie groups

$$1 \to T \to A \to M \to 1, \tag{5.1.28}$$

where M is a compact complex torus and $T \cong (\mathbf{C}^*)^t$ for some $t \in \mathbf{N}$. Such a short exact sequence is called a *presentation* for the semi-torus A.

Similarly, one defines a presentation for a semi-abelian variety with an abelian variety M.

The same complex semi-torus M may admit several such exact sequences as (5.1.28) which may be quite different. For instance, let τ be an arbitrary complex number with $\Im\tau > 0$. Let Λ be the discrete subgroup of $\mathbf{C}^2$ generated by

$$\begin{pmatrix}1\\0\end{pmatrix}, \quad \begin{pmatrix}0\\1\end{pmatrix}, \quad \begin{pmatrix}i\\\tau\end{pmatrix}. \tag{5.1.29}$$

Then $A = \mathbf{C}^2/\Lambda$ is a complex semi-torus and the natural projection of $\mathbf{C}^2 = \mathbf{C} \times \mathbf{C}$ onto the first and the second factors induce respectively exact sequences

of the forms

$$0 \to \mathbf{C}^* \to A \to \mathbf{C}/\langle 1, i\rangle_{\mathbf{Z}} \to 0,$$
$$0 \to \mathbf{C}^* \to A \to \mathbf{C}/\langle 1, \tau\rangle_{\mathbf{Z}} \to 0.$$

There are also examples where even the algebraic dimension $a(M)$ of M jumps. Set $\mathbf{Z}[i] = \{m + ni : m, n \in \mathbf{Z}\} = \langle 1, i\rangle_{\mathbf{Z}}$ and let $E = \mathbf{C}/\mathbf{Z}[i]$. For $\tau \in \mathbf{C}$ we define a compact complex torus A_τ as the quotient of $\mathbf{C}^2$ by the lattice

$$\begin{pmatrix}1\\0\end{pmatrix}, \quad \begin{pmatrix}i\\0\end{pmatrix}, \quad \begin{pmatrix}0\\1\end{pmatrix}, \quad \begin{pmatrix}\tau\\i\end{pmatrix}.$$

The projection $(z_1, z_2) \mapsto z_1$ induces a fibration of A_τ where both base and fiber are biholomorphic to $E = \mathbf{C}/\mathbf{Z}[i]$. Hence $a(A_\tau) \geqq 1$. Now $a(A_\tau) = 2$ iff A_τ is an abelian variety. By Poincaré's complete reducibility theorem this is the case iff A_τ is isogenous to $E \times E$. An easy explicit calculation shows that such an isogeny exists iff $\tau \in \mathbf{Q}[i]$. Therefore we see

$$a(A_\tau) = \begin{cases} 2 & \text{if } \tau \in \mathbf{Q}[i], \\ 1 & \text{if } \tau \notin \mathbf{Q}[i]. \end{cases}$$

Now, let $\alpha \notin \mathbf{Q}[i]$ and let M be the semi-torus given as the quotient of $\mathbf{C}^3$ by the discrete subgroup generated by

$$\begin{pmatrix}1\\0\\0\end{pmatrix}, \quad \begin{pmatrix}0\\1\\0\end{pmatrix}, \quad \begin{pmatrix}i\\i\\0\end{pmatrix}, \quad \begin{pmatrix}0\\0\\1\end{pmatrix}, \quad \begin{pmatrix}0\\\alpha\\i\end{pmatrix}.$$

Then $(z_1, z_2, z_3) \mapsto (z_2, z_3)$ (resp. $(z_1, z_2, z_3) \mapsto (z_1, z_3)$) induces a presentation of M with a two-dimensional compact complex torus of algebraic dimension 1 (resp. 2).

5.1.6 Presentations of Semi-abelian Varieties

Semi-abelian varieties admit exactly one algebraic presentation.

Proposition 5.1.30 *Let G be a semi-abelian variety, let $L_i \cong (\mathbf{C}^*)^{n_i}$ for some $n_1, n_2 \in \mathbf{N}$ and let A_i $(i = 1, 2)$ be abelian varieties such that there exist short exact sequences of algebraic groups*

$$1 \to L_i \xrightarrow{j_i} G \xrightarrow{\pi_i} A_i \to 1.$$

Then $j_1(L_1) = j_2(L_2)$ and $A_1 \cong A_2$.

Proof Every morphism from $\mathbf{C}^*$ to an abelian variety is constant. Hence $\pi_i \circ j_k$ are constant for all $i \in \{1, 2\}$, $k \in \{1, 2\}$. This implies the assertions. □

5.1.7 Inequivalent Algebraic Structures

There are complex semi-abelian varieties which are not isomorphic as algebraic groups although they are isomorphic as complex Lie groups. To see this, consider the example described by (5.1.29). This is a complex semi-torus A admitting two different presentations. Each presentation

$$1 \to \mathbf{C}^* \to A \xrightarrow{\pi_i} E_i \to 1$$

can be used to define an algebraic structure on A such that $\pi_i : A \to E_i$ becomes an algebraic morphism. In the example given by (5.1.29) there are two such presentations for which E_1 and E_2 are non-isogenous elliptic curves. Observe that every algebraic morphism from $\mathbf{C}^*$ to an elliptic curve is constant. Because of this reason, for a fixed algebraic structure on A at most one of the projection maps π_i can be an algebraic morphism if the E_i are non-isogenous elliptic curves. It follows that there are at least two non-isomorphic structures as semi-abelian varieties on the semi-torus defined by (5.1.29).

5.1.8 Choice of Presentation

Let $A = \mathbf{C}^g/\Gamma$ be a semi-torus. Then $K = \langle\Gamma\rangle_{\mathbf{R}}/\Gamma$ is the maximal compact subgroup. How can we choose a presentation? We have to find an embedding of $(\mathbf{C}^*)^d$ into A as a closed Lie subgroup I such that A/I is compact. Let V denote the maximal complex vector subspace of $\hat{K} = \langle\Gamma\rangle_{\mathbf{R}}$. Then we have to find $\mathbf{C}$-linearly independent elements $\gamma_1, \ldots, \gamma_d \in \Gamma$ (where d is the real codimension of $\hat{K}$ in $\mathbf{C}^g$), such that $\langle\gamma_1, \ldots, \gamma_d\rangle_{\mathbf{C}} \cap V = \{0\}$.

Let A be a semi-torus with presentation

$$1 \to L = \left(\mathbf{C}^*\right)^g \to A \to M \to 1.$$

Given a holomorphic or regular function f on A we may regard its restriction to a fiber of $\pi : A \to M$ and its Laurent development there. This yields

Theorem 5.1.31 *Under the above assumption, let f be a holomorphic function on A and let $\chi = \mathrm{Hom}(L, \mathbf{C}^*) \cong \mathbf{Z}^g$. Then*

$$f(x) = \sum_{v \in \chi} f_v(x).$$

Here, for each v there are an associated $\mathbf{C}^*$-principal bundle $E_v \to M$, an associated line bundle L_v and a section $s_v \in \Gamma(M, L_v)$ such that $f_v(x) = s_v(v(x))$; this sum is finite if A is a semi-abelian variety, the presentation is algebraic and f is regular.

Corollary 5.1.32 *Let A be a semi-abelian variety. Then each holomorphic function on A can be approximated by regular functions with respect to the topology of locally uniform convergence.*

Remark 5.1.33 This is not true for arbitrary commutative algebraic groups. There is an example of an algebraic group on which every regular function is constant although it is biholomorphic to $\mathbf{C}^* \times \mathbf{C}^*$ (Example 5.1.44).

Corollary 5.1.34 *Let A be a semi-abelian variety and let H be the set of all $g \in A$ such that $f(g) = f(e)$ for every holomorphic function f on A. Then H is an algebraic subgroup.*

5.1.9 Construction of Semi-tori via Presentations

Let us fix a compact complex torus T and a natural number n. We want to construct a semi-torus G for which there is a presentation

$$1 \to \left(\mathbf{C}^*\right)^g \to G \to T \to 1.$$

Theorem 5.1.35 *Let $L = (\mathbf{C}^*)^g$ and $U = (S^1)^g \subset L$. For a holomorphic L-principal bundle $E \to T$ the following properties are equivalent*:

(i) *The bundle is topologically trivial.*
(ii) *The bundle admits a flat holomorphic connection.*
(iii) *There exists a group homomorphism $\rho : \pi_1(T) \to U$ such that $G \to T$ is isomorphic to $\tilde{T} \times_\rho L \to T$. Here $\tilde{T}$ denotes the universal covering of T and $\tilde{T} \times_\rho L$ denotes the quotient of $\tilde{T} \times L$ under the equivalence relation for which two pairs (x, p) and (y, q) are equivalent iff there exists an element $\gamma \in \pi_1(T)$ such that $y = \gamma \cdot x$ and $q = \rho(\gamma^{-1})(p)$.*
(iv) *There exists a complex Lie group structure on E for which $\pi : E \to T$ is a morphism of complex Lie groups which coincides with the given group structure on $\pi^{-1}(e_T) \cong L$.*

Proof The equivalence of the first two properties is due to Hodge theory. The existence of a flat connection allows the lifting of holomorphic vector fields. A group structure yields a short exact sequence of Lie algebras which splits (because everything is commutative) and this splitting induces a flat connection. □

This shows that there exist many semi-tori and many semi-abelian varieties: For a compact complex torus T the exponential sequence yields an exact sequence of cohomology groups,

$$0 \to H^1(T, \mathbf{Z}) \to H^1(T, \mathscr{O}) \to H^1\left(T, \mathscr{O}^*\right) \to H^2(T, \mathbf{Z}) \to \cdots,$$

induced from $0 \to \mathbf{Z} \to \mathscr{O}_T \to \mathscr{O}_T^* \to 0$, where $\mathscr{O}^*$ denotes the sheaf of all units of the structure sheaf $\mathscr{O}_T$. Here $H^1(T, \mathscr{O}^*) \cong \mathrm{Pic}(T)$ parameterizes $\mathbf{C}^*$-principal bundles over T. The groups $H^k(T, \mathbf{Z})$ are countable and $H^1(T, \mathscr{O})$ is a complex vector space of dimension $\dim T$ for every compact complex torus. Thus $\mathrm{Pic}(T)$ is a complex Lie group of dimension $\dim T$, i.e., there are many $\mathbf{C}^*$-principal bundles on T and as a consequence many semi-tori arising as extensions of T by $\mathbf{C}^*$ resp. $(\mathbf{C}^*)^k$.

5.1.10 Morphisms and GAGA[1]

Some maps are automatically group homomorphisms:

Theorem 5.1.36 *Let G be a semi-torus without non-constant holomorphic functions, let H be a complex Lie group, and let $f : G \to H$ be a holomorphic map with $f(e_G) = e_H$. Then f is a group homomorphism.*

There is an algebraic analogue which holds even if there are non-constant holomorphic functions:

Theorem 5.1.37 *Let G and H be semi-abelian varieties over a field k of characteristic zero and let $f : G \to H$ be a morphism of k-varieties with $f(e_G) = e_H$. Then f is a morphism of algebraic groups.*

We prove both theorems together.

Proof First we discuss the case where G is compact. Here we can prove the statement in the same way for both the analytic and the algebraic category. We define an auxiliary map $\zeta : G \times G \to H$ given by

$$\zeta(x, y) = f(x) - f(y) - f(x - y). \tag{5.1.38}$$

This map measures to which degree f fails to be a homomorphism of groups: f is a group homomorphism if and only if ζ is the constant map with value e_H. The condition $f(e_G) = e_H$ implies that $\zeta(x, e_G) = e_H = \zeta(e_G, x)$ for all $x \in G$. Now let Ω be a Stein open (resp. an affine Zariski open) neighborhood of e_H in H. Then $\zeta^{-1}(H \setminus \Omega)$ is closed in $G \times G$. Because G is compact, the projection from $G \times G$ onto its first factor is a proper map. It follows that there is an open neighborhood W of e_G in G such that $\zeta(w, x) \in \Omega$ for all $w \in W$, $x \in G$. Since G is compact, while Ω is Stein, it is clear that the map $x \mapsto \zeta(w, x)$ must be constant for each $w \in W$. Since $\zeta(w, e_G) = e_H$, we may deduce that $\zeta(w, x) = e_H$ for all $w \in W$ and $x \in G$. By the identity principle, we may deduce that $\zeta \equiv e_H$ which is equivalent to f being a homomorphism of groups.

It remains to discuss the non-compact case in which we will need different techniques for the two categories (analytic vs. algebraic).

Let us first discuss the algebraic case. Let L be the maximal connected linear subgroup of G. Then $L \cong G_m^g$ (possibly after applying some finite field extension) and G/L is an abelian variety. Similarly, there is a maximal connected linear subgroup M of H and the quotient H/M is an abelian varieties. An abelian variety cannot contain rational curves. Therefore every morphism from the multiplicative group G_m to an abelian variety must be constant. For this reason f induces a mor-

[1] Géometrie Algébraique et Geometrie Analytique (phrase due to J.P. Serre).

phism F between the abelian varieties A and B such that we have a commutative diagram:

$$\begin{array}{ccc} G & \xrightarrow{f} & H \\ \downarrow & & \downarrow \\ G/L & \xrightarrow{F} & H/M \end{array}$$

Now F inherits the property $F(e) = e$; therefore F is a morphism of algebraic groups. Defining ζ as in (5.1.38), the fact that F is a morphism of algebraic groups implies that the image of ζ is contained in $M \cong G_m^r$. The semi-abelian variety G can be decomposed as a direct product $G \cong G_m^d \times G_0$ such that every regular function on G_0 is constant (Proposition 5.1.12 and Corollary 5.1.34). Hence ζ is constant along the $G_0 \times G_0$-orbits in $G \times G$ and therefore induced by a map $\zeta_0 : L \times L \to M$. Recall that (after some finite field extension) $L \cong G_m^g$ and $M \cong G_m^r$. Now every algebraic morphism from G_m to G_m is given as $z \mapsto az^n$ for some $a \in k$, $n \in \mathbf{Z}$. But the condition $\zeta_0(e, x) = e$ forces $a = 1$ and every morphism of the form $z \mapsto z^n$ is a homomorphism of groups.

Finally, we deal with the analytic case. Again we use the auxiliary map $\zeta : G \times G \to H$. The universal covering $\tilde{H}$ of H is a Stein manifold (quote). For each $g \in G$ we consider the map $\zeta_g : G \to H$ given by $\zeta_g(x) = \zeta(g, x)$. Each map ζ_g is homotopic to $\zeta_e \equiv e$. Hence each ζ_g induces the zero homomorphism between the fundamental groups $\pi_1(G)$ and $\pi_1(H)$. This implies that each ζ_g can be lifted to a map $\tilde{\zeta}_g : G \to \tilde{H}$. Then $\tilde{\zeta}_g$ must be constant, because $\tilde{H}$ is Stein and G does not carry non-constant holomorphic functions. It follows that

$$\zeta(g, x) = \zeta_g(x) = \zeta_g(e) = \zeta(g, e) = e$$

for all $g, x, \in G$. Hence f must be a group homomorphism. □

Corollary 5.1.39 *If two semi-abelian varieties are isomorphic as algebraic varieties, then they are isomorphic as algebraic groups.*

Let G be a group acting on a set X.

Definition 5.1.40 We say that the G-action on X is *effective* iff for every $g \in G \setminus \{e\}$ there exists an element $x \in X$ with $g \cdot x \neq x$.

Proposition 5.1.41 *Let A be a semi-torus. Let G denote the group of all automorphisms of A as a complex Lie group. Then G is countable and the natural action on $H_1(A, \mathbf{Z})$ is effective.*

Proof We realize A as a quotient $\mathbf{C}^g/\Gamma$ (where Γ is a discrete subgroup). Then every $\phi \in G$ lifts to an automorphism of $\mathbf{C}^g$ (because $\mathbf{C}^g$ is the universal covering of A or because $\mathbf{C}^g$ can be identified with the Lie algebra of A). Evidently ϕ must fix Γ. Hence the statement holds, since $\Gamma \cong \pi_1(A) \cong H_1(A, \mathbf{Z})$ and $\mathbf{C}^g = \langle \Gamma \rangle_{\mathbf{C}}$. □

For compact complex algebraic spaces there is the GAGA principle which states that every analytic object is already algebraic. In our context this means the following.

Theorem 5.1.42 *Let A be an abelian variety defined over $\mathbf{C}$. Then every closed complex Lie subgroup of A is algebraic and every holomorphic map from A to a complex algebraic variety Z is a morphism of algebraic varieties.*

There are some GAGA-type statements for non-compact Lie groups. For example, if U is a unipotent complex Lie group, then U and every connected complex Lie subgroup of U are algebraic. If ϕ is a holomorphic Lie group homomorphism from $\mathbf{C}^*$ to $\mathbf{C}^*$, then $\phi(z) = z^n$ for some $n \in \mathbf{N}$. In particular, such a ϕ is algebraic.

However, for general semi-abelian varieties most GAGA-style assertions fail. As we have already seen in Sect. 5.1.7, semi-abelian varieties which are isomorphic as complex Lie groups need not be isomorphic as algebraic groups. Still, there are some positive results.

Theorem 5.1.43 *Every holomorphic Lie group homomorphism from a reductive complex algebraic group to $\mathbf{C}^*$ is already algebraic.*

Proof Let H be a reductive complex algebraic group and let H' be its commutator group. Then every Lie group homomorphism from H to $\mathbf{C}^*$ fibers through H/H' and $S = H/H'$ is a commutative reductive algebraic group, i.e. a semi-abelian variety. Let I be the group of all $s \in S$ such that $f(s) = f(e)$ for all $s \in S$. Then I is an algebraic subgroup (Corollary 5.1.34). Furthermore, S/I is holomorphically separable, hence $S/I \cong (\mathbf{C}^*)^d$. Thus it suffices to consider the case $H \cong (\mathbf{C}^*)^d$ and in this case the assertion is clear. □

In contrast, there are many holomorphic group homomorphisms from $\mathbf{C}^*$ to semi-abelian varieties which are not algebraic: For instance, due to the inclusion $\mathbf{Z} \subset \mathbf{Z}[i]$ there is a natural holomorphic map from $\mathbf{C}/\mathbf{Z} \cong \mathbf{C}^*$ onto the elliptic curve $\mathbf{C}/\mathbf{Z}[i]$ which cannot be algebraic, because its kernel is an infinite discrete subgroup. There is a striking example of Serre illustrating the failure of GAGA for general non-compact algebraic groups.

Example 5.1.44 (Serre's Example; see Hartshorne [70], Chap. VI, Example 3.2, p. 232) There exists a complex algebraic group G fitting into a short exact sequence

$$1 \to (\mathbf{C}, +) \to G \to E \to 1,$$

where E is an elliptic curve, such that $G \cong \mathbf{C}^* \times \mathbf{C}^*$ as complex Lie group.

The complex Lie group G is a semi-torus, but G is not a semi-abelian variety, because it contains an algebraic subgroup isomorphic to $(\mathbf{C}, +)$. In particular, G and $\mathbf{C}^* \times \mathbf{C}^*$ are isomorphic as complex Lie groups, but not isomorphic as algebraic groups. One can also show that every regular function on the algebraic variety G is constant although G is Stein as a complex manifold.

5.2 Reductive Group Actions

Let X be a complex space with structure sheaf $\mathcal{O}_X$ (reduced as in Convention (xvi)). Let $\mathfrak{m}_p \subset \mathcal{O}_{X,p}$ be the maximal ideal at $p \in X$.

Proposition 5.2.1 *Let G be a complex Lie group acting effectively on an irreducible complex space X. Then every isotropy group $G_p = \{g \in G;\, g(p) = p\}$ (for $p \in X$) is holomorphically separable.*

Proof There is an induced G_p-action on the local ring $\mathcal{O}_{X,p}$ of germs of holomorphic functions about $p \in X$. If G_p is not holomorphically separable, there are two distinct elements $g, h \in G_p$ such that $f(g) = f(h)$ for every holomorphic function f on G_p. It follows that $\rho(g) = \rho(h)$ for every finite-dimensional representation $\rho : G \to GL_n(\mathbf{C})$. It follows that $g^{-1}h$ acts trivially on $\mathcal{O}_{X,p}/\mathfrak{m}_p^k$ for every $k \in \mathbf{N}$. By the identity principle this would imply that the $g^{-1}h$ induces the identity map on X. However, the G-action is assumed to be effective. Thus $f(g) = f(h)$ for every holomorphic function on G_p implies $g^{-1}h = e$, i.e., it implies $g = h$. □

Corollary 5.2.2 *Let G be a complex Lie group acting effectively on an irreducible complex space X. Assume that every holomorphic function on G is constant and* $\dim G > 0$. *Then the G-action on X has no fixed point (i.e. $G \cdot p \neq \{p\}$, $^\forall p \in X$).*

Corollary 5.2.3 *Let A be a semi-abelian variety acting algebraically and effectively on a complex algebraic variety X. Let $p \in X$. Then the connected component G_p^0 of the isotropy group $G_p = \{g \in G;\, g(p) = p\}$ which contains $\{e\}$ is isomorphic to some $(\mathbf{C}^*)^d$ with $d \in \mathbf{Z}^+$.*

In particular, the isotropy group G_p is reductive.

Proof The group G_p^0 is an algebraic subgroup of A and therefore a semi-abelian variety due to Proposition 5.1.26. In particular, G_p^0 is a semi-torus and therefore reductive (Proposition 5.1.8). Now Proposition 5.2.1 implies that G_p^0 is holomorphically separable. But every holomorphically separable semi-torus is isomorphic to some $(\mathbf{C}^*)^d$ (Corollary 5.1.16). □

Lemma 5.2.4 *Let I be a closed complex Lie subgroup of $G = (\mathbf{C}^*)^g$. Let $K = (S^1)^g$ be the maximal compact subgroup of G and let H be the smallest closed complex Lie subgroup of G containing $I \cap K$. Assume that G/I is holomorphically separable. Then $H = I$.*

Proof By construction $I \cap K$ is the maximal compact subgroup of I. Hence I/H contains no compact subgroup and therefore $I/H \cong \mathbf{Z}^r \times \mathbf{C}^s$ for some $r, s \in \mathbf{N}$. Then $\dim G/I = g - s$ and $b_1(G/I) = g + r$. Now

$$g + r = b_1(G/I) \leqq \dim_{\mathbf{C}}(G/I) = g - s,$$

because G/I is holomorphically separable (see Corollary 5.1.17). Therefore $r = s = 0$ and consequently $I = H$. □

Corollary 5.2.5 *Let G be a reductive complex Lie group acting effectively on an irreducible complex space X with a fixed point $p \in X$. Then the natural G-action on the Zariski tangent space $\mathbf{T}(X)_p$ is effective.*

Proof Let H denote the set of all $g \in G$ inducing the identity map on $\mathbf{T}(X)_p$. There are canonical isomorphisms,

$$\mathbf{T}(X)_p \cong \left(\mathfrak{m}_p/\mathfrak{m}_p^2\right)^*, \qquad S^k\left(\mathfrak{m}_p/\mathfrak{m}_p^2\right) \cong \mathfrak{m}_p^k/\mathfrak{m}_p^{k+1}.$$

Hence H acts trivially on each quotient space $\mathfrak{m}_p^k/\mathfrak{m}_p^{k+1}$. Now G is reductive and the G-action on $\mathscr{O}_{X,p}$ stabilizes each $\mathfrak{m}_p^k$. It follows that for each $N \in \mathbf{N}$ there is an isomorphism of G-modules between $\mathscr{O}_{X,p}/\mathfrak{m}_p^N$ and $\mathbf{C} \oplus \bigoplus_{k<N} \mathfrak{m}_p^k/\mathfrak{m}_p^{k+1}$. It follows that H acts trivially on every quotient $\mathscr{O}_{X,p}/\mathfrak{m}_p^N$. But then H acts trivially on $\mathscr{O}_{X,p}$, which by the identity principle implies the triviality of the H-action on X. Therefore $H = \{e\}$. □

Lemma 5.2.6 *Let K be a compact real Lie group acting on a complex space X by biholomorphic transformations. Let $p \in X$ be a fixed point of the action. Then there exist an open K-invariant neighborhood W of p and a K-equivariant biholomorphic map F from W onto a closed analytic subset of an open neighborhood of the origin of the Zariski tangent space $\mathbf{T}(X)_p$.*

Proof We choose a K-invariant open neighborhood W of p on X such that W is biholomorphic to a closed analytic subset of the unit ball in some $\mathbf{C}^N$. We may assume that $\mathbf{T}(X)_p = \mathbf{T}(\mathbf{C}^N)_p$. Using any biholomorphic map F_0 from $\mathbf{C}^N$ onto $\mathbf{T}(X)_p$, we obtain an injective holomorphic map $F_1 : W \to \mathbf{T}(X)_p$ by restriction. Now we define

$$F(x) = \int_{g \in K} g^{-1} F_1(gx) d\mu(g),$$

where $d\mu$ is the Haar measure on K. Then $F : W \to \mathbf{C}^g \cong \mathbf{T}(X)_p$ is a K-invariant holomorphic map which has still maximal rank at p. Hence, by shrinking W, we may assume that $F : W \to \mathbf{T}(X)_p$ is injective. □

Theorem 5.2.7 *Let $G = (\mathbf{C}^*)^g$ act effectively with a fixed point p on a complex manifold X with $g = \dim X$. Then there exists a G-invariant open subset U of X containing p and an injective holomorphic G-equivariant map ϕ from U to an open subset of $\mathbf{T}(X)_p$ with $\phi(p) = 0$.*

Proof Let $K = (S^1)^g$ be the maximal compact subgroup of G. Using Lemma 5.2.6 we obtain a K-invariant open neighborhood W of p on X and a K-equivariant injective holomorphic map $F_1 : W \to \mathbf{T}(X)_p \cong \mathbf{C}^g$.

Set $U = G \cdot W$. We want to extend F_1 to a map F defined on all of U via

$$F(g \cdot x) = g \cdot F_1(x).$$

The problem is to verify that this map is well-defined and injective. For this, we have to show that $g \cdot x = \tilde{g} \cdot \tilde{x}$ iff

$$g \cdot F_1(x) = \tilde{g} \cdot F_1(\tilde{x}), \quad {}^{\forall}g, {}^{\forall}\tilde{g} \in G,\ x, \tilde{x} \in W.$$

Since G acts effectively on $\mathbf{T}(X)_p$ (Corollary 5.2.5), after a suitable change of coordinates, the action is given by

$$(g_1, \ldots, g_g) : (v_1, \ldots, v_g) \mapsto (g_1 v_1, \ldots, g_n v_g).$$

By shrinking W again, we may assume that $F_1(W)$ is a complete Reinhardt domain, i.e., that

$$(a_1 v_1, \ldots, a_g v_g) \in F_1(W)$$

for all $(v_1, \ldots, v_g) \in F_1(W)$ and $|a_i| \leqq 1$ $(1 \leqq i \leqq g)$. We define $G_- = \{(g_1, \ldots, g_g) \in G; |g_i| \leqq 1\}$. Also we define closed analytic subsets $A, A' \subset G \times W \times W$ as follows:

$$A = \big\{(g, x, y); g \cdot x = y\big\}, \qquad A' = \big\{(g, x, y); g \cdot F_1(x) = F_1(y)\big\}.$$

Since F_1 is injective and K-equivariant, both sets have the same intersection with $K \times W \times W$. Now $F_1(W)$ being a complete Reinhardt domain implies that $B' = A' \cap (G_- \times W \times W)$ is connected. Therefore B' is contained in the analytic Zariski closure of $A' \cap (K \times W \times W)$. It follows that $B' \subset A$, because

$$A' \cap (K \times W \times W) = A \cap (K \times W \times W) \subset A.$$

Therefore, $g \cdot x = \tilde{g} \cdot \tilde{x}$ iff

$$g \cdot F_1(x) = \tilde{g} \cdot F_1(\tilde{x}), \quad {}^{\forall}g, {}^{\forall}\tilde{g} \in G_-,\ x, \tilde{x} \in W. \tag{5.2.8}$$

Observe that (5.2.8) holds for a pair $(g, \tilde{g}) \in G \times G$ iff it holds for $(hg, h\tilde{g})$ (for any given choice of $h \in G$). For every pair $(g, \tilde{g}) \in G \times G$ there exists an element $h \in G$ such that $(hg, h\tilde{g}) \in G_- \times G_-$. Therefore (5.2.8) holds for every $g, \tilde{g} \in G$. □

Corollary 5.2.9 *Let $H = (\mathbf{C}^*)^g$ act effectively on a compact g-dimensional complex manifold X such that each H-orbit contains an H-fixed point in its closure. Then X and the H-action on X are algebraic.*

Proof For each fixed point there is an open neighborhood which is H-equivariantly isomorphic to a Zariski open subset in a complex vector space on which H acts linearly. Hence the statement holds. □

Corollary 5.2.10 *Let $H = (\mathbf{C}^*)^g$ act effectively on a compact g-dimensional complex manifold X such that each H-orbit contains an H-fixed point in its closure. Then there are only finitely many orbits in X.*

Theorem 5.2.11 *Let A be a semi-torus of dimension g acting effectively on a compact complex manifold X of dimension g. Assume that all isotropy groups are reductive. Then there are only finitely many orbits.*

Proof Fix $p \in X$. The isotropy group A_p is reductive, but also holomorphically separable (Corollary 5.2.1). Therefore its connected component is isomorphic to some $(\mathbf{C}^*)^d$. Using local linearization (Lemma 5.2.6) we deduce that there are only finitely many orbits in a neighborhood of p. From compactness the statement follows. □

Lemma 5.2.12 *Let W be a Stein complex space on which a reductive complex Lie group H acts. Let V be a complex vector space, let $Z \subset W$ be an invariant closed analytic subset and let $f_0 : Z \to V$ be an H-invariant holomorphic map. Then f_0 extends to an H-invariant holomorphic map $f : W \to V$.*

Proof Let $f_1 : W \to V$ be an arbitrary (i.e., not necessarily invariant) holomorphic extension (which exists, because W is Stein). Let K be a maximal compact subgroup of T with normalized Haar measure $d\mu$. Then we can define f by

$$f(x) = \int_K f_1(k \cdot x) d\mu(k).$$

□

Lemma 5.2.13 *Let W be an irreducible affine variety on which a reductive commutative complex Lie group $T = (\mathbf{C}^*)^g$ acts with finitely many orbits. Let S_i, $i = 1, 2$ be two T-orbits. Then either $S_1 = S_2$ or for any choice of points $p_i \in S_i$ $(i = 1, 2)$ the isotropy groups of T at p_1 and p_2 have different Lie algebras.*

Remark 5.2.14 Because T is commutative, all points in the same orbit have the same isotropy group. Hence the choice of $p_i \in S_i$ $(i = 1, 2)$ does not matter. In the sequel, given an orbit S_i the Lie algebra of the isotropy group of a point $p \in S_i$ is simply called "the isotropy Lie algebra of the orbit S_i".

Proof Let S_1 and S_2 be two orbits with the same isotropy Lie algebra. Note that this implies $\dim S_1 = \dim S_2$. Let Y denote the closure of $S_1 \cup S_2$ in W. Let H denote the connected identity component of the isotropy group for S_1. By our assumption, H acts trivially on Y. Now we choose a holomorphic embedding $i : Y \hookrightarrow \mathbf{C}^N$. Since H acts trivially on Y, this map i is H-invariant. Thus we can extend it to an H-invariant holomorphic map $F : W \to \mathbf{C}^N$ (using Lemma 5.2.12). We observe that

$$\dim F(Y) = \dim i(S_1) = \dim T - \dim H.$$

Since F is H-invariant,

$$\dim F(\Omega) \leqq \dim T - \dim H$$

for every T-orbit $\Omega \subset W$. Recall that we assumed that the number of T-orbits is finite. Therefore the above inequality implies $\dim F(W) \leqq \dim T - \dim H$. It follows that $\dim F(Y) = \dim F(W)$. Since W is irreducible and $Y \subset W$, it follows that $F(Y) = i(Y) \cong Y$ is irreducible. Therefore $S_1 = S_2$. □

Lemma 5.2.15 *Let A be a semi-abelian variety acting effectively on an algebraic variety X. Let L be the maximal connected linear subgroup of A and let $x \in X$. Then L contains the connected identity component of the isotropy group $A_x = \{a \in A; a \cdot x = x\}$.*

Proof Note that A_x acts linearly on each vector space $\mathcal{O}_{X,x}/\mathfrak{m}_x^k$. Because the A-action is supposed to be effective, these actions cannot all be trivial. Hence A_x is linear and its connected component is contained in L. □

5.3 Semi-toric Varieties

5.3.1 Toric Varieties

Here we collect some basic facts on toric varieties which we will need.

An equivariant algebraic compactification of $T = (\mathbf{C}^*)^t$ is called a *toric variety*. Here we are interested mainly in toric varieties which are non-singular, compact and moreover projective. For a general theory of toric varieties see Oda [85], Fulton [93], Kempf–Knudsen–Mumford–Saint-Donat [73].

Let $\bar{T}$ be a projective smooth equivariant compactification of $T = (\mathbf{C}^*)^t$. Then $\bar{T}$ is simply connected (Proposition 5.3.10) and the homology $H_*(\bar{T}, \mathbf{Z})$ is generated (as $\mathbf{Z}$-module) by the fundamental classes of closures of T-orbits (see Oda [85], Chap. 2). In particular, the Chern classes of the tangent bundle are linear combinations of the fundamental classes of orbit closures.

There are only finitely many T-orbits (Theorem 5.2.11), and each T-invariant divisor has only simple normal crossings (Lemma 5.3.8). Every ample line bundle on the smooth toric variety $\bar{T}$ is already very ample (see Oda [85], Corollary 1.15). Note that $H^1(\bar{T}, \mathcal{O}_{\bar{T}}) = 0$ because $\bar{T}$ is simply-connected. Hence every topologically trivial line bundle on $\bar{T}$ is already holomorphically trivial. Furthermore the T-action on $\mathrm{Pic}(\bar{T})$ is trivial and every line bundle is given by a T-invariant divisor (see Oda [85], Proposition 2.4). The Nakai–Moishezon criterion for ampleness of line bundles specializes to the following "toric Nakai criterion" (see Oda [85], Theorem 2.18):

Theorem 5.3.1 *A divisor D on a smooth toric variety $\bar{T}$ is ample if and only if $D \cdot C > 0$ for every T-invariant curve C on $\bar{T}$.*

There are only countably many non-isomorphic compact toric varieties. This is a consequence of the following fact: Toric varieties can be classified by certain combinatorial data, called *fans* (cf. Oda [85]). Set $N = \mathrm{Hom}(\mathbf{C}^*, T)$. Then a *fan* for an equivariant T-compactification is a collection Δ of convex cones in $N \otimes \mathbf{Q}$ of a certain type. Each cone $\sigma \in \Delta$ corresponds to a T-orbit.

We will need the following result (see Sumihiro [75], Sect. 3):

Lemma 5.3.2 *Let Y be a normal toric variety with algebraic torus T. Then every point of Y admits a Zariski open invariant affine neighborhood W.*

5.3.2 Semi-toric Varieties

Here we introduce the concept of a "*semi-toric variety*". This concept contains as special cases the following:

- Compact complex tori.
- Toric varieties.
- Smooth equivariant algebraic compactifications of semi-abelian varieties.

Definition 5.3.3 A compact complex space X together with an action of a semi-torus A on X is called a *semi-toric variety* if all of the following conditions are fulfilled:

(i) There is a dense open orbit of A on which A acts freely. (By abuse of language we will identify this orbit with A.)
(ii) There is a "presentation"

$$1 \to L \to A \xrightarrow{\pi} M \to 1$$

of the semi-torus A (where $L \cong (\mathbf{C}^*)^g$ and M is a compact complex torus).
(iii) The Lie group homomorphism $\pi : A \to M$ extends to an A-equivariant holomorphic map $\bar{\pi} : X \to M$.
(iv) For each point $p \in M$ the fiber $X_p = \bar{\pi}^{-1}(p)$ is a *toric variety*, i.e., X_p is admits the structure of a smooth algebraic variety such that the L-action on X_p is algebraic.

A *possibly singular semi-toric variety* is defined in exactly the same way except that we do not require X_p to be smooth.

Smoothness Assumption If not stated otherwise, semi-toric varieties are always *smooth*.

Proposition 5.3.4 *Every smooth* (*resp. normal*) *algebraic equivariant compactification $\bar{A}$ of a semi-abelian variety A is a smooth* (*resp. possibly singular*) *semi-toric variety.*

Proof Let $\iota : A \hookrightarrow \bar{A}$ be the inclusion map and let L be the maximal connected linear algebraic subgroup of A. Then there is a short exact sequence of algebraic groups

$$1 \to L \to A \xrightarrow{\pi} M \to 1,$$

where M is an abelian variety. The morphism $\pi : A \to M$ extends to an rational map $\bar{\pi}$ from $\bar{A}$ to M. However, since M is a compact complex torus, every meromorphic map to M is already holomorphic. Thus $\bar{\pi}$ is a morphism. Since $\pi : A \to M$ is a group homomorphism, it is clear that $\bar{\pi}$ is A-equivariant. □

In contrast, there are smooth equivariant compactifications of complex semi-tori which are not semi-toric varieties in the sense of Definition 5.3.3 above. See Example 5.3.27.

5.3.3 Key Properties of Semi-toric Varieties

Proposition 5.3.5 *Let $\bar{A}$ be a semi-toric variety with an action of a semi-torus A. Then the following properties hold*:

(i) *Every isotropy group of the A-action on $\bar{A}$ is isomorphic to some $(\mathbf{C}^*)^s$ $(s \in \mathbf{Z}^+)$.*
(ii) *There are only finitely many A-orbits in $\bar{A}$.*
(iii) *$\bar{A}$ is a compact Kähler manifold.*
(iv) *The map $\mu : A \times A \to A$ defining the group multiplication on A extends to a meromorphic map $\hat{\mu} : \bar{A} \times \bar{A} \to \bar{A}$.*
(v) *The projection map $\bar{A} \to M$ induced by the presentation realizes $\bar{A}$ as a fiber bundle which is topologically trivial and also locally holomorphically flat.*

Proof (i) This is a direct consequence of the definition.
(ii) This follows from Theorem 5.2.11.
(iii) See Theorem 5.3.21 below.
(iv) Let Γ_A (resp. Γ_L) denote the graph of the map defining the group structure, i.e.,

$$\Gamma_A = \left\{(g, h, x) \in A^3;\, g \cdot h = x\right\}$$

and

$$\Gamma_L = \left\{(g, h, x) \in L^3;\, g \cdot h = x\right\}.$$

Furthermore, let $\bar{\Gamma}$ be the Zariski closure of Γ_L in $\bar{L}^3$ (note that L and $\bar{L}$ are assumed to be algebraic). Now we define a subset $G \subset \bar{A}^3$ as follows: We take the image G of $\Gamma_A \times \bar{\Gamma}$ under the natural map from $A^3 \times \bar{L}^3$ to $\bar{A}^3$. Then one checks easily that G is compact, hence it is a closed analytic subset and moreover it is the closure of the graph of the map defining the group structure on A. Thus G defines a meromorphic map $\hat{\mu}$ as desired.

(v) The bundle $\pi : \bar{A} \to M$ is the associated fiber bundle to the principal bundle $A \to M$. Hence the assertions follow from the corresponding properties of $A \to M$. □

Lemma 5.3.6 *Let X be a possibly singular, but normal irreducible complex space. Let A be a semi-torus and let T be a semi-torus, too. Assume that A is acting on both X and T.*

Let $\pi : \tilde{X} \to X$ be an equivariant desingularization and let $a : \tilde{X} \to T$ be an equivariant holomorphic map to a semi-torus T. Then there exists an equivariant holomorphic map $f : X \to T$ such that $a = f \circ \pi$.

Proof Because X is assumed to be normal, the meromorphic map defined by $a \circ \pi^{-1}$ from X to the semi-torus T is already a holomorphic map from X to T. Since π^{-1} is biholomorphic on an open dense subset of X, a being equivariant implies that f is equivariant as well. □

Theorem 5.3.7 *Let X be a possibly singular, but normal complex algebraic variety with a semi-torus A acting with an open orbit. Then every point $x \in X$ admits a Zariski open neighborhood W of x in X such that the following property holds*:

"Two points $y, z \in W$ are contained in the same A-orbit if and only if they have the same isotropy Lie algebra with respect to the A-action."

Proof We may assume that the A-action on X is effective. Let L be the maximal connected linear subgroup of A and $T = A/L$. Due to Lemma 5.3.6 there is a surjective equivariant morphism $\alpha : X \to T$ (which is the Albanese variety of a desingularization of X).

Now A-invariant subsets of X correspond to L-invariant subsets of a fiber F of this morphism $\alpha : X \to T$. Moreover, due to Lemma 5.2.15 every isotropy Lie algebra for the A-action is contained in the Lie algebra $Lie(L)$ and therefore coincides with the isotropy Lie algebra for the L-action. Therefore there is no loss in generality in assuming that $L = A$. Then X is a normal toric variety and every point $x \in X$ admits an invariant affine neighborhood (Lemma 5.3.2). Due to Lemma 5.2.11 there are only finitely many orbits. Hence the statement follows from Lemma 5.2.13. □

Lemma 5.3.8 *Let A be a semi-torus acting on a complex manifold X with an open orbit. Assume that all isotropy groups are reductive. Let D be a reduced effective A-invariant divisor on X. Then D is a divisor of normal crossing type.*

Proof Let $p \in X$. By assumption the isotropy group

$$A_p = \{a \in A; a \cdot p = p\}$$

is a reductive subgroup of the semi-torus A. Let $A(p)$ denote the A-orbit through the point p. Since A is commutative and acts on X with an open orbit, we have $n = \dim A = \dim X$. It follows that $A(p)$ has dimension $d = n - \dim A_p$. Let K de-

note the maximal compact subgroup of A_p. Its connected component of the neutral element is isomorphic to $(S^1)^g$ where $g = n - d = \dim A_p$. Due to Lemma 5.2.6 there exists an open K-invariant neighborhood W of p and a K-equivariant biholomorphic map from W onto an open subset of $\mathbf{T}(X)_p \cong \mathbf{C}^n$. Since $\dim A(p) = d$, the vector subspace

$$\left(\mathbf{T}(X)_p\right)^K = \left\{v \in \mathbf{T}(X)_p; g \cdot v = v, {}^\forall g \in K\right\}$$

has dimension at least d. On the other hand, K acts effectively on $\mathbf{T}(X)_p$ (Corollary 5.2.5). This implies that $\mathbf{T}(X)_p \cong V_0 \oplus \mathbf{C}^g$ where $K = (S^1)^g$ acts trivially on the d-dimensional vector space V_0 and via the map $K \times \mathbf{C}^g \to \mathbf{C}^g$ given by

$$(a_1, \ldots, a_g) : (v_1, \ldots, v_g) \mapsto (a_1 v_1, \ldots, a_g v_g).$$

Thus every K-invariant hypersurface of $\mathbf{T}(X)_p$ is a union of coordinate hyperplanes. In particular, every K-invariant hypersurface of $\mathbf{T}(X)_p$ is of normal crossing type. By making use of the isomorphism between W and an open subset of $\mathbf{T}(X)_p$ it follows that D is of normal crossing type near p. Hence the statement follows, since p was arbitrary. □

Corollary 5.3.9 *On a semi-toric variety every invariant reduced effective divisor is of normal crossing type.*

Proposition 5.3.10 *Let X be a g-dimensional complex manifold on which $G = (\mathbf{C}^*)^g$ acts effectively with a fixed point p. Then X is simply-connected.*

Proof Theorem 5.2.7 implies that there is an open orbit Ω. Every continuous map $f : S^1 \to X$ may be deformed into a map whose image is contained in Ω, because $X \setminus \Omega$ is of real codimension two in X. Now Ω is contained in every open invariant neighborhood of p. Hence by Theorem 5.2.7, there is an open subset U of X with $\Omega \subset U$ and a G-equivariant biholomorphic map between U and an open subset of $\mathbf{T}(X)_p$ which contains the origin 0. But, for the standard action of $(\mathbf{C}^*)^g$ on $\mathbf{C}^g$ the origin is an attractive fixed point. It follows that we can use the group action to deform the given loop into a constant one. Hence $\pi_1(U) = \{1\}$ and consequently X is simply-connected. □

Lemma 5.3.11 *Let G be a commutative complex Lie group acting with a dense connected open orbit Ω on a complex space X and acting transitively on a complex space Y. Let $\pi : X \to Y$ be a surjective equivariant holomorphic map. Then for every $y \in Y$ and every G-orbit $Z \subset X$ the intersection with the fiber $Z \cap \pi^{-1}(y)$ is connected.*

Proof Fix $y \in Y$ and define $F = \pi^{-1}(y)$. Let H denote the isotropy group $H = \{g \in G; g(y) = y\}$. There is a one-to-one correspondence between G-orbits in X and H-orbits in F. Note that $W = \Omega \cap F$ is a dense open H-orbit in F. Choose $p \in W$ and let $I = \{h \in H; h(p) = p\}$. Because H is commutative, the group I acts

trivially on W. Since W is dense in F, the action of I on F is trivial. Let H^0 denote the identity connected component of H. Then $H^0 \cdot I = H$, because $W \cong H/I$ is connected. Thus the fact that I acts trivially on F implies that the H-orbits on F coincide with the H^0-orbits. In particular, they are connected. Thus $Z \cap F$ which is an H-orbit must be connected. □

For the special case where Y is a point one obtains the following.

Corollary 5.3.12 *If a commutative complex Lie group acts on a complex space with an open dense connected orbit, then every orbit is connected.*

5.3.4 Quasi-algebraic Subgroups

Proposition 5.3.13 *Let $\bar{A}$ be a semi-toric variety with presentation*

$$1 \to L \to A \xrightarrow{\pi} M \to 1.$$

Let H be a subgroup of A such that the closure $\bar{H}$ of H in $\bar{A}$ is a closed analytic subset of $\bar{A}$ with $\bar{H} \cap A = H$. Then we have the following properties:

(i) *H is a closed complex Lie subgroup of A.*
(ii) *$\pi(H) = \pi(\bar{H})$.*
(iii) *$\pi(H)$ is a compact complex torus.*
(iv) *For every $x \in M$ the fiber $\bar{H}_x = \{p \in \bar{H}; \bar{\pi}(p) = x\}$ is the closure of $H_x = \{p \in H; \pi(p) = x\}$.*
(v) *$H \cap L$ is an algebraic subgroup of L.*
(vi) *H is a semi-torus.*
(vii) *The quotient Lie group $B = A/H$ is a semi-torus.*
(viii) *There exists a semi-toric variety $\bar{B}$ acting such that the projection map $\pi : A \to A/H$ extends to a meromorphic map from $\bar{A}$ to $\bar{B}$.*

Proof (i) It is an immediate consequence of $H = \bar{H} \cap A$ that H is a closed complex Lie subgroup of A.

(ii) Since $\bar{\pi} : \bar{A} \to M$ is proper, we know that $\pi(\bar{H})$ is compact and in particular closed. It follows that $\bar{\pi}(\bar{H})$ is the closure of $\pi(H)$ in M. Since $\pi(H)$ is a subgroup of M, its closure is a group as well. Hence $\bar{\pi}(\bar{H})$ is a compact complex subtorus of M. Since the fibers of $H \to \pi(H)$ are contained in the fibers of $\bar{H} \to \bar{\pi}(\bar{H})$ and furthermore $\dim H = \dim \bar{H}$, it is clear that $\dim \pi(H) = \dim \bar{\pi}(\bar{H})$.

Together with $\pi(H)$ being a dense Lie subgroup of $\bar{\pi}(\bar{H})$ this implies that $\pi(H) = \bar{\pi}(\bar{H})$.

(iii) Thus $\pi(H)$ is a compact complex torus.

(iv) This assertion is obviously true for a generic fiber. Due to the transitive action of H on $\pi(H)$ it follows that it is true for every fiber.

(v) This follows from Theorem 2.2.8, because $\bar{L}$ is algebraic.

(vi) To verify that H is a semi-torus, we regard the short exact sequence

$$1 \to (H \cap L)^0 \to H \to H/(H \cap L)^0 \to 1.$$

Here $(H \cap L)^0 \cong (\mathbf{C}^*)^s$, because $H \cap L$ is an algebraic subgroup of $L \cong (\mathbf{C}^*)^t$. Moreover $H/(H \cap L)^0$ is a compact complex torus, because $H/(H \cap L)^0 \to H/(H \cap L) \cong \pi(H)$ is a finite covering and $\pi(H)$ is a compact complex torus. Thus H is a semi-torus.

(vii) A/H is Hausdorff, because H is closed. Due to Corollary 5.1.11 the quotient $B = A/H$ is a semi-torus.

(viii) Note that

$$1 \to (L \cap H) \to H \to \pi(H) \to 1$$

is a presentation for H. Let Z be a smooth algebraic compactification of $L/(L \cap H)$ and let $A_1 \hookrightarrow \bar{A}_1$ be the semi-toric variety obtained as fiber product $A \times_{L/(L\cap H)} Z$. The algebraicity of L and Z implies that the holomorphic quotient map from L to $L/(L \cap H)$ extends to a rational map from $\bar{L}$ to Z. This in turn implies that the holomorphic quotient map from A to $A/(L \cap H)$ extends to a meromorphic map from $\bar{A}$ to $\bar{A}_1$.

The H-action on $\bar{A}$ induces an action of the compact complex torus $H/(H \cap L) \cong \pi(H)$ on $\bar{A}_1$. Recall that for actions of a *compact* complex Lie group on a normal complex space, there always exists a geometric quotient which is again a normal complex space (see H. Holmann [63], Satz 20, p. 358). We define $\bar{B}$ (resp. B) as the quotient of $\bar{A}_1$ (resp. A_1) by the action of $H/(H \cap L)$. □

Definition 5.3.14 Let A be a complex Lie group with an equivariant compactification $A \hookrightarrow \bar{A}$. A subgroup $H \subset A$ is called "*quasi-algebraic*" if there exists a closed analytic subset $H' \subset \bar{A}$ such that $H = H' \cap A$ (equivalently: if the topological closure of H in $\bar{A}$ is an analytic subset).

Let A be a semi-torus compactified by a semi-toric variety $\bar{A}$. Then due to Proposition 5.3.13 every quasi-algebraic subgroup of A is again a semi-torus. The notion of "*quasi-algebraic subgroup*" naturally generalizes that of an *algebraic subgroup*, as we see by the following statement.

Proposition 5.3.15 *Let G be a complex algebraic group with an algebraic compactification $G \hookrightarrow \bar{G}$. Then a complex Lie subgroup H of G is algebraic iff it is quasi-algebraic with respect to $G \hookrightarrow \bar{G}$.*

Proof Due to Chow's Theorem every closed analytic subset of the compact algebraic variety $\bar{G}$ is already an algebraic subvariety. Therefore every quasi-algebraic subgroup is an algebraic subgroup. The other direction is trivial. □

Proposition 5.3.16 *Let G be a complex Lie group with equivariant compactification $\iota : G \hookrightarrow \bar{G}$. Assume that the map $\mu : G \times G \to G$ defining the group structure on G extends to a meromorphic map $\hat{\mu} : \bar{G} \times \bar{G} \to \bar{G}$. Let $\bar{Z}$ be a closed analytic*

subset of $\bar{G}$ and define $Z = G \cap \bar{Z}$. Then

$$\mathrm{St}_G(Z) = \{g \in G;\, g(Z) = Z\}$$

is a quasi-algebraic subgroup of G.

Proof Let $\Gamma \subset \bar{G} \times \bar{G} \times \bar{G}$ be the graph of the meromorphic map $\hat{\mu}$. Let $\mathrm{pr}_1 : \bar{G} \times \bar{G} \times \bar{G} \to \bar{G}$ denote the projection onto the first factor. Then

$$\{g \in \bar{G};\, \bar{\mu}(g, z) \cap \bar{Z} \neq \emptyset\} = \mathrm{pr}_1\big(\Gamma \cap (\{g\} \times \{z\} \times \bar{Z})\big)$$

is a closed analytic subset (for every $z \in Z$) because pr_1 is a proper holomorphic map. Now the assertion follows, because $\mathrm{St}_G(Z) = \bar{H} \cap G$ with

$$\bar{H} = \bigcap_{z \in Z} \{g \in \bar{G};\, \bar{\mu}(g, z) \cap \bar{Z} \neq \emptyset\}.$$

□

Proposition 5.3.17 *Let G be a complex Lie group with equivariant compactification $\iota : G \hookrightarrow \bar{G}$. Assume that the map $\mu : G \times G \to G$ defining the group structure on G extends to a meromorphic map $\hat{\mu} : \bar{G} \times \bar{G} \to \bar{G}$.*

Let $\bar{Z}$ be a closed analytic subset of $\bar{G}$. Let H be the subgroup of G generated by all $x \cdot y^{-1}$ with $x, y \in Z = \bar{Z} \cap G$. Then H is a quasi-algebraic subgroup.

Proof Without loss of generality we may assume that $\bar{Z}$ is the closure of Z in $\bar{G}$. We define recursively a sequence of subsets W_n of $\bar{G}$ as follows: $W_1 = \bar{Z}$, $W_{k+1} = \hat{\mu}(W_k \times W_k)$ $(k > 0)$. Then W_k is an ascending sequence of closed irreducible subsets of $\bar{G}$. For dimension reasons it will stabilize, i.e., there is a number N such that $W_k = W_N$ for all $k \geqq N$. Since

$$H = G \cap \left(\bigcup_k W_k\right) = G \cap W_N,$$

the assertion follows. □

5.3.5 Compactifiable Groups and Kähler Condition

The notions of a semi-toric variety and quasi-algebraic subgroups can be regarded as the special cases of a more general concept, namely that of "meromorphic" resp. "compactifiable" group actions. Here we briefly discuss this context.

Definition 5.3.18 Let G be a complex Lie group with equivariant compactification $G \hookrightarrow \bar{G}$. A holomorphic G-action on a compact complex space X is said to be *compactifiable* (with respect to $G \hookrightarrow \bar{G}$) if the map defining the action $\mu : G \times X \to X$ extends to a meromorphic map from $\bar{G} \times X$ to X, in other words, if there is a closed analytic subset $Z \subset \bar{G} \times X \times X$ such that for every $g \in G$, $(g, x, y) \in Z$ iff $\mu(g, x) = y$.

In this language we have seen that on a semi-toric variety the corresponding semi-torus acts compactifiably. The notion "*compactifiable action*" generalizes the notion of an algebraic action to complex Lie groups which are not necessarily algebraic:

Proposition 5.3.19 *Let G be a complex algebraic group acting holomorphically on a compact algebraic variety X. Let $G \hookrightarrow \bar{G}$ be an algebraic compactification. Then the action of G on X is algebraic if and only if it is compactifiable with respect to $G \hookrightarrow \bar{G}$.*

Proof If the action is algebraic, it is obviously compactifiable. Conversely, by GAGA (Chow's Theorem) every closed analytic subset of $\bar{G} \times X \times X$ is algebraic. Hence, if the action is compactifiable, the graph of the defining map $\mu : G \times X \to X$ is algebraic, which implies that μ is algebraic. □

The concept of "being compactifiable" was introduced by Fujiki [78b][2] who proved the following.

Theorem 5.3.20 *Let X be a compact complex manifold which is bimeromorphic to a Kähler manifold. Then the connected component of the automorphism group* $\mathrm{Aut}(X)$ *is a complex Lie group acting compactifiably on X.*

Here $\mathrm{Aut}(X)$ is compactified as follows: We embed $\mathrm{Aut}(X)$ into a cycle space of $X \times X$ by mapping each automorphism to its graph. Because X is assumed to be bimeromorphic to a Kähler manifold, this cycle space has compact irreducible components. Thus we obtain a compactification of $\mathrm{Aut}(X)$. See Fujiki [78b] for details.

Theorem 5.3.21 *Let $\bar{A}$ be a (smooth) semi-toric variety with presentation*

$$1 \to T \to A \to M \to 1.$$

Then there exists a decomposition of the tangent bundle $\mathbf{T}(\bar{A}) = H \oplus V$ such that

(i) *H and V are vector subbundles of $\mathbf{T}(\bar{A})$;*
(ii) *$V = \mathrm{Ker}\, d_{\bar{\pi}}$;*
(iii) *H and V are involutive; i.e. if $X, Y \in \Gamma(\bar{A}, H) \subset \Gamma(\bar{A}, \mathbf{T}(\bar{A}))$, then $[X, Y] \in \Gamma(\bar{A}, H)$ and similarly for V;*
(iv) *There is a Kähler form ω on $\bar{A}$ for which H and V are orthogonal and such that the restriction of ω to H is a pull-back of a Kähler form on M.*

Proof The T-principal bundle $A \to M$ can be defined using locally constant transition functions with values in the maximal compact subgroup $U \cong (S^1)^g$ of

[2] However, Fujiki used the term "meromorphic" where we use "compactifiable".

$T \cong (\mathbf{C}^*)^g$. In other words, M is covered by a collection of open subsets W_i and there are locally constant maps $\zeta_{ij} : W_i \cap W_j \to U$ such that

$$\bar{A} \cong \left(\bigsqcup_i (W_i \times \bar{T}) \right) \Big/ \sim,$$

where $(x, t_i) \sim (x', t_j)$ for $x \in W_i$ and $x' \in W_j$ iff $x = x' \in M$ and $t_i = \zeta_{ij}(x) \cdot t_j$. Now $\mathbf{T}(\bar{A})$ decomposes as a direct sum in a "horizontal" subbundle H and a "vertical" subbundle V using this local trivializations, because the transition functions are locally constant. Furthermore H and V are involutive.

Now let μ be a Kähler form on $\bar{T}$. Then we can average with respect to the Haar measure $d\lambda$ on U:

$$\hat{\mu} = \int_{k \in U} k^* \mu d\lambda(k).$$

Thus we obtain a U-invariant Kähler form on $\bar{T}$. Since the transition functions ζ_{ij} take values in U, this yields a closed $(1,1)$-form ω' on $\bar{A}$ which is positive on V and zero on H and such that $\omega'(X, Y) = 0$ for $X \in \Gamma(\bar{A}, H)$ and $Y \in \Gamma(\bar{A}, V)$.

Finally, we choose a Kähler form α on the compact complex torus M and define ω'' as the pull-back of α and a Kähler form ω on $\bar{A}$ as $\omega = \omega' + \omega''$. □

Theorem 5.3.22 *Let A be a semi-torus with presentation*

$$1 \to T \to A \to M \to 1,$$

and let $\bar{A}$ be the equivariant compactification of A induced by an algebraic *compactification $T \hookrightarrow \bar{T}$; i.e., $\bar{A}$ is a semi-toric variety. Then*

(i) *the Albanese variety of $\bar{A}$ is isomorphic to M (cf. Sect. 4.5);*
(ii) $\pi_1(\bar{A}) \cong \pi_1(M)$.

Proof Since toric varieties are simply-connected, it is clear that $\pi_1(\bar{A}) \cong \pi_1(M)$. Now every holomorphic map from a compact simply-connected complex space to a compact complex torus $\mathbf{C}^g/\Lambda$ is constant, because the condition "simply-connected" implies that such a map lifts to a map with values in the universal covering $\mathbf{C}^g$ which must be constant. Hence $\mathrm{Alb}(\bar{A}) \cong M$. □

Theorem 5.3.23 *Let $\bar{A}$ be a semi-toric variety and let Z be a closed irreducible analytic subset of A such that its closure $\bar{Z}$ in $\bar{A}$ is an analytic subset. Then Z is a translate of a subgroup iff $\bar{q}(Z) = \dim Z$.*

Proof We recall that $\bar{q}(Z)$ equals the dimension of the quasi-Albanese variety of Z. If Z is a translate of a subgroup, then it is its own quasi-Albanese variety, implying $\bar{q}(Z) = \dim Z$. If Z is not a translate of a subgroup, we regard the subgroup B of A generated by all $p - q$ $(p, q \in Z)$. This is a quasi-algebraic subgroup due to Proposition 5.3.17 and in particular a semi-torus. Hence we obtain a surjection from the quasi-Albanese variety of Z onto B, implying $\bar{q}(Z) \geqq \dim B > \dim Z$. □

Proposition 5.3.24 *Let $\bar{A}$ be a semi-toric variety and let Z be a subvariety of A whose closure in $\bar{A}$ is a closed analytic subset. Then $\mathrm{St}(Z)^0$ and $A/\mathrm{St}(Z)^0$ are semi-tori, and the projection $A \to A/\mathrm{St}(Z)^0$ extends to a meromorphic map from $\bar{A}$ to an equivariant compactification of $A/\mathrm{St}(Z)^0$.*

Proof Due to Proposition 5.3.16 the stabilizer $\mathrm{St}(Z)$ is a "quasi-algebraic" subgroup of A in the sense of Definition 5.3.14. Hence the assertions follow from Proposition 5.3.13. □

In Winkelmann [04] one can find the following characterization of semi-toric varieties.

Theorem 5.3.25 *Let X be a compact Kähler manifold and let D be an s.n.c. divisor such that $\Omega^1_X(\log D)$ is a globally trivial bundle. Then there exists a semi-toric variety $\bar{A}$ with open orbit A such that $\bar{A} \cong X$ and $D = \bar{A} \setminus A$.*

5.3.6 Examples of Non-semi-toric Varieties

The following are examples of compactifications of complex semi-tori which are not semi-toric varieties.

Example 5.3.26 Let $A \hookrightarrow \bar{A}$ be an equivariant compactification and let $Z \subset \partial A = \bar{A} \setminus A$ be a submanifold which is not A-invariant. Then blowing up $\bar{A}$ along Z yields a compactification of A which is not equivariant. In particular, the space obtained by blowing up $\bar{A}$ along Z is not a semi-toric variety.

Example 5.3.27 Let $\lambda, \mu \in \mathbf{C}$ with $|\lambda|, |\mu| > 1$. Now $\{(\lambda^n, \mu^n); n \in \mathbf{Z}\} = \Lambda$ is an infinite cyclic subgroup of $\mathbf{C}^* \times \mathbf{C}^*$ and $X = (\mathbf{C}^2 \setminus \{(0,0)\})/\Lambda$ is a smooth equivariant compactification of the semi-torus $(\mathbf{C}^* \times \mathbf{C}^*)/\Lambda$. This quotient manifold X is an example of a *Hopf surface*. It is not Kähler and its Albanese variety is trivial. There is no presentation

$$1 \to \mathbf{C}^* \to A \to E \to 1$$

for which the projection map $\pi : A \to E$ extends to a meromorphic map on $\bar{A}$. Furthermore $A \hookrightarrow \bar{A}$ is an equivariant compactification of the complex Lie group A, but it does not define a "meromorphic structure" in the sense of Fujiki [78b]: The map $x \mapsto x^{-1}$ does *not* extend to a meromorphic map from $\bar{A}$ to $\bar{A}$.

Example 5.3.28 The semi-torus $A = \mathbf{C}^* \times \mathbf{C}^*$ contains a closed subgroup isomorphic to $\mathbf{C}$:

$$H = \left\{\left(e^t, e^{it}\right); t \in \mathbf{C}\right\}.$$

This realizes A as a $\mathbf{C}$-principal bundle over an elliptic curve E which can be compactified to a $\mathbf{P}^1$-bundle $\bar{A}$ over E. The logarithmic tangent bundle is not trivial

and the isotropy group at a point in the boundary $\bar{A} \setminus A$ is not a semi-torus; it is isomorphic to the additive group $\mathbf{C}$.

See Lescure–Meersseman [02] for more about non-algebraic and non-Kähler equivariant compactifications.

5.4 Jet Bundles over Semi-toric Varieties

Let $\bar{A}$ be a *semi-toric variety*. The A-invariant divisor ∂A has only simple normal crossings (Lemma 5.3.8). We have the logarithmic tangent bundle $\mathbf{T}(\bar{A}; \log \partial A)$ and logarithmic jet bundles $J_k(\bar{A}; \log \partial A)$, $k \in \mathbf{Z}^+$, over $\bar{A}$ (cf. Sect. 4.6.3).

By definition, $J_0(\bar{A}; \log \partial A) = \bar{A}$, and for $k = 1$ we have

$$J_1(\bar{A}; \log \partial A) \cong \mathbf{T}(\bar{A}; \log \partial A). \tag{5.4.1}$$

Moreover, there is a natural bundle morphism,

$$J_k(\bar{A}; \log \partial A) \to J_k(\bar{A}). \tag{5.4.2}$$

We regard the k-jets as maps from $S_k = \operatorname{Spec} \mathbf{C}\{t\}/(t^{k+1})$ to $\bar{A}$. Let $\mathfrak{m}(k)$ denote the maximal ideal (t) of $\mathbf{C}\{t\}/(t^{k+1})$.

Proposition 5.4.3 *Let $\bar{A}$ be a semi-toric variety (Definition 5.3.3). Then the logarithmic jet bundle $J_k(\bar{A}; \log \partial A)$ is trivialized as*

$$J_k(\bar{A}; \log \partial A) \cong \bar{A} \times \big(\mathfrak{m}(k) \otimes \operatorname{Lie} A\big).$$

Proof We first show that the logarithmic tangent bundle is trivialized by the A-fundamental vector fields. Due to Proposition 5.3.5 (ii), all the isotropy groups are reductive. Therefore the action of every isotropy group is linearizable in some open neighborhood (Lemma 5.2.6). This implies that for every point on $\bar{A}$ we can find a system of local coordinates in which the A-fundamental vector fields are simply

$$\frac{\partial}{\partial z_1}, \dots, \frac{\partial}{\partial z_k}, z_{k+1}\frac{\partial}{\partial z_{k+1}}, \dots, z_n \frac{\partial}{\partial z_n},$$

for some $k, n \in \mathbf{N}$, where $n - k$ equals the dimension of the isotropy group. Hence we have

$$\mathbf{T}(\bar{A}; \log \partial A) \cong \bar{A} \times \operatorname{Lie}(A). \tag{5.4.4}$$

For $k \geqq 2$, (5.4.4) induces a global trivialization of $J_k(\bar{A}; \log \partial A)$ in the following way: For $p \in A$ and $\alpha = \sum_i \alpha_i \otimes v_i \in \mathfrak{m}(k) \otimes \operatorname{Lie}(A)$ we define a map (germ) from $\operatorname{Spec} \mathbf{C}\{t\}/(t^{k+1})$ by

$$\alpha : t \mapsto \exp\left(\sum_i \alpha_i v_i\right) \cdot p.$$

A calculation in local coordinates shows that this gives a trivialization

$$J_k(\bar{A}; \log \partial A)|_A \cong A \times \big(\mathfrak{m}(k) \otimes \mathrm{Lie}(A)\big). \qquad \square$$

Now fix a point $p \in \bar{A}$ and consider the induced map

$$\mathfrak{m}(k) \otimes \mathrm{Lie}(A) \to J_k(\bar{A}; \log \partial A)_p.$$

Let V be a complex vector space and let E be a vector space of linear mappings from V to $\mathrm{Lie}(A)$. Then we obtain a map $\mathfrak{m}(k) \otimes V \times L \to J_k(\bar{A}; \log \partial A)_p$ induced by the natural pairing $V \times E \to \mathrm{Lie}(A)$. Observe that this map $\mathfrak{m}(k) \otimes V \times E \to J_k(\bar{A}; \log \partial A)_p$ is polynomial of degree $\leqq k$ in E.

Corollary 5.4.5 *A smooth semi-toric variety is log-parallelizable; i.e., the logarithmic tangent bundle as well as the bundle of logarithmic* 1*-forms* $\Omega^1_{\bar{A}}(\log(\bar{A} \setminus A))$ *are holomorphically trivial vector bundles on* $\bar{A}$.

5.5 Line Bundles on Toric Varieties

5.5.1 Ample Line Bundles

We first investigate ample line bundles on toric varieties.

Lemma 5.5.1 *Let* $\bar{T}$ *be a smooth projective toric variety. Then there exists a number* $k \in \mathbf{N}$ *such that for every ample line bundle* H *and every* T*-invariant reduced hypersurface* D *on* $\bar{T}$*, the line bundle* $H^k \otimes L(-D)$ *is ample.*

Proof There are only finitely many T-invariant reduced hypersurfaces $D_1, \ldots, D_r$ on $\bar{T}$ and only finitely many T-invariant curves $C_1, \ldots, C_s$. We choose k such that $k > D_i \cdot C_j$ for all $i \in \{1, \ldots, r\}$, $j \in \{1, \ldots, s\}$. Then $(kc_1(H) - D_i) \cdot C_j > 0$ for all i, j. By the toric Nakai criterion (Theorem 5.3.1) it follows that $H^k \otimes L(-D_i)$ is ample for every i. $\square$

Remark 5.5.2 The interesting point here is that k can be chosen independently of H and D.

Lemma 5.5.3 *Let* L *be an ample line bundle on a smooth compact toric variety* $\bar{T}$ *and let* p *be a point in* $\bar{T}$*. Then there exists an effective* T*-invariant divisor* D *with* $p \notin \mathrm{Supp}\, D$ *and* $L \cong L(D)$.

Proof On a smooth toric variety every ample line bundle is already very ample (Oda [85], Corollary 2.15). Thus there are sections $\tau \in H^0(\bar{T}, L)$ with $\tau(p) \neq 0$. Now consider the T-action on the $H^0(\bar{T}, L)$. This is a linear action of the commutative reductive group $T \cong (\mathbf{C}^*)^t$ and thus completely diagonalizable. Then by the

general theory of linear algebraic groups (cf., e.g., Borel [91]) there is a "character" χ, i.e., a Lie group homomorphism $\chi : T \to \mathbf{C}^*$ and a section $\sigma \in H^0(\bar{T}, L)$ such that $\sigma(p) \neq 0$ and $\sigma(t \cdot x) = \chi(t)\sigma(x)$ for all $x \in \bar{T}$ and $t \in T$. Now let $D = (\sigma)$, the divisor defined by σ. □

Remark 5.5.4 Let D be a divisor on a compact complex torus X. Due to the triviality of the tangent bundle of X the Riemann–Roch formula for the line bundle $L(D)$ specializes to

$$\chi\big(L(D)\big) = \sum_k (-1)^k \dim H^k\big(X, L(D)\big) = D^{\dim X}.$$

If D is in addition ample, using the Kodaira Vanishing Theorem we obtain

$$\dim \Gamma\big(X, L(D)\big) = \chi\big(L(D)\big) = D^{\dim X}.$$

We have the following.

Theorem 5.5.5 *Let L be an ample line bundle on a compact complex torus X. Then $\dim \Gamma(X, L) > 0$, and L^k is globally spanned for all $k \geqq 2$. Furthermore L^k is very ample for all $k \geqq 3$.*

Proof See, e.g., Weil [58], Chap. VI; Birkenhake–Lange [04], Chap. 4 (Sect. 1 for the first assertion (on the existence of sections), Prop. 1.6 for L^k being globally spanned for $k \geq 2$ and Cor. 5.8 for the statement on very ampleness). □

Theorem 5.5.6 *Let $X = \mathbf{C}^g/\Gamma$ be a compact complex torus and let L be a line bundle over X. Then L is topologically trivial if and only if there is a group homomorphism $\rho : \Gamma \to S^1$ defining L via $L = \mathbf{C}^g \times \mathbf{C}/\sim$, where $(v, s) \sim (w, t)$ iff $w - v \in \Gamma$ and $t = \rho(w - v)s$.*

Proof Due to Hodge theory (see e.g. Voisin [02]) there is an isomorphism

$$H^1(X, \mathbf{R}) \otimes_{\mathbf{R}} \mathbf{C} \to H^1(X, \mathscr{O}_X) \oplus H^0\big(X, \Omega^1_X\big)$$

such that complex conjugation on the left-hand side interchanges the two summands on the right-hand side. That implies that (with respect to this isomorphism) we have

$$H^1(X, \mathbf{R}) \cap H^0\big(X, \Omega^1_X\big) = \{0\}.$$

As a consequence the natural homomorphism from $H^1(X, \mathbf{R})$ to $H^1(X, \mathscr{O}_X)$ is surjective. Standard results from algebraic topology ("Universal coefficient theorem", see e.g. Spanier [66], Theorem 5.5.3) tell us that

$$\mathrm{Hom}\big(\pi_1(X), G\big) \cong H^1(X, G)$$

for every commutative group G. Now consider the exponential sequence. It yields the following commutative diagram of cohomology groups:

$$\begin{array}{ccccc} H^1(X,\mathbf{R}) & \longrightarrow & H^1(X,S^1) & \xrightarrow{\gamma} & H^2(X,\mathbf{Z}) \\ \downarrow{\scriptstyle\alpha} & & \downarrow{\scriptstyle\mu} & & \| \\ H^1(X,\mathscr{O}_X) & \xrightarrow{\beta} & H^1(X,\mathscr{O}_X^*) = \mathrm{Pic}(X) & \xrightarrow{c} & H^2(X,\mathbf{Z}) \end{array}$$

Since $\pi_1(T) \cong \Gamma$ is a free abelian group, every group homomorphism from $\pi_1(T)$ to S^1 can be lifted to a group homomorphism with values in $\mathbf{R}$. As a consequence, $\gamma = 0$.

An element $\eta \in H^1(X, \mathscr{O}_X^*)$ defines a topologically trivial line bundle iff $c(\eta) = 0$. Now $\operatorname{Ker} c = \operatorname{Im} \beta$. Hence η defines a topologically trivial line bundle iff there is an element $\omega \in H^1(X, \mathscr{O}_X)$ with $\beta(\omega) = \eta$. Using $\gamma = 0$ and the bijectivity of α this in turn is equivalent to the existence of an element $\xi \in H^1(X, S^1)$ with $\mu(\xi) = \eta$. Hence the assertion follows. □

Let a Lie group G act on a space X, and let $Y \subset X$. Set the *stabilizer* of Y by

$$\mathrm{St}_G(Y) = \big\{ g \in G;\, g(Y) = Y \big\}, \tag{5.5.7}$$

and denote by $\mathrm{St}_G(Y)^0$ the identity component of $\mathrm{St}_G(Y)$. Similarly, if $L \to X$ is a holomorphic line bundle over a complex space X and G is a subgroup of the biholomorphic automorphism $\mathrm{Aut}(X)$, we set

$$\mathrm{St}_G(L) = \big\{ g \in G;\, g^*L \cong L \big\},$$

and denote by $\mathrm{St}_G(L)^0$ the identity connected component of $\mathrm{St}_G(L)$. In the above notation, the subscript G will be omitted when the context is clear.

Lemma 5.5.8 *Let M be a compact complex torus and let D be an effective divisor on M. Then* $\mathrm{St}(D)^0 = \mathrm{St}(L(D))^0$.

Proof Sections in $L(D)$ induce a meromorphic map $\phi_{L(D)} : M \to P = \mathbf{P}(\Gamma(M, L(D))^*)$. There is a natural action of $\mathrm{St}(L(D))/\mathrm{St}(D)$ on P such that $\phi_{L(D)}$ is equivariant. As an isotropy group of the M-action on the Picard variety $\mathrm{Pic}(M)$, the group $\mathrm{St}(L(D))$ is a closed subgroup of M. The closedness of $\mathrm{St}(D)$ is obvious. Thus $\phi_{L(D)}$ induces a holomorphic linear action of a compact complex torus on a projective space. But such an action is necessarily trivial. Hence the $\mathrm{St}(L(D))^0/\mathrm{St}(D)^0$ must be trivial. □

Proposition 5.5.9 *Let D be an effective divisor on a compact complex torus M. The line bundle $L = L(D)$ on M is ample iff* $\mathrm{St}(L)$ *is discrete.*

Proof By the preceding lemma, the discreteness of $\mathrm{St}(L)$ implies that of $\mathrm{St}(D)$. Now $\mathrm{St}(D)^0 = B$ is a compact complex subtorus of M. If $\dim B > 0$, then the line bundle $L(D)$ is the pull-back of a line bundle on M/B and therefore not ample.

Now suppose that $\mathrm{St}(D)$ is discrete. The tangent bundle $\mathbf{T}(M)$ is trivial, $\mathbf{T}(M) \cong M \times \mathbf{C}^g$, and $\mathrm{St}(D)$ being discrete imply that for every vector $v \in \mathbf{C}^g$ there is a point $p \in D$ with $v \notin \mathbf{T}(D)_p$ (because otherwise $\mathrm{St}(D)$ would contain the one-parameter group corresponding to V). From this we may deduce that for every subvariety $Z \subset M$ there are points $g_1, \ldots, g_d \in M$ ($d = \operatorname{codim} Z$) such that Z and the translates $g_i^* D$ intersect transversally. It follows that the intersection number $Z \cdot D^d$ is positive. Via the Nakai–Moishezon criterion this implies that D is ample. □

5.5.2 *Leray Spectral Sequence*

Let A be a semi-torus with presentation

$$0 \to \left(\mathbf{C}^*\right)^g \longrightarrow A \xrightarrow{\pi} M \to 0$$

and let $\bar{A}$ be a semi-toric compactification with corresponding projection

$$\bar{\pi} : \bar{A} \to M.$$

Let $\mathscr{S}$ be a coherent analytic sheaf (see, e.g., Grauert–Remmert [84]) on $\bar{A}$ (e.g., the structure sheaf or an ideal sheaf of a closed analytic subset). Then we have the *Leray spectral sequence* for the fibration $\bar{\pi} : \bar{A} \to M$,

$$H^{p+q}(\bar{A}, \mathscr{S}) \quad \Leftarrow \quad E_1^{p,q} = H^p\left(M, R^q \bar{\pi}_* \mathscr{S}\right).$$

(See Voisin [02], II, 4.1.3.) Now, let us assume that $\mathscr{S}$ is one of the sheaves Ω_X^k. Let F denote a fiber of $\bar{\pi} : \bar{A} \to M$. Assume further that there is a natural number k_0 with $H^k(F, \mathscr{S}) = \{0\}$ for $k \neq k_0$. Then $R^k \bar{\pi}_* \mathscr{S} = 0$ for $k \neq k_0$. As a consequence, the Leray spectral sequence degenerates and we obtain

$$H^m(\bar{A}, \mathscr{S}) \cong H^{m-k_0}\left(M, R^{k_0} \pi_* \mathscr{S}\right).$$

As a special case we obtain

Proposition 5.5.10 *Let $\bar{A}$ be a semi-toric variety. Then*

$$H^1(\bar{A}, \mathscr{O}_A) \cong H^1(M, \mathscr{O}_M)$$

and the natural A-action on $H^1(\bar{A}, \mathscr{O}_A)$ is trivial.

Proof The Leray spectral sequence implies that $H^1(E, \mathscr{O}_E) \cong H^1(B, \mathscr{O}_B)$ for every locally trivial holomorphic fiber bundle $E \to B$, where the fiber F is a compact

complex space with $H^1(F, \mathscr{O}_F) = \{0\}$. Since $H^1(\bar{T}, \mathscr{O}_{\bar{T}}) = \{0\}$ for every smooth toric variety, $H^1(\bar{A}, \mathscr{O}_{\bar{A}}) \cong H^1(M, \mathscr{O}_M)$. As a connected group, A acts trivially on the cohomology groups $H^*(\bar{A}, \mathbf{C})$. Recall that $\bar{A}$ is a compact Kähler manifold (Theorem 5.3.21). Due to the Hodge decomposition there is a natural surjective map from $H^{p+q}(\bar{A}, \mathbf{C})$ to $H^q(\bar{A}, \Omega^p_{\bar{A}})$ for all (p, q). It follows that A acts trivially on all the Dolbeault cohomology groups, thus in particular on $H^1(\bar{A}, \mathscr{O}_{\bar{A}})$. □

For a compact complex torus the next is a direct consequence of Kodaira's vanishing theorem, since the canonical bundle on a compact complex torus is trivial.

Theorem 5.5.11 *Let L be an ample line bundle on a semi-toric variety $\bar{A}$. Then $H^k(\bar{A}, L) = \{0\}$ for all $k \geqq 1$.*

Proof Let F be a fiber of $\bar{\pi} : \bar{A} \to M$. Then $L|_F$ is ample, and by standard results on toric varieties we have $H^k(F, L|_F) = \{0\}$ for $k \geqq 1$ and $\Gamma(F, L|_F) = \sum_\chi \mathbf{C}_\chi$ where $\mathbf{C}_\chi$ is the (one-dimensional) space of sections transforming like the character χ. Using this decomposition, we see that $\mathscr{R}^0\bar{\pi}_* L$ decomposes as a direct sum of ample line bundles on M, while $\mathscr{R}^k\bar{\pi}_* L = 0$ for $k > 0$. The claim now follows from the Leray spectral sequence (see Voisin [02], II, 4.1.3). □

5.5.3 Decomposition of Line Bundles

Let A be a semi-torus with presentation

$$0 \to T = \left(\mathbf{C}^*\right)^t \to A \xrightarrow{\pi} M \to 0 \tag{5.5.12}$$

compactified by a semi-toric variety $\bar{A}$.

Lemma 5.5.13 *There is a closed semi-positive $(1, 1)$-form Ω on $\bar{A}$ such that*

(i) *Ω is positive definite on every fiber of $\bar{A} \to M$,*
(ii) *$\Omega^t|_Z \equiv 0$ $(t = \dim T)$ for any A-invariant divisor Z.*

Proof Using Theorem 5.3.21, we have a decomposition of $T_{\bar{A}}$ into a direct sum of involutive subbundles H and V. Moreover there is a Kähler form ω for which H and V are orthogonal. We define Ω via

$$\Omega(v, w) = \begin{cases} \omega(v, w) & \text{if } v, w \in V, \\ 0 & \text{if } v \in H \text{ or } w \in H. \end{cases}$$

Then Ω is semi-positive and closed (closed, because H is involutive and $d\omega = 0$). The fibers of $\bar{A} \to M$ are of dimension t and every A-invariant divisor on $\bar{A}$ restricts to a divisor on each fiber. Hence the second assertion follows. □

Lemma 5.5.14 *Let Z be a divisor on $\bar{A}$ such that $Z \cap A$ is effective. Assume that there are a line bundle $L_0 \in \mathrm{Pic}(M)$ and a divisor E with $\mathrm{Supp}\, E \subset \partial A$ and the associated line bundle $L(E)$, satisfying $L(Z) \otimes \bar{\pi}^* L_0^{-1} \cong L(E)$. Then $c_1(L_0) \geqq 0$.*

Proof Assume the contrary. Recall that M is a compact complex torus with universal covering $\pi_0 : \mathbf{C}^m \to M$ (with $m = \dim M$). We may regard the Chern class $c_1(L_0)$ as a bilinear form on the vector space $\mathbf{C}^m$. Since $c_1(L_0)$ is not semi-positive definite, there is a vector $v \in \mathbf{C}^m$ with $c_1(L_0)(v, v) < 0$. Set $W = \{w \in \mathbf{C}^m : c_1(L_0)(v, w) = 0\}$. Let μ be a semi-positive $(1, 1)$-form on $\mathbf{C}^m$ such that $\mu(v, \cdot) \equiv 0$ and $\mu|_{W \times W} > 0$. Let Ω be as in Lemma 5.5.13, and consider the $(n-1, n-1)$-form ω on $\bar{A}$ given by

$$\omega = \Omega^t \wedge \bar{\pi}^* \mu^{m-1}, \quad m = \dim M. \tag{5.5.15}$$

By construction we have $\omega \wedge \bar{\pi}^* c_1(L_0) < 0$.

Let $Z = Z' + Z''$ so that Z' is effective and no component of Z' is contained in ∂A, and $\mathrm{Supp}\, Z'' \subset \partial A$. By the Poincaré duality (Theorem 2.3.29),

$$\int_{\bar{A}} c_1\big(L(Z)\big) \wedge \omega = \int_Z \omega.$$

Since $\omega \wedge c_1(L(E)) = 0$, we have

$$\int_{\bar{A}} c_1\big(L(Z)\big) \wedge \omega = \int_{\bar{A}} \bar{\pi}^* c_1(L_0) \wedge \omega < 0.$$

On the other hand,

$$\int_Z \omega = \int_{Z'} \omega + \int_{Z''} \omega.$$

Note that $\int_{Z'} \omega \geqq 0$, because Z' is effective and $\omega \geqq 0$, and that $\int_{Z''} \omega = 0$, because $\mathrm{Supp}\, Z'' \subset \partial A$, and Ω^t vanishes on ∂A by Lemma 5.5.13 (ii). Thus we deduced a contradiction. □

We have the following decomposition of line bundles over $\bar{A}$.

Proposition 5.5.16 (i) *For every line bundle L on $\bar{A}$ there exist a line bundle L_0 on M and an A-invariant divisor D on $\bar{A}$ such that $L = \bar{\pi}^* L_0 \otimes L(D)$.*

(ii) *Let $L_1, L_2 \in \mathrm{Pic}(M)$ and let D_1, D_2 be A-invariant divisors on $\bar{A}$. Assume that*

$$\bar{\pi}^* L_1 \otimes L(D_1) \cong \bar{\pi}^* L_2 \otimes L(D_2).$$

Then there is a topologically trivial line bundle $H \in \mathrm{Pic}^0(M)$ such that $L_1 = L_2 \otimes H$ and $\bar{\pi}^* H \cong L(D_2 - D_1)$.

Proof (i) Recall that for the toric variety $\bar{T}$ the cohomology group $H^2(\bar{T}, \mathbf{Z})$ is generated by the fundamental classes of invariant divisors. Therefore there exists

an A-invariant divisor D on $\bar{A}$ such that $L \otimes L(D)$ is topologically trivial on $\bar{T}$. For a normal toric variety $\bar{T}$ the cohomology group $H^1(\bar{T}, \mathscr{O}_{\bar{T}})$ vanishes (Oda [85], Corollary 2.8). As a consequence for a line bundle on $\bar{T}$ topological triviality implies holomorphic triviality. Hence, $L \otimes L(D)$ is trivial along the $\bar{\pi}$-fibers. Therefore $R^0\bar{\pi}_*\mathscr{O}(L \otimes L(D))$ is an invertible coherent sheaf on M and thus $L = \bar{\pi}^*L_0 \otimes L(D)$ for some line bundle $L_0 \in \mathrm{Pic}(M)$.

(ii) Since $(D_1 - D_2) \cap A = \emptyset$ and $L(D_1 - D_2) \cong \bar{\pi}^*(L_2 \otimes L_1^{-1})$, we may apply Lemma 5.5.14 to $Z = D_1 - D_2$ and $L_0 = L_2 \otimes L_1^{-1}$. Hence $c_1(L_2 \otimes L_1^{-1}) \geqq 0$. Similarly, $c_1(L_1 \otimes L_2^{-1}) \geqq 0$. Thus $c_1(L_2 \otimes L_1^{-1}) = 0$. Because $H^2(M, \mathbf{Z})$ is torsion-free (M is a torus), $L_2 \otimes L_1^{-1}$ is topologically trivial. By the same arguments as in (i), this implies that this bundle comes from M. □

Example 5.5.17 In general, the decomposition $L = \pi^*L_0 \otimes L(D)$ given by the above proposition is not unique.

Let E be an elliptic curve and let $A \to E$ be a $\mathbf{C}^*$-principal bundle which is topologically trivial, but not holomorphically trivial. Then we get a semi-toric variety $\bar{A}$ by adding a zero section Z and an infinity section S to this $\mathbf{C}^*$-bundle. Consider $D = Z - S$. Then D is an A-invariant divisor. On the other hand, $L(D)$ is a line bundle which is topologically trivial on $\bar{A}$. Every topologically trivial line bundle on $\bar{A}$ is a pull-back of a line-bundle L_0 on E, because $H^1(\bar{A}, \mathscr{O}_{\bar{A}}) \cong H^1(E, \mathscr{O}_E)$. This bundle L_0 cannot be holomorphically trivial, for the holomorphic triviality of L_0 (and thereby of $L(D)$) would imply the existence of a meromorphic function on $\bar{A}$ with zero divisor Z and pole divisor H. Such a function would give a holomorphic trivialization of the $\mathbf{C}^*$-principal bundle $A \to E$.

Proposition 5.5.18 *Let L_0 be a line bundle on M and let D be an A-invariant divisor on $\bar{A}$. Let $L = \bar{\pi}^*L_0 \otimes L(D)$. Then L is ample on $\bar{A}$ if and only if both L_0 and $D \cap \bar{T}$ are ample.*

Proof The ampleness of L_0 follows from that of L by arguments similar to those in the proof of Lemma 5.5.14. Furthermore, $L|_F \cong L(D)|_{\bar{T}}$. Hence L being ample implies that both L_0 and $D \cap \bar{T}$ are ample.

For the converse, recall that ampleness is equivalent to being positive. Since $\bar{A} \to T$ is a topologically trivial bundle, the positivities of $c_1(L_0)$ and $c_1(D|_{\bar{T}})$ imply the positivity and therefore the ampleness of L. □

5.5.4 Global Span and Very Ampleness

We keep the notation as above.

Proposition 5.5.19 *Let $\bar{A}$ be a semi-toric variety, and let L be an ample line bundle on $\bar{A}$. Then L^2 is spanned by global sections.*

Proof Fix $p \in \bar{A}$. We want to show that there is a section $\sigma \in \Gamma(\bar{A}, L^2)$ such that $\sigma(p) \neq 0$. Since L is ample, $L \cong \pi^* L_0 \otimes L(D)$ where L_0 is an ample line bundle on M and D is an A-invariant divisor such that $D|_F$ is ample (Proposition 5.5.18). Using Lemma 5.5.3 there is an effective A-invariant divisor D' with $p \notin |D'|$ such that D and D' are linearly equivalent on each fiber of $\bar{\pi} : \bar{A} \to M$. From Proposition 5.5.16 (ii), it follows that $L(D - D') = \bar{\pi}^* H$ for some topologically trivial line bundle H on M. Now $L_0 \otimes H$ is ample on the abelian variety M. Therefore $(L_0 \otimes H)^2$ is spanned by global sections on M. On the other hand, $L(2D')$ has a section not vanishing at p. Combined, these facts imply that

$$L^2 \cong \bar{\pi}^*(L_0 \otimes H)^2 \otimes L\big(2D'\big)$$

has a section which does not vanish at p. □

For every ample line bundle L on a projective manifold X there exists a number m, depending on both X and the polynomial $P_L(k) = \chi(X, L^k)$ such that L^m is very ample (Matsusaka's Theorem, see Matsusaka [72]). It is well-known that m can be chosen as "3" if X is an abelian variety (Theorem 5.5.5). Here we prove a similar result for the case where X is a smooth equivariant compactification of a semi-abelian variety.

Theorem 5.5.20 *Let L be an ample line bundle on a semi-toric variety $\bar{A}$. Then L^3 is very ample on $\bar{A}$.*

Proof By Propositions 5.5.16 and 5.5.18 we know that $L = \pi^* L_0 \otimes L(D)$, where L_0 is ample on M and D is an A-invariant divisor on $\bar{A}$ such that $D \cap F$ is ample on F for every fiber F of $\bar{\pi}$. Note that L_0^3 is very ample on M (see Theorem 5.5.5).

Claim 5.5.21 *Let $p \in M$ and $F = \bar{\pi}^{-1}(p)$. Then every section $\sigma \in H^0(F, L^3)$ extends to a section of L^3 on $\bar{A}$.*

Recall that $F \cong \bar{T}$ is a toric variety. Hence $L|_F$ being ample implies that $L|_F$ is already very ample. Furthermore $H^1(F, \mathcal{O}_F) = \{0\}$. As a consequence the connected Lie group T acts trivially on $\mathrm{Pic}(T)$. For this reason $L|_F$ is T-invariant and we obtain a T-action on $H^0(F, L)$. Similarly, we obtain a T-action on $H^0(F, L^3)$. Let $\sigma \in H^0(F, L^3)$. Then there are characters $\chi_i : T \to \mathbf{C}^*$ and sections $\sigma_i \in H^0(F, L^3)$ such that $\sigma = \sum_i \sigma_i$ and $t^*\sigma_i = \chi_i(t)\sigma_i$ for all $t \in T$. Now the divisor (σ_i) determined by σ_i is T-invariant on F for every i. These divisors extend to A-invariant divisors $D_i = A \cdot (\sigma_i)$ on $\bar{A}$ and σ_i extend to sections $\tilde{\sigma}_i \in H^0(\bar{A}, L(D_i))$. Now $L(3D - D_i)$ is topologically trivial for each i. Hence there are topologically trivial line bundles $H_i' \in \mathrm{Pic}^0(M)$ such that $L(3D - D_i) \cong \bar{\pi}^* H_i'$. Since $\mathrm{Pic}^0(M)$ is a divisible group, there are line bundles $H_i \in \mathrm{Pic}^0(M)$ with $H_i^3 \cong H_i'$. The ampleness of $L_0 \in \mathrm{Pic}(M)$ implies that $L_0 \otimes H_i$ is also ample and that consequently $(L_0 \otimes H_i)^3 \cong L_0^3 \otimes H_i'$ is very ample. It follows that $L_0^3 \otimes H_i'$ admits a section not vanishing at p. Hence there exists a section

$\zeta_i \in H^0(\bar{A}, \bar{\pi}^*(L_0^3 \otimes H_i'))$ such that ζ_i has no zero on F. Now

$$\tilde{\sigma}_i \otimes \zeta_i \in H^0\big(\bar{A}, \bar{\pi}^*\big(L_0^3 \otimes H_i'\big) \otimes L(D_i)\big) \cong H^0\big(\bar{A}, L^3\big).$$

Thus each σ_i and consequently also their sum $\sigma = \sum_i \sigma_i$ extend to a section of L^3 over $\bar{A}$. The claim is proved.

If we now combine this claim with the facts that L_0^3 is very ample on M and that $L|_F$ is very ample on each fiber F, we easily verify that L^3 is very ample on $\bar{A}$. □

Lemma 5.5.22 *Let $\bar{T}$ be a toric variety. Then there exists a number l_0 such that for every ample line bundle L over $\bar{T}$ the bundle $L^{l_0} \otimes K_{\bar{T}}^{-1}$ is ample, too.*

Proof There are finitely many T-stable curves $C_1, \ldots, C_r$ (this follows from Theorem 5.2.11). By the toric Nakai criterion (Theorem 5.3.1) a line bundle L on $\bar{T}$ is ample if and only if $\deg(L|_{C_j}) > 0$ for all $j \in \{1, \ldots, r\}$. It follows that every l_0 with $\deg(K_{\bar{T}}^{-1}|_{C_j}) > -l_0$, $1 \leqq j \leqq r$, has the desired property. □

Proposition 5.5.23 *Let $A \hookrightarrow \bar{A}$ and $\bar{T}$ be as above. Then there exists a number l depending only on the toric variety $\bar{T}$ such that for every ample line bundle L on $\bar{A}$, the line bundle $L^l \otimes K_{\bar{A}}^{-1}$ is ample.*

Proof By Propositions 5.5.16 and 5.5.18 there is an ample line bundle L_0 on M and an A-invariant divisor D on $\bar{A}$ such that $D|_F$ is ample for every fiber F of $\bar{\pi}$. If $\{v_1, \ldots, v_m\}$ is a basis for $\mathrm{Lie}(A)$, then

$$K_{\bar{A}}^{-1} = (v_1 \wedge \cdots \wedge v_m).$$

Here, one recalls that $(\bullet)$ stands for the divisor defined by the section.

Thus $K_{\bar{A}}^{-1} = \partial A$. In particular, the anti-canonical bundle $K_{\bar{A}}^{-1}$ is induced by an A-invariant divisor. The adjunction formula implies $K_{\bar{A}}^{-1}|_F = K_F^{-1}$. Let l_0 be a number as in Lemma 5.5.22. Then $l_0 D|_F - K_{\bar{T}}$ is an ample divisor on the fiber F of $\bar{A} \to M$. Thus $l_0 D - K_{\bar{A}}$ is an A-invariant divisor on $\bar{A}$ whose restriction to a fiber is ample. On the other hand, $(L_0)^{l_0}$ is ample on M. Thus

$$(L_0)^{l_0} \otimes L(l_0 D - K_{\bar{A}}) \cong L^{l_0} \otimes K_{\bar{A}}^{-1}$$

is ample on $\bar{A}$ by Proposition 5.5.18. □

Corollary 5.5.24 *For every toric variety $\bar{T}$ there exists a number l such that for every semi-abelian variety A with smooth equivariant compactification $A \hookrightarrow \bar{A}$ of type $\bar{T}$ and every ample line bundle L on $\bar{A}$ the tensor power L^l is very ample and moreover $h^0(L^k) = \chi(L^k)$, $k \geqq l$.*

Proof We may choose l such that L^l is very ample (Proposition 5.5.20) and that furthermore $L^l \otimes K_{\bar{A}}^{-1}$ is ample (Proposition 5.5.23). By the Kodaira Vanishing Theorem, the ampleness of $L^k \otimes K_{\bar{A}}^{-1}$ with $k \geqq l$ implies $h^0(L^k) = \chi(L^k)$. □

Remark 5.5.25 For line bundles on the semi-tori themselves (as opposed to line bundles defined on some equivariant compactification) similar results are due to Takayama (see Takayama [01]).

5.5.5 Stabilizer and Bigness

We continue to use the same notation as above.

Theorem 5.5.26 *Let $\bar{A}$ be a semi-toric variety and let D be a reduced effective divisor on $\bar{A}$ such that $D \cap A \neq \emptyset$. Assume that* $\mathrm{St}(D)$ *is finite. Then $L(D)$ is big on $\bar{A}$ and A is semi-abelian.*

Proof We consider presentation (5.5.12). Due to Proposition 5.5.16 there is a line bundle L_0 on M and an A-invariant divisor E on $\bar{A}$ such that $L(\bar{D}) \cong L(E) \otimes \bar{\pi}^* L_0$. By Proposition 5.5.9 the finiteness of $\mathrm{St}(D)$ implies the ampleness of L_0. Therefore M is abelian and so A is semi-abelian.

Now consider the T-action. Evidently E is T-invariant. Since T acts only along the fibers of $\bar{\pi} : \bar{A} \to M$, the line bundle $\pi^* L_0$ is also T-invariant. It follows that for every $g \in T$ the pull-back $g^* D$ is linearly equivalent to D.[3] Next we define sets S_x for $x \in A$ as follows:

$$S_x = \bigcap_{g \in T : g(x) \in D} g^* D.$$

By this definition we know that for every $y \notin S_x$ there is a section σ in $L(D)$ such that $\sigma(x) = 0 \neq \sigma(y)$. From the definition it follows furthermore that S_x is an algebraic subvariety of A. Using the A-invariant trivialization of the tangent bundle $\mathbf{T}A \cong A \times \mathrm{Lie}(A)$ we can identify $\mathbf{T}(S_x)_x$ with a vector subspace of $\mathrm{Lie}(A)$. In this identification we obtain

$$\begin{aligned} \mathbf{T}(S_x)_x &= \bigcap_{g \in T : g(x) \in D} g^* \mathbf{T} D \\ &= \bigcap_{g \in T : g(x) \in D} \mathbf{T}(D)_{g(x)} \\ &= \bigcap_{y \in \bar{\pi}^{-1}(\bar{\pi}(x)) \cap D} \mathbf{T}(D)_y. \end{aligned}$$

Thus $\mathbf{T}(S_x)_x$ depends only on $\bar{\pi}(x)$. Let $F_x = \bar{\pi}^{-1}(\bar{\pi}(x))$. Then $F_x \cap S_x$ has the same tangent space (regarded as a vector subspace of $\mathrm{Lie}(A)$ at every point of

[3] Actually, $g^* D \sim D$ holds for every $g \in T$ and every $T \cong (\mathbf{C}^*)^s$-action on a projective manifold. This can be deduced from the fact that the Picard variety of a projective manifold contains no rational curves.

$F_x \cap S_x$. It follows that $F_x \cap S_x$ is an orbit under the action of a Lie subgroup of T. On the other hand, $F_x \cap S_x$ is an algebraic subvariety. Therefore $F_x \cap S_x$ is an orbit of an algebraic subgroup of T. A priori this subgroup may depend on the point x. However, $T \cong (\mathbf{C}^*)^s$ contains only countably many algebraic subgroups. For this reason it follows that this algebraic subgroup must be the same for almost all points $x \in A$. Thus there is an algebraic subgroup $H \subset T$ such that each connected component of $S_x \cap F_x$ is an H-orbit for almost all $x \in A$. But this implies that D is invariant under H. Since $\mathrm{St}(D)$ is finite, so is H. Thus $S_x \to A_0$ is generically finite for almost all $x \in A$. It follows that there are sections $\sigma_0, \ldots, \sigma_t$ of $L(D)$ such that for general $x \in A$ the meromorphic map

$$\alpha : p \mapsto \big[\sigma_0(p), \ldots, \sigma_t(p)\big] \in \mathbf{P}^t(\mathbf{C})$$

is holomorphic near x with the restricted map $\alpha|_{F_x}$ having maximal rank. Combined with the ampleness of L_0 this implies that D is big. □

Remark 5.5.27 The condition $\dim \mathrm{St}(D) = 0$ implies ampleness for a divisor on a compact complex torus. For a general semi-toric variety, however, we can only deduce bigness. We cannot deduce ampleness. For instance, let $\bar{A} = \mathbf{P}^2(\mathbf{C})$ and let D be a smooth curve of degree at least 4. Then by the genus formula ($g = (d-1)(d-1)/2$) we know that the genus of the curve D is at least 3. It follows that its automorphism group is finite. This easily implies that $\mathrm{St}(D)$ is finite. Furthermore $L(D)$ is ample on $\mathbf{P}^2(\mathbf{C})$. If we now blow up $\mathbf{P}^2(\mathbf{C})$ at a point $p \in \mathbf{P}^2(\mathbf{C}) \setminus D$, then we obtain a new toric variety Y. Now the strict transform of D cannot be ample on Y, because it does not intersect the exceptional divisor of the blow-up. On the other hand, the stabilizer is still finite.

There is a result of an opposite direction:

Proposition 5.5.28 *Let D be an effective divisor on a semi-toric variety $\bar{A}$ such that $L(D)$ is big. Then the connected component of $\mathrm{St}(D)$ is contained in the maximal linear connected subgroup T of A.*

Proof The stabilizer group $\mathrm{St}(D)$ acts naturally on the projective space $P(\Gamma(\bar{A}, L(kD))^*)$ for each $k \in \mathbf{N}$. Since D is big, there is a number k for which this action is effective. Thus $\mathrm{St}(D)$ can be embedded in some $PSL_N(\mathbf{C})$ and therefore is linear. □

Lemma 5.5.29 *Let D be an effective divisor on A and let $\bar{D}$ be its closure in an equivariant compactification $\bar{A}$ of A. Let $L_0 \in \mathrm{Pic}(M)$ and let E be an A-invariant divisor on $\bar{A}$ such that $L(\bar{D}) \cong L(E) \otimes \bar{\pi}^* L_0$. Assume that $\mathrm{St}(D)^0$ is contained in T. Then L_0 is ample on M.*

Proof By Lemma 5.5.14 we obtain $c_1(L_0) \geqq 0$. We may regard $c_1(L_0)$ as a bilinear form on a vector space V which can be interpreted as the Lie algebra $\mathrm{Lie}(M)$ or

the dual $\Omega^1(M)^*$ of the cotangent bundle over M. Assume that L_0 is not ample. Then there is a vector $v \in V \setminus \{0\}$ such that $c_1(L_0)|_{\mathbf{C}v} \equiv 0$. Choose a direct sum decomposition (orthogonal with respect to $c_1(L_0)$), $V = \mathbf{C}v \oplus V'$ and let ω be a $(1,1)$-form which is positive on V', but annihilates $\mathbf{C}v$. Then $c_1(L) \wedge \omega^{g-1} = 0$, where $g = \dim M = \dim V$. Let Ω be a $(1,1)$-form on $\bar{A}$ which is positive along the fibers of $\bar{A} \to M$ as constructed in Lemma 5.5.13. Then

$$0 = \int_{\bar{A}} \Omega^s \wedge \pi^*\big(c_1(L_0) \wedge \omega^{g-1}\big) = \int_D \Omega^s \wedge \pi^*\big(\omega^{g-1}\big).$$

By the construction of ω this implies that v is everywhere tangent to D. But in this case $v \in \mathrm{Lie}(M)$ is in the Lie algebra of the stabilizer $\mathrm{St}(D)$. This is a contradiction. □

5.6 Good Position and Stabilizer

5.6.1 Good Position

Definition 5.6.1 Let Z be a closed analytic subset of pure codimension k in a complex space X. Let G be a complex Lie group acting on X. As usual, given a subgroup $H \subset G$, the set of fixed points for the H-action on X is denoted by X^H. We say that a closed analytic subset Z of X is in *good position* (with respect to the G-action) if $Z \cap X^H$ is of pure codimension k in X^H for every connected complex Lie subgroup $H \subset G$. (If $\dim X^H < k$, we require $Z \cap X^H$ to be empty.)

Proposition 5.6.2 *Let Z be a closed algebraic subvariety of pure codimension k in an algebraic variety X on which an algebraic group G acts. Then Z is in good position if and only if $Z \cap X^H$ is of pure codimension k in X^H for every connected algebraic subgroup $H \subset G$.*

Proof Let $\bar{X}$ be an equivariant compactification (which exists by a result of Sumihiro [75]). Let H be a connected complex Lie subgroup of G and let $\bar{H}$ denote its Zariski closure, i.e., $\bar{H}$ is the smallest algebraic subgroup of G containing H. Then $X^H = \bar{X}^H \cap X$. By the GAGA principle, $\bar{X}^H$ is an algebraic subvariety of $\bar{X}$. Hence

$$I = \big\{g \in G;\, g \cdot x = x, {}^\forall x \in \bar{X}^H\big\}$$

is an algebraic subgroup of G. By this construction we have $X^I = X^H$ and $H \subset I$. Since I is an algebraic subgroup of G, it follows that $\bar{H} \subset I$. Now $X^H = X^{\bar{H}}$ follows from $H \subset \bar{H} \subset I$ and $X^H = X^I$. □

Remark 5.6.3 If Z is a subvariety in good position on a complex space X with respect to the action of a complex Lie group G, and $H \subset G$ is a complex Lie subgroup, then Z is also in good position with respect to the H-action.

Proposition 5.6.4 *Let D be a hypersurface on a complex space X on which a commutative complex Lie group G acts with finitely many orbits. Then D is in good position iff D does not contain any closed G-orbit.*

Proof Because G is commutative, points in the same G-orbit have the same isotropy groups. Since furthermore there are only finitely many orbits, it is clear that for every G-orbit W there is a closed connected complex Lie subgroup H of G such that W is an open subset of X^H. Therefore D is not in good position if D contains a G-orbit. Now assume that D does not contain a closed G-orbit. Due to the finiteness of the number of orbits every closure of an orbit contains a closed orbit. Hence in this case D cannot contain any G-orbit. But due to commutativity of G, for every subgroup $H \subset G$ we know that X^H is a G-invariant set and therefore is a finite union of closures of G-orbits. Hence we have the statement. □

Corollary 5.6.5 *Let $\bar{A}$ be a semi-toric variety. An effective divisor D on $\bar{A}$ is* in good position *if its support contains no closed orbit.*

Corollary 5.6.6 *An effective divisor D on a toric variety X is in good position iff its support contains no fixed point.*

Proposition 5.6.7 *Let G be a complex Lie group acting on a complex manifold X with finitely many orbits. Let Z be a subvariety in good position and let C be a G-invariant submanifold. Let $\pi : \hat{X} \to X$ be the blow-up along C.*

Then $\pi^{-1}Z$ is in good position. Furthermore $\pi^{-1}Z$ is the closure of $\pi^{-1}(Z \setminus C)$ in $\hat{X}$. (*In other words, the strict and total transforms of Z under the blow-up agree and are in good position.*)

Proof First note that there is a G-action on $\hat{X}$, because the center of blow-up C is G-invariant. For every $H \subset G$ the fixed point set $\hat{X}^H$ maps surjectively onto X^H. This implies the assertion on good position. Furthermore the condition of good position implies that no irreducible component of $\pi^{-1}Z$ may be contained in the exceptional divisor $\pi^{-1}C$. Hence the last assertion of the proposition is deduced. □

Example 5.6.8 Take a classical case where A is the complement of $n+1$ hyperplanes H_j of $X = \mathbf{P}^n(\mathbf{C})$ in general position. Then $G \cong (\mathbf{C}^*)^n$ acts on X with open orbit A. Let $D = H_{n+2}$ be an $(n+2)^{\text{th}}$ hyperplane of $\mathbf{P}^n(\mathbf{C})$. Then D is a divisor in good position on $X = \mathbf{P}^n(\mathbf{C})$ with respect to the G-action iff all H_j, $1 \leqq j \leqq n+2$, are in general position.

5.6.2 Good Position and Choice of Compactification

We investigate how the property of being in "good position" relates to the choice of the compactification.

Theorem 5.6.9 *Let $\bar{D}$ be a divisor on a semi-toric variety $\bar{A}$ such that the stabilizer group* $\mathrm{St}(D)$ *is discrete. Then the following assertions hold*:

(i) *$\bar{A}$ is an algebraic compactification of a semi-abelian variety A.*
(ii) *There is a (possibly singular) algebraic equivariant compactification $\bar{A}_1$ of A such that the closure of D in $\bar{A}_1$ is an ample Cartier divisor which is in good position.*
(iii) *There is a smooth algebraic equivariant compactification $\bar{A}_2$ of A such that the closure of D in $\bar{A}_2$ is big and in good position.*

We first show an auxiliary lemma.

Lemma 5.6.10 *Let $\bar{A}$ be a toric variety where $A = (\mathbf{C}^*)^d$. Let $\chi(A) \cong \mathbf{Z}^d$ be the character group of A, i.e., the group of all group homomorphisms from A to $\mathbf{C}^*$. Let $f_0, \ldots, f_s \in \chi(A)$. Let $\phi : \bar{A} \to \mathbf{P}^s(\mathbf{C})$ be an embedding whose restriction to A is given by*

$$\phi : x \mapsto \big[f_0(x), \ldots, f_s(x)\big].$$

Let D be a divisor on A given by $\sum_{j=0}^{s} a_j f_j = 0$ with $a_j \in \mathbf{C}$. Let Ω be the convex hull spanned by the $(f_j)_j$ in $\chi(A) \otimes \mathbf{R} \simeq \mathbf{R}^d$.

Then D is in good position in $\phi(\bar{A})$ iff $a_j \neq 0$ whenever f_j is a corner of Ω.

Proof The divisor D is in good position if it contains no fixed point for the A-action on $\bar{A}$ (Corollary 5.6.6). By projecting on a lower dimensional hyperplane we may assume that the monomials f_j are all distinct. Then every A-fixed point in the closure of $\phi(A)$ in $\mathbf{P}^s(\mathbf{C})$ must be of the form $[x_0, \ldots, x_s]$ with all x_i except one being zero. Put

$$e_m = [0, \ldots, \overset{m\text{-th}}{1}, \ldots, 0] \in \mathbf{P}^s(\mathbf{C}), \quad 0 \leqq m \leqq s.$$

Let k be an index such that f_k is a corner of the convex hull of the set of all f_j. Then there exists a one-parameter subgroup $H \cong \mathbf{C}^*$ of A such that $\lim |f_k/f_j| \to \infty$ for $t \to \infty$ in $t \in H \cong \mathbf{C}^*$. This implies that there exits a limit

$$\lim_{t \to \infty} \phi(t \cdot x) = e_k.$$

Hence $e_k \in \overline{\phi(A)}$. Now $e_k \notin \bar{D}$ implies $a_k \neq 0$.

Conversely, if f_k is not a corner of this convex hull, then there are real numbers $\lambda_j \geqq j$ such that

$$\log |f_k| = \sum_{j \neq k} \lambda_j \log |f_j|$$

which easily implies that $e_k \notin \overline{\phi(A)}$. □

Now we start the proof of Theorem 5.6.9.

Proof We start with showing (ii). We follow Vojta [96], Sect. 2.4.2.

First we discuss the case where $A \cong (\mathbf{C}^*)^g$. Let $z = (z_1, \ldots, z_g)$ be the standard coordinates on A. For a multi-index $I = (n_1, \ldots, n_g)$ we use the notation

$$z^I = \prod_j z_j^{n_j}.$$

Let the divisor D be defined by a function $f = \sum a_I z^I$. The *Newton polyhedron* H associated to D is defined as the convex hull of the set $\{I; a_I \neq 0\} \subset \mathbf{R}^g$. Let S denote the set of integral points in the Newton polyhedron, i.e., $S = H \cap \mathbf{Z}^g$. The assumption $\mathrm{St}(D) = 0$ implies that H is not contained in any hyperplane. It follows that H has non-empty interior. Upon replacing D by nD and thereby f by f^n and H by nH we may therefore assume that the set of all $s - s'$ with $s, s' \in S$ generates $\mathbf{Z}^g$ as a group. We consider the map $\phi : A \to \mathbf{P}^N(\mathbf{C})$ defined by

$$\phi : z \mapsto \left[z^{I_0}, \ldots, z^{I_N}\right], \qquad S = \{I_0, \ldots, I_N\}.$$

Since S generates $\mathbf{Z}^g$ as group, ϕ_L is an embedding. With

$$f = \sum_{j=0}^{N} a_{I_j} z^{I_j}$$

we define a divisor D' on $\mathbf{P}^N(\mathbf{C})$ by

$$D' = \left\{ [x_0, \ldots, x_N] \in \mathbf{P}^N(\mathbf{C}); \sum_{j=0}^{N} a_{I_j} x^j = 0 \right\}.$$

Then $D' \cap \phi(A) = D$ implies that the divisor D on $\phi(A)$ extends to an ample divisor on the closure $\overline{\phi(A)}$. Furthermore, Lemma 5.6.10 implies that D is in good position.

Next let us discuss the general case, where A is a semi-torus with presentation

$$1 \to L \to A \xrightarrow{\pi} M \to 1.$$

Consider the projection $\bar{\pi} : \bar{A} \to M$ and let $D_p = (\pi^{-1} p) \cap D$ for $p \in M$. Then $\mathrm{St}(D_p)$ is quasi-algebraic due to Proposition 5.3.16, and hence a semi-torus (Proposition 5.3.13). Since there are only countably many sub-semi-tori of A (Corollary 5.1.9), there is one sub-semi-torus B of A and a Zariski dense subset Ω of M such that $\mathrm{St}(D_p)^0 = B$ for all $p \in \Omega$. Since we assume $\mathrm{St}(D) = \{0\}$, we have $B = \{0\}$. Thus we have a point $p \in M$ where we can apply the preceding considerations, and deduce that there is a (possibly singular) toric variety $A_p \subset Z$ such that the closure of D_p in Z is ample and in good position. We now define $\bar{A}$ as the quotient of $A \times Z$ under the equivalence relation

$$(p, x) \cong \left(p', x'\right) \quad \Longleftrightarrow \quad {}^{\exists} g \in L : p = gp',\ x' = gx.$$

Let $\rho : A \times Z \to \bar{A}$ denote the quotient map. Then every A-orbit in $\bar{A}$ intersects $\rho(\{p\} \times Z)$. Hence the assertion about good position is clear. The ampleness of the

closure of D in $\bar{A}$ follows from Proposition 5.5.18. This ampleness implies that $\bar{A}$ is algebraic, completing the proof of the assertions (i) and (ii).

In order to obtain statement (iii), we just take an equivariant desingularization (cf. Hironaka [64a], [64b]; Bierstone–Milman [97]). This does not change the property of being in "good position" (Proposition 5.6.7), but the ampleness may get lost. However, in any case the closure of D is at least big due to Theorem 5.5.26. □

Example 5.6.11 We cannot hope to get one equivariant compactification $\bar{A}$ such that simultaneously $\bar{A}$ is smooth *and* the closure of D is ample. Consider the divisor D on $\mathbf{C}^* \times \mathbf{C}^*$ defined by

$$D = \left\{(z, w) \in \left(\mathbf{C}^*\right)^2; z^2 w + w^2 z + 1 = 0\right\}.$$

Assume that there exists a smooth equivariant compactification $\bar{A}$ of $A = \mathbf{C}^* \times \mathbf{C}^*$ such that the closure $\bar{D}$ of D in $\bar{A}$ is ample and in good position. From the theory of toric varieties we know that in this case D is already very ample and that the line bundle $L(D)$ is stabilized by A. It follows that there is an equivariant morphism ϕ from $\bar{A}$ to some projective space $\mathbf{P}^N(\mathbf{C})$ (namely, $P(\Gamma(\bar{A}, L(\bar{D}))^*)$) such that D is the pull-back of a linear hyperplane in $\mathbf{P}^N(\mathbf{C})$. Because ϕ is equivariant, we can write $\phi|_A$ as

$$\phi : (z, w) \mapsto \left[p_0(z, w), p_1(z, w), \ldots, p_N(z, w)\right]$$

for some monomials $p_j(z, w)$ $(0 \leq j \leq N)$. Recall that D is the pull-back of a linear hyperplane. Hence without loss of generality

$$\phi : (z, w) \mapsto \left[1, z^2 w, z w^2, p_2(z, w), \ldots, p_N(z, w)\right].$$

The condition of D being in good position can now be used to deduce that all the monomials $p_j(z, w)$ must lie in the Newton polyhedron associated to $z^2 w + z w^2 + 1$ (Lemma 5.6.10). There is only one more monomial in this polyhedron, namely zw. We observe that repeated occurrence of one and the same monomial may be ruled out by projecting to some lower dimensional projective space. Thus ϕ is reduced to either

$$\phi(z, w) = \left[1, z^2 w, z w^2\right]$$

or

$$\phi(z, w) = \left[1, z^2 w, z w^2, zw\right].$$

However, in the first case ϕ is not injective, while in the second case the image of ϕ is not smooth: The image is the hypersurface of $\mathbf{P}^3(\mathbf{C})$ given by

$$\phi(\bar{A}) = \left\{[x_0, x_1, x_2, x_3]; x_0 x_1 x_2 = x_3^3\right\}$$

which has singularities at $[1, 0, 0, 0]$, $[0, 1, 0, 0]$ and $[0, 0, 1, 0]$.

The following is an immediate consequence of Theorem 5.6.9.

Proposition 5.6.12 *Let Z be a reduced subvariety of A and let $\bar{Z}$ be its closure in a smooth equivariant compactification $\bar{A}$ of A. If $\mathrm{St}(Z)^0 = \{0\}$, then there exists a smooth equivariant compactification of A in which Z is in good position.*

Theorem 5.6.13 *Let D be a divisor on a semi-torus A with analytic closure $\bar{D}$ in a semi-toric variety $\bar{A}$ and let $B = A/\mathrm{St}(D)^0$. Then there exists semi-toric varieties $A \hookrightarrow \hat{A}$ and $B \hookrightarrow \bar{B}$ such that*

(i) *there is a bimeromorphic holomorphic and equivariant map $\tau : \hat{A} \to \bar{A}$;*
(ii) *the closure of D in $\hat{A}$ is in good position;*
(iii) *there is a holomorphic equivariant map $\rho : \hat{A} \to \bar{B}$ such that the generic fiber is a closure of an $\mathrm{St}(D)^0$-orbit;*
(iv) *the closure of the $D' = D/\mathrm{St}(D)^0$ is a divisor on $\bar{B}$ which is big and in good position.*

Proof From Theorem 5.6.9 (iii) we know that there exists a smooth equivariant compactification $B \hookrightarrow \bar{B}$ such that property (iv) of the statement holds. We obtain a dominant meromorphic equivariant map $\bar{\pi}$ from $\bar{A}$ to $\bar{B}$. We may blow up $\hat{A} \to \bar{A}$ equivariantly to obtain a *holomorphic map* $\hat{\pi}$ onto $\bar{B}$. Let $\hat{D}$ denote the closure of D in $\hat{A}$. Observe that $\hat{\pi}$ maps closed A-orbits onto closed B-orbits in $\bar{B}$. Hence $\pi(D)$ being in good position in $\bar{B}$ and Corollary 5.6.5 imply that the closure of D in $\hat{A}$ is a divisor in good position.

The other claimed properties are obvious from the construction. □

Proposition 5.6.14 *Let $\hat{B}$ be a semi-toric variety with semi-torus B. Let X be an irreducible complex space which can be realized as a B-principal bundle $\pi : X \to Y$ over a complex space Y. Define $\hat{X}$ as the quotient of $X \times_B \hat{B}$, i.e., as the quotient under the equivalence relation*

$$(x, b) \sim \left(x', b'\right) \quad \Longleftrightarrow \quad {}^{\exists} g \in B : x' = gx,\ b = gb'.$$

Let E be an irreducible closed analytic subset of $\hat{X}$ such that

$$\mathrm{St}(E) = \left\{b \in B; b(E) = E\right\}$$

is discrete.

Then there exists a B-equivariant blow-up $B^\dagger$ of $\hat{B}$ such that the B-orbits in $\hat{B}$ induce a stratification of $X^\dagger \overset{\mathrm{def}}{=} X \times_B B^\dagger$ with the following properties:

(i) *There are finitely many strata Γ_λ which are B-invariant locally closed analytic subsets.*
(ii) *The closure of $E \cap X$ in $X^\dagger$ contains none of the strata Γ_λ.*
(iii) *$\Gamma_\lambda \cong X/B_x$ $(x \in \Gamma_\lambda)$ where $B_x = \{b \in B; b \cdot x = x\}$ is the isotropy group at x.*
(iv) *The open subset X of $X^\dagger$ coincides with one of the strata Γ_λ.*

Proof Let $\hat{\pi} : \hat{X} \to Y$ denote the natural extension of $\pi : X \to Y$, i.e., the map obtained by sending the equivalence class of an element $(x, p) \in X \times \hat{B}$ to $\pi(x)$.

For $y \in Y$, set $E_y = \{p \in E; \pi(p) = y\}$. We observe that $\bar{E}_y = (\hat{\pi}^{-1}y) \cap \bar{E}$ for almost all $y \in \pi(E)$.

For each $y \in Y$ the stabilizer $\mathrm{St}(E_y)^0$ is a semi-torus (Proposition 5.3.16). Since there are only countably many sub-semi-tori of B (Corollary 5.1.9), none of which stabilizes E, we see that the stabilizer of E_y must be discrete for a generic point $y \in Y$. Thus we may invoke Proposition 5.6.12 and deduce that there exists an equivariant blow-up $B^\dagger \to \hat{B}$ such that E_y is in good position in $B^\dagger$ for a generic point $y \in Y$. Now we define $X^\dagger$ as $X \times_B B^\dagger$. The strata Γ_λ are defined by associating to each B-orbit Ω_λ in $B^\dagger$ the set

$$\Gamma_\lambda \stackrel{\mathrm{def}}{=} X \times_B \Omega_\lambda.$$

Now E_y being in good position in $B^\dagger$ implies that the closure of E in $X^\dagger$ contains none of the strata Γ_λ. The construction together with Theorem 5.2.11 implies that there are only finitely many strata Γ_λ. The other claimed properties are immediate. □

5.6.3 Regular Subgroups

Let A be a semi-torus. We give a definition of regular subgroups.

Definition 5.6.15 A closed complex Lie subgroup $H \subset A$ is said to be *regular* (with respect to the equivariant compactification $A \hookrightarrow \bar{A}$) if the quotient $A \to A/H$ extends holomorphically to a quotient $\bar{A} \to \bar{A}/H$.

Example 5.6.16 For $T = (\mathbf{C}^*)^t \hookrightarrow \mathbf{P}^t(\mathbf{C})$ there is no non-trivial connected *regular* subgroup of T. But we consider the case where $T \hookrightarrow \bar{T}$ is given by $(\mathbf{C}^*)^t \to (\mathbf{P}^1)^t$. Then we have the following. For a connected complex Lie subgroup $H \subset T$ the following are equivalent:

(i) The quotient $T \to T/H$ extends to a holomorphic quotient map $\bar{T} \to \bar{T}/H$.
(ii) There is a subset $I \subset \{1, \ldots, t\}$ such that

$$H = \left\{(z_1, \ldots, z_t) \in T; z_i = 1, {}^\forall i \in I\right\}.$$

We will prove the following proposition.

Proposition 5.6.17 *Let $\bar{A}$ be a semi-toric variety with $T = (\mathbf{C}^*)^t \hookrightarrow (\mathbf{P}^1)^t = \bar{T}$, and let D be an effective divisor on $\bar{A}$ in good position. Then there exists a finite unramified covering $\mu' : M' \to M$ with the following property*: $\mathrm{St}(\mu^* D)$ *is regular in A', where $\mu : A' \to A$ is the finite covering of A induced by μ'; i.e., $A' = A \times_M M'$.*

Remark 5.6.18 Note that μ extends holomorphically to a unramified covering of the compactification $\bar{A}$, $\bar{\mu} : \bar{A}' \to \bar{A}$.

Proof First, if D is invariant under one of the t direct factors of $T = (\mathbf{C}^*)^t$, we take the corresponding quotient. Thus we may assume that $\mathrm{St}(D) \cap T$ does not contain any of the t coordinate factors of T.

Assume that $\dim \mathrm{St}(D) \cap T > 0$. Let I be a subgroup of $\mathrm{St}(D) \cap T$ isomorphic to $\mathbf{C}^*$. Then there are integers $n_1, \ldots, n_t$ such that

$$I = \left\{ \left(w^{n_1}, \ldots, w^{n_t} \right); w \in \mathbf{C}^* \right\}.$$

By re-arranging indices and coordinate changes of type, $z_i \mapsto \frac{1}{z_i}$, we may assume that there is a natural number q such that $n_i > 0$ for $i \leqq q$ and $n_i = 0$ for $i > q$. Let $T = T_1 \times T_2$ with

$$T_1 = \left\{ (u_1, \ldots, u_{q-1}, \underbrace{1, \ldots, 1}_{t-q}); u_i \in \mathbf{C}^* \right\} \subset T,$$

$$T_2 = \left\{ (\underbrace{1, \ldots, 1}_{q}, u_{q+1}, \ldots, u_t); u_i \in \mathbf{C}^* \right\} \subset T.$$

Then $I \subset T_1$. Consider $\lambda : A \to A/T_1$. If $\lambda(D) \neq A/T_1$, then D would be T_1-invariant and in particular would be invariant under the coordinate factor groups contained in T_1. Since this was ruled out, we have $\lambda(D) = A/T_1$. Now observe that for every $u = (u_1, \ldots, u_t) \in \mathbf{C}^t \subset (\mathbf{P}^1(\mathbf{C}))^t$ we have

$$\lim_{w \to 0} \left(w^{n_1}, \ldots, w^{n_t} \right) \cdot u = (0, \ldots, 0, u_{q+1}, u_{q+2}, \ldots, u_t).$$

Hence it follows from $I \subset \mathrm{St}(D)$ and $\lambda(D) = A/T_1$ that the divisor D contains the A-invariant subset of $\bar{A}$ corresponding to

$$\{0\}^q \times \left(\mathbf{P}^1(\mathbf{C}) \right)^{t-q} \subset \bar{A}.$$

But D was assumed to be in good position, i.e., we assumed that D contains no orbits. Thus $T \cap \mathrm{St}(D)$ is zero-dimensional, and hence finite. As a consequence, $\mathrm{St}(D)$ is compact. After a finite covering, $\mathrm{St}(D)$ maps injectively in M and therefore is regular. □

5.6.4 More Facts on Semi-tori

A theorem of Kawamata [80] describes abelian subvarieties contained in a given subvariety of an abelian variety. It was generalized to the case of semi-abelian varieties by Noguchi [81a]. Here we formulate it for semi-tori.

Theorem 5.6.19 *Let T be a semi-torus and let $T \hookrightarrow \bar{T}$ be an equivariant compactification. Let $\bar{Z}$ be a closed analytic subset of $\bar{T}$ and let W be the union of all positive-dimensional orbits of sub-semi-tori of T which are contained in $Z = \bar{Z} \cap T$.*

Then there is a closed analytic subset $\bar{W}$ of $\bar{T}$ such that $W = \bar{W} \cap T$. Moreover each irreducible component of W has a positive-dimensional stabilizer group.

Proof Let $\mathbf{P} = P(\mathrm{Lie}\, T)$. Then we define subsets $Y_k \subset \bar{Z} \times \mathbf{P}$ by $(p, [v]) \in Y_k$ iff the $\exp(v)$-orbit touches Z at p with order at least k. This is a closed analytic subset, in fact $(p, [v]) \in Y_k$ iff $(v)^i(f)_p = 0$ for all $f \in \mathscr{I}\langle \bar{Z} \rangle_p$ and $i \leqq k$. Now let V_k denote the image of the projection of Y_k onto $\bar{Z}$. Set $V = \bigcap_{k \in \mathbf{N}} V_k$. Then V is a closed analytic set in $\bar{T}$ and $W = Z \cap T$.

Finally, let W_0 denote an irreducible component of W. We want to show that W_0 has a positive-dimensional stabilizer group. This is a consequence of the fact that a given semi-torus admits only countably many sub-semi-tori (Corollary 5.1.9). □

For the Kawamata Structure Theorem it is necessary to consider not arbitrary analytic subvarieties Z, i.e., to consider only those subvarieties Z of T whose closure $\bar{Z}$ in some equivariant compactification $\bar{T}$ of T is again analytic.

Example 5.6.20 We claim: *There is a closed analytic subset Z of $T = (\mathbf{C}^*)^8$ such that the stabilizer is trivial although Z is foliated by curves each of which is an orbit of a subgroup of T.*

For this phenomenon to be possible, evidently these leaves cannot be orbits of one fixed subgroup; instead for each leaf there must be a subgroup varying together with the leaf in question.

As the first step we observe that for every $\alpha \in \mathbf{C} \setminus \mathbf{R}$ the map

$$z \mapsto \left(\left|e^z\right|, \left|e^{\alpha z}\right|\right) = \left(e^{\Re z}, e^{\Re(\alpha z)}\right)$$

defines a bijective group homomorphism from $\mathbf{C}$ to $(\mathbf{R}_{>0})^2$, where $\mathbf{R}_{>0}$ denotes the group of positive numbers. It follows that for every $\alpha \in \mathbf{C} \setminus \mathbf{R}$ we obtain a closed embedding $\phi_\alpha : \mathbf{C} \to (\mathbf{C}^*)^2$ given by

$$\phi_\alpha : z \mapsto \left(e^z, e^{\alpha z}\right).$$

Next recall that every open Riemann surface can be embedded into $\mathbf{C}^3$ as a closed complex submanifold (due to the theory of Stein manifolds, e.g., cf. Hörmander [89], Chap. V, Sect. 5.3). Hence there is a closed holomorphic embedding

$$\tau : H^+ := \{z \in \mathbf{C};\, \Im z > 0\} \hookrightarrow \mathbf{C}^3.$$

As a consequence, we can embed H^+ into $(\mathbf{C}^*)^6$ as follows:

$$\eta = (\phi_i)^3 \circ \tau : H^+ \hookrightarrow \mathbf{C}^3 \hookrightarrow \left(\mathbf{C}^*\right)^6.$$

Combining η with the family ϕ_α, we obtain a closed embedding of $H^+ \times \mathbf{C}$ into $(\mathbf{C}^*)^8$:

$$H^+ \times \mathbf{C} \ni (\alpha, t) \overset{\psi}{\mapsto} \left(\eta(\alpha), e^t, e^{\alpha t}\right) \in \left(\mathbf{C}^*\right)^8.$$

By construction, the closed analytic subset $Z := \psi(H^+ \times \mathbf{C})$ admits a foliation by leaves $C_\alpha = \psi(\{\alpha\} \times \mathbf{C})$. Each C_α is an orbit of a one-dimensional subgroup of

$T = (\mathbf{C}^*)^6$, namely

$$H_\alpha = \{(1, \ldots, 1, e^t, e^{\alpha t}); t \in \mathbf{C}\}.$$

Since H_α depends on α, it is clear that the stabilizer group of Z in T is discrete.

The (logarithmic) Kodaira dimension of ananlytic subspaces of a semi-torus A which have compactifications in $\bar{A}$ can be characterized by using the dimension of the stabilizer group.

Proposition 5.6.21 *Let A be a semi-torus, let Z be an irreducuble analytic subset of A with compactification $\bar{Z} \subset \bar{A}$, let D be a Cartier divisor on Z and set $X = Z \setminus D$. Then*

$$\bar{\kappa}(X) + \dim \mathrm{St}(X)^0 = \dim X.$$

In particular, X is of logarithmic general type iff $\mathrm{St}(X)$ is discrete. Furthermore $\bar{\kappa}(X) = 0$ iff X is an orbit of $\mathrm{St}(X)$ and $D = \emptyset$.

Proof We consider the Ueno fibration $\pi : X \to Y$ (cf. Iitaka [77] and its Remark at p. 185). A very general fiber F of π has logarithmic Kodaira dimension zero. For its closure $\bar{F}$ in A we have $\bar{\kappa}(\bar{F}) \leq \bar{\kappa}(F) \leq 0$. Since $\bar{F} \subset A$ implies $\bar{\kappa}(\bar{F}) \neq -\infty$, we have $\bar{\kappa}(\bar{F}) = 0$ which implies that $\bar{F}$ is a sub-semi-torus of A. If $D \cap \bar{F} \neq \emptyset$, then $\bar{\kappa}(F) > 0$ (see Iitaka [77]). Thus F must be a sub-semi-torus of A. Because there are only countably many such sub-semi-tori (Corollary 5.1.9), we may conclude that there is one sub-semi-torus $B \subset A$ such that every general fiber of π is a B-orbit. Because Z is closed, it follows that Z is B-invariant. Now D is a hypersurface. Therefore either D is B-invariant or $B \cdot D = \{b \cdot x; b \in B, x \in D\}$ contains an open subset of Z. In the second case the union of all B-orbits in Z with empty intersection with D is not Zariski dense. This leads to a contradiction with the fact that a general fiber of π is a B-orbit. Thus D must be B-invariant. it follows that $B \subset \mathrm{St}(X)$. Hence

$$\bar{\kappa}(X) + \dim \mathrm{St}(X)^0 \geqq \dim Y + \dim B = \dim X.$$

On the other hand, $\mathrm{St}(X)$ induces a fibration $A \to A/\mathrm{St}(X)$. Applying the easy addition law of Iitaka [77] yields an inequality in the opposite direction. Hence we have that $\bar{\kappa}(X) + \dim \mathrm{St}(X)^0 = \dim X$. □

Theorem 5.6.22 (Kawamata [81], Theorem 27) *Let A be a semi-abelian variety. Let X be a normal algebraic variety. Let $\pi : X \to A$ be a finite morphism. Then $\bar{\kappa}(X) \geqq 0$ and there are a semi-abelian subvariety B of A, finite étale Galois covers $\tilde{X} \to X$ and $\tilde{B} \to B$, and a normal algebraic variety Y such that*

(i) *there is a finite morphism from Y to the quotient A/B,*
(ii) *$\tilde{X}$ is a fiber bundle over Y with fiber $\tilde{B}$ and with translations by $\tilde{B}$ as structure group,*
(iii) *$\bar{\kappa}(Y) = \dim Y = \bar{\kappa}(X)$.*

In the special case where $\bar{\kappa}(X) = 0$, this result takes the following form.

Theorem 5.6.23 (Kawamata [81], Theorem 26) *Let X be a normal algebraic variety, let A be a semi-abelian variety and let $\pi : X \to A$ be a surjective finite morphism. If $\bar{\kappa}(X) = 0$, then X is a semi-abelian variety and π is étale.*

Cf. Abramovich [94] and Abramovich–Oort [00] for some similar and related results to those in this subsection.

Chapter 6
Entire Curves in Semi-abelian Varieties

The value distribution of entire curves f from $\mathbf{C}$ into semi-abelian varieties A is studied here. We first study the order functions and the structure of the Zariski closures of their jet images $J_k(f)(\mathbf{C})$. Then we will prove the Second Main Theorem for $J_k(f), k \geqq 0$. For better understanding we begin with the simplest case where A is compact. Then we will prove the Second Main Theorem in the case of semi-abelian A with counting functions $N_{k_0}(r, f^*D)$. Making use of these results we finally establish the Second Main Theorem for $J_k(f)$ with counting functions truncated to level one; this is the best case.

The Second Main Theorem with truncation level one has a number of interesting applications; in particular, we will apply it to the algebraic degeneracy problem for entire curves into algebraic varieties.

6.1 Order Functions

Let M be a semi-torus. Then there is a short exact sequence of complex Lie groups, called "presentation" (see Sect. 5.1.5),

$$0 \to \left(\mathbf{C}^*\right)^p \to M \xrightarrow{\eta} M_0 \to 0, \tag{6.1.1}$$

where M_0 is a compact complex torus. Taking the universal coverings of (6.1.1), one gets

$$0 \to \mathbf{C}^p \to \mathbf{C}^n \to \mathbf{C}^m \to 0,$$

and an additive discrete subgroup Λ of $\mathbf{C}^n$ such that

$$\begin{aligned} &\pi : \mathbf{C}^n \to M = \mathbf{C}^n/\Lambda, \\ &\pi_0 : \mathbf{C}^m = \left(\mathbf{C}^n/\mathbf{C}^p\right) \to M_0 = \left(\mathbf{C}^n/\mathbf{C}^p\right)/\left(\Lambda/\left(\Lambda \cap \mathbf{C}^p\right)\right), \\ &\left(\mathbf{C}^*\right)^p = \mathbf{C}^p/\left(\Lambda \cap \mathbf{C}^p\right). \end{aligned} \tag{6.1.2}$$

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We fix a linear complex coordinate system

$$x = (x', x'') = (x'_1, \ldots, x'_p, x''_1, \ldots, x''_m)$$

on $\mathbf{C}^n$ such that $\mathbf{C}^p \cong \{x''_1 = \cdots = x''_m = 0\}$ and

$$\Lambda \cap \mathbf{C}^p = \mathbf{Z}\begin{pmatrix}1\\ \vdots\\ 0\end{pmatrix} + \cdots + \mathbf{Z}\begin{pmatrix}0\\ \vdots\\ 1\end{pmatrix}.$$

The covering mapping $\mathbf{C}^p \to (\mathbf{C}^*)^p$ is given by

$$\begin{pmatrix}x_1\\ \vdots\\ x_p\end{pmatrix} \in \mathbf{C}^p \to \begin{pmatrix}e^{2\pi i x_1}\\ \vdots\\ e^{2\pi i x_p}\end{pmatrix} = \begin{pmatrix}u_1\\ \vdots\\ u_p\end{pmatrix} \in (\mathbf{C}^*)^p.$$

We may regard $\eta : M \to M_0$ as a flat $(\mathbf{C}^*)^p$-principal fiber bundle. By a suitable change of coordinates $(x'_1, \ldots, x'_p, x''_1, \ldots, x''_m)$ the discrete group Λ is generated over $\mathbf{Z}$ by the column vectors of a matrix of the following type:

$$\begin{pmatrix}1 & \cdots & 0 & \\ \vdots & \ddots & \vdots & A\\ 0 & \cdots & 1 & \\ & O & & B\end{pmatrix}, \tag{6.1.3}$$

where A is a $(p, 2m)$-matrix and B is a $(m, 2m)$-matrix. Due to Theorem 5.1.35 the matrix-valued transition functions of the flat $(\mathbf{C}^*)^p$-principal fiber bundle $\eta : M \to M_0$ are locally constant with values of the form

$$\begin{pmatrix}a_1 & 0 & \cdots & 0\\ 0 & a_2 & \cdots & 0\\ \vdots & \vdots & \ddots & \vdots\\ 0 & 0 & \cdots & a_p\end{pmatrix}, \quad |a_1| = \cdots = |a_p| = 1. \tag{6.1.4}$$

Let $T \hookrightarrow \bar{T}$ be an algebraic smooth equivariant compactification of $T = (\mathbf{C}^*)^p$. For example, we may take $(\mathbf{C}^*)^p = (\mathbf{P}^1(\mathbf{C}) \setminus \{0, \infty\})^p \hookrightarrow (\mathbf{P}^1(\mathbf{C}))^p$. This gives us an equivariant compactification of $\bar{M}$ of M such that $\eta : M \to M_0$ extends to a flat $\bar{T}$-fiber bundle

$$\bar{\eta} : \bar{M} \to M_0.$$

Set

$$\partial M = \bar{M} \setminus M,$$

which is a divisor on $\bar{M}$ with only simple normal crossings (cf. Lemma 5.3.8).

Let Ω be a Kähler form on $\bar{M}$ as constructed in Theorem 5.3.21 (iv). Then Ω decomposes as $\Omega = \Omega_1 + \bar{\eta}^*\Omega_2$, where Ω_1 is induced by a Kähler form on $\bar{T}$ and Ω_2 is a Kähler form on the compact complex torus M_0.

In the sequel we always consider a complex semi-torus M with a *fixed* presentation as in (6.1.1) and with the discrete subgroup Λ satisfying (6.1.2).

Let $f : \mathbf{C} \to M$ be an entire curve. We regard f as an entire curve in $\bar{M}$ equipped with the Kähler form Ω, and define the order function by

$$T_f(r, \Omega) = \int_1^r \frac{dt}{t} \int_{\Delta(t)} f^*\Omega.$$

The order of f is defined by

$$\rho_f = \varlimsup_{r\to\infty} \frac{\log T_f(r, \Omega)}{\log r}.$$

Remark 6.1.5 It should be noticed that the order ρ_f is independent of the presentation (6.1.1) and the choice of compactification $\bar{M}$ (see Theorem 6.1.13 below).

Let $\tilde{f} : \mathbf{C} \to \mathbf{C}^n$ be the lift of f to the universal covering $\mathbf{C}^n \to M$, and set

$$\tilde{f}(z) = \big(F_1(z), \dots, F_p(z), G_1(z), \dots, G_m(z)\big),$$

where $F_i(z)$ and $G_j(z)$ are entire functions. By the base change from M_0 to the universal covering $\pi_0 : \mathbf{C}^m \to M_0$ applied to the fiber bundle $M \to M_0$, we have

$$M \times_{M_0} \mathbf{C}^m \cong \big(\mathbf{C}^*\big)^p \times \mathbf{C}^m, \qquad \bar{M} \times_{M_0} \mathbf{C}^m \cong \bar{T} \times \mathbf{C}^m.$$

Set

$$\hat{M} = \bar{T} \times \mathbf{C}^m.$$

Then $\hat{M}$ is the universal covering of $\bar{M}$, and therefore $\tilde{f}$ induces a lifting $\hat{f}$ of $f : \mathbf{C} \to M \hookrightarrow \bar{M}$,

$$\hat{f} : z \in \mathbf{C} \to \big(e^{2\pi i F_1(z)}, \dots, e^{2\pi i F_p(z)}, G_1(z), \dots, G_m(z)\big) \in \big(\mathbf{C}^*\big)^p \times \mathbf{C}^m \subset \hat{M}.$$

Set

$$\hat{f}_{(1)} : z \in \mathbf{C} \to \big(e^{2\pi i F_1(z)}, \dots, e^{2\pi i F_p(z)}\big) \in \big(\mathbf{C}^*\big)^p,$$
$$\hat{f}_{(2)} : z \in \mathbf{C} \to \big(G_1(z), \dots, G_m(z)\big) \in \mathbf{C}^m.$$

By definition we have

$$T_f(r, \Omega) = T_{\hat{f}_{(1)}}(r, \Omega_1) + T_{\hat{f}_{(2)}}(r, \Omega_2). \tag{6.1.6}$$

If $\bar{T} = (\mathbf{P}^1(\mathbf{C}))^p$, by the Shimizu–Ahlfors Theorem 1.1.19 we have

$$T_{\hat{f}_{(1)}}(r, \Omega_1) = \sum_{j=1}^{p} T\big(r, e^{2\pi i F_j}\big) + O(1). \tag{6.1.7}$$

By Jensen's formula (Lemma 1.1.5) we have

(6.1.8)

$$\begin{aligned} T_{\hat{f}_{(2)}}(r, \Omega_2) &= \int_1^r \frac{dt}{t} \int_{\Delta(t)} \frac{i}{2\pi} \partial\bar{\partial} \sum_{j=1}^{m} \big|G_j(z)\big|^2 \\ &= \frac{1}{4\pi} \int_0^{2\pi} \left(\sum_{j=1}^{m} \big|G_j\big(re^{i\theta}\big)\big|^2 \right) d\theta - \frac{1}{4\pi} \int_0^{2\pi} \left(\sum_{j=1}^{m} \big|G_j\big(e^{i\theta}\big)\big|^2 \right) d\theta. \end{aligned}$$

Theorem 6.1.9 *Let the notation be as above. Assume that $f : \mathbf{C} \to M$ has a finite order ρ_f. Then $F_j(z), 1 \leqq j \leqq p$, are polynomials of degree at most ρ_f, and $G_k, 1 \leqq k \leqq m$, are polynomials of degree at most $\rho_f/2$; moreover, at least one of F_j has degree ρ_f, or at least one of G_k has degree $\rho_f/2$.*

Proof Let $\varepsilon > 0$ be an arbitrary positive number. Then there is a $r_0 > 0$ such that

$$T_f(r, \Omega) \leqq r^{\rho_f + \varepsilon}, \quad r \geqq r_0.$$

It follows from (6.1.6)–(6.1.8) that for $r \geqq r_0$

$$T_{\hat{f}_{(1)}}(r, \Omega_1) \leqq r^{\rho_f + \varepsilon}, \tag{6.1.10}$$

$$\frac{1}{2\pi} \int_0^{2\pi} \left(\sum_{j=1}^{m} \big|G_j\big(re^{i\theta}\big)\big|^2 \right) d\theta \leqq r^{\rho_f + \varepsilon}. \tag{6.1.11}$$

Using the complex Poisson integral (1.1.15), we have

$$F_j(z) = \frac{i}{2\pi} \int_{\{|\zeta|=R\}} \frac{\zeta + z}{\zeta - z} \Im F(\zeta)\, d\theta + \Re F(0).$$

Therefore, using Theorems 1.1.16 and 1.1.17 with $1 < r < R < R'$ we obtain

(6.1.12)

$$\begin{aligned} &\max_{|z|=r} \big|F_j(z)\big| \\ &\quad \leqq \frac{R+r}{R-r} \max_{|\zeta|=R} \big|\Im F_j(\zeta)\big| + \big|\Re F_j(0)\big| \\ &\quad \leqq \frac{R+r}{R-r} \Big(\max_{|\zeta|=R} \big(\Im F_j(\zeta)\big)^+ + \max_{|\zeta|=R} \big(\Im\big(-F_j(\zeta)\big)\big)^+ \Big) + \big|\Re F_j(0)\big| \\ &\quad \leqq \frac{R+r}{R-r} \cdot \frac{R'+R}{R'-R} \cdot \frac{1}{2\pi} \big(m\big(R', e^{-2\pi i F_j}\big) + m\big(R', e^{2\pi i F_j}\big) \big) + \big|\Re F_j(0)\big| \end{aligned}$$

(continued)

$$\leqq \frac{R+r}{R-r}\cdot\frac{R'+R}{R'-R}\cdot\frac{1}{\pi}\big(T\big(R', e^{2\pi i F_j}\big)+O(1)\big)+O(1).$$

It follows from (6.1.10), (6.1.7), and (6.1.12) applied with $R=2r$ and $R'=3r$ that there is a positive constant C such that

$$\max_{|z|=r}\big|F_j(z)\big| \leqq Cr^{\rho_f+\varepsilon}.$$

Therefore, $F_j(z)$ is a polynomial of degree at most ρ_f.

Expand $G_j(z)=\sum_{\nu=0}^{\infty} c_{j\nu}z^{\nu}$. Then one gets

$$\frac{1}{2\pi}\int_0^{2\pi}\left(\sum_{j=1}^{m}\big|G_j\big(re^{i\theta}\big)\big|^2\right)d\theta=\sum_{j=1}^{m}\sum_{\nu=0}^{\infty}|c_{j\nu}|^2r^{2\nu}.$$

It follows from (6.1.11) that

$$\sum_{j=1}^{m}\sum_{\nu=1}^{\infty}|c_{j\nu}|^2r^{2\nu}\leqq r^{\rho_f+\varepsilon},\quad r\geqq r_0.$$

Hence, $c_{j\nu}=0$ for all $\nu>\rho_f/2$ and $1\leqq j\leqq m$. We see that $G_j(z)$, $1\leqq j\leqq m$, are polynomials of degree at most $\rho_f/2$. The remaining part is clear. □

In terms of Lie group theory we obtain the following characterization of entire curves of finite order ρ.

Theorem 6.1.13 *Let M be an n-dimensional complex semi-torus with the above compactification $\bar{M}$. Let $\mathrm{Lie}(M)$ be the Lie algebra of M and let $\exp:\mathrm{Lie}(M)\to M$ be the exponential map. Let $f:\mathbf{C}\to M$ be an entire curve. Then f is of finite order considered as an entire curve in $\bar{M}$ if and only if there is a polynomial map $P:\mathbf{C}\to\mathrm{Lie}(M)\cong\mathbf{C}^n$ such that $f=\exp\circ P$, and hence the property of f being of finite order is independent of the choice of the compactification $\bar{M}\supset M$.*

Remark 6.1.14 Different compactifications are not necessarily birationally equivalent (see e.g. Example 5.1.44). Therefore the above theorem is not a consequence of Theorem 2.5.18.

Lemma 6.1.15 *Let X be a compact Kähler manifold, let L be a hermitian line bundle on X, and let $\sigma_1,\sigma_2\in H^0(X,L)$ with $\sigma_1\not\equiv 0$. Let $f:\mathbf{C}\to X$ be an entire curve such that $f(\mathbf{C})\not\subset\mathrm{Supp}(\sigma_1)$. Then we have*

$$T\left(r,\frac{\sigma_2}{\sigma_1}\circ f\right)\leqq T_f\big(r,c_1(L)\big)+O(1).$$

Proof It follows from the definition that

$$N\left(r,\left(\frac{\sigma_2}{\sigma_1}\circ f\right)_{\infty}\right)\leqq N\big(r,f^*(\sigma_1)\big).$$

Moreover, we have

$$m\left(r, \frac{\sigma_2}{\sigma_1} \circ f\right) = \frac{1}{2\pi} \int_{\{|z|=r\}} \log^+ \frac{\|\sigma_2 \circ f\|}{\|\sigma_1 \circ f\|} \, d\theta$$
$$\leqq \frac{1}{2\pi} \int_{\{|z|=r\}} \log^+ \frac{1}{\|\sigma_1 \circ f\|} \, d\theta + O(1).$$

Together with the First Main Theorem this yields the assertion. □

6.2 Structure of Jet Images

Let A be a semi-abelian variety of dimension n, let $f : \mathbf{C} \to A$ be an entire curve, and let $J_k(f) : \mathbf{C} \to J_k(A)$ be its k-th jet lift. The A-action on itself by left multiplication lifts canonically to an A-action on each $J_k(A)$. There is a trivialization

$$J_k(A) \cong A \times \mathbf{C}^{nk}$$

of the jet bundle $J_k(A)$ such that the natural A-action is given by $a : (x, v) \mapsto (x + a, v)$ for $x \in A$, $v \in \mathbf{C}^{nk}$. We denote by $X_k(f)$ the Zariski closure of $J_k(f)(\mathbf{C})$ in $J_k(A) \cong A \times \mathbf{C}^{nk}$.

We are interested in the structure of $X_k(f)$. In particular, we show that its stabilizer group is always positive-dimensional.

6.2.1 Image of f (Case $k = 0$)

The next theorem is a direct consequence of Theorems 4.8.17 and 5.3.23, but is fundamental in the theory of entire curves into semi-abelian varieties.

Theorem 6.2.1 *Let $f : \mathbf{C} \to A$ be an entire curve. Then the Zariski closure $X_0(f)$ of $f(\mathbf{C})$ in A is a translate of a subgroup of A.*

It is noted that this is equivalent to the Logarithmic Bloch–Ochiai Theorem 4.8.17.

6.2.2 Jet Projection Method

Basically, we follow the arguments of Noguchi [98]. The key idea is to consider the jets of jets.

Lemma 6.2.2 *Let X be an irreducible complex algebraic variety with $p \in X$. Then there exists an algebraically non-degenerate holomorphic map f from the unit disk $\Delta(1)$ to X with $f(0) = p$.*

Proof Using a desingularization (see Hironaka [64a], [64b]; Hironaka–Milman [97]), there is no loss of generality in assuming that X is smooth. Let $m = \dim X$. Let $\alpha_1, \ldots, \alpha_m$ be $\mathbf{Q}$-linearly independent positive real numbers. Let $\Delta(2) = \{z \in \mathbf{C} : |z| < 2\}$ and fix an open embedding $\iota : \Delta(2)^m \hookrightarrow X$ such that $p \in \iota(\Delta(2)^m)$. Recall that there exists an isomorphism $\zeta : \Delta(1) \cong H^+ = \{z \in \mathbf{C} : \Im z > 0\}$. We define a map $\phi : H^+ \cup \mathbf{R} \to \Delta(2)^m \subset X$ by

$$z \mapsto \left(e^{\alpha_1 iz}, \ldots, e^{\alpha_m iz}\right).$$

Because the real numbers α_i are $\mathbf{Q}$-linearly independent, $\phi(\mathbf{R})$ is dense in the real torus $(S^1)^m$. Hence $(S^1)^m$ is contained in the closure of $\phi(H^+)$. Next we choose a holomorphic automorphism ψ of $\Delta(2)^m$ with $\psi(\phi(\zeta(0))) = p$ and define $f = \psi \circ \phi \circ \zeta$.

Because $(S^1)^m$ and therefore also $\psi((S^1)^m)$ is totally real and of real dimension m, it follows that no closed analytic subset of X except X itself can contain $\iota(\psi(\phi(H^+)))$. Thus $f(\Delta(1))$ must be Zariski dense. □

Remark 6.2.3 Regarded as a map from $\mathbf{C}$ to $\mathbf{C}^n$ the map ϕ defined in the above proof is an example of an entire curve from $\mathbf{C}$ to $\mathbf{C}^n$ whose image $\phi(\mathbf{C})$ is not contained in any proper analytic subset of $\mathbf{C}^n$. See Winkelmann [05] for more results in this direction.

Let $X \subset A \times \mathbf{C}^m$ be an irreducible algebraic subvariety. As in (5.5.7) we have the stabilizer group (with respect to the natural A-action) by

$$\mathrm{St}_A(X) = \left\{a \in A; (x + a, y) \in X \text{ for } {}^\forall(x, y) \in X \subset A \times \mathbf{C}^m\right\},$$

and the identity component $\mathrm{St}_A(X)^0$. Then $\mathrm{St}_A(X)^0$ is a connected closed algebraic subgroup of A.

For $l \in \mathbf{N}$, let $J_l(X)$ be the l-th jet space of X and let ρ_l denote the *jet projection*,

$$\rho_l : J_l\left(A \times \mathbf{C}^m\right) \cong A \times \mathbf{C}^{nl+m(l+1)} \to J_l\left(A \times \mathbf{C}^m\right)/A \cong \mathbf{C}^{nl+m(l+1)}.$$

Lemma 6.2.4 *Let X, A, m, ρ_l be as above. Let $V = \mathrm{Lie}(\mathrm{St}_A(X))$. Then for every sufficiently large l we have*

$$\mathrm{Ker}(d\rho_l|_{J_l(X)})_y = V$$

for general points y of $J_l(X)$. (Here V is regarded as subspace of $\mathrm{Lie}(A) \subset \mathbf{T}(J_l(X))_y$.)

Proof We fix arbitrarily a point $y_0 \in X$. There is an algebraically non-degenerate holomorphic map $f : \Delta(1) \to X$ with $f(0) = y_0$ by Lemma 6.2.2. Let $J_l(f) : \Delta(1) \to J_l(X)$ be the l-th jet lift, and set

$$y_l = J_l(f)(0) \in J_l(X), \quad l = 0, 1, 2, \ldots.$$

We are going to show that there is a number l_0 satisfying the condition: For every $l \geqq l_0$, the differential at y_l

$$d(\rho_l|_{J_l(X)})_{y_l} : \mathbf{T}\big(J_l(X)\big)_{y_l} \to \mathbf{T}\big(\mathbf{C}^{nl+m(l+1)}\big)_{\rho_l(y_l)}$$

is injective.

Let $x_0 \in A$ be the base point of y_l by the projection to the base space, $J_l(X) \to A$. First we note that

$$\mathbf{T}\big(J_l(X)\big)_{y_l} \subset \mathbf{T}\big(A \times \mathbf{C}^{nl+m(l+1)}\big)_{y_l} \cong \mathbf{T}(A)_{x_0} \oplus \mathbf{C}^{nl+m(l+1)}.$$

Because of the definition of $\rho_l|_{J_l(X)}$, we have the kernel,

$$\operatorname{Ker} d(\rho_l|_{J_l(X)})_{y_l} \subset \mathbf{T}(A)_{x_0}.$$

Since

$$\operatorname{Ker} d(\rho_l|_{J_l(X)})_{y_l} \supset \operatorname{Ker} d(\rho_{l+1}|_{J_{l+1}(X)})_{y_{l+1}},$$

there is a number l_0 such that

$$\operatorname{Ker} d(\rho_l|_{J_l(X)})_{y_l} = \operatorname{Ker} d(\rho_{l_0}|_{J_{l_0}(X)})_{y_{l_0}} \quad \text{for all } l \geqq l_0.$$

Suppose $\operatorname{Ker} d(\rho_{l_0}|_{J_{l_0}(X)})_{y_{l_0}} \neq \{0\}$. Take a vector $v \in \operatorname{Ker} d(\rho_{l_0}|_{J_{l_0}(X)})_{y_{l_0}} \setminus \{0\}$. Let $h \in \mathscr{I}\langle X\rangle_{y_0}$ be a germ of holomorphic functions in the ideal sheaf of X at y_0. Then v, considered as a vector field on A, satisfies

$$\left.\frac{d^l}{dz^l}\right|_{z=0} vh\big(f(z)\big) = 0, \quad l = 0, 1, 2, \ldots.$$

Therefore, $z \mapsto vh(f(z))$ has zero power series development, i.e. $vh(f(z)) \equiv 0$ in a neighborhood of 0. Since h was chosen arbitrarily, we obtain $vh(f(z)) \equiv 0$ near 0 for all $h \in \mathscr{I}\langle X\rangle_{y_0}$. Hence v is tangent to X at every point of $f(\Delta(1))$. Since $f(\Delta(1))$ is Zariski dense in X, v is everywhere tangent to X, so that X is invariant by the action of the one-parameter subgroup generated by v.

Hence

$$\operatorname{Ker} d(\rho_{l_0}|_{J_{l_0}(X)})_{y_{l_0}} \subset V.$$

On the other hand, $v \in \operatorname{Lie}(\operatorname{St}(X))$ immediately implies that v is everywhere tangent to $J_l(X)$ for all l. Thus

$$\operatorname{Ker} d(\rho_{l_0}|_{J_{l_0}(X)})_y \supset V, \quad {}^{\forall}y \in J_{l_0}(X).$$

A semicontinuity argument for the dimension of $\operatorname{Ker} d(\rho_{l_0}|_{J_{l_0}(X)})$ now completes the proof. □

By Lemma 6.2.4 we immediately have the following.

Lemma 6.2.5 *Let A be a semi-abelian variety, and let $g : \mathbf{C} \to A \times \mathbf{C}^m$ be an entire curve. Let X be the Zariski closure of $g(\mathbf{C})$. Let l be as in Lemma 6.2.4.*

Then for every $\mathrm{St}_A(X)^0$-invariant rational function ϕ on X the induced meromorphic function $\phi \circ g$ on $\mathbf{C}$ is algebraic over the field extension of $\mathbf{C}$ generated by all the components of $\rho_l \circ J_l(g)$.

Theorem 6.2.6 *Let A be a semi-abelian variety of dimension n, and let $f : \mathbf{C} \to A$ be a non-constant entire curve. Let X_k be the Zariski closure of $(J_k f)(\mathbf{C})$ and let $B = \mathrm{St}_A(X_k)$.*

(i) *Let $q : A \to A/B$ be the natural quotient map. Then $T_{q\circ f}(r) = S_f(r)$.*
(ii) $\dim B > 0$.
(iii) *Assume that f is of finite order. Then there exists a point $a \in A$, and a subvariety $W_k \subset \mathbf{C}^{nk}$ such that $X_k = (B + a) \times W_k$.*
(iv) *If A is a simple abelian variety, then there is a subvariety $W_k \subset \mathbf{C}^{nk}$ such that $X_k = A \times W_k$.*

Proof (i) Let $\{\phi_j\}$ be a transcendental basis of the rational function field of A/B over $\mathbf{C}$. Let $T_f(r)$ be the order function of f regarded as an entire curve in a projective algebraic compactification $\bar{A}$ of A with respect to an ample line bundle over $\bar{A}$.

The preceding Lemma 6.2.5 implies that $(q \circ f)^*\phi_j$ is algebraic over the rational function field generated by the components of $\rho \circ J_l(J_k(f))$. It follows from Lemma 4.7.1 that Nevanlinna's order function satisfies

$$T\big(r, (q \circ f)^*\phi_j\big) = S_f(r).$$

Then Theorem 2.5.18 implies that

$$T_{q\circ f}(r) = S_f(r).$$

This proves the first statement.

(ii) The assumption $\dim B = 0$ would imply $T_f(r) = S_f(r)$, which is impossible.

(iii) Let $F : \mathbf{C} \to (A/B) \times \mathbf{C}^{nk} \cong J_k(A)/B$ be a holomorphic map induced by $J_k(f) : \mathbf{C} \to J_k(A) \cong A \times \mathbf{C}^{nk}$. Let $F_1 : \mathbf{C} \to A/B$ denote its first component. Then F_1 is the composition of $f : \mathbf{C} \to A$ with the natural projection q from A to A/B.

Assume that f is of finite order. Let $\tilde{f} : \mathbf{C} \to \mathbf{C}^n$ denote the lift of f to the universal covering $\mathbf{C}^n$ of A. Then the lift $\tilde{f}$ is a polynomial map by Theorem 6.1.13, and hence $\rho_l \circ J_l(F)$ are polynomial maps, too. Lemma 6.2.5 above now yields that for every rational function ϕ on the semi-abelian variety A/B the induced meromorphic function $\phi \circ F_1$ is algebraic over the rational function field $\mathbf{C}(z)$ in z. It follows from Lemma 2.5.15 that Nevanlinna's order function fulfills $T_{\phi\circ F_1}(r) = O(\log r)$. By Theorem 1.1.26 this implies that $\phi \circ F_1$ is a rational function. Since this is true for every rational function ϕ on A/B, it follows that $F_1 : \mathbf{C} \to A/B$ is an algebraic morphism. But every algebraic morphism from $\mathbf{C}$ to a semi-abelian variety is constant. Thus the projection map from $J_k(A)$ to A/B maps X_k to a point. This implies that $X_k = (B + a) \times W_k$ for some subvariety $W_k \subset J_k(A)/A$ and $a \in A$.

(iv) If A is a simple abelian variety, then by the above (ii) $B = A$, so that the required assertion follows. □

6.2.3 A Counter-Example

In some literatures a statement stronger than that of Theorem 6.2.6 has been claimed.

It was claimed in Siu–Yeung [96], Theorem 2.2 for the case of an abelian variety A and in Noguchi [98], Proposition 1.8(ii) that $X_k(f)$ is a direct product of a translate of a semi-abelian subvariety of A with a subvariety of $\mathbf{C}^{nk}$. Unfortunately, there was a gap in these proofs; this was pointed out by P. Vojta in a private communication. But the proofs of the main theorems there can be easily fixed to work (cf. Remark 6.3.6 (iii)).

The key problem in these proofs was the following:

For $k > 0$ and $0 < \dim B < \dim A$, the quotient $J_k(A)/B$ is strictly larger than $J_k(A/B)$. Therefore one cannot use induction on $\dim A$ *to prove that $X_k(f)$ has a product decomposition as in the case where f is of finite order.*

In fact not only were the proofs deficient, the statement itself ($X_k(f)$ being a direct product) is wrong, as has been pointed in Noguchi–Winkelmann [03].

Below we describe the relevant counter-example.

Proposition 6.2.7 *There exist a three-dimensional complex abelian variety A and an algebraically non-degenerate entire curve $f : \mathbf{C} \to A$ such that $X_1(f)(\subset J_1(A) \cong A \times \mathbf{C}^3)$ is* not *a direct product inside $J_1(A)$; i.e., there do not exist subvarieties $C \subset A$, $W \subset \mathbf{C}^3$ such that $X = C \times W$.*

Proof Let C be a non-singular cubic curve in $\mathbf{P}^2(\mathbf{C})$. Then C is an elliptic curve, i.e., a one-dimensional compact complex torus. For a point $x = [x_0, x_1, x_2] \in \mathbf{P}^2(\mathbf{C})$ we define $x^\perp = \{[z_0, z_1, z_2]; \sum_i z_i x_i = 0\}$. We define a surface $\bar{S} \subset C \times \mathbf{P}^2(\mathbf{C})$ by

$$\bar{S} = \left\{(x, z) \in \mathbf{P}^2(\mathbf{C}) \times \mathbf{P}^2(\mathbf{C}); z \in x^\perp \text{ and } x \in C\right\}.$$

Note that the projection $C \times \mathbf{P}^2(\mathbf{C}) \to C$ induces the structure of a holomorphic fiber bundle on $\bar{S}$, with fiber $\mathbf{P}^1(\mathbf{C})$ and base C.

Fix $p \in \mathbf{P}^2(\mathbf{C}) \setminus C$. Now for every $x \in C$, the sets $p^\perp$ and $x^\perp$ are two different lines in $\mathbf{P}^2(\mathbf{C})$. In particular, $x^\perp \cap p^\perp$ contains exactly one point. Define $S = \bar{S} \setminus (C \times p^\perp)$. Then S is a closed algebraic subvariety of $C \times \mathbf{C}^2 \cong C \times (\mathbf{P}^2(\mathbf{C}) \setminus p^\perp)$. Let $\iota : S \to C \times \mathbf{C}^2$ denote this embedding. On the other hand, by the projection on the first factor, S can be realized as a holomorphic fiber bundle with fiber $\mathbf{C}$ and base C. Let $\pi : \tilde{S} \to S$ be the universal covering. Then $\tilde{S} \overset{\varphi}{\cong} \mathbf{C}^2$, because $\mathbf{C}$ is the universal covering of C and every $\mathbf{C}$-bundle over $\mathbf{C}$ is trivial.

Let $g : \mathbf{C} \to \mathbf{C}^2$ be a holomorphic map such that no proper analytic subset of $\mathbf{C}^2$ contains $g(\mathbf{C})$ (cf. Remark 6.2.3). Let $G = (G_1, G_2) : \mathbf{C} \to C \times \mathbf{C}^2$ be the holomorphic map given by the compositions

$$\mathbf{C} \xrightarrow{g} \mathbf{C}^2 \overset{\varphi}{\cong} \tilde{S} \xrightarrow{\pi} S \xrightarrow{i} C \times \mathbf{C}^2.$$

Note that $\iota(S)$ is the Zariski closure of $G(\mathbf{C})$ in $C \times \mathbf{C}^2$.

Choose a lattice Λ in $\mathbf{C}^2$ such that $B = \mathbf{C}^2/\Lambda$ is a simple complex abelian variety. Denote the natural projection by $\tau : \mathbf{C}^2 \to B$. Let $A = C \times B$. Now we are in a position to define the desired entire curve $f : \mathbf{C} \to A$: Let $f = (f_1, f_2) : \mathbf{C} \to C \times B$ with $f_1 = G_1$ and

$$f_2(x) = \tau\left(\int_0^x G_2(t)\,dt\right).$$

We have to verify that f has the desired properties. By construction $f(\mathbf{C})$ is not contained in (a translate of) one of the two factors of the product $C \times B$. On the other hand, the Zariski closure of $f(\mathbf{C})$ must be (a translate of) a subtorus and B is simple. Therefore $f(\mathbf{C})$ is Zariski dense in A.

Next we consider the first jet lift

$$J_1(f) : \mathbf{C} \to J_1(A) \cong A \times \mathbf{C}^3 \cong C \times B \times \mathbf{C} \times \mathbf{C}^2.$$

By construction we have

$$J_1(f) = \big(G_1, f_2, G_1', G_2\big).$$

Let $q : (x_1, x_2; v_1, v_2) \mapsto (x_1, v_2)$ denote the projection on the first and fourth factor. By our construction the closure of $q(J_1(f)(\mathbf{C}))$ in $C \times \mathbf{C}^2$ coincides with $\iota(S)$. In particular, it is not a direct product of subvarieties of the two factors. It follows that the Zariski closure of $J_1(f)$ in $J_1(A) = A \times \mathbf{C}^3$ cannot be a direct product of a subvariety of A with a subvariety of $\mathbf{C}^3$. □

Remark 6.2.8 The projection map p_2 from $\iota(S) \subset C \times \mathbf{C}^2$ onto the second factor $\mathbf{C}^2$ maps $\iota(S)$ surjectively onto $\mathbf{C}^2$. As a result, for a holomorphic map $G : \mathbf{C} \to \iota(S)$ with a Zariski dense image in $\iota(S)$ the induced map $G_2 = p_2 \circ G : \mathbf{C} \to \mathbf{C}^2$ has a Zariski dense image. Therefore in the above construction $G_2 : \mathbf{C} \to \mathbf{C}^2$ cannot be a polynomial map. As a consequence f_2 and therefore $f : \mathbf{C} \to A$ is not an entire curve of finite order. Note also that $\mathrm{St}_A(X) = \{0\} \times B$.

6.3 Compact Complex Tori

6.3.1 Entire Curves

Here we establish a Second Main Theorem for entire curves $f : \mathbf{C} \to A$ into a compact complex torus A and an effective divisor D on A. From this a conjecture of S. Lang (Corollary 6.3.5 (ii)) will be deduced.

Theorem 6.3.1 (The Second Main Theorem; Noguchi–Winkelmann–Yamanoi [00], [02]) *Let $L \to A$ be a line bundle on a compact complex torus A and let $D \in |L|$.*

For every entire curve $f : \mathbf{C} \to A$ *such that* $f(\mathbf{C}) \not\subset \operatorname{Supp} D$ *there is a number* $k_0 \in \mathbf{N}$ *such that*

$$T_f(r, L) = N_{k_0}\big(r, f^*D\big) + S_f(r).$$

Proof By taking a quotient, $A \xrightarrow{\lambda} A' := A/\operatorname{St}(D)^0$, we have an effective divisor $D' = \lambda(D)$ on the compact complex torus A' such that $\operatorname{St}(D')^0 = \{0\}$; hence A' is an abelian variety and D' is ample (Proposition 5.5.9). Note that

$$\begin{aligned} \lambda \circ f(\mathbf{C}) &\not\subset \operatorname{Supp} D', \\ T_f(r, L) &= T_{\lambda\circ f}\big(r, L\big(D'\big)\big), \\ N_k\big(r, f^*D\big) &= N_k\big(r, (\lambda \circ f)^*D'\big), \quad 1 \leqq k \leqq \infty. \end{aligned}$$

Therefore we may assume that A is an abelian variety and $\operatorname{St}(D)^0 = \{0\}$.

Furthermore, by Theorem 6.2.1 the Zariski closure $X_0(f)$ of the image of f is a translate of an abelian subvariety A_0 of A. By translation we may assume that $X_0(f) = A_0$. The restriction $D_0 = D|_{A_0}$ is an ample divisor on A_0 and as $f : \mathbf{C} \to A_0$, $T_f(r, L(D_0)) = T_f(r, L)$ and $f^*D_0 = f^*D$. Therefore we may assume that $A = A_0$, i.e., f is algebraically non-degenerate.

Let $\dim A = n$ and let $X_k(f) \subset J_k(A)$ be the Zariski closure of the image of the k-jet lift $J_k(f)$ of f, as before. Let I_k denote the restriction of the jet projection $p_k : J_k(A) \to \mathbf{C}^{nk}$ to $X_k(f)$. Note that I_k is proper. Therefore $Y_k = I_k(X_k(f))$ is an irreducible algebraic subset of $\mathbf{C}^{nk}$. To continue the proof we need the key lemma below.

Lemma 6.3.2 *There is a number* $k_0 \in \mathbf{N}$ *such that for* $k \geqq k_0$

$$I_k\big(X_k(f)\big) \cap I_k\big(J_k(D)\big) \neq I_k\big(X_k(f)\big) = Y_k.$$

Proof It suffices to show that there is a number $k \in \mathbf{N}$ satisfying

$$I_k\big(J_k(D)\big) \not\ni I_k\big(J_k(f)(0)\big).$$

Assume the converse. Then there is a sequence of points $a_k \in A$, $k = 1, 2, \ldots$, such that

$$a_k + J_k(f)(0) \in J_k(D),$$

because the I_k-fibers are A-orbits for the natural A-action on $J_k(A) \cong A \times (\mathbf{C}^n)^k$. The compactness of A implies that a subsequence of the sequence (a_k) converges to an element $a \in A$.

Then the entire curve defined by

$$g : z \in \mathbf{C} \to a - f(0) + f(z) \in A$$

satisfies that $J_k(g)(0) \in J_k(D)$ for all $k \geqq 0$. This means that $g(\mathbf{C}) \subset D$, and hence that f is algebraically degenerate; a contradiction was deduced. □

Continuation of the Proof of Theorem 6.3.1 Let $\{U_\lambda\}$ be an affine covering of A such that on every U_λ the divisor D is defined by some regular function, i.e., $D \cap U_\lambda = \{\sigma_\lambda = 0\}$. The defining equations of $J_k(D)|_{U_\lambda}$ are given by

$$\sigma_\lambda = d\sigma_\lambda = \cdots = d^k\sigma_\lambda = 0.$$

By Lemma 6.3.2 there is $k = k_0$ for which there is a polynomial P on $\mathbf{C}^{nk}$ satisfying

$$P|_{Y_k} \not\equiv 0, \qquad P|_{J_k(D)} \equiv 0.$$

On each U_λ one obtains the following equation:

$$a_{\lambda 0}\sigma_\lambda + \cdots + a_{\lambda k}d^k\sigma_\lambda = {I_k}^* P|_{U_\lambda}. \tag{6.3.3}$$

Here $a_{\lambda j}$ are polynomials in jet coordinates with coefficients of rational holomorphic functions on U_λ restricted on $J_k(A)|_{U_\lambda}$.

Using a hermitian metric on the line bundle $L(D)$, we have functions $\rho_\lambda \in C^\infty(U_\lambda)$ such that $\rho_\lambda > 0$ and $\frac{|\sigma_\lambda|}{\rho_\lambda} = \frac{|\sigma_\mu|}{\rho_\mu}$ on $U_\lambda \cap U_\mu$. Therefore

$$\rho_\lambda a_{\lambda 0} + \rho_\lambda a_{\lambda 1}\frac{d\sigma_\lambda}{\sigma_\lambda} + \cdots + \rho_\lambda a_{\lambda k}\frac{d^k\sigma_\lambda}{\sigma_\lambda} = \frac{\rho_\lambda}{\sigma_\lambda}{I_k}^* P|_{U_\lambda}.$$

Substituting $J_k(f)(z)$, $f(z) \in U_\lambda$ to the above equation, we have

(6.3.4)

$$\left|\rho_\lambda\big(f(z)\big)a_{\lambda 0}\big(f(z)\big) + \rho_\lambda\big(f(z)\big)a_{\lambda 1}\big(f(z)\big)\frac{\frac{d}{dz}\sigma_\lambda(f(z))}{\sigma_\lambda(f(z))} + \cdots \right.$$

$$\left. + \rho_\lambda\big(f(z)\big)a_{\lambda k}\big(f(z)\big)\frac{\frac{d^k}{dz^k}\sigma_\lambda(f(z))}{\sigma_\lambda(f(z))}\right| \cdot \frac{1}{|P(I_k(J_k(f)(z)))|} = \frac{|\rho_\lambda(f(z))|}{|\sigma_\lambda(f(z))|}.$$

Let $\{\tau_\lambda\}$ be a partition of unity subordinated to the covering $\{U_\lambda\}$. Then

$$\frac{1}{\|\sigma(f(z))\|} \leqq \frac{1}{|P(I_k(J_k(f)))|} \times \sum_\lambda \left\{\tau_\lambda\rho_\lambda|a_{\lambda 0}| + \cdots + \tau_\lambda\rho_\lambda|a_{\lambda k}|\left|\frac{\frac{d^k}{dz^k}\sigma_\lambda(f(z))}{\sigma_\lambda(f(z))}\right|\right\}.$$

Since $a_{\lambda j}$ are polynomials in jet coordinates with coefficients of holomorphic functions on U_λ, Lemma 4.7.1 yields that

$$m_f(r, D) \leqq O\left(\sum_{\lambda, 1\leqq j\leqq k} m\left(r, \frac{(\sigma_\lambda \circ f)^{(j)}}{\sigma_\lambda \circ f}\right)\right) + m\left(r, \frac{1}{P(I_k(J_k(f)))}\right) + S_f(r).$$

Since σ_λ is a rational function on A, $d^j\sigma_\lambda/\sigma_\lambda$ is a logarithmic jet differential carrying logarithmic poles on zeros and poles of σ_λ. It follows from Lemma 4.7.1 that

$$m\left(r, \frac{(\sigma_\lambda \circ f)^{(j)}}{\sigma_\lambda \circ f}\right) = S_f(r).$$

Moreover the First Main Theorem 1.1.17 and Lemma 4.7.1 imply that

$$m\left(r, \frac{1}{P(I_k(J_k(f)))}\right) \leqq T\big(r, P\big(I_k\big(J_k(f)\big)\big)\big) + O(1) = S_f(r).$$

Combining all these, we obtain $m_f(r, D) = S_f(r)$.

It is inferred from Lemma 4.7.1 and (6.3.4) that

$$\begin{aligned} N\big(r, f^*D\big) - N_k\big(r, f^*D\big) &\leqq N\big(r, \big(P\big(I_k\big(J_k(f)\big)\big)\big)_0\big) \\ &\leqq T\big(r, P\big(I_k\big(J_k(f)\big)\big)\big) + O(1) = S_f(r). \end{aligned}$$

This with the First Main Theorem 2.3.31 implies that

$$\begin{aligned} T_f(r, L) &= N\big(r, f^*D\big) + m_f(r, D) \\ &= N_k\big(r, f^*D\big) + S_f(r). \end{aligned}$$

□

Corollary 6.3.5 *Let f and D be as in Theorem 6.3.1.*

(i) $\delta_{k_0}(f, D) = 0$.

(ii) (Lang's Conjecture) *If D is ample and $f(\mathbf{C}) \cap D = \emptyset$, then f is constant.*

Proof (i) This is clear by definition.

(ii) The assumption, Theorems 6.3.1 and 2.5.4 immediately imply the constancy of f. □

Remark 6.3.6 (i) (Example) The number k_0 in Theorem 6.3.1 depends on f and D in general, even if D is required to be reduced. For example, with $\mathbf{Z}[i] = \mathbf{Z} + i\mathbf{Z}$ we set $E = \mathbf{C}/\mathbf{Z}[i]$, $\Gamma = (\mathbf{Z}[i])^2$ and $A = E^2 = \mathbf{C}^2/\Gamma$. Consider the following entire curve:

$$f : z \in \mathbf{C} \to \big[(z, z^2)\big] \in A.$$

Here $[\cdot]$ stands for the Γ-equivalence class. Let D be a reduced divisor on A such that it has a cusp of order p at the point $e_A = [(0, 0)]$. Put $L = L(D)$. Since $f^{-1}e_A = \Lambda$,

$$f^*D \geqq p\Lambda.$$

Thus, for an arbitrary k_0 we take $p > k_0$ so that

$$N\big(r, f^*D\big) - N_{k_0}\big(r, f^*D\big) \geqq (p - k_0)r^2\big(1 + o(1)\big).$$

On the other hand, $T_f(r, L) = r^4(1 + o(1))$ and therefore

$$T_f(r, L) - N_{k_0}\big(r, f^*D\big) \geqq (p - k_0)r^2\big(1 + o(1)\big).$$

The right-hand side cannot be bounded by $S_f(r)$ from above.

This serves also as a counter-example to Lemma 4 of the paper cited below,[1] where the same problem was dealt with.

(ii) While the truncation level k_0 cannot be improved to 1 as long as we keep $S_f(r)$ as the small term in the Second Main Theorem, it is in fact possible to achieve truncation level 1 if we are ready to replace the small term $S_f(r)$ by a slightly bigger term. See Sect. 6.5 below.

(iii) The above Lang's Conjecture stated in Corollary 6.3.5 (ii) was first proved by Siu–Yeung [96]. But note that there is a gap in the proof of the "Generalized Bloch's Theorem" and it does not hold in general (see Remark 6.3.8); the proof of Lang's Conjecture itself may be fixed. They used the Wronskian formed by the associated theta function and the theory of differential equations in an elaborate way, which is different to the proof given above. Another proof based on the jet projection method, jets of jet bundles and the theta Wronskian was given by Noguchi [98], dealing with semi-abelian varieties as well.

The statement is as follows:

Theorem 6.3.7 *Let D be an algebraic reduced divisor on a semi-abelian variety A. Let $f : \mathbf{C} \to A \setminus D$ be an entire curve. Then the image $f(\mathbf{C})$ is contained in a translate B of a semi-abelian subvariety of A such that $B \cap D = \emptyset$.*

The proof is now immediate by Theorems 6.2.1 and 6.4.1. Cf. Tiba [13] for an application of this theorem for the Kobayashi hyperbolicity problem.

(iv) Let A and D be as in Corollary 6.3.5 (ii). Then every entire curve $f : \mathbf{C} \to A \setminus D$ is constant, but it is not yet proved that $A \setminus D$ is Kobayashi hyperbolic (cf. Chap. 5). In this sense the title of the paper Siu–Yeung [96] seems to be causing a confusion. Kobayashi hyperbolicity of $A \setminus D$ does hold under the additional condition that D is Kobayashi hyperbolic (see Theorem 7.3.8). In general it is an open problem if $A \setminus D$ is Kobayashi hyperbolic.

(v) The problem to prove the Second Main Theorem 6.3.1 for entire curves into abelian varieties was dealt with in other papers.[2,3] Unfortunately, those proofs are incomplete or based on an incorrect lemma. The reason why their proof failed is rather involved, and will be discussed in detail in Remark 6.3.8 below.

Hence Noguchi–Winkelmann–Yamanoi [00], [02] were the first to give a proof of the Second Main Theorem for entire curves in abelian varieties. Later, in Siu–Yeung [03] they improved the level k_0 to be dependent only on D.

Remark 6.3.8 (Cf. Noguchi–Winkelmann–Yamanoi [02], Example (5.36)) Siu and Yeung (see footnote 3) claimed a Second Main Theorem for abelian A of dimension n. The most essential part of their proof was their Lemma 2 (see footnote 3), but the claimed assertion is unfortunately mathematically incorrect. We recall it.

[1] Kobayashi, R., Holomorphic curves into algebraic subvarieties of an abelian variety, Int. J. Math. **2** (1991), 711–724.

[2] Kobayashi, R., Holomorphic curves in Abelian varieties: The second main theorem and applications, Jpn. J. Math. **26** (2000), 129–152.

[3] Siu, Y.-T. and Yeung, S.-K., Defects for ample divisors of Abelian varieties, Schwarz lemma, and hyperbolic hypersurfaces of low degrees, Am. J. Math. **119** (1997), 1139–1172.

Lemma 2 (Siu–Yeung, p. 1147 (see footnote 3)) *Let A be an n-dimensional Abelian variety and let D be an ample divisor on A. Let $f : \mathbf{C} \to A$ be an entire curve with Zariski dense image, and define Y_k as in Lemma 6.4.4. Let $\delta \geqq 1$ and $k \geqq n$ be arbitrarily fixed integers. Then for every positive integer q there exists a positive integer $m_0(Y_k, \delta, q)$ depending on Y_k, δ, q (and A and D) such that for $m \geqq m_0(Y_k, \delta, q)$ there exists an $(L(D))^{\delta}$-valued holomorphic k-jet differential ω on A of weight m whose restriction to $A \times Y_k$ is not identically zero and which vanishes along $J_k(D) \cap (A \times Y_k)$ to order at least q. In particular, from the definition of Y_k one knows that ω is not identically zero on $J_k(f)$.*

In the proof of their Second Main Theorem they applied this lemma, taking $\delta = 1$ and $k \geqq n$ fixed, and increasing $q \to \infty$.

Assume for a moment that A is simple. Then $X_k(f) = A \times W_k \subset J_k(A) \cong A \times \mathbf{C}^{nk}$, and $W_k = Y_k$ in Lemma 2 above. Hence it is clear that the variety W_k must be allowed to be quite arbitrary. In fact, in the proof of Lemma 2 the fact that $A \times Y_k$ was defined to be the Zariski closure of the k-jet lifting of $f : \mathbf{C} \to A$ with Zariski dense image was not used at all, and hence Lemma 2 should be true for arbitrary non-empty subvariety $Y_k \subset \mathbf{C}^{nk}$ of an arbitrary abelian variety A with ample divisor D if the proof were correct. This is a very different point from our proof (cf. the proof of Lemma 6.3.2). But, we then deduce some contradictory conclusions as follows.

(a) We take an ample divisor D on A such that it contains a translate of a non-trivial Abelian subvariety A' (cf. Noguchi–Ochiai [90], Example (6.4.13) for such an example). Let $g : \mathbf{C} \to A'$ be a one-parameter subgroup with Zariski dense image. We regard g as an entire curve in A, and set $f(z) = g(z) + a$ with $a \in A \setminus D$. Then f is an entire curve such that $f(\mathbf{C}) \not\subset D$, and Y_k consists of only one point for every $k \geqq 1$. We obtain $A \times Y_k \cong A$. Through this isomorphism, we have that

$$A' \subset J_k(D) \cap (A \times Y_k) \subsetneqq A.$$

Let $\mathscr{I}_k = \mathscr{I}\langle J_k(D) \cap (A \times Y_k)\rangle$ denote the ideal sheaf of $J_k(D) \cap (A \times Y_k)$ $(\subset A)$. Note that any jet differential of any weight m restricted to $A \times Y_k$ is reduced to a jet differential of weight 0, for Y_k consists of one point. Then Lemma 2 above should imply that for all $q \geqq 1$

$$H^0\big(A, \mathscr{O}\big(\big(L(D)\big)^{\delta}\big) \otimes \mathscr{I}_k^q\big) \neq \{0\}.$$

Since $A' \subset J_k(D) \cap (A \times Y_k)$, the ideal sheaf $\mathscr{I} = \mathscr{I}\langle A'\rangle$ contains $\mathscr{I}_k$. Therefore,

$$H^0\big(A, \mathscr{O}\big(\big(L(D)\big)^{\delta}\big) \otimes \mathscr{I}^q\big) \neq \{0\}, \quad {}^{\forall}q \geqq 1,$$

and hence

$$\dim H^0\big(A, \mathscr{O}\big(\big(L(D)\big)^{\delta}\big)\big) = \infty$$

would follow, where δ had been fixed. This is clearly absurd. This observation implies that the Zariski denseness of the image $f(\mathbf{C})$ in A must be used essentially.

(b) We also observe that the aforementioned Lemma 2 is not valid even for $f : \mathbf{C} \to A$ with Zariski dense image, and moreover that k cannot be fixed as stated in Lemma 2. Let $k \geqq n$ be any fixed number. Let $f : \mathbf{C} \to A$ be a one-parameter subgroup with Zariski dense image. Let D be an ample divisor on A containing the zero $0 \in A$ such that $f(\mathbf{C})$ is tangent highly enough to D at 0 so that $J_k f(0) \in J_k(D)$. Let $\mathfrak{m}_0$ be the maximal ideal sheaf of the structure sheaf $\mathscr{O}_A$ at 0. Since Y_k consists of only one point, $A \times Y_k \cong A$, and through this isomorphism $0 \in J_k(D) \cap (A \times Y_k)$. Therefore we have that $\mathscr{I}_k = \mathscr{I}\langle J_k(D\rangle \cap (A \times Y_k)) \subset \mathfrak{m}_0$. As in (a), Lemma 2 should imply that

$$H^0\big(A, \mathscr{O}\big(\big(L(D)\big)^\delta\big)\otimes\mathfrak{m}_0^q\big) \supset H^0\big(A, \mathscr{O}\big(\big(L(D)\big)^\delta\big)\otimes\mathscr{I}_k^q\big) \neq \{0\}, \quad {}^\forall q \geqq 1.$$

Thus we would obtain

$$\dim H^0\big(A, \mathscr{O}\big(\big(L(D)\big)^\delta\big)\big) = \infty;$$

this is a contradiction.

(c) The reason of the contradictions observed in (a) and (b) with respect to Lemma 2 above comes from their use of the semicontinuity theorem for a non-flat family of coherent ideal sheaves. They used a deformation technique of the given ample divisor D. That is, taking a generic small deformation family $D(t), t \in \Delta(1)$ on A with $D(0) = D$, they considered the family of ideals, $\{(\mathscr{I}\langle J_k(D(t)) \cap (A \times Y_k)\rangle)^q\}_{t\in\Delta(1)}$ for $k \geqq n$ and $q \geqq 1$. More precisely, they worked on the compactification $\overline{J_k(A)} = A \times \mathbf{P}^{nk}(\mathbf{C})$ of $J_k(A) \cong A \times \mathbf{C}^{nk}$. Let $\overline{J_k(D(t))}$ (resp. $\bar{Y}_k$) denote the closure of $J_k(D(t))$ (resp. Y_k) in $\overline{J_k(A)}$ (resp. $\mathbf{P}^{nk}(\mathbf{C})$). To apply the semicontinuity theorem of the dimension of cohomology groups, one needs the flatness of the family, $\{\mathscr{I}\langle\overline{J_k(D(t))} \cap (A \times \bar{Y}_k)\rangle\}_{t\in\Delta(1)}$. In general, the constructed family $\{\mathscr{I}\langle\overline{J_k(D(t))} \cap (A \times \bar{Y}_k)\rangle\}_{t\in\Delta(1)}$ may not be flat, since there may be a "jump" of the supports of those ideals. This fact tells us the difficulty to apply the deformation technique to obtain the Second Main Theorem in general. Because of its own interest we give such an example in what follows.

Let $E = \mathbf{C}/\mathbf{Z}[i]$ be an elliptic curve and set $A = E \times E$. Let (x, y) be a local flat coordinate system of A, and define an entire curve $f : \mathbf{C} \to A$ by

$$f : z \in \mathbf{C} \to (z, \alpha z) \in A,$$

where α is an irrational number. Then the image $f(\mathbf{C})$ is Zariski dense in A. The 2-jet lifting of f is given by

$$J_2(f)(z) = \big((z, \alpha z), (1, \alpha), (0, 0)\big) \in J_2(A) \cong A \times \mathbf{C}^2 \times \mathbf{C}^2.$$

Thus,

$$Y_2 = \big((1, \alpha), (0, 0)\big).$$

Let L be a sufficiently ample line bundle over A such that L carries a global holomorphic section $\sigma(x, y)$ whose germ at $(0, 0)$ is written as

$$-y^2 + x^3 + \alpha^2 x^2 + x^4 G(x, y).$$

Let D be the divisor defined by the zero locus of σ. By an easy computation one sees

$$J_2(D)_{(0,0)} = \big\{(0,0)\big\} \times \mathbf{C}(1, \pm\alpha) \times \mathbf{C}^2.$$

Therefore we have

$$J_2(D) \cap (A \times Y_2) \ni \big((0,0), (1,\alpha), (0,0)\big).$$

For small $t \in \mathbf{C}$ we consider a generic deformation $D(t)$ defined by

$$-y^2 + x^3 + \alpha^2 x^2 + x^4 G(x,y) + tH(x,y) = 0.$$

We look for a point $(x_0, y_0) \in D(t)$ near $(0,0)$ with $t \neq 0$ such that $(x_0, y_0) \times (1,\alpha) \times (0,0) \in J_2(D(t)) \cap (A \times Y_2)$. First we have

$$-y_0^2 + x_0^3 + \alpha^2 x_2^2 + x_0^4 G(x_0, y_0) + tH(x_0, y_0) = 0. \tag{6.3.9}$$

Set

$$\begin{aligned}\phi(z) = &-(y_0 + \alpha z)^2 + (x_0 + z)^3 + \alpha^2 (x_0 + z)^2 \\ &+ (x_0 + z)^4 G(x_0 + z, y_0 + \alpha z) + tH(x_0 + z, y_0 + \alpha z).\end{aligned}$$

Then one gets

$$\begin{aligned}\phi'(z) = &-2\alpha(y_0 + \alpha z) + 3(x_0 + z)^2 + 2\alpha^2 (x_0 + z) \\ &+ (x_0 + z)^3 G_1(x_0 + z, y_0 + \alpha z) + tH_x(x_0 + z, y_0 + \alpha z) \\ &+ t\alpha H_y(x_0 + z, y_0 + \alpha z),\end{aligned}$$

where G_1 is a naturally defined holomorphic function. Hence,

$$\begin{aligned}\phi'(0) = &-2\alpha y_0 + 3x_0^2 + 2\alpha^2 x_0 + x_0^3 G_1(x_0, y_0) \\ &+ tH_x(x_0, y_0) + t\alpha H_y(x_0, y_0) = 0.\end{aligned} \tag{6.3.10}$$

Taking the second derivative, we have

$$\begin{aligned}\phi''(z) = &\, 6(x_0 + z) + (x_0 + z)^2 G_2(x_0 + z, y_0 + \alpha z) + tH_{xx}(x_0 + z, y_0 + \alpha z) \\ &+ 2t\alpha H_{xy}(x_0 + z, y_0 + \alpha z) + t\alpha^2 H_{yy}(x_0 + z, y_0 + \alpha z),\end{aligned}$$

and so

$$\begin{aligned}\phi''(0) = &\big(6 + x_0 G_2(x_0, y_0)\big)x_0 \\ &+ \big(H_{xx}(x_0, y_0) + 2\alpha H_{xy}(x_0, y_0) + \alpha H_{yy}(x_0, y_0)\big)t = 0.\end{aligned}$$

We may assume that for (x_0, y_0) close to $(0,0)$

$$6 + x_0 G_2(x_0, y_0) \neq 0.$$

Thus we may write

$$x_0 = t\psi(t, y_0).$$

We substitute this to (6.3.10), and get

$$-2\alpha y_0 + 3t^2\psi^2(t, y_0) + 2\alpha^2 t\psi(t, y_0) + t^3\psi^3(t, y_0)G_1\big(t\psi(t, y_0), y_0\big) + tH_x\big(t\psi(t, y_0), y_0\big) + t\alpha H_y\big(t\psi(t, y_0), y_0\big) = 0.$$

Therefore we have

$$y_0 = t\lambda(t), \qquad x_0 = t\psi\big(t, t\lambda(t)\big) = t\mu(t).$$

Then we substitute these to (6.3.9), and obtain

$$-t^2\lambda^2(t) + t^3\mu^3(t) + \alpha^2 t^2\mu^2(t) + t^4\mu^4(t)G\big(t\mu(t), t\lambda(t)\big) + tH\big(t\mu(t), t\lambda(t)\big) = 0.$$

Since $t \neq 0$, we have

$$-t\lambda^2(t) + t^2\mu^3(t) + \alpha^2 t\mu^2(t) + t^3\mu^4(t)G\big(t\mu(t), t\lambda(t)\big) + H\big(t\mu(t), t\lambda(t)\big) = 0. \tag{6.3.11}$$

We assume a generic condition, $H(0, 0) \neq 0$; (6.3.11) is not trivial. Thus, t satisfying (6.3.11) is isolated, and cannot approach 0.

It follows that there is a neighborhood $U \subset J_2(A)$ of $((0, 0), (1, \alpha), (0, 0)) \in J_2(D(0)) \cap (A \times Y_2)$ such that for every small $t \neq 0$,

$$J_2\big(D(t)\big) \cap (A \times Y_2) \cap U = \emptyset.$$

Therefore the ideal family $\{\mathscr{I}\langle\overline{J_k(D(t))} \cap (A \times \bar{Y}_k)\rangle\}_{t\in\Delta(1)}$ is not flat.

6.3.2 Applications to Differentiably Non-degenerate Maps

As an application of the method of the proof of Theorem 6.3.1 we extend Theorem 3.2.2 for D carrying any singularities in the case of compact tori. We follow the notation in Chaps. 2 and 3.

Let $(z_1, \ldots, z_m)$ be the standard coordinate system of $\mathbf{C}^m$ and define multi-indices and partial differentiations of order k as usual:

$$\lambda = (\lambda_1, \ldots, \lambda_n), \quad \lambda_i \in \mathbf{Z}^+,$$
$$k = |\lambda| = \lambda_1 + \cdots + \lambda_n,$$
$$\partial^\lambda = \frac{\partial^k}{(\partial z_1)^{\lambda_1} \cdots (\partial z_n)^{\lambda_n}}.$$

Lemma 6.3.12 *Let $g \not\equiv 0$ be a meromorphic function on $\mathbf{C}^m$. For a partial derivative $\partial^\lambda g$ of order $|\lambda| > 0$ the following holds*:

(i) $m(r, \partial^\lambda g/g) = S(r, g)$ (*cf.* (1.2.4) *for notation* $S(r, g)$).
(ii) $T(r, \partial^\lambda g) \leqq (|\lambda| + 1)T(r, g) + S(r, g)$.

The proof is similar to that of Lemma 4.2.9 by making use of Lemma 3.1.1 and the detail is left to the readers.

Let N be a complex projective algebraic manifold and let D be a reduced divisor on N. Let ω be a logarithmic k-jet differential over N along D. Let $f : \mathbf{C}^m \to N$ be a meromorphic mapping such that $f(\mathbf{C}^m) \not\subset D$. Out of the indeterminacy locus $I(f)$ of f we obtain a holomorphic mapping

$$f_* : J_k\big(\mathbf{C}^m \setminus I(f)\big) \cong \big(\mathbf{C}^m \setminus I(f)\big) \times \mathbf{C}^{mk} \to J_k(N).$$

Take a point $w \in \mathbf{C}^{mk}$. Since $\operatorname{codim} I(f) \geqq 2$, the meromorphic function

$$\xi(z) = \omega\big(f_*(z, w)\big)$$

extends meromorphically on $\mathbf{C}^m$.

Lemma 6.3.13 *With the notation as above,* $m(r, \xi) = S_f(r)$.

The proof is similar to that of Lemma 4.7.1 through Lemma 6.3.12.

Now we let A be an n-dimensional compact torus. Let $\mathbf{C}^n \to A$ be the universal covering. Take a meromorphic mapping $f : \mathbf{C}^m \to A$. As $\dim I(f) \leqq m - 2$, $\mathbf{C}^m \setminus I(f)$ is simply-connected. Therefore the lifting $\tilde{f}|_{\mathbf{C}^m \setminus I(f)} : \mathbf{C}^m \setminus I(f) \to \mathbf{C}^n$ is obtained. By Theorem 2.2.6 (ii) $\tilde{f}|_{\mathbf{C}^m \setminus I(f)}$ extends holomorphically over all $\mathbf{C}^m$. Thus f is holomorphic everywhere.

Theorem 6.3.14 (Noguchi [04]) *Let A be as above. Let $L \to A$ be a line bundle and let $D \in |L|$. For every differentiably non-degenerate holomorphic mapping $f : \mathbf{C}^m \to A$ $(m \geqq n)$*

$$T_f(r, L) \leqq N_n\big(r, f^*D\big) + S_f(r),$$
$$\delta_n(f, D) = 0.$$

Proof By the same reason as at the beginning of the proof of Theorem 6.3.1, we may assume that A is an abelian variety and D is ample.

By (4.6.3) f induces a holomorphic mapping

$$f_* : J_n\big(\mathbf{C}^m\big) \cong \mathbf{C}^m \times \mathbf{C}^{mn} \to J_n(A) \cong A \times \mathbf{C}^{n^2}.$$

Since f is differentiably non-degenerate, so is f_*, and in particular it is algebraically non-degenerate. Since the jet projection

$$I_n : J_n(A) \cong A \times \mathbf{C}^{n^2} \to \mathbf{C}^{n^2}$$

is proper, $I_n(J_n(D)) \subset \mathbf{C}^{n^2}$ is an algebraic subset of dimension at most $(n + 1) \times (n - 1) = n^2 - 1$. Therefore there are a point $(z_0, w) \in \mathbf{C}^m \times \mathbf{C}^{mn} \cong J_n(\mathbf{C}^m)$ and a

polynomial P in jet coordinates of $J_n(A) \cong A \times \mathbf{C}^{n^2}$ such that

$$P|_{I_n(J_n(D))} \equiv 0, \qquad P\big(I_n\big(f_*(z_0, w)\big)\big) \not\equiv 0.$$

With notation similarly to (6.3.3) this implies the following equation:

$$a_{\lambda 0}\sigma_\lambda + \cdots + a_{\lambda k} d^k \sigma_\lambda = {I_n}^* P|_{U_\lambda}. \tag{6.3.15}$$

For the holomorphic mapping

$$J_n(f)_w : z \in \mathbf{C}^m \to f_*(z, w) \in J_n(A),$$

$P(I_n(J_n(f)_w(z))) \not\equiv 0$. The rest of the proof is the same as in that after (6.3.4). □

Remark It is noted that in Theorem 6.3.14 there is no condition on the singularities of D. If D is assumed to be of simple normal crossings, Theorem 3.2.2 implies a stronger estimate

$$T_f(r, L) = N_1\big(r, f^*D\big) + S_f(r).$$

6.4 Semi-tori: Truncation Level k_0

We first establish a Second Main Theorem for entire curves into semi-tori, where the counting functions are truncated to level $k_0 \geqq 1$ and the remainder term is $S_f(r)$.

Theorem 6.4.1 *Let A be a semi-torus with equivariant compactification $\bar{A}$. Let $f : \mathbf{C} \to A$ be an entire curve with Zariski dense image in $\bar{A}$. Let D be an effective divisor on A which extends to a divisor $\bar{D}$ on $\bar{A}$.*

Then, possibly after changing the compactification $\bar{A}$ which depends only on D and independent of f, there is a positive integer $k_0 = k_0(f, D)$ depending on f and D such that

$$T_f\big(r, c_1(\bar{D})\big) = N_{k_0}\big(r, f^*D\big) + S_f\big(r, c_1(\bar{D})\big).$$

If the order ρ_f of f is finite, then k_0 can be chosen to depend only on the order ρ_f and D.

In particular, $\delta(f, \bar{D}) = \delta_{k_0}(f, \bar{D}) = 0$.

Reduction Let $q : A \to A/\operatorname{St}(D)^0$ be the quotient map. For $f : \mathbf{C} \to A$ we have

$$f^*D = (q \circ f)^*\big(D/\operatorname{St}(D)^0\big).$$

Then by Theorems 5.5.26 and 5.6.13 we may assume from the beginning that

Assumption 6.4.2 (i) A is semi-abelian with algebraic presentation

$$0 \to \big(\mathbf{C}^*\big)^p \to A \to A_0 \to 0,$$

and $\bar{A}$ is a equivariant smooth projective algebraic compactification,

(ii) $\mathrm{St}(D)^0 = \{0\}$,
(iii) $\bar{D}$ is big on $\bar{A}$ and in good position.

See Example 6.4.20 and Proposition 6.4.15 that show the necessity of $\bar{D}$ being in "good position".

Here we need the notion of logarithmic jet spaces due to Noguchi [86] (see Sect. 4.6.3). Since ∂A has only normal crossings, we have the logarithmic k-th jet bundle $J_k(\bar{A}; \log \partial A)$ over $\bar{A}$ along ∂A, and a morphism

$$\psi_k : J_k(\bar{A}; \log \partial A) \to J_k(\bar{A})$$

such that the sheaf of germs of holomorphic sections of $J_k(\bar{A}; \log \partial A)$ is isomorphic to that of logarithmic k-jet fields (see Sect. 4.6.3); there, the "sub-bundle" $J_k(\bar{A}; \log \partial A)$ of $J_k(\bar{A})$ should be understood in this way). Because of the flat structure of the logarithmic tangent bundle $\mathbf{T}(\bar{A}; \log \partial A)$,

$$J_k(\bar{A}; \log \partial A) \cong \bar{A} \times \mathbf{C}^{nk}.$$

Let

$$\begin{aligned} &\pi_1 : J_k(\bar{A}; \log \partial A) \cong \bar{A} \times \mathbf{C}^{nk} \to \bar{A}, \\ &\pi_2 : J_k(\bar{A}; \log \partial A) \cong \bar{A} \times \mathbf{C}^{nk} \to \mathbf{C}^{nk} \end{aligned} \tag{6.4.3}$$

be the first and the second projections. For a k-jet $y \in J_k(\bar{A}; \log \partial A)$ we call $\pi_2(y)$ the *jet part* of y.

Let $x \in \bar{D}$ and let $\sigma = 0$ be a local defining equation of $\bar{D}$ about x. For a germ $g : (\mathbf{C}, 0) \to (A, x)$ of a holomorphic mapping we denote its k-jet by $j_k(g)$ and write

$$d^j\sigma(g) = \frac{d^j}{d\zeta^j}\bigg|_{\zeta=0} \sigma\big(g(\zeta)\big).$$

We set

$$\begin{aligned} J_k(\bar{D})_x &= \big\{ j_k(g) \in J_k(\bar{A})_x;\ d^j\sigma(g) = 0, 1 \leqq j \leqq k \big\}, \\ J_k(\bar{D}) &= \bigcup_{x \in D} J_k(\bar{D})_x, \\ J_k(\bar{D}; \log \partial A) &= \psi_k^{-1} J_k(\bar{D}). \end{aligned}$$

Then $J_k(\bar{D}; \log \partial A)$ is a subspace of $J_k(\bar{A}; \log \partial A)$, which depends in general on the embedding $\bar{D} \hookrightarrow \bar{A}$ (cf. Sect. 4.6.3 and Noguchi [86]). Note that $\pi_2(J_k(\bar{D}; \log \partial A))$ is an algebraic subset of $\mathbf{C}^{nk}$, because π_2 is proper. Let $J_k(f) : \mathbf{C} \to J_k(\bar{A}; \log \partial A) = \bar{A} \times \mathbf{C}^{nk}$ be the k-th jet lift of f. We define

$$Y_k = \pi_2\big(X_k(f)\big).$$

Let $\pi : \mathbf{C}^n \to A$ be the universal covering and let

$$\tilde{f} : z \in \mathbf{C} \to \big(\tilde{f}_1(z), \ldots, \tilde{f}_n(z)\big) \in \mathbf{C}^n$$

be a lift of f. Assume that f is of finite order. Then $\tilde{f}(z)$ is a vector-valued polynomial by Theorem 6.1.9. Note that every non-constant polynomial map from $\mathbf{C}$ to $\mathbf{C}^n$ is proper, and hence the image is an algebraic subset. It follows that

$$Y_k = \overline{\big\{\big(\tilde{f}'(z), \ldots, \tilde{f}^{(k)}(z)\big) z \in \mathbf{C}\big\}} = \big\{\big(\tilde{f}'(z), \ldots, \tilde{f}^{(k)}(z)\big), z \in \mathbf{C}\big\},$$

and hence $\dim Y_k \leqq 1$. Thus we deduced the following lemma.

Lemma 6.4.4 *Let the notation be as above. If $f : \mathbf{C} \to A$ is of finite order, then $\dim Y_k \leqq 1$ and for every point $w_k \in Y_k$ there is a point $a \in \mathbf{C}$ with $\pi_2 \circ J_k(f)(a) = w_k$.*

Now, the following is the key.

Lemma 6.4.5 *Let the notation be as above.*

(i) *Suppose that f is of finite order ρ_f. Then there is a number $k_0 = k_0(\rho_f, D)$ such that*

$$\pi_2\big(J_k(\bar{D}; \log \partial A)\big) \cap Y_k = \emptyset, \quad k \geqq k_0.$$

(ii) *Suppose that f is of infinite order. Then there is a number $k_0 = k_0(f, D)$ such that*

$$\pi_2\big(J_k(\bar{D}; \log \partial A)\big) \cap Y_k \neq Y_k, \quad k \geqq k_0.$$

Proof (i) By making use of (6.4.3) we have the projection $p_{k,l} : \mathbf{C}^{nk} \to \mathbf{C}^{nl}$ for $k \geqq l$ induced from the canonical projection $J_k(\bar{A}; \log \partial A) \to J_l(\bar{A}; \log \partial A)$. For a subset or a point E_k of $\mathbf{C}^{nk}$ and $l \leqq k$ we write $E_{k,l} = p_{k,l}(E_k)$.

We see first by Theorem 6.1.9 that $\rho_f \in \mathbf{Z}^+$, and $\tilde{f}(z)$ is a vector-valued polynomial of degree $\leqq \rho_f$. Thus, Y_k is of form

$$Y_k = (Y_{k,\rho_f}, \underbrace{O, \ldots, O}_{k-\rho_f}).$$

Set $W'_k = Y_k \cap \pi_2(J_k(\bar{D}; \log \partial A))$. Then we have

$$W'_k = \big(W'_{k,\rho_f}, \underbrace{O, \ldots, O}_{k-\rho_f}\big).$$

Assume that the present assertion fails. Then, by the Noetherian property of algebraic subsets, there is a point $\xi_{\rho_f} \in \bigcap_{k=\rho_f}^{\infty} W'_{k,\rho_f}$ such that, setting $\xi_k = (\xi_{\rho_f}, \underbrace{O, \ldots, O}_{k-\rho_f}) \in \mathbf{C}^{nk}$, we have

$$\xi_k \in \pi_2\big(J_k(\bar{D}; \log \partial A)\big), \quad {}^{\forall}k \geqq \rho_f.$$

We identify ξ_k with a logarithmic k-jet field on $\bar{A}$ along ∂A (see Sect. 4.6.3). Set $S_k = \pi_1(J_k(\bar{D}; \log \partial A) \cap \pi_2^{-1}(\xi_k))$. Then,

$$\bar{D} \supset S_{\rho_f} \supset S_{\rho_f+1} \supset \cdots,$$

which stabilize to $S_0 = \bigcap_{k=\rho_f}^{\infty} S_k \neq \emptyset$. Let $x_0 \in S_0$. If $x_0 \in A$, it follows from Lemma 6.4.4 that there are points $a \in \mathbf{C}$ and $y_0 \in A$ such that

$$f(a) + x_0 + y_0 \in D,$$
$$\left.\frac{d^k}{dz^k}\right|_{z=a} \sigma\big(f(z)\big) = 0, \quad {}^{\forall}k \geqq 1,$$

where σ is a local defining function of D about $f(a) + x_0 + y_0$. Therefore

$$f(\mathbf{C}) + x_0 + y_0 \subset D,$$

and hence this contradicts the Zariski denseness of $f(\mathbf{C})$ in A. This finishes the proof in the case of $x_0 \in A$.

Suppose now that $x_0 \in \bar{A} \setminus A$. Let $\partial A = \bigcup B_j$ be the Whitney stratification by A-orbits, and let $x_0 \in B_q$. Let B be the stratum of B_q containing x_0. Then B itself is a semi-abelian variety such that

$$0 \to \big(\mathbf{C}^*\big)^{p-q} \to B \to A_0 \to 0.$$

Let H denote the isotropy group of A in x_0. The action of $H \cong (\mathbf{C}^*)^q$ can be linearized near x_0. We choose local coordinates $z = (z', z'')$ about x_0 such that

(i) there is an open neighborhood W of x_0 which is mapped biholomorphically by (z', z'') onto an open neighborhood of $(0, 0)$ in $\mathbf{C}^q \times \mathbf{C}^r$;
(ii) $z'(x_0) = 0$;
(iii) $h \cdot (z', z'') = (hz', z'') = (h_1 z'_1, \ldots, h_q z'_q, z'')$ for all $h = (h_1, \ldots, h_q) \in (\mathbf{C}^*)^q$ and $z = (z', z'') \in W$ with $h \cdot (z', z'') \in W$;
(iv) for every $a \in A$ there is a $h \in H$ and an element $v \in \mathbf{C}^r$ such that $a \cdot (z', z'') = (hz', z'' + v)$ whenever $z, a \cdot z \in W$.

We assume

$$\pi_2\big(X_k(f)\big) \subset \pi_2\big(J_k(\bar{D}; \log \partial A)\big), \quad {}^{\forall}k \geqq 1.$$

Then there are $a_k \in A$ such that

$$\big(J_k(f_{a_k})\big)(0) \in J_k(\bar{D}; \log \partial A), \quad {}^{\forall}k \geqq 1$$

for $f_a(t) = a + f(t)$.

We assume that $f_{a_k}(0)$ converges to x_0. The description in local coordinates yields that $a_k = (b_k, c_k)$ with $\lim b_k = 0$ and $\lim c_k = c \in \mathbf{C}^r$ for sufficiently large k. A calculation in these local coordinates shows that the sequence of maps

$$(f_{a_k}) = \big(f_{a_k}^{(1)}, f_{a_k}^{(2)}\big)$$

converges uniformly to

$$g = \big(0, f_c^{(2)}\big)$$

in a neighborhood of 0. Therefore,

$$\big(J_k(g)\big)(0) \in J_k(\bar{D}), \quad {}^{\forall}k \in \mathbf{Z}^+. \tag{6.4.6}$$

Now consider the "region of attraction" Ω defined as the set of all $p \in \bar{A}$ for which $\lim_{h\to 0} h \cdot p$ exists and is contained in B. Since Ω is an A-invariant subset of $\bar{A}$ containing an open subset (namely an open neighborhood of x_0), it contains all of A. By using $f(\mathbf{C}) \subset A$ it follows that the sequence of maps f_{a_k} converges to an entire curve $g : \mathbf{C} \to \bar{A}$. Moreover, the maps ϕ_k given by $x \mapsto a_k \cdot x$ converge to a map ψ which in local coordinates as described above is given as $\psi : (x', x'') \mapsto (0, c + x'')$. From (6.4.6) it follows that $g(\mathbf{C}) \subset \bar{D}$.

On the other hand, as $g = \psi \circ f$ and $\psi(A) \subset B$, the property of f having a Zariski dense image implies that $\psi \circ f$ has a Zariski dense image in $\psi(A)$. But ψ is equivariant and A acts transitively on B. Therefore $\psi(A) = B$ and $g(\mathbf{C}) = \psi(f(\mathbf{C}))$ must be Zariski dense in B, contradicting $g(\mathbf{C}) \subset \bar{D} \cap B \subsetneqq B$.

The order of the tangency of f and the above used g with $\bar{D}$ is bounded as $\tilde{f}$ runs over all vector-valued polynomials of order at most ρ_f such that $f(\mathbf{C})$ is Zariski dense in A. Hence there is such a number k_0 depending only on ρ_f and D.

(ii) Assume contrarily that $\pi_2(J_k(\bar{D}; \log \partial A)) \cap Y_k = Y_k$ for all $k \geqq 1$. Since $\pi_2 \circ J_k(f)(0) \in Y_k$, ${}^{\forall}k \geqq 1$, we apply the same argument as in (i) with setting $\xi_k = \pi_2 \circ J_k(f)(0)$. Then we deduce a contradiction that f has no Zariski dense image. □

Lemma 6.4.7 *Let $L \to \bar{A}$ be an ample line bundle. Then*

$$m_f(r, \bar{D}) = S_f(r, L) = S_f\big(r, c_1(\bar{D})\big). \tag{6.4.8}$$

Proof Let $\{U_j\}$ be an affine open covering of $\bar{A}$ such that the line bundle $L(\bar{D})$ is locally trivial over every U_j. We take a regular function σ_j on every U_j such that σ_j is a defining function of $\bar{D} \cap U_j$; i.e.,

$$(\sigma_j) = D|_{U_j}.$$

Let us now fix a hermitian metric $\|\cdot\|$ on $L(\bar{D})$. Then there are positive smooth functions h_j on U_j such that

$$\|\sigma(x)\| = \frac{|\sigma_j(x)|}{h_j(x)}, \quad x \in U_j,$$

which is a well-defined function on $\bar{A}$.

Assume that f is of finite order. By Lemma 6.4.5 there is a number $k_0 \in \mathbf{N}$ and regular functions b_{ji}, $0 \leqq i \leqq k_0$, on $U_j \times Y_{k_0}$ such that

$$b_{j0}\sigma_j + b_{j1} d\sigma_j + \cdots + b_{jk_0} d^{k_0}\sigma_j = 1. \tag{6.4.9}$$

Here every b_{ji} is expressed as

$$b_{ji} = \sum_{\text{finite}} b_{jil\beta_l}(x) w_l^{\beta_l},$$

where $b_{jil\beta_l}(x)$ are regular functions on U_j and w_l are restrictions of coordinate functions of $\mathbf{C}^{nk_0}$ to Y_{k_0}. Thus we infer that in every U_j

$$\frac{1}{\|\sigma\|} = \frac{h_j}{|\sigma_j|} = \left| h_j b_{j0} + h_j b_{j1} \frac{d\sigma_j}{\sigma_j} + \cdots + h_j b_{jk_0} \frac{d^{k_0}\sigma_j}{\sigma_j} \right|. \tag{6.4.10}$$

Take relatively compact open subsets $U_j' \Subset U_j$ (in the sense of differential topology) so that $\bigcup U_j' = \bar{A}$. For every j there is a positive constant C_j such that for $x \in U_j'$

$$h_j |b_{ji}| \leqq \sum_{\text{finite}} h_j \left| b_{jil\beta_l}(x) \right| \cdot |w_l|^{\beta_l} \leqq C_j \sum_{\text{finite}} |w_l|^{\beta_l}.$$

Thus, after making C_j larger if necessary, there is a number $d_j > 0$ such that for $f(z) \in U_j'$

$$h_j\big(f(z)\big)\left|b_{ji}\big(J_{k_0}(f)(z)\big)\right| \leqq C_j \left(1 + \sum_{1 \leqq l \leqq n, 1 \leqq k \leqq k_0} \left|\tilde{f}_l^{(k)}(z)\right|\right)^{d_j}.$$

We deduce that

$$\begin{aligned} \frac{1}{\|\sigma(f(z))\|} \leqq & \sum_{j=1}^{N} C_j \left(1 + \sum_{1 \leqq l \leqq n, 1 \leqq k \leqq k_0} \left|\tilde{f}_l^{(k)}(z)\right|\right)^{d_j} \\ & \times \left(1 + \left|\frac{d\sigma_j}{\sigma_j}\big(J_1(f)(z)\big)\right| + \cdots + \left|\frac{d^{k_0}\sigma_j}{\sigma_j}\big(J_{k_0}(f)(z)\big)\right|\right). \end{aligned}$$

Hence one gets

$$\begin{aligned} m_f(r, \bar{D}) &= \frac{1}{2\pi} \int_0^{2\pi} \log^+ \frac{1}{\|\sigma(f(re^{i\theta}))\|}\, d\theta \\ &\leqq O\Bigg(\sum_{1 \leqq l \leqq n, 1 \leqq k \leqq k_0} \frac{1}{2\pi} \int_0^{2\pi} \log^+ \left|\tilde{f}_l^{(k)}\big(re^{i\theta}\big)\right| d\theta \\ &\quad + \sum_{1 \leqq j \leqq N, 1 \leqq k \leqq k_0} \frac{1}{2\pi} \int_0^{2\pi} \log^+ \left|\frac{d^k\sigma_j}{\sigma_j}\big(J_k(f)\big(re^{i\theta}\big)\big)\right| d\theta \Bigg) + O(1). \end{aligned} \tag{6.4.11}$$

Recall that σ_j are rational functions on $\bar{A}$. By Lemmas 4.2.9 and 6.1.15 we see that

$$\frac{1}{2\pi}\int_0^{2\pi}\log^+\left|\frac{d^k\sigma_j}{\sigma_j}\big(J_k(f)(re^{i\theta})\big)\right|d\theta = m\left(r,\frac{(\sigma_j\circ f)^{(k)}}{\sigma_j\circ f}\right) = O(\log r).$$

This combined with (6.4.11) implies that $m_f(r,\bar{D}) = O(\log r)$; this completes the proof in the case of finite order.

Assume that f is of infinite order. It follows from Lemma 6.4.5 (ii), that there exists a polynomial function $R(w)$ in $w \in Y_{k_0}$ such that

$$\pi_2\big(J_k(\bar{D};\log\partial A)\big)\cap Y_k \subset \big\{w\in Y_{k_0};\, R(w)=0\big\} \neq Y_{k_0}.$$

We regard R as a regular function on every $U_j\times Y_{k_0}$. Then we have the following equation on every $U_j\times Y_{k_0}$ with coefficients similar to those of (6.4.9):

$$b_{j0}\sigma_j + b_{j1}d\sigma_j + \cdots + b_{jk_0}d^{k_0}\sigma_j = R. \tag{6.4.12}$$

Then, after the same arguments as in the case of finite order, we have that for $f(z)\in U_j'$

(6.4.13)

$$\begin{aligned}\frac{1}{\|\sigma(f(z))\|} &= \frac{1}{|R(\tilde{f}'(z),\dots,\tilde{f}^{(k_0)}(z))|}\\ &\quad\times\left|h_jb_{j0}+h_jb_{j1}\frac{d\sigma_j}{\sigma_j}+\cdots+h_jb_{jk_0}\frac{d^{k_0}\sigma_j}{\sigma_j}\right|\\ &\leqq \frac{1}{|R(\tilde{f}'(z),\dots,\tilde{f}^{(k_0)}(z))|}\sum_{j'=1}^{N}C_{j'}\left(1+\sum_{1\leqq l\leqq n,1\leqq k\leqq k_0}\big|\tilde{f}_l^{(k)}(z)\big|\right)^{d_{j'}}\\ &\quad\times\left(1+\left|\frac{d\sigma_{j'}}{\sigma_{j'}}\big(J_1(f)(z)\big)\right|+\cdots+\left|\frac{d^{k_0}\sigma_{j'}}{\sigma_{j'}}\big(J_{k_0}(f)(z)\big)\right|\right).\end{aligned}$$

It follows that

$$\begin{aligned}m_f(r,\bar{D}) &\leqq \frac{1}{2\pi}\int_0^{2\pi}\log^+\frac{1}{\|\sigma(f(re^{i\theta}))\|}\,d\theta + O(1)\\ &\leqq m\left(r,\frac{1}{R(\tilde{f}',\dots,\tilde{f}^{(k_0)})}\right)+O\Bigg(\sum_{1\leqq l\leqq n,1\leqq k\leqq k_0}m\big(r,\tilde{f}_l^{(k)}\big)\\ &\quad+\sum_{1\leqq j\leqq N,1\leqq k\leqq k_0}m\left(r,\frac{d^k\sigma_j}{\sigma_j}\circ J_k(f)\right)\Bigg)+O(1)\\ &\leqq T\big(r,R\big(\tilde{f}',\dots,\tilde{f}^{(k_0)}\big)\big)\\ &\quad+O\Bigg(\sum_{l,k,j}m\big(r,\tilde{f}_j^{(k)}\big)+m\left(r,\frac{(\sigma_j\circ f)^{(k)}}{\sigma_j\circ f}\right)\Bigg)+O(1).\end{aligned}$$

This combined with Lemmas 4.2.9 and 6.1.15 implies that

$$m_f(r, \bar{D}) = S_f(r, L).$$

Since $\bar{D}$ is big, Proposition 2.5.23 implies that

$$S_f\big(r, c_1(\bar{D})\big) = S_f(r, L).$$

This finishes the proof. □

Now we give:

Proof of Theorem 6.4.1 We keep the notation used above.

Thanks to Lemma 6.4.7 the only things we still have to show are the statements on the truncation, i.e., the bounds on $N(r, f^*D) - N_{k_0}(r, f^*D)$.

Observe that $\operatorname{ord}_z f^*D > k$ if and only if $J_k(f)(z) \in J_k(\bar{D}; \log \partial A)$. Therefore, if f is of finite order, Lemma 6.4.5 (i) implies that

$$N\big(r, f^*D\big) = N_{k_0}\big(r, f^*D\big).$$

In the case where f is of infinite order we infer from (6.4.13) that

$$\operatorname{ord}_z f^*D - \min\big\{\operatorname{ord}_z f^*D, k_0\big\} \leqq \operatorname{ord}_z\big(R\big(\tilde{f}', \ldots, \tilde{f}^{k_0}\big)\big)_0.$$

Thus we have after integration that

$$N\big(r, f^*D\big) - N_{k_0}\big(r, f^*D\big) \leqq N\big(r, \big(R\big(\tilde{f}', \ldots, \tilde{f}^{(k_0)}\big)\big)_0\big).$$

It follows that

$$\begin{aligned} N\big(r, \big(R\big(\tilde{f}', \ldots, \tilde{f}^{(k_0)}\big)\big)_0\big) &\leqq T\big(r, R\big(\tilde{f}', \ldots, \tilde{f}^{(k_0)}\big)\big) + O(1) \\ &\leqq O\Bigg(\sum_{1 \leqq l \leqq n, 1 \leqq k \leqq k_0} T\big(r, \tilde{f}_l^{(k)}\big)\Bigg) \\ &= S_f(r, L), \end{aligned}$$

Hence,

$$N\big(r, f^*D\big) \leqq N_{k_0}\big(r, f^*D\big) + S_f\big(r, c_1(\bar{D})\big).$$

The proof is completed. □

Corollary 6.4.14 *Let M be a complex torus and let $f : \mathbf{C} \to M$ be an entire curve. Let D be an effective divisor on M such that* $\operatorname{Supp} D \not\supset f(\mathbf{C})$. *Then we have the following.*

(i) *Suppose that f is of finite order ρ_f. Then there is a positive integer $k_0 = k_0(\rho_f, D)$ such that*

$$T_f\big(r, c_1(D)\big) = N_{k_0}\big(r, f^*D\big) + O(\log r).$$

(ii) *Suppose that f is of infinite order. Then there is a positive integer $k_0 = k_0(f, D)$ such that*

$$T_f\big(r, c_1(D)\big) = N_{k_0}\big(r, f^*D\big) + S_f\big(r, c_1(D)\big).$$

In particular, $\delta(f, D) = \delta_{k_0}(f, D) = 0$ in both cases.

Proof Since the Zariski closure of $f(\mathbf{C})$ is a translate of a complex subtorus of M (Theorem 6.2.1), we may assume that $f(\mathbf{C})$ is Zariski dense. Furthermore, by taking the quotient by the stabilizer group $\mathrm{St}(D)$ of D, we may assume that D is ample (Proposition 5.5.9). Hence this statement is a special case of Theorem 6.4.1. □

Proposition 6.4.15 *Let M be a complex semi-torus and let D be an effective divisor on M such that its topological closure $\bar{D}$ is a divisor on an equivariant compactification $\bar{M}$. Assume that $\bar{D}$ is not in* good position. *Then there exists an entire curve $f : \mathbf{C} \to M$ of an arbitrarily given integral order $\rho \geqq 2$ in general, and $\rho \geqq 1$ in the case of affine M, such that $f(\mathbf{C})$ is Zariski dense in M and $\delta(f, \bar{D}) > 0$.*

Proof Let

$$1 \to \big(\mathbf{C}^*\big)^p \to M \to M_0 \to 1$$

be a representaion of M, and let $\bar{M}$ be the equivariant compactification associated with $(\mathbf{C}^*)^p \hookrightarrow (\mathbf{P}^1(\mathbf{C}))^p$. Let $\hat{M} = (\mathbf{P}^1(\mathbf{C}))^p \times \mathbf{C}^m \to \bar{M}$ (resp. $\mathbf{C}^m \to M_0$) be the universal covering of $\bar{M}$ (resp. M_0), and $\hat{D} \subset \hat{M}$ the preimage of $\bar{D}$. We may assume that

$$\big\{(\infty)\big\}^p \times \mathbf{C}^m \subset \hat{D}.$$

Let $c_1, \ldots, c_p$ be $\mathbf{Q}$-linear independent real numbers with

$$0 < c_1 < c_2 < \cdots < c_p. \tag{6.4.16}$$

Let $\rho \geqq 2$ or $\rho \geqq 1$ be such an integer as assumed in the proposition, and set

$$\hat{f} : z \mapsto \big(\big[1, e^{c_1 z^\rho}\big], \big[1, e^{c_2 z^\rho}\big], \ldots, \big[1, e^{c_p z^\rho}\big], L(z)\big), \tag{6.4.17}$$

where $L : \mathbf{C} \to \mathbf{C}^m$ is a linear map such that the image $L(\mathbf{C})$ in M_0 is Zariski dense. Moreover, by a generic choice of c_j and L we have that $f(\mathbf{C})$ is Zariski dense in M. Let

$$U_i \Subset V_i \Subset M_0$$

be a finite collection of relatively compact holomorphically convex open subsets of M_0 such that there are sections $\mu_i : V_i \xrightarrow{\sim} \hat{V}_i \subset \mathbf{C}^m$ and such that the U_i cover M_0. Set $\hat{U}_i = \mu_i(U_i)$.

For every i the restricted divisor $\hat{D}|((\mathbf{P}^1(\mathbf{C}))^p \times \hat{V}_i)$ is defined by a homogeneous polynomial P_{i0} of multi-degree $(d_1, \ldots, d_p)$, where the coefficients are holomorphic functions on V_i. Let P_i denote the associated inhomogeneous polynomial.

Then P_i is a polynomial of multi-degree $(d_1, \dots, d_p)$. Due to $\{\infty\}^p \times \mathbf{C}^m \subset \hat{D}$, P_i does not carry the highest degree monomial, $u_1^{d_1} \cdots u_p^{d_p}$.

Recall that $\bar{M} = \hat{M}/\Lambda_0$ with Λ_0, a lattice in $\mathbf{C}^m$ acting on $\hat{M}$ via

$$\lambda : \big(u_1, \dots, u_p, x''\big) \mapsto \lambda \cdot \big(u, x''\big) = \big(\beta_1(\lambda)u_1, \dots, \beta_p(\lambda)u_p, x'' + \lambda\big),$$

where $\beta : \Lambda_0 \to (S^1)^p$ is a group homomorphism into the product of $S^1 = \{|z| = 1; z \in \mathbf{C}^*\}$.

Together with (6.4.17) and (6.4.16) it follows that there is a constant $C > 0$ such that

$$\big|P_i\big(\lambda \cdot \hat{f}(z)\big)\big| \leqq C\big|e^{(\sum_j d_j c_j)z^\rho - c_1 z^\rho}\big| \tag{6.4.18}$$

for all $\lambda \in \Lambda_0$ and $z \in \mathbf{C}$ with $\Re z^\rho > 0$ and $\lambda \cdot \hat{f}(z) \in (\mathbf{P}^1(\mathbf{C}))^p \times \hat{U}_i$. Note that for every $z \in \mathbf{C}$ there exists an element $\lambda \in \Lambda_0$ and an index i such that $\lambda \cdot \hat{f}(z) \in (\mathbf{P}^1(\mathbf{C}))^p \times \hat{U}_i$. Then there is a constant $C' > 0$ such that

$$\big\|\sigma(x)\big\|^2 \leqq C' \frac{|P_i(\lambda \cdot x)|^2}{\prod_j (1 + |u_j|^2)^{d_j}} \tag{6.4.19}$$

for all $x \in \hat{M}$, $\lambda \in \Lambda_0$ with $\lambda \cdot x \in U_i$. From (6.4.18) and (6.4.19) it follows that for $\Re z^\rho > 0$

$$\begin{aligned}\big\|\sigma\big(f(z)\big)\big\|^2 &\leqq C'C^2 \frac{|e^{(\sum_j d_j c_j)z^\rho - c_1 z^\rho}|^2}{\prod_j (1 + |e^{2c_j z^\rho}|)^{d_j}} \leqq C'C^2 \frac{|e^{(\sum_j d_j c_j)z^\rho - c_1 z^\rho}|^2}{\prod_j |e^{2c_j d_j z^\rho}|} \\ &= C'C^2\big|e^{-c_1 z^\rho}\big|^2 = C'C^2 e^{-2c_1 \Re z^\rho}.\end{aligned}$$

Hence,

$$\log^+ \frac{1}{\|\sigma(f(z))\|} \geqq c_1 \Re z^\rho + O(1)$$

for all $z \in \mathbf{C}$ with $\Re z^\rho > 0$. Therefore,

$$\begin{aligned} m_f(r, \bar{D}) &= \frac{1}{2\pi} \int_{\{|z|=r\}} \log \frac{1}{\|\sigma(f(z))\|}\, d\theta \\ &= \frac{1}{2\pi} \int_{\{|z|=r\}} \log^+ \frac{1}{\|\sigma(f(z))\|}\, d\theta + O(1) \\ &\geqq \frac{1}{2\pi} \int_{\{|z|=r;\, \Re z^\rho > 0\}} \log^+ \frac{1}{\|\sigma(f(z))\|}\, d\theta + O(1) \\ &= \frac{1}{2\pi} \int_{\{|z|=r\}} c_1 \cdot \big(\Re z^\rho\big)^+\, d\theta + O(1) \\ &= \frac{1}{2\pi} \int_0^{2\pi} c_1 r^\rho \cos^+ \rho\theta\, d\theta + O(1) \\ &= \frac{c_1}{\pi} r^\rho + O(1). \end{aligned}$$

On the other hand, one deduces easily from (6.4.17) that $T_f(r, c_1(\bar{D})) = O(r^\rho)$. Hence,

$$\delta(f, \bar{D}) = \varliminf_{r\to\infty} \frac{m_f(r, \bar{D})}{T_f(r, c_1(\bar{D}))} > 0. \qquad \square$$

We will now give an explicit example with $\mathrm{St}(D)^0 = \{0\}$.

Example 6.4.20 Let A be the semi-abelian variety $A = \mathbf{C}^* \times \mathbf{C}^*$, compactified by $\mathbf{P}^1(\mathbf{C}) \times \mathbf{P}^1(\mathbf{C})$ with a pair of homogeneous coordinates, $([x_0, x_1], [y_0, y_1])$. For a pair of natural numbers (m, n) with $m < n$, let $\bar{D}$ be the divisor given by

$$\bar{D} = \big\{\big([x_0, x_1], [y_0, y_1]\big);\ y_0^n x_1 + y_0^{n-m} y_1^m x_0 + y_1^n x_0 = 0\big\}.$$

Set $D = \bar{D} \cap A$. Note that $\mathrm{St}(D)^0 = \{0\}$. Moreover, $\bar{D}$ is not in good position since $\bar{D} \ni ([1, 0], [1, 0])$. Let c be a positive irrational real number such that

$$0 < cm < 1 < cn. \tag{6.4.21}$$

Let $f : \mathbf{C} \to A$ be the entire curve given by

$$f : z \mapsto \big(\big[1, e^z\big], \big[1, e^{cz}\big]\big).$$

Let Ω_i, $i = 1, 2$, be the Fubini–Study metric forms of the two factors of $(\mathbf{P}^1(\mathbf{C}))^2$. Then $c_1(\bar{D}) = \Omega_1 + n\Omega_2$. By an easy computation one obtains

$$T_f\big(r, c_1(\bar{D})\big) = \frac{1 + nc}{\pi} r + O(1). \tag{6.4.22}$$

Thus, $\rho_f = 1$, and the image $f(\mathbf{C})$ is Zariski dense in A, because c is irrational.

We compute $N(r, f^*D)$ as follows. Note the following identity for divisors on $\mathbf{C}$:

$$f^*D = \big(e^z + e^{mcz} + e^{ncz}\big)_0. \tag{6.4.23}$$

We consider an entire curve g in $\mathbf{P}^2(\mathbf{C})$ with the homogeneous coordinate system $[w_0, w_1, w_2]$ defined by

$$g : z \in \mathbf{C} \to \big[e^z, e^{mcz}, e^{ncz}\big] \in \mathbf{P}^2(\mathbf{C}).$$

By computing the Wronskian of e^z, e^{mcz} and e^{ncz} one sees that they are linearly independent over $\mathbf{C}$; that is g is linearly non-degenerate. Let $T_g(r)$ be the order function of g with respect to the Fubini–Study metric form on $\mathbf{P}^2(\mathbf{C})$. It follows that

$$\begin{aligned} T_g(r) &= \frac{1}{4\pi} \int_{\{|z|=r\}} \log\big(|e^z|^2 + |e^{mcz}|^2 + |e^{ncz}|^2\big)\, d\theta + O(1) \\ &= \frac{1}{4\pi} \int_{\{|z|=r\}} \log\big(1 + |e^{(mc-1)z}|^2 + |e^{(nc-1)z}|^2\big)\, d\theta + O(1). \end{aligned} \tag{6.4.24}$$

If $\Re z \geqq 0$ (resp. $\leqq 0$), $|e^{(mc-1)z}| \leqq 1$ (resp. $\geqq 1$) and $|e^{(nc-1)z}| \geqq 1$ (resp. $\leqq 1$). Therefore, if $z = re^{i\theta}$ and $\Re z \geqq 0$,

$$\begin{aligned}\log\bigl(1 + \bigl|e^{(mc-1)z}\bigr|^2 + \bigl|e^{(nc-1)z}\bigr|^2\bigr) &= 2\log^+\bigl|e^{(nc-1)z}\bigr| + O(1)\\ &= 2(nc-1)r\cos\theta + O(1).\end{aligned}$$

If $z = re^{i\theta}$ and $\Re z \leqq 0$,

$$\begin{aligned}\log\bigl(1 + \bigl|e^{(mc-1)z}\bigr|^2 + \bigl|e^{(nc-1)z}\bigr|^2\bigr) &= 2\log^+\bigl|e^{(mc-1)z}\bigr| + O(1)\\ &= 2(mc-1)r\cos\theta + O(1).\end{aligned}$$

Combining these with (6.4.24), we have

$$T_g(r) = \frac{(n-m)c}{\pi}r + O(1). \tag{6.4.25}$$

We consider the following four lines H_j, $1 \leqq j \leqq 4$, of $\mathbf{P}^2(\mathbf{C})$ in general position:

$$H_j = \{w_{j-1} = 0\}, \quad 1 \leqq j \leqq 3, \qquad H_4 = \{w_0 + w_1 + w_2 = 0\}.$$

Noting that g is linearly non-degenerate and has a finite order (in fact, $\rho_g = 1$), we infer from Cartan's Second Main Theorem (see Corollary 4.2.15) that

$$T_g(r) \leqq \sum_{j=1}^{4} N_2\bigl(r, g^*H_j\bigr) + O(\log r). \tag{6.4.26}$$

Since $N_2(r, g^*H_j) = 0$, $1 \leqq j \leqq 3$, we deduce from (6.4.26), (6.4.25) and (4.5.5) that

$$N(r, g^*H_4) = \frac{(n-m)c}{\pi}r + O(\log r).$$

By (6.4.23), $N(r, g^*H_4) = N(r, f^*D)$, and so

$$N\bigl(r, f^*D\bigr) = \frac{(n-m)c}{\pi}r + O(\log r). \tag{6.4.27}$$

It follows from (6.4.22) and (6.4.27) that

$$\delta(f, \bar{D}) = \frac{1+mc}{1+nc}. \tag{6.4.28}$$

By elementary calculations one shows that $\operatorname{ord}_z f^*D \geqq 2$ implies

$$(mc-1)\bigl(e^{cz}\bigr)^m + (nc-1)\bigl(e^{cz}\bigr)^n = 0.$$

Furthermore, $f(z) \in D$ if and only if $e^z + e^{mcz} + e^{ncz} = 0$. Combined, these two relations imply that there is a finite subset $S \subset \mathbf{C}^2$ such that $\operatorname{ord}_z f^*D \geqq 2$ implies

$(e^z, e^{cz}) \in S$. Since $z \mapsto (e^z, e^{cz})$ is injective, it follows that $\{z; \operatorname{ord}_z f^*D \geqq 2\}$ is a finite set. Therefore,

$$\begin{aligned} N_1\big(r, f^*D\big) &= N\big(r, f^*D\big) + O(\log r), \\ \delta_1(f, \bar{D}) &= \delta(f, \bar{D}) = \frac{1+mc}{1+nc}. \end{aligned} \tag{6.4.29}$$

Let $c' > 1$ be an irrational number, and set

$$c = 1/c', \qquad m = \big[c'\big], \qquad n = \big[c'\big] + 1,$$

where $[c']$ denotes the integral part of c'. Then m, n and c satisfy (6.4.21), and by (6.4.28)

$$\delta(f, \bar{D}) = \frac{1 + [c']/c'}{1 + ([c'] + 1)/c'} \to 1 \quad \big(c' \to \infty\big).$$

Thus $\delta(f, \bar{D})$ $(=\delta_1(f, \bar{D})$ by (6.4.29)) takes values arbitrarily close to 1.

Remark 6.4.30 The most essential part of the proof of Theorem 6.4.1 is the following estimate of the proximity function (see Lemma 6.4.7):

$$m_f(r, \bar{D}) = O(\log r) \text{ or } O\big(\log T_f\big(r, c_1(\bar{D})\big)\big) + O(\log r)\|_E. \tag{6.4.31}$$

The notion of logarithmic jet spaces due to Noguchi [86] (see Sect. 4.6.3) also plays a crucial role (cf. Dethloff–Lu [01] for an extension to the case of directed jets). We then use the jet projection method developed by Noguchi–Ochiai [90], Chap. 6, Sect. 3 (Ochiai–Noguchi [84]) (cf. Noguchi [77b], [81a], [98]).

Remark 6.4.32 In Noguchi [96], the first author proved that for D, not necessarily in good position, an entire curve $f : \mathbf{C} \to A$, omitting D, has no Zariski dense image, and is contained in a translate of a semi-abelian subvariety which has no intersection with D. What was proved in Noguchi [96] applied to $f : \mathbf{C} \to A$ with Zariski dense image yields that there is a positive constant κ such that

$$\kappa T_f\big(r, c_1(\bar{D})\big) \leqq N_1\big(r, f^*D\big) + S_f\big(r, c_1(\bar{D})\big), \tag{6.4.33}$$

provided that $\mathrm{St}(D)^0 = \{0\}$. The above κ may, in general, be very small because of the method of the proof. One needs more detailed properties of $J_k(D)$ to get an optimal Second Main Theorem than to get (6.4.33). This is why we need to assume $\bar{D}$ *being in good position* in Assumption 6.4.2 (iii).

Remark 6.4.34 It is an interesting problem to see if the truncation level k_0 of the counting function $N_{k_0}(r, f^*D)$ in Theorem 6.4.1 can be taken as a function only in $\dim A$. By the above proof, it would be sufficient to find a natural num-

ber k such that $\pi_2(J_k(\bar{D}; \log \bar{A} \cap \partial A)) \cap Y_k \neq Y_k$. Note that $\dim \pi_2(J_k(\bar{D}; \log \bar{A})) \leqq \dim J_k(\bar{D}; \log \bar{A}) = (n-1)(k+1)$. Thus, if $\dim Y_k > (n-1)(k+1)$ we may set $k_0 = k$. For example, if $J_n(f)(\mathbf{C})$ is Zariski dense in $J_n(A)$, then $\dim Y_n = n^2$. Since $\dim \pi_2(J_n(\bar{D}; \log \partial A)) = n^2 - 1$, we may set $k_0 = n$ (cf. Theorem 6.3.14).

6.5 Semi-abelian Varieties: Truncation Level 1

6.5.1 Truncation Level 1

In the previous section we established a Second Main Theorem for an entire curve f in a semi-torus with truncated counting function $N_{k_0}(r, f^*D)$ (Theorem 6.4.1). It is an interesting and important problem to decrease the truncation level k_0 as low as possible, and the best is, of course, $k_0 = 1$. The truncation level is essential in particular for various applications (cf. Sect. 6.6 and Corvaja–Noguchi [12]). By Example (i) of Remark 6.3.6, however, this is impossible as far as the remainder term $S_f(r)$ in Theorems 6.3.1 and 6.4.1 is kept as

$$O(\log r) + O\big(\log T_f(r)\big)\|.$$

In this section we are going to establish a Second Main Theorem with the best truncation level $k_0 = 1$, allowing the remainder term to be "$\varepsilon T_f(r)\|$".

Let $f : \mathbf{C} \to A$ be an entire curve in a semi-abelian variety A. We use the same notation as in Sect. 6.2; e.g., $X_k(f)$ denotes the Zariski closure of the image of the k-jet lift of f in the k-jet space $J_k(A)$ over A. The aim of this section is to prove the following.

Theorem 6.5.1 (The Second Main Theorem; Noguchi–Winkelmann–Yamanoi [08]) *Let $f : \mathbf{C} \to A$ be an algebraically non-degenerate entire curve in a semi-abelian variety A.*

(i) *Let Z be an algebraic reduced subvariety of $X_k(f)$ $(k \geqq 0)$. Then there exists a compactification $\bar{X}_k(f)$ of $X_k(f)$ such that*

$$T(r, \omega_{\bar{Z}, J_k(f)}) \leqq N_1\big(r, J_k(f)^*Z\big) + \varepsilon T_f(r)\|_\varepsilon, \quad {}^\forall\varepsilon > 0, \tag{6.5.2}$$

where $\bar{Z}$ is the closure of Z in $\bar{X}_k(f)$.

(ii) *Moreover, if $\operatorname{codim}_{X_k(f)} Z \geqq 2$, then*

$$T(r, \omega_{\bar{Z}, J_k(f)}) \leqq \varepsilon T_f(r)\|_\varepsilon, \quad {}^\forall\varepsilon > 0. \tag{6.5.3}$$

(iii) *In the case when $k = 0$ and Z is an effective divisor D on A, the compactification $\bar{A}$ of A can be chosen as smooth, equivariant with respect to the A-action, and independent of f; furthermore, (6.5.2) takes the form*

$$T_f\big(r, c_1(\bar{D})\big) \leqq N_1\big(r, f^*D\big) + \varepsilon T_f\big(r, c_1(\bar{D})\big)\|_\varepsilon, \quad {}^\forall\varepsilon > 0. \tag{6.5.4}$$

Even to prove (6.5.4) we need to establish a Second Main Theorem for the jet lifts $J_k(f)$ $(k \geqq 2)$ of f. The proof is rather long, and we proceed step by step.

6.5.2 The Second Main Theorem for Jet Lifts

Let A be a semi-abelian variety of dimension n and let T be the maximal affine subgroup of A. Then $T \cong (\mathbf{C}^*)^t$ and there is an exact sequence of rational homomorphisms

$$0 \to T \to A \to A_0 \to 0,$$

where A_0 is an abelian variety. Let $\bar{A}$ be a smooth equivariant compactification of A. Set $\partial A = \bar{A} \setminus A$ and let $J_k(\bar{A}; \log \partial A)$ be the logarithmic k-jet bundle along ∂A (cf. Sect. 5.4). Then A acts on $J_k(\bar{A}; \log \partial A)$ and there is an equivariant trivialization

$$J_k(\bar{A}; \log \partial A) \cong \bar{A} \times J_{k,A},$$

where A acts trivially on the second factor $J_{k,A} = \mathbf{C}^{kn}$. Let $\bar{J}_{k,A}$ be a projective compactification of $J_{k,A}$. With the trivial action of A on $\bar{J}_{k,A}$ and the usual action on A (by translations) and $\bar{A}$ this yields an A-equivariant compactification

$$\bar{J}_k(\bar{A}; \log \partial A) = \bar{A} \times \bar{J}_{k,A}$$

of $J_k(A)$ with an open A-invariant subset

$$\tilde{J}_k(A) = A \times \bar{J}_{k,A}.$$

For example, we may set $\bar{J}_{k,A} = \mathbf{P}^{nk}(\mathbf{C})$ or $\bar{J}_{k,A} = (\mathbf{P}^n(\mathbf{C}))^k$. Then $J_k(A) = J_k(\bar{A}; \log \partial A)|_A$ is a Zariski open subset of $\bar{J}_k(\bar{A}; \log \partial A)$ and

$$J_k(A) \cong A \times J_{k,A}.$$

We set

$$J_k^{\mathrm{reg}}(\bar{A}; \log \partial A) = \left\{ j_k(g) \in J_k(\bar{A}; \log \partial A);\ j_1(g) \neq 0 \right\} \cong \bar{A} \times J_{k,A}^{\mathrm{reg}},$$
$$J_k^{\mathrm{reg}}(A) = J_k^{\mathrm{reg}}(\bar{A}; \log \partial A)|_A \cong A \times J_{k,A}^{\mathrm{reg}},$$

of which elements are called *regular jets*.

Let $f : \mathbf{C} \to A$ be an entire curve and $J_k(f) : \mathbf{C} \to J_k(A)$ be the k-jet lift of f. We denote by $X_k(f)$ (resp. $\tilde{X}_k(f)$) the Zariski closure of the image $J_k(f)(\mathbf{C})$ in $J_k(A)$ (resp. $\tilde{J}_k(A)$):

$$X_k(f) \subset J_k(A), \qquad \tilde{X}_k(f) \subset \tilde{J}_k(A). \tag{6.5.5}$$

With an ample or big line bundle $L \to \bar{A}$ we write

$$S_f(r) = S_f(r, L).$$

We extend Theorem 6.4.1 for jet lifts:

Theorem 6.5.6 *Let $f : \mathbf{C} \to A$ be an algebraically non-degenerate an entire curve. Let Z be a reduced subvariety of $X_k(f)$. Then there exists a natural number l_0 and a compactification $\bar{X}_k(f)$ of $X_k(f)$ such that for the closure $\bar{Z}$ of Z in $\bar{X}_k(f)$*

$$m_{J_k(f)}(r, \bar{Z}) = S_f(r), \tag{6.5.7}$$

$$T(r, \omega_{\bar{Z}, J_k(f)}) = N_{l_0}\big(r, J_k(f)^* Z\big) + S_f(r). \tag{6.5.8}$$

Proof The basic idea of the proof is the same as that of the proof of Theorem 6.4.1.

We extend the subvariety Z to the closure in $\tilde{X}_k(f)$ which is denoted by the same Z. We first prove (6.5.7) and (6.4.8). Set $B = \mathrm{St}(X_k(f))^0$. Then we have the quotient maps:

$$q^B : A \to A/B = C,$$

$$q_k^B : J_k(A) \to J_k(A)/B \cong C \times J_{k,A},$$

$$\tilde{q}_k^B : \tilde{J}_k(A) \to C \times \bar{J}_{k,A}.$$

By Theorem 6.2.6 and Lemma 6.2.5

$$\dim B > 0, \qquad T_{q_k^B \circ J_k(f)}(r) = S_f(r). \tag{6.5.9}$$

Setting $\tilde{Y}_k = \tilde{X}_k(f)/B$, we have a quotient map:

$$\tilde{\pi}_k : \tilde{X}_k(f) \to \tilde{Y}_k \subset C \times \bar{J}_{k,A}.$$

Let $\bar{B}$ be a smooth equivariant compactification of B. Define $\hat{A}$, $\hat{X}_k(f)$, $\hat{Z}$, etc. as the partial compactifications of A, $\tilde{X}_k(f)$, Z, etc. as in Sect. 5.6. We then have proper maps,

$$\hat{q}_k^B : \hat{A} \times \bar{J}_{k,A} \to C \times \bar{J}_{k,A},$$

$$\hat{\pi}_k = \hat{q}_k^B\big|_{\hat{X}_k(f)} : \hat{X}_k(f) \to \tilde{Y}_k \subset C \times \bar{J}_{k,A},$$

whose fibers are isomorphic to $\bar{B}$.

There are two cases, $B \subset \mathrm{St}(Z)$ and $B \not\subset \mathrm{St}(Z)$, which we consider separately.

(a) Suppose that $B \subset \mathrm{St}(Z)^0$. Set $\hat{W} = \hat{\pi}_k(\hat{Z}) = \hat{Z}/B$. Then $\hat{W}$ has at least codimension one in $\tilde{Y}_k$. Let $T \cong (\mathbf{C}^*)^t$ be the maximal affine subgroup of A and let S be that of B. Then S is a subgroup of T and there is a splitting, $T \cong S \times S'$. Take an equivariant compactification $\bar{S}'$ of S' and set

$$\bar{A} = \hat{A} \times_{S'} \bar{S}'.$$

Then $\bar{A}$ is an equivariant compactification of A and $\hat{A}$. We have an algebraic exact sequence

$$0 \to S' \to C \to C_0 \to 0,$$

where C_0 is an abelian variety, and an equivariant compactification $\bar{C} = C \times_{S'} \bar{S}'$.

Thus $\hat{q}_k^B$ extends to

$$\bar{q}_k^B : \bar{A} \times J_{k,A} \to \bar{C} \times J_{k,A}.$$

Let $\bar{X}_k(f)$ (resp. $\bar{Y}_k$, $\bar{W}$) be the closure of $\hat{X}_k(f)$ (resp. $\hat{Y}_k$, $\hat{W}$) in $\bar{A} \times \bar{J}_{k,A}$ (resp. $\bar{C} \times \bar{J}_{k,A}$). Thus we have the restriction

$$\bar{\pi}_k = \bar{q}_k^B\big|_{\bar{X}_k(f)} : \bar{X}_k(f) \to \bar{Y}_k.$$

Note that $\bar{\pi}_k$ is surjective and

$$\bar{W} \neq \bar{Y}_k. \tag{6.5.10}$$

It follows from Theorem 2.4.9 (ii) and (6.5.9) that

$$\begin{aligned} m_{J_k(f)}(r, \bar{Z}) &\leqq m_{\bar{\pi}_k \circ J_k(f)}(r, \bar{W}) + O(1) \\ &= O\big(T_{\bar{\pi}_k \circ J_k(f)}(r)\big) = S_f(r). \end{aligned} \tag{6.5.11}$$

(b) Suppose that $B \not\subset \mathrm{St}(Z)^0$. We set

$$B' = B \cap \mathrm{St}(Z)^0, \qquad Z' = Z/B', \qquad q\tilde{X}'_k(f) = \tilde{X}_k(f)/B',$$
$$A' = A/B', \qquad B'' = B/B'.$$

Moreover, we define W as the image of Z under the quotient

$$\tilde{X}'_k(f) \to \tilde{X}'_k(f)/B'' = \tilde{Y}_k.$$

We have the following commutative diagram and quotient maps:

$$\begin{array}{ccccc} Z & & \underset{\text{(codim=1)}}{\subsetneqq} & \tilde{X}_k(f) \subset A \times \bar{J}_{k,A} \\ \downarrow & & & \downarrow & \downarrow q_k^{B'} \\ Z' & & \underset{\text{(codim=1)}}{\subsetneqq} & \tilde{X}'_k(f) \subset A' \times \bar{J}_{k,A} \\ \downarrow \tilde{\pi}'_k|_{Z'} & & & \downarrow \tilde{\pi}'_k & \downarrow q_k^{B''} \\ W & & \subset & \tilde{Y}_k & \subset C \times \bar{J}_{k,A}. \end{array}$$

Note that

$$\mathrm{St}\big(X'_k(f)\big)^0 = B'', \qquad \mathrm{St}\big(Z'\big)^0 \cap B'' = \{0\}. \tag{6.5.12}$$

Let $\bar{B}''$ be a smooth equivariant compactification of B''. We have

$$\hat{A}' = A' \times_{B''} \bar{B}'',$$
$$\hat{\partial} A' = \hat{A}' \setminus A',$$

(continued)

$$\hat{X}'_k(f) = \tilde{X}'_k(f) \times_{B''} \bar{B}'',$$
$$\hat{Z}' = \bar{Z}' \quad \big(\text{the closure of } Z' \text{ in } \hat{X}'_k(f)\big),$$
$$\hat{\partial} X'_k(f) = \hat{X}'_k(f) \setminus \tilde{X}'_k(f).$$

Note that the boundary divisor $\hat{\partial} A'$ has only normal crossings (Lemma 5.3.8). We obtain proper maps

$$\begin{array}{ccccc} \hat{Z}' & & \subsetneqq \hat{X}'_k(f) & \subset & \hat{A}' \times \bar{J}_{k,A} \\ \downarrow \hat{\pi}'_k|_{\hat{Z}'} & & \downarrow \hat{\pi}'_k & & \downarrow \hat{q}_k^{B''} \\ \hat{W} & \subset & \tilde{Y}_k & \subset & C \times \bar{J}_{k,A}\,, \end{array}$$

where $\hat{W} = \hat{\pi}'_k(\hat{Z}')$. By Proposition 5.6.14 we have a blow-up

$$\psi : \hat{X}_k'^{\dagger}(f) \to \hat{X}'_k(f)$$

with center in $\hat{\partial} X'_k(f)$, the strict transform $\hat{Z}'^{\dagger}$ of $\hat{Z}'$ and the boundary

$$\Gamma = \hat{X}_k'^{\dagger}(f) \setminus \tilde{X}'_k(f)$$

with stratification $\Gamma = \bigcup_\lambda \Gamma_\lambda$, such that

$$\Gamma_\lambda \cong \tilde{X}'_k(f)/\mathrm{Iso}_x\big(B''\big) \quad (x \in \Gamma_\lambda), \tag{6.5.13}$$

$$\Gamma_\lambda \cap \hat{Z}'^{\dagger} \neq \Gamma_\lambda. \tag{6.5.14}$$

Here, if $k = 0$, we use Proposition 5.6.12 in place of Proposition 5.6.14, and deduce the stated property for $\bar{A}$.

Let $\psi_{*l} : J_l(\hat{X}_k'^{\dagger}(f); \log \Gamma) \to J_l(\hat{X}'_k(f); \log \hat{\partial} X'_k(f))$ be the morphism naturally induced by ψ. We consider a sequence of morphisms

$$\begin{aligned} J_l\big(\hat{Z}'^{\dagger}; \log \Gamma\big) \subset J_l\big(\hat{X}_k'^{\dagger}(f); \log \Gamma\big) &\overset{\psi_{*l}}{\to} J_l\big(\hat{X}'_k(f); \log \hat{\partial} X'_k(f)\big) \\ &\hookrightarrow J_l\big(\hat{A}' \times \bar{J}_{k,A}; \log\big(\hat{\partial} A' \times \bar{J}_{k,A}\big)\big) \\ &\cong J_l\big(\hat{A}'; \log \hat{\partial} A'\big) \times J_l(\bar{J}_{k,A}) \\ &\cong \hat{A}' \times J_l(J_{k,A'}) \times J_l(\bar{J}_{k,A}) \\ &\overset{\text{proj.}}{\longrightarrow} J_l(J_{k,A'}) \times J_l(\bar{J}_{k,A}). \end{aligned}$$

Thus we have a morphism

$$\beta_l : J_l\big(\hat{X}_k'^{\dagger}(f); \log \Gamma\big) \to J_l(J_{k,A'}) \times J_l(\bar{J}_{k,A}).$$

Let $p_l : J_l(\hat{X}_k'^{\dagger}(f)) \to \hat{X}_k'^{\dagger}(f)$ be the projection to the base space. Hence we obtain a proper morphism

$$\gamma_l = \big(\hat{\pi}_k' \circ \psi \circ p_l\big) \times \beta_l : J_l\big(\hat{X}_k'^{\dagger}(f); \log \Gamma\big) \to \tilde{Y}_k \times J_l(J_{k,A'}) \times J_l(\bar{J}_{k,A}).$$

Claim 6.5.15 *For some $l_0 \geqq 1$,*

$$\gamma_{l_0}\big(J_{l_0}\big(\hat{Z}'\big)\big) \neq \gamma_{l_0}\big(J_{l_0}\big(\hat{X}_k'(f)\big)\big).$$

Assume contrarily that $\gamma_l(J_l(\hat{Z}')) = \gamma_l(J_l(\hat{X}_k'(f)))$ for all $l \geqq 1$. Then for every $z \in \mathbf{C}$

$$J_l\big(q_1^{B'} \circ J_k(f)\big)(z) \in \gamma_l\big(J_l\big(\hat{Z}'^{\dagger}; \log \Gamma\big)\big). \tag{6.5.16}$$

Fix $z_0 \in \mathbf{C}$. Then $\hat{\pi}_k \circ J_k(f)(z_0) \in \tilde{Y}_k$ and we set

$$\xi_l = J_l\big(q_1^{B'} \circ J_k(f)\big)(z_0) \in \gamma_l\big(J_l\big(\hat{Z}'^{\dagger}; \log \Gamma\big)\big), \quad l \geqq 1.$$

Set $\Xi_l = \gamma_l^{-1}(\xi_l)$ for $l \geqq 0$. Then the restriction $p_l|_{\Xi_l}$ is proper and $p_l|_{\Xi_l} : \Xi_l \to p_l(\Xi_l)$ is an isomorphism. We set

$$\Lambda_l = p_l(\Xi_l), \quad l = 1, 2, \ldots.$$

The sequence of $\Lambda_l \supset \Lambda_{l+1}$, $l = 1, 2, \ldots$ terminates to $\Lambda_\infty = \Lambda_{l_0} = \Lambda_{l_0+1} = \cdots$ $(\subset \hat{X}_k'^{\dagger}(f))$ for some l_0. Then $\Lambda_\infty \neq \emptyset$. If $\Lambda_\infty \cap \tilde{X}_k'(f) \neq \emptyset$, there is an element $a \in A'$ such that

$$a \cdot \big(J_l\big(q_1^{B'} \circ J_k(f)\big)(z_0)\big) \in J_l\big(Z'\big), \quad {}^\forall l \geqq 0.$$

By the identity principle we deduce that $a \cdot \tilde{X}_k'(f) \subset Z'$; this is absurd.

Now assume that $\Lambda_\infty \cap \Gamma \neq \emptyset$. There is a point $x_0 \in \Lambda_\infty \cap \Gamma$ such that

$$(x_0, \xi_l) \in J_l\big(\hat{Z}'^{\dagger}\big)_{x_0}, \quad l \geqq 1.$$

Let Γ_{λ_0} be the boundary stratum containing x_0. Let

$$\alpha : \tilde{X}_k'(f) \to \tilde{X}_k'(f)/\mathrm{Iso}_{x_0}\big(B''\big) \cong \Gamma_{\lambda_0}$$

be the quotient map. Then there exists an element $a_0 \in A$ such that

$$a \cdot \big(\alpha \circ q_1^{B'} \circ J_k(f)(z)\big) \in \Gamma_{\lambda_0} \cap \hat{Z}'^{\dagger}$$

in a neighborhood of z_0 and hence for all $z \in \mathbf{C}$. Hence a contradiction follows from this, (6.5.14) and the image $J_k(f)(\mathbf{C})$ being Zariski dense in $X_k(f)$. This proves Claim 6.5.15.

We infer (6.5.8) as in the proof of Theorem 6.4.1 with a modification as follows. Let $\bar{X}_k(f) = \bigcup_\alpha U_\alpha$ be a finite affine covering, and let $\sigma_{\alpha\nu}$ be the defining functions

of $Z \cap U_\alpha$. Then by Claim 6.5.15 there is a rational function η on $\tilde{Y}_k \times J_{l_0}(J_{k,A'} \times J_{l_0}(\bar{J}_{k,A}))$, regarded as a rational function on $J_{l_0}(X_k(f))$ such that

$$\eta \circ J_{l_0}\big(q_1^{B'} \circ J_k(f)\big)(z) \not\equiv 0, \quad z \in \mathbf{C},$$
$$\eta|_{U_\alpha} = \sum_\nu \sum_{0 \leqq j \leqq l_0} a_{\alpha\nu j} d^j \sigma_{\alpha\nu},$$

where the coefficients $a_{\alpha\nu j}$ are jet differentials on U_α. Then we have

$$\eta|_{U_\alpha} = \sum_\nu \sigma_{\alpha\nu} \sum_{0 \leqq j \leqq l_0} a_{\alpha\nu j} \frac{d^j \sigma_{\alpha\nu}}{\sigma_{\alpha\nu}},$$
$$|\eta|_{U_\alpha}| \leqq \left(\sum_\nu |\sigma_{\alpha\nu}|^2\right)^{1/2} \left(\sum_\nu \left(\sum_{0 \leqq j \leqq l_0} |a_{\alpha\nu j}| \left|\frac{d^j \sigma_{\alpha\nu}}{\sigma_{\alpha\nu}}\right|\right)^2\right)^{1/2},$$
$$\frac{1}{(\sum_\nu |\sigma_{\alpha\nu}|^2)^{1/2}} \leqq \frac{1}{|\eta|_{U_\alpha}|} \left(\sum_\nu \left(\sum_{0 \leqq j \leqq l_0} |a_{\alpha\nu j}| \left|\frac{d^j \sigma_{\alpha\nu}}{\sigma_{\alpha\nu}}\right|\right)^2\right)^{1/2}.$$

We therefore deduce from (6.5.9) that

$$m_{J_k(f)}(r, \bar{Z}) = S_f(r),$$
$$N\big(r, J_k(f)^* Z\big) = N_{l_0}\big(r, J_k(f)^* Z\big) + S_f(r).$$

Combining these with the First Main Theorem 2.4.9, we obtain (6.5.7) and (6.5.8). □

6.5.3 Higher Codimensional Subvarieties of $X_k(f)$

Let $f : \mathbf{C} \to A$ be an entire curve in a semi-abelian variety A. We use the same notation, $X_k(f)$, $\mathrm{St}(X_k(f))^0$, etc. as in the previous subsection.

The purpose of this subsection is to prove the following.

Theorem 6.5.17 *Let $f : \mathbf{C} \to A$ be an entire curve and let $Z \subset X_k(f)$ be a subvariety of $\mathrm{codim}_{X_k(f)} Z \geqq 2$. Then there is a compactification $\bar{X}_k(f)$ such that for the closure $\bar{Z}$ of Z in $\bar{X}_k(f)$*

$$T(r, \omega_{\bar{Z} J_k(f)}) \leqq \varepsilon T_f(r)\|_\varepsilon, \quad {}^\forall \varepsilon > 0.$$

In particular,

$$N\big(r, J_k(f)^* Z\big) \leqq \varepsilon T_f(r)\|_\varepsilon, \quad {}^\forall \varepsilon > 0. \tag{6.5.18}$$

It suffices to prove Theorem 6.5.17 for irreducible Z. Hence, we assume throughout this subsection that Z is *irreducible*. Since the Zariski closure of $f(\mathbf{C})$ is a translate of a semi-abelian subvariety of A (Theorem 6.2.1) we also assume all through the present subsection that

6.5.19 $f : \mathbf{C} \to A$ is algebraically *non-degenerate*.

Our proof naturally divides into three steps (a)–(c). Before going to discuss the details, we give an outline of the proof.

(a) First, we reduce the case to the one that A admits a splitting $A = B \times C$ where B and C are semi-abelian varieties of positive dimensions such that

$$B \subset \mathrm{St}\big(X_l(f)\big)^0 \quad \text{for all } l \geqq 0 \tag{6.5.20}$$

and the composition of f and the second projection $q^B : A \to A/B = C$ satisfies

$$T_{q^B \circ f}(r) = S_f(r). \tag{6.5.21}$$

By this reduction, we may assume that the variety $X_l(f)$ has splitting $X_l(f) = B \times (X_l(f)/B)$ for all $l \geqq 0$.

We also make a reduction such that the image of Z under the second projection $\pi_k : X_k(f) \to X_k(f)/B$ has a Zariski dense image. Hence by the assumption $\mathrm{codim}_{X_k(f)} Z \geqq 2$, we may assume $\mathrm{codim}_{\pi_k^{-1}(x)} Z \cap \pi_k^{-1}(x) \geqq 2$ for general $x \in X_k(f)/B$.

(b) To prepare for the next step, we construct a certain auxiliary subscheme.

(c) The third step is the main part of the proof. We shall construct auxiliary divisors $F_l \subset \bar{B} \times (X_{k+l}(f)/B)$ for all $l \geqq 0$ with the following properties:

(i) $(l+1)N_1(r, J_k(f)^*Z) \leqq N(r, J_{k+l}(f)^*F_l) + \varepsilon T_f(r)\|_\varepsilon$, $^\forall\varepsilon > 0$,

(ii) $T_{J_{k+l}(f)}(r, L(F_l)) \leqq n(l)T_{\gamma\circ f}(r, L(\bar{D}_B)) + \varepsilon T_f(r, L(\bar{D}))\|_\varepsilon$, $^\forall\varepsilon > 0$, where $\gamma : A \to B$ is the first projection, $\bar{D}$ (resp. $\bar{D}_B$) is an ample divisor on $\bar{A}$ (resp. $\bar{B}$), and $n(l)$ is a positive integer such that $\lim_{l\to\infty} n(l)/l = 0$.

(d) Finally, by (i) and (ii) above we have

$$\begin{aligned} N_1\big(r, J_k(f)^*Z\big) &\leqq \frac{1}{l+1} N\big(r, J_{k+l}(f)^*F_l\big) + \frac{\varepsilon}{l+1} T_f\big(r, L(\bar{D})\big)\|_\varepsilon \\ &\leqq \frac{n(l)}{l+1} T_{\gamma\circ f}\big(r, L(\bar{D}_B)\big) + \frac{\varepsilon}{l+1} T_f\big(r, L(\bar{D})\big)\|_\varepsilon \end{aligned}$$

for all $\varepsilon > 0$ and all integer $l \geqq 0$. Since $n(l)/l \to 0$ $(l \to \infty)$, we have

$$N_1\big(r, J_k(f)^*Z\big) \leqq \varepsilon\big(T_{\gamma\circ f}\big(r, L(\bar{D}_B)\big) + T_f\big(r, L(\bar{D})\big)\big)\big\|_\varepsilon, \quad {}^\forall\varepsilon > 0.$$

Since $T_{\gamma\circ f}(r, L(\bar{D}_B)) = O(T_f(r, L(\bar{D})))$, the proof is completed.

(a) **Reduction**. Let $f : \mathbf{C} \to A$ be as above. Let

$$I_k : \hat{X}_k(f) (\hookrightarrow \hat{A} \times J_{k,A}) \to J_{k,A}$$

be the jet projection. It follows from Lemma 4.7.1 that

$$T_{I_k\circ J_k(f)}(r) = S_f(r). \tag{6.5.22}$$

We need the following.

Lemma 6.5.23 *Let the notation be as above. Let* $G = \bigcap_{l\geqq 0} \mathrm{St}(X_l(f))^0$ *and let* $q^G : A \to A/G$ *be the quotient map. Then*

$$T_{q^G\circ f}(r) = O\big(T_{I_k\circ J_k(f)}(r)\big)\big(= S_f(r)\big).$$

Proof This follows from (6.5.22) in combination with Theorem 6.2.6 and Lemma 6.2.5. □

Lemma 6.5.24 *Let* $B \subset A$ *be a semi-abelian subvariety. Put* $B' = B \cap (\bigcap_{l\geqq 0} \mathrm{St}(X_l(f))^0)$. *Let* $q^B : A \to A/B$ *and* $q^{B'} : A \to A/B'$ *be the quotient mappings. Then we have*

$$T_{q^{B'}\circ f}(r) = O\big(T_{q^B\circ f}(r)\big) + S_f(r).$$

Proof We write $G = \bigcap_{l\geqq 0} \mathrm{St}(X_l(f))^0$. Taking the natural embedding $A/B' \to (A/B) \times (A/G)$, we see that

$$T_{q^{B'}\circ f}(r) = O\big(T_{q^B\circ f}(r) + T_{q^G\circ f}(r)\big).$$

Thus the claim follows from Lemma 6.5.23. □

Lemma 6.5.25 *Let* A *and* A' *be semi-abelian varieties with a surjective homomorphism* $p : A \to A'$. *Let* $g : \mathbf{C} \to A'$ *be an entire curve. Then we have an entire curve* $\hat{g} : \mathbf{C} \to A$ *such that* $p \circ \hat{g} = g$ *and*

$$T_{\hat{g}}(r, \omega) = T_g\big(r, \omega'\big)$$

for suitably chosen Kähler forms ω *and* ω' *on the compactifications* $\bar{A}$ *and* $\bar{A}'$, *respectively.*

Proof Let

$$0 \to T \to A \to A_0 \to 0,$$
$$0 \to T' \to A' \to A'_0 \to 0,$$

be the algebraic presentations, where T and T' (resp. A_0 and A'_0) are algebraic tori (resp. abelian varieties) (cf. Sect. 5.1.6). Then the morphism p induces surjections

$$p_1 = p|_T : T \to T', \qquad p_2 : A_0 \to A'_0.$$

Set $n = \dim A$ and $n' = \dim A'$. Let $\varpi : \tilde{A} \cong \mathbf{C}^n \to A$ and $\varpi' : \tilde{A}' \cong \mathbf{C}^{n'} \to A'$ be the universal coverings. Then there is a surjective linear homomorphism $\tilde{p} : \tilde{A} \to \tilde{A}'$

induced by p, and there are direct sum representations as $\mathbf{C}$-vector spaces,

$$\tilde{A} = V \oplus V_0, \qquad \tilde{A}' = V' \oplus V_0',$$

such that p_1 (resp. p_2) induces $\tilde{p}_1 : V \to V'$ (resp. $\tilde{p}_2 : V_0 \to V_0'$). Since $\tilde{p}_j$ are surjective, there are subspaces $W \subset V$ and $W_0 \subset V_0$ such that

$$\tilde{p}_1|_W : W \to V', \qquad \tilde{p}_2|_{W_0} : W_0 \to V_0',$$

are isomorphisms. Let $q_1 : (\tilde{p}_1|_W)^{-1} : V' \to W$, $q_2 : (\tilde{p}_2|_{W_0})^{-1} : V_0' \to W_0$, and set

$$q = (q_1, q_2) : \tilde{A}' = V' \oplus V_0' \to W \oplus W_0 \hookrightarrow V \oplus V_0 = \tilde{A}.$$

Let $\tilde{g} : \mathbf{C} \to \tilde{A}'$ be a lifting of g. Define $\hat{g} = \varpi \circ q \circ \tilde{g} : \mathbf{C} \to A$. We see by Theorem 5.3.21 (iv) with its proof that there is a Kähler form ω (resp. ω') on a compactification $\bar{A}$ of A (resp. $\bar{A}'$ of A') satisfying $q^*\varpi^*\omega = \varpi'^*\omega'$. Therefore, $g^*\omega' = \hat{g}^*\omega$, so that $T_g(r, \omega') = T_{\hat{g}}(r, \omega)$. □

Next we are going to reduce our proof to the case such that $A = B \times C$ and that B and C are semi-abelian subvarieties satisfying (6.5.20) and (6.5.21). Let $\mathscr{B}$ be the set of all semi-abelian subvarieties $B \subset A$ such that

$$T_{q^B \circ f}(r) = S_f(r).$$

Observe that $\mathrm{St}(X_l(f))^0 \in \mathscr{B}$ due to Theorem 6.2.6. Since $\mathrm{St}(X_l(f))^0$ is a decreasing sequence of algebraic groups, there is a number m such that

$$\mathrm{St}\big(X_m(f)\big)^0 = \bigcap_{l \geqq 0} \mathrm{St}\big(X_l(f)\big)^0.$$

Hence $\bigcap_{l \geqq 0} \mathrm{St}(X_l(f))^0 \in \mathscr{B}$. In particular, $\mathscr{B} \neq \emptyset$, because $\dim \mathrm{St}(X_l(f))^0 > 0$ for all $l \geqq 0$ (Theorem 6.2.6 (ii)). Let $B \in \mathscr{B}$ be a minimal element of $\mathscr{B}$; i.e., if $B' \subset B$ and $B' \in \mathscr{B}$, then $B' = B$. If $B_i \in \mathscr{B}$, $i = 1, 2$, we deduce from Lemma 6.5.24 that $B_1 \cap B_2 \in \mathscr{B}$. Thus we get

$$B \subset \bigcap_{l \geqq 0} \mathrm{St}\big(X_l(f)\big)^0.$$

Put $C = A/B$ and let $q^B : A \to C$ be the quotient map. By Lemma 6.5.25 we may take an entire curve $g : \mathbf{C} \to A$ such that $q^B \circ g = q^B \circ f$ and

$$T_g(r) = S_f(r). \tag{6.5.26}$$

We may assume that the Zariski closure of the image $g(\mathbf{C})$ is a semi-abelian subvariety $C' \subset A$ (Theorem 6.2.1). Define the semi-abelian variety A' by $A' = \{(c, a) \in$

$C' \times A; q^B(c) = q^B(a)\}$. We obtain the following commutative diagram:

$$\begin{array}{ccc} A' & \xrightarrow{p_2} & A \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle q^B} \\ C' & \xrightarrow{q^B|_{C'}} & C \end{array}$$

The inclusion map $\iota : C' \to A$ yields a map $\tau : C' \to A'$ defined by $\tau(x) = (x, \iota(x))$. Note that this morphism τ is a section for $p_1 : A' \to C'$. Hence this bundle is trivial, i.e. $A' \cong C' \times B$ and $A'/B = C'$.

Put $\tilde{f} = g \times f : \mathbf{C} \to A'$. Then by (6.5.26) we have

$$T_f(r) = O\big(T_{\tilde{f}}(r)\big), \qquad T_{\tilde{f}}(r) = O\big(T_f(r)\big), \tag{6.5.27}$$

$$T_{p_1 \circ \tilde{f}}(r) = S_{\tilde{f}}(r). \tag{6.5.28}$$

The group variety B acts on $A' \subset C' \times A$ via

$$b : (c, a) \mapsto (c, a + b).$$

We define B' as the set of all $b \in B$ such that the induced automorphism of $J^l(A')$ stabilizes $X_l(\tilde{f})$ for every $l \geqq 0$.

Let $p_1' : A' \to A'/B'$ be the natural quotient map. By Lemma 6.5.24 and (6.5.28), we have

$$T_{p_1' \circ \tilde{f}}(r) = S_{\tilde{f}}(r). \tag{6.5.29}$$

Put $q^{B'} : A \to A/B'$ be the quotient map. Then we have

$$T_{q^{B'} \circ f}(r) = O\big(T_{p_1' \circ \tilde{f}}(r)\big). \tag{6.5.30}$$

Hence by (6.5.27), (6.5.29) and (6.5.30) we conclude $B' \in \mathscr{B}$. Since B is minimal in $\mathscr{B}$, we get $B' = B$. By the definition of B' we have $B \subset \bigcap_{l \geqq 0} \mathrm{St}(X_l(\tilde{f}))^0$. Let $p_{2,k} : X_k(\tilde{f}) \to X_k(f)$ be the morphism induced from $p_2 : A' \to A$. Set

$$\tilde{Z} = p_{2,k}^{-1}(Z) \subset X_k(\tilde{f}).$$

Note that

$$N_1\big(r, J_k(f)^* Z\big) = N_1\big(r, J_k(\tilde{f})^* \tilde{Z}\big)$$

and that (6.5.27) holds.

For the reduction we need $\mathrm{codim}_{X_k(\tilde{f})} \tilde{Z} \geqq 2$. By Lemma 6.5.24 we see that

$$B \subset \left(\bigcap_{l \geqq 0} \mathrm{St}\big(X_l(f)\big)^0\right) \cap \left(\bigcap_{l \geqq 0} \mathrm{St}\big(X_l(\tilde{f})\big)^0\right).$$

Thus $p_{2,l} : X_l(\tilde{f}) \to X_l(f)$ is B-equivariant, and induces a morphism

$$p_{2,l}^B : X_l(\tilde{f})/B \to X_l(f)/B.$$

Let $\pi_l : X_l(f) \to X_l(f)/B$ be the quotient map. Then it follows from (6.5.21) and (6.5.22) that

$$T_{\pi_l \circ J_l(f)}(r) = S_f(r). \tag{6.5.31}$$

If the image $\pi_k(Z)$ is not Zariski dense in $X_k(f)/B$, there is a Cartier divisor H on $X_k(f)/B$ containing $\pi_k(Z)$. Then, making use of (6.5.31) and the natural embedding $X_k(f)/B \hookrightarrow (A/B) \times J_{k,A}$ we get

$$N_1\big(r, J_k(f)^* Z\big) \leqq N\big(r, \big(\pi_k \circ J_k(f)\big)^* H\big) = O\big(T_{\pi_k \circ J_k(f)}(r)\big) = S_f(r). \tag{6.5.32}$$

Therefore the proof of Theorem 6.5.17 is finished in this case.

We assume that $\pi_k(Z)$ is Zariski dense in $X_k(f)$, and has a relative dimension at most $\dim B - 2$. Therefore the relative dimension of $\tilde{Z} \to X_k(\tilde{f})/B$ is at most $\dim B - 2$, and hence $\operatorname{codim}_{X_k(\tilde{f})} \tilde{Z} \geqq 2$. Therefore, by replacing A by A', C by C', f by $\tilde{f}$ and Z by $p_2^{-1}(Z)$, we may reduce our problem to the desired situation (6.5.20) and (6.5.21). Therefore we assume the following in the sequel: There is a non-trivial semi-abelian subvariety $B \subset A$ satisfying

$$\begin{aligned} &B \subset \textstyle\bigcap_{l \geqq 0} \mathrm{St}\big(X_l(f)\big), \\ &T_{q^B \circ f}(r) = S_f(r), \\ &A \cong B \times (A/B), \\ &\pi_k(Z) \text{ is Zariski dense in } X_k(f)/B, \end{aligned} \tag{6.5.33}$$

where $q^B : A \to A/B$ is the quotient map.

(b) **A special non-reduced subscheme**. We will use a technique of "thickening" of a submanifold in a certain direction developed by Yamanoi [04b], [04c].

First some algebraic preparations are given. Let R be a commutative ring, let I be an ideal of R, and let d be a derivation of R. Let d^s $(s \in \mathbf{Z}^+)$ denote the s-fold iteration of d, i.e., $d^s = d \circ \cdots \circ d$ (s-times). Then we can define

$$I_r = \big\{ f \in R; d^s(f) \in I, 0 \leqq {}^\forall s < r \big\}. \tag{6.5.34}$$

One easily verifies the following properties.

Lemma 6.5.35 (i) *I_r is an ideal of R for every $r \in \mathbf{N}$.*
(ii) *$fg \in I_{r+1}$ for all $f \in I_r$, $g \in I$.*
(iii) *I_r/I_{r+1} is an R/I-module.*
(iv) *The radicals* $\mathrm{rad}(I)$ *and* $\mathrm{rad}(I_r)$ *agree.*

Later, we will apply this to jet spaces. For this we will need another algebraic preparation.

Proposition 6.5.36 *Let B_k $(k \in \mathbf{N})$ be a sequence of increasing k-algebras (i.e. $B_k \subset B_{k+1}, {}^\forall k \in \mathbf{N}$), let $B = \bigcup_k B_k$, let δ be a derivation on B (with values in B), and let I be an ideal of B.*

Assume that there are algebra homomorphisms $\pi_k : B \to B_k$ such that $\pi_k|_{B_k} = \mathrm{id}_{B_k}$, and furthermore, assume the following:

(i) $\delta(\operatorname{Ker}\pi_k) \subset \operatorname{Ker}\pi_k, {}^\forall k \in \mathbf{N}$;
(ii) $(\delta - (\pi_k \circ \delta))(I) \subset I$.

Put $\delta_k = \pi_k \circ \delta$, and

$$I_r = \big\{f \in B; \delta^s(f) \in I, 0 \leqq {}^\forall s < r\big\},$$
$$I'_{r,k} = \big\{f \in B_k; \delta_k^s(f) \in I, 0 \leqq {}^\forall s < r\big\}.$$

Then

$$I'_{r,k} = I_r \cap B_k, \quad k \in \mathbf{N}, r \in \mathbf{Z}^+.$$

Proof We first note that $\pi_k \circ \delta_k$ restricted to B_k is a derivation of B_k with values in B_k. We show

Claim 1 $\pi_k \circ \delta^s \circ \pi_k = \pi_k \circ \delta^s, {}^\forall k \in \mathbf{N}, {}^\forall s \in \mathbf{Z}^+$.

For $s = 0$ this follows because $\pi_k|_{B_k} = \mathrm{id}_{B_k}$ implies $\pi_k \circ \pi_k = \pi_k$. To verify this claim in general we consider

$$\big(\pi_k \circ \delta^s \circ \pi_k - \pi_k \circ \delta^s\big)(f) = \pi_k \circ \delta^s\big(\pi_k(f) - f\big), \quad f \in B.$$

Note that $\pi_k \circ \pi_k = \pi_k$ implies $(\pi_k(f) - f) \in \operatorname{Ker}\pi_k$ for all f. It follows that $\delta^s(\pi_k(f) - f) \in \operatorname{Ker}\pi$. Consequently,

$$\big(\pi_k \circ \delta^s \circ \pi_k - \pi_k \circ \delta^s\big)(f) = 0,$$

which proves the claim.

Let us now fix a number k. We prove the assertion $I'_{r,k} = I_r \cap B_k$ by induction on r. For $r = 0$ the statement is trivial. Thus we assume $I'_{r,k} = I_r \cap B_k$ and want to show under this assumption that an element $f \in B_k$ is contained in $I'_{r+1,k}$ iff $f \in I_{r+1} \cap B_k$. Now we observe that for every $f \in I'_{r,k} = I_r \cap B_k$ we have $\delta^{r-1}(f) \in I$. Combined with $(\pi_k \circ \delta - \delta)(I) \subset I$, this implies

$$\delta_k^r(f) - \delta^r(f) = (\pi \circ \delta - \delta)\big(\delta^{r-1}(f)\big) \in I.$$

Hence $\delta_k^r(f) \in I$ implies $\delta^r(f) \in I$ and vice versa. Therefore $I'_{r+1,k} = I_{r+1} \cap B_k$. □

Translated into a geometric context, Lemma 6.5.35 yields

Proposition 6.5.37 *Let N be a complex submanifold of a complex manifold M and let V be a holomorphic vector field on M. Let $\mathscr{I}$ denote the ideal sheaf of N. Then M admits non-reduced subschemas N_k which are defined by ideal sheaves $\mathscr{I}_k$*

defined by

$$f \in (\mathscr{I}_k)_x \iff V^s(f) \in \mathscr{I}, \quad 0 \leqq {}^\forall s < k$$

such that the support of N_k $(=(N_k)_{\mathrm{red}})$ *is precisely* N.

A vector field V on a manifold M with submanifold N is said to be *transversal to* N in a point $p \in N$ if $V_p \notin \mathbf{T}(N)_p$. In this case we can always choose local coordinates $x_1, \ldots, x_n$ around p such that N is the common zero set of $x_1, \ldots, x_d$ (with $d = \dim M - \dim N$) and $V = \frac{\partial}{\partial x_1}$. Using such local coordinates the ideal sheaves $\mathscr{I}_k$ are generated by

$$x_1^k, x_2, \ldots, x_d.$$

Thereby the following is easily verified.

Proposition 6.5.38 *Let* M, N, V, $\mathscr{I}$ *and* $\mathscr{I}_k$ *be defined as before and assume that* V *is transversal to* N *in a point* $p \in N$. *Then we have the following*:

(i) $\mathscr{I}_k/\mathscr{I}_{k+1}$ *is a free* $\mathscr{O}_N$*-module sheaf of rank* 1 *near* p.
(ii) *If* $\iota : N \to N_k$ *denotes the natural embedding and* $\rho : N_k \to N$ *is a holomorphic map with* $\rho \circ \iota = \mathrm{id}_N$, *then* ρ *is a finite morphism and* $\rho_* \mathscr{O}_{N_k}$ *is locally a free* $\mathscr{O}_N$*-module of rank* k *in the neighborhood of each point* $p \in N$ *in which the vector field* V *is transversal*.

We will next apply all this theory to jet spaces. Given an algebraic variety X, let $J_k(X)$ denote the k-th jet space. By construction there are natural projection morphisms from $J_{k+1}(X)$ to $J_k(X)$ for every $k \in \mathbf{N}$. Hence we can form the inverse limit, $J_\infty(X) := \lim_{k\to\infty} J_k(X)$.

A function $J_\infty(X) \to \mathbf{C}$ is said to be regular (resp. holomorphic) if it is factored through a regular (resp. holomorphic) map defined on some $J_k(X)$. An entire curve $\mathbf{C} \to X$ induces a map $J_\infty(f) : \mathbf{C} \to J_\infty(X)$. Given a system of local coordinates $(x_1, \ldots, x_n)$ on X, we obtain induced coordinate functions $x_j^{(k)}$ on $J_\infty(X)$. Now we can define a vector field on $J_\infty(X)$ as

$$V = \sum_{i,k} x_i^{(k+1)} \frac{\partial}{\partial x_i^{(k)}}$$

such that

$$\big(J_\infty(f)\big)_* \left(\frac{\partial}{\partial z}\right) = V_{(J_\infty(f))(z)}, \quad {}^\forall z \in \mathbf{C}.$$

There is a natural embedding j of X into each $J_k(X)$ (and therefore also into $J_\infty(X)$) by mapping each point $x \in X$ to the jet defined by the constant map with value x.

Now let C be a submanifold of X and let $\mathscr{I}$ denote the ideal sheaf of $j(C)$ in $J_\infty(X)$. Define ideal sheaves $\mathscr{I}_k$ as in (6.5.34), i.e., by taking the sheaf of germs of

functions f for which

$$V^s(f) \in \mathscr{I}, \quad 0 \leq {}^{\forall}s \leq k \tag{6.5.39}$$

holds.

Let $\mathscr{I}_{r,k}$ denote the ideal sheaf on J_k obtained from $\mathscr{I}_r$ via pull-back. Proposition 6.5.36 implies that $\mathscr{I}_{r,k}$ can be defined on $J_k(X)$ as in (6.5.39) by using the vector field

$$V_k = \sum_{i,j<k} x_i^{(j+1)} \frac{\partial}{\partial x_i^{(j)}}$$

Hence from Proposition 6.5.38 we obtain

Proposition 6.5.40 *Let A be a semi-abelian variety with a semi-abelian subvariety C, and let $k, l \in \mathbf{N}$. Define $M = J_{k+l}(A) \times C \times J_{k,A}$, let $\lambda : M \to J_{k+l}(A)$ denote the projection on the first component, and set*

$$T = \big\{(v, c, w) \in M;\, p_{k+l,k}(v) = (c, w)\big\}.$$

Then there exists a closed subscheme $\mathscr{T} \subset M$ with the following properties:

(i) $\operatorname{Supp} \mathscr{T} = T$.

(ii) *The restriction $\lambda' = \lambda|_{\mathscr{T}} : \mathscr{T} \to J_{k+l}(A) \cong A \times J_{k+l,A}$ is a finite morphism onto its image $C \times J_{k+l,A}$. Furthermore the restriction of the direct image sheaf $\lambda'_*(\mathscr{O}_{\mathscr{T}})$ to $C \times J^{\mathrm{reg}}_{k+l,A}$ is a rank $l+1$ locally free $\mathscr{O}_{C \times J^{\mathrm{reg}}_{k+l,A}}$-module.*

(iii) *Let $f : \mathbf{C} \to A$ be an entire curve with $f(a) \in C$ $(a \in \mathbf{C})$ and define*

$$\gamma : z \mapsto \big(J_{k+l}(f)(z), f(a)\big).$$

Then

$$\operatorname{ord}_a \gamma^* \mathscr{T} \geqq l + 1.$$

(c) **Auxiliary divisor**. Let the notation and the assumption be as above. Set $C = A/B$. We have by (6.5.33)

$$A \cong B \times C. \tag{6.5.41}$$

Then it naturally induces

$$X_l(f) \cong B \times \big(X_l(f)/B\big) \quad (l \geqq 0).$$

Let $\bar{B}$ be an equivariant compactification of B and set $\hat{X}_l(f) = \bar{B} \times (X_l(f)/B)$. Let

$$\hat{\gamma}_l : \hat{X}_l(f) \to \bar{B},$$

$$\hat{\pi}_l : \hat{X}_l(f) \to X_l(f)/B$$

be the natural projections.

We denote by Z^{ns} the set of non-singular points of Z.

Lemma 6.5.42 *Let $L \to \bar{B}$ be an ample line bundle. Then there is a sequence of natural numbers $n(1), n(2), n(3), \ldots$ satisfying the following*:

(i) $\lim_{l\to\infty} \frac{n(l)}{l} = 0$.

(ii) *There exist effective Cartier divisors $F_l \subset \hat{X}_{k+l}(f)$ and line bundles M_l on $X_{k+l}(f)/B$ such that F_l is defined by a non-zero element of*

$$H^0\big(\hat{X}_{k+l}(f), \hat{\gamma}_{k+l}^* L^{\otimes n(l)} \otimes (\hat{\pi}_{k+l})^* M_l\big)$$

and that for every point $a \in \mathbf{C}$ with $J_k(f)(a) \in Z^{\mathrm{ns}}$

$$\mathrm{ord}_a\, J_{k+l}(f)^* F_l \geqq l + 1.$$

Proof Let $r_1 : Z^\dagger \to \bar{Z}$ be a desingularization of $\bar{Z}$ such that r_1 gives an isomorphism over Z^{ns}. Put $Y_k = X_k(f)/B$. Consider the sequence of morphisms

$$Z^{\mathrm{ns}} \overset{r_0}{\hookrightarrow} Z^\dagger \overset{r_1}{\longrightarrow} \bar{Z} \overset{r_2}{\hookrightarrow} \hat{X}_k(f) \overset{\hat{\pi}_k}{\longrightarrow} Y_k. \tag{6.5.43}$$

Here r_0, $r_1 \circ r_0$ are open immersions and r_2 is a closed immersion. Put the composition of morphisms to be

$$r = \hat{\pi}_k \circ r_2 \circ r_1 : Z^\dagger \longrightarrow Y_k.$$

Let Y_k^{fl} be a Zariski open subset of Y_k such that Y_k^{fl} is non-singular and the fibers of $r : Z^\dagger \to Y_k$ over Y_k^{fl} are all of the same dimension $\dim Z^\dagger - \dim Y_k$. Then the restriction of the family $r : Z^\dagger \to Y_k$ to Y_k^{fl} is a flat family.

Consider the pull-back of the sequence of morphisms (6.5.43) by the natural projection $B \times Y_k \to Y_k$:

$$B \times Z^{\mathrm{ns}} \overset{s_0}{\hookrightarrow} B \times Z^\dagger \overset{s_1}{\longrightarrow} B \times \bar{Z} \overset{s_2}{\hookrightarrow} B \times \hat{X}_k(f) \overset{s_3}{\longrightarrow} B \times Y_k.$$

Again put the composition of these morphisms to be $s = s_3 \circ s_2 \circ s_1 : B \times Z^\dagger \to B \times Y_k$. Then s maps as

$$s : (a, z) \in B \times Z^\dagger \to \big(a, r(z)\big) \in B \times Y_k.$$

Let L be an ample line bundle on $\bar{B}$ and set

$$\phi : (a, w) \in B \times \hat{X}_k(f) \to a + \gamma_k(w) \in \bar{B}. \tag{6.5.44}$$

Let $L_1^\dagger$ be the line bundle on $B \times Z^\dagger$ which is the pull-back of L by the composition of morphisms

$$B \times Z^\dagger \overset{s_2 \circ s_1}{\longrightarrow} B \times \hat{X}_k(f) \overset{\phi}{\longrightarrow} \bar{B}.$$

Since the restriction of s over $B \times Y_k^{\mathrm{fl}}$ (i.e., $s|_{B\times Y_k^{\mathrm{fl}}} : B \times (Z^\dagger|_{Y_k^{\mathrm{fl}}}) \to B \times Y_k^{\mathrm{fl}}$) is a flat family, the semicontinuity theorem of Hartshorne [77], p. 288, implies that there

is a Zariski open subset $U_n \subset B \times Y_k^{\mathrm{fl}}$ $(n > 0)$ such that $H^0((B \times Z^\dagger)|_P, L_{1,P}^{\dagger\otimes n})$ are all $\mathbf{C}$-vector spaces of the same dimension for $P \in U_n$. Here $(B \times Z^\dagger)|_P$ denotes the fiber of the morphism $s : B \times Z^\dagger \to B \times Y_k$ over $P \in B \times Y_k$, and $L_{1,P}^{\dagger\otimes n}$ is the induced line bundle. Put

$$G_n = \dim H^0\big(\big(B \times Z^\dagger\big)|_P, L_{1,P}^{\dagger\otimes n}\big).$$

Since the intersection $\bigcap_{n\geqq 1} U_n$ is non-empty, putting $(a, w) \in \bigcap_{n\geqq 1} U_n$ and replacing L by the pull-back by the morphism

$$B \ni x \mapsto x + a \in B,$$

we may assume $a = 0 \in B$.

Now for a positive integer $l > 0$, let $\mathscr{T}_l^\dagger \subset A \times J_{k+l,A} \times C \times J_{k,A}$ be the closed subscheme, and let $\lambda' : \mathscr{T}_l^\dagger \to C \times J_{k+l,A}$ be the morphism obtained in Proposition 6.5.40. Then λ' has the following properties:

(i) λ' is finite;
(ii) the direct image sheaf $\lambda'_* \mathscr{O}_{\mathscr{T}_l^\dagger}$ is locally generated by $l + 1$ elements as $\mathscr{O}_{C\times J_{k+l,A}}$ module on $C \times J_{k+l,A}^{\mathrm{reg}}$;
(iii) λ' induces an isomorphism of the underlying topological spaces of $\mathscr{T}_l^\dagger$ and $C \times J_{k+l,A}$.

Since Y_{k+l} is a Zariski closed subset of $C \times J_{k+l,A}$, we denote by $\sigma_{k+l} : Y_{k+l} \to C$ (resp. $\eta_{k+l} : Y_{k+l} \to J_{k+l,A}$) the restriction of the first projection $C \times J_{k+l,A} \to C$ (resp. the second projection $C \times J_{k+l,A} \to J_{k+l,A}$) to C. We have the closed immersion

$$\begin{aligned} B \times Y_{k+l} \times Y_k &\hookrightarrow B \times C \times J_{k+l,A} \times C \times J_{k,A} \\ &\cong A \times J_{k+l,A} \times C \times J_{k,A}, \end{aligned} \tag{6.5.45}$$

where the first inclusion is given by

$$\begin{aligned} B \times Y_{k+l} \times Y_k \ni \big(b, v, v'\big) &\mapsto \big(b, \sigma_{k+l}(v), \eta_{k+l}(v), \sigma_k\big(v'\big), \eta_k\big(v'\big)\big) \\ &\in B \times C \times J_{k+l,A} \times C \times J_{k,A}, \end{aligned}$$

and the second isomorphism is given by

$$\begin{aligned} B \times C \times J_{k+l,A} \times C \times J_{k,A} \ni \big(b, c, u, c', u'\big) &\mapsto \big((b, c), u, c', u'\big) \\ &\in A \times J_{k+l,A} \times C \times J_{k,A}. \end{aligned}$$

Let $\mathscr{S}_l \subset B \times Y_{k+l} \times Y_k$ be the closed subscheme obtained by the pull-back of $\mathscr{T}_l^\dagger$ by (6.5.45). Let $q : \mathscr{S}_l \to Y_{k+l}$ be the composition with the second projection $B \times Y_{k+l} \times Y_k \to Y_{k+l}$. We put

$$Y_{k+l}^{\mathrm{reg}} = Y_{k+l} \cap \big(C \times J_{k+l,A}^{\mathrm{reg}}\big),$$

which is the Zariski open subset of Y_{k+l}. Then by the above properties of λ', we have the corresponding properties for q:

(i) q is finite;
(ii) the direct image sheaf $q_*\mathscr{O}_{\mathscr{S}_l}$ is locally generated by $l+1$ elements as an $\mathscr{O}_{Y_{k+l}}$-module on Y_{k+l}^{reg};
(iii) q gives an isomorphism of underlying topological spaces of $\mathscr{S}_l$ and Y_{k+l}.

We consider the following commutative diagram (6.5.46) obtained by the base change of (6.5.43) with a sequence of morphisms

$$\mathscr{S}_l \hookrightarrow B \times Y_{k+l} \times Y_k \to B \times Y_k \to Y_k.$$

Here $B \times Y_{k+l} \times Y_k \ni (a, w, w') \mapsto (a, w') \in B \times Y_k$ is the natural projection.

$$\begin{array}{ccccccc}
\mathscr{Z}_l^{\mathrm{ns}} & \longrightarrow & B \times Y_{k+l} \times Z^{\mathrm{ns}} & \longrightarrow & B \times Z^{\mathrm{ns}} & \longrightarrow & Z^{\mathrm{ns}} \\
\downarrow{\scriptstyle u_0} & & \downarrow{\scriptstyle t_0} & & \downarrow{\scriptstyle s_0} & & \downarrow{\scriptstyle r_0} \\
\mathscr{Z}_l^{\dagger} & \longrightarrow & B \times Y_{k+l} \times Z^{\dagger} & \longrightarrow & B \times Z^{\dagger} & \longrightarrow & Z^{\dagger} \\
\downarrow{\scriptstyle u_1} & & \downarrow{\scriptstyle t_1} & & \downarrow{\scriptstyle s_1} & & \downarrow{\scriptstyle r_1} \\
\mathscr{Z}_l & \xrightarrow{v'} & B \times Y_{k+l} \times \bar{Z} & \longrightarrow & B \times \bar{Z} & \longrightarrow & \bar{Z} \\
\downarrow{\scriptstyle u_2} & & \downarrow{\scriptstyle t_2} & & \downarrow{\scriptstyle s_2} & & \downarrow{\scriptstyle r_2} \\
\cdot & \longrightarrow & B \times Y_{k+l} \times \hat{X}_k(f) & \longrightarrow & B \times \hat{X}_k(f) & \longrightarrow & \hat{X}_k(f) \\
\downarrow{\scriptstyle u_3} & & \downarrow{\scriptstyle t_3} & & \downarrow{\scriptstyle s_3} & & \downarrow{\scriptstyle \hat{\pi}_k} \\
\mathscr{S}_l & \xrightarrow{v} & B \times Y_{k+l} \times Y_k & \longrightarrow & B \times Y_k & \longrightarrow & Y_k
\end{array} \tag{6.5.46}$$

Let $\mathscr{L}_l^{\dagger}$ be the line bundle on $\mathscr{Z}_l^{\dagger}$ obtained by the pull-back of $L_1^{\dagger}$ by the morphisms in diagram (6.5.46) above. Let $\mathscr{S}_{l,n}$ be the non-empty Zariski open subset of $\mathscr{S}_l$ obtained by the inverse image of U_n. Since $\dim H^0((B \times Z^{\dagger})|_P, L_{1,P}^{\dagger\otimes n}) = G_n$ for $P \in U_n$, the direct image sheaf $s_* L_1^{\dagger\otimes n}$ is a locally free sheaf of rank G_n on U_n and the natural map

$$s_* L_1^{\dagger\otimes n} \otimes \mathbf{C}(P) \to H^0\big(\big(B \times Z^{\dagger}\big)\big|_P, L_{1,P}^{\dagger\otimes n}\big)$$

is an isomorphism for $P \in U_n$. This follows from the Theorem of Grauert (see Hartshorne [77], p. 288), since U_n is reduced and irreducible. Here $s : B \times Z^{\dagger} \to B \times Y_k$ is the natural map; i.e., $s = s_3 \circ s_2 \circ s_1$. Let u be the morphism $u : \mathscr{Z}_l^{\dagger} \to \mathscr{S}_l$ obtained by the composition $u = u_3 \circ u_2 \circ u_1$, where u_1, u_2, u_3 are the morphisms in diagram (6.5.46) above. Then the natural map

$$u_* \mathscr{L}_l^{\dagger\otimes n} \otimes \mathbf{C}(P) \to H^0\big(\mathscr{Z}_l^{\dagger}|_P, \mathscr{L}_{l,P}^{\dagger\otimes n}\big)$$

is also surjective, so an isomorphism on $P \in \mathscr{S}_{l,n}$. This follows by the Theorem of Cohomology and Base Change (Hartshorne [77], p. 290). Hence $u_*\mathscr{L}_l^{\dagger\otimes n}$ is locally generated by G_n elements as an $\mathscr{O}_{\mathscr{S}_l}$-module on $\mathscr{S}_{l,n} \subset \mathscr{S}_l$. Let $Y_{k+l,n} = q(\mathscr{S}_{l,n})$ be a non-empty Zariski open subset of Y_{k+l} (note that the underlying topological spaces of $\mathscr{S}_l$ and Y_{k+l} are the same). Then by the above properties of q, the direct image sheaf $(q \circ u)_*\mathscr{L}_l^{\dagger\otimes n}$ is locally generated by $(l+1)G_n$ elements as an $\mathscr{O}_{Y_{k+l}}$-module on $Y_{k+l,n} \cap Y_{k+l}^{\mathrm{reg}}$. Here, note that Y_{k+l}^{reg} is non-empty (otherwise f must be constant) and Y_{k+l} is irreducible. Hence $Y_{k+l,n} \cap Y_{k+l}^{\mathrm{reg}}$ is also non-empty.

Now look at the following commutative diagram:

$$\begin{array}{ccccccc}
\mathscr{Z}_l^{\mathrm{ns}} & & & & & & \\
\downarrow{\scriptstyle u_0} & & & & & & \\
\mathscr{Z}_l^{\dagger} & \xrightarrow{t_2\circ v'\circ u_1} & B \times Y_{k+l} \times \hat{X}_k(f) & \xrightarrow{\psi} & \bar{B} \times Y_{k+l} & \xrightarrow{\rho} & \bar{B} \\
\downarrow{\scriptstyle q\circ u} & & \downarrow{\scriptstyle \text{2nd proj}} & & \downarrow{\scriptstyle \tau} & & \\
Y_{k+l} & = & Y_{k+l} & = & Y_{k+l}, & &
\end{array}$$

where ρ is the first projection, τ is the second projection and ψ is the morphism

$$\psi : B \times Y_{k+l} \times \hat{X}_k(f) \ni (a, v, w) \mapsto \big(a + \gamma_k(w), v\big) \in \bar{B} \times Y_{k+l}.$$

Since $(\rho \circ \psi \circ t_2 \circ v' \circ u_1)^* L = \mathscr{L}_l^{\dagger}$, we have a natural morphism

$$\tau_*\rho^* L^{\otimes n} = H^0\big(\bar{B}, L^{\otimes n}\big) \otimes_{\mathbf{C}} \mathscr{O}_{Y_{k+l}} \to (q \circ u)_*\mathscr{L}_l^{\dagger\otimes n}. \tag{6.5.47}$$

Here, note that $\rho \circ \psi = \phi \circ \beta$, where $\beta : B \times Y_{k+l} \times \hat{X}_k(f) \to B \times \hat{X}_k(f)$ is the morphism in the diagram (6.5.46) and ϕ was defined by (6.5.44).

Now put $I_n = \dim_{\mathbf{C}} H^0(\bar{B}, L^{\otimes n})$. Then there is a positive integer n_0 and positive constants C_1, C_2 such that

$$I_n > C_1 n^{\dim \bar{B}}, \quad G_n < C_2 n^{\dim \bar{B} - 2} \quad \text{for } n > n_0.$$

Here note that $G_n = \dim_{\mathbf{C}} H^0(B \times Z^{\dagger}|_P, L_{1,P}^{\dagger\otimes n})$ for $P \in \cap_{n \geqq 1} U_n$, and $B \times Z^{\dagger}|_P = s^{-1}(P)$ has dimension $\leqq \dim \bar{B} - 2$, for $\operatorname{codim}_{\hat{X}_k(f)} \bar{Z} \geqq 2$ and $\hat{\pi}_k \circ r_2 : \bar{Z} \to Y_k$ is dominant. Hence for a positive integer l, we can take a positive integer $n(l)$ (e.g. $\sim l^{3/4}$) such that

$$I_{n(l)} > (l+1)G_{n(l)}, \quad \lim_{l\to\infty} \frac{n(l)}{l} = 0.$$

Let $\mathscr{F}$ be the kernel of (6.5.47) for $n = n(l)$:

$$0 \to \mathscr{F} \to \tau_*\rho^* L^{\otimes n(l)} \to (q \circ u)_*\mathscr{L}_l^{\dagger\otimes n(l)} \text{ (exact)}.$$

Then we have $\mathscr{F} \neq 0$. By taking the tensor product of a sufficiently ample line bundle M_l on Y_{k+l} with $\mathscr{F}$, we may assume that $H^0(Y_{k+l}, \mathscr{F} \otimes M_l) \neq 0$. Since we have

$$\begin{aligned} H^0(Y_{k+l}, \mathscr{F} \otimes M_l) &\subset H^0\big(Y_{k+l}, \big(\tau_* \rho^* L^{\otimes n(l)}\big) \otimes M_l\big) \\ &= H^0\big(Y_{k+l}, \tau_*\big(\rho^* L^{\otimes n(l)} \otimes \tau^* M_l\big)\big) \\ &= H^0\big(\bar{B} \times Y_{k+l}, \rho^* L^{\otimes n(l)} \otimes \tau^* M_l\big), \end{aligned}$$

we may take a divisor $F_l \subset \bar{B} \times Y_{k+l}$ which is defined by a non-zero global section of $H^0(Y_{k+l}, \mathscr{F} \otimes M_l)$. Then we have

$$\mathscr{Z}_l^{\mathrm{ns}} \subset \psi^* F_l.$$

Here note that $\mathscr{Z}_l^{\mathrm{ns}} \subset \mathscr{Z}_l$ is an open immersion and $\mathscr{Z}_l \overset{t_2 \circ v'}{\hookrightarrow} B \times Y_{k+l} \times \hat{X}_k(f)$ is a closed subscheme.

Using the decomposition $A = B \times C$, we let $f_B : \mathbf{C} \to B$ be the entire curve obtained by the composition of f and the first projection $A \to B$, and let $f_C : \mathbf{C} \to C$ be the entire curve obtained by the composition of f and the second projection $A \to C$.

Now let $a \in \mathbf{C}$ be a point such that $J_k(f)(a) \in Z^{\mathrm{ns}}$. Put $\tilde{f} : \mathbf{C} \to B \times Y_{k+l} \times \hat{X}_k(f)$ as

$$\tilde{f}(z) = \big(f_B(z) - f_B(a), \hat{\pi}_{k+l} \circ J_{k+l}(f)(z), J_k(f)(a)\big).$$

Then we have

$$\tilde{f}(\mathbf{C}) \subset B \times Y_{k+l} \times Z, \qquad \tilde{f}(a) \in \operatorname{Supp} \mathscr{Z}_l^{\mathrm{ns}}, \qquad \psi \circ \tilde{f} = J_{k+l}(f),$$

where the last equality holds under the identification $\bar{B} \times Y_{k+l} = \hat{X}_{k+l}(f)$.

Since v' is the base change of v in (6.5.46) and $\tilde{f}$ factors through t_2, we have

$$\operatorname{ord}_a \tilde{f}^* \mathscr{Z}_l = \operatorname{ord}_a (t_3 \circ \tilde{f})^* \mathscr{S}_l;$$

hence by the construction of $\mathscr{S}_l$ and Proposition 6.5.40, we have

$$\operatorname{ord}_a \tilde{f}^* \mathscr{Z}_l = \operatorname{ord}_a \big(J_{k+l}(f) - f(a)\big)^* \mathscr{T}^{\dagger}_{l,(J_k(f)-f(a))(a)} \geqq l + 1.$$

Therefore,

$$\operatorname{ord}_a J_{k+l}(f)^* F_l = \operatorname{ord}_a \tilde{f}^* \psi^* F_l \geqq \operatorname{ord}_a \tilde{f}^* \mathscr{Z}_l^{\mathrm{ns}} = \operatorname{ord}_a \tilde{f}^* \mathscr{Z}_l \geqq l + 1.$$

Here, note that we consider F_l as the divisor on $\hat{X}_{k+l}(f)$ by the identification $B \times Y_{k+l} \cong \hat{X}_{k+l}(f)$, and τ correspond to π_{k+l} by this identification. □

(d) **The end of the proof**. By (6.5.7) and (6.5.8) it suffices to show

$$N_{l_0}\big(r, J_k(f)^* Z\big) \leqq \varepsilon T_f(r) \|_{\varepsilon}, \quad {}^{\forall}\varepsilon > 0.$$

Note that

$$N_{l_0}\big(r, J_k(f)^*Z\big) \leqq l_0 N_1\big(r, J_k(f)^*Z\big).$$

Furthermore, it suffices to prove

$$N_1\big(r, J_k(f)^*Z^{\mathrm{ns}}\big) \leqq \varepsilon T_f(r)\|_\varepsilon, \quad {}^\forall\varepsilon > 0. \tag{6.5.48}$$

For we have

$$N_1\big(r, J_k(f)^*Z\big) = N_1\big(r, J_k(f)^*Z^{\mathrm{ns}}\big) + N_1\big(r, J_k(f)^*\big(Z \setminus Z^{\mathrm{ns}}\big)\big)$$

and the second term of the right-hand side is estimated to be at most "$\varepsilon T_f(r)\|_\varepsilon$" by induction on dimension of Z. Here note that $\dim Z > \dim(Z \setminus Z^{\mathrm{ns}})$.

It follows from Lemma 6.5.42 and (6.5.31) that

$$\begin{aligned}(l+1)N_1\big(r, J_k(f)^*Z^{\mathrm{ns}}\big) &\leqq N\big(r, J_{k+l}(f)^*F_l\big) \leqq T_{J_{k+l}(f)}\big(r, L(F_l)\big)\\ &= n(l)T_{\gamma_{k+l}\circ J_{k+l}(f)}(r, L) + T_{\pi_{k+l}\circ J_{k+l}(f)}(r, M_l)\\ &\leqq n(l)T_{f_B}(r, L) + S_f(r).\end{aligned} \tag{6.5.49}$$

Using $\lim_{l\to\infty} n(l)/(l+1) = 0$ and $T_{f_B}(r, L) = O(T_f(r, L(\bar{D})))$, we obtain (6.5.48) and our Theorem 6.5.17.

6.5.4 *Proof of Theorem 6.5.1*

(a) Let the notation be as in Theorem 6.5.1. The case of $\operatorname{codim}_{X_k(f)} Z \geqq 2$ was finished by Theorem 6.5.17. Therefore we assume in the rest of this section that Z is a reduced Weil divisor D on $X_k(f)$.

Set $B = \mathrm{St}(X_{k+1}(f))^0$, which has a positive dimension (cf. (6.5.9)).

Lemma 6.5.50 *Assume that D is irreducible and $B \not\subset \mathrm{St}(D)^0$. Taking an embedding $X_{k+1}(f) \hookrightarrow J_1(X_k(f))$, we have*

$$\operatorname{codim}_{X_{k+1}(f)}\big(X_{k+1}(f) \cap J_1(D)\big) \geqq 2.$$

Proof Let $k = 0$. It is first noted that $J_1(A)$ is the holomorphic tangent bundle $\mathbf{T}(A)$ over A, and $X_1(f) \subset \mathbf{T}(A)$.

Assume that $\operatorname{codim}_{X_1(f)}(X_1(f) \cap J_1(D)) = 1$. Let E be an irreducible component of codimension 1 of $X_1(f) \cap J_1(D)$. Let $\pi_1 : X_1(f) \to A$ be the natural projection. Then E is an irreducible component of $X_1(f) \cap \pi_1^{-1}(D)$ and $\overline{\pi_1(E)} = D$.

Now $\overline{\pi_1(E)} = D$ combined with $B \not\subset \mathrm{St}(D)^0$ implies that B cannot stabilize E. Therefore $B \cdot E$ (resp. $B \cdot D$) contains an open subset of $X_1(f)$ (resp. A). In fact, since B and E are algebraic, $B \cdot E$ contains a B-invariant Zariski open subset Ω of $X_1(f)$.

Let $p = f(z_0) \in f(\mathbf{C})$ be a point with the properties:

(i) The orbit $B \cdot p$ intersects $D \setminus \mathrm{Sing}(D)$ transversely in a point q;
(ii) $J_1(f)(z_0) \in \Omega$.

Then we choose an analytic one-dimensional disk $\Delta \subset B$ which contains the unit element e_B of B, and we choose a non-empty open subset U of the non-singular part D^{ns} of D containing q such that

(i) the map $\phi : \Delta \times U \hookrightarrow A$ given by $\phi(b, u) = b \cdot u$ is an open embedding,
(ii) the subbundle $\cup_{\zeta \in \Delta} \mathbf{T}(\{\zeta\} \times U) \subset \mathbf{T}(\Delta \times U)$ with $\mathbf{T}(\{\zeta\} \times U) \cong \mathbf{T}(U)$ gives rise to a holomorphic foliation on $\phi(\Delta \times U) \subset A$.

Consider $\hat{f}(z) = b \cdot f(z + z_0)$ with $b \in B$ such that $b \cdot p = q$ and $p = f(z_0)$. Note that $\hat{f}(0) = b \cdot p = q$ and $J_1(\hat{f})(0) = b \cdot J_1(f)(z_0) \in \Omega$. Then there is an open neighborhood W of 0 in $\mathbf{C}$ such that $J_1(\hat{f})(z) \in \Omega$ for all $z \in W$. Since $\Omega \subset B \cdot E \subset B \cdot J_1(D)$, it follows that $\hat{f}'(z)$ is tangent to the leaves of the above defined foliation for all $z \in W$. By the identity principle this implies $\hat{f}(\mathbf{C}) = b \cdot f(\mathbf{C}) \subset D$, which is absurd, since f is algebraically non-degenerate.

The proof for $k \geqq 1$ is similar to the above. □

(b) *Proof of Theorem 6.5.1.* Let $D = \sum_i D_i$ be the irreducible decomposition. By making use of Theorem 6.5.6 we have

$$
\begin{aligned}
(6.5.51) \qquad T(r, \omega_{\bar{D}, J_k(f)}) &\leqq N_{k_0}\big(r, J_k(f)^* D\big) + S_f(r) \\
&\leqq N_1\big(r, J_k(f)^* D\big) + k_0 \sum_{i<j} N_1\big(r, J_k(f)^*(D_i \cap D_j)\big) \\
&\quad + k_0 \sum_i N_1\big(r, J_{k+1}(f)^* J_1(D_i)\big) + S_f(r).
\end{aligned}
$$

Since $\mathrm{codim}_{X_k(f)} D_i \cap D_j \geqq 2$ for $i \neq j$, it follows from Theorem 6.5.17 that

$$
k_0 \sum_{i<j} N_1\big(r, J_k(f)^*(D_i \cap D_j)\big) \leqq \varepsilon T_f(r) \|_\varepsilon, \quad {}^\forall \varepsilon > 0.
$$

Note that $J_{k+1}(f)^* J_1(D_i) = J_{k+1}(f)^*(X_{k+1}(f) \cap J_1(D_i))$. If $B \subset \mathrm{St}(D_i)^0$, then the image of D_i by $X_k(f) \to X_k(f)/B$ is contained in a divisor on $X_k(f)/B$. Then as in (6.5.11) we infer that

$$
N_1\big(r, J_{k+1}(f)^* J_1(D_i)\big) \leqq N\big(r, J_k(f)^* D_i\big) \leqq S_f(r).
$$

Suppose that $B \not\subset \mathrm{St}(D_i)^0$. It follows from Lemma 6.5.50 and Theorem 6.5.17 that

$$
\begin{aligned}
N_1\big(r, J_{k+1}(f)^* J_1(D)\big) &\leqq N_1\big(r, J_{k+1}(f)^*\big(X_{k+1}(f) \cap J_1(D_i)\big)\big) \\
&\leqq \varepsilon T_f(r) \|_\varepsilon, \quad {}^\forall \varepsilon > 0.
\end{aligned}
$$

Combining these with (6.5.51), we obtain

$$T(r,\omega_{\bar{D},J_k(f)}) \leqq N_1\big(r,J_k(f)^*D\big)+\varepsilon CT_f(r)\|_{\varepsilon},\quad {}^{\forall}\varepsilon>0,$$

where C is a positive constant independent of ε. Now the proof of Theorem 6.5.1 is completed. □

Comments Theorem 6.5.1 for abelian varieties with $k=0$ was proved by Yamanoi [04b]. Subsequently, it was proved for semi-abelian varieties and general k by Noguchi–Winkelmann–Yamanoi [08].

To deal with semi-abelian varieties the main difficulties are caused by the following two facts:

(i) Semi-abelian varieties are not compact and need some good compactifications.
(ii) There is no Poincaré reducibility theorem for semi-abelian varieties.

To overcome these difficulties we needed some new methods.

6.6 Applications

Here we give several applications of the Second Main Theorem 6.5.1, some of which resolves conjectures due to S. Lang and M. Green in 1960s and 70s.

6.6.1 Algebraic Degeneracy of Entire Curves

Let X be an algebraic variety X which may be non-compact in general. If X is compact, we denote by $\kappa(X)$ (resp. $q(X)$) the Kodaira dimension (resp. irregularity) of X; in general, we write $\bar{\kappa}(X)$ (resp. $\bar{q}(X)$) for the log Kodaira dimension (resp. log irregularity) of X. If $\bar{\kappa}(X)=\dim X$, X is said to be of log general type. We call X an *irregular variety* if $\bar{q}(X)>0$.

If $\bar{q}(X)>\dim X$, then every entire curve $f:\mathbf{C}\to X$ is algebraically degenerate by the Logarithmic Bloch–Ochiai Theorem 4.8.17. Here we discuss the case $\bar{q}(X)=\dim X$. If the quasi-Albanese map α from X to its quasi-Albanese variety is not dominant, then again the Logarithmic Bloch–Ochiai Theorem implies that every entire curve in X is algebraically degenerate. Thus the interesting case is the case where the quasi-Albanese map is dominant and generically finite.

If α is proper and $\bar{\kappa}(X)=0$ in addition to our assumption $\bar{q}(X)=\dim X$, then Kawamata's Theorem 5.6.23 implies that X is birational to a semi-abelian variety and therefore admits entire curves which are algebraically non-degenerate. This leaves the following case.

Problem If $\bar{q}(X)=\dim X$ and $\bar{\kappa}(X)>0$, is every entire curve $f:\mathbf{C}\to X$ algebraically degenerate?

Here, we give a complete positive answer in the case where α is proper.

Theorem 6.6.1 (Noguchi–Winkelmann–Yamanoi [07]) *Let X be a complex algebraic variety and let $\pi : X \to A$ be a finite morphism onto a semi-abelian variety A. Let $f : \mathbf{C} \to X$ be an arbitrary entire curve. If $\bar{\kappa}(X) > 0$, then f is algebraically degenerate.*

Moreover, the normalization of the Zariski closure of $f(\mathbf{C})$ is a semi-abelian variety which is a finite étale cover of a translate of a proper semi-abelian subvariety of A.

Corollary 6.6.2 *Let X be a complex algebraic variety whose quasi-Albanese map is a proper map. Assume that $\bar{\kappa}(X) > 0$ and $\bar{q}(X) \geqq \dim X$. Then every entire curve $f : \mathbf{C} \to X$ is algebraically degenerate.*

The proof of Theorem 6.6.1 will be carried out as follows:

(i) By Kawamata's Theorem 5.6.22 it is readily reduced to the case of log general type.
(ii) For step (iii) below we need rather delicate constructions of a compactification of X and its non-singular model which specifies the log canonical divisor (cf. Theorem 6.6.6).
(iii) Assuming the algebraic non-degeneracy of f, we then apply Theorems 6.5.1 and 6.5.17 to conclude a contradiction.

The Compactification

Let X be a normal algebraic variety, let A be a semi-abelian variety and let $\pi : X \to A$ be a finite morphism. We need a good compactification.

Before constructing such a compactification we remark that given a finite morphism from a *normal* variety X to a smooth variety A we may desingularize X in order to get a generically finite morphism between smooth varieties. Conversely, given a generically finite morphism $p : \tilde{X} \to A$ between smooth varieties, the Stein factorization gives us a normal variety X together with a finite morphism from X to A and a proper connected morphism from $\tilde{X}$ to X which is a desingularization of X.

Lemma 6.6.3 *Let $\pi : X \to A$ be a finite morphism from a normal variety X with singular locus S onto a smooth variety A. Let R denote the ramification divisor of the restriction of π to $X \setminus S$. Then S is contained in the closure of R.*

Proof Let $p \in X \setminus \bar{R}$. Then there are small connected open Stein neighborhoods V of p in $X \setminus \bar{R}$ and W of $\pi(p)$ in A such that π restricts to a finite morphism from V to W. We may assume that W is simply-connected. Because X is normal, the codimension of $\pi(S) \cap W$ is at least two. Hence $W \setminus \pi(S)$ is simply-connected. It follows that there is a section $\sigma : W \setminus \pi(S) \to V$ which (again because of $\operatorname{codim} \pi(S) \geqq 2$) extends to all of W. This yields a biholomorphic map between V and W. Since $W \subset A$ is smooth, we deduce $p \notin S$. □

(a) Simple Compactification

We begin with a simple compactification of X.

Proposition 6.6.4 *Let $\pi : X \to A$ be a finite surjective morphism between normal varieties and let $A \hookrightarrow \bar{A}$ be a normal compactification of A. Then there exists a unique normal compactification $X \hookrightarrow \bar{X}$ such that $\bar{X}$ is normal and π extends to a finite morphism $\bar{\pi} : \bar{X} \to \bar{A}$.*

Proof Let $\Gamma \subset X \times A$ be the graph of π. Choose a compactification $X \hookrightarrow \tilde{X}$ and let $\hat{\Gamma}$ be the closure of Γ in $\bar{A} \times \tilde{X}$. Then $\bar{X}$ is obtained by first normalizing $\hat{\Gamma}$ and then taking the Stein factorization of the projection onto $\bar{A}$. It is easy to deduce the uniqueness from the assumption of X being normal. □

(b) Better Compactification

We then improve the compactification.

Proposition 6.6.5 *Let π be a finite morphism from a normal algebraic variety X to a semi-abelian variety A. Let R denote the set of all non-singular points $p \in X$ at which π is ramified. Let $A \hookrightarrow \bar{A}$ be a smooth equivariant compactification and let ω be an invariant log canonical form on $\bar{A}$.*

Then there exists a compactification $X \hookrightarrow \bar{X}$ and a proper morphism $\bar{\pi} : \bar{X} \to \bar{A}$ such that

(i) *$\bar{X} \setminus \bar{R}$ is smooth and $\bar{X} \setminus (X \cup \bar{R})$ is an s.n.c. divisor on $\bar{X} \setminus \bar{R}$ (where $\bar{R}$ denotes the closure of R in $\bar{X}$),*
(ii) *$\bar{\pi}|_X = \pi$,*
(iii) *Let $\tilde{\omega} = \bar{\pi}^* \omega \in \Omega^d(\bar{X}; \log \partial X)$ where $d = \dim X$. Then $\tilde{\omega}$ has poles along all divisorial components of $\bar{X} \setminus (X \cup \bar{R})$.*

Proof Due to Lemma 6.6.3 the singular locus S of X is contained in the closure of R. We let D be a divisor on A containing $\pi(R)$. Then D contains $\pi(S)$, too.

(0) To prove the assertion we use the following strategy:

- We define a class of "admissible compactifications" of X.
- We show, using Lemma 6.6.4, that there exists an admissible compactification.
- For each admissible compactification we define an "indicator function" $\zeta : \partial A \to \mathbf{N}$ which measures the presence of singularities outside of $\bar{R}$.
- Using the theory of toroidal embeddings, we show that we can blow up admissible compactification in such a way that we stay inside the category of admissible compactification, but decrease the indicator function.
- We verify that after finitely many steps the indicator function vanishes and that then we have found a compactification as desired.

(1) A compactification $X \hookrightarrow X'$ is "*admissible*" if the following conditions are satisfied:

(i) X' is normal.
(ii) The map $\pi : X \to A$ extends to a proper holomorphic map $\pi' : X' \to \bar{A}$ with $(\pi')^{-1}(A) = X$.
(iii) For each point $p \in X' \setminus \bar{R}$ there is an open neighborhood U of $\pi(p) = q$ in $\bar{A}$ such that the connected component Ω of $(\pi')^{-1}(U) \setminus \bar{R}$ which contains p admits a biholomorphic map $\psi : \Omega \to W$ into an open subset W of a toric variety Z.
(iv) Let G denote the algebraic torus $(\mathbf{C}^*)^g$ acting on the toric variety Z. Then map $\pi \circ \psi^{-1} : W \to \bar{A}$ is (locally) equivariant for some holomorphic Lie group homomorphism with discrete fibers from G to A.

Condition (iii) could be rephrased by saying that $X \hookrightarrow X'$ should be locally a "toroidal embedding" in the sense of Kempf–Knudsen–Mumford–Saint-Donat [73] except at $\bar{R}$.

(2) Using Lemma 6.6.4, we obtain a normal compactification $X \hookrightarrow X_1$ such that $\pi : X \to A$ extends to a finite morphism $\pi_0 : X_1 \to \bar{A}$. Let $p \in X' \setminus \bar{R}$, $q = (\pi')(p)$. Let U be an open neighborhood of p in $X' \setminus \bar{R}$ and define $\Omega = (\pi')(U)$. Since ∂A is an s.n.c. divisor, we can shrink these open neighborhoods to obtain local coordinates z_i on Ω such that $z_i(q) = 0$ and such that ∂A is the zero locus of $z_1 \cdots z_k$ for some k. By shrinking Ω we may assume that Ω is biholomorphic to a polydisc with $\Omega \cap A \cong (\Delta^*)^k \times \Delta^{n-k}$. Let $i : \Delta^* \to \mathbf{C}^*$ be the standard injection and $j : \Delta \to \mathbf{C}^*$ any open embedding (e.g. $j(z) = z + 2$). We obtain $\xi = (i^k, j^{n-k}) : \Omega \cap A \hookrightarrow (\mathbf{C}^*)^n = G$. Now $\pi_1(G) \cong \mathbf{Z}^n$ and ξ induces an embedding of $\pi_1(\Omega \cap A) \cong \mathbf{Z}^k$ into $\mathbf{Z}^n \cong \pi_1(G) \cong \mathbf{Z}^n$ as $\xi_*(\pi_1(\Omega \cap A)) = \mathbf{Z}^k \times \{0\}^{n-k}$. Thus for each subgroup Γ of fixed finite index d of $\pi_1(\Omega \cap A)$ we can choose a corresponding subgroup of the same finite index in $\pi_1(G) \cong \mathbf{Z}^n$, namely $\xi_*(\Gamma) \times \mathbf{Z}^{n-k}$. Therefore the unramified covering $\pi^{-1}(\Omega \cap A) \to \Omega \cap A$ extends via ξ to an unramified covering $G_1 \to G$ of G. Now G_1 is again an algebraic group and as algebraic group isomorphic to $(\mathbf{C}^*)^n$. Since the procedure of normal compactification as in Proposition 6.6.4 is canonical, we can embed $\pi^{-1}(\Omega) \to \Omega$ into a finite morphism $\bar{G}_1 \to \bar{G} = (\mathbf{P}^1)^n$. Now $\bar{G}_1$ is a toric variety since the G_1 action on itself extends to the boundary $\partial G_1 \cong \partial G$ via $G_1 \to G$. This proves that $X \hookrightarrow X_1$ is admissible.

(3) We are looking for admissible compactifications which are smooth outside the closure of R. Thus given an admissible compactification $X \hookrightarrow \bar{X}$ with $\pi : \bar{X} \to \bar{A}$, we define our indicator function $\zeta : \partial A \to \mathbf{N}$ as follows: $\zeta(p)$ denotes the number of connected components of the fiber $\pi^{-1}(p)$ which intersect $\mathrm{Sing}(\bar{X}) \setminus \bar{R}$. Evidently this function ζ vanishes iff $\bar{X} \setminus \bar{R}$ is smooth.

(4) We suppose to be given an admissible compactification $\bar{X}$ with indicator function ζ. Since the level sets $\{z \in \partial A; \zeta(z) = c\}$ $(c \in \mathbf{N})$ of ζ are constructible sets, it makes sense to define $d_\zeta = \dim\{z; \zeta(z) \neq 0\}$ and to define a number n_ζ which is the generic value of ζ on $\{z; \zeta(z) \neq 0\}$. We choose a generic point $p \in \{z \in \partial A; \zeta(z) \neq 0\}$ and a point $q \in \pi^{-1}(p) \cap \mathrm{Sing}(\bar{X}) \setminus \bar{R}$. Since the compactification is admissible, there is an isomorphism $\phi : U \to W$ where U is a neighbor-

hood of the connected component of $\pi^{-1}(p)$ containing q and W is an open neighborhood of p in a toric variety Z. By the theory of toroidal embeddings (see e.g. Kempf–Knudsen–Mumford–Saint-Donat [73]) there exists an equivariant desingularization $\tilde{Z} \to Z$, and thus a desingularization of U.

The problem is to extend this blow-up of U to a blow-up of $\bar{X}$. Because the blow-up of Z is equivariant and the number of G-orbits in Z is finite, it is given by a blow-up of invariant strata (=closures of G-orbits in Z). Now $\bar{X}$ admits a natural stratification induced by the A-action on $\bar{A}$. In fact, there is a local A-action on $\bar{X} \setminus \bar{R}$ which gives this stratification. In order to extend the blow-up, we need to extend its center, and this means: Given a closed invariant subvariety $Q \subset Z$ we need to show that either $M \cap U = \phi^{-1}(Q)$ or $M \cap U = \emptyset$ for each stratum M of $\bar{X}$.[4] We prove this indirectly. So let us assume that this property fails. Then $\phi(M \cap U)$ intersects several G-orbits some of which are contained in Q and some of which are not. Since M is one stratum, $\pi(M)$ is one A-orbit in $\bar{A}$ which implies that for each point in M we have the same isotropy Lie algebra. If we have chosen U sufficiently small, we can conclude from Proposition 5.3.7 that $\pi(M \cap U)$ is connected. Let $N = \pi^{-1}(\pi(M))$. Using Lemma 5.3.11 we may deduce that $M \cap U$ is connected, as desired.

(5) We recall that we introduced an indicator function ζ with associated numbers d_ζ and n_ζ. For a generic point $p \in \mathrm{Sing}(\bar{X}) \setminus \bar{R}$ the above considerations show that there is an appropriate blow-up yielding an other admissible compactification which is smooth at (the preimage of) p. By the definition of d_ζ and n_ζ this means that given an admissible compactification we can always blow up $\bar{X}$ so that either d_ζ decreases or n_ζ decreases while d_ζ is kept fixed. Thus we can strictly decrease the value of $(d_\zeta, n_\zeta) \in \mathbf{N}^2$ where $\mathbf{N}^2$ is endowed with the lexicographic order. It follows that $(d_\zeta, n_\zeta) = (0, 0)$ after finitely many steps. But $(d_\zeta, n_\zeta) = (0, 0)$ implies the vanishing of $\zeta : \partial A \to \mathbf{N}$ which in turn implies that $\bar{X}$ is smooth outside $\bar{R}$.

Thus we have established: There exists an admissible compactification $X \hookrightarrow X'$ such that X' is smooth outside the closure of R.

(6) By the definition of admissibility $X' \setminus \bar{R}$ is locally a toroidal embedding. This implies that $\partial X' \setminus \bar{R}$ is an s.n.c. divisor in $X' \setminus \bar{R}$. Furthermore, the logarithmic tangent bundle on a smooth toric variety is trivial (Proposition 5.4.3). Hence conditions (iii) and (iv) of "admissibility" imply the statement about the poles of $\tilde{\omega}$. □

(c) The Best Compactification

Here we obtain the best compactification to be used.

Theorem 6.6.6 *Let $\pi : X \to A$ be a finite morphism from a normal variety X onto a semi-abelian variety A. Let $\bar{A}$ be a smooth equivariant compactification of A. Let*

[4] In the language of Kempf–Knudsen–Mumford–Saint-Donat [73], here we prove that the embedding of $X \setminus R$ into $\bar{X} \setminus \bar{R}$ is a toroidal embedding "*without self-intersection*".

D denote the critical locus of π, i.e. the closure of the set of all $\pi(z)$ where z is a smooth point and $d\pi : \mathbf{T}(X)_z \to \mathbf{T}(A)_{\pi(z)}$ fails to have the full rank. Then there exist

(a) *a desingularization $\tau : \tilde{X} \to X$ and a smooth compactification $j : \tilde{X} \hookrightarrow \hat{X}$ such that the boundary $\partial\hat{X} = \hat{X} \setminus j(\tilde{X})$ is an s.n.c. divisor,*
(b) *a proper holomorphic map $\psi : \hat{X} \to \bar{A}$ such that $\psi \circ j = \pi \circ \tau$ with $\psi^{-1}(A) = \tilde{X}$,*
(c) *an effective divisor Θ on $\hat{X}$,*
(d) *a subvariety $\hat{S} \subset \hat{X}$,*

such that

(i) *Θ is linearly equivalent to the log canonical divisor $K_{\hat{X}} + \partial\hat{X}$ of $\tilde{X}$,*
(ii) *$\psi(\mathrm{Supp}\,\Theta) \subset \bar{D}$,*
(iii) *the image $\psi(\hat{S})$ has at least codimension two in A,*
(iv) *for every holomorphic curve $f : \Delta \to A$ from a disk in $\mathbf{C}$ with lifting $F : \Delta \to \tilde{X}$ and for $z \in F^{-1}(\mathrm{Supp}\,\Theta \setminus \hat{S})$ we have the following inequality of multiplicities*:

$$\mathrm{mult}_z\, F^*\Theta \leqq \mathrm{mult}_z\, f^*D - 1. \tag{6.6.7}$$

Proof Let $\mathrm{Sing}(X)$ be the singular locus of X and let R be the ramification divisor of π restricted to $X \setminus \mathrm{Sing}(X)$. We apply Proposition 6.6.5 and obtain a first compactification X' of X and an extension $\pi' : X' \to \bar{A}$ of $\pi : X \to A$. We recall that $X' \setminus \bar{R}$ is smooth (assertion (i) of Proposition 6.6.5). Thus we can use Hironaka desingularization (Hironaka [64a], [64b], see also Bierstone–Milman [97]) to desingularize X' without changing $X' \setminus \bar{R}$. We obtain a desingularization $\tau : \hat{X} \to X'$ which restricts to a desingularization $\tau_0 : \tilde{X} \to X$.

Set

$$\Theta = \left(\psi^*\omega\right) + \psi^{-1}\partial A.$$

Then the properties (i) and (ii) are satisfied.

Let S_1 denote the image of the set of singular points of X by π, let S_2 be the set of singularities of D, and let S_3 be the image of the set of singularities of $\mathrm{Supp}\,\Theta|_{\tilde{X}}$ by ψ. Then $\mathrm{codim}\, S_j \geqq 2$ $(1 \leq j \leq 3)$. Set $S = S_1 \cup S_2 \cup S_3$ and $\hat{S} = \psi^{-1}(\bar{S})$. Then $\mathrm{codim}\, \hat{S} \geqq 2$ and (iii) is satisfied.

Let $p \in \mathrm{Supp}\,\Theta|_{\tilde{X}} \setminus \hat{S}$. Then $q = \pi(p) \in D \setminus S$. Because of the construction there exist local coordinate systems $(x_1, \ldots, x_n)$ about p and $(y_1, \ldots, y_n)$ about q such that locally

$$y_1 = x_1^k \quad (k \in \mathbf{N}, k \geqq 2), \qquad y_j = x_j \quad (2 \leqq j \leqq n),$$
$$\Theta = (k-1)\{x_1 = 0\}, \qquad D = \{y_1 = 0\}.$$

Set $F = (F_1, \ldots, F_n)$ and $f = (f_1, \ldots, f_n)$. Let ν denote the multiplicity of zero of F_1 at z. Note that $\mathrm{mult}_z\, F^*\Theta = (k-1)\nu$ and $\mathrm{mult}_z\, f^*D = k\nu$. Thus $(k-1)\nu \leqq k\nu - 1$ and so (6.6.7) holds. □

Remark 6.6.8 Let the notation be as in Theorem 6.6.6. Assume that X is of log general type. We claim

$$\mathrm{St}(D)^0 = \{0\}. \tag{6.6.9}$$

Suppose that it is not the case. Then $A \setminus D$ is also $\mathrm{St}(D)^0$-invariant. Then, setting $U = X \setminus \pi^{-1}D$, we have a sequence of surjective morphisms

$$U \xrightarrow{\pi|_U} A \setminus D \longrightarrow \left(A/\mathrm{St}(D)^0\right) \setminus \left(D/\mathrm{St}(D)^0\right)$$

induced from the restriction of π and the quotient map. Let

$$\mu : U \longrightarrow \left(A/\mathrm{St}(D)^0\right) \setminus \left(D/\mathrm{St}(D)^0\right)$$

be the composed morphism. Then every connected component Z of the fibers of μ is a finite étale cover over $\mathrm{St}(D)^0$. Since $\bar{\kappa}(Z) = 0$, $\bar{\kappa}(U) \leqq \dim U - 1$ (Iitaka [77], Theorem 4). Then we have a contradiction, $\bar{\kappa}(X) \leqq \dim X - 1$. From (6.6.9) it follows that $\bar{D}$ is big on $\bar{A}$ (Theorem 5.5.26).

(d) Proof of Theorem 6.6.1

The following is an essential case.

Theorem 6.6.10 *Let X be a normal variety of log general type and $\pi : X \to A$ be a finite morphism onto a semi-abelian variety. Then every entire curve $g : \mathbf{C} \to X$ is algebraically degenerate.*

Proof We use the desingularization and compactification obtained by Theorem 6.6.6 and follow the notation there.

Assume that there exists an algebraically non-degenerate entire curve $g : \mathbf{C} \to X$. Since g is algebraically non-degenerate and $\tilde{X} \to X$ is birational, we can lift g to an algebraically non-degenerate entire curve $F : \mathbf{C} \to \tilde{X}$. Set $f = \pi \circ g$. We are going to deduce a contradictory estimate for the order function $T_F(r)$ of F.

Let $\bar{A}$ be an equivariant compactification of A such that $\bar{D}$ is in good position; that is, $\bar{D}$ contains no A-orbit in $\bar{A}$. Since $\bar{D}$ is in good position in $\bar{A}$, it follows from Theorems 2.4.9, 6.5.1 and 6.6.6 (ii), (iii) that

$$m_F(r, \Theta) \leqq m_f(r, \bar{D}) = S_f(r). \tag{6.6.11}$$

Since Θ is big, one infers from (2.5.20) that

$$T_f(r) \sim T_F(r). \tag{6.6.12}$$

Combining this with (6.6.11), one gets

$$m_F(r, \Theta) = S_F(r). \tag{6.6.13}$$

Theorem 6.6.6 (iv) implies

$$N\big(r, F^*\Theta\big) \leqq N\big(r, F^*\hat{S}\big) + N\big(r, f^*D\big) - N_1\big(r, f^*D\big). \tag{6.6.14}$$

Now $\psi(\hat{S})$ is of codimension at least two in A. Therefore we can infer from Theorem 6.5.17 that

$$N\big(r, F^*\hat{S}\big) \leqq N\big(r, f^*(\psi_*\hat{S})\big) \leqq \varepsilon T_f(r)\|_\varepsilon. \tag{6.6.15}$$

By virtue of (6.5.4) we have

$$\begin{aligned} N\big(r, f^*D\big) - N_1\big(r, f^*D\big) &\leqq T_f\big(r, L(\bar{D})\big) - N_1\big(r, f^*D\big) \\ &\leqq \varepsilon T_f(r)\|_\varepsilon, \quad {}^\forall\varepsilon > 0. \end{aligned}$$

The combination of this with (6.6.12) yields

$$N\big(r, f^*D\big) - N_1\big(r, f^*D\big) \leqq \varepsilon T_F(r)\|_\varepsilon, \quad {}^\forall\varepsilon > 0. \tag{6.6.16}$$

Now one infers from (6.6.14)–(6.6.16) that

$$N\big(r, F^*\Theta\big) \leqq \varepsilon T_F(r)\|_\varepsilon, \quad {}^\forall\varepsilon > 0. \tag{6.6.17}$$

Note that $T_F(r) \sim T_F(r, L(\Theta)) \to \infty$ as $r \to \infty$. Hence, the First Main Theorem 2.4.9, (6.6.13) and (6.6.17) lead to a contradiction:

$$T_F(r) \leqq \varepsilon T_F(r)\|_\varepsilon, \quad {}^\forall\varepsilon > 0. \qquad \square$$

Proof of Theorem 6.6.1 Let the notation be as in Theorem 6.6.1. Assume that $\bar{\kappa}(X) > 0$ and $f : \mathbf{C} \to X$ is algebraically non-degenerate. By lifting f to the normalization of X we may assume further that X is normal.

We use Kawamata's Theorem 5.6.22 and the notation there. Since $\tilde{X} \to X$ is étale, we can lift f to an entire curve $\tilde{f} : \mathbf{C} \to \tilde{X}$. By Theorem 6.6.10 the composed map of $\tilde{f}$ with $\tilde{X} \to Y$ must be algebraically degenerate, because $\dim Y = \bar{\kappa}(X) > 0$ due to our assumption. This implies that f itself is algebraically degenerate; this is a contradiction.

Now let Z be the normalization of the Zariski closure of $f(\mathbf{C})$ in X. Then $\bar{\kappa}(Z) = 0$ by what we have just proved, and the last statement follows from Kawamata's Theorem 5.6.23. $\square$

Remark 6.6.18 (i) The case where X is compact was proved by Yamanoi [04b], Corollary 3.1.14.

(ii) If the properness condition for the quasi-Albanese mapping is dropped, Dethloff–Lu [07] gave a counter-example of a surface X with $\bar{q}(X) = 2$ and $\bar{\kappa}(X) = 1$, for which Theorem 6.6.1 does not hold.

(iii) For *Brody* curves (see Definition 7.6.1) into a log *surface* S with $\bar{\kappa}(S) = \bar{q}(S) = 2$, Dethloff–Lu [07] showed some algebraic degeneracy theorem for entire curves in S.

(iv) Lu–Winkelmann [12] and Noguchi–Winkelmann–Yamanoi [13] proved the algebraic degeneracy of entire curves in a quasi-projective algebraic variety X with $\bar{\kappa}(X) = \bar{q}(X) = \dim X$, but without the properness condition for the quasi-Albanese mapping.

6.6.2 Kobayashi Hyperbolicity

Here we give an application for Kobayashi hyperbolic manifolds. As a direct consequence of Theorem 6.6.1 we have:

Theorem 6.6.19 *Let A be a semi-abelian variety with smooth equivariant compactification $A \hookrightarrow \bar{A}$ and let $\bar{X}$ be a projective variety with a finite morphism $\pi : \bar{X} \to \bar{A}$. Then $X = \pi^{-1}A$ is hyperbolic and hyperbolically embedded into $\bar{X}$ unless there exists a semi-abelian subvariety $B \subset A$, a positive-dimensional orbit $B(p) \subset \bar{A}$, an étale cover $\rho : C \to B(p)$ from a semi-abelian variety C and a morphism $\tau : C \to \bar{X}$ such that $\rho = \pi \circ \tau$.*

Proof By the Brody–Green Theorem 7.2.13 either X is hyperbolic and hyperbolically embedded into $\bar{X}$ or there is a non-constant entire curve in one of the strata of the natural stratification on $\bar{X}$, the one induced by the stratification on $\bar{A}$ which is given by the A-orbits. Using this, the statement follows from Theorem 6.6.1. □

Theorem 6.6.20 *Let X be a normal algebraic variety which admits a finite morphism π onto a simple abelian variety A. Then either X is Kobayashi hyperbolic or X itself is an abelian variety.*

Proof Since A is assumed to be simple, there does not exist any non-trivial (semi-) abelian subvariety. Hence every entire curve $f : \mathbf{C} \to X$ must be constant unless $\pi : X \to A$ is an étale cover by Theorem 6.6.1. If every holomorphic map $f : \mathbf{C} \to X$ is constant, then X is Kobayashi hyperbolic by Brody's Theorem 7.2.8 in the next chapter. If π is étale, then X is an abelian variety, too. □

Remark 6.6.21 (i) The case of $\dim X = \kappa(X) = 2$ was proved by C.G. Grant [86].

(ii) If A is not simple, it is not sufficient to assume the Kobayashi hyperbolicity of the ramification locus R of π to obtain the Kobayashi hyperbolicity of X, as shown by the next proposition.

Proposition 6.6.22 *There exists an abelian surface A with a smooth ample hyperbolic curve $D \subset A$ and a smooth projective surface X with a finite covering $\pi : X \to A$ with ramification locus R such that $R = \pi^{-1}D$ and X is not Kobayashi hyperbolic.*

Proof Let E be an elliptic curve with line bundles H and H' of degree 2 and 3 respectively. Then $\phi_H : E \to \mathbf{P}^1$, while $\phi_{H'}$ embeds E into $\mathbf{P}^2$ as a cubic curve C. Let $Y = \mathbf{P}^1 \times C$. Furthermore let Z be the union of all $\{p\} \times C$ for points $p \in \mathbf{P}^1$ over

which $\phi_H : E \to \mathbf{P}^1$ is ramified. Choose an even number d. Then using Bertini's theorem we obtain a hypersurface $L \subset Y$ such that L is ample, smooth, with $L \cap Z$ being smooth too, and such that the bidegree is $(1, d)$. Let $\tau : E \times E \to Y = \mathbf{P}^1 \times C$ be given by $\tau(x_1, x_2) = (\phi_H(x_1), \phi_{H'}(x_2))$. Then $D = \tau^* L$ is a smooth ample divisor (smoothness of D can be deduced from the conditions that both L and $L \cap Z$ are smooth) and by construction $L(D) = p_1^* L(H) \otimes p_2^* L(dH')$ where $p_i : E \times E \to E$ are the respective projections. Since D is smooth and ample on $E \times E$, it is a curve of genus larger than one and therefore Kobayashi hyperbolic. By construction there is a divisor D_0 such that $2D_0$ is linearly equivalent to D. Now, by the usual cyclic covering method there is a surface X with a two-to-one covering $\pi : X \to A = E \times E$ which is precisely ramified over D. More precisely, by taking squares fiber-wise there is a morphism from (the total space of) $L(D_0)$ to $L(D)$. We let σ denote a section of $L(D)$ whose zero-divisor is D and define $X = \{(x, t) \in L(D_0); t^2 = \sigma(x)\}$. Now $d(t^2 - s(x)) = 2tdt - ds$ in local coordinates; hence X is smooth if D is smooth.

We claim that X is not Kobayashi hyperbolic. For each $q \in E$, let $E_q = E \times \{q\} \subset A$. Then $D \cap E_q$ is a divisor of degree 2 which is linearly equivalent to H. If in some neighborhood of q there are holomorphic functions a, b with values in E such that $\{a(p), b(p)\} = D \cap E_p$ for all p in this neighborhood, then $a + b$ is constant (because all $D \cap E_p$ are linearly equivalent to H). On the other hand, a and b cannot be constant. Hence $a - b$ is non-constant. If we now define an equivalence relation $\sim$ on E via $z \sim -z$, we obtain a globally well-defined holomorphic non-constant map $E \to E/\sim$ locally given by $p \mapsto [a(p) - b(p)]$. As a non-constant holomorphic map between compact Riemann surfaces ($\mathbf{P}^1 \cong E/\sim$), this map must be surjective. It follows that there exists a point $q \in E$ such that $D \cap E_q = 2\{r\}$ for some $r \in E_q$. Fix $F = E_q \subset A$ and define $F' = \pi^{-1} F \subset X$. Then $F' \to F$ is ramified at exactly one point. In local coordinates x for the base and t for the fiber of $L(D_0|_F)$ this implies $F' = \{(x, t); t^2 = g(x)\}$ for some holomorphic function g vanishing at the chosen base point r of order 2. Then $g(x) = x^2 e^{h(x)}$ for some holomorphic function h and consequently F' locally decomposes into two irreducible components given by $t = \pm x e^{h(x)/2}$. For each of the two components the projection onto F is unramified.

Thus, if $\hat{F}$ denotes the normalization of F' then the naturally induced map $\hat{F} \to F$ is an unramified covering. It follows that $\hat{F}$ is an elliptic curve, and therefore it follows that X contains an image of an elliptic curve under a non-constant holomorphic map. In particular, X is not Kobayashi hyperbolic. □

Remark 6.6.23 It is noted that Theorem 6.1 in Noguchi–Winkelmann–Yamanoi [02] was not correct; the above proposition gives a counter-example.

6.6.3 Complements of Divisors in Projective Space

Following the conjectures of Kobayashi, divisors of large degree in the projective space should (if sufficiently generic) be hyperbolic and have hyperbolic comple-

ments. It turns out that hyperbolicity of the complement of a divisor becomes more accessible if one assumes that the divisor has many irreducible components. In fact we have the following.

Theorem 6.6.24 *Let D be an s.n.c. divisor in $\mathbf{P}^n(\mathbf{C})$ with at least $n+2$ irreducible components. Then every entire curve in $\mathbf{P}^n(\mathbf{C}) \setminus D$ is algebraically degenerate.*

This is immediately verified by the Logarithmic Bloch–Ochiai Theorem 4.8.17, because $\bar{q}(\mathbf{P}^n(\mathbf{C}) \setminus D) \geqq n+1 > n$. If the number of irreducible components does not exceed $n+1$, we need some condition on the degree. For $(\mathbf{C}^*)^n$ can be realized as the complement of $n+1$ linear hypersurfaces in general position in $\mathbf{P}^n(\mathbf{C})$ and obviously $(\mathbf{C}^*)^n$ admits many algebraically non-degenerate entire curves.

Theorem 6.6.25 *Let $E_i, 1 \leqq i \leqq q$, be smooth hypersurfaces of the complex projective space $\mathbf{P}^n(\mathbf{C})$ of dimension n such that $E = \sum E_i$ is an s.n.c. divisor. Assume that*

(i) $q \geqq n+1$.
(ii) $\deg E \geqq n+2$.

Then every entire curve $f : \mathbf{C} \to \mathbf{P}^n(\mathbf{C}) \setminus E$ is algebraically degenerate.

Proof If $q \geqq n+2$, then the logarithmic irregularity $\bar{q}(\mathbf{P}^n(\mathbf{C}) \setminus E) > n$, and hence this is immediate from Theorem 4.8.17.

Assume that $q = n+1$. We observe that $K_{\mathbf{P}^n(\mathbf{C})} + E$ is ample, because $\deg(K_{\mathbf{P}^n(\mathbf{C})} + E) \geqq (-n-1) + n + 2 = 1$. Thus E is an s.n.c. divisor for which $K_{\mathbf{P}^n(\mathbf{C})} + E$ is ample, and therefore $\mathbf{P}^n(\mathbf{C}) \setminus E$ is of log general type.

Let d be the l.c.m. of $d_i = \deg E_i, 1 \leqq i \leqq n+1$. Define $k_i = d/d_i$. Let $[x_0, \ldots, x_n]$ be a homogeneous coordinate system of $\mathbf{P}^n(\mathbf{C})$. Let $P_i(x_0, \ldots, x_n)$ be a homogeneous polynomial defining E_i. Now we define a morphism $\bar{\pi}$ from $\mathbf{P}^n(\mathbf{C})$ to $\mathbf{P}^n(\mathbf{C})$ as follows:

$$\bar{\pi} : x \to \left[P_1^{k_1}(x), \ldots, P_{n+1}^{k_{n+1}}(x)\right].$$

Then $\bar{\pi}$ is a finite morphism which restricts to a finite morphism from $\mathbf{P}^n(\mathbf{C}) \setminus E$ to $A = \{[y_0, \ldots, y_n] \in \mathbf{P}^n(\mathbf{C}); y_0 \cdot y_1 \cdots y_n \neq 0\}$. Since A is biholomorphic to $(\mathbf{C}^*)^n$, we may now use Theorem 6.6.1 and deduce that every holomorphic map $f : \mathbf{C} \to \mathbf{P}^n(\mathbf{C}) \setminus E$ is algebraically degenerate. □

Corollary 6.6.26 *Let $f : \mathbf{C} \to \mathbf{P}^n(\mathbf{C})$ be an entire curve and let $[x_0, \ldots, x_n]$ be the homogeneous coordinate system of $\mathbf{P}^n(\mathbf{C})$. Assume that f omits hyperplanes given by*

$$x_i = 0, \quad 1 \leqq i \leqq n, \tag{6.6.27}$$

and a hypersurface defined by

$$x_0^q + \cdots + x_n^q = 0, \quad q \geqq 2.$$

Then f is algebraically degenerate.

Proof This is just a special case of Theorem 6.6.25. □

Remark 6.6.28 (i) The case in which E_i, $1 \leqq i \leqq n$, are hyperplanes and $\deg E_{n+1} \geqq 2$ is due to E. Borel [1897] (cf. Corollary 4.2.15).

(ii) Grauert [89] dealt with the case where $n = 2$ and E_i, $1 \leqq i \leqq 3$, are three smooth quadrics, followed by the papers of Dethloff–Schumacher–Wong [95a], [95b].

(iii) In the case of $n = 2$ the above theorem is optimal in two senses:

a. the number of irreducible components is 3;
b. the total degree is 4.

The case of 4 lines is already due to E. Borel [1897] or to Theorem 4.8.17. If $\deg D \leqq 3$, then there exists a differentiably non-degenerate mapping $g : \mathbf{C}^2 \to \mathbf{P}^2(\mathbf{C}) \setminus D$, and hence there are many algebraically non-degenerate entire curves $f : \mathbf{C} \to \mathbf{P}^2(\mathbf{C}) \setminus D$ (cf. Buzzard–Lu [00]).

(iv) M. Green [74] proved Corollary 6.6.26 in the case of $n = d = 2$ for f of finite order, and left it as a conjecture for general f without order condition. Thus, his conjecture was confirmed affirmatively.

6.6.4 Strong Green–Griffiths Conjecture

In Griffiths [72] (Problem E) he posed:

Problem 6.6.29 Let X be a complex projective algebraic variety with ample canonical bundle. Then is every entire curve in X algebraically degenerate?

In Lang [86] (Conjectures 5.5 and 5.8) this was strengthened as follows.

Conjecture 6.6.30 (Green–Griffiths) *Let X be a complex projective algebraic variety of general type. Then every entire curve in X is algebraically degenerate.*

Futhermore, there exists a closed proper algebraic subset $E \subsetneqq X$ such that E contains the image $f(\mathbf{C})$ of every non-constant entire curve $f : \mathbf{C} \to X$.

The smallest subset of the above E is called the *exceptional subset* of X.

If X in the above conjecture is a subvariety of a semi-abelian variety, then the exceptional subset of X is completely known by Theorem 4.8.17, Propositon 5.6.21 and Theorem 5.6.19 (cf. Kawamata [80]; Noguchi [81a]).

Here we deal with this "*strong Green–Griffiths Conjecture* 6.6.30" for surfaces.

Theorem 6.6.31 (Noguchi–Winkelmann–Yamanoi [07]) *Let A be a semi-abelian surface, and let X be a smooth surface of log general type. Assume that there exists a proper finite morphism $\pi : X \to A$. Then there are only finitely many non-hyperbolic curves C on X; moreover any nonconstant entire curve $f : \mathbf{C} \to X$ has the image contained in one of such C's.*

Proof The ramification locus R of π is the set of all points $x \in X$ for which the differential $d\pi_x : \mathbf{T}(X)_x \to \mathbf{T}(A)_{\pi(x)}$ is not surjective and we denote by R^* the set of all point $x \in X$ for which $d\pi_x$ has exactly rank one. Set $D = \pi(R)$. Since X is of log general type, $\mathrm{St}(D)^0$ is trivial by Remark 6.6.8. We choose a smooth equivariant compactification $A \hookrightarrow \bar{A}$ in which D is in good position (cf. Sect. 5.6). Then we choose a compactification $\bar{X}$ of X such that $\pi : X \to A$ extends to a finite morphism from $\bar{X}$ to $\bar{A}$ (see Proposition 6.6.4). Notice that there are only finitely many A-orbits in $\bar{A}$ (see Theorem 5.2.11).

Now we observe the following: If E is a semi-abelian subvariety of A for which the fixed point set $\bar{A}^E$ in $\bar{A}$ is larger than $\bar{A}^A$, then E is the connected component of the isotropy group of A at a point in a one-dimensional A-orbit in $\bar{A}$. Since A is commutative, all points in the same orbit have the same isotropy group. Therefore the finiteness of the number of A-orbits implies that there are only finitely many semi-abelian subvarieties $E \subset A$ with $\bar{A}^E \neq \bar{A}^A$.

If C is a non-hyperbolic curve on X, then C is either an elliptic curve or $\mathbf{C}^*$, because a morphism from $\mathbf{C}$ or $\mathbf{P}^1$ to a semi-abelian variety must be constant. Hence the image of C by π is necessarily an orbit in A of a semi-abelian subvariety of A. Thus it suffices to show that there exists only finitely many such semi-abelian subvarieties of A over which we can find a non-hyperbolic curve on X.

Thus we have to investigate semi-abelian subvarieties $E \subset A$ of dimension one with an orbit $E(q) \subset A$ such that $\pi^{-1}(E(q))$ contains a non-hyperbolic curve. We may therefore assume that $\bar{A}^E = \bar{A}^A$. Now E is either an elliptic curve or $\mathbf{C}^*$ and in both cases $C \to E$ is an unramified covering. In particular, $d\pi_x$ maps $\mathbf{T}(C)_x$ surjectively on $\mathbf{T}(E)_{\pi(x)}$ for every $x \in C$. Therefore $C \cap R \subset R^*$. We claim furthermore that $\bar{C} \cap \bar{R} \subset C \cap R$. Indeed, if $C \neq \bar{C}$, then $\bar{C} = \mathbf{P}^1$ and $\pi(\bar{C} \setminus C) \subset \partial A$. Therefore $\bar{C} \cap R = C \cap R$. Since D is in good position, we have $\pi(\bar{R}) \cap \bar{A}^A = \emptyset$. Hence $\bar{C} \cap \bar{R} \cap \partial X = \emptyset$. Together, these arguments yield

$$\bar{C} \cap \bar{R} \subset C \cap R \subset C \cap R^*.$$

Next we define a "Gauss map" on $\overline{R^*}$ by setting

$$\gamma : x \mapsto (\text{Image of } d\pi_x) \subset \mathbf{P}(\mathrm{Lie}\, A).$$

It is readily verified that γ is a rational map, thus it extends to a morphism from the closure $R' = \overline{R^*}$ to $\mathbf{P}^1 \cong \mathbf{P}(\mathrm{Lie}\, A)$. Let R_1 denote the union of irreducible components of R' on which γ is locally constant. Then each irreducible component K of R_1 maps onto an orbit of a one-dimensional algebraic subgroup H of A in $\bar{A}$. The value of the Gauss map is evidently Lie H. But then $x \in K \cap E$ implies that $E = H$ and that there is an E-orbit inside D which is excluded. Therefore $C \cap R_1 = \emptyset$.

Let R_2 denote the union of all irreducible components of R' along which γ is nowhere locally constant. Let $x \in C \cap R_2$. A calculation in local coordinates shows that $\mathrm{mult}_x(C, R) = \mathrm{mult}_x \gamma + 1$. Therefore $\deg(C \cap R) \leqq 2 \deg \gamma$. In this way we obtain a universal bound for the degree of $\bar{C}$ with respect to the big divisor $\bar{R}$. Therefore all such curves C are contained in a finite number of families.

Now fix a one-dimensional semi-abelian subvariety E and consider all such curves $\bar{C}$ for which $\pi(C)$ is an orbit of the fixed algebraic subgroup E of A. Then $\pi(C)$ is an E-orbit containing $\pi(R_2 \cap C)$. Therefore $\pi(C)$ is uniquely determined by $C \cap R_2$ provided that the latter is not empty. Now $\{x \in R_2; \gamma(x) = [\mathrm{Lie}\, E]\}$ is finite and the number of curves with empty intersection with $\bar{R}$ is finite as well, since $\bar{R}$ is big. Thus we have finished the proof that there are only finitely many such curves.

The last assertion for entire curves follows from Theorem 6.6.1 combined with the result obtained above. □

6.6.5 *Lang's Questions on Theta Divisors*

In the last paragraph of Chap. 3 of Lang [66] he raised the following two interesting questions.

Question 6.6.32 Does an algebraically non-degenerate one-parameter subgroup of an abelian variety intersect with a hyperplane section?

Question 6.6.33 Is the intersection of the image of an algebraically non-degenerate one-parameter subgroup of an abelian variety and a hyperplane section infinite?

Six years later, J. Ax [72] (cf. [70], too) took this problem and proved:

Theorem 6.6.34 *Let θ be a reduced theta function on $\mathbf{C}^m$ with respect to a lattice $\Gamma \subset \mathbf{C}^m$. Let L be a one-dimensional affine subspace of $\mathbf{C}^m$. Then either $(\theta|L)$ is constant or it has an infinite number of zeros*:

$$\left|\left\{(\theta|L) = 0\right\} \cap \Delta(r)\right| \sim r^2.$$

Let $f : \mathbf{C} \to A$ be a one-parameter subgroup of an abelian variety A. In the case where $f(\mathbf{C})$ is Zariski dense in A, Ax [72] in fact proved the following estimate:

$$0 < \varliminf_{r\to\infty} \frac{n(r, f^*D)}{r^2} \leqq \varlimsup_{r\to\infty} \frac{n(r, f^*D)}{r^2} < \infty. \tag{6.6.35}$$

This is equivalent to

$$0 < \varliminf_{r\to\infty} \frac{N(r, f^*D)}{r^2} \leqq \varlimsup_{r\to\infty} \frac{N(r, f^*D)}{r^2} < \infty. \tag{6.6.36}$$

The deduction of (6.6.36) from (6.6.35) is clear, so we show the opposite. Assume (6.6.36). Then

$$n\big(r, f^*D\big) \leqq \frac{1}{\log 2} \int_r^{2r} \frac{n(t, f^*D)}{t}\, dt \leqq \frac{1}{\log 2} N\big(2r, f^*D\big) = O\big(r^2\big).$$

For the estimate from below we take a number, $0 < \delta < 1$. We get

$$n\big(r, f^*D\big) \geqq \frac{1}{-\log\delta}\int_{\delta r}^{r}\frac{n(t, f^*D)}{t}\,dt = \frac{1}{-\log\delta}\big(N\big(r, f^*D\big) - N\big(\delta r, f^*D\big)\big).$$

By assumption there are two positive constants $C_1 < C_2$ such that

$$N\big(r, f^*D\big) - N\big(\delta r, f^*D\big) \geqq C_1 r^2 - C_2(\delta r)^2 = \big(C_1 - C_2\delta^2\big)r^2.$$

Putting $\delta = \sqrt{C_1/(2C_2)}$, we have $N(r, f^*D) - N(\delta r, f^*D) \geqq \frac{C_1}{2}r^2$, so that $n(r, f^*D) \geqq \frac{C_1}{-2\log\delta}r^2$.

By making use of Theorems 6.3.1 or 6.5.1, we are able to give a very precise growth estimate of such intersections.

Theorem 6.6.37 *Let A be a compact complex torus. Let $f : \mathbf{C} \to A$ be a one-parameter analytic subgroup in A with $a = f'(0)$. Let D be an effective divisor on A with the Riemann form $H(\cdot, \cdot)$. Then we have*

$$N\big(r, f^*D\big) = H(a, a)\pi r^2 + O(\log r),$$
$$N\big(r, f^*D\big) = \big(1 + o(1)\big)N_1\big(r, f^*D\big) = \big(H(a, a)\pi + o(1)\big)r^2.$$

Proof Note that $i\partial\bar\partial H(w, w)$ represents the first Chern class $c_1(L(D))$. It follows from Theorem 6.3.1 that

$$\begin{aligned} N\big(r, f^*D\big) &= T_f\big(r, L(D)\big) + O(\log r) \\ &= \int_0^r \frac{dt}{t}\int_{\Delta(t)} i\,H(a, a)\,dz\wedge d\bar z + O(\log r) \\ &= H(a, a)\pi r^2 + O(\log r). \end{aligned}$$

The remaining estimates follow from this and Theorem 6.5.1 (iii). □

Lang's Question 6.6.32 was readily extended for entire curves in abelian varieties (Griffiths [72]) and in semi-abelian varieties (Noguchi [77b]). Question 6.6.32 has been referred to as "Lang's Conjecture", and was solved as mentioned in Sect. 6.3 (see Corollary 6.3.5 and Remark 6.3.6).

On the other hand, Lang's Question 6.6.33 on the intersections in the the image space has not been studied very much in the value distribution theory; it deals mainly with the inverse images of analytic maps. In this sense, Question 6.6.33 being generalized to entire curves as Question 6.6.32 raises a new problem in the value distribution theory.

Lang's Question 6.6.33 extended to entire curves into abelian or semi-abelian variety was affirmatively resolved.

Theorem 6.6.38 (Corvaja–Noguchi [12]) *Let A be a semi-abelian variety, and let D be an algebraic effective reduced divisor on A with $\mathrm{St}(D)^0 = \{0\}$. Let $f : \mathbf{C} \to A$ be an algebraically non-degenerate entire curve.*

Then, there exists at least one irreducible component D' of D such that the intersection set $f(\mathbf{C}) \cap D'$ is Zariski dense in D'; in particular, if $n \geqq 2$, $f(\mathbf{C}) \cap D$ is infinite.

Proof This is a simple application of Theorem 6.5.1. Suppose that there is no such irreducible component D'. Then there is a proper algebraic subset $Z \subset D$ such that $\mathrm{codim}_A Z \geqq 2$ and $f(\mathbf{C}) \cap D \subset f(\mathbf{C}) \cap Z$. Then we get

$$N_1\big(r, f^*D\big) \leqq N_1\big(r, f^*Z\big) \leqq N\big(r, f^*Z\big).$$

It follows from Theorem 6.5.1 that there is a compactification $\bar{A}(\supset \bar{D})$ of A and

$$\begin{aligned} T_f\big(r, L(\bar{D})\big) &\leqq N_1\big(r, f^*D\big) + \varepsilon T_f\big(r, L(\bar{D})\big)\big\|_\varepsilon, \\ N\big(r, f^*Z\big) &\leqq \varepsilon T_f\big(r, L(\bar{D})\big)\big\|_\varepsilon. \end{aligned}$$

Therefore, combining all the above estimates we have that

$$T_f\big(r, L(\bar{D})\big) \leqq \varepsilon T_f\big(r, L(\bar{D})\big)\big\|_\varepsilon,$$

which is a contradiction. □

6.6.6 Algebraic Differential Equations

Let A be a semi-abelian variety as above and let $X \subset J_k(A)$ be an irreducible algebraic subvariety. We consider the existence problem of an algebraically non-degenerate entire curve $f : \mathbf{C} \to A$ such that $J_k(f)(\mathbf{C}) \subset X$ and $J_k(f)(\mathbf{C})$ is Zariski dense in X. This is a problem of a system of algebraic differential equations described by the equations defining the subvariety X.

The first necessary condition for the existence of such a solution f is that $\mathrm{St}(X)^0 \neq \{0\}$ (cf. (6.5.9)). Now we assume the existence of f. Then we take a big line bundle $L \to X$ and a section $\sigma \in H^0(X, L)$ which defines a reduced divisor on X. We fix a trivialization

$$J_k(f)^*L \cong \mathbf{C} \times \mathbf{C}, \tag{6.6.39}$$

and regard $J_k(f)^*\sigma$ as an entire function.

Proposition 6.6.40 *Let the notation be as above. Then there is no entire function $\psi(z)$ such that every zero of $\psi(z)$ has degree $\geqq 2$ and*

$$J_k(f)^*\sigma(z) = \psi(z), \quad z \in \mathbf{C}. \tag{6.6.41}$$

In particular, there is no entire function $\psi(z)$ satisfying

$$J_k(f)^*\sigma(z) = \big(\psi(z)\big)^q, \quad z \in \mathbf{C}, \tag{6.6.42}$$

where $q \geqq 2$ is an integer.

Remark 6.6.43 The property given by (6.6.41) or (6.6.42) is independent of the choice of the trivialization (6.6.39).

Proof Suppose that there is an entire function $\psi(z)$ satisfying (6.6.41) or (6.6.42). Then it follows that

$$N_1\big(r, J_k(f)^*D\big) \leqq \frac{1}{2}N\big(r, J_k(f)^*D\big).$$

Combining this with Theorem 6.5.1, we infer the following contradiction:

$$T_{J_k(f)}(r, L) \leqq \frac{1}{2}T_{J_k(f)}(r, L) + \varepsilon T_{J_k(f)}(r, L)\|_\varepsilon. \qquad \square$$

Comments Kobayashi Conjecture 7.1.8 and Green–Griffiths Conjecture 6.6.30 have been the central guidelines and the motivations in the value distribution theory. The first will be discussed in the next chapter; here we give some supplements on the second. The both conjectures are related to the problem to establish a Second Main Theorem for entire curves in algebraic varieties.

(i) The Logarithmic Bloch–Ochiai Theorem 4.8.17 for irregular varieties may be considered as a partial answer to the Green–Griffiths Conjecture 6.6.30, since it is reduced to the case of varieties of log general type, which is the most essential. There are other types of results which were not dealt with and will be briefly mentioned.

By a purely potential theoretic method Eremenko–Sodin [92] proved a Second Main Theorem for entire curves in $\mathbf{P}^n(\mathbf{C})$ *without truncations*, which implies a defect relation bounded by $2n$. Min Ru [04], [09] obtained a Second Main Theorem for algebraically non-degenerate entire curves in a projective algebraic variety of dimension n without truncations, which yields a defect relation bounded by $n+1$. It is interesting to know that his proof is modeled after that of Diophantine approximation due to Corvaja–Zannier [04b], and Evertse–Ferretti [02], [08]. These Second Main Theorems do not have applications to the Green–Griffiths Conjecture 6.6.30, so far.

Aihara–Noguchi [91] obtained a Second Main Theorem for holomorphic maps from $\mathbf{C}^m$ into a compactification of a quotient of a bounded symmetric domain, where the counting function was truncated to level one.

M. McQuillan [01] obtained a Second Main Theorem for entire curves in semi-abelian A by a method, different to those mentioned above and ours, which was modeled after Vojta's on Diophantine approximation (cf. Vojta [96]), where the counting function was not truncated at any level.

(ii) Noguchi [77a] obtained an algebraic degeneracy of entire curves by negative Finsler pseudo-metrics, which was later generalized to jet differential metrics, e.g.,

by Green–Griffiths [80] and Lu–Yau [90]. Lu–Yau proved the Green–Griffiths Conjecture 6.6.30 in the case of $c_1^2 > 2c_2$, and McQuillan [98] for the case of $c_1^2 > c_2$ (cf. Brunella [99]); this result was extended to the logarithmic case by El Goul [03].

E. Rousseau [06] proved strong the Green–Griffiths Conjecture for entire curves into a generic projective hypersurface $X \subset \mathbf{P}^4(\mathbf{C})$ of degree $\leqq 593$. Diverio–Merker–Rousseau [10] showed the strong Green–Griffiths Conjecture 6.6.30 for a generic smooth hypersurface $X \subset \mathbf{P}^{n+1}(\mathbf{C})$ of degree $d \geqq 2^{n^5}$.

Let X be a complex algebraic variety with $\dim X = \bar{q}(X)$. In Winkelmann [11] there is a complete description under which conditions there exists an algebraically non-degenerate entire curve for the case where $\dim X = 2$. In this case it turns out that existence of an algebraically non-degenerate entire curve is equivalent to being special in the sense of Campana (see Campana [04]).

For $\dim X > 2$ Lu–Winkelmann [12] proved that the existence of an algebraically non-degenerate entire curve implies being special.

Chapter 7
Kobayashi Hyperbolicity

We describe a way to construct Kobayashi hyperbolic algebraic varieties, and one to construct Kobayashi hyperbolic projective hypersurfaces. Here we will effectively use the theory of entire curves given in the previous chapters. The results of this chapter can be proved so far now only through Nevanlinna theory of entire curves.

7.1 Kobayashi Pseudodistance

For general references of this chapter confer Kobayashi [70], [98], Lang [87], Noguchi–Ochiai [90] (Ochiai–Noguchi [84]).

For two points $z, w \in \Delta(1)$ of the unit disk $\Delta(1)$ of the complex plane $\mathbf{C}$ with center at the origin we define the hyperbolic distance, also called Poincaré's distance by

$$d_{\Delta(1)}(z,w) = \log \frac{1+|\lambda|}{1-|\lambda|}, \quad \lambda = \frac{z-w}{1-\bar{w}z}. \tag{7.1.1}$$

It is easy to show that $d_{\Delta(1)}$ is complete and invariant with respect to the biholomorphic automorphism group $\mathrm{Aut}(\Delta(1))$ of $\Delta(1)$.

In general we let M be a complex manifold. For two points $x, y \in M$ we can always take finitely many holomorphic curves $f_\nu : \Delta(1) \to M$ and points $z_\nu \in \Delta(1)$, $1 \leqq \nu \leqq l$, such that

$$x = f_1(0), \qquad f_\nu(z_\nu) = f_{\nu+1}(0), \quad 1 \leqq \nu \leqq l-1, \qquad f_l(z_l) = y.$$

The sequence $\{(f_\nu, z_\nu)\}$ is called a *holomorphic chain* connecting x and y.[1] For x, y we set

[1] Actually, given two points x, y on a complex manifold X there always exists a chain of length one, i.e., a holomorphic map $f : \Delta(1) \to X$ with $x, y \in F(\Delta(1))$ (see Winkelmann [05]). However, in any case we need to consider for the triangle inequality.

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$$d_M(x, y) = \inf\left\{\sum_{\nu=1}^{l} d_{\Delta(1)}(0, z_\nu)\right\}, \tag{7.1.2}$$

where the infimum is taken over all possible holomorphic chains connecting x and y. The following properties are easily checked:

$$d_M(x, y) = d_M(y, x) \geqq 0, \quad {}^\forall x, y \in M,$$
$$d_M(x, z) \leqq d_M(x, y) + d_M(y, z), \quad {}^\forall x, y, z \in M.$$

In general "$d_M(x, y) = 0$" does not imply "$x = y$". For instance

$$d_{\mathbf{C}^m} \equiv 0.$$

We call d_M the *Kobayashi pseudodistance*.

Using Schwarz's Lemma, one can show that the Kobayashi pseudodistance on the unit disk $\Delta(1)$ coincides with the Poincaré distance.

Definition 7.1.3 If d_M is a (resp. complete) distance function, M is said to be (resp. complete) Kobayashi hyperbolic.

In the same way as above we may define the Kobayashi pseudodistance and the Kobayashi hyperbolicity for connected complex spaces with singularities.

Note that $d_M : M \times M \to \mathbf{R}$ is continuous and if M is Kobayashi hyperbolic, the topology defined by d_M coincides with the underlying differential topology of M (Barth [72]). For a holomorphic mapping $\Phi : M \to N$ between complex manifolds

$$d_M(x, y) \geqq d_N\big(\Phi(x), \Phi(y)\big), \quad x, y \in M.$$

This is called the distance-decreasing principle or property and is a very fundamental property of d_M. From this it follows that if N is Kobayashi hyperbolic, the set $\mathrm{Hol}(M, N)$ of holomorphic mappings from M into N is equicontinuous. Therefore, if d_N is complete Kobayashi hyperbolic, $\mathrm{Hol}(M, N)$ is a normal family due to the Ascoli–Arzelà Theorem. It also follows that d_M is invariant with respect to the biholomorphic automorphism group $\mathrm{Aut}(M)$.

There is also an infinitesimal version of the Kobayashi pseudodistance. This is the "Kobayashi–Royden pseudometric". Cf. Kobayashi [98] and Noguchi–Ochiai [90] for details. It is defined as follows: For a complex manifold M and a tangent vector $v \in \mathbf{T}(M)_p$ we set

$$F_M(v) = \inf\left\{|a|; df\left(a\frac{\partial}{\partial z}\right)_0 = v\right\},$$

where f runs over all holomorphic maps $f : \Delta(1) \to M$ with $f(0) = p$. By Royden [71] the function $F_M : \mathbf{T}(M) \to \mathbf{R}$ is upper semi-continuous and for the unit disk

$\Delta(1)$, $F_{\Delta(1)}$ coincides with the Poincaré metric,

$$F_{\Delta(1)}\left(\left(a\frac{\partial}{\partial z}\right)_z\right) = \frac{|a|}{(1-|z|^2)}.$$

For a real tangent vector $\xi \in T(M)_x$ we write $\xi = v + \bar{v}$ with a holomorphic tangent $v \in \mathbf{T}(M)_x$ and set

$$F_M(\xi) = 2F_M(v).$$

The Kobayashi pseudodistance d_M is the "integrated version" of the infinitesimal Kobayashi–Royden pseudometric F_M,

$$d_M(x, y) = \inf \int_0^1 F_M\big(\gamma'(t)\big)dt,$$

where the infimum is taken over all piecewise differentiable paths $\gamma : [0, 1] \to M$ with $\gamma(0) = x$ and $\gamma(1) = y$.

We assume that M is a relatively compact locally closed complex submanifold embedded into a complex manifold X.

Definition 7.1.4 The embedding $M \hookrightarrow X$ is called a *hyperbolic embedding* if M is Kobayashi hyperbolic and for arbitrary distinct two points $x, y \in \partial M$, $x \neq y$ there are respectively neighborhoods $U, V \subset X$ of x and y in X satisfying

$$\inf\big\{d_M\big(x', y'\big);\, x' \in M \cap U,\, y' \in M \cap V\big\} > 0.$$

In this case we also say that M is hyperbolically embedded into X.

Theorem 7.1.5 *Let X be a compact complex manifold, let D be a reduced divisor, and let $M = X\backslash D$. Assume that M is hyperbolically embedded in X. Then M is complete hyperbolic.*

Proof Assume the opposite. Let $\{p_\nu\}_\nu$ be a Cauchy sequence for d_M which does not converge inside M. Since X is compact, the sequence $\{p_\nu\}_\nu$ converges to a point $p \in D$ (after replacing $\{p_\nu\}_\nu$ by a suitably chosen subsequence). Now let U be an open neighborhood of p in X such that

(i) U is biholomorphic to some polydisc $\mathrm{P}\Delta = (\Delta(1))^n$,
(ii) in an open neighborhood of $\bar{U}$ the divisor D is defined as the zero set of a holomorphic function f.

We may assume that $p_\nu \in U$ for all $\nu \in \mathbf{N}$. Since the closure $\bar{U}$ of U in X is compact, the function f is bounded on U. Let $R = \sup\{|f(z)|;\, z \in U\}$. Then

$$U \setminus D \cong \big\{(x, z) \in \mathrm{P}\Delta \times \big(\Delta(1) \setminus \{0\}\big);\, f(x) = Rz\big\}.$$

Closed complex submanifolds of complete hyperbolic complex spaces are complete hyperbolic. Hence it follows that $U\backslash D$ is *complete* hyperbolic.

Consider now

$$c = \inf\big\{d_M(p_\nu, q);\, \nu \in \mathbf{N}, q \in \partial U \backslash D\big\}.$$

The assumption that M is hyperbolically embedded in X implies that $c > 0$. Now there are two possibilities for any disk chain connecting p_ν and p_μ $(\nu, \mu \in \mathbf{N})$: Either the disk chain stays inside $U \backslash D$, or it passes through the boundary ∂U. As a consequence:

$$d_M(p_\nu, p_\mu) \geq \min\big\{c, d_{U\backslash D}(p_\nu, p_\mu)\big\}.$$

But $c > 0$ and the sequence $\{p_\nu\}$ cannot be a Cauchy sequence for $d_{U\backslash D}$ because $U \backslash D$ is complete and $\lim p_\nu \notin U \backslash D$. It follows that $\{p_\nu\}$ cannot be a Cauchy sequence for d_M. □

The property of being hyperbolically embedded implies a conspicuous result, e.g.:

Theorem 7.1.6 (Kobayashi [70]) *Let M be hyperbolically embedded into X. Then any holomorphic mapping $f : \Delta(1)\backslash\{0\} \to M$ extends holomorphically to $\tilde{f} : \Delta(1) \to X$.*

Theorem 7.1.7 (Noguchi [88]) *Let $\bar{N}$ (resp. $\bar{X}$) be a compact complex manifold (resp. space), and let N (resp. X) be a Zariski open subset of $\bar{N}$ (resp. $\bar{X}$). Assume that*

(i) *$\partial N = \bar{N} \setminus N$ is an s.n.c. divisor;*
(ii) *X is hyperbolic and hyperbolically embedded into $\bar{X}$.*

Then we have the following.

(i) *The moduli space $\mathrm{Hol}(N, X)$ of all holomorphic mappings from N into X is a Zariski open subset of a compact analytic subspace of $\mathrm{Hol}(\bar{N}, \bar{X})$ and the evaluation mapping*

$$\Phi : (f, x) \in \mathrm{Hol}(N, X) \times N \to f(x) \in X$$

is holomorphic and holomorphically extends to $\bar{\Phi} : \overline{\mathrm{Hol}(N, X)} \times \bar{N} \to \bar{X}$.

(ii) (Universal property) *Let T be a complex space and let $\psi : T \times N \to X$ be a holomorphic mapping. Then the natural mapping*

$$t \in T \to \psi(t, \cdot) \in \mathrm{Hol}(N, X)$$

is holomorphic.

The assumption and Theorem 7.1.5 implies that the above X is complete hyperbolic. Cf. Noguchi–Ochiai [90] Chap. 1, and Kobayashi [98] Chaps. 5, 6 for more results in this direction.

Notice that $M = \mathbf{P}^1(\mathbf{C})\backslash\{3 \text{ points}\}$ is hyperbolically embedded into $X = \mathbf{P}^1(\mathbf{C})$ (see Corollary 7.3.5). Thus this theorem is a generalization of the Big Picard Theorem.

It is immediately seen that bounded domains in $\mathbf{C}^m$ are Kobayashi hyperbolic and hyperbolically embedded into $\mathbf{C}^m$.

Compact Riemann surfaces are hyperbolic if and only if they are of genus $g \geqq 2$. This can be seen via uniformization theorem but also via Schwarz's Lemma and curvature considerations (Kobayashi [70]). Non-compact Riemann surfaces are always hyperbolic except for $\mathbf{C}$ and $\mathbf{C}^*$. In this chapter we will consider how to obtain Kobayashi hyperbolic manifolds which are compact or complements of reduced divisors of compact complex manifolds in dimension greater than one.

The research of this direction was motivated by the following conjecture due to Kobayashi [70].

Conjecture 7.1.8 (Kobayashi Conjecture) (i) *A generic hypersurface X of $\mathbf{P}^m(\mathbf{C})$ of degree d is Kobayashi hyperbolic for large d ($d \geqq 2m-1$, $m \geqq 3$).*

(ii) *For larger $d \geqq 2m+1$ the complement $\mathbf{P}^m(\mathbf{C})\backslash X$ is hyperbolically embedded into $\mathbf{P}^m(\mathbf{C})$.*

In $\mathbf{P}^2(\mathbf{C})$ a smooth curve C of degree $d \geqq 4$ has genus $g = \frac{(d-1)(d-2)}{2} \geqq 3$, so that it is Kobayashi hyperbolic. This confirms (i) for $m = 2$.

It was not specified what the "generic" meant precisely. Anyway, it is clear that not *all* hypersurfaces of any degree are hyperbolic. For instance, consider the Fermat hypersurface

$$F_{d,m} = \left\{ [x_0, \ldots, x_m]; \sum_i x_i^d = 0 \right\}.$$

Regardless of the degree d, a non-constant entire curve from $\mathbf{C}$ to $F_{d,m}$ is given by

$$z \mapsto \left[z, e^{\pi i/d} z, 1, e^{\pi i/d} : 0, \ldots, 0 \right].$$

Hence for every $d \in \mathbf{N}$ and $m \geq 3$, $F_{m,d}$ is a non-hyperbolic hypersurface of degree d.

7.2 Brody's Theorem

7.2.1 Brody's Reparametrization

The purpose of this section is to deduce the theorem of Brody (Theorem 7.2.8 below).

Since $d_{\mathbf{C}} \equiv 0$, the distance-decreasing property of the Kobayashi pseudodistance implies immediately that every holomorphic map from $\mathbf{C}$ to a hyperbolic complex

manifold must be constant. This raises the question whether, given a complex manifold X, conversely the property "*every holomorphic map from* $\mathbf{C}$ *to* X *be constant*" implies that X is hyperbolic.

The theorem of Brody gives a positive answer for *compact* complex manifolds. Its proof is based on the "reparametrization lemma" of Brody, which says the following.

Lemma 7.2.1 *Let (Y, h) be a hermitian complex manifold and let X be a relatively compact locally closed submanifold. Let $f_k : \Delta(1) \to X$ be a sequence of holomorphic maps such that $\lim_{k\to\infty} \|(df_k)_0\| = \infty$, where $\|(df_k)_0\|$ stands for the operator norm of the exterior differential with respect to the Poincaré metric on $\Delta(1)$ and h. Let ε_k be a sequence of positive numbers with $\lim_{k\to\infty} \varepsilon_k = 0$.*

Then there exist a sequence of positive numbers r_k with $\lim r_k = \infty$ and a sequence of automorphisms $\phi_k \in \mathrm{Aut}(\Delta(1))$ such that there is a subsequence of $g_k : \Delta(r_k) \to X$ given by

$$g_k : z \mapsto f_k\left(\frac{1}{1+\varepsilon_k}\phi_k\left(\frac{z}{r_k}\right)\right), \quad k = 1, 2, \ldots,$$

which converges locally uniformly to a non-constant entire curve $f : \mathbf{C} \to Y$ with

$$\sup_{z\in\mathbf{C}} \left\|(df)_z\right\|_e < \infty,$$

where $\|(df)_z\|_e$ denotes the operator norm with respect to the Euclidean norm on $\mathbf{C}$ and h.

Proof Consider

$$\hat{f}_k : z \in \Delta(1) \to f_k\left(\frac{z}{1+\varepsilon_k}\right).$$

Then

$$\lim_{z\to\partial\Delta(1)} \left\|(d\hat{f}_k)_z\right\| = 0, \quad {}^\forall k.$$

Therefore $z \mapsto \|(d\hat{f}_k)_z\|$ assumes a maximum at some point $q_k \in \Delta(1)$. We choose an automorphism ϕ_k of the unit disk $\Delta(1)$ such that $\phi_k(0) = q_k$. Finally, we set $r_k = \|(d\hat{f}_k)_{q_k}\|$. Then we can apply the Ascoli–Arzelà Theorem. □

There is a variant of this result due to Zalcman [98].

Theorem 7.2.2 *Let (Y, h) be a hermitian complex manifold and let X be a relatively compact open submanifold. Let $f_k : \Delta(1) \to X$ be a sequence of holomorphic maps from the unit disk $\Delta(1)$ to a compact complex manifold X. Assume that $\{f_k\}$ is not a normal family.*

Then there exist sequences of complex numbers $a_k, b_k \in \mathbf{C}$ such that for each $z \in \mathbf{C}$ there is a number N such that $a_k z + b_k \in \Delta(1)$ for all $k \geqq N$. Moreover

a subsequence of

$$g_k : z \mapsto f_k(a_k z + b_k)$$

converges locally uniformly to a non-constant entire curve $f : \mathbf{C} \to X$ *such that the operator norm* $\|df\|_e$ *with respect to the Euclidean metric on* $\mathbf{C}$ *and a hermitian metric on* X *is bounded.*

Proof Choose $0 < r^* < r < 1$ such that f_k is a non-normal family on $\Delta(r^*)$. Fix a hermitian metric on X and define $f^{\#}(z) = \|(df)_z\|_e$ with respect to this metric on X and the Euclidean metric on $\mathbf{C}$. ($f^{\#}$ is classically known as the *spherical derivative in the case where* $X = \mathbf{P}^1$ *and the chosen metric is the Fubini–Study metric.*) Choose $z_k \in \Delta(r)$ such that

$$z \mapsto \left(1 - \frac{|z|^2}{r^2}\right) f_k^{\#}(z)$$

assumes the maximum at z_k. Let M_k denote this maximum, i.e.

$$M_k = \left(1 - \frac{|z_k|^2}{r^2}\right) f_k^{\#}(z_k).$$

It follows that

$$\frac{M_k}{f_k^{\#}(z_k)} = \frac{(r - |z_k|)(r + |z_k|)}{r^2}.$$

Define affine-linear maps

$$\lambda_k : \zeta \mapsto z_k + \frac{\zeta}{f_k^{\#}(z_k)}, \quad k = 1, 2, \ldots.$$

Let $0 < \varepsilon < 1$ be any fixed. An easy calculation shows the following:

$$\big|z_k - \lambda_k(\zeta)\big| < \varepsilon\big(r - |z_k|\big) \quad \text{for } |\zeta| < \varepsilon \frac{r^2}{r + |z_k|} M_k. \tag{7.2.3}$$

Therefore in this case, $|\lambda_k(\zeta)| < r$. We set

$$g_k(\zeta) = f_k \circ \lambda_k(\zeta), \quad |\zeta| < \varepsilon \frac{r^2}{r + |z_k|} M_k.$$

Let us consider the derivative. Assume that ζ is a given complex number such that $w = \lambda_k(\zeta) \in \Delta(r)$. Then we have

$$\begin{aligned} g_k^{\#}(\zeta) = f_k^{\#}(w)\big|\lambda_k'(\zeta)\big| &= \frac{f_k^{\#}(w)}{f_k^{\#}(z_k)} \leqq \frac{r^2 - |z_k|^2}{r^2 - |w|^2} \\ &\leqq \frac{(r - |z_k|)(r + |z_k|)}{(r - |w|)(r + |w|)}, \\ g_k^{\#}(0) &= 1. \end{aligned} \tag{7.2.4}$$

By (7.2.3) we have that $|w - z_k| < \varepsilon(r - |z_k|)$, so that

$$r - |w| \geqq r - |z_k| - |w - z_k| > r - |z_k| - \varepsilon\big(r - |z_k|\big) \tag{7.2.5}$$
$$= (1 - \varepsilon)\big(r - |z_k|\big),$$

$$r + |w| \geqq r + |z_k| - |w - z_k| > r + |z_k| - \varepsilon\big(r - |z_k|\big). \tag{7.2.6}$$

It follows from (7.2.4)–(7.2.6) that

$$\begin{aligned} g_k^{\#}(\zeta) &\leqq \frac{1}{(1-\varepsilon)} \cdot \frac{r + |z_k|}{r + |z_k| - \varepsilon(r - |z_k|)} \\ &\leqq \frac{1}{(1-\varepsilon)} \cdot \frac{r + |z_k|}{r(1-\varepsilon) + |z_k|(1+\varepsilon)} \\ &\leqq \frac{1}{(1-\varepsilon)^2}, \qquad |\zeta| < \varepsilon \frac{r^2}{r + |z_k|} M_k. \end{aligned}$$

Thus

$$g_k^{\#}(\zeta) \leqq \frac{1}{(1-\varepsilon)^2}, \qquad |\zeta| < \varepsilon \frac{r^2}{r + |z_k|} M_k.$$

This implies that we can use the Ascoli–Arzelà Theorem to extract a subsequence convergent to $f : \mathbf{C} \to X$. It then follows that

$$f^{\#}(0) = 1, \qquad f^{\#}(\zeta) \leqq 1, \quad {}^{\forall}\zeta \in \mathbf{C}. \qquad \square$$

Now we are in a position to prove Brody's Theorem.

Theorem 7.2.7 (Brody [78]) *Let X be a complex manifold and let Y be a locally closed analytic subset. Assume that Y is relatively compact in X and that there does not exist an entire curve $f : \mathbf{C} \to X$ with $f(\mathbf{C}) \subset \bar{Y}$. Then Y is hyperbolically embedded into X.*

Proof We fix an hermitian metric $\| \ \|$ on X. Define

$$C = \sup\left\{ \left\| f_*\left(\frac{\partial}{\partial z}\right)_0 \right\|; f : \Delta(1) \to Y \right\}.$$

If $C = \infty$, Lemma 7.2.1 implies the existence of a non-constant entire curve $f : \mathbf{C} \to \bar{Y}$.

Let us assume $C < \infty$. Let $F_{\Delta(1)}$ denote the Poincaré metric on the unit disk as before. It is invariant under all automorphisms of the unit disk. Let $f : \Delta(1) \to X$ be a holomorphic map and $q \in \Delta(1)$. Then there is an automorphism α of $\Delta(1)$ with $\alpha(0) = q$. Now

$$C \geqq \left\| (f \circ \alpha)_*\left(\frac{\partial}{\partial z}\right)_0 \right\| = \| f_* v \|$$

for $v \in \mathbf{T}(\Delta(1))_q$ with $F_{\Delta(1)}(v) = 1$. Therefore

$$\|f_* v\| \leqq C F_{\Delta(1)}(v)$$

for all $q \in \Delta(1)$, $v \in \mathbf{T}(\Delta(1))_q$. If d_h denotes the distance function on X induced by the hermitian metric $\| \ \|_h$, it follows that

$$d_X(x, y) \geqq \frac{1}{C} d_h(x, y), \quad {}^\forall x, {}^\forall y \in \bar{Y}.$$

This implies that Y is hyperbolically embedded in X. □

Theorem 7.2.8 *Let X be a compact complex manifold. X is Kobayashi hyperbolic if and only if there is no non-constant entire curve $f : \mathbf{C} \to X$.*

Proof If there is such a non-constant entire curve, X cannot be hyperbolic. The converse follows from Theorem 7.2.7. □

For non-compact manifolds the situation is more complicated. For certain homogeneous complex manifolds there are analogous results (namely, if the group acting transitively is solvable (Winkelmann [90]) or if there is an invariant hermitian metric (Kodama [83]), but in general the absence of entire curves does not imply hyperbolicity. Eisenman was the first to give an example (see Kobayashi [70]) of a non-compact complex manifold which is not hyperbolic, although it does not admit any non-constant holomorphic map from $\mathbf{C}$. Here we give a different example. For this we need the auxiliary lemma below.

Lemma 7.2.9 *Let S be a smooth surface and let $\pi : \hat{S} \to S$ be the blow-up in one point $p \in S$ with exceptional divisor $E = \pi^{-1} p \subset \hat{S}$. Let Z be a finite subset of such E's. Then $\hat{S} \setminus Z$ is not hyperbolic.*

Proof There is an open neighborhood W of E in $\hat{S}$ which is biholomorphic to

$$\big\{\big(z, w, [x, y]\big) \in \mathbf{C}^2 \times \mathbf{P}^1(\mathbf{C}); |z|^2 + |w|^2 < 1, zy = wx\big\}.$$

We choose a number, $0 < r < \frac{1}{2}$ such that $|x| \geq 2r|y|$ for every $(0, 0, [x, y]) \in Z \setminus \{(0, 0, [0, 1])\}$. We will deduce a bound for

$$d_{\hat{S}}\big(\big(0, 0, [r, 1]\big), \big(0, 0, [-r, 1]\big)\big]$$

by giving a disk chain. Fix $0 < \varepsilon < \frac{1}{2}$, and define

$$f_1(t) = \left(\frac{1}{2}t, \frac{1}{2}tr, [r, 1]\right),$$

$$f_2(t) = \left(\frac{1}{2}\varepsilon, \frac{r}{2}(\varepsilon - 2t), \big[r(\varepsilon - 2t), \varepsilon\big]\right),$$

(continued)

$$f_3(t) = \left(\frac{1}{2}(\varepsilon - t), \frac{r}{2}(t - \varepsilon), [-r, 1]\right).$$

Note that $f_1(\varepsilon) = f_2(0)$ and $f_2(\varepsilon) = f_3(0)$. Hence this yields a holomorphic chain which implies

$$d_{\hat{S}}\big((0, 0, [r, 1]), (0, 0, [-r, 1])\big) \leq 3d_{\Delta(1)}(0, \varepsilon), \quad {}^{\forall}\varepsilon > 0,$$
$$\Rightarrow \quad d_{\hat{S}}\big((0, 0, [r, 1]), (0, 0, [-r, 1])\big) = 0.$$

Therefore $\hat{S}$ is not hyperbolic. □

Now we can give the example announced before.

Proposition 7.2.10 (Example) *Let S be a compact hyperbolic complex surface (e.g., the product of two compact Riemann surfaces of genus larger than one) and let $\bar{X}$ denote S blown up in one point $p \in S$. Let E denote the exceptional divisor and let $M \subset E$ be a subset of cardinality three.*

Then $X = \bar{X} \setminus M$ is not Kobayashi hyperbolic, although every entire curve $f : \mathbf{C} \to X$ is constant.

Proof Every non-constant entire curve $f : \mathbf{C} \to X$ has its image contained in $E \setminus M$, because S is hyperbolic. But every map to $E \setminus M \cong \mathbf{P}_1 \setminus \{0, 1, \infty\}$ is constant due to Picard's Theorem. On the other hand, X is not hyperbolic because of Lemma 7.2.9. □

There is a variant of Brody's Theorem for manifolds compactified by adding a divisor at infinity. This was obtained by M. Green.

Lemma 7.2.11 *Let $\bar{X}$ be a complex manifold with a reduced divisor D. If a sequence of holomorphic maps $f_\nu : \Delta(1) \to \bar{X} \setminus D$ converges locally uniformly to a holomorphic map $f : \Delta(1) \to \bar{X}$, then either $f(\Delta(1)) \subset D$ or $f(\Delta(1)) \cap D = \emptyset$.*

Proof This is a consequence of Hurwitz' Theorem, because locally every divisor on a manifold is given as the zero set of one holomorphic function. □

Combining this fact with Brody's Theorem 7.2.8 one obtains the following hyperbolicity criterion due to Green [77]; we will state it in a form more general.

Definition 7.2.12 A family of divisors $\{D_i\}$ (or the divisor $\sum_i D_i$) on a complex manifold X is said to be in *general position* if every intersection of arbitrary k elements of $\{\operatorname{Supp} D_i\}$ has everywhere codimension k; in the case of $k > \dim X$, the intersection is empty.

When $D_i \subset \mathbf{P}^n(\mathbf{C})$ are hyperplanes, the above definition coincides with Definition 4.1.1. If $\{D_i\}$ is in general position, then it is independent in supports.

Theorem 7.2.13 *Let X be a compact complex manifold. Let D_i, $1 \leqq i \leqq l$ $(<\infty)$, be hypersurfaces of X such that $D = \sum D_i$ is in general position. Then $M = X \setminus \bigcup_{i=1}^{l} D_i$ is hyperbolically embedded into X if and only if the following condition is satisfied*:

"For every partition $\{1, \ldots, l\} = I \cup J$ of indices with $I \cap J = \emptyset$ there is no non-constant entire curve

$$f : \mathbf{C} \to \bigcap_{i \in I} D_i \setminus \bigcup_{j \in J} D_j."$$

Remark 7.2.14 The family $\{D_i\}$ of divisors with the above property is said to be in *hyperbolic configuration*.

Proof If M is not hyperbolically embedded into X, there are points $p, q \in X$ and sequences $p_\nu, q_\nu \in M$ such that $\lim p_\nu = p$, $\lim q_\nu = q$, $\lim d_M(p_\nu, q_\nu) = 0$ but $p \neq q$. Let $\| \ \|_h$ be a hermitian metric on X. If d_h denotes the distance function defined by $\| \ \|_h$, then

$$\lim_{\nu \to \infty} d_h(p_\nu, q_\nu) = d_h(p, q) > 0.$$

Hence there is no constant $C > 0$ such that $d_M(x, y) \geq C d_h(x, y)$ for all $x, y \in M$. By the definition of the Kobayashi pseudodistance it follows that there is a sequence of holomorphic maps $f_\nu : \Delta(1) \to M$ with $\overline{\lim}_{\nu \to \infty} \|(df_\nu)_0\| = \infty$, where $\|(df_\nu)_0\|$ denotes the operator norm with respect to the Poincaré metric on $\Delta(1)$ and the hermitian metric $\| \ \|_h$ on M. Using Brody's Theorem 7.2.8 it follows that there is a non-constant entire curve $f : \mathbf{C} \to X$ arising from these holomorphic maps f_ν. By Lemma 7.2.11 above we know that for each i either $f(\mathbf{C}) \subset D_i$ or $f(\mathbf{C}) \cap D_i = \emptyset$. Let I denote the set of all indices i with $f(\mathbf{C}) \subset D_i$ and let J denote the set of all i with $f(\mathbf{C}) \cap D_i = \emptyset$. Then

$$f(\mathbf{C}) \subset \bigcap_{i \in I} D_i \setminus \bigcup_{j \in J} D_j.$$

Conversely let us assume that there is such an entire curve

$$f : \mathbf{C} \to \bigcap_{i \in I} D_i \setminus \bigcup_{j \in J} D_j.$$

Using Siu's theorem on Stein neighborhoods (see Siu [76]) we know that the graph $G = \{(z, f(z)); z \in \mathbf{C}\}$ admits a Stein neighborhood base in $\mathbf{C} \times (X \setminus \bigcup_{j \in J} D_j)$. Note that the pull-backs $f^* D_i$ define trivial line bundles on $\mathbf{C}$. It follows that there exists a Stein open neighborhood W of G in $\mathbf{C} \times (X \setminus \bigcup_{j \in J} D_j)$ such that the divisors $D_i' = W \cap (\mathbf{C} \times (D_i \setminus \bigcup_{j \in J} D_j))$ are principal divisors on W, defined by some holomorphic function ϕ_i. Since $D = \sum D_i$ is assumed to be an s.n.c. divisor, we may deduce that the 1-forms $d\phi_i$ are linearly independent at every point on G.

Therefore $(d\phi_1, \ldots, d\phi_l)$ defines a surjective holomorphic vector bundle homomorphism from $\mathbf{T}(W)|_G$ to the trivial bundle $\mathscr{O}_G^l$. Since $G \cong \mathbf{C}$, this homomorphism splits, i.e. there is a section $\sigma \in \Gamma(G, \mathbf{T}(W))$ with $d\phi_i(\sigma) \equiv 1$ for all i.

We extend the holomorphic map $\gamma : \mathbf{C} \to W$ given by $\gamma(t) = (t, f(t))$ to a holomorphic map $F : \Omega \to W$ with $\mathbf{C} \times \{0\} \subset \Omega \subset \mathbf{C}^2$ such that

(i) $F(t; 0) = \gamma(t)$ for all $t \in \mathbf{C}$,
(ii) $F_*(v, v_2)_{(t,0)} = (\gamma_* v_1, v_2 \sigma(\gamma(t)))$ for all $t \in \mathbf{C}$,
(iii) $\partial(\phi_i \circ F)/\partial z_2 \equiv 1$ on $\mathbf{C} \times \{0\} \subset \Omega$ for any $i \in I$.

We fix two points $a, b \in \mathbf{C}$ with $f(a) \neq f(b)$. Let $p = f(a)$ and let $q = f(b)$. For every $R > 0$ we may choose a small number ε such that

$$\overline{\Delta(R)} \times \overline{\Delta(\varepsilon)} \subset \Omega,$$

because Ω is open and contains $\mathbf{C} \times \{0\}$. Therefore we can choose a sequence $\varepsilon_\nu > 0$ such that

(i) $\lim \varepsilon_\nu = 0$,
(ii) $(a, \varepsilon_\nu), (b, \varepsilon_\nu) \in \Omega$ for all $\nu \in \mathbf{N}$,
(iii) $\overline{\Delta(\nu)} \times \overline{\Delta(\varepsilon_\nu)} \subset \Omega$ for all $\nu \in \mathbf{N}$,
(iv) $\phi_i \circ F$ has no zero on $\overline{\Delta(\nu)} \times \{\varepsilon_\nu\}$ for any $i \in I$, $\nu \in \mathbf{N}$.

Let $\pi : W \to X$ denote the natural projection map. We define holomorphic maps $f_\nu : \Delta(n) \to M$ as follows:

$$f_\nu : z \mapsto \pi\big(F(z, \varepsilon_\nu)\big).$$

Define $p_\nu = f_\nu(a)$, $q_\nu = f_\nu(b)$.

By construction we have $\lim p_\nu = p$, $\lim q_\nu = q$ and

$$d_M(p_\nu, q_\nu) = d_M\big(f_\nu(a), f_\nu(b)\big) \leqq d_{\Delta(\nu)}(a, b), \quad {}^\forall\nu \in \mathbf{N}.$$

Since $\lim_{\nu\to\infty} d_{\Delta(\nu)}(a, b) = 0$, we obtain $\lim_{\nu\to\infty} d_M(p_\nu, q_\nu) = 0$. Thus M is not hyperbolically embedded in X. □

7.2.2 Hyperbolicity as an Open Property

Theorem 7.2.15 *Let $\pi : X \to Y$ be a proper holomorphic map between complex spaces. For $y \in Y$ let X_y denote the fiber: $X_y = \{x \in X; \pi(x) = y\}$. Then*

$$\{y \in Y;\ X \text{ is hyperbolic}\}$$

is an open subset of Y.

Proof Let $y_n \in Y$ with $\lim y_n = q \in Y$. Assume that all the X_{y_n} are non-hyperbolic. We have to show that X_q is not hyperbolic. Since π is proper, every X_y $(y \in Y)$ is compact. By Brody's Theorem 7.2.8 non-hyperbolicity of the X_{y_ν} implies that there is a non-constant entire curve $f_\nu : \mathbf{C} \to X_{y_\nu}$ for every ν. Now let W be a relatively compact open neighborhood of q in Y. We may assume that W contains all the points y_ν. Then $V = \pi^{-1}(W)$ is relatively compact in X (because π is proper) and we may use Brody's reparametrization Lemma 7.2.1 to deduce that there is a non-constant entire curve $f : \mathbf{C} \to X_q$. Thus X_q is not hyperbolic. □

Applying the above theorem to the universal family of hypersurfaces of a fixed degree over some projective space yields the important consequence below.

Corollary 7.2.16 *Let $\mathscr{D}_d$ be the space of all hypersurfaces of $\mathbf{P}^n(\mathbf{C})$ of degree d. Then the subset $\{X \in \mathscr{D}_d;\ X$ is Kobayashi hyperbolic$\}$ is open in the sense of differential topology.*

It is an open question whether this open set is in fact Zariski-open.

7.3 Kobayashi Hyperbolic Manifolds

Here we present a contribution related to Kobayashi Conjecture 7.1.8 (ii).

A divisor D on a complex space X is said to be locally **Q**-*Cartier* if for every $x \in X$ there is a non-zero divisor meromorphic function ϕ on a neighborhood U of x with divisor $(\phi) = D|_U$; if $U = M$, D is said to be **Q**-Cartier. If X is compact, D is **Q**-Cartier if it is locally **Q**-Cartier.

Remark 7.3.1 Note that Theorem 7.2.13 holds for the supports of **Q**-Cartier divisors which are in general position.

Lemma 7.3.2 *Let X be a projective algebraic variety which may be singular, and let D_j, $1 \leqq j \leqq l$, be **Q**-Cartier ample divisors. Assume that $\{D_j\}_{j=1}^l$ is in general position and*

$$l \geqq 2 \dim X + r\big(\{D_j\}\big).$$

Then any entire curve in $X \setminus \operatorname{Supp} \sum_{j=1}^l D_j$ is constant.

Proof We use the same notation D_j for $\operatorname{Supp} D_j$, unless confusion occurs.

Let $f : \mathbf{C} \to X \setminus \sum_j D_j$ be an entire curve, and denote by W the Zariski closure of $f(\mathbf{C})$ in X. Assume that $\dim W > 0$. Now, we are going to deduce a contradiction.

Set $n = \dim X$ and $k = \dim W$.

(a) $k = 1$. In this case, we claim that the cardinality $|W \cap \sum_j D_j|$ is at least three. Since D_j are ample, $W \cap \sum_j D_j \neq \emptyset$. Let $P_1 \in W \cap D_1$. By the assumption the number of D_j $(1 \leqq j \leqq l)$ such that $D_j \ni P_1$ is at most n. After re-ordering,

we may assume that $P_1 \notin D_j$ for $n+1 \leqq j \leqq l$. Let $P_2 \in W \cap D_{n+1}$. Then $P_2 \neq P_1$. By the same reason as above, we may assume that $P_2 \notin W \cap D_j$ for $2n+1 \leqq j \leqq l$. Let $P_3 \in W \cap D_{2n+1}$. Then P_i, $i = 1, 2, 3$, are distinct, so that $|W \cap (\sum_{j=1}^{l} D_j| \geqq 3$. Then f must be constant; this is a contradiction.

(b) $k \geqq 2$. We set $r_0 = r(\{D_j\})$. By changing the indices we may assume that $c_1(D_1), \ldots, c_1(D_{r_0})$ form a maximal linearly independent system of $\{c_1(D_j)\}_{j=1}^{l}$ modulo torsion. We may assume that $l = 2n + r_0$. For $r_0 + 1 \leqq j \leqq r_0 + 2n$ there are integers, $a_j \neq 0$ and a_{ji}, such that

$$a_j c_1(D_j) + \sum_{i=1}^{r_0} a_{ji} c_1(D_i) = 0 \quad \text{(mod torsion)}.$$

Restricting this to W, and taking a resolution $\pi : \tilde{W} \to W$ of singularities, we have

$$a_j c_1(\tilde{D}_j) + \sum_{i=1}^{r_0} a_{ji} c_1(\tilde{D}_i) = 0 \quad \text{(mod torsion)}, \tag{7.3.3}$$

where $\tilde{D}_j = \pi^*(D_i|_W)$. Then, by Lemma 4.9.6 there exist logarithmic 1-forms ω_j on $\tilde{W}$ with poles, exactly $\tilde{D}_j \cup \sum_{1 \leqq i \leqq r_0,\, a_{ji} \neq 0} \tilde{D}_i$. If $D_j|_W$, $j \in J$, for some $J \subset \{r_0 + 1, \ldots, l\}$ are independent in supports on W, then $\{\tilde{D}_j\}_{j \in J}$ is independent in supports on $\tilde{W}$, so that ω_j, $j \in J$, are linearly independent and hence $\bar{q}(W \setminus (\sum_{i=1}^{r_0} D_i + \sum_{j \in J} D_j)) \geqq |J|$: This property will be essential in the arguments below.

By the ampleness of D_j the algebraic subset $W \cap \sum_{i=1}^{r_0} D_i$ is not empty and of pure dimension $k-1$. Let $E_1 \subset W \cap \sum_{i=1}^{r_0} D_i$ be an irreducible component of it. Then $\dim E_1 = k-1$ and there is some $D_{j_1} \supset E_1$ ($1 \leqq j_1 \leqq r_0$). The general position assumption for $\{D_j\}$ implies that

$$\left|\{j; r_0 + 1 \leqq j \leqq r_0 + 2n,\, D_j \supset E_1\}\right| \leqq n - k.$$

After re-ordering we have that $D_{r_0+1} \not\supset E_1$. Let $E_2 \subset E_1 \cap D_{r_0+1}$ ($\subsetneqq E_1 \subset D_{j_1}$) be an irreducible component of dimension $k-2$. After re-ordering, $D_{r_0+2} \not\supset E_2$.

In this way by induction we can find sequences of irreducible subvarieties E_h, $1 \leqq h \leqq k-1$, and divisors D_{r_0+h} such that

(i) E_h is an irreducible component of $W \cap (\sum_{i=1}^{r_0} D_i) \cap D_{r_0+1} \cap \cdots \cap D_{r_0+h-1}$ of dimension $k-h$;
(ii) $D_{r_0+h} \not\supset E_h$.

Then at $h = k$, $E_k = \{P_0\}$ is a set of one point. There is a neighborhood U of P_0 in W such that the divisors $U \cap \sum_{i=1}^{r_0} D_i, U \cap D_{r_0+1}, \ldots, U \cap D_{r_0+k-1}$ are independent in supports. Therefore, the $k-1$ logarithmic 1-forms ω_j, $r_0 + 1 \leqq j \leqq k-1$, are linearly independent.

Since there are already k divisors D_j with $1 \leqq j \leqq r_0 + k - 1$ containing P_0, $|\{j; r_0 + k - 1 < j \leqq r_0 + 2n, D_j \ni P_0\}| \leqq n - k$. After re-ordering we have that

$$P_0 \notin D_j, \quad r_0 + n \leqq j \leqq r_0 + 2n.$$

Therefore there are $n+1$ such j's. Notice that $\bigcap_{j=r_0+n}^{2n} D_j = \emptyset$. Hence, there is a D_{j_2} with $r_0 + n \leqq j_2 \leqq r_0 + 2n$ such that $W \cap (\sum_{i=1}^{r_0} D_i) \cap D_{j_2} \neq W \cap (\sum_{i=1}^{r_0} D_i)$. We may suppose $j_2 = r_0 + n$. Let F_1 be a $(k-2)$-dimensional irreducible component of $W \cap (\sum_{i=1}^{r_0} D_i) \cap D_{r_0+n}$. Again, there is some D_j with $r_0 + n < j \leqq r_0 + 2n$ such that $F_1 \cap D_j \neq F_1$; let such j be $r_0 + n + 1$. Then there exist two logarithmic 1-forms ω_k and ω_{k+1} such that the polar locus of ω_k (resp. ω_{k+1}) is exactly $D_k + \sum_{1 \leqq i \leqq r_0,\, a_{r_0+ki} \neq 0} D_j$ (resp. $D_{k+1} + \sum_{1 \leqq i \leqq r_0,\, a_{r_0+k+1i} \neq 0} D_i$), so that ω_k and ω_{k+1} are linearly independent. (Here, in fact, we can find k linearly independent such logarithmic 1-forms.)

Since the polar divisors of ω_j, $1 \leqq j \leqq k+1$ (in fact, $\leqq 2k-1$), are independent in supports, they are linearly independent. Therefore, we see that $\bar{q}(W \setminus \sum_{j=1}^{r_0+2n} D_j) \geqq k+1$ (in fact, $\geqq 2k-1$). The logarithmic Bloch-Ochiai Theorem 4.8.17 implies the algebraic degeneracy of $f : \mathbf{C} \to W$; this is absurd. □

Theorem 7.3.4 *Let $\{D_j\}_{j=1}^{l}$ be a family of ample divisors on a projective algebraic variety X of dimension n, which is in general position. Assume that*

$$l \geqq 2n + r\big(\{D_j\}_{j=1}^{l}\big),$$

or, more strongly that $l \geqq 2n + \operatorname{rank}_{\mathbf{Z}} \mathrm{NS}(X)$ (cf. (4.9.2)). Then, the open variety $X \setminus \operatorname{Supp} \sum_{j=1}^{l} D_j$ is complete hyperbolic and hyperbolically embedded into X.

Proof We are concerned only in the supports of D_j, and so we use the same D_j for the supports, unless confusion occurs.

Let $\{1, \ldots, l\} = I \cup J$ be any disjoint decomposition of indices with $0 \leqq |I| \leqq n-1$. Let Y' be an irreducible component of $\bigcap_{i \in I} D_i \setminus \bigcup_{j \in J} D_j$, where $Y' = X \setminus \sum_{j=1}^{l} D_j$ in the case of $|I| = 0$, and set $Y = \bar{Y}'$. Then the family $\{D_j|_Y\}_{j \in J}$ of the restricted divisors on Y is in general position. Since $r(\{D_i\}_{i=1}^{l}) \geqq r(\{D_j|_Y\}_{j \in J})$, it is immediate that

$$\begin{aligned} |J| &= l - |I| \geqq 2n - |I| + r\big(\{D_i\}_{i=1}^{l}\big) \\ &\geqq 2\big(n - |I|\big) + r\big(\{D_j|_Y\}_{j \in J}\big) \\ &= 2\dim Y + r\big(\{D_j|_Y\}_{j \in J}\big). \end{aligned}$$

Therefore, by Lemma 7.3.2 there is no non-constant entire curve in Y': $\{D_j\}_{j=1}^{l}$ is in hyperbolic configuration on X. By Theorem 7.2.13 $X \setminus \sum_{j=1}^{l} D_j$ is hyperbolically embedded into X, and the completeness follows from Theorem 7.1.5. □

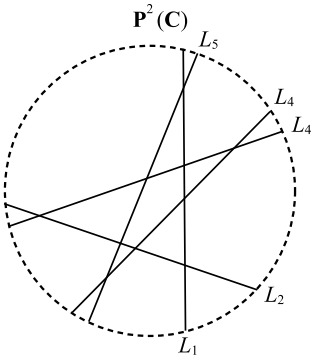


Fig. 7.1 Five lines in hyperbolic configuration

Corollary 7.3.5 (i) *Let* H_i, $1 \leqq i \leqq l$, *be hyperplanes of* $\mathbf{P}^n(\mathbf{C})$ *in general position. If* $l \geqq 2n+1$, *then* $\mathbf{P}^n(\mathbf{C}) \setminus \sum_i H_i$ *is complete Kobayashi hyperbolic and hyperbolically embedded into* $\mathbf{P}^n(\mathbf{C})$.

(ii) *Let* D_i, $1 \leqq i \leqq l$, *be hypersurfaces of* $\mathbf{P}^n(\mathbf{C})$ *in general position. If* $l \geqq 2n+1$, $\mathbf{P}^n(\mathbf{C}) \setminus \sum_i D_i$ *is complete Kobayashi hyperbolic and hyperbolically embedded into* $\mathbf{P}^n(\mathbf{C})$.[2]

In the case above, $r(\{D_i\}) = 1$.

Remark In (i) above it was H. Fujimoto [72a] who first noticed $\mathbf{P}^n(\mathbf{C}) \setminus \sum_{i=1}^{l} D_i$ ($l \geqq 2n+1$) to be Kobayashi hyperbolic. In the same paper he proved the Big Picard Theorem for $\mathbf{P}^n(\mathbf{C}) \setminus \sum_{i=1}^{l} D_i$, which is an important consequence of the hyperbolic embedding. Afterwards the same was proved by M. Green [77]. Before the Kobayashi hyperbolicity was defined, there had been related works from the viewpoint of the extensions of the Big Picard Theorem and the theory of normal families (Bloch [26a]; H. Cartan [28]; Dufresnoy [44]).

As for (ii), Eremenko [99] gave another proof based on the potential theory.

Thus for large l we see at some extent when $X \setminus \sum_{i=1}^{l} D_i$ is Kobayashi hyperbolic. For irreducible $D \subset X$, however, it is not easy to find Kobayashi hyperbolic $X \setminus D$.

Theorem 7.3.6 (Zaidenberg [89]) *If* $d \geqq 5$, *there is a non-singular curve* C *of* $\mathbf{P}^2(\mathbf{C})$ *of degree* d *such that* $\mathbf{P}^2(\mathbf{C}) \setminus C$ *is hyperbolically embedded into* $\mathbf{P}^2(\mathbf{C})$.

Proof We first show the case of $d = 5$. Take five lines $L_i \subset \mathbf{P}^2(\mathbf{C})$, $1 \leqq i \leqq 5$, in general position (Fig. 7.1).

[2]This statement was claimed in Babets, V.A., Picard-type theorems for holomorphic mappings, Siberian Math. J. **25** (1984), 195–200. In Theorem 1 the case of $f(\mathbf{C}) \subset D$ by the notation there was not dealt with, and it was stated that the quasi-Albanese variety is the product of the Albanese variety and $(\mathbf{C}^*)^t$; this is not true in general.

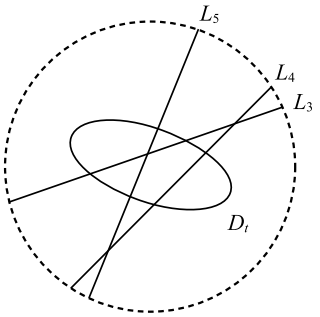


Fig. 7.2 Merging two lines to a quadric D_t: Keeping hyperbolic configuration

By Corollary 7.3.5 (i), $\mathbf{P}^2(\mathbf{C})\backslash\sum L_i$ is hyperbolically embedded into $\mathbf{P}^2(\mathbf{C})$. Keeping L_3, L_4, L_5, we deform L_1 and L_2 to a non-singular quadric by a deformation family D_t, $D_0 = L_1 + L_2$ (Fig. 7.2).

We want to show that for some $\delta > 0$, $\{D_t, L_3, L_4, L_5\}$ is in hyperbolic configuration for $|t| < \delta$. Otherwise, there is a sequence of points $t_\nu \to 0, t_\nu \neq 0$ such that $\{D_{t_\nu}, L_3, L_4, L_5\}$ is not in hyperbolic configuration. Since every member of D_{t_ν}, L_3, L_4, L_5 is isomorphic to $\mathbf{P}^1(\mathbf{C})$ and intersects all the other members at distinct three or more points, Theorem 7.2.13 implies the existence of a sequence of non-constant entire curves

$$f_\nu : \mathbf{C} \to \mathbf{P}^2(\mathbf{C})\backslash(D_{t_\nu} + L_3 + L_4 + L_5), \quad \nu = 1, 2, \ldots.$$

By Lemma 7.2.1 there is a subsequence,

$$g_\nu : \Delta(\nu) \to \mathbf{P}^2(\mathbf{C})\backslash(D_{t_\nu} + L_3 + L_4 + L_5), \quad \nu = 1, 2, \ldots$$

which is convergent uniformly on compact subsets to a non-constant entire curve $g : \mathbf{C} \to \mathbf{P}^2(\mathbf{C})$. We consider the divisor Θ on $\mathbf{P}^2(\mathbf{C}) \times \Delta(\delta)$ given by $D_t = \Theta \cap (\mathbf{P}^2(\mathbf{C}) \times \{t\})$, $|t| < \delta$. By Hurwitz' Theorem ("The limit of zero-free holomorphic functions convergent uniformly on compact subsets is either zero-free or constantly vanishing") either $g(\mathbf{C}) \subset \Theta_0$ or $g(\mathbf{C}) \cap \Theta_0 = \emptyset$ for every irreducible component Θ_0 of Θ. In our situation this implies that $g(\mathbf{C}) \subset L_1 + L_2$. Therefore either $g(\mathbf{C}) \subset L_1$ or $g(\mathbf{C}) \subset L_2$. Since $g(\mathbf{C}) \cap (L_3 + L_4 + L_5) = \emptyset$, g has an image contained in $\mathbf{P}^1(\mathbf{C})$ minus three distinct points. This is absurd.

We take a small $t \neq 0$ such that $D_1 = D_t$ is non-singular and $\{D_1, L_3, L_4, L_5\}$ is in hyperbolic configuration. Next we deform $L_3 + L_4$ to a non-singular quadric. By the same argument as above there is a small deformation of $L_3 + L_4$ which is a non-singular quadric D_2, such that $\{D_1, D_2, L_5\}$ is in hyperbolic configuration (Fig. 7.3). Take a point $\mathrm{P} \in D_2 \cap L_5$. We take a small deformation family of elliptic curves $E_t, E_0 = D_2 + L_5$ of $D_2 + L_5$ fixing P such that every E_t, $t \neq 0$ has a node at P and non-singular at all the other points (Fig. 7.4). We claim that for a very small $t \neq 0$, $\{D_1, E_t\}$ is a hyperbolic configuration. Deny the claim. Then the same argument as above yields a non-constant entire curve $f : \mathbf{C} \to \mathbf{P}^2(\mathbf{C})$ such that either $f(\mathbf{C}) \subset D_1\backslash(D_2 + L_5)$ or $f(\mathbf{C}) \subset (D_2 + L_5)\backslash D_1$. Since D_1 and

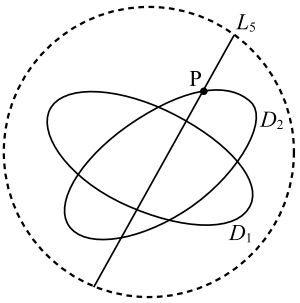


Fig. 7.3 Merging other two lines to a quadric D_2: Keeping hyperbolic configuration

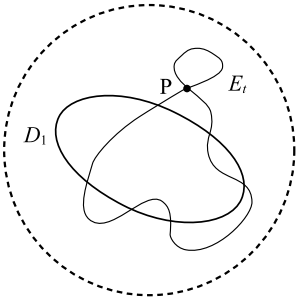


Fig. 7.4 Merging the last line and D_2 to an elliptic E_t: Keeping hyperbolic configuration

$D_2 + L_5$ intersect each other at distinct six points, it is necessary to have that $f(\mathbf{C}) \subset (D_2 + L_5) \backslash D_1$. Since D_2 and D_1 intersect each other at distinct four points, it does not happen that $f(\mathbf{C}) \subset D_2$. Therefore $f(\mathbf{C}) \subset L_5 \backslash D_1$ is the only remaining case. It is noted that L_5 and D_2 intersect each other at two points other than P. If there is a point $z_0 \in \mathbf{C}$ with $f(z_0) = \mathrm{P}$, the construction of f and Hurwitz' Theorem imply that $f(\mathbf{C}) \subset D_2 \cap L_5$. Thus f must be constant; this is a contradiction. We see that f omits P. Then $f(\mathbf{C})$ is contained in a domain of $\mathbf{P}^1(\mathbf{C})$ minus distinct three points, and so a contradiction follows again. In this way we obtain a cubic D_3 carrying a node at $\mathrm{P} \notin D_1$ and non-singular at all other points such that $D_1 + D_3$ has only normal crossings at intersections and $\{D_1, D_3\}$ is in hyperbolic configuration.

We choose three points $\mathrm{Q}_1, \mathrm{Q}_2, \mathrm{Q}_3 \in D_1 \cap D_3$. Because a line can intersect D_1 in only two points, these three points are not on one line. Consider the quintic W' defined by $x^3z^2 - y(y^2 - x^2)^2 = 0$. It has three singularities, namely $[1, -1, 0]$, $[1, 1, 0]$ and $[0, 0, 1]$. Since these three points are not collinear, there is an automorphism ϕ of $\mathbf{P}^2(\mathbf{C})$ moving $\mathrm{Q}_1, \ldots, \mathrm{Q}_3$ to the singularities of W'. Therefore $D_1 \cup D_3$ can be deformed into a quintic W_t with nodal singularities at $\mathrm{Q}_1, \ldots, \mathrm{Q}_3$ (Fig. 7.5). We observe that the local irreducible components of W_t nearby each Q_i correspond to the irreducible components of $D_1 \cup D_2$. Using this fact and similar arguments as before we may deduce that there is a $t > 0$ such that $\mathbf{P}^2(\mathbf{C}) \setminus W_t$ is hyperbolic. The genus of the normalization of W_t can be determined as $g = (5-1)(5-2)/2 - 3 =$

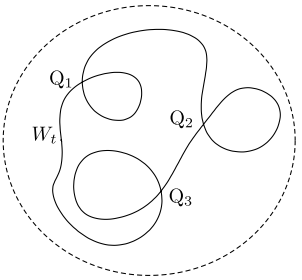


Fig. 7.5 Quintic W_t with three nodes in hyperbolic configuration

$3 \geq 2$, because W_t is defined by a polynomial of degree 5 and has 3 nodal singularities (and no other singularities). Thus both W_t and its complement are hyperbolic.

This fact in turn implies that for each small deformation of $\hat{W}$ of W_t both $\hat{W}$ and $\mathbf{P}^2(\mathbf{C}) \setminus \hat{W}$ are hyperbolic. Since a generic deformation of any hypersurface is smooth, we have completed the proof for the case $d = 5$.

Finally, in the case of $d > 5$ we have the non-singular quintic F in hyperbolic configuration obtained above and moreover $d-5$ lines L_i, $1 \leqq i \leqq d-5$ in general position, such that $F + \sum L_i$ is of simple normal crossing type. We take a small deformation family C_t, $C_0 = F + \sum L_i$ of $F + \sum L_i$ such that C_t are non-singular for $t \neq 0$. Suppose that there is no small $t \neq 0$ with $\{C_t\}$ in hyperbolic configuration. Then there is a non-constant entire curve $h : \mathbf{C} \to \mathbf{P}^2(\mathbf{C})$ such that $h(\mathbf{C}) \subset F + \sum L_i$. Since F is Kobayashi hyperbolic, there is some L_i with $h(\mathbf{C}) \subset L_i \setminus F$. Since the number of the intersection points of L_i and F is five, h must be constant, and hence we get a contradiction. Thus it is proved that there exists a non-singular curve $C \subset \mathbf{P}^2(\mathbf{C})$ of degree d such that $\mathbf{P}^2(\mathbf{C}) \setminus C$ is complete Kobayashi hyperbolic and hyperbolically embedded into $\mathbf{P}^2(\mathbf{C})$. □

The above deformation from $L_1 + \cdots + L_5$ to C_t is very elaborate, and it does not suffice to take an arbitrary deformation to a non-singular curve as the following example shows.

Remark 7.3.7 (Zaidenberg [89]) Let x_0, x_1, x_2 be the homogeneous coordinates of $\mathbf{P}^2(\mathbf{C})$, and define $L_1 + \cdots + L_5$ by

$$P(x) = x_0 x_1 x_2 (x_0 + x_1 + x_2)(x_0 + 2x_1 + 3x_2) = 0.$$

Take a non-singular quintic

$$Q(x) = x_0^5 + x_1^4 x_2 + x_2^5 = 0,$$

and set $P_t(x) = P(x) + tQ(x)$, $t \in \mathbf{C} \cup \{\infty\}$. For every t the line $L = \{x_2 = 0\}$ intersects $C_t = \{P_t = 0\}$ only at one point $a = (0, 1, 0)$. That is,

$$\mathbf{P}^2(\mathbf{C}) \setminus C_t \supset L \setminus \{a\} \cong \mathbf{C}.$$

Thus $\mathbf{P}^2(\mathbf{C}) \setminus C_t$ cannot be Kobayashi hyperbolic.

Concerning hyperbolicity of subspaces of a given complex manifolds, a rather well understood case is that of compact complex tori. This is shown in the theorem below.

Theorem 7.3.8 (Green [78]) *Let A be a compact complex torus.*

(i) *A closed complex subspace X of A is Kobayashi hyperbolic if and only if X contains no translation of a positive dimensional complex subtorus.*
(ii) *Let D be a reduced divisor on A. If D contains no translation of a positive dimensional complex subtorus of A, then $A\backslash D$ is hyperbolically embedded into A.*

Proof (i) Suppose that X is not Kobayashi hyperbolic. By Theorem 7.2.8 there exists a non-constant entire curve $f : \mathbf{C} \to X$. By Theorem 6.2.1 the Zariski closure of the image of f is a translation of a positive dimensional abelian subvariety of A. This is absurd.

(ii) Theorem 6.2.1 and the assumption imply that every entire curve $f : \mathbf{C} \to D$ is constant. By Theorem 7.2.13 it suffices to show that an entire curve $f : \mathbf{C} \to A\backslash D$ is always a constant one. The condition on D implies that $\mathrm{St}(D)^0 = \{0\}$. Hence Corollary 6.3.5, (ii) implies the constancy of f. □

Remark 7.3.9 In the above proof we used a general result obtained in Chap. 6, namely Theorem 6.2.1, which states that the Zariski closure of the image of an arbitrary holomorphic map from $\mathbf{C}$ to a compact complex torus must be a translate of a subtorus.

There is a more elementary way to prove the above Theorem 7.3.8 (used by Green [78]). Namely, for showing (i) it suffices to consider only entire curves which are given as translates of one-parameter subgroups. This is seen as follows. By making use of Lemma 7.2.1 the entire curve $f : \mathbf{C} \to X$ in Theorem 7.2.8 has a bounded derivative $\|f_*(\partial/\partial z)_z\|$. The lift of such a $f : \mathbf{C} \to (X \hookrightarrow)A$ to the universal covering space $\mathbf{C}^{\dim A}$ of A must be an affine-linear function. Since the Zariski closure of a one-parameter subgroup is an abelian subvariety, (i) follows immediately. In (ii) we have, however, to prove that a translation of a one-parameter subgroup necessarily intersects D (Ax [72]). Hence the proof of (ii), unlike that of (i), allows no such simplification.

Example 7.3.10 An ample divisor D may contain a translation of a positive dimensional abelian subvariety. This may happen even if D is irreducible as can be seen by the following example which is due to Ueno (Ochiai [77]). Let E_j, $1 \leqq j \leqq 4$, be non-singular elliptic curves of degree 3 of $\mathbf{P}^2(\mathbf{C})$. Set

$$A = E_1 \times \cdots \times E_4 \subset \left(\mathbf{P}^2(\mathbf{C})\right)^4.$$

Let $[u_{j0}, u_{j1}, u_{j2}]$ be the homogeneous coordinate system of each $\mathbf{P}^2(\mathbf{C})$. Let $H_j(u_{10}, u_{11}, u_{12}, \ldots, u_{32})$, $0 \leqq j \leqq 2$, be a homogeneous polynomial of degree l_j

in each (u_{j0}, u_{j1}, u_{j2}), and set

$$D = \left\{ (\ldots, [u_{j0}, u_{j1}, u_{j2}], \ldots) \in \left(\mathbf{P}^2(\mathbf{C})\right)^4; \sum_{j=0}^{2} H_j \cdot u_{j4}^p = 0 \right\}.$$

If H_j are generally taken, D is non-singular, and by Lefschetz' Theorem $q(D) = 4$. Writing

$$\{H_j = 0; 0 \leqq j \leqq 2\} \cap (E_1 \times E_2 \times E_3) = \{p_1, \ldots, p_s\},$$

we have $\bigcup_{i=1}^{s} \{p_i\} \times E_4 \subset D$.

We do not know whether $A \setminus D$ is Kobayashi hyperbolic.

As for deformations we know the following theorem (see also Theorem 7.2.15 above).

Theorem 7.3.11 *Let X be an m-dimensional complex projective algebraic manifold and let D_i, $1 \leqq i \leqq l$, be ample reduced divisors on X in general position. Assume that all the divisors D_i are hyperbolic and $l \geqq 2m + r(\{D_i\})$. Let $E_t, t \in \Delta(1)$, be a local deformation family of the divisor $E_0 = \sum_{i=1}^{l} D_i$.*

Then there is a small $\delta > 0$ such that for every $t \in \Delta(\delta)$ the divisor E_t is Kobayashi hyperbolic and $X \setminus E_t$ is hyperbolically embedded into X.

Proof We first show that for an enough small arbitrary t the complex subspace E_t is Kobayashi hyperbolic. Indeed, $E_0 = \sum D_i$ is hyperbolic by assumption and the set of all t for which E_t is hyperbolic forms an open set due to Theorem 7.2.15.

Next assume that there is a non-constant entire curve $f_\nu : \mathbf{C} \to X \setminus E_{t_\nu}$. Then, in the same way one gets a non-constant entire curve $g : \mathbf{C} \to X \setminus \sum_{i=1}^{l} D_i$. Since $l \geqq 2m + r(\{D_i\})$, this contradicts Theorem 7.3.4. □

The above theorem means that if the boundary of $M = X \setminus \sum D_i$ is Kobayashi hyperbolic, the property of being hyperbolically embedded is locally stable (cf. Zaidenberg [89]).

7.4 Kobayashi Hyperbolic Projective Hypersurfaces

In this section we deal with the Kobayashi Conjecture 7.1.8 (i).

In dimension one this conjecture is easily seen to be true. Indeed, let X be a non-singular curve of $\mathbf{P}^2(\mathbf{C})$ of degree d. Then the genus of X is $g = (d-1)(d-2)/2$. Therefore if $d = 1, 2$, X is isomorphic to $\mathbf{P}^1(\mathbf{C})$, and if $d = 3$, X is an elliptic curve. In these cases, X is not Kobayashi hyperbolic. If $d \geqq 4$, $g \geqq 3$ and so X is Kobayashi hyperbolic.

For $n = 3$, Brody–Green [77] gave an example of such projective hypersurface of even degree $\geqq 50$, and Demailly–El Goul [00] solved affirmatively the conjecture for a large degree.

In general it is already rather difficult to find a projective hypersurface that contains no rational curve nor elliptic curve, and at least the existence of such a hypersurface was proved by Voisin [96]. Here the problem is more non-trivial since transcendental entire curves $f : \mathbf{C} \to X$ must be dealt with. In the present section we describe the construction of Kobayashi hyperbolic projective hypersurface obtained in Masuda–Noguchi [96]. Afterwards those of lower degrees were obtained (Shirosaki [98]; Fujimoto [01]; Shiffman–Zaidenberg [02a], [02b]; etc.). The lowest degree for which a hyperbolic projective hypersurface in $\mathbf{P}^3(\mathbf{C})$ has been obtained so far is *six* by Duval [04].

The examples obtained in this section carry a finiteness property of unit points of an algebraic number field, and so there is some interest from the arithmetic viewpoint.

We begin with some algebraic preparation. We consider finitely many monomials

$$M_j(z_1, \ldots, z_n) = z_1^{\alpha_{j1}} \cdots z_n^{\alpha_{jn}}, \quad 1 \leqq j \leqq s.$$

Here the exponents α_{jk} are non-negative rationals. For a moment we normalize the degree to be 1:

$$\alpha_{jk} \in \mathbf{Q}, \quad \alpha_{jk} \geqq 0, \quad \sum_{k=1}^{n} \alpha_{jk} = 1.$$

If $\alpha_{j\lambda} > 0$ and $z_\lambda = 0$ is substituted, then we set $M_j = 0$.

Definition 7.4.1 A finite family $\{M_j(z_1, \ldots, z_n)\}_{j=1}^{s}$ of monomials is said to be *admissible* if for arbitrary indices $1 \leqq j_\nu < k_\nu \leqq s$, $1 \leqq \nu \leqq l$, satisfying

$$1 \leqq j_1 < j_2 < \cdots < j_l \leqq s,$$
$$\{j_1, \ldots, j_l, k_1, \ldots, k_l\} = \{1, \ldots, s\} \quad \text{(as sets)}$$

the matrix

$$\begin{pmatrix} \alpha_{j_1 1} - \alpha_{k_1 1} & \cdots & \alpha_{j_l 1} - \alpha_{k_l 1} \\ \vdots & & \vdots \\ \alpha_{j_1 n} - \alpha_{k_1 n} & \cdots & \alpha_{j_l n} - \alpha_{k_l n} \end{pmatrix} \tag{7.4.2}$$

has rank$(n-1)$.

Let $\{z_{\lambda_1}, \ldots, z_{\lambda_l}\}$ be a subset of variables. We put $z_\nu = 0$ for the variables not contained in the above subset, and then the monomials containing that variable with positive exponent reduces to 0. Excluding 0 monomial, we obtain the subset of the remaining monomials

$$\left\{M'_j(z_{\lambda_1}, \ldots, z_{\lambda_l})\right\}. \tag{7.4.3}$$

Definition 7.4.4 (i) If the family (7.4.3) is admissible as monomials of variables $(z_{\lambda_1}, \ldots, z_{\lambda_l})$, $\{M_j\}$ is said to be admissible with respect to $(z_{\lambda_1}, \ldots, z_{\lambda_l})$.

(ii) $\{M_j(z_1,\dots,z_n)\}_{j=1}^s$ is said to be *k-admissible* if it is admissible with respect to arbitrarily chosen k variables $(z_{\lambda_1},\dots,z_{\lambda_l})$, $1 \leqq l \leqq k$.

Lemma 7.4.5 *For a given family $\{M_j(z_1,\dots,z_n)\}_{1\leqq j\leqq s}$ of monomials of degree 1 there is a family $\{N_k(z_1,\dots,z_n)\}_{1\leqq k\leqq t}$ of monomials with positive exponents to all variables such that $\{M_1,\dots,M_s,N_1,\dots,N_t\}$ is admissible.*

Proof We choose, inductively on t, monomials $N_k(z_1,\dots,z_n) = z_1^{\beta_{k1}}\cdots z_n^{\beta_{kn}}$, $1 \leqq k \leqq t$, such that for any chosen indices $1 \leqq k_j < l_j \leqq t$ $(1 \leqq j \leqq \lambda \leqq t)$, where k_j, $1 \leqq j \leqq \lambda$, are distinct,

$$\operatorname{rank}\begin{pmatrix} \beta_{k_1 1}-\beta_{l_1 1} & \cdots & \beta_{k_\lambda 1}-\beta_{l_\lambda 1} \\ \vdots & & \vdots \\ \beta_{k_1 n}-\beta_{l_1 n} & \cdots & \beta_{k_\lambda n}-\beta_{l_\lambda n} \end{pmatrix}$$
$$= \operatorname{rank}\begin{pmatrix} \beta_{k_1 1}-\beta_{l_1 1} & \cdots & \beta_{k_\lambda 1}-\beta_{l_\lambda 1} \\ \vdots & & \vdots \\ \beta_{k_1 n-1}-\beta_{l_1 n-1} & \cdots & \beta_{k_\lambda n-1}-\beta_{l_\lambda n-1} \end{pmatrix}$$

is maximal.

The case of $t = 1$ is trivial. Assume that $t-1$ monomials N_k are chosen as above. We are going to choose the t-th monomial N_t so that the stated condition is satisfied. By the induction hypothesis it suffices to show that for indices with $\{k_1,\dots,k_\lambda,l_1,\dots,l_{\lambda-1}\} \not\ni t$ the following matrix has maximal rank:

$$\begin{pmatrix} \beta_{k_1 1}-\beta_{l_1 1} & \cdots & \beta_{k_{\lambda-1} 1}-\beta_{l_{\lambda-1} 1} & \beta_{k_\lambda 1}-\beta_{t1} \\ \vdots & & \vdots & \vdots \\ \beta_{k_1 n-1}-\beta_{l_1 n-1} & \cdots & \beta_{k_{\lambda-1} n-1}-\beta_{l_{\lambda-1} n-1} & \beta_{k_\lambda n-1}-\beta_{tn-1} \end{pmatrix}. \tag{7.4.6}$$

By the induction hypothesis the matrix consisting of the first $(\lambda-1)$ columns of (7.4.6) is maximal. Therefore there are finitely many affine hyperplanes of $\mathbf{Q}^{n-1}$ such that for a rational vector $(\beta_{t1},\dots,\beta_{tn-1})$ taken outside them, satisfying $\beta_{tj} > 0$, $\sum_{j=1}^{n-1}\beta_{tj} < 1$, the matrix (7.4.6) has maximal rank. Thus we put $\beta_{tn} = 1 - \sum_{j=1}^{n-1}\beta_{tj} > 0$.

Now we take $t = s + 2(n-1)$. Then the matrix (7.4.2) formed from the exponent vectors of $\{M_j, N_k\}$ contains at least the following $(n-1)$ vector of type

$$\begin{pmatrix} \beta_{k_\nu 1}-\beta_{l_\nu 1} \\ \vdots \\ \beta_{k_\nu n}-\beta_{l_\nu n} \end{pmatrix}, \quad k_\nu < l_\nu.$$

These are linearly independent. Therefore $\{M_j, N_k\}$ is an admissible family. □

Lemma 7.4.7 *Let $\{M_j(z_1,\dots,z_n)\}_{j=1}^{s}$ be a k-admissible family of monomials of degree 1. Suppose that $k \leqq n-1$. Then there are monomials $N_k(z_1,\dots,z_n)$, $1 \leqq k \leqq t$, of degree 1 such that $\{M_1,\dots,M_s,N_1,\dots,N_t\}$ is $(k+1)$-admissible.*

Proof Take arbitrary $(k+1)$ variables $z_{\lambda_1},\dots,z_{\lambda_{k+1}}$. As in (7.4.3) we get $\{M'_j(z_{\lambda_1},\dots,z_{\lambda_{k+1}})\}_{j=1}^{s'}$. By Lemma 7.4.5 there are monomials $N'_k(z_{\lambda_1},\dots,z_{\lambda_{k+1}})$ of degree 1 with all positive rational exponents such that $\{M'_j,N'_k\}$ is admissible with respect to $z_{\lambda_1},\dots,z_{\lambda_{k+1}}$. Therefore $\{M_j,N'_k\}$ is k-admissible, and admissible with respect to $(z_{\lambda_1},\dots,z_{\lambda_{k+1}})$.

For other $(k+1)$ variables $z_{\mu_1},\dots,z_{\mu_{k+1}}$ we get $\{M''_j\}$ as $\{M'_j\}$. In the same way as above we take a family $\{N''_k\}$ of monomials such that $\{M''_j,N''_k\}$ is admissible with respect to the variables $z_{\mu_1},\dots,z_{\mu_{k+1}}$. Then $\{M_j,N'_k,N''_k\}$ is k-admissible and admissible with respect to the variables $(z_{\lambda_1},\dots,z_{\lambda_{k+1}})$ and $(z_{\mu_1},\dots,z_{\mu_{k+1}})$. Therefore, repeating this procedure, we obtain a family of monomials that is admissible with respect to all sets of $(k+1)$ variables, and so which is $(k+1)$-admissible. □

The family $\{z_1,\dots,z_n\}$ of monomials is clearly 2-admissible. Applying Lemma 7.4.7 to this, we obtain the following.

Theorem 7.4.8 *Let $n \geqq 2$ be an arbitrary natural number. Then there exists an n-admissible family $\{M_j(z_1,\dots,z_n)\}_{j=1}^{s}$ of monomials of degree 1.*

Let $M_j = z_1^{a_{j1}}\cdots z_n^{a_{jn}}$, $1 \leqq j \leqq s$, be distinct monomials of degree d. Let $\{1,\dots,s\} = \bigcup_{\nu=1}^{t} I_\nu$ be a partition of indices. Suppose that every $|I_\nu| \geqq 2$. Let I_ν consist of

$$j_{\nu 1} < \cdots < j_{\nu p_\nu}.$$

Then we form the following pairs:

$$j_{\nu\mu} < j_{\nu p_\nu}, \quad 1 \leqq \mu \leqq p_\nu - 1.$$

We denote by $\{j_\tau < k_\tau\}_{\tau=1}^{l}$ the set of the pairs formed from all I_ν in this way. By making use of these, we define the following matrix:

$$R\big(\{M_j\};\{I_\nu\}\big) = \begin{pmatrix} a_{j_1 1} - a_{k_1 1} & \cdots & a_{j_l 1} - a_{k_l 1} \\ \vdots & & \vdots \\ a_{j_1 n} - a_{k_1 n} & \cdots & a_{j_l n} - a_{k_l n} \end{pmatrix}. \tag{7.4.9}$$

Taking an arbitrary choice of indices $1 \leqq \lambda_1 < \cdots < \lambda_{n'} \leqq n$ we set $z_i = 0$ for the indices i with $i \notin \{\lambda_\nu\}_{\nu=1}^{n'}$. Then, excluding the zero monomial from $\{M_j(z_1,\dots,z_n)\}_{1\leqq j\leqq s}$, we get a family of non-zero monomials $\{M_{j_k}(z_{\lambda_1},\dots,z_{\lambda_{n'}})\}_{1\leqq k\leqq s'}$. For a partition $\{1,\dots,s'\} = \bigcup_{\xi=1}^{t} I'_\xi$, $|I'_\xi| \geqq 2$ we set

$$R\big(\{M_j\};\{\lambda_\nu\}_{\nu=1}^{n'},\{I'_\xi\}\big) = R\big(\{M_{j_k}\};\{I'_\xi\}\big).$$

Now let $M_j = z_1^{a_{j1}} \cdots z_n^{a_{jn}}$, $a_{jk} \in \mathbf{Z}^+$, $1 \leqq j \leqq s$, be distinct monomials of degree l, and let d be a natural number. Let X be a hypersurface of $\mathbf{P}^n(\mathbf{C})$ defined by the next equation:

$$X: \quad c_1 M_1^d + \cdots + c_s M_s^d = 0, \quad c_j \in \mathbf{C}^*. \tag{7.4.10}$$

Lemma 7.4.11 *Let the notation be as above. Assume that $d > s(s-2)$. Then X is not Kobayashi hyperbolic if and only if there are some indices $1 \leqq \lambda_1 < \cdots < \lambda_{n'} \leqq n$, inducing the above-formed family of monomials $\{M_{j_k} = z_{\lambda_1}^{a_{j_k\lambda_1}} \cdots z_{\lambda_{n'}}^{a_{j_k\lambda_{n'}}};\ 1 \leqq k \leqq s'\}$, and a partition of indices $\{1, \ldots, s'\} = \bigcup I'_\xi$, $|I'_\xi| \geqq 2$ satisfying*

$$\operatorname{rank} R\big(\{M_j\}; \{\lambda_\nu\}_{\nu=1}^{n'}, \big\{I'_\xi\big\}\big) < n' - 1 \tag{7.4.12}$$

and constants $A_{\lambda_\nu} \in \mathbf{C}^*$, $1 \leqq \nu \leqq n'$, such that for I'_ξ

$$\sum_{k \in I'_\xi} c_{j_k} \big(A_{\lambda_1}^{a_{j_k\lambda_1}} \cdots A_{\lambda_{n'}}^{a_{j_k\lambda_{n'}}}\big)^d = 0. \tag{7.4.13}$$

Proof Assume that X is not Kobayashi hyperbolic. It follows from Theorem 7.2.8 that there is a non-constant entire curve

$$f = [f_1, \ldots, f_n] : \mathbf{C} \to X.$$

Here f_i are entire functions without common zero. For the sake of simplicity we may assume without loss of generality the order of indices such that $f_i \not\equiv 0$, $1 \leqq i \leqq n'$, $f_i \equiv 0$, $n' < i \leqq n$. For the indices $\lambda_\nu = \nu$, $1 \leqq \nu \leqq n'$ we get the above-formed family of monomials $\{M_{j_k}(z_1, \ldots, z_{n'})\}_{k=1}^{s'}$. Hence we have

$$c_{j_1} M_{j_1}^d \circ f + \cdots + c_{j_{s'}} M_{j_{s'}}^d \circ f = 0.$$

By Theorem 4.2.16 there is a partition of indices, $\{1, \ldots, s'\} = \bigcup I'_\xi$, $\#I'_\xi \geqq 2$ such that for every $i, k \in I'_\xi$

$$M_{j_i}^d \circ f(z) = b_{ik} M_{j_k}^d \circ f(z), \quad b_{ik} \in \mathbf{C}^*.$$

Therefore the vector $(\log f_1(z), \ldots, \log f_{n'}(z))$ is a solution of the following linear equations:

$$(y_1, \ldots, y_{n'}) R\big(\big\{M_j^d\big\}; \{\nu\}_{\nu=1}^{n'}, \big\{I'_\xi\big\}\big) = (\ldots, \log b_{ik}, \ldots). \tag{7.4.14}$$

By the construction the vector $(1, \ldots, 1)$ is always a solution of

$$(y_1, \ldots, y_{n'}) R\big(\big\{M_j^d\big\}; \{\nu\}_{\nu=1}^{n'}, \big\{I'_\xi\big\}\big) = (0, \ldots, 0). \tag{7.4.15}$$

Thus the rank r_0 of $R(\{M_j^d\}; \{\nu\}_{\nu=1}^{n'}, \{I'_\xi\})$ is at most $n' - 1$. If $r_0 = n' - 1$, then every f_i is a constant multiple of f_1, and so f is constant; this is absurd. We see

$r_0 < n' - 1$. Let $(p_1, \ldots, p_{n'})$ be a non-trivial solution of (7.4.14) and set $A_\nu = e^{p_\nu}$, $1 \leqq \nu \leqq n'$. Then by Theorem 4.2.16 (iii) we see that $\{A_\nu\}$ satisfies (7.4.13).

We show the converse. For the sake of simplicity we assume that $1 \leqq \lambda_\nu = \nu \leqq n'$. Taking a solution $(q_1, \ldots, q_{n'})$ of (7.4.15) linearly independent of $(1, \ldots, 1)$, we set

$$\begin{aligned} &f_\nu(z) = A_\nu e^{q_\nu z}, \quad 1 \leqq \nu \leqq n', \\ &f_\nu(z) = 0, \quad n' < \nu \leqq n. \end{aligned} \tag{7.4.16}$$

Then we get a non-constant entire curve $f = [f_1, \ldots, f_n]$ with values in X, and hence X is not Kobayashi hyperbolic. □

Let Z be the set of all points

$$\left\{\left((c_j), (A_i)\right) \in \left(\mathbf{C}^*\right)^s \times \mathbf{P}^{n-1}(\mathbf{C})\right\}$$

defined by (7.4.13) associated with a partition of indices of those variables appearing in Lemma 7.4.11. Then Z is an algebraic subset. Let $\pi : (\mathbf{C}^*)^s \times \mathbf{P}^{n-1}(\mathbf{C}) \to (\mathbf{C}^*)^s$ be the projection to the first factor. We obtain the following algebraic subset:

$$\Sigma = \pi(Z) \subset \left(\mathbf{C}^*\right)^s. \tag{7.4.17}$$

For a given family $\{M_j(z_1, \ldots, z_n) = z_1^{\alpha_{j1}} \cdots z_n^{\alpha_{jn}}\}_{j=1}^s$ of monomials of degree 1 we take the smallest natural number l such that all $l\alpha_{j\nu}$ are integral.

Let X be the hypersurface of $\mathbf{P}^{n-1}(\mathbf{C})$ of degree l defined by

$$X: \quad c_1 M_1^{ld} + \cdots + c_s M_s^{ld} = 0, \quad c_j \in \mathbf{C}^*, \ d \in \mathbf{Z}, \ d > 0. \tag{7.4.18}$$

We take the algebraic subset $\Sigma \subset (\mathbf{C}^*)^s$ defined by (7.4.17) with respect to $\{M_j\}$.

Theorem 7.4.19 *Let the notation be as above. Assume that $d > s(s-2)$.*

(i) *X is Kobayashi hyperbolic if and only if $(c_j) \in (\mathbf{C}^*)^s \setminus \Sigma$.*
(ii) *It is necessary and sufficient for $\Sigma = \emptyset$ that $\{M_j(z_1, \ldots, z_n)\}_{j=1}^s$ is n-admissible.*
(iii) *There is a number $d(n-1)$ such that for every $d \geqq d(n-1)$ there is a Kobayashi hyperbolic hypersurface of $\mathbf{P}^{n-1}(\mathbf{C})$ of degree d.*

Proof There remains to prove only (iii). We are going to show that the set of d such that there exists a Kobayashi hyperbolic hypersurface of $\mathbf{P}^{n-1}(\mathbf{C})$ of degree d forms an additive semi-group. From the proof of the existence of an n-admissible family it follows that for a sufficient large positive number p there is an n-admissible family $\{N_j = z_1^{\alpha_{j1}} \cdots z_n^{\alpha_{jn}}\}_{j=1}^t$ of monomials of degree 1 with non-negative rational exponents which are written as $\alpha_{jk} = q_{jk}/p$, $q_{jk} \in \mathbf{Z}^+$. Then it follows from (ii) above that for $d > d_0 = t(t-2)$ the hypersurface

$$Y: \quad P = N_1^{pd} + \cdots + N_t^{pd} = 0$$

is Kobayashi hyperbolic. For a sufficiently large positive number p' co-prime to p there exists similarly an n-admissible family of monomials of degree 1, $\{N'_j = z_1^{\alpha'_{j1}} \cdots z_n^{\alpha'_{jn}}\}_{j=1}^{t'}$, $\alpha'_{jk} = q'_{jk}/p'$, $q'_{jk} \in \mathbf{Z}^+$. For $d' > d'_0 = t(t-2)$

$$Y' : \quad P' = N_1'^{p'd'} + \cdots + N_{t'}^{p'd'} = 0$$

is Kobayashi hyperbolic. We set

$$E = \left\{ pd + p'd'; d, d' \in \mathbf{Z}, d > d_0, d' > d'_0 \right\}.$$

Since p and p' are co-prime and E is an additive semi-group, there is a number $d(n-1)$ such that

$$\left\{ l \in \mathbf{Z}; l \geqq d(n-1) \right\} \subset E.$$

Take arbitrarily $l \in \mathbf{Z}$, $l \geqq d(n-1)$ and write $l = pd + p'd'$, $d > d_0$, $d' > d'_0$. There are homogeneous polynomials P and P' of degrees d and d' respectively whose zeros define Kobayashi hypersurfaces Y and Y' respectively. (Here the zero-locus is only concerned and the multiplicities do not matter.) Taking an irreducible homogeneous polynomial Q of degree l such that $\{Q = 0\}$ is non-singular, we form a pencil

$$Y_\sigma : \quad (1-\sigma) P P' + \sigma Q = 0, \quad \sigma \in \mathbf{C}.$$

Outside a finite set of points of $\mathbf{C}$ Y_σ is non-singular.

Suppose that there is a sequence of non-zero points $\sigma_\nu \to 0$ such that Y_{σ_ν} is not Kobayashi hyperbolic for every ν. By Theorem 7.2.13 there are non-constant entire curves $f_\nu : \mathbf{C} \to Y_{\sigma_\nu}$. By Lemma 7.2.1 we may assume that $\| f_{\nu *}(\partial/\partial z)_0 \| = 1$, and $\| f_{\nu *}(\partial/\partial z)_z \| \leqq 1$, $z \in \mathbf{C}$. Here $\| \cdot \|$ denotes the length with respect to an arbitrary hermitian metric, e.g., the Fubini–Study metric on $\mathbf{P}^{n-1}(\mathbf{C})$. Therefore $f_\nu : \mathbf{C} \to \mathbf{P}^{n-1}(\mathbf{C})$ form a normal family. Taking a subsequence $\{f_\nu\}$ converging uniformly on compact subsets, we obtain the limit

$$g : \mathbf{C} \to Y \text{ or } Y', \qquad \left\| g_*(\partial/\partial z)_0 \right\| = 1.$$

This contradicts the Kobayashi hyperbolicity of Y or Y'. Therefore there is a $\delta > 0$ such that Y_σ, $0 < |\sigma| < \delta$ are all Kobayashi hyperbolic. □

7.5 Hyperbolic Embedding into Complex Projective Space

In this section we consider the Kobayashi Conjecture 7.1.8 (ii) on complements of hypersurfaces in the projective space. In the preceding section we proved that there is a number $d(n)$ such that for every $d \geqq d(n)$ there is a Kobayashi hyperbolic hypersurface of $\mathbf{P}^n(\mathbf{C})$ of degree d (Theorem 7.4.19).

Here we apply the method of the construction of the preceding section in order to obtain the Kobayashi hyperbolicity of the complements of certain hypersurfaces as well (see Theorem 7.5.4 (iii) below).

In the sequel "monomial" means a monomial with non-negative rational exponents.

Lemma 7.5.1 *There is an n-admissible family $\{M_j(z_1,\dots,z_n)\}_{j=1}^{s}$ of monomials of degree 1 such that $\{M_1,\dots,M_s,z_{n+1}\}$ is $(n+1)$-admissible.*

Proof By Theorem 7.4.8 there is an n-admissible family $\{M_j\}$ with respect to the variables $(z_1,\dots,z_n)$. Here we change the construction a little bit. In Lemma 7.4.5 we can add monomials N_k so that the rank of the matrix (7.4.6) being deleted an arbitrarily chosen one N_j is $n-1$. In this way we modify the constructions in the proofs of Lemma 7.4.8 and Theorem 7.4.9.

Now we arbitrarily choose variables $(z_{\lambda_1},\dots,z_{\lambda_t})$, $\lambda_1<\cdots<\lambda_t$. We put all the other variables $z_\nu=0$ except for z_{n+1}, and then obtain a family $\{M'_1,\dots,M'_{s'},z_{n+1}\}$ from $\{M_j\}$. We set

$$M'_j=z_{\lambda_1}^{\alpha_{j1}}\cdots z_{\lambda_t}^{\alpha_{jt}},\quad 1\leqq j\leqq s'.$$

We take indices $1\leqq j_\nu<k_\nu\leqq s'$, $1\leqq\nu\leqq l-1$, $1\leqq k_l\leqq s'$ so that

$$j_1<\cdots<j_l,$$

with

$$\{j_1,\dots,j_{l-1},k_1,\dots,k_l\}=\{1,\dots,s'\}\quad\text{(as sets)}.$$

By the modified construction the matrix

$$\begin{pmatrix} 0 & \cdots & 0 & 1\\ \alpha_{j_1 1}-\alpha_{k_1 1} & \cdots & \alpha_{j_{l-1}1}-\alpha_{k_{l-1}1} & -\alpha_{k_l 1}\\ \vdots & & \vdots & \vdots\\ \alpha_{j_1 t}-\alpha_{k_1 t} & \cdots & \alpha_{j_{l-1}t}-\alpha_{k_{l-1}t} & -\alpha_{k_l t}\end{pmatrix}$$

has rank t. Therefore $\{M'_1,\dots,M'_{s'},z_{n+1}\}$ is admissible, and $\{M_1,\dots,M_s,z_{n+1}\}$ is $(n+1)$-admissible. □

Let $\{M_1(z_1,\dots,z_n),\dots,M_s(z_1,\dots,z_n),z_{n+1}\}$ be a family of monomials of degree 1. Let l be the least natural number such that the exponents of M_j^l are all integral. Let d be a natural number and define a hypersurface X of $\mathbf{P}^{n-1}(\mathbf{C})$ by

$$X:\quad c_1M_1^{ld}+\cdots+c_sM_s^{ld}=0,\quad c_j\in\mathbf{C}^*. \tag{7.5.2}$$

Let Y be a hypersurface of $\mathbf{P}^n(\mathbf{C})$ defined by

$$Y:\quad c_1M_1^{ld}+\cdots+c_sM_s^{ld}+z_{n+1}^{ld}=0,\quad c_j\in\mathbf{C}^*. \tag{7.5.3}$$

We define an algebraic subset $\Xi_1 \subset (\mathbf{C}^*)^s$ for (7.5.2) as in (7.4.17), and an algebraic subset $\Xi_2 \subset (\mathbf{C}^*)^s$ for (7.5.3) with normalization $c_{s+1} = 1$ as in (7.4.17). Set $\Xi = \Xi_1 \cup \Xi_2$.

Theorem 7.5.4 *Let the notation be as above. Assume that*

$$d > s(s-1). \tag{7.5.5}$$

(i) *X is Kobayashi hyperbolic and $\mathbf{P}^{n-1}(\mathbf{C}) \setminus X$ is hyperbolically embedded into $\mathbf{P}^{n-1}(\mathbf{C})$ iff $(c_j) \in (\mathbf{C}^*)^s \setminus \Xi$.*

(ii) *$\Xi = \emptyset$ iff $\{M_j, z_{n+1}\}$ is $(n+1)$-admissible.*

(iii) *There is a number $d'(n-1)$ such that for every number $d \geqq d'(n-1)$ there is a non-singular Kobayashi hyperbolic hypersurface X of $\mathbf{P}^{n-1}(\mathbf{C})$ of degree d with the complement $\mathbf{P}^{n-1}(\mathbf{C}) \setminus X$ being hyperbolically embedded into $\mathbf{P}^{n-1}(\mathbf{C})$.*

Proof (i) By Theorem 7.4.19 (i) X is Kobayashi hyperbolic iff $(c_j) \notin \Xi_1$. By Theorem 7.2.13 it suffices to show that there is a non-constant entire curve $f : \mathbf{C} \to \mathbf{P}^{n-1}(\mathbf{C}) \setminus X$ iff $(c_j) \in \Xi_2$. Let f such an entire curve. Then $F(z) = \sum_{j=1}^{s} c_j M_j^{ld}(f(z))$ is a zero-free entire function. Setting $f_{n+1} = (-F(z))^{1/ld}$, we have that $g(z) = [f_1(z), \ldots, f_n(z), f_{n+1}(z)] \in Y$. Noting f_{n+1} to be zero-free, we deduce from (7.5.5), Corollary 4.2.15, Theorem 4.2.16, and Lemma 7.4.11 that $(c_j) \in \Xi_2$.

Conversely, if $(c_j) \in \Xi_2$, then the construction in (7.4.16) yields a non-constant entire curve $f(z) = [f_1(z), \ldots, f_{n+1}(z)] \in Y$ such that f_{n+1} is zero-free. Hence $[f_1(z), \ldots, f_n(z)] \in \mathbf{P}^{n-1}(\mathbf{C}) \setminus X$, and so $\mathbf{P}^{n-1}(\mathbf{C}) \setminus X$ is not Kobayashi hyperbolic.

(ii) The proof is similar to that of Theorem 7.4.19 (ii).

(iii) The proof is similar to that of Theorem 7.4.19 (iii). □

Examples In what follows we give examples of Kobayashi hyperbolic projective hypersurfaces obtained by the construction shown in Sects. 7.4 and 7.5.

(a) (Noguchi) In $\mathbf{P}^3(\mathbf{C})$ we set

$$X_1 : \quad z_1^{3d} + \cdots + z_4^{3d} + t(z_1 z_2 z_3)^d = 0, \quad t \in \mathbf{C}^*, \tag{7.5.6}$$
$$d > 7 \ (\deg X_1 = 3d \geqq 24).$$

$$X_2 : \quad z_1^{4d} + \cdots + z_4^{4d} + t(z_1 z_2 z_3 z_4)^d = 0, \quad t \in \mathbf{C}^*, \tag{7.5.7}$$
$$d > 6 \ (\deg = 4d \geqq 28).$$

Here deg stands for degree. These are defined by 4-admissible families. If we simply apply Theorem 7.4.19 to X_1, we have $d > 5 \cdot 3 = 15$; in this case, however, we decrease 15 to 7.

Let $f : \mathbf{C} \to X_1$ be a non-constant entire curve. Let $f = [f_1, \ldots, f_4] : \mathbf{C} \to \mathbf{P}^3(\mathbf{C})$ be a reduced representation. Without loss of generality we may assume that

$f_1 f_2 f_3 \not\equiv 0$. Applying Corollary 4.2.15 to

$$g_1 = \left[f_1^{3d}, \dots, f_4^{3d}\right] : \mathbf{C} \to \mathbf{P}^3(\mathbf{C}),$$

we see that if $d > 7$, $f_1^{3d}, \dots, f_4^{3d}$ are linearly dependent. Let the non-trivial linear relation be

$$c_1 f_1^{3d} + \cdots + c_4 f_4^{3d} = 0, \quad (c_1, \dots, c_4) \neq 0. \tag{7.5.8}$$

Since $3d > 3 \cdot 7 > 4 \cdot 2$ Theorem 4.2.16 implies that either all mutual ratios of $f_1, \dots, f_4$ are constant, or at least two of $f_1, \dots, f_4$ have a constant ratio. In the first case f is a constant map. In the latter case, for instance, we suppose the ratio of f_3 and f_4 is constant. Substituting it into (7.5.8) we get

$$f_1^{3d} + f_2^{3d} + c_3 f_3^{3d} + t(f_1 f_2 f_3)^d = 0, \quad c_3 \in \mathbf{C}. \tag{7.5.9}$$

If $c_3 \neq 0$, we apply the above argument to

$$g_2 = \left[f_1^{3d}, \dots, f_3^{3d}\right] : \mathbf{C} \to \mathbf{P}^2(\mathbf{C}).$$

Then at least two of f_j, $1 \leqq j \leqq 3$, have a constant ratio. This with (7.5.9) implies that every ratio of f_j, $1 \leqq j \leqq 3$, is constant. Lastly, we see by (7.5.6) that every ratios of f_j, $1 \leqq j \leqq 4$, is constant, so that f is constant.

In the same way we deduce that X_2 is Kobayashi hyperbolic.

It follows from the above two examples that

$$d(3) \leqq 54.$$

For a later purpose we add two more examples.

$$X_3 : \quad z_1^{4d} + \cdots + z_4^{4d} + t\left(z_1^2 z_2 z_3\right)^d = 0, \tag{7.5.10}$$
$$d > 6 \ (\deg X_2 = 4d \geqq 28), \ t \in \mathbf{C}^*.$$

This is defined by a 4-admissible family.

$$X_4 : \quad z_1^{2d} + \cdots + z_4^{2d} + t_1(z_1 z_2)^d + t_2(z_2 z_3)^d = 0, \tag{7.5.11}$$
$$d > 6 \cdot 4 = 24 \ (\deg X_4 = 2d \geqq 50), \ (t_1, t_2) \in \left(\mathbf{C}^*\right)^2.$$

The family of monomials used in the above equation is not 4-admissible, but for the parameter (t_j) belonging to some non-empty Zariski open subset X_4 is Kobayashi hyperbolic; in fact it is the case for $t_1 = t_2 = 1$.

(b) For $n \geqq 3$ it is not easy to prove that there exist any Kobayashi hyperbolic hypersurfaces in $\mathbf{P}^n(\mathbf{C})$ at all. Brody–Green [77] were the first to prove that $\mathbf{P}^3(\mathbf{C})$ contains a Kobayashi hyperbolic hypersurface.

For $n \geqq 4$ the existence of a Kobayashi hyperbolic hypersurface was first proved by Masuda–Noguchi [96] with explicit examples in $\mathbf{P}^n(\mathbf{C})$, $n = 4, 5$. Here we describe some of them using the methods described above.

It seems to be hard to check by hand whether the families of monomials used in the sequel are admissible. We used "MATHEMATICA". For readers interested in the programming details the program code is attached at the end of this chapter.

We take a hypersurface of $\mathbf{P}^4(\mathbf{C})$ defined by

$$X_5: \quad z_1^d + \cdots + z_5^d + t_1\left(z_1^2 z_2\right)^{d/3} + t_2\left(z_2^2 z_3\right)^{d/3} + t_3\left(z_3^2 z_4\right)^{d/3} + t_4\left(z_4^2 z_1\right)^{d/3} = 0, \quad t_j \in \mathbf{C}^*,\ d = 3e \geqq 192. \tag{7.5.12}$$

This is not defined by a 5-admissible family. For example, if $t_j = 1$, $1 \leqq j \leqq 4$, then X_5 admits the following non-constant entire curve

$$f(z) = \left[z \exp(\pi i/e), \exp(\pi i/e), z, 1, 0\right] \in X_5, \quad z \in \mathbf{C},$$

where e is the number used in (7.5.12). For general (t_j), however, X_5 is Kobayashi hyperbolic; e.g., X_5 is Kobayashi hyperbolic for $(t_j) = (-1, -1, 1, 1)$.

Note that $\deg X_5$ is a multiple of 3. In view of Theorem 7.4.19 (iii) the following example of degree 4 is of interest:

$$z_1^d + \cdots + z_5^d + t_1\left(z_3 z_4^2 z_5\right)^{d/4} + t_2\left(z_1 z_2^2 z_5\right)^{d/4} + t_3\left(z_1 z_2 z_3^2\right)^{d/4} = 0, \quad t_j \in \mathbf{C}^*,\ d = 4e \geqq 196. \tag{7.5.13}$$

This is Kobayashi hyperbolic for $t_1 = t_2 = t_3 = 1$.

We set

$$X_6: \quad z_1^d + \cdots + z_5^d + t_1\left(z_1^3 z_2\right)^{d/4} + t_2\left(z_2^3 z_3\right)^{d/4} + t_3\left(z_3^3 z_4\right)^{d/4} + t_4(z_4 z_1)^{d/2} = 0, \quad t_j \in \mathbf{C}^*,\ d = 4e \geqq 256. \tag{7.5.14}$$

This is defined by a 5-admissible family. Therefore X_6 is Kobayashi hyperbolic for arbitrary non-zero t_j.

(c) We give now examples of Kobayashi hyperbolic hypersurfaces in $\mathbf{P}^5(\mathbf{C})$:

$$X_7: \quad z_1^d + \cdots + z_6^d + t_1\left(z_1 z_2^3\right)^{d/4} + t_2\left(z_2 z_3^3\right)^{d/4} + t_3\left(z_3 z_4^3\right)^{d/4} + t_4\left(z_4 z_5^3\right)^{d/4} + t_5\left(z_5 z_1^3\right)^{d/4} + t_6(z_1 z_3)^{d/2} + t_7(z_2 z_4)^{d/2} + t_8(z_3 z_5)^{d/2} + t_9(z_4 z_1)^{d/2} = 0, \quad t_j \in \mathbf{C}^*,\ d = 4e \geqq 784. \tag{7.5.15}$$

This is not defined by a 6-admissible family, but for $t_1 = -1$ and other $t_j = 1$ X_7 is Kobayashi hyperbolic.

The next one is neither defined by a 6-admissible family, but X_8 with all $t_j = 1$ is Kobayashi hyperbolic:

$$X_8: \quad z_1^d + \cdots + z_6^d + t_1\left(z_1 z_2^4\right)^{d/5} + t_2\left(z_2^2 z_3^3\right)^{d/5} + t_3\left(z_3^2 z_4^3\right)^{d/5} + t_4\left(z_4 z_5^4\right)^{d/5} + t_5\left(z_5^3 z_1^2\right)^{d/5} + t_6\left(z_1^4 z_3\right)^{d/5} + t_7\left(z_2 z_4^4\right)^{d/5} + t_8\left(z_3^3 z_5^2\right)^{d/5} = 0, \quad t_j \in \mathbf{C}^*,\ d = 5e \geqq 845. \tag{7.5.16}$$

The following is similar; with $t_1 = -1$ and other $t_j = 1$ it is Kobayashi hyperbolic:

(7.5.17)

$$\begin{aligned} X_9: \quad & z_1^d + \cdots + z_6^d + t_1\big(z_1 z_2^3\big)^{d/4} + t_2\big(z_2 z_3^3\big)^{d/4} + t_3\big(z_3 z_4^3\big)^{d/4} + t_4\big(z_4 z_5^3\big)^{d/4} \\ & + t_5\big(z_5 z_1^3\big)^{d/4} + t_6(z_1 z_3)^{d/2} + t_7(z_2 z_4)^{d/2} + t_8(z_3 z_5)^{d/2} + t_9(z_1 z_4)^{d/2} \\ & + t_{10}(z_2 z_5)^{d/2} = 0, \quad t_j \in \mathbf{C}^*,\ d = 4e \geqq 900. \end{aligned}$$

The next is a Kobayashi hyperbolic hypersurface defined by a 6-admissible family:

(7.5.18)

$$\begin{aligned} X_{10}: \quad & z_1^d + \cdots + z_6^d + t_1\big(z_1 z_2^4\big)^{d/5} + t_2\big(z_2 z_3^4\big)^{d/5} + t_3\big(z_3 z_4^4\big)^{d/5} + t_4\big(z_4 z_5^4\big)^{d/5} \\ & + t_5\big(z_5 z_1^4\big)^{d/5} + t_6\big(z_1^2 z_3^3\big)^{d/5} + t_7\big(z_2^2 z_4^3\big)^{d/5} + t_8\big(z_3^2 z_5^3\big)^{d/5} + t_9\big(z_4^2 z_1^3\big)^{d/5} \\ & + t_{10}\big(z_5^2 z_2^3\big)^{d/5} = 0, \quad t_j \in \mathbf{C}^*,\ d = 5e \geqq 1125. \end{aligned}$$

(d) We now give examples as stipulated by Theorem 7.5.4, i.e., examples of Kobayashi hyperbolic hypersurfaces where the complement is hyperbolically embedded.

In $\mathbf{P}^2(\mathbf{C})$ the first example was given by Azukawa–Suzuki [80]. Later Nadel [89b] and Zaidenberg gave others. We have already described the construction of Zaidenberg (see Theorem 7.3.6).

In general dimension the existence was first proved by Masuda–Noguchi [96], to which the following examples are due.

In $\mathbf{P}^3(\mathbf{C})$ we set

$$\begin{aligned} Y_1 &= X_1 \cap \{z_4 = 0\}, \quad \deg Y_1 = 3e \geqq 21 \quad \text{(cf. (7.5.6))}, \\ Y_3 &= X_3 \cap \{z_4 = 0\}, \quad \deg Y_2 = 4e \geqq 24 \quad \text{(cf. (7.5.10))}, \\ Y_4 &= X_4 \cap \{z_4 = 0\}, \quad \deg Y_3 = 2e \geqq 42 \quad \text{(cf. (7.5.11))}. \end{aligned}$$

We identify $\mathbf{P}^2(\mathbf{C})$ with the hyperplane $z_4 = 0$. Then $\mathbf{P}^2(\mathbf{C}) \setminus Y_i$ $(i = 1, 3, 4)$ is complete Kobayashi hyperbolic, and hyperbolically embedded into $\mathbf{P}^2(\mathbf{C})$ (cf. Theorem 7.2.13).

Similarly, we set

$$\begin{aligned} Y_5 &= X_5 \cap \{z_5 = 0\}, \qquad d = 3e \geqq 171, \\ Y_6 &= X_6 \cap \{z_5 = 0\}, \qquad d = 4e \geqq 228. \end{aligned}$$

We identify $\mathbf{P}^3(\mathbf{C})$ with the hyperplane $\{z_5 = 0\}$ of $\mathbf{P}^4(\mathbf{C})$. Then $\mathbf{P}^3(\mathbf{C}) \setminus Y_i$ $(i = 5, 6)$ is complete Kobayashi hyperbolic, and is hyperbolically embedded into $\mathbf{P}^3(\mathbf{C})$. Put

$$\begin{aligned} Y_7 &= X_7 \cap \{z_6 = 0\}, \qquad d = 4e \geqq 732, \\ Y_8 &= X_8 \cap \{z_6 = 0\}, \qquad d = 5e \geqq 785, \\ Y_9 &= X_9 \cap \{z_6 = 0\}, \qquad d = 4e \geqq 844, \\ Y_{10} &= X_{10} \cap \{z_6 = 0\}, \qquad d = 5e \geqq 1055. \end{aligned}$$

These are examples such that $\mathbf{P}^4(\mathbf{C}) \setminus Y_i$ is complete Kobayashi hyperbolic, and is hyperbolically embedded into $\mathbf{P}^4(\mathbf{C})$.

7.6 Brody Curves and Yosida Functions

We begin with the definition.

Definition 7.6.1 Let X be a compact complex space. An entire curve $f : \mathbf{C} \to X$ is called a *Brody curve* if

$$\sup_{z \in \mathbf{C}} \left\| (df)_z \right\|_e < \infty,$$

where $\|(df)_z\|_e$ denotes the operator norm with respect to the Euclidean metric on $\mathbf{C}$ and an arbitrary hermitian metric on X (the choice does not matter due to the compactness of X).

Using this notion, Brody's Theorem 7.2.8 may be reformulated as follows.

Theorem 7.6.2 *For a compact complex manifold the following conditions are equivalent*:

(i) *There exists a non-constant entire curve* $f : \mathbf{C} \to X$.
(ii) *There exists a non-constant Brody curve* $f : \mathbf{C} \to X$.
(iii) *There is a sequence of holomorphic maps* $f_\nu : \Delta(1) \to X$, $\nu = 1, 2, \ldots$ *with*

$$\overline{\lim_{\nu \to \infty}} \left\| (df_\nu)_0 \right\|_e = +\infty.$$

(iv) X *is not Kobayashi hyperbolic*.

Proposition 7.6.3 *Let X be a compact complex algebraic variety and let $f : \mathbf{C} \to X$ be an algebraic morphism. Then f is a Brody curve.*

Proof Due to algebraicity, f extends to a holomorphic map $\tilde{f} : \mathbf{P}^1(\mathbf{C}) \to X$. Let $\| \ \|_{\mathrm{FS}}$ denote the Fubini–Study metric on $\mathbf{P}^1(\mathbf{C})$ and let $\| \ \|_h$ be a hermitian metric on X. Due to compactness of $\mathbf{P}^1(\mathbf{C})$ there is a constant $C > 0$ such that $\|d\tilde{f}(v)\|_h \leq C\|v\|_{\mathrm{FS}}$ for all $z \in \mathbf{P}^1(\mathbf{C})$ and all $v \in \mathbf{T}(\mathbf{P}^1(\mathbf{C}))_z$. Now

$$\|v\|_{\mathrm{FS}} = \frac{1}{1 + |z|^2} \|v\|_e, \quad {}^\forall z \in \mathbf{C},\ v \in \mathbf{T}(\mathbf{C})_z = \mathbf{T}\big(\mathbf{P}^1(\mathbf{C})\big)_z.$$

In particular, $\| \ \|_{\mathrm{FS}} \leq \| \ \|_e$. Hence the statement follows. □

Example 7.6.4 Let $T = \mathbf{C}^g / \Gamma$ be a compact complex torus. An entire curve $f : \mathbf{C} \to T$ is a Brody curve if and only if it lifts to an affine-linear map from $\mathbf{C}$ to $\mathbf{C}^g$.

As for the relation between Brody curves and order functions, we first have the following.

Proposition 7.6.5 *Let $f : \mathbf{C} \to (X, \omega)$ be a Brody curve, where ω denotes some hermitian metric form on X. Then the order ρ_f (cf. (2.5.19)) of f is at most two; in fact, we have a better estimate,*

$$\overline{\lim_{r\to\infty}} \frac{T_f(r,\omega)}{r^2} < \infty.$$

Proof Let $f^*\omega = a(z)\frac{i}{2}dz \wedge d\bar{z}$ $(a(z) \geqq 0)$. The Brody assumption implies that $a(z) \leqq C$ for some constant C. Therefore

$$T_f(r,\omega) \leqq \int_1^r \frac{dt}{t}\int_{\Delta(t)} C\frac{i}{2}dz \wedge \bar{z}d \leqq \frac{\pi C}{2}r^2. \tag{7.6.6}$$

□

In one variable, a meromorphic function f on $\mathbf{C}$ is called a *Yosida function* iff it is a Brody curve considered as holomorphic map from $\mathbf{C}$ to $\mathbf{P}^1(\mathbf{C})$. This class of meromorphic functions was introduced much earlier by Yosida [34] than Brody and has been studied for a long time.

The structure of Yosida functions are rather complicated:

7.6.7 (i) The exponential function e^z is Yosida and has order one.

(ii) $\phi_t : z \mapsto e^z + \frac{z}{tz+1}$ is Yosida iff $t \neq 0$. Thus the condition of being a Yosida function is not a closed condition.

(iii) $ze^z + z$ is not Yosida although it can be written as a product of two Yosida functions (z and $e^z + 1$) as well as sum of two Yosida functions (ze^z and z).

(iv) Weierstrass' p-function $\wp(z)$ is Yosida and has order two.

(v) However, a theorem of Clunie and Hayman [66] states that an entire Yosida function $f : \mathbf{C} \to \mathbf{C}$ has order $\rho_f \leqq 1$.

The assertions (i) and (iv) are easily seen; (ii) and (iii) are due to Winkelmann.[3]

7.6.1 Growth Conditions and Yosida Functions

In general, there is no characterization of Yosida functions or Brody curves by growth conditions on the order function. More precisely, if an upper-bound on the characteristic function is not strong enough to force the entire function to be a polynomial, then it is not strong enough to force the function to be a Yosida function.

Theorem 7.6.8 *Let $\rho : \mathbf{R}^+ \to \mathbf{R}^+$ be an increasing continuous function. Assume that there exists a transcendental meromorphic function $f(z)$ with $T(r, f) \leqq \rho(r)$*

[3] J. Winkelmann, On meromorphic functions which are Brody curves, arXiv:0709.3929.

for all $r \geqq r_0$ *with some* $r_0 \in \mathbf{R}^+$. *Then there exists an entire function* $g : \mathbf{C} \to \mathbf{C}$ *with* $T(r, g) \leqq \rho(r)$ *such that* $g : \mathbf{C} \to \mathbf{P}^1(\mathbf{C})$ *is* not *a Yosida function.*

The remainder of this section is devoted to the proof of this theorem. We follow Winkelmann (see footnote 3, p. 322). We use the notation $f^{\#}$ for the "spherical derivative", i.e.,

$$f^{\#}(z) = \frac{|f'(z)|}{1 + |f(z)|^2}.$$

First we derive an auxiliary result.

Theorem 7.6.9 *Let* $\{a_k\}_{k=1}^{\infty}$ *be a sequence of complex numbers such that there is a number* $\lambda > 1$ *fulfilling*

$$|a_{k+1}| > \lambda |a_k| > 0, \quad {}^{\forall}k \in \mathbf{N}. \tag{7.6.10}$$

Let D *be the divisor defined by* $D = \sum_{k\in\mathbf{N}} \{a_k\}$ (*reduced, i.e. all multiplicities being one*). *Let* H *denote the convex hull of the closure of the set of accumulation points of the sequence* $\{\frac{a_k}{|a_k|}\}_{k\in\mathbf{N}}$.

Then the following statements hold true.

(i) *If* 0 *is contained in the interior of* H, *then there does not exist any Yosida function with* D *as zero divisor.*

(ii) *If* 0 *is not contained in* H, *then there does exist a Yosida function with* D *as zero divisor.*

Basically we show that for such a divisor D the function

$$F(z) = \prod_{a\in|D|} \left(1 - \frac{z}{a}\right)$$

is not a Yosida function and that the only way to obtain a Yosida function f with D as a zero divisor is via the approach

$$f(z) = e^{\alpha z} F(z) = e^{\alpha z} \prod_{a\in|D|} \left(1 - \frac{z}{a}\right) \quad (\alpha \in \mathbf{C}).$$

However, as we will see, the additional factor $e^{\alpha z}$ is helpful only if $e^{\alpha z}$ is rapidly converging to zero along the sequence a_k. This in turn is possible only if the divisor is concentrated in one half-plane.

Proof We begin with the proof of statement (i). Condition (7.6.10) implies the absolute convergence of

$$\sum_{k=1}^{\infty} \frac{1}{|a_k|},$$

which implies the convergence of

$$\sum_{k>k_0} \log\left(1-\frac{|z|}{|a_k|}\right) \quad \left(k_0=\max\{k;\,|a_k|\leqq|z|\}\right)$$

and then implies the absolute convergence, uniform on compact subsets, of

$$F(z)=\prod_{k=1}^{\infty}\left(1-\frac{z}{a_k}\right). \tag{7.6.11}$$

Thus an arbitrary entire function f with divisor D can be written as

$$f(z)=F(z)e^{g(z)}, \tag{7.6.12}$$

where $F(z)$ is defined by (7.6.11), while $g(z)$ is an entire function. □

Claim 7.6.13 *There is a constant $C>0$ (depending only on λ) such that*

$$\left|\prod_{k\geqq n}\left(1-\frac{z}{a_k}\right)\right|\geqq C$$

for every $n\in\mathbf{N}$ and z with $|z|\sqrt{\lambda}<|a_n|$.

To prove Claim 7.6.13, we observe that

$$\left|\prod_{k\geqq n}\left(1-\frac{z}{a_k}\right)\right|\geqq C=\prod_{l\geqq 0}(1-\lambda^{-l-\frac{1}{2}}).$$

We emphasize that C is independent of n. We need the second claim.

Claim 7.6.14 *For every $K>0$ and $m\in\mathbf{N}$ there is a natural number $N\geqq m$ (depending on K and m) such that*

$$\left|\prod_{k=m}^{n}\left(1-\frac{z}{a_k}\right)\right|\geqq K$$

for all $n\geqq N$ and z with $|z|\geqq\sqrt{\lambda}|a_n|$.

To prove the second claim it suffices to note that

$$\left|\prod_{k=m}^{n}\left(1-\frac{\zeta}{a_k}\right)\right|\geqq\prod_{k=m}^{n}\left(\left|\frac{\zeta}{a_k}\right|-1\right)\geqq\prod_{l=0}^{n-m}(\lambda^{l+\frac{1}{2}}-1)$$

and

$$\lim_{s\to\infty}\prod_{l=0}^{s}\left(\lambda^{l+\frac{1}{2}}-1\right)=\infty.$$

Due to Claims 7.6.13 and 7.6.14 above we know that there is a number $M\in\mathbf{N}$ such that

$$|F(z)|=\left|\prod_{k=1}^{\infty}\left(1-\frac{z}{a_k}\right)\right|\geqq 1 \tag{7.6.15}$$

for all $n\geqq M$ and all $z\in\mathbf{C}$ with

$$\sqrt{\lambda}|a_n|<|z|<\frac{|a_{n+1}|}{\sqrt{\lambda}}.$$

We use the notation

$$M_f(r)=\max_{|z|=r}\log^+|f(z)|,\quad r>0$$

for an entire function f.

If we set

$$r_n=\sqrt{|a_{n+1}a_n|}\in\left(\sqrt{\lambda}|a_n|,\frac{|a_{n+1}|}{\sqrt{\lambda}}\right),$$

we can now deduce from (7.6.12) and (7.6.15) that

$$M_{\exp(g)}(r_n)=\max_{|z|=r_n}\Re^+\big(g(z)\big)\leqq M_f(r_n).$$

Hence, if f is Yosida and therefore of order at most one (cf. 7.6.7 (v)), we can deduce that $\exp(g(z))$ is likewise of order at most one, which in turn implies that g is a polynomial of degree 1 (by Theorem 6.1.9), i.e., g is affine-linear. Thus we may assume $g(z)=\alpha z+\beta$ with $\alpha,\beta\in\mathbf{C}$. There is no loss of generality in assuming $\beta=0$. Now the condition that 0 is contained in the interior of H implies that there is a subsequence a_{n_j} with

$$\left|\exp\big(g(a_{n_j})\big)\right|\geqq 1,\quad {}^\forall j\geqq 1$$

because $|e^w|=e^{\Re w}$.

Next we calculate $|f^\#(a_{n_j})|$. Since $f(a_k)=F(a_k)=0$ for every k, we have

$$f^\#(a_k)=\left|f'(a_k)\right|=\left|F'(a_k)e^{g(a_k)}\right|,\quad {}^\forall k\in\mathbf{N}$$

and therefore

$$f^\#(a_{n_j})\geqq\left|F'(a_{n_j})\right|,\quad {}^\forall j\in\mathbf{N}.$$

Thus it suffices to show that

$$\lim_{k\to\infty}\left|F'(a_k)\right|=+\infty$$

in order to deduce that f is not Yosida. Now

$$\left|F'(a_n)\right| = \frac{1}{|a_n|}\left|\prod_{k\neq n}\left(1-\frac{a_n}{a_k}\right)\right|,$$

and

$$\lim_{n\to\infty}\frac{1}{|a_n|}\left|1-\frac{a_n}{a_1}\right| = \frac{1}{|a_1|}.$$

By Claim 7.6.13

$$\left|\prod_{k>n}\left(1-\frac{a_n}{a_k}\right)\right| \geqq C$$

and by Claim 7.6.14 there is a number $N \in \mathbf{N}$ for each $K > 0$ such that

$$\left|\prod_{1\leqq k<n}\left(1-\frac{a_n}{a_k}\right)\right| \geqq K, \quad {}^\forall n \geqq N.$$

Combined, these assertions show that for each $K > 0$ there is a number N such that

$$\left|F'(a_n)\right| \geqq \frac{C}{2|a_0|}K, \quad {}^\forall n \geqq N.$$

Thus $\overline{\lim}\,|f^\#(a_n)| = +\infty$ and f is not Yosida. This proves statement (i).

Next we deal with statement (ii). We assume that 0 is not contained in H. From the definition of H it follows that there are numbers $M_0 \in \mathbf{N}$, $\eta \in S^1 = \{z \in \mathbf{C};\ |z| = 1\}$ and $\mu \in (0, 1)$ such that

$$\Re(\eta a_k) \geqq \mu|a_k|, \quad {}^\forall k \geqq M_0.$$

Let $S = \sum_k \frac{1}{|a_k|}$ (the convergence of this infinite sum follows from (7.6.10)). We choose a complex number $\alpha = -r\eta$ such that r is a positive real number with $r\mu > S$. We claim that

$$f(z) = e^{\alpha z}\prod_k\left(1-\frac{z}{a_k}\right)$$

defines a Yosida function.

We start with choosing some more constants

$$0 < \gamma < \frac{2}{1+\lambda^{-1}} - 1, \qquad C = \inf_k \gamma|a_k|.$$

Next we observe that the above choices imply

$$\Re(\alpha a_k) + S|a_k| = -r\Re(\eta a_k) + S|a_k| \leqq (-r\mu + S)|a_k| < 0, \quad {}^\forall k \geqq M_0.$$

It follows that there is a constant $M \in \mathbf{N}$ such that

$$\Re(\alpha z) + S|z| < 0 \tag{7.6.16}$$

for every complex number z for which there exists a natural number $k \geqq M$ with $|z - a_k| \leqq C$.

Claim 7.6.17 *There is a constant $K_1 > 0$ such that*

$$\left|\frac{f'(z)}{f(z)}\right| < K_1$$

for all $z \in \{z \in \mathbf{C};\, |z - a_k| \geqq C,\ {}^{\forall}k \in \mathbf{N}\}$.

Before we prove this claim, we need another claim.

Claim 7.6.18 *For $k \neq l$ we have*

$$\{z;\, |z - a_k| \leqq \gamma |a_k|\} \cap \{z;\, |z - a_l| \leqq \gamma |a_l|\} = \emptyset.$$

Proof of Claim 7.6.18 Let w be a point in the intersection. Then $|w - a_k| \leqq \gamma |a_k|$ and $|w - a_l| \leqq \gamma |a_l|$, which imply

$$|a_k - a_l| \leqq |w - a_k| + |w - a_l| \leqq \gamma \big(|a_k| + |a_l|\big).$$

We may assume that $l = k + d$ for some $d > 0$. Then $|a_l| \geqq \lambda^d |a_k|$ and therefore

$$|a_k - a_l| \geqq \big(1 - \lambda^{-d}\big)|a_l|.$$

Together one gets

$$\big(1 - \lambda^{-d}\big)|a_l| \leqq \gamma \big(|a_k| + |a_l|\big) \leqq \gamma \big(\lambda^{-d} + 1\big)|a_l|$$

and therefore

$$\frac{2}{1 + \lambda^{-d}} - 1 = \frac{1 - \lambda^{-d}}{\lambda^{-d} + 1} \leqq \gamma.$$

Hence

$$\frac{2}{1 + \lambda^{-1}} - 1 \leqq \frac{2}{1 + \lambda^{-d}} - 1 \leqq \gamma,$$

which contradicts the choice of γ. Therefore there cannot be a point w in the intersection. This proves Claim 7.6.18.

Our choice of f implies

$$\frac{f'(z)}{f(z)} = \alpha + \sum_k \frac{1}{z - a_k}.$$

For each $z \in \mathbf{C}$ and $k \in \mathbf{N}$ with $|z - a_k| \geqq \gamma |a_k|$ we have

$$\left| \frac{1}{z - a_k} \right| \leqq \frac{1}{\gamma |a_k|}.$$

For a given $z \in \mathbf{C}$ there is at most one index k with $|z - a_k| \leqq \gamma |a_k|$ (see Claim 7.6.18 above). Thus,

$$\left| \frac{f'(z)}{f(z)} \right| = \left| \alpha + \sum_k \frac{1}{z - a_k} \right| \leqq |\alpha| + \frac{1}{C} + \sum_k \frac{1}{\gamma |a_k|}$$

if $|z - a_k| \geqq C$ for all $k \in \mathbf{N}$. This proves Claim 7.6.17 with

$$K_1 = |\alpha| + \frac{1}{C} + \sum_k \frac{1}{\gamma |a_k|} = |\alpha| + \frac{1}{C} + \frac{S}{\gamma}.$$

As a next step, we deduce an upper-bound for $|f'|$ on

$$W_2 = \left\{ z \in \mathbf{C};\ {}^{\exists} k \geqq M,\ |z - a_k| \leqq C \right\}.$$

Fix a number $k \in \mathbf{N}$. Then

$$f(z) = e^{\alpha z} \left(1 - \frac{z}{a_k} \right) \prod_{l \neq k} \left(1 - \frac{z}{a_l} \right),$$

and

$$f'(z) = -\frac{1}{a_k} e^{\alpha z} \prod_{l \neq k} \left(1 - \frac{z}{a_l} \right) + \left(\alpha + \sum_{l \neq k} \frac{1}{z - a_l} \right) f(z).$$

In order to get the desired bounds we start as follows:

$$\begin{aligned} \left| \prod_{l \neq k} \left(1 - \frac{z}{a_l} \right) \right| &= \prod_{l \neq k} \left| \left(1 - \frac{z}{a_l} \right) \right| \\ &\leqq \prod_{l \neq k} \left(1 + \left| \frac{z}{a_l} \right| \right) = \exp\left(\sum_{l \neq k} \log \left(1 + \left| \frac{z}{a_l} \right| \right) \right) \\ &\leqq \exp\left(\sum_{l \neq k} \left| \frac{z}{a_l} \right| \right) = \exp\left(|z| \sum_{l \neq k} \frac{1}{|a_l|} \right) \leqq e^{S|z|}. \end{aligned}$$

Hence, for $|z - a_k| \leqq C$,

$$|f(z)| \leqq e^{\Re(\alpha z)} \frac{C}{|a_k|} e^{S|z|}.$$

We also observe that if

$$|z - a_k| \leqq C = \inf_j \gamma |a_j|,$$

then $|z - a_k| \leqq \gamma |a_k|$ implying (due to Claim 7.6.18) that

$$\left| \frac{1}{z - a_l} \right| \leqq \frac{1}{\gamma |a_l|}, \quad {}^{\forall} l \neq k.$$

Thus $|z - a_k| \leqq C$ implies

$$\left| \sum_{l \neq k} \frac{1}{z - a_l} \right| \leqq \frac{1}{\gamma} \sum_{l \neq k} \frac{1}{|a_l|} \leqq \frac{S}{\gamma}.$$

Combined, these imply

$$\begin{aligned} \left| f'(z) \right| &\leqq \frac{1}{|a_k|} e^{\Re(\alpha z)} e^{S|z|} + \left(|\alpha| + \frac{S}{\gamma} \right) e^{\Re(\alpha z)} \frac{C}{|a_k|} e^{S|z|} \\ &= e^{(\Re(\alpha z) + S|z|)} \frac{1}{|a_k|} \left(1 + C|\alpha| + \frac{CS}{\gamma} \right). \end{aligned} \tag{7.6.19}$$

Therefore, by (7.6.16)

$$\begin{aligned} f^{\#}(z) \leqq \left| f'(z) \right| &\leqq \frac{1}{|a_k|} \left(1 + C|\alpha| + \frac{CS}{\gamma} \right) \\ &\leqq \frac{1 + C|\alpha| + \frac{CS}{\gamma}}{|a_M|} \stackrel{\text{def}}{=} K_2 \end{aligned}$$

for all $z \in W_2 = \{z;\ {}^{\exists} k \geqq M, |z - a_k| \leqq C\}$.

We have also seen that there is a constant $K_1 > 0$ such that

$$f^{\#}(z) \leqq \frac{1}{2} \left| \frac{f'(z)}{f(z)} \right| \leqq K_1$$

for all $z \in W_1 = \{z; |z - a_k| \geqq C, {}^{\forall} k \geqq 1\}$. Hence $f^{\#}(z) \leqq \max\{K_1, K_2\}$ for all $z \in W_1 \cup W_2$. Because $\mathbf{C} \setminus (W_1 \cup W_2)$ is contained in a finite union of bounded disks and therefore relatively compact in $\mathbf{C}$, it follows that $f^{\#}$ is bounded on $\mathbf{C}$, i.e., f is a Yosida function. □

Our next goal is to show that no bound on the characteristic function $T(r, f)$ forces an entire function f to be Yosida except when this bound is strong enough to force f to be a polynomial.

In view of Theorem 7.6.9 the crucial point is to verify that for every such bound there exists an entire function f fulfilling this condition on $T(r, f)$ and fulfilling simultaneously the condition of (i) of Theorem 7.6.9.

Theorem 7.6.20 *Let $\rho : [1, \infty) \to (0, \infty)$ be a continuous increasing function.*

(i) *If $\underline{\lim}_{r\to\infty} \frac{\rho(r)}{\log r} < \infty$, then every entire function f with $T(r, f) \leqq \rho(r)$ for all $r \geqq 1$ must be a polynomial.*

(ii) *If $\underline{\lim}_{r\to\infty} \frac{\rho(r)}{\log r} = \infty$, then there exists an entire function f which is not Yosida and such that $T(r, f) \leqq \rho(r)$ for all $r \geqq 1$.*

Proof (i) This is an immediate consequence of Theorem 1.1.26.

(ii) As a first preparation, we observe that

$$\prod_{k\geqq 0}\left(1 - \frac{1}{2^{2k+1}}\right) \geqq 1 - \sum_{k\geqq 0} \frac{1}{2^{2k+1}} = \frac{1}{3}. \tag{7.6.21}$$

We will construct $f(z)$ as follows: We choose a sequence $c_k \in \mathbf{C}$ with $|c_{k+1}| > 4|c_k| \geqq 4$ for all k ($\lambda = 4$ in (7.6.10)), and then define

$$P_k(z) = 3\prod_{j=1}^{k}\left(1 - \frac{z}{c_j}\right).$$

The conditions on the c_k ensure that $P_k(z)$ converges locally uniformly to an entire function $f(z)$ whose zeros are precisely the points c_k.

Let $k \in \mathbf{N}$ and let z be a complex number with $2|c_k| \leqq |z| \leqq \frac{1}{2}|c_{k+1}|$. Then

$$\left|1 - \frac{z}{c_j}\right| \geqq 1$$

for all $j \leqq k$. For $j > k$ then conditions $|c_{l+1}| > 4|c_l|$ imply

$$\left|\frac{z}{c_j}\right| \leqq \frac{1}{2^{2(j-k-1)+1}}$$

which in turn implies by (7.6.21)

$$|P_j(z)| \geqq 1$$

for all j and all z with $2|c_k| \leqq |z| \leqq \frac{1}{2}|c_{k+1}|$. As a consequence,

$$m(r, 1/f) = 0, \quad 2|c_k| \leqq r \leqq \frac{1}{2}|c_{k+1}|$$

and hence

$$T(r, 1/f) = N\big(r, (1/f)_\infty\big), \quad 2|c_k| \leqq r \leqq \frac{1}{2}|c_{k+1}|.$$

Therefore,

$$T(r, 1/f) = \sum_{j=1}^{k} \log\left|\frac{r}{c_j}\right| \leqq k \log r$$

for all such r. Similarly, one obtains $T(r, 1/f) = 0$ for $r \leqq \frac{1}{2}|c_1|$.

It is noted that we are dealing with the sequence

$$2|c_{k-1}| < \frac{1}{2}|c_k| < 2|c_k| < \frac{1}{2}|c_{k+1}|, \quad k = 1, 2, \ldots.$$

Now we take $r \in [\frac{1}{2}|c_k|, 2|c_k|]$. Since $T(r, 1/f)$ is increasing, we have

$$\begin{aligned} T(r, 1/f) &\leqq T\big(2|c_k|, 1/f\big) = N\big(2|c_k|, (1/f)_\infty\big) \\ &= \sum_{1 \leqq j \leqq k} \log\left|\frac{2c_k}{c_j}\right| \leqq k \log\big(2|c_k|\big) \\ &\leqq k \log(4r). \end{aligned}$$

Summarizing, we have shown that

$$T(r, f) + O(1) = T(r, 1/f) \leqq k \log(4r) \tag{7.6.22}$$

for all r with $r \leqq 2|c_k|$.

Thus it suffices to choose the c_k such that $k \log(4r) \leqq \rho(r)$ for all $r \in [2|c_{k-1}|, 2|c_k|]$. This is possible: We assumed $\underline{\lim}_{t\to\infty} \frac{\rho(t)}{\log t} = \infty$. Hence for each $k \in \mathbf{N}$ there is a constant R_k such that $k \log(4r) \leqq \rho(r)$ for all $r \geqq R_k$. Now it suffices to choose the c_k, $k = 1, 2, \ldots$ so that $2|c_{k-1}| \geqq R_k$ (in addition to the other conditions $|c_k| \geqq 4|c_{k-1}| \geqq 4$). Since $O(1)$-term in (7.6.22) is negligible in the growth estimate, we have established that $T(r, f)$ satisfies the required growth condition.

Finally, we note that in our construction we choose the c_k such that $|c_{k+1}| \geqq 4|c_k|$. Furthermore we may choose the c_k such that the set

$$\left\{\frac{c_k}{|c_k|}; k \in \mathbf{N}\right\}$$

is dense in $S^1 = \{z \in \mathbf{C}; |z| = 1\}$. Then Theorem 7.6.9 implies that such an entire function f is *not* Yosida. □

The statement of Theorem 7.6.8 is now an immediate consequence of Theorem 7.6.20.

7.6.2 Characterizing Brody Maps into Tori

In the case of tori we have a complete characterization of a Brody curve f by the order ρ_f (cf. (1.1.28), (2.5.19)).

Proposition 7.6.23 *Let T be a compact complex torus. Then an entire curve $f : \mathbf{C} \to T$ is a Brody curve iff $\rho_f \leqq 2$.*

Proof If f is Brody, the desired inequality follows from Proposition 7.6.5 above. Conversely, inequality (7.6.6) implies $\rho_f \leqq 2$. Due to Theorem 6.1.9 it follows that f is induced by an affine-linear map $\tilde{f} : \mathbf{C} \to \mathbf{C}^g$ ($g = \dim T$). Such an affine-linear map is easily seen to be a Brody map (cf. Example 7.6.4). □

7.6.3 Brody Curves with Prescribed Points in the Image

Let X be a compact complex manifold with two distinct points p, q such that $d_X(p, q) = 0$. Then Brody's Theorem 7.2.8 guarantees the existence of a non-constant Brody curve $f : \mathbf{C} \to X$. However, the image of $f(\mathbf{C})$ may not contain p or q. The following result was proved in Winkelmann [07].

Theorem 7.6.24 *There exists a three-dimensional abelian variety A containing a curve C such that the compact complex manifold X arising as the blow-up of A along C has the following properties*:

(i) *The Kobayashi pseudodistance on X vanishes identically.*
(ii) *There exists an entire curve $f : \mathbf{C} \to X$ with dense image in the sense of differentiable topology.*
(iii) *There exists a hypersurface $E \subset X$ which contains the image of every non-constant Brody curve $f : \mathbf{C} \to X$.*

In particular, this is a compact complex manifold X with vanishing Kobayashi pseudodistance d_X where we cannot find a Brody curve through every point.

However, the following question remains unanswered.

Question 7.6.25 Let X be a compact complex manifold with two distinct point p, q such that $d_X(p, q) = 0$. Does this imply the existence of a non-constant entire curve $f : \mathbf{C} \to X$ with $p \in \overline{f(\mathbf{C})}$?

Remark 7.6.26 We cannot hope to find a single entire curve $f : \mathbf{C} \to X$ with $p \in \overline{f(\mathbf{C})}$ *and* $q \in \overline{f(\mathbf{C})}$. For instance, let S_0 be a hyperbolic surface, let S_1 be the blow-up of S_0 at one point $x \in S_0$ and let $E_1 \subset S_1$ denote the exceptional divisor. Let S denote the surface obtained from S_1 by blowing up a point on the exceptional divisor E_1. Now S contains two rational curves which intersect at one point. By construction every holomorphic map $f : \mathbf{C} \to S$ has its image contained in one of this two curves. On the other hand, $d_S(p, q) = 0$ whenever p and q lie on the union of the two rational curves.

Question 7.6.27 Let X be a compact complex manifold with $d_X \equiv 0$. Does this imply the existence of a non-constant entire curve $f : \mathbf{C} \to X$ with Zariski dense image?

Note that Theorem 7.6.24 above shows that there are compact complex manifolds X with $d_X \equiv 0$ such that no Brody curve has a Zariski dense image.

7.6.4 Ahlfors' Currents

Let $f : \mathbf{C} \to X$ be a non-constant entire curve to a compact complex manifold X. To such an entire curve one can associate "Ahlfors currents". These currents have proved to be quite useful. For example, they play an important role in the work of McQuillan on the Green–Griffiths Conjecture 6.6.30 for surfaces (McQuillan [98]).

The idea is to take the current of integration over the image. Of course, due to non-compactness of $\mathbf{C}$ and the possibly transcendental character of the map f this is not so easy.

Fix a hermitian metric form ω on X. For every $r > 0$ we may define

$$T_r(\phi) = \frac{1}{\int_{\Delta(r)} f^*\omega} \int_{\Delta(r)} f^*\phi$$

for a differentiable $(1, 1)$-form ϕ on X. This current, however, depends on r and is not necessarily closed. We underline that (due to the theorem of Banach–Alaoglu) for every given sequence $r_k \to \infty$ we may choose a subsequence such that

$$\lim_{k\to\infty} T_{r_k}(\phi)$$

converges for any $(1, 1)$-form ϕ on X. For the limit current to be closed we would need (with $V_r = \int_{\Delta(r)} f^*\omega$):

$$0 = \lim_{n\to\infty} T_{r_n}(d\psi) = \lim_{n\to\infty} \frac{1}{V_{r_n}} \int_{\partial\Delta(r_n)} f^*\psi = \lim_{n\to\infty} \frac{1}{V_{r_n}} \int_{\Delta(r_n)} f^*(d\psi)$$

for every 1-form ψ. This is not true for an arbitrary sequence r_n. However, we will see that one can choose a sequence $r_k \to \infty$ such that this property holds. This can be done by proving that there always exists a sequence of positive real numbers $r_k \to +\infty$ such that

$$\lim_{r\to\infty} \frac{\text{length}(f(\partial\Delta(r_k)))}{\text{vol}(f(\Delta(r_k)))} = 0.$$

The first step is the lemma below.

Lemma 7.6.28 *Let $\phi : [0, \infty) \to \mathbf{R}^+$ be a continuous function. Let $a, c > 0$ and let $0 < \rho < R$. Assume that*

$$\phi(r) \geqq c \int_0^r \big(\phi(t)\big)^2 dt, \quad \rho \leqq r \leqq R,$$

and

$$\int_0^\rho \big(\phi(t)\big)^2 dt \geqq a.$$

Then

$$R - \rho < 2 + \max\left\{0, \frac{2 - ac^2}{a^2c^4}\right\}.$$

Proof We start by observing that

$$\phi(r) \geqq c\int_0^r \big(\phi(t)\big)^2 dt \geqq c\int_0^\rho \big(\phi(t)\big)^2 dt \geqq ca, \quad \rho \leqq r \leqq R,$$

which in turn implies

$$\begin{aligned}\phi(r) &\geqq c\int_0^r \big(\phi(t)\big)^2 dt \geqq ca + c\int_\rho^r \big(\phi(t)\big)^2 dt \\ &\geqq ca + c(r-\rho)(ca)^2, \quad \rho \leqq r \leqq R.\end{aligned}$$

We want to choose a number $p > 0$ such that $\phi(t) \geqq 2/c$ for all $t \in [p, R]$. It suffices to choose $p \geqq \rho$ such that

$$ca + c(p-\rho)(ca)^2 \geqq \frac{2}{c} \quad \Longleftrightarrow \quad p - \rho \geqq \left(\frac{2}{c} - ac\right)\frac{1}{a^2c^3}.$$

Thus we set

$$p = \rho + \max\left\{0, \frac{2 - ac^2}{a^2c^4}\right\}.$$

What we would like to show is $R < p + 2$.

Suppose that $R \geqq p + 2$. We define a sequence, $p_k < R$, recursively as $p_0 = p$, $p_{k+1} = p_k + 2^{-k}$. We will verify by induction that

$$\phi(t) \geqq 2^{k+1}\frac{1}{c}, \quad p_k \leqq t \leqq R. \tag{7.6.29}$$

Indeed, the case of $k = 0$ holds by definition, and we assume that this is true until k. Let $p_{k+1} = p_k + 2^{-k} \leqq r \leqq R$. Then

$$\phi(t) \geqq c\int_{p_k}^{p_{k+1}} \big(\phi(t)\big)^2 dt \geqq c2^{-k}\left(\frac{2^{k+1}}{c}\right)^2 = \frac{1}{c}2^{k+2}.$$

This proves (7.6.29). Now $\lim_{k\to\infty} p_k = p + 2 \leqq R$ and $\lim_{k\to\infty} \phi(p_k) = +\infty$; this is absurd. □

Let X be a complex manifold equipped with a hermitian metric form ω. For an entire curve $f : \mathbf{C} \to X$ we set

$$f^*\omega = \big(\alpha(z)\big)^2 \frac{i}{2} dz \wedge d\bar{z}, \quad \alpha(z) \geqq 0,\ z \in \mathbf{C}.$$

Note that $(\alpha(z))^2$ is C^∞ and $\alpha(z)$ is at least continuous. For each $r > 0$, let $L_f(r)$ denote the length of $f(\partial\Delta(r))$, i.e.,

$$L_f(r) = \int_0^{2\pi} \alpha\big(re^{i\theta}\big) r d\theta,$$

and let $A_f(r)$ denote the area of $f(\Delta(r))$,

$$A_f(r) = \int_{\Delta(r)} f^*\omega = \int_0^r r dr \int_0^{2\pi} \big(\alpha\big(re^{i\theta}\big)\big)^2 d\theta.$$

Proposition 7.6.30 *Let $f : \mathbf{C} \to X$ be a non-constant entire curve, and let the notation be as above. Then*

$$\varliminf_{r\to\infty} \frac{L_f(r)}{A_f(r)} = 0.$$

Proof Assume the converse. Then there exists a positive number $\varepsilon > 0$ such that

$$L_f(r) \geqq \varepsilon A_f(r), \quad {}^\forall r \geqq 1.$$

Schwarz's inequality for the L^2-norm implies that

$$\left(\int_0^{2\pi} \alpha\big(re^{i\theta}\big) d\theta\right)^2 \leqq 2\pi \int_0^{2\pi} \big(\alpha\big(re^{i\theta}\big)\big)^2 d\theta.$$

Hence,

$$\int_0^r \frac{1}{t^2}\big(L_f(t)\big)^2 t dt \leqq 2\pi A_f(r) \leqq \frac{2\pi}{\varepsilon} L_f(r), \quad r \geqq 1$$

and therefore

$$\int_1^r \big(L_f(t)\big)^2 d\log t \leqq \frac{2\pi}{\varepsilon} L_f(r), \quad r \geqq 1.$$

Now, set $s = \log t$, $q = \log r$ and $\phi(s) = L_f(e^s)$. Then we have

$$\int_0^q \big(\phi(s)\big)^2 ds \leqq \frac{2\pi}{\varepsilon}\phi(q), \quad q \geqq 0.$$

Such a function $\phi(q)$ cannot exist by Lemma 7.6.28 above. □

Now we can formulate the statement on the existence of Ahlfors' currents:

Definition 7.6.31 Let X be a compact complex manifold equipped with a hermitian metric form ω. Let $f : \mathbf{C} \to X$ be an entire curve. An "*Ahlfors current* for f" is

a positive closed current T on X such that there is some divergent sequence $\{r_k\}$ of positive real numbers satisfying

$$T(\phi) = \lim_{n\to\infty} \frac{1}{A_f(r_k)} \int_{\Delta(r_k)} f^*\phi$$

for an arbitrary $(1,1)$-from ϕ on X.

Theorem 7.6.32 *Let X be a compact complex manifold. For every entire curve there exists an* Ahlfors current.

Duval [08] showed the following.

Theorem 7.6.33 *Let T be an Ahlfors current on a compact complex manifold X. Let $K \subset X$ be a compact subset such that $T(\chi_U \omega) > 0$ for every open neighborhood U of K in X. (Here χ_U denotes the characteristic function of U.) Then there exists an entire curve $f : \mathbf{C} \to X$ whose image $f(\mathbf{C})$ intersects K with positive volume.*

A related result states that if such a K happens to be a complex compact Riemann surface, then its genus is zero or one.

Appendix: Program for Mathematica

```
llcm[li_]:=
Block[{g,i},
For[g=1;i=1,i<=Length[li],i++,g=LCM[g,li[[i]]]];Return[g]
]

deg[e_]:=
Block[{le,a,t},
le=Length[e];
a=Table[Apply[Plus,e[[t]]],{t,le}];
Print["deg=",llcm[a],"e >= ", llcm[a] (le (le-2)+1]
]

ncr[n_,r_]:=
Block[{r1,a,c,i},
If[r==0,Return[{{}}]];
r1=r+1;a=Table[0,{r1}];
c={};i=r1;a[[i]]=n+1;
While[True,
   If[i==1,c=Prepend[c,Drop[a,-1]],i--;a[[i]]=a[[i+1]]];
   While[True,
     If[a[[i]]==i,i++;If[i==r1,Return[c]],a[[i]]--;Break[]]
   ]
]
]
```

```
cu[f_]:=
Block[{g,i},
For[g=f;i=1,i<=r+1,i++,
     While[(g/.x[[i]]->0)==0,g=Expand[g/x[[i]]]]
];Return[g]
]

hyp[e_,co_]:=
Block[{n,m1,n1,m,z,x,cc,t,ed,el,er,er2,i,j,fz,f0,b,c,rv,
c2,d,d2,na,k,n2,n3,n22,n33,i1,i2,i3,a,ng,rr,lz,ii,jj,zz,
p,q,ta,ZZ,gz},
n=Length[e];m1=Length[e[[1]]];n1=n+1;m=m1-1;
z={X,Y,Z,U,V,W,P,Q,R,S};
x={A,B,C,D,F,G,H,J,K,L};
cc=Table[co[[t]]->1,{t,Length[co]}];
ed=Table[Apply[Plus,e[[t]]],{t,n}];el=llcm[ed];
er=Table[el/ed[[t]] Drop[e[[t]],-1],{t,n}];
er2=Table[er[[i]]-er[[j]],{i,n},{j,n}];
fz=Sum[co[[i]] Product[z[[j]]^er[[i,j]],{j,m}],{i,n}];
f0=Table[0,{m}];
b=c=rv=Table[0,{n1}];c2=d=d2=Table[0,{n}];rv[[n1]]={};
na=0;
For[k=0,k<=n/3,k++,
If[Mod[n-3 k,2]==1,Continue[]];
n2=(n-3 k)/2;n3=k;n22=2 n2;n33=3 n3;
For[i=0,i<n3,i++,i1=3 i+1;i2=i1+1;i3=i2+1;
   c[[i1]]=i1-3;c[[i2]]=i1;c[[i3]]=i2;
   d[[i1]]=0;d[[i2]]=d[[i3]]=i1;
   d2[[i2]]=i1-1;d2[[i3]]=i2
];
For[i=0,i<n2,i++,i1=n33+2 i+1;i2=i1+1;
   c[[i1]]=i1-2;c[[i2]]=i1;
   d[[i1]]=0;d[[i2]]=i1;d2[[i2]]=i1-1
];
c[[1]]=n1;c[[n33+1]]=n1;d2[[2]]=n1;
a=Table[0,{n}];
b[[1]]=1;a[[1]]=1;If[k>0,c2[[1]]=n22,c2[[1]]=0];
i=1;
Label[la1];
i++;
If[i==n1,Goto[la3]];
For[j=b[[c[[i]]]]+1,j<=n,j++,If[a[[j]]==0,Break[]]];
If[j==n1,Goto[la4]];ng=1;
Label[la2];
b[[i]]=j;a[[j]]=i;
(*If[i==3,Print[Take[b,3]]];*)
If[d[[i]]==0&&i<n33,If[ng==1,c2[[i]]=c2[[i-3]],c2[[i]]--]];
If[d[[i]]==0,Goto[la1]];
rv[[i]]=RowReduce[Append[rv[[d2[[i]]]],er2[[j,b[[d[[i]]]]]]]];
For[r=1,r<=Length[rv[[i]]],r++,If[rv[[i,r]]==f0,Break[]]];r--;
If[r!=m,Goto[la1]];i++;Goto[la4];
Label[la3];
rr=rv[[n]];
For[r=1,r<=Length[rr],r++,If[rr[[r]]==f0,Break[]]];r--;
If[r==m,Goto[la4]];
```

```
na++;
For[p={};q=b;ii=1,ii<=n3,ii++,p=Append[p,Take[q,3]];q=Drop[q,3]];
For[ii=1,ii<=n2,ii++,p=Append[p,Take[q,2]];q=Drop[q,2]];
Print[na,p];
For[lz={};ii=1,ii<=r,ii++,
   For[jj=1,jj<=m,jj++,If[rr[[ii,jj]] !=0,Break[]]];
   zz=x[[jj]] Product[z[[t]]^(-rr[[ii,t]]),{t,jj+1,m}];
   lz=Append[lz,z[[jj]]->zz]
];
p=Table[llcm[Table[Denominator[rr[[ii,jj]]],{ii,r}]],{jj,m}];
ta=Append[Table[z[[t]]^(1/p[[t]]),{t,m}],ZZ];
gz=Collect[PowerExpand[(fz/.lz) ZZ+1],ta];
Table[cu[gz[[t,1]]],{t,2,Length[gz]}]>>tmp;
If[r<m-1||Length[gz]-1<=r,Print[(gz-1)/.ZZ->1];Return[-1]];
Label[la4];
i--;
If[i==0,Goto[la5]];
a[[b[[i]]]]=0;
If[d[[i]]==0&&c2[[i]]==0,j=n1,
   For[j=b[[i]]+1,j<=n,j++,If[a[[j]]==0,Break[]]]
];ng=0;
If[j==n1,Goto[la4],Goto[la2]];
Label[la5];
];
Return[na]
]

hyperb[e_]:=
Block[{n,m,tm,na,coe,i,nc,j,nj,njc,ej,co,k,ek,l,hy},
n=Length[e];m=Length[e[[1]]];tm=Table[t,{t,m}];na=0;
coe=Table[ToExpression["a"<>ToString[t]],{t,n}];
For[i=4,i<=m,i++,nc=ncr[m,i];
   For[j=1,j<=Length[nc],j++,nj=nc[[j]];njc=Complement[tm,nj];
     For[ej={};co={};k=1,k<=n,k++,ek=e[[k]];
       For[l=m-i,l>=1,l--,
         If[ek[[njc[[l]]]]==0,ek=Delete[ek,njc[[l]]],Break[]]
       ];
       If[l==0,ej=Append[ej,ek];co=Append[co,coe[[k]]]]
     ];
     Print[nj,"s'=",Length[ej]];
     If[Length[ej]>1,hy=hyp[ej,co];
       If[hy==-1,Print["maybe non-hyperbolic"];Return[],na=na+hy]
     ]
   ]
];
If[na==0,Print["n-admissible"],Print["generically hyperbolic"]]
]

eq[cc_]:= (* list of alg eq *)
Block[{cc2,t,da,da2},
cc2=Table[ToExpression["a"<>ToString[t]]->cc[[t]],{t,Length[cc]}];
da=ReadList["tmp"];da2=Union[da/.cc2];Print["#=",Length[da2]];
Return[da2]
]
```

```
alg[ta_,tn_]:= (* ta-->tn *)
Block[{s,t,n,i,a},
x={A,B,C,D,F,G,H,J,K,L};
s=N[NSolve[Table[l[[t]]==0,{t,Length[l]}][[ta]],x]];
For[n=0;i=1,i<=Length[s],i++,
   If[(a=N[Abs[l[[tn]]/.s[[i]]]])<0.01,n++;Print[a,s[[i]]]]
];
If[n==0,Print["no solutions"]]
]
```

An example follows to use the above program to check (7.5.12). Vectors correspond to the exponents of z_5, z_1, z_2, z_3, z_4, in that order.

```
e={
{1,0,0,0,0},
{0,1,0,0,0},
{0,0,1,0,0},
{0,0,0,1,0},
{0,0,0,0,1},
{0,2,1,0,0},
{0,0,2,1,0},
{0,0,0,2,1},
{0,1,0,0,2}
};
deg[e];
OpenWrite["tmp"];Close["tmp"];
hyperb[e]

   deg=3e>=192
   {1, 2, 3, 4}s'=6
   {1, 2, 3, 5}s'=6
   {1, 2, 4, 5}s'=6
   {1, 3, 4, 5}s'=6
   {2, 3, 4, 5}s'=8
   1{{1, 3}, {2, 4}, {5, 7}, {6, 8}}
   {1, 2, 3, 4, 5}s'=9
   1{{1, 2, 4}, {3, 5}, {6, 8}, {7, 9}}
   2{{1, 3, 5}, {2, 4}, {6, 8}, {7, 9}}
   3{{1, 6, 8}, {2, 4}, {3, 5}, {7, 9}}
   4{{1, 7, 9}, {2, 4}, {3, 5}, {6, 8}}
   generically hyperbolic

eq[{}]

   #=5

   {{a5 + a3*B^3, A*a9 + a7*B^2, a8 + A^2*a6*B, A^3*a2 + a4},
      {a5 + a3*C^3, a9*B + a7*C^2, a8 + a6*B^2*C, A^3*a1 + a4 + a2*B^3},
      {a5 + a3*C^3, a9*B + a7*C^2, A^3*a1 + a8 + a6*B^2*C, a4 + a2*B^3},
      {a5 + a3*C^3, A^3*a1 + a9*B + a7*C^2, a8 + a6*B^2*C, a4 + a2*B^3},
      {A^3*a1 + a5 + a3*C^3, a9*B + a7*C^2, a8 + a6*B^2*C, a4 + a2*B^3}}

dd=eq[{1,1,1,1,1,1,1,1,1}];
```

```
   #=5

l=dd[[1]]

   {1 + B^3, A + B^2, 1 + A^2*B, 1 + A^3}

alg[{1,4},2]

 3.14018 10^-16 {B -> 0.5 - 0.866025 I, A -> 0.5 + 0.866025 I}
 4.00297 10^-16 {B -> -1. + 3.33067 10^-16 I,
                 A -> -1. + 3.33067 10^-16 I}
 1.24127 10^-16 {B -> 0.5 + 0.866025 I, A -> 0.5 - 0.866025 I}

alg[{1,2,4},3]

 8.9509 10^-16 {B -> -1. + 1.11022 10^-16 I, A -> -1. + 3.33067 10^-16 I}

dd=eq[{1,1,1,1,1,-1,-1,1,1}];

   #=5

l=dd[[1]]

   {1 + B^3, A - B^2, 1 - A^2*B, 1 + A^3}

alg[{1,4},2]

   no solutions

l=dd[[2]]

   {1 + C^3, B - C^2, 1 - B^2*C, 1 + A^3 + B^3}

alg[{1,2},3]

   no solutions

l=dd[[3]]

   {1 + C^3, B - C^2, 1 + A^3 - B^2*C, 1 + B^3}

alg[{1,4},2]

   no solutions

l=dd[[4]]

   {1 + C^3, A^3 + B - C^2, 1 - B^2*C, 1 + B^3}

alg[{1,4},3]

   no solutions

l=dd[[5]]

   {1 + A^3 + C^3, B - C^2, 1 - B^2*C, 1 + B^3}

alg[{2,4},3]

   no solutions
```

Chapter 8
Nevanlinna Theory over Function Fields

We develop here Nevanlinna theory described in Sects. 4.1 and 4.2 for holomorphic curves over algebraic function fields. This is understood as an approximation theory of algebraic functions by algebraic functions. Vojta [87] formulated Diophantine approximation theory from the viewpoint of Nevanlinna theory, noticing their analogies. From that viewpoint Nevanlinna theory is an approximation theory of complex numbers by transcendental meromorphic functions. This brought a new viewpoint to the both theories and has activated their research. The theory over algebraic function fields is considered to be situated in the middle of them.

8.1 Lang's Conjecture

Here a function field means an algebraic function field over $\mathbf{C}$. In the present chapter we describe Nevanlinna theory over an algebraic function field, but there lies the Kobayashi hyperbolicity behind it. We discuss the problem briefly in this section. Cf. S. Lang [91] for more details.

In general we consider the N-dimensional projective space $\mathbf{P}^N_F$ over a field F. Take a homogeneous coordinate system $[x_0, \ldots, x_N]$. We set

$$\mathbf{P}^N(F) = \left\{[x_0, \ldots, x_N];\, x_j \in F,\ {}^{\exists}x_k \neq 0\right\}.$$

Let V be an algebraic subvariety of $\mathbf{P}^N_F$ defined over F. It is standard to consider a point of V once over the algebraic closure $\bar{F}$ of F, but here we take a set $V(F)$ of F-solutions of V,

$$V(F) = V \cap \mathbf{P}^N(F),$$

whose element is called an *F-rational point* of V.

Based on Mordell's Conjecture (Mordell [22]), S. Lang [74] proposed the following daring conjecture.

Conjecture 8.1.1 (Lang's Conjecture) *Let F be an algebraic number field (i.e., a finite extension field over $\mathbf{Q}$) and let V be a projective algebraic variety defined*

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over F. Assume that for some embedding $F \hookrightarrow \mathbf{C}$, $V_{\mathbf{C}}$ given by V as complex space is Kobayashi hyperbolic. Then $V(F)$ is a finite set.

For an analogue over algebraic function fields (i.e., of relative case) he proposed in the same paper

Conjecture 8.1.2 (Lang [74]) *The analogue of Conjecture 8.1.1 over algebraic function fields should hold. In particular, in the flat case (product structure), there are only finitely many surjective holomorphic maps from a compact complex manifold onto a compact Kobayashi hyperbolic manifold.*

When V is one-dimensional, $V_{\mathbf{C}}$ is Kobayashi hyperbolic if and only if the genus of V is greater than or equal to 2. In this case Lang's Conjecture 8.1.1 is equivalent to Mordell's Conjecture, and it was proved by Faltings [83b]. As for Mordell's Conjecture, S. Lang [60] raised a problem to deal with Mordell's Conjecture not only over algebraic number fields but also over algebraic function fields. In view of the finiteness of rational points the case of a one-dimensional function field is essential. Let R be an algebraic curve defined over an algebraically closed field k with characteristic zero. The analogue of Mordell's Conjecture (Faltings' Theorem) over an algebraic function field is as follows.

Let $\pi : \mathscr{X} \to R$ be a projective family of algebraic curves over R, and let $S \subset R$ be a finite subset such that over the restriction $\mathscr{X}|_{R\setminus S} = \pi^{-1}(R \setminus S)$ the differential $d\pi$ is non-degenerate, and the fibers $\mathscr{X}_t$, $t \in R \setminus S$ are irreducible. One may think of $\mathscr{X}$ as a projective algebraic curve defined over the rational function field $k(R)$ of R. In this case a $k(R)$-rational point of $\mathscr{X}$ is naturally identified with a rational section $\sigma : R \to \mathscr{X}$, $\pi \circ \sigma = \mathrm{id}$.

Theorem 8.1.3 *Let the genus of $\mathscr{X}_t$, $t \in R \setminus S$, be g and assume that $g \geqq 2$. If there exist infinitely many rational sections of $\mathscr{X} \to R$, there exist a non-empty Zariski open subset $R' \subset R$ and an isomorphism $\mathscr{X}|_{R'} \cong R' \times \mathscr{X}_t$ $(t \in R')$ such that all but for finitely many rational sections are reduced to constant sections.*

There are now known three direct proofs by different methods besides the indirect approach of Shafarevich–Parshin, which will be discussed below:

(i) The first proof was given by Y. Manin [63], but later on an incomplete part was found in the proof, which was filled in by Coleman [90] (cf. Chai [91]). Y. Manin used differential equations.
(ii) H. Grauert [65] gave another proof based on algebraic geometry.
(iii) J. Noguchi [85a] gave a rather simple proof of Theorem 8.1.3 based on the Kobayashi hyperbolicity (see Theorem 8.1.9).

It is natural to think of the higher dimensional version of Theorem 8.1.3 in view of Lang's Conjecture 8.1.1. A higher dimensional notion of the higher genus of algebraic curves is the ampleness of cotangent bundles, equivalently, the (weak) negativity of tangent bundles by Grauert [62]. It is noted that a compact complex

manifold with ample cotangent bundle is Kobayashi hyperbolic (Kobayashi [75], [98], Theorem 3.6.21).

H. Grauert's method was extended to the higher dimensional case.

Theorem 8.1.4 (Noguchi [81b]) *Let K be an algebraic function field of a projective algebraic variety R over an algebraically closed field k of* $\operatorname{char} k = 0$. *Let $\mathscr{X}$ be a smooth projective algebraic variety over K. Assume that the cotangent bundle of $\mathscr{X}$ is ample. Then one of the following holds*:

(i) *The set $\mathscr{X}(K)$ of K-rational points of $\mathscr{X}$ is contained in a proper algebraic subset of $\mathscr{X}$.*
(ii) *There exists a variety $\mathscr{X}_0$ defined over k such that $\mathscr{X} \cong \mathscr{X}_0 \times R$ and $\mathscr{X}(K) \setminus \mathscr{X}(k)$ is finite.*

On the other hand, Shafarevich [63] proposed the following conjecture.

Conjecture 8.1.5 (Shafarevich's Conjecture) *Let F be an algebraic number field and let $\mathscr{O}_F$ be the ring of algebraic integers of F. Let $S \subset \operatorname{Spec}(\mathscr{O}_F)$ be a finite subset and fix an integer $g \geqq 2$. Then there are at most finitely many algebraic curves of genus g or g-dimensional abelian varieties defined over F with good reduction outside S.*

For this conjecture the analogue over algebraic function fields was first considered. Parshin [68] proved the analogue over an algebraic function field in the case of $S = \emptyset$. Moreover he showed that Shafarevich's Conjecture implies Mordell's Conjecture over algebraic function fields and algebraic number fields as well. Then Arakelov [71] proved the analogue of Shafarevich's Conjecture over algebraic function fields in the general case of S. There is a counter-example which is due to Serre to the analogue of Shafarevich's Conjecture for abelian varieties defined over algebraic function fields in general, but Faltings [83a] gave a necessary and sufficient condition for that analogue for abelian varieties to hold. Then in the same year Faltings proved Mordell's Conjecture by solving Shafarevich's Conjecture. Afterwards Vojta [91] gave a proof modeled after his function field case, and Bombieri [90] gave also another proof based on Diophantine approximations.

It is interesting to learn that the motivation of Shafarevich's Conjecture 8.1.5 was to provide a good problem in algebraic geometry as an analogue of Hermite's theorem over number fields.

There are a number of results related to the analogues of Mordell's Conjecture and Shafarevich's Conjecture over algebraic function fields (Mumford [75]; Noguchi [81b], [85a], [88]; Imayoshi–Shiga [88]; Miyano–Noguchi [91]; Saito–Zucker [91]; Zaidenberg [90]; Makoto Suzuki [94]; etc.).

In this way it is worthwhile and interesting by itself to deal with problems related to rational points over algebraic function fields.

Now we return to Conjecture 8.1.2 of S. Lang. The following finiteness theorem due to Kobayashi–Ochiai [75] was also a result motivated by Conjecture 8.1.2.

Theorem 8.1.6 *There are at most finitely many surjective meromorphic mappings from a compact complex manifold onto a compact complex manifold of general type.*

Let $\mathscr{Y}$ be an irreducible complex space, and let $\eta : \mathscr{Y} \to R$ be a proper surjection. Assume the existence of a finite subset $S \subset R$ such that on the restriction denoted by

$$R' = R \setminus S, \qquad \mathscr{Y}' = \eta^{-1}(R'), \qquad \eta' = \eta|_{\mathscr{Y}'}$$

$d\eta'$ is non-degenerate everywhere on $\mathscr{Y}'$ and $\mathscr{Y}_t = \eta'^{-1}(t)$ $(t \in R')$ is a connected complex manifold. We consider the following condition to think of Lang's Conjecture 8.1.1.

Condition 8.1.7 (i) For every $t \in R'$, $\mathscr{Y}_t$ is Kobayashi hyperbolic.

(ii) For every $t \in S$, there is a neighborhood $U \ni V \ni t$ such that $\mathscr{Y}'|_{V\setminus\{t\}}$ is hyperbolically embedded into $\mathscr{Y}|_U$.

Proposition 8.1.8 (Noguchi [85a]) *Let $\eta : \mathscr{Y} \to R$ be as above with $\dim \mathscr{Y}_t = 1$. If the genus of $\mathscr{Y}_t$ is not less than 2, then there is a compactification $\bar{\mathscr{Y}}'$ for which Condition 8.1.7 (ii) holds.*

Theorem 8.1.9 (Noguchi [85a], [92]) *Let the notation be as above. We assume Condition 8.1.7 (i), and in the case of $\dim \mathscr{Y}_t \geqq 2$, we further assume Condition 8.1.7 (ii).*

Then the set of meromorphic section of $\eta' : \mathscr{Y}' \to R'$ forms a compact complex space Γ and the evaluation mapping

$$\Phi : (s, t) \in \Gamma \times R' \to s(t) \in \mathscr{Y}'$$

is meromorphic. In particular, if there is a point $t_0 \in R'$ with $\{s(t_0); s \in \Gamma\}$ being Zariski dense in $\mathscr{Y}'_{t_0}$, then there is an irreducible component $\Gamma_0 \subset \Gamma$ such that

$$\Phi|_{\Gamma_0 \times R'} : \Gamma_0 \times R' \to \mathscr{Y}'$$

is bimeromorphic such that after lifting to the normalizations of the domain and the image spaces it gives rise to a holomorphic isomorphism.

In the proof of the above theorem we obtained the following finiteness theorem, which was used there: Combined, these give an affirmative answer to Conjecture 8.1.2.

Theorem 8.1.10 (Noguchi [92]) *Let X and Y be compact complex spaces, and assume that Y is Kobayashi hyperbolic. Then there are at most finitely many meromorphic surjections $f : X \to Y$.*

Cf. Noguchi [85a], Horst [90], and Makoto Suzuki [94] for closely related results.

As seen in the above the Kobayashi hyperbolicity is related to various kinds of finiteness theorems in view of the theory over function fields. As shown in Sect. 7.1,

Nevanlinna theory is very effective in deducing the Kobayashi hyperbolicity. On the other hand, Diophantine approximation theory is an important tool to derive finiteness theorems for the rational solutions of Diophantine equations (Schmidt [80], [91]; Lang [83]; Bombieri [90]; Hindry–Silberman [00]; Waldschmidt [00]). Vojta [87] noticed the analogy between the approximation theory of algebraic numbers due to Thue–Siegel–Roth–Schmidt (Diophantine approximation) and Nevanlinna theory, and formulated it (Vojta's Conjecture). This brought a new viewpoint to both of the theories, and produced new results (Lang [91]; Ru–Wong [91]; Faltings [91]; Vojta [96], [99]; Noguchi–Winkelmann [02]; Noguchi [03a], [03b], [09]; Corvaja–Zannier [04a]; Corvaja–Noguchi [12]; etc.). The theory over algebraic function fields is considered to be situated in the middle of the two theories.

The plan here is to describe the theory over algebraic function fields to make clear the connection of Nevanlinna theory developed up to the preceding chapter with the theory of algebraic numbers, which will be dealt with in the next chapter.

8.2 Nevanlinna–Cartan Theory over Function Fields

In the present section we mainly follow Noguchi [97]. The more general cases of higher dimension and positive characteristic should be conferred to the references at the end of this book.

Let R be a compact Riemann surface (i.e., a complex projective algebraic smooth curve). We denote by $\mathbf{C}(R)$ the rational function field over R. For $u \in \mathbf{C}(R)$ and $a \in R$ we take a holomorphic coordinate t with a as the origin and write

$$u(t) = t^d v(t), \quad d \in \mathbf{Z},\ v(0) \neq 0.$$

Then we set

$$\operatorname{ord}_a u = d,$$

which is called the *order* of u at a, and is well-defined independently of the choice of such t.

Let $L \to R$ be a line bundle over R and fix a hermitian metric $\|\cdot\|$ in L. Let $\sigma \in \Gamma_{\mathrm{rat}}(R, L)$, $\sigma \not\equiv 0$ be a non-zero rational section, where Γ_{rat} denotes the set of all rational sections. The Chern form ω_L of $(L, \|\cdot\|)$ is given by

$$\omega_L = dd^c \log \frac{1}{\|\sigma(x)\|^2}, \quad \sigma(x) \neq 0, \infty. \tag{8.2.1}$$

The Chern form ω_L is a well-defined $C^\infty(1,1)$-form on R. Let D denote the divisor determined by σ. We write $\operatorname{ord}_a D = \operatorname{ord}_a \sigma$. We set

$$\begin{aligned} N(D) &= \sum_{a \in R} \operatorname{ord}_a D, \\ N_k(D) &= \sum_{a \in R} \min\{k, \operatorname{ord}_a D\}, \quad k \in \mathbf{N}. \end{aligned} \tag{8.2.2}$$

The number $N(D)$ is usually denoted by $\deg D$, but for comparison with Nevanlinna theory we use the notation $N(D)$ of the counting function. As observed in Chap. 2, D defines a $(1,1)$ current on R, whose coefficients are Radon measures; here it is a finite sum of Dirac measures with integral coefficients.

Let $R = \bigcup U_\alpha$ be a local trivialization covering of L with $L|_{U_\alpha} \cong U_\alpha \times \mathbf{C}$, and let $\{\xi_{\alpha\beta}\}$ be the associated transition function system. If non-negative functions $\psi_\alpha \geqq 0$ on U_α satisfy

(i) $\psi_\alpha = |\xi_{\alpha\beta}|^2 \psi_\beta$, $U_\alpha \cap U_\beta \neq \emptyset$,
(ii) $\log \psi_\alpha$ is locally integrable,

then $\psi = \{\psi_\alpha\}$ is called a *singular metric* in L. For $\sigma = \{\sigma_\alpha\} \in \Gamma_{\mathrm{rat}}(R, L)$ a function $\|\sigma\|_\psi^2 = |\sigma_\alpha|^2/\psi_\alpha$ on R is defined and $\log \|\sigma\|_\psi$ is integrable over R. As a current on R, the *curvature current* of ψ is defined by

$$\omega_\psi = dd^c[\log \psi_\alpha].$$

Lemma 8.2.3 *Let the notation be as above. Then the following current equation holds.*

(i) $dd^c[\log \frac{1}{\|\sigma(x)\|^2}] = \omega_L - D$.
(ii) $\int_R \omega_L = N(D)$.
(iii) $\int_R \omega_\psi = \int_R \omega_L$.

Proof (i) This follows from Theorem 2.2.16 (cf. Lemma 2.3.28, too).

(ii) This is obtained by integrating the both sides of the current equation of (i) over R.

(iii) There is an integrable function ϕ on R such that $\omega_\psi - \omega_L = dd^c[\phi]$. Therefore

$$\int_R 1 \cdot \omega_\psi - \int_R 1 \cdot \omega_L = \int_R \phi dd^c 1 = 0. \qquad \square$$

Set

$$\deg L = \int_R \omega_L.$$

This is independent of the choice of the hermitian metric in L. By Lemma 8.2.3

$$\deg L = N(D).$$

Let $x = [x_0, \ldots, x_m]$ be a homogeneous coordinate system of m-dimensional complex space $\mathbf{P}^m(\mathbf{C})$ and consider a linear form

$$F = \sum_{j=0}^{m} a_j x_j, \quad a_j \in \mathbf{C}.$$

Letting x_j, $0 \leqq j \leqq m$, be elements of $\mathbf{C}(R)$ without common zero, we consider the counting function $N((F(x))_0)$ of zeros of $F(x)$. We define a divisor D_0 by

$$D_0 = -\sum_{a \in R} \min_j \{\mathrm{ord}_a\, x_j\} \cdot a. \tag{8.2.4}$$

Then D_0 is an effective divisor on R and we have the line bundle $L = L(D_0)$ over R determined by D_0.

There is a section $\sigma_0 \in H^0(R, L)$ whose divisor is $(\sigma_0) = D_0$. Put $\sigma_j = x_j \sigma_0$, $1 \leqq j \leqq m$. Then $\sigma_j \in H^0(R, L)$ and they have no common zero. We write

$$\mathrm{Ht}(x) = \mathrm{Ht}\big((\sigma_j)\big) = \deg L. \tag{8.2.5}$$

The function $\mathrm{Ht}(x)$ corresponds to the order function in Nevanlinna theory, and to the height function in Diophantine approximation theory.

Theorem 8.2.6 *Let the notation be as above, and assume that $F(x) \not\equiv 0$. Then*

$$\mathrm{Ht}(x) = N\Big(\big(F(x)\big)_0\Big) = N\Big(\big(F(\sigma_0, \ldots, \sigma_m)\big)\Big).$$

Proof From the construction we deduce that $F(\sigma_j) \in H^0(R, L)$ and $(F(x))_0 = (F(\sigma_j))$. Thus the required equality follows from Lemma 8.2.3. □

For the sake of notational simplicity we write

$$N\big(F(\sigma_j)\big) = N\Big(\big(F(\sigma_j)\big)\Big), \qquad N_k\big(F(\sigma_j)\big) = N_k\Big(\big(F(\sigma_j)\big)\Big).$$

Let σ_j, $0 \leqq j \leqq l$, be holomorphic sections of $L \to R$ and let (U, z) be a holomorphic local coordinate neighborhood of R. Suppose that $L|_U \cong U \times \mathbf{C}$ and $\sigma_j|_U$ is given by a holomorphic function σ_{jU} on U. We define their Wronskian by

$$W_{(U,z)}\big((\sigma_j)\big) = \begin{vmatrix} \sigma_{0U} & \cdots & \sigma_{lU} \\ \frac{d}{dz}\sigma_{0U} & \cdots & \frac{d}{dz}\sigma_{lU} \\ \vdots & \cdots & \vdots \\ \frac{d^l}{dz^l}\sigma_{0U} & \cdots & \frac{d^l}{dz^l}\sigma_{lU} \end{vmatrix}. \tag{8.2.7}$$

Let (U', z') be another holomorphic local coordinate and let σ_j be given by $\sigma'_{jU'}$. Similarly to (8.2.7) we define $W_{(U',z')}$. Assume that $U \cap U' \neq \emptyset$ and the transition function of L is given by

$$\sigma_{jU} = \xi_{UU'} \cdot \sigma_{jU'}.$$

Inductively using this relation and $\frac{d}{dz} = \frac{dz'}{dz}\frac{d}{dz'}$, we easily see that

$$W_{(U,z)}\big((\sigma_j)\big) = \xi_{UU'}^{l+1} W_{(U',z')}\big((\sigma_j)\big)\left(\frac{dz'}{dz}\right)^{\frac{l(l+1)}{2}}.$$

Therefore we may set

$$W\big((\sigma_j)\big) = W_{(U,z)}\big((\sigma_j)\big)(dz)^{\frac{l(l+1)}{2}},$$

which is an $\frac{l(l+1)}{2}$-differential with values in L^{l+1}. This is called the *Wronskian* of (σ_j). We set

$$\Delta\big((\sigma_j)\big) = \frac{W((\sigma_j))}{\sigma_0 \cdots \sigma_l} \in \Gamma_{\mathrm{rat}}\big(R, K_R^{\frac{l(l+1)}{2}}\big),$$

which is a rational $\frac{l(l+1)}{2}$-differential and is called the *logarithmic Wronskian*. Here K_R denotes the canonical bundle over R. Let g denote the genus of R. Then

$$\deg K_R = 2g - 2.$$

Summarizing the above, we have the following.

Lemma 8.2.8 *Let $\sigma_j \in H^0(R, L)$, $0 \leqq j \leqq l$, be holomorphic sections.*

(i) $W((\sigma_j)) \in H^0(R, L^{l+1} \otimes K_R^{\frac{l(l+1)}{2}})$.
(ii) σ_j, $0 \leqq j \leqq l$, *are linearly independent if and only if* $W((\sigma_j)) \not\equiv 0$.
(iii) *Let A be a complex square $(l+1)$-matrix. If* ${}^t(\tau_0, \ldots, \tau_l) = A\,{}^t(\sigma_0, \ldots, \sigma_l)$,

$$W\big((\tau_j)\big) = (\det A) \cdot W\big((\sigma_j)\big).$$

(iv) $\Delta((\sigma_j)) \in \Gamma_{\mathrm{rat}}(R, K_R^{\frac{l(l+1)}{2}})$. *If* σ_j, $0 \leqq j \leqq l$, *are linearly independent,*

$$N\big(\Delta(\sigma_j)\big) = l(l+1)(g-1).$$

On R, which is compact, there is no term corresponding to $m(r, F)$ in Nevanlinna theory and the order function is identical to the counting function. Thus the First Main Theorem coincides with the Second Main Theorem with non-truncated counting function (Theorem 8.2.6). The Second Main Theorem with truncated counting function $N_k(F(\sigma_j))$ carries some implication which does not finish there. It is impossible to bound $\mathrm{Ht}(x)$ by only one $N_k(F(\sigma_j))$ but possible by using several of them, which is the Second Main Theorem in this case. Put

$$F_i = \sum_{j=0}^{m} a_{ij} x_j, \quad 1 \leqq i \leqq q,\ a_{ij} \in \mathbf{C}.$$

If the hyperplanes defined by these linear equations are in general position (resp. n-subgeneral position), we say that F_i, $1 \leqq i \leqq q$ are in *general position* (resp. *n-subgeneral position*).

Theorem 8.2.9 *Let F_i, $1 \leqq i \leqq q$, be linear forms of $\mathbf{P}^m(\mathbf{C})$ in general position, and let $L \to R$ be a line bundle. Suppose that $\sigma_j \in H^0(R, L)$, $0 \leqq j \leqq m$, have no*

common zero. Let l be the dimension of the linear subspace spanned by the image of $\sigma = [\sigma_0, \ldots, \sigma_m] : R \to \mathbf{P}^m(\mathbf{C})$. Then

$$(q - 2m + l - 1)\,\mathrm{Ht}(\sigma) \leqq \sum_{i=1}^{q} N_l\big(F_i(\sigma)\big) + C(l, m, g).$$

Here

$$C(l, m, g) = \begin{cases} -m(l+1), & g = 0, \\ l(2m - l + 1)(g - 1), & g \geqq 1. \end{cases}$$

In particular, if $l = m$,

$$(q - m - 1)\,\mathrm{Ht}(\sigma) \leqq \sum_{i=1}^{q} N_m\big(F_i(\sigma)\big) + m(m+1)(g-1). \tag{8.2.10}$$

Proof Being restricted F_i to $\mathbf{P}^l(\mathbf{C})$ ($\subset \mathbf{P}^m(\mathbf{C})$), $\sigma : R \to \mathbf{P}^l(\mathbf{C})$ is linearly non-degenerate, and F_i, $1 \leqq i \leqq q$, are in m-subgeneral position. The maximal linearly independent system of $\{\sigma_j\}$ is denoted for the sake of simplicity by $\{\sigma_0, \ldots, \sigma_l\}$. We may assume that $q > 2m - l + 1$. Set $Q = \{1, \ldots, q\}$. There are Nochka's weights $\omega(i) > 0$ for F_i ($i \in Q$) satisfying (i)–(iv) of Theorem 4.1.10 and Nochka's constant $\tilde{\omega} = \max \omega(i)$.

We take an open set $U \subset R$ so that there is a trivialization $L|_U \cong U \times \mathbf{C}$. On U, σ_j, $0 \leqq j \leqq m$, are represented by holomorphic functions σ_{jU}. We put

$$\sigma_U = (\sigma_{0U}, \ldots, \sigma_{mU}), \qquad \|\sigma_U\| = \left(\sum_{j=0}^{m} |\sigma_{jU}|^2\right)^{1/2}.$$

By the similar computation in the proof of Lemma 4.2.3 one gets

$$\|\sigma_U\|^{\tilde{\omega}(q-2m+l-1)} = \phi \frac{\prod_{i \in Q} |F_i(\sigma_U)|^{\omega(i)}}{|W(\sigma_{0U}, \ldots, \sigma_{lU})|} \cdot \left\{ \sum_{P \subset Q, |P| = l+1} \big|\Delta\big((F_i(\sigma_U), i \in P)\big)\big| \right\}. \tag{8.2.11}$$

Here $\phi \geqq 0$ is a bounded function on R, and $\log \phi$ is integrable. Taking the logarithm of (8.2.11) and applying dd^c as currents, we see that $dd^c[\log \|\sigma_U\|]$ is defined globally on R. It follows from Lemmas 8.2.3 and 8.2.8 that

(i) $\int_R 2dd^c[\log \|\sigma_U\|^{\tilde{\omega}(q-2m+l-1)}] = \tilde{\omega}(q - 2m - l + 1)\mathrm{Ht}(\sigma)$,
(ii) $\int_R 2dd^c[\log \phi] = 0$,
(iii) $\int_R 2dd^c[\log \sum_{P \subset Q, |P| = n+1} |\Delta((F_i(\sigma_U), i \in P))|] = l(l+1)(g-1)$.

By the same computation as in Lemma 4.2.6 we have

$$\sum_{i\in Q}\omega(i)\big(F_i(\sigma_U)\big)-\big(W(\sigma_U)\big)\leqq\sum_{i\in Q}\omega(i)\sum_{a\in U}\min\big\{\mathrm{ord}_a\,F_i(\sigma_U),l\big\}\cdot a.$$

Hence we infer that

$$\tilde{\omega}(q-2m+l-1)\mathrm{Ht}(\sigma)\leqq\sum_{i\in Q}\omega(i)N_l\big(F_i(\sigma)\big)+l(l+1)(g-1).$$

By $\omega(j)\leqq\tilde{\omega}$

$$(q-2m+l-1)\mathrm{Ht}(\sigma)\leqq\sum_{i\in Q}N_l\big(F_i(\sigma)\big)+\frac{l(l+1)(g-1)}{\tilde{\omega}}. \tag{8.2.12}$$

It follows from Theorem 4.1.10 (iii) that if $g=0$,

$$(q-2m+l-1)\mathrm{Ht}(\sigma)\leqq\sum_{i\in Q}N_l\big(F_i(\sigma)\big)-m(l+1),$$

and if $g\geqq 1$,

$$(q-2m+l-1)\mathrm{Ht}(\sigma)\leqq\sum_{i\in Q}N_l\big(F_i(\sigma)\big)+l(2m-l+1)(g-1).$$

□

8.3 Borel's Identity and Unit Equations

We consider the following simple equation

$$x_1+x_2+\cdots+x_n=0. \tag{8.3.1}$$

Let (x_j) be a solution in some ring or a field. If $\sum_{j\in J}x_j\neq 0$ for every proper subset $J\subset Q=\{1,\ldots,n\}$, (x_j) is called a *non-degenerate solution*.

In function theory it was initiated by E. Borel [1897] to take x_j from the units of the ring of entire functions, i.e., entire functions without zero. In that case (8.3.1) is called classically Borel's identity (cf. the Historical Supplement of Sect. 4.2).

On the other hand, when x_j are taken from the units of the integral ring $\mathcal{O}_F$ of an algebraic number field F, (8.3.1) is called a *unit equation*, and in the case of $n=3$ it was first dealt with by C.L. Siegel [26]. By making use of it he proved the finiteness of $\mathcal{O}_F$-integral solutions of an affine algebraic curve of genus $\geqq 1$.

In this section we consider (8.3.1) not only for units. Being related to this problem, there is a deep conjecture due to Masser–Oesterlé.

Conjecture 8.3.2 (*abc*-Conjecture; Oesterlé [88], Sect. 3; cf. Granville–Tucker [02]) *We consider mutually prime integers a,b,c satisfying*

$$a+b+c=0. \tag{8.3.3}$$

For given $\varepsilon > 0$ there exists a constant $C_\varepsilon > 0$ such that for a, b, c satisfying (8.3.3),

$$\max\{|a|, |b|, |c|\} \leqq C_\varepsilon \Bigg(\prod_{\substack{p>1 \text{ prime} \\ p|abc}} p \Bigg)^{1+\varepsilon}. \tag{8.3.4}$$

Since $|c| \leqq 2\max\{|a|, |b|\}$ by the assumption, (8.3.4) is equivalent to the following estimate with another constant C'_ε:

$$\log \max\{|a|, |b|\} \leqq (1+\varepsilon) \sum_{\substack{p>1 \text{ prime} \\ p|abc}} \log p + C'_\varepsilon. \tag{8.3.5}$$

We regard $[a, b]$ as a point of $\mathbf{P}^1$. Then $c = -a - b$ is a linear form on $\mathbf{P}^1$, and the linear forms a, b and c are in general position.

Let $S \subset R$ be a finite subset and let $|S|$ denote the cardinality.

Definition 8.3.6 An element $x \in \mathbf{C}(R)$ is said to be *S-integral* if x has no pole outside S. Moreover, x is said to be an *S-unit* if it has no zero nor pole outside S.

Let a, b, c be S-integral elements of the rational function field $\mathbf{C}(R)$ satisfying (8.3.1). They are said to be co-prime if every pair of them has no common zero nor pole. For a, b, c we take D_0 and $\sigma_0 \in H^0(R, L(D_0))$ as in (8.2.4). For $a\sigma_0, b\sigma_0, c\sigma_0$ we apply Theorem 8.2.9 with $l = m = 1$, $q = 3$ and then have

$$\begin{aligned} \mathrm{Ht}\big((a, b)\big) &= \mathrm{Ht}(a, b, c) \\ &\leqq N_1\big((a\sigma_0)_0\big) + N_1\big((b\sigma_0)_0\big) + N_1\big((c\sigma_0)_0\big) + C(1, 1, g) \\ &\leqq N_1\big((a)_0\big) + N_1\big((b)_0\big) + N_1\big((c)_0\big) + 2|S| + C(1, 1, g). \end{aligned}$$

That is, an analogue of the abc-Conjecture holds over an algebraic function field with $\varepsilon = 0$. This was first shown by Stothers [81] and then later (independently) by Mason [84]. Based on this result, Masser and Oesterlé formulated the above abc-Conjecture (Oesterlé [88], Sect. 3).

Fix a line bundle $L \to R$. Let $\sigma_j \in H^0(R, L)$, $1 \leqq j \leqq n$, be holomorphic sections without common zero. Then

$$\mathrm{Ht}\big((\sigma_j)\big) = \deg L.$$

Moreover, when they satisfy (8.3.1), i.e.,

$$\sigma_1 + \cdots + \sigma_n = 0, \tag{8.3.7}$$

and (σ_j) is non-degenerate, we investigate the estimate to bound $\mathrm{Ht}((\sigma_j))$. For the purpose we prepare several lemmas.

For a moment we do not assume that $(x_j) = (\sigma_j)$ is a non-degenerate solution of (8.3.1). Let $\mathscr{L}$ be the family of linear sums $L = \sum_{j=1}^n c_j x_j$ in x_j with coefficients

in $\mathbf{C}$ such that $L = 0$ after substituting $x_j = \sigma_j$, $1 \leqq j \leqq n$. A subset $I \subset Q$, $I \neq \emptyset$ is said to be *minimal* if σ_i, $i \in I$, are linearly dependent and σ_i, $i \in I'$, are linearly independent for every proper subset $I' \subsetneqq I$. If I is minimal, there exists a linear combination $L_I = \sum_{i \in I} c_i x_i$ belonging to $\mathscr{L}$. Then every coefficient $c_i \neq 0$, and they are unique up to non-zero constant multiples. We call it a *minimal form*.

Lemma 8.3.8 *The above minimal forms generate* $\mathscr{L}$.

Proof We show that an arbitrary $L \in \mathscr{L}$ is written as a linear combination of minimal forms. We denote by l the length of L, that is, the number of non-zero coefficients, and use the induction on l. If $l = 1$, then $L = 0$ and the claim is trivial. Suppose that $l > 1$ and that all the cases up to $l - 1$ hold. Changing the order of indices if necessary, we may have

$$L = \sum_{i=1}^{l} c_i x_i, \quad c_i \neq 0.$$

If $I = \{1, \ldots, l\}$ is minimal, the proof finishes. Otherwise, there is some $1 \leqq k < l$ such that

$$L' = c'_1 x_1 + \cdots + c'_k x_k \in \mathscr{L}, \quad c'_1 \neq 0.$$

Set $L'' = c'_1 L - c_1 L'$. Then $L', L'' \in \mathscr{L}$ and the lengths of them are less than or equal to $l - 1$. Then it suffices to apply the induction hypothesis. □

Lemma 8.3.9 *Assume* (8.3.7) *and that there is a subset* $I \subset Q$ *satisfying* $\sum_{i \in I} \sigma_i \neq 0$. *Then there is a minimal subset* $J \subset Q$ *such that* $J \cap I \neq \emptyset$ *and* $J \cap (Q \setminus I) \neq \emptyset$.

Proof Since $L = \sum_{i=1}^{n} x_i \in \mathscr{L}$, it follows from Lemma 8.3.8 that there are several minimal forms L_J satisfying

$$L = \sum L_J. \tag{8.3.10}$$

Suppose that the present lemma does not hold. Then every J used in (8.3.10) is contained in I or $Q \setminus I$. Setting variables $x_i = 0$ ($i \in Q \setminus I$) in (8.3.10), we have

$$\sum_{i \in I} x_i = \sum_{J \subset I} c_J L_J.$$

Thus $\sum_{i \in I} \sigma_i = 0$ and this is a contradiction. □

Lemma 8.3.11 *Assume that* σ_i, $i \in Q$, *give a non-degenerate solution of* (8.3.7), *but there is a linearly dependent proper subfamily. Then there is a decomposition of* Q *to non-empty mutually disjoint subsets*

$$Q = I_1 \cup \cdots \cup I_k, \quad k \geqq 2,$$

and furthermore there exist non-empty subsets

$$J_1 \subset I_1, J_2 \subset I_1 \cup I_2, \ldots, J_{k-1} \subset I_1 \cup \cdots \cup I_{k-1},$$

such that

$$I_1, J_1 \cup I_2, \ldots, J_{k-1} \cup I_k$$

are all minimal.

Proof Take arbitrarily a minimal $I_1 \subset Q$. By the hypothesis Q is not minimal and $I_1 \neq Q$. From the non-degeneracy we infer that $\sum_{i \in I_1} \sigma_i \neq 0$. Then Lemma 8.3.9 implies a minimal I_2' which has non-empty intersections with both I_1 and $Q \setminus I_1$. Set $I_2 = I_2' \cap (Q \setminus I_1)$ and $J_1 = I_2' \cap I_1$. If $I_1 \cup I_2 = Q$, the proof ends. Assume that $I_1 \cup I_2 \neq Q$. Applying Lemma 8.3.9 to $I = I_1 \cup I_2$, we obtain a minimal I_3', and in the same way I_3 and J_2. Repeating this process, we obtain the required subsets in finitely many steps. □

Theorem 8.3.12 *Let $x_i \in \mathbf{C}(R)$, $1 \leqq i \leqq n$, be co-prime S-integral elements which give rise to a non-degenerate solution of* (8.3.1). *Then*

$$\mathrm{Ht}\big((x_i)\big) \leqq \sum_{i=1}^{n} N_{n-2}\big((x_i)_0\big) + (n-1)|S| + (n-1)(n-2)(g-1)^+.$$

Proof For x_i we take D_0 and $\sigma_0 \in H^0(R, L(D_0))$ as in (8.2.4), and set

$$\sigma_i = x_i \sigma_0, \quad 1 \leqq i \leqq n.$$

By the assumption σ_i have no common zero. If σ_i, $1 \leqq i \leqq n-1$, are linearly independent, we apply (8.2.10) to the case of $m = n-2, q = n$ to deduce that

$$\begin{aligned}\mathrm{Ht}\big((\sigma_i)\big) &\leqq \sum_{i=1}^{n} N_{n-2}(\sigma_i) + (n-1)(n-2)(g-1)^+ \\ &\leqq \sum_{i=1}^{n} N_{n-2}\big((x_i)_0\big) + (n-1)|S| + (n-1)(n-2)(g-1)^+.\end{aligned}$$

Hence the required inequality is obtained.

In what follows we assume that σ_i, $1 \leqq i \leqq n-1$, are linearly dependent. The assumption of Lemma 8.3.11 is hence satisfied. Take I_i, J_j as in Lemma 8.3.11. Then $|I_1| \geqq 2$ and there is a non-trivial linear relation

$$\sum_{i \in I_1} c_i \sigma_i = 0. \tag{8.3.13}$$

Since I_1 is minimal, all $c_i \neq 0$ for $i \in I_1$. Taking a holomorphic local coordinate z of R, we may represent there σ_i by holomorphic functions, being denoted by the

same notation. Then we differentiate (8.3.13) in z up to order $|I_1|-1$, and write

$$\sum_{i\in I_1} c_i \frac{\sigma_i^{(h)}}{\sigma_i}\sigma_i = 0, \quad 1 \leqq h \leqq |I_1|-1.$$

Similarly for $J_1 \cup I_2$ by differentiating $|I_2|$-times, we obtain

$$\sum_{i\in I_2\cup J_1} c_i \frac{\sigma_i^{(h)}}{\sigma_i}\sigma_i = 0, \quad 1 \leqq h \leqq |I_2|.$$

We carry this process in order up to $J_{k-1}\cup I_k$. Then we have $(|I_1|-1)+|I_2|+\cdots+|I_k| = n-1$ simultaneous equations with respect to σ_i, $1 \leqq i \leqq n$. Let Δ_i be the determinant of the coefficients matrix excluding the row coefficients of σ_i. Then Δ_i is a rational section of $K_R^{(n-1)(n-2)/2}$. By the construction

$$\sigma_i \Delta_j = \pm\sigma_j \Delta_i. \tag{8.3.14}$$

Assume that $i \in I_1$ and set $I_1' = I_1 \setminus \{i\}$. Let $\Delta(I_1'), \Delta(I_2), \ldots, \Delta(I_k)$ be the logarithmic Wronskians of σ_i with indices in $I_i', I_2, \ldots, I_k$ respectively. Then we see that

$$\Delta_i = c_i \Delta\big(I_1'\big)\Delta(I_2)\cdots\Delta(I_k) \not\equiv 0.$$

We infer from this and (8.3.14) that $\Delta_i \not\equiv 0$, $1 \leqq i \leqq n$. Again by (8.3.14) we have

$$\mathrm{Ht}\big((x_i)\big) = \mathrm{Ht}\big((\Delta_i)\big).$$

Calculating the orders of poles of Δ_i, we get

$$\begin{aligned}\mathrm{Ht}\big((\Delta_i)\big) &\leqq \sum_{i=1}^{n} N_{n-2}\big((\sigma_i)\big) + \frac{(n-1)(n-2)}{2}(2g-2)^+ \\ &\leqq \sum_{i=1}^{n} N_{n-2}\big((x_i)_0\big) + (n-1)|S| + (n-1)(n-2)(g-1)^+.\end{aligned}$$

□

Corollary 8.3.15 *If x_i are S-units in Theorem 8.3.12,*

$$\mathrm{Ht}\big((x_i)\big) \leqq \big(n^2-n-1\big)|S| + (n-1)(n-2)(g-1)^+.$$

Proof This is immediate from the following:

$$\begin{aligned}\mathrm{Ht}\big((x_i)\big) &\leqq \sum_{i=1}^{n} N_{n-2}\big((x_i)_0\big) + (n-1)|S| + (n-1)(n-2)(g-1)^+ \\ &\leqq n(n-2)|S| + (n-1)|S| + (n-1)(n-2)(g-1)^+.\end{aligned}$$

□

Theorem 8.3.12 and Corollary 8.3.15 are due to Voloch [85] and Brownawell–Masser [86].

8.4 Generalized Borel's Theorem and Applications

We use the same notation as in the preceding section. Let $x_1, \dots, x_s$ ($s \geqq 2$) be variables. We consider the following Diophantine equation of Fermat type with coefficients $a_i \in \mathbf{C}(R)^*$ ($=\mathbf{C}(R) \setminus \{0\}$), $1 \leqq i \leqq s$:

$$a_1 x_1^d + \cdots + a_s x_s^d = 0, \quad d \in \mathbf{N}. \tag{8.4.1}$$

Lemma 8.4.2 *Let $x_i \in \mathbf{C}(R)$. Assume that $a_i x_i^d$, $1 \leqq i \leqq s-1$, are linearly independent over $\mathbf{C}$. Then*

$$\big(d - s(s-2)\big)\,\mathrm{Ht}\big((x_i)\big) \leqq s(s-1)\,\mathrm{Ht}\big((a_i)\big) + (s-1)(s-2)(g-1)^+.$$

Proof For $a_i x_i^d$, $1 \leqq i \leqq s-1$, we take D_0 and $\sigma_0 \in H^0(R, L(D_0))$ with $(\sigma_0) = D_0$ as in (8.2.4). Then the holomorphic mapping

$$t \in R \to \big[a_1(t)x_1^d(t)\sigma_0(t), \dots, a_{s-1}(t)x_{s-1}^d(t)\sigma_0(t)\big] \in \mathbf{P}^{s-2}(\mathbf{C})$$

is linearly non-degenerate. Let $[u_1, \dots, s_{s-2}] \in \mathbf{P}^{s-2}(\mathbf{C})$ be a homogeneous coordinate system. Then the linear forms

$$u_1, \dots, u_{s-1}, -u_1 - \cdots - u_{s-1}$$

are in general position. Theorem 8.2.9 applied with $q = s$, $m = l = s-2$ yields

$$\begin{aligned}
\mathrm{Ht}\big((a_i x_i^d)_{1\leqq i\leqq s-1}\big) &\leqq \sum_{i=1}^{s} N_{s-2}\big(a_i x_i^d \sigma_0\big) + (s-1)(s-2)(g-1)^+ \\
&\leqq s(s-2)\,\mathrm{Ht}\big((x_i)_{1\leqq i\leqq s-1}\big) + s(s-2)\,\mathrm{Ht}\big((a_i)_{1\leqq i\leqq s-1}\big) \\
&\quad + (s-1)(s-2)(g-1)^+ \\
&\leqq s(s-2)\,\mathrm{Ht}\big((x_i)_{1\leqq i\leqq s}\big) + s(s-2)\,\mathrm{Ht}\big((a_i)_{1\leqq i\leqq s}\big) \\
&\quad + (s-1)(s-2)(g-1)^+.
\end{aligned}$$

On the other hand, one gets

$$\begin{aligned}
\mathrm{Ht}\big((a_i x_i^d)_{1\leqq i\leqq s-1}\big) &= \mathrm{Ht}\big((a_i x_i^d)_{1\leqq i\leqq s}\big) \\
&\geqq \mathrm{Ht}\big((x_i^d)_{1\leqq i\leqq s}\big) - \mathrm{Ht}\big((a_i^{-1})_{1\leqq i\leqq s}\big) \\
&\geqq \mathrm{Ht}\big((x_i^d)_{1\leqq i\leqq s}\big) - s\,\mathrm{Ht}\big((a_i)_{1\leqq i\leqq s}\big).
\end{aligned}$$

Therefore the required inequality is deduced. □

Theorem 8.4.3 *Assume that $x_i \in \mathbf{C}(R)^*$ satisfy (8.4.1) and the following holds*:

$$d > s(s-2) + s(s-1)\,\mathrm{Ht}\big((a_i)\big) + (s-1)(s-2)(g-1)^+. \tag{8.4.4}$$

Then there is a partition of indices $\{1,\dots,s\}=\bigcup_{\nu=1}^{t} I_\nu$ *such that*

(i) $|I_\nu|\geqq 2$, $1\leqq \nu\leqq t$,
(ii) *for arbitrary* $j,k\in I_\nu$, x_j/x_k *is constant*,
(iii) $\sum_{i\in I_\nu} a_i x_i^d=0$, $1\leqq \nu\leqq t$.

Proof We use the induction in $s\geqq 2$. In the case of $s=2$,

$$\frac{a_1}{a_2}=-\left(\frac{x_2}{x_1}\right)^d.$$

If the above right-hand side is not constant,

$$\mathrm{Ht}\big((a_i)\big)=d\,\mathrm{Ht}\big((x_i)\big)\geqq d.$$

This contradicts the assumption (8.4.4). Thus it holds for $I_1=\{1,2\}$.

Suppose that $s>2$ and the statement holds up to $s-1$. From Lemma 8.4.2 it follows that $a_1x_1^d,\dots,a_{s-1}x_{s-1}^d$ are linearly dependent. Changing the indices if necessary, we may have the following non-trivial linear relation:

$$a_1x_1^d+c_2a_2x_2^d+\cdots+c_ra_rx_r^d=0,\quad 2\leqq r\leqq s-1,\ c_j\in\mathbf{C}^*. \tag{8.4.5}$$

We infer from (8.4.1) and (8.4.5) that there are some $c_j'\in\mathbf{C}$ satisfying

$$c_2'a_2x_2^d+\cdots+c_r'a_rx_r^d+a_{r+1}x_{r+1}^d+\cdots+a_sx_s^d=0. \tag{8.4.6}$$

Now the assertion follows from the induction hypothesis applied to (8.4.5) and (8.4.6). □

We take an n-admissible family $\{M_j(z_1,\dots,z_n)\}_{j=1}^{s}$ of monomials in variables $z_1,\dots,z_n$ constructed in Sect. 7.4. We consider a Diophantine equation in variables $[z_1,\dots,z_n]\in\mathbf{P}^{n-1}$ given by

$$\sum_{j\ 1}^{s} a_jM_j^d(z_1,\dots,z_n)=0,\quad a_j\in\mathbf{C}(R)^*. \tag{8.4.7}$$

Theorem 8.4.8 *Let* $[z_1,\dots,z_n]\in\mathbf{P}^{n-1}$, $z_j\in\mathbf{C}(R)$ *be a solution of* (8.4.7).

(i) *If* $d>s(s-2)$, *there is a constant* $C(s,g,\mathrm{Ht}((a_j)))\geqq 0$ *such that*

$$\mathrm{Ht}\big((z_j)\big)\leqq C\big(s,g,\mathrm{Ht}\big((a_j)\big)\big).$$

(ii) *If* d *satisfies* (8.4.4), $\mathrm{Ht}((z_j))=0$; *that is, the continued ratio of* z_j *is constant*.

Proof (i) From Definition 7.4.4 of an n-admissible family we may assume, without loss of generality to show (i) and (ii), that every $z_j\in\mathbf{C}(R)^*$. Write $z=(z_j)$. By a change of indices there is a number $r\in\mathbf{N}$ such that

$$\big\{a_jM_j^d(z_1,\dots,z_n)\big\}_{j=1}^{r} \text{ is a maximal linearly independent system.} \tag{8.4.9}$$

For every $r < k \leqq s$ there exist uniquely an $I(k) \subset \{1, \dots, r\}$ and $c_j \in \mathbf{C}^*$, $j \in I(k)$ such that

$$\sum_{j \in I(k)} c_j a_j M_j^d(z) + a_k M_k^d(z) = 0. \tag{8.4.10}$$

By Lemma 8.4.2

$$\begin{aligned} &\big(d - \big(|I(k)| + 1\big)\big(I(k) - 1\big)\big) \operatorname{Ht}\big((M_j, M_k)_{j \in I(k)}\big) \\ &\quad \leqq \big(|I(k)| + 1\big)|I(k)| \operatorname{Ht}\big((a_j)\big) + |I(k)|\big(|I(k)| - 1\big)(g-1)^+. \end{aligned}$$

Setting

$$A = s(s-1) \operatorname{Ht}\big((a_j)\big) + (s-1)(s-2)(g-1)^+, \tag{8.4.11}$$

we have

$$\operatorname{Ht}\big((M_j, M_k)_{j \in I(k)}\big) \leqq A. \tag{8.4.12}$$

We substitute

$$a_k M_k^d(z) = -\sum_{j \in I(k)} c_j a_j M_j^d(z), \quad r < k \leqq s$$

obtained in (8.4.10) to (8.4.7). We then deduce from (8.4.9) that

$$\bigcup_{k=r+1}^{s} I(k) = \{1, \dots, r\}.$$

We put $I = \{1, \dots, s\}$, and call the index subset $I(k) \cup \{k\}$ a *string* of I. We say that $i, j \in I$ are connected if there are strings $C_1, \dots, C_l$ satisfying

$$i \in C_1, \qquad j \in C_l, \qquad C_h \cap C_{h+1} \neq \emptyset, \quad 1 \leqq h \leqq l-1.$$

Note that $l \leqq s - r \leqq s - 1$, and that $i, j \in I$ being connected clearly defines an equivalence relation in I. We classify I by that equivalence relation:

$$I = \bigcup_{\nu=1}^{t} I_\nu.$$

Set $k_\nu = \max I_\nu$ $(> r)$. Taking $j \in I_\nu \setminus \{k_\nu\}$, we consider the all possible pairs (j, k_ν) $(1 \leqq \nu \leqq t)$. By a change of indices we may take $1 \leqq j_1 < \cdots < j_h < s$ and j_μ, $k_\mu \leqq s$, $1 \leqq \mu \leqq h$, so that j_μ and k_μ are connected, and as sets

$$\{j_1, \dots, j_h, k_1, \dots, k_l\} = I.$$

Now it follows from (8.4.12) that

$$\operatorname{Ht}\big((M_{j_\mu})(z), (M_{k_\mu})(z)\big) = \operatorname{Ht}\big((M_{j_\mu})(z)/(M_{k_\mu})(z), 1\big) \leqq (s-1)A.$$

Writing $M_j(z_1,\dots,z_n)=z_1^{\alpha_{j1}}\cdots z_n^{\alpha_{jn}}$, we get

$$\begin{aligned}\operatorname{ord}_a\left(\frac{M_{j_\mu}}{M_{k_\mu}}\right)&=\sum_{i=1}^{n}(\alpha_{j_\mu i}-\alpha_{k_\nu i})\operatorname{ord}_a z_i\\&=\sum_{i=1}^{n-1}(\alpha_{j_\mu i}-\alpha_{k_\nu i})(\operatorname{ord}_a z_i-\operatorname{ord}_a z_n)\end{aligned}$$

for $a\in R$. The rank of the matrix in the above right-hand side is $n-1$ by (7.4.2), and so there are $\beta_{i\mu(\lambda)}$ such that

$$\operatorname{ord}_a z_i-\operatorname{ord}_a z_n=\sum_{\lambda=1}^{n-1}\beta_{i\mu(\lambda)}\operatorname{ord}_a\left(\frac{M_{j_\mu(\lambda)}}{M_{k_\mu(\lambda)}}\right).$$

Setting $C=\max\{|\beta_{i\mu}|\}$, we obtain

$$\sum_{a\in R}|\operatorname{ord}_a z_i-\operatorname{ord}_a z_n|\leqq 2(n-1)(s-1)CA.$$

Therefore $\operatorname{Ht}(z_i/z_n,1)\leqq(n-1)(s-1)CA$, which implies

$$\operatorname{Ht}(z_1,\dots,z_n)=\operatorname{Ht}\left(\frac{z_1}{z_n},\dots,\frac{z_{n-1}}{z_n},1\right),$$

$$\sum_{i=1}^{n-1}\operatorname{Ht}(z_i/z_n,1)\leqq(n-1)^2(s-1)CA.$$

(ii) By the definition of an n-admissible family we may assume without loss of generality that $z_i\in\mathbf{C}(R)^*$, $1\leqq i\leqq n$. By the assumption for d we may use Theorem 8.4.3. Thus there is a partition of indices $\{1,\dots,s\}=\bigcup_{\nu=1}^{t}I_\nu$ such that for every $j,k\in I_\nu$

$$\frac{M_j^d(z)}{M_k^d(z)}=b_{jk}\in\mathbf{C}^*.$$

Using the same notation as in (7.4.9), we see in the same way as in (7.4.14) that

$$(\log z_1,\dots,\log z_{n'})R\left(\left\{M_j^d\right\};\{I_\nu\}_{\nu=1}^{t}\right)=(\dots,\log b_{jk},\dots).$$

Since the rank of $R(\{M_j^d\};\{I_\nu\}_{\nu=1}^{t})$ is $n-1$, the continued ratio of $(z_1,\dots,z_n)$ is constant. □

The next corollary is a direct consequence of Theorem 8.4.8.

Corollary 8.4.13 (i) *If $d>s(s-2)$, the set of solutions (z_i) of (8.4.7) in $\mathbf{C}(R)$ form a projective algebraic variety.*

(ii) *If d satisfies* (8.4.4) *and a_j, $1 \leqq j \leqq s$, are linearly independent, then* (8.4.7) *does not have a solution in* $\mathbf{C}(R)$.

Proof (i) Let X be the algebraic subvariety of $R \times \mathbf{P}^{n-1}(\mathbf{C})$ defined by (8.4.7), and let $\pi : X \to R$ be the projection. Then (z_j) is identified with a section $\sigma : R \to X$. Since $\mathrm{Ht}((z_j))$ is uniformly bounded, the area of $\sigma(R) \subset X \subset R \times \mathbf{P}^{n-1}(\mathbf{C})$ with respect to the product hermitian metric on $R \times \mathbf{P}^{n-1}(\mathbf{C})$ is uniformly bounded. By Chow's Theorem (the theory of Chow coordinates) such σ's form a projective algebraic variety.

(ii) This is immediate from Theorem 8.4.8 (ii). □

Remark 8.4.14 Theorem 8.4.8 was obtained by Noguchi [97]; cf. Wang [96a], [96b] for related results. Based on a hypersurface defined by (8.4.7), Sarnak–Wang [95] constructed an interesting example from the viewpoint of number theory.

Chapter 9
Diophantine Approximation

The approximation theory of algebraic numbers by rationals or algebraic numbers occupies an important part in Diophantine approximation theory. After C.F. Osgood's observation, P. Vojta formulated an analogue of Nevanlinna theory and Diophantine approximation theory and proposed the so-called Vojta conjecture in Diophantine approximation theory which corresponds to Nevanlinna's Second Main Theorem. In this chapter, referring the proofs of results of the Diophantine approximation to other literature, we formulate them in view of Nevanlinna theory which has been discussed up to the present point. As an application we will prove some theorems on rational points which are analogous to those obtained in Chaps. 7 and 8; the analogy will be observed not only in the statements but also in their proofs.

9.1 Valuations

9.1.1 Definition and the Basic Properties

Let k be a field in general. If a function $|x| \in [0, \infty)$ in $x \in k$ satisfies the following conditions, it is called a *valuation* on k, and the pair $(k, |\cdot|)$ or simply k is called a valuation field:

Definition 9.1.1 (i) $|x| = 0$ if and only if $x = 0$.
(ii) $|xy| = |x| \cdot |y|$, $x, y \in k$.
(iii) $|x + y| \leqq C \max\{|x|, |y|\}$, $x, y \in k$, where $C > 0$ is a constant.

If (iii) is satisfied with $C = 1$, it is called a *non-archimedean valuation* otherwise, it is called an *archimedean valuation*.

A valuation such that $|x| = 1$ identically for $x \neq 0$ and $|0| = 0$ is called the trivial valuation.

Here we describe the fundamental properties of valuations without proofs. Those may be found for instance in Fujisaki [75], Morita [99], Nagata [67], Lang [65], etc.

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Two valuations $|\cdot|_1, |\cdot|_2$ are said to be equivalent if there exists a constant $c > 0$ satisfying

$$|x|_1 = |x|_2^c, \quad x \in k.$$

Theorem 9.1.2 (i) *Two valuations* $|\cdot|_1$ *and* $|\cdot|_2$ *are equivalent if and only if the following holds*:

$$|x|_1 < 1 \iff |x|_2 < 1.$$

(ii) *For every valuation there is an equivalent valuation* $|\cdot|$ *that satisfies the triangle inequality*; *i.e.*, $|x + y| \leqq |x| + |y|$ *for all* $x, y \in k$.

Let k be a number field. We form a set M_k of valuations, taking one representative from every equivalence class of all non-trivial valuations over k. Let M_k^∞ denotes the subset consisting of all archimedean valuations of M_k, and let M_k^0 denote the subset of all non-archimedean valuations of M_k; naturally,

$$M_k = M_k^0 \cup M_k^\infty.$$

Often, we want to regard the absolute value under consideration as a function, and then we write $v(x)$ instead of $|x|$. We will normalize all the elements of M_k in what follows, so that what is called the product formula holds (see (9.1.6), (9.1.18)).

When $v(x)$ is a non-archimedean valuation over k,

$$\mathcal{O}_v = \left\{x \in k;\, v(x) \leqq 1\right\}$$

forms a subring of k and is an integral domain. It is called the *valuation ring* associated to v. The ring $\mathcal{O}_v$ carries a unique maximal ideal

$$\mathfrak{m}_v = \left\{x \in k;\, v(x) < 1\right\}.$$

When $k = \mathbf{Q}$, the absolute value $|x| = \max\{x, -x\}$ satisfies Definition 9.1.1 (iii) with $C = 2$, but not with $C = 1$. Hence this is an archimedean valuation, and we write $v = \infty$ or $|\cdot|_\infty$ for it.

Take a rational prime $p > 1$. For an element $x \in \mathbf{Q}$ we set

$$x = p^\nu y, \quad y \in \mathbf{Q},\ \nu \in \mathbf{Z},$$

where the denominator and the numerator of y have no p-factor. Then we set

$$\operatorname{ord}_p x = \nu, \tag{9.1.3}$$

and

$$|x|_p = p^{-\operatorname{ord}_p x}. \tag{9.1.4}$$

Then $|x|_p$ gives rise to a non-archimedean valuation, called a *p-adic valuation* of $\mathbf{Q}$.

Theorem 9.1.5 *Every non-trivial valuation on* $\mathbf{Q}$ *is equivalent either to the absolute value or to a p-adic valuation.*

Therefore $M_{\mathbf{Q}} = \{p > 1; \text{rational prime}\} \cup \{\infty\}$. These valuations satisfy the *product formula*:

$$\prod_{v \in M_{\mathbf{Q}}} |x|_v = 1, \quad x \in \mathbf{Q}^*. \tag{9.1.6}$$

Equivalently one has

$$\log |x|_\infty = \sum_{p \in M_{\mathbf{Q}}^0} (\mathrm{ord}_p\, x) \log p, \quad x \in \mathbf{Q}^*. \tag{9.1.7}$$

Every $v \in M_k$ defines a distance on k and hence a topology on k. It can be deduced from Theorem 9.1.2 (i) that two valuations are mutually equivalent if and only if they define the same topology.

If the distance on k defined by v is complete, v is called a complete valuation, and the pair (k, v), or simply k is called a complete valuation field. When it is not complete, one takes its completion and denotes it by k_v.

Theorem 9.1.8 (i) *In general the completion k_v of a field k with respect to a non-trivial valuation v is a field, and v is uniquely extended to a valuation $\hat{v}$ on k_v such that $\hat{v}(x) = v(x)$, $^\forall x \in k$. If v is non-archimedean, so is $\hat{v}$.*

(ii) *If v is non-archimedean and k is algebraically closed, then k_v is algebraically closed.*

The completion of $\mathbf{Q}$ with respect to the archimedean valuation $v = \infty$ is $\mathbf{R}$; its algebraic closure is $\mathbf{C}$ which is complete.

Every element x of the completion $\mathbf{Q}_p$ of $\mathbf{Q}$ with respect to the p-adic valuation is uniquely represented in the following form:

$$x = a_N p^N + a_{N+1} p^{N+1} + \cdots, \quad N \in \mathbf{Z},\ a_N \neq 0,$$

where $a_i \in \{0, 1, \ldots, p-1\}$. In this case,

$$|x|_p = p^{-N}.$$

The completion of the algebraic closure $\bar{\mathbf{Q}}_p$ of $\mathbf{Q}_p$ is denoted by $\mathbf{C}_p$. It follows from Theorem 9.1.8 (ii) that $\mathbf{C}_p$ is algebraically closed.

Let v be a non-trivial non-archimedean valuation on a field k. The valuation ring $\mathscr{O}_v$ associated to v is noetherian if and only if the image $v(k^*)$ is an infinite cyclic subgroup of the multiplicative group $\mathbf{R}_+^* = \{x \in \mathbf{R};\, x > 0\}$, and in this case v is said to be *discrete*.

If v is discrete, the maximal ideal $\mathfrak{m}_v$ of $\mathscr{O}_v$ gives rise to a monic ideal $\mathfrak{m}_v = (\pi_v)$, and the element π_v is called a *uniformizer*, and any ideal of $\mathscr{O}_v$ is written as $\{0\}$ or $\mathfrak{m}_v^n$ $(n \in \mathbf{Z}^+)$.

When k is an algebraic number field, every non-trivial non-archimedean valuation on it is discrete.

9.1.2 Extensions of Valuations

Here we describe the extensions of valuations for a given field extension k'/k. Let v (resp. v') be a valuation on k (resp. k'). If the restriction $v'|_k = v$, v' is said to be a valuation over v and is written as

$$v'|v.$$

Theorem 9.1.9 *Let (k, v) be a complete valuation field and let k' be a finite field extension of k. Then v (resp. a valuation w equivalent to v) is uniquely extended to a valuation v' on k' (resp. w' equivalent to w). Moreover, (k', v') is a complete valuation field.*

Let v be a non-trivial non-archimedean valuation on k. Let $\hat{v}$ be the extended valuation on the completion k_v. By Theorem 9.1.9 $\hat{v}$ is extended to a valuation $\bar{\hat{v}}$ on the algebraic closure $\bar{k}_v$ of k_v. Applying this for $k = \mathbf{Q}$ and a rational prime $p > 1$, we equip $\mathbf{C}_p$ with the complete valuation over the p-adic valuation on $\mathbf{Q}$.

Let k'/k be a finite field extension. We embed k' into the algebraic closure $\bar{k}_v$ of k_v by a k-isomorphism, and restrict $\bar{\hat{v}}$ to the image. Thus an extended valuation on k' over v is obtained. The number of k-isomorphisms of k' into $\bar{k}_v$ is equal to the separable degree $[k' : k]_s$.

The *ramification index* of an extension $v'|v$ is defined by

$$e\big(v'|v\big) = \big[v'\big(k'^*\big) : v\big(k^*\big)\big]$$

and its *residue index* is defined by

$$f\big(v'|v\big) = [\mathscr{O}_{v'}/\mathfrak{m}_{v'} : \mathscr{O}_v/\mathfrak{m}_v].$$

If v is discrete, we have

$$e\big(v'|v\big) f\big(v'|v\big) = \big[k'_{v'} : k_v\big]. \tag{9.1.10}$$

Theorem 9.1.11 *Let v be a discrete valuation on k and let k' be a finite separable extension of k. Let $v'_1, \ldots, v'_h$ be all of mutually inequivalent valuations on k' which are over v. Then*

$$\sum_{i=1}^{h} e\big(v'_i|v\big) f\big(v'_i|v\big) = \sum_{i=1}^{h} \big[k'_{v'_i} : k_v\big] = \big[k' : k\big].$$

9.1.3 Normalized Valuations

In what follows, let k be an algebraic number field and set

$$d = [k : \mathbf{Q}].$$

We are going to extend the archimedean and p-adic valuations on $\mathbf{Q}$ to those on k so that the product formula (9.1.6) is preserved.

(i) The Case of Archimedean Valuations

Let $\{\sigma_j\}_{j=1}^{d}$ be the set of embeddings of k into $\mathbf{C}$ over $\mathbf{Q}$. Every σ_j defines a valuation

$$|x|_{\sigma_j} = \left|\sigma_j(x)\right|, \quad x \in k,$$

and an arbitrary non-trivial archimedean valuation on k is equivalent to one of them. Changing the indices if necessary, we may assume that $\sigma_1, \ldots, \sigma_r$ are all those of real embeddings, that is, $\sigma_j(k) \subset \mathbf{R}$, $1 \leqq j \leqq r$, and the others, that is, complex embeddings which always have the complex conjugates, are denoted by $\sigma_{r+1}, \ldots, \sigma_{r+s}, \sigma_{r+s+1} = \bar{\sigma}_{r+1}, \ldots, \sigma_{r+2s} = \bar{\sigma}_{r+s}$. The valuations $|\cdot|_{\sigma_{r+j}}$ and $|\cdot|_{\bar{\sigma}_{r+j}}$ are mutually equivalent for each $1 \leqq j \leqq s$, so that

$$M_k^{\infty} = \{\sigma_j;\, 1 \leqq j \leqq r+s\}.$$

We set

$$d_{\sigma_j} = \begin{cases} 1, & \sigma_j \text{ is real,} \\ 2, & \sigma_j \text{ is complex.} \end{cases}$$

For an element $\sigma \in M_k^{\infty}$ the *normalized valuation* is defined by

$$\|x\|_{\sigma} = |x|_{\sigma}^{d_{\sigma}}.$$

Let $\alpha \in k$. We denote the minimal polynomial of α over $\mathbf{Z}$ by

$$P(X) = a_0 X^{d(\alpha)} + a_1 X^{d(\alpha)-1} + \cdots + a_{d(\alpha)}. \tag{9.1.12}$$

By definition $a_0 \neq 0$ and $a_0, \ldots, a_{d(\alpha)}$ are mutually co-prime. Setting $\alpha_j = \sigma_j(\alpha)$, we have by the relation of coefficients and roots

$$\begin{aligned} \prod_{\sigma \in M_{\mathbf{Q}(\alpha)}^{\infty}} \|\alpha\|_{\sigma} &= \prod_{j=1}^{d(\alpha)} |\alpha_j| = \left|\frac{a_{d(\alpha)}}{a_0}\right|, \\ \prod_{\sigma \in M_{\mathbf{Q}(\alpha)}^{\infty}} \max\bigl\{1, \|\alpha\|_{\sigma}\bigr\} &= \prod_{j=1}^{d(\alpha)} \max\bigl\{1, |\alpha_j|\bigr\}. \end{aligned} \tag{9.1.13}$$

This leads to

$$\prod_{w|v,\ w \in M_k^{\infty}} \|\alpha\|_w = \|\alpha\|_v^{[k:\mathbf{Q}(\alpha)]}, \quad \alpha \in k,\ v \in M_{\mathbf{Q}(\alpha)}^{\infty}. \tag{9.1.14}$$

(ii) The Case of Non-archimedean Valuations

Let $p > 1$ be a rational prime and let $\alpha \in k$. Take an element $w \in M_k^0$, $w|p$. By making use of the norm function $N_{k_w/\mathbf{Q}_p}(\alpha) \in \mathbf{Q}_p$ and the extension degree $d_w = [k_w : \mathbf{Q}_p]$, w is given by

$$|\alpha|_w = \left|N_{k_w/\mathbf{Q}_p}(\alpha)\right|_p^{1/d_w}.$$

The *normalized valuation* of w is defined by

$$\|\alpha\|_w = |\alpha|_w^{d_w} = \left|N_{k_w/\mathbf{Q}_p}(\alpha)\right|_p.$$

Let (9.1.12) be the minimal polynomial of α over $\mathbf{Z}$. Then Theorem 9.1.11 implies that

$$\sum_{w|p,\ w\in M_k^0} d_w = d, \tag{9.1.15}$$

$$N_{k/\mathbf{Q}}(\alpha) = \prod_{w|p,\ w\in M_k^0} N_{k_w/\mathbf{Q}_p}(\alpha) = \left((-1)^d \frac{a_{d(\alpha)}}{a_0}\right)^{[k:\mathbf{Q}(\alpha)]}. \tag{9.1.16}$$

Therefore,

$$\begin{aligned}\prod_{w|p,\ w\in M_k^0} \|\alpha\|_w &= \prod_{w|p,\ w\in M_k^0} \left|N_{k_w/\mathbf{Q}_p}(\alpha)\right|_p \\ &= \left|N_{k/\mathbf{Q}}(\alpha)\right|_p = \left|\frac{a_{d(\alpha)}}{a_0}\right|_p^{[k:\mathbf{Q}(\alpha)]}.\end{aligned} \tag{9.1.17}$$

Thus, the following *product formula* is deduced from (9.1.17), (9.1.14) and (9.1.6):

$$\prod_{w\in M_k} \|\alpha\|_w = 1 \quad (\alpha \in k^*). \tag{9.1.18}$$

For any finite extension k'/k and any $v \in M_k$ we have

$$\prod_{w\in M_{k'},\ w|v} \|\alpha\|_w = \|\alpha\|_v^{[k':k]}, \quad \alpha \in k. \tag{9.1.19}$$

We deduce other useful identities by making use of the embeddings of $\mathbf{Q}(\alpha)$ into $\mathbf{C}_p$. Let $v \in M_{\mathbf{Q}(\alpha)}$, $v|p$. Let $\alpha_i^{(p)}$, $1 \leqq i \leqq d(\alpha)$, be the roots of $P(X) = 0$ in $\mathbf{C}_p$ with counting multiplicities. An embedding $\tau_i : k \to \mathbf{C}_p$, $\tau_i(\alpha) = \alpha_i^{(p)}$ defines a valuation

$$|x|_{\tau_i} = \left|\tau_i(x)\right|_p, \quad x \in \mathbf{Q}(\alpha)$$

on k and the following holds:

$$\prod_{v|p,\ v\in M_{\mathbf{Q}(\alpha)}} \|\alpha\|_v = \prod_{i=1}^{d(\alpha)} \left|\alpha_i^{(p)}\right|_p = \left|\frac{a_{d(\alpha)}}{a_0}\right|_p, \tag{9.1.20}$$

$$\prod_{v|p,\ v\in M_{\mathbf{Q}(\alpha)}} \max\{1, \|\alpha\|_v\} = \prod_{i=1}^{d(\alpha)} \max\{1, \left|\alpha_i^{(p)}\right|_p\}. \tag{9.1.21}$$

Let $\mathscr{O}_k$ denote the ring of integers of k; i.e., $\mathscr{O}_k$ consists of those $\alpha \in k$ such that its minimal polynomial $P(X)$ over $\mathbf{Z}$ is of the form

$$P(X) = X^h + a_1 X^{h-1} + \cdots + a_h, \quad h = \deg_{\mathbf{Q}} \alpha,\ a_i \in \mathbf{Z}.$$

We can give more detailed information using the fact that $\mathscr{O}_k$ is a Dedekind domain. An $\mathscr{O}_k$-module $\mathfrak{a} \subset k$ is called a *fractional ideal* of k if there exists an element $\alpha \in k^*$ such that $\alpha\mathfrak{a} \subset \mathscr{O}_k$. For a non-zero fractional ideal $\mathfrak{a}$, its inverse $\mathfrak{a}^{-1} := \{x \in k;\ x\mathfrak{a} \subset \mathscr{O}_k\}$ is also a fractional ideal and $\mathfrak{a} \cdot \mathfrak{a}^{-1} = \mathscr{O}_k$. The next theorem is fundamental.

Theorem 9.1.22 *For every non-zero fractional ideal $\mathfrak{a} \subset k$ there exist finitely many prime ideals $\mathfrak{p}_i$, $1 \leqq i \leqq l$, and $\operatorname{ord}_{\mathfrak{p}_i} \mathfrak{a} \in \mathbf{Z}$, determined uniquely up to the order of the prime ideals such that*

$$\mathfrak{a} = \prod_{i=1}^{l} \mathfrak{p}_i^{\operatorname{ord}_{\mathfrak{p}_i} \mathfrak{a}}.$$

Let $\alpha \in k^*$ be an arbitrary element. We apply Theorem 9.1.22 to the fractional ideal $\alpha\mathscr{O}_k$. Writing $\operatorname{ord}_{\mathfrak{p}_i} \alpha = \operatorname{ord}_{\mathfrak{p}_i} \alpha\mathscr{O}_k$, we have

$$\alpha\mathscr{O}_k = \prod_{\mathfrak{p}_i} \mathfrak{p}_i^{\operatorname{ord}_{\mathfrak{p}_i} \alpha}.$$

The quotient $\mathscr{O}_k/\mathfrak{p}_i$ is a finite set whose cardinality is denoted by $N_{k/\mathbf{Q}}(\mathfrak{p}_i)$ and is called the *norm*. Let $\mathfrak{p}$ be one of $\mathfrak{p}_i$'s and set

$$\|\alpha\|_{\mathfrak{p}} = \left(N_{k/\mathbf{Q}}(\mathfrak{p})\right)^{-\operatorname{ord}_{\mathfrak{p}} \alpha}. \tag{9.1.23}$$

There is a unique rational prime $p > 1$ such that $\mathfrak{p} \cap \mathbf{Z} = \mathbf{Z} \cdot p$. Set

$$|\alpha|_{\mathfrak{p}} = \|\alpha\|_{\mathfrak{p}}^{1/\operatorname{ord}_{\mathfrak{p}} p}. \tag{9.1.24}$$

Then $|p|_{\mathfrak{p}} = p^{-1}$, $|\cdot|_{\mathfrak{p}}$ is a valuation over $|\cdot|_p$ and $\|\cdot\|_{\mathfrak{p}}$ is the normalized valuation of $|\cdot|_{\mathfrak{p}}$. We use p and $\mathfrak{p}$ for the associated valuations. Let $e(\mathfrak{p}|p)$ denote the ramification index of the extension $\mathfrak{p}|p$ and let $f(\mathfrak{p}|p)$ denote the residue index. Then

$$N_{k/\mathbf{Q}}(\mathfrak{p}) = p^{f(\mathfrak{p}|p)}, \qquad \|p\|_{\mathfrak{p}} = p^{-e(\mathfrak{p}|p) f(\mathfrak{p}|p)}.$$

Every non-archimedean valuation $v \in M_k^0$ on k defines a prime ideal $\mathfrak{p}_v = \mathscr{O}_k \cap \mathfrak{m}_v = \{x \in \mathscr{O}_k;\, v(x) < 1\}$ of $\mathscr{O}_k$. This gives a bijection between M_k^0 and the collection of non-zero prime ideals of $\mathscr{O}_k$. We write $\operatorname{ord}_v \alpha = \operatorname{ord}_{\mathfrak{p}_v} \alpha$. Rewriting (9.1.23), one gets

$$\|\alpha\|_v = \big(N_{k/\mathbf{Q}}(\mathfrak{p}_v)\big)^{-\operatorname{ord}_v \alpha}. \tag{9.1.25}$$

For a finite extension k'/k

$$\prod_{w|v,\ w\in M_{k'}} \|\alpha\|_w = \|\alpha\|_v^{[k':k]}, \quad \alpha \in k,\ v \in M_k. \tag{9.1.26}$$

9.2 Heights

For a given rational number $x \in \mathbf{Q}$ the first idea to measure its magnitude may be the ordinary absolute value $|x|$, but $|x|$ does not represent well the complexity of x. By making use of co-prime integers $p, q \in \mathbf{Z}$, $p \neq 0$ we write $x = q/p$ and set

$$H(x) = \max\big\{|p|, |q|\big\}, \tag{9.2.1}$$

which is considered to measure the complexity of x better than $|x|$. We call $H(x)$ the *height* of x.

There are several approaches to define the height of an algebraic number $\alpha \in \bar{\mathbf{Q}}(\subset \mathbf{C})$. We give two of them here.

(a) (**Minimal polynomials**) Let $P(X) \in \mathbf{Z}[X]$ be the minimal polynomial of α over $\mathbf{Z}$:

$$P(X) = a_0 X^d + a_1 X^{d-1} + \cdots + a_d. \tag{9.2.2}$$

Here $a_0 \neq 0$ and $\gcd(a_0, \ldots, a_d) = 1$. Then

$$H(\alpha) = \max\big\{|a_j|, 0 \leqq j \leqq d\big\} \tag{9.2.3}$$

is called the *height* of α. For $\alpha \in \mathbf{Q}$, this definition coincides with (9.2.1).

For instance, if $\alpha = \sqrt{2/3}$, then

$$P(X) = 3X^2 - 2$$

is the minimal polynomial of α over $\mathbf{Z}$, and hence $H(\sqrt{2/3}) = 3$.

(b) (**Valuations**) Let k be an algebraic number field. For $\alpha \in k$ the quantities

$$\begin{aligned} H_k(\alpha) &= \prod_{v\in M_k} \max\big\{1, \|\alpha\|_v\big\}, \\ h(\alpha) &= \frac{1}{[k:\mathbf{Q}]} \log H_k(\alpha) = \frac{1}{[k:\mathbf{Q}]} \sum_{v\in M_k} \log^+ \|\alpha\|_v \end{aligned} \tag{9.2.4}$$

are both called *heights*; $h(\alpha)$ is also called the *logarithmic height*.

Remark 9.2.5 (i) By the product formula (9.1.18) we have

$$H_k(\alpha) = H_k\big(\alpha^{-1}\big), \qquad h(\alpha) = h\big(\alpha^{-1}\big), \quad \alpha \in k^*. \tag{9.2.6}$$

(ii) By (9.1.19) $h(\alpha)$ is independent of the choice of k containing α.

We will show that the height $H(\alpha)$ defined by (9.2.3) is essentially the same as $H_k(\alpha)$ defined by (9.2.4) (see Lemma 9.2.8 below).

Lemma 9.2.7 *Let $P(X)$ given by (9.2.2) be the minimal polynomial of $\alpha \in k$ over $\mathbf{Z}$. Let $\alpha_i^{(p)}$, $1 \leqq i \leqq d$, be the roots of $P(X) = 0$ in $\mathbf{C}_p$ with counting the multiplicities.. Then we have*

$$|a_0|_p \prod_{i=1}^{d} \max\big\{1, |\alpha_i^{(p)}|_p\big\} = |a_0|_p \prod_{v|p,\ v\in M_k} \max\big\{1, \|\alpha\|_v\big\} = 1.$$

Proof Changing the indices if necessary, we may assume that $|\alpha_1^{(p)}|_p \leqq \cdots \leqq |\alpha_d^{(p)}|_p$. By the relations between roots and coefficients

$$\frac{a_i}{a_0} = (-1)^i \sum_{1\leqq s_1<\cdots<s_i\leqq d} \alpha_{s_1}^{(p)} \cdots \alpha_{s_i}^{(p)}.$$

Case (a). Suppose that $|\alpha_i^{(p)}|_p \leqq 1$ for all $1 \leqq i \leqq d$. Then the above implies $|a_i|_p \leqq |a_0|_p$. Since $a_j \in \mathbf{Z}$ $(0 \leqq j \leqq d)$ are co-prime, $\max\{|a_j|_p; 0 \leqq j \leqq d\} = 1$, and hence

$$1 = \max\big\{|a_j|_p; 0 \leqq j \leqq d\big\} \leqq |a_0|_p \leqq 1.$$

Thus the required equation follows.

Case (b). Suppose that there is a number $1 \leqq h \leqq d$ satisfying

$$|\alpha_1^{(p)}|_p \leqq \cdots \leqq |\alpha_{h-1}^{(p)}|_p \leqq 1 < |\alpha_h^{(p)}|_p \leqq \cdots \leqq |\alpha_d^{(p)}|_p.$$

Since

$$\left|\frac{a_i}{a_0}\right|_p \leqq \max\big\{|\alpha_{s_1}^{(p)} \cdots \alpha_{s_i}^{(p)}|_p\big\} \leqq |\alpha_h^{(p)}|_p \cdots |\alpha_d^{(p)}|_p = \left|\frac{a_{d-h+1}}{a_0}\right|_p,$$

one gets

$$\max\left\{\left|\frac{a_i}{a_0}\right|_p; 1 \leqq i \leqq d\right\} = \left|\frac{a_{d-h+1}}{a_0}\right|_p = |\alpha_h^{(p)}|_p \cdots |\alpha_d^{(p)}|_p = \prod_{i=1}^{d} \max\big\{1, |\alpha_i^{(p)}|_p\big\}.$$

Therefore,

$$\max\big\{|a_i|_p; 1 \leqq i \leqq d\big\} = |a_0|_p \prod_{i=1}^{d} \max\big\{1, |\alpha_i^{(p)}|_p\big\} \geqq |a_0|_p.$$

It follows that

$$|a_0|_p \prod_{i=1}^{d} \max\{1, |\alpha_i^{(p)}|_p\} = \max\{|a_i|_p; 0 \leqq i \leqq d\} = 1.$$

□

Lemma 9.2.8 *Let d be the degree of $\alpha \in k$. Then*

$$\frac{1}{d}\log H(\alpha) - \log 2 \leqq h(\alpha) \leqq \frac{1}{d}\log H(\alpha) + \frac{\log(d+1)}{2d}.$$

Proof Let the notation be the same as in Lemma 9.2.7. We infer from Lemma 9.2.7 and the product formula (9.1.18) that

$$|a_0| = \prod_{v \in M_k^0} \max\{1, \|\alpha\|_v\}.$$

Let $\alpha_1, \ldots, \alpha_d$ be the roots of $P(X) = 0$ in $\mathbf{C}$ with counting multiplicities. Then by the above equation one obtains

(9.2.9)

$$|a_0| \prod_{i=1}^{d} \max\{1, |\alpha_i|\} = \prod_{v \in M_k^0} \max\{1, \|\alpha\|_v\} \prod_{v \in M_k^\infty} \max\{1, \|\alpha\|_v\} = H_k(\alpha).$$

Hence we see that

$$\left|\frac{a_i}{a_0}\right| = \left| \sum_{1 \leqq s_1 < \cdots < s_i \leqq d} \alpha_{s_1} \cdots \alpha_{s_i} \right| \leqq \sum_{1 \leqq s_1 < \cdots < s_i \leqq d} \prod_{j=1}^{i} \max\{1, |\alpha_{s_j}|\}$$

$$\leqq 2^d \prod_{i=1}^{d} \max\{1, |\alpha_i|\} = \frac{2^d}{|a_0|} H_k(\alpha).$$

Thus

$$\frac{1}{d}\log H(\alpha) - \log 2 \leqq \frac{1}{d}\log H_k(\alpha) = h(\alpha).$$

On the other hand, by making use of (9.2.9), $P(X) = a_0 \prod_{i=1}^{d}(X - \alpha_i)$ and Lemma 2.5.14, we have

$$dh(\alpha) = \log\left(|a_0| \prod_{j=1}^{d} \max\{1, |\alpha_j|\}\right) = \log|a_0| + \sum_{j=1}^{d} \log^+ |\alpha_j|$$

$$= \log|a_0| + \sum_{j=1}^{d} \frac{1}{2\pi} \int_0^{2\pi} \log|e^{i\theta} - \alpha_j| d\theta$$

(continued)

$$
\begin{aligned}
&= \frac{1}{2\pi}\int_0^{2\pi} \log\left|a_0 \prod_{j=1}^{d}\left(e^{i\theta}-\alpha_j\right)\right| d\theta = \frac{1}{2\pi}\int_0^{2\pi} \log\left|P\left(e^{i\theta}\right)\right| d\theta \\
&= \frac{1}{2}\frac{1}{2\pi}\int_0^{2\pi} \log\left|P\left(e^{i\theta}\right)\right|^2 d\theta \\
&\leqq \frac{1}{2}\log\left(\frac{1}{2\pi}\int_0^{2\pi}\left|a_0 e^{id\theta}+\cdots+a_d\right|^2 d\theta\right) \\
&= \frac{1}{2}\log\left(|a_0|^2+\cdots+|a_d|^2\right) \leqq \frac{1}{2}\log\left\{\left(\max\left\{|a_j|\right\}\right)^2(d+1)\right\} \\
&= \log H(\alpha) + \frac{\log(d+1)}{2}.
\end{aligned}
$$

From these the required second inequality is deduced. □

We denote by $|A|$ the cardinality of a set A as before.

Theorem 9.2.10 *For every $d \in \mathbf{N}$ and $r > 0$ we have*

$$\left|\left\{\alpha \in \bar{\mathbf{Q}};\, \left[\mathbf{Q}(\alpha):\mathbf{Q}\right] \leqq d,\, h(\alpha) \leqq r\right\}\right| \leqq d\left(2e^{dr}+1\right)^{d+1}.$$

Proof It follows from Lemma 9.2.8 that $H(\alpha) \leqq 2e^{dr}$. Therefore the number of possible minimal polynomials of those given α over $\mathbf{Z}$ is at most $(2e^{dr}+1)^{d+1}$. Hence the number of α is at most $d(2e^{dr}+1)^{d+1}$. □

It should be noted that (9.1.23) contains more information about α than just its height $H(\alpha)$. Taking this into count, we introduce the following quantities.

We take and fix an algebraic number field k and a finite subset $S \subset M_k$ with $S \supset M_k^{\infty}$. For a number $\lambda \in \mathbf{N}$ or $\lambda = \infty$ we define the *λ-truncated counting function* by

$$
\begin{aligned}
N_\lambda(x;S) &= \frac{1}{[k:\mathbf{Q}]}\sum_{v\in M_k\setminus S}\left(\min\{\lambda, -\operatorname{ord}_v x\}\right)^+ \log N_{k/\mathbf{Q}}(\mathfrak{p}_v), \\
N(x;S) &= N_\infty(x;S) = \frac{1}{[k:\mathbf{Q}]}\sum_{v\in M_k\setminus S}\log^+\|x\|_v.
\end{aligned}
\tag{9.2.11}
$$

We define the approximation function for ∞ by

$$m(x;S) = \frac{1}{[k:\mathbf{Q}]}\sum_{v\in S}\log^+\|x\|_v. \tag{9.2.12}$$

It is noted that $N(x;S)$ and $m(x;S)$ are independent of the choice of k containing x.

Fix an arbitrary $a \in k$. Then for every $x \in k \setminus \{a\}$ one obtains

$$
\begin{aligned}
&\log^+\|x-a\|_v \leqq \log^+\|x\|_v + \log^+\|a\|_v, \quad v \in M_k^0, \\
&\log^+\|x-a\|_v \leqq \log^+\|x\|_v + \log^+\|a\|_v + d_v\log 2, \quad v \in M_k^\infty,
\end{aligned}
$$

where $d_v = 1$ if $k_v = \mathbf{R}$ and $d_v = 2$ if $k_v = \mathbf{C}$. Since $\sum_{v \in M_k^\infty} d_v = [k : \mathbf{Q}]$, we obtain

$$\big|h(x-a) - h(x)\big| \leqq h(a) + \log 2.$$

For $a = \infty$ we set

$$N\left(\frac{1}{x-\infty}; S\right) = N(x; S), \qquad m\left(\frac{1}{x-\infty}; S\right) = m(x; S).$$

From the above and (9.2.6) we obtain the following.

Theorem 9.2.13 (Analogue of the First Main Theorem) *For $a \in k \cup \{\infty\}$*

$$h(x) = N\left(\frac{1}{x-a}; S\right) + m\left(\frac{1}{x-a}; S\right) + C(a, x), \quad x \in k \setminus \{a\}.$$

Here $C(\infty, x) = 0$, and for $a \neq \infty$, $|C(a, x)| \leqq h(a) + \log 2$.

Remark 9.2.14 Cf. Nevanlinna's First Main Theorem 1.1.17.

Definition 9.2.15 (i) An element $x \in k$ is said to be *S-integral* or an *S-integer* if $|x|_v \leqq 1$ for every $v \in M_k \setminus S$. We denote by $\mathcal{O}_{kS}$ the ring of S-integers of k.

(ii) An element $x \in k$ is called an *S-unit* if $|x|_v = 1$ for every $v \in M_k \setminus S$. The multiplicative group of S-units of k is denoted by $\mathcal{O}_{kS}^*$.

Notice that

$$\begin{aligned} &N(x; S) = 0, && x \in \mathcal{O}_{kS}, \\ &N(x; S) = N(x^{-1}; S) = 0, && x \in \mathcal{O}_{kS}^*. \end{aligned}$$

We fix a homogeneous coordinate system $[x_0, \ldots, x_n]$ of the n-dimensional projective space $\mathbf{P}_k^n$ over k. A point $\mathrm{x} = [x_0, \ldots, x_n]$ with $x_j \in k$, $0 \leqq j \leqq n$, is called a *k-rational point* of $\mathbf{P}_k^n$, and the set of all k-rational points of $\mathbf{P}_k^n$ is denoted by $\mathbf{P}^n(k)$. Let $x = (x_0, \ldots, x_n)$ be an $(n+1)$-dimensional vector and let V be a subvariety defined by a system of homogeneous polynomials with coefficients in k:

$$\sigma_\lambda(x_0, \ldots, x_n) = 0, \quad \lambda \in \Lambda.$$

We denote by $V(k)$ the set of all points $\mathrm{x} \in \mathbf{P}^n(k)$ such that $\sigma_\lambda(\mathrm{x}) = 0$, $\lambda \in \Lambda$, and call it the k-rational point set of V.

The (*multiplicative*) *height* of $\mathrm{x} = [x_0, \ldots, x_n] \in \mathbf{P}^n(k)$ with respect to k and the (*logarithmic*) *height* of x are defined respectively by

$$\begin{aligned} \mathrm{H}_k(\mathrm{x}) &= \prod_{v \in M_k} \max\big\{\|x_j\|_v; 0 \leqq j \leqq n\big\}, \\ \hat{\mathrm{h}}(\mathrm{x}) &= \frac{1}{[k : \mathbf{Q}]} \log \mathrm{H}_k(\mathrm{x}). \end{aligned} \tag{9.2.16}$$

By the product formula (9.1.18) $\mathrm{H}_k(\mathrm{x})$ and $\hat{\mathrm{h}}(\mathrm{x})$ are defined over $\mathbf{P}^n(k)$. By (9.1.26) $\hat{\mathrm{h}}(\mathrm{x})$ is independent of the choice of k containing x_j, $0 \leqq j \leqq n$.

If $n = 1$ and $\mathrm{x} = [1, x] \in \mathbf{P}^1(k)$, then

$$\hat{\mathrm{h}}(\mathrm{x}) = h(x). \tag{9.2.17}$$

In view of definition (9.2.16) $\hat{\mathrm{h}}(\mathrm{x})$ is an analogue of Cartan's order function (cf. (2.5.11)).

Theorem 9.2.18 *For $r > 0, d > 0$ and $n \in \mathbf{N}$ there is a constant $C(r, d, n) > 0$ such that*

$$\left|\left\{\mathrm{x} = [x_0, \ldots, x_n] \in \mathbf{P}^n(\bar{\mathbf{Q}}); \left[\mathbf{Q}(x_i) : \mathbf{Q}\right] \leqq d, \hat{\mathrm{h}}(\mathrm{x}) \leqq r\right\}\right| \leqq C(r, d, n).$$

In particular, for a number field k with $d = [k : \mathbf{Q}]$ and $r > 0$ we have

$$\left|\left\{\mathrm{x} \in \mathbf{P}^n(k); \hat{\mathrm{h}}(\mathrm{x}) \leqq r\right\}\right| \leqq C(r, d, n).$$

Proof Let $\mathrm{x} = [x_0, \ldots, x_n] \in \mathbf{P}^n(\bar{\mathbf{Q}})$. Then some $x_j \neq 0$ and so for instance we may assume $x_0 \neq 0$. Then we let $x_0 = 1$. Since $h(x_j) \leqq \hat{\mathrm{h}}(\mathrm{x}) \leqq r$ by assumption, Theorem 9.2.10 implies the finiteness of such x_j's. Therefore the finiteness of such x's follows. It follows from Theorem 9.2.10 that the bound of the number depends only on r, d, n. □

Let D be an effective divisor of degree d on $\mathbf{P}^n_k$ defined by a homogeneous polynomial $\sigma(x_0, \ldots, x_n) = 0$ with coefficients in k. For $\mathrm{x} = [x_0, \ldots, x_n] \in \mathbf{P}^n(k)$ we set

$$\left\|\sigma(\mathrm{x})\right\|_v = \frac{\|\sigma(x_0, \ldots, x_n)\|_v}{(\max\{\|x_j\|_v; 0 \leqq j \leqq n\})^d}, \quad v \in M_k. \tag{9.2.19}$$

We write $\sigma(\mathrm{x}) = \sum_{|\nu|=d} c_\nu x^\nu$ with $c_\nu \in \mathscr{O}_k$, where $\nu = (\nu_0, \ldots, \nu_n)$, $\nu_j \geqq 0$, $|\nu| = \sum \nu_j$, and $x^\nu = (x_0)^{\nu_0} \cdots (x_n)^{\nu_n}$. Take an arbitrary $v \in M_k^0$. One gets

$$\left\|\sigma(\mathrm{x})\right\|_v \leqq \max\left\{\left\|c_\nu x^\nu\right\|_v\right\} \leqq \max\left\{\|c_\nu\|_v\right\}\left(\max\left\{\|x_j\|_v; 0 \leqq j \leqq n\right\}\right)^d.$$

Since $c_\nu \in \mathscr{O}_k$, $\|c_\nu\|_v \leqq 1$. Therefore

$$\left\|\sigma(\mathrm{x})\right\|_v \leqq \left(\max\left\{\|x_j\|_v; 0 \leqq j \leqq n\right\}\right)^d.$$

For $v \in M_k^\infty$, $k_v = \mathbf{R}$ or $k_v = \mathbf{C}$, and hence the compactness of projective space implies that there is a constant $c \geqq 1$ independent of x and v satisfying $\|\sigma(\mathrm{x})\|_v \leqq c$. We set $c_v = 1$ for $v \in M_k^0$, and $c_v = c$ for $v \in M_k^\infty$. Then we have

$$\left\|\sigma(\mathrm{x})\right\|_v \leqq c_v, \quad v \in M_k. \tag{9.2.20}$$

Now we define the following quantities for $\mathrm{x} \in \mathbf{P}^n(k) \setminus D$:

$$
\begin{aligned}
m(\mathrm{x}; S, D) &= \frac{1}{[k:\mathbf{Q}]} \sum_{v \in S} \log \frac{1}{\|\sigma(\mathrm{x})\|_v} \geqq -\log c, \\
N(\mathrm{x}; S, D) &= \frac{1}{[k:\mathbf{Q}]} \sum_{v \in M_k \setminus S} \log \frac{1}{\|\sigma(\mathrm{x})\|_v} \geqq 0.
\end{aligned}
\tag{9.2.21}
$$

In analogy to Nevanlinna theory, we may view $m(\mathrm{x}; S, D)$ as a proximity (approximation) function of x for D, and $N(\mathrm{x}; S, D)$ as a counting function.

Remark 9.2.22 For a given D, $m(\mathrm{x}; S, D)$ and $N(\mathrm{x}; S, D)$ depend on the choice of σ, but the differences of these quantities for different choices of σ are bounded functions in $\mathrm{x} \in \mathbf{P}^n(k) \setminus D$ which we denote by $O(1)$.

We study the relation of the quantities $m(x; S)$ and $N(x; S)$ for $x \in k$ introduced in the previous section with those defined in the present section. Let $a \in k$, $\mathrm{x} = [x_0, x_1] = [1, x] \in \mathbf{P}^1(k)$. Writing $a = a_0/a_1$ with $a_i \in \mathcal{O}_k$, one gets a linear form $\sigma_a = a_0 x_0 - a_1 x_1$. Note that $[1, a] = \{\sigma_a = 0\}$. A simple computation implies

$$
\begin{aligned}
m\left(\frac{1}{x-a}; S\right) &= m\big(\mathrm{x}; S, [1, a]\big) + O(1), \\
N\left(\frac{1}{x-a}; S\right) &= N\big(\mathrm{x}; S, [1, a]\big) + O(1).
\end{aligned}
\tag{9.2.23}
$$

For $\lambda \in \mathbf{N} \cup \{\infty\}$ we define the *truncated counting function* to level λ by

$$
\begin{aligned}
N_\lambda(\mathrm{x}; S, D) = \frac{1}{[k:\mathbf{Q}]} \sum_{v \in M_k \setminus S} \min\big\{\lambda, \big(&\operatorname{ord}_v \sigma(x_0, \ldots, x_n) \\
&- d \max\{\operatorname{ord}_v x_j; 0 \leqq j \leqq n\}\big)^+\big\} \log N_{k/\mathbf{Q}}(\mathfrak{p}_v), \\
N_\infty(\mathrm{x}; S, D) = N(\mathrm{x}; S, D).&
\end{aligned}
\tag{9.2.24}
$$

This depends on the choice of σ by $O(1)$-term. The product formula (9.1.18) implies the following.

Theorem 9.2.25 (Analogue of the First Main Theorem) *For an effective divisor D on $\mathbf{P}^n_k$ we have*

$$
d\hat{\mathrm{h}}(\mathrm{x}) = m(\mathrm{x}; S, D) + N(\mathrm{x}; S, D) + O(1), \quad \mathrm{x} \in \mathbf{P}^n(k) \setminus D.
$$

Definition 9.2.26 A family $C = \{C(v)\}_{v \in M_k}$ with $C(v) \in \mathbf{R}$ is called an *M_k-constant* if $C(v) = 0$ except for finitely many $v \in M_k$.

Let V be a projective algebraic variety defined over k. Let $L \to V$ be a line bundle defined over k. In what follows everything is defined over k. Let $\phi, \psi \in H^0(V, L)$ be regular sections. For $\mathrm{x} \in V(k)$ and $v \in M_k$ the values $\|\phi(\mathrm{x})\|_v$, $\|\psi(\mathrm{x})\|_v$ are not defined, but the comparison of the order, which is larger or smaller, makes sense.

Lemma 9.2.27 *Let $L \to V$ be a line bundle and let $\phi_0, \ldots, \phi_n \in H^0(V, L)$ be regular sections over V such that*

$$\phi = [\phi_0, \ldots, \phi_n] : V \to \mathbf{P}_k^n$$

is a regular embedding. Then for every $\phi_{n+1} \in H^0(V, L)$ there is an M_k-constant $\{C(v)\}$ such that

$$0 \leqq \log \frac{\max\{\|\phi_j(\mathrm{x})\|_v; 0 \leqq j \leqq n+1\}}{\max\{\|\phi_j(\mathrm{x})\|_v; 0 \leqq j \leqq n\}} \leqq C(v), \quad \mathrm{x} \in V(k).$$

Proof For $v \in M_k^\infty$ we have $k_v = \mathbf{R}$ or $\mathbf{C}$, and so the assertion follows from the compactness of V.

Let $v \in M_k^0$. Take an arbitrary $\mathrm{x} \in V(k)$. Suppose

$$\max\{\|\phi_j(\mathrm{x})\|_v; 0 \leqq j \leqq n\} = \|\phi_{j_0}(\mathrm{x})\|_v.$$

Then $\|\phi_j(\mathrm{x})/\phi_{j_0}(\mathrm{x})\|_v \leqq 1$, $0 \leqq j \leqq n$. Since ϕ is giving an embedding, we can express ϕ_{n+1}/ϕ_{j_0} on the affine open subset $\{\phi_{j_0} \neq 0\}$ by the following polynomial:

$$\frac{\phi_{n+1}}{\phi_{j_0}} = \sum c_{j_0\lambda_0\cdots\lambda_n} \left(\frac{\phi_0}{\phi_{j_0}}\right)^{\lambda_0} \cdots \left(\frac{\phi_n}{\phi_{j_0}}\right)^{\lambda_n}, \quad c_{j_0\lambda_0\cdots\lambda_n} \in k.$$

Therefore

$$\left\| \frac{\phi_{n+1}(\mathrm{x})}{\phi_{j_0}(\mathrm{x})} \right\|_v \leqq \max \|c_{j_0\lambda_0\cdots\lambda_n}\|_v.$$

Thus it suffices to set $C(v) = \log \max\{\|c_{j_0\lambda_0\cdots\lambda_n}\|_v; j_0, \lambda_0, \ldots, \lambda_n\}$. □

Now assume $L \to V$ to be an ample line bundle. Take a large number $l_0 \in \mathbf{N}$ such that L^l, $l \geqq l_0$ is very ample. Using bases $\phi_0, \ldots, \phi_n$ of $H^0(V, L^l)$, one obtains an embedding

$$\phi = [\phi_0, \ldots, \phi_n] : V \longrightarrow \mathbf{P}_k^n. \tag{9.2.28}$$

We define a height function on $V(k)$ with respect to L by

$$\hat{\mathrm{h}}(\mathrm{x}; L) = \frac{1}{l} \hat{\mathrm{h}}\big(\phi(\mathrm{x})\big), \quad \mathrm{x} \in V(k). \tag{9.2.29}$$

Lemma 9.2.30 (i) *$\hat{\mathrm{h}}(\mathrm{x}; L)$ depends on $l > 0$ and embedding (9.2.28), but it is uniquely determined up to addition with $O(1)$, i.e., a bounded function on $V(k)$.*

(ii) *Let $L' \to V$ be another ample line bundle. Then*

$$\hat{\mathrm{h}}\big(\mathrm{x}; L \otimes L'\big) = \hat{\mathrm{h}}(\mathrm{x}; L) + \hat{\mathrm{h}}\big(\mathrm{x}; L'\big) + O(1).$$

Proof (i) Take $l \geqq l_0$ as above, and fix it for a moment. Let

$$\psi = [\psi_0, \ldots, \psi_n] : V \to \mathbf{P}_k^n \tag{9.2.31}$$

be an embedding given by another choice of bases $\psi_0, \ldots, \psi_n$ of $H^0(V, L^l)$. Since ψ_j is a linear sum of ϕ_i and vice versa, there is an M_k-constant $\{C(v)\}$ such that for $\mathrm{x} \in V(k)$, $v \in M_k$

$$\left|\log \max\left\{\left\|\phi_i(\mathrm{x})\right\|_v; 0 \leqq i \leqq n\right\} - \log \max\left\{\left\|\psi_i(\mathrm{x})\right\|_v; 0 \leqq i \leqq n\right\}\right| \leqq C(v).$$

Hence

$$\left|\hat{\mathrm{h}}\big(\phi(\mathrm{x})\big) - \hat{\mathrm{h}}\big(\psi(\mathrm{x})\big)\right| \leqq \frac{\sum_v C(v)}{[k : \mathbf{Q}]}.$$

Let $d \in \mathbf{N}$. The family $\{\sigma_j\}_{j=0}^N$ of monomials of degree d in $\phi_0, \ldots, \phi_n$ gives an embedding of V into a projective space $\mathbf{P}_k^N$, and the following holds:

$$\log \max_i\left\{\left\|\phi_i(\mathrm{x})\right\|_v\right\} = \frac{1}{d} \log \max_j\left\{\left\|\sigma_i(\mathrm{x})\right\|_v\right\}, \quad \mathrm{x} \in V(k),\ v \in M_k. \tag{9.2.32}$$

Let σ be an embedding of V into a projective space given by a choice of bases of $H^0(V, L^{dl})$. It follows from (9.2.32) and Lemma 9.2.27 that

$$d\hat{\mathrm{h}}\big(\phi(\mathrm{x})\big) = \hat{\mathrm{h}}\big(\sigma(\mathrm{x})\big) + O(1). \tag{9.2.33}$$

Take another $l' \geqq l_0$. Let ϕ' be a projective embedding of V given by bases of $H^0(V, L^{l'})$ and let τ be that given by bases of $H(V, L^{ll'})$. By (9.2.33)

$$l'\hat{\mathrm{h}}\big(\phi(\mathrm{x})\big) = \hat{\mathrm{h}}\big(\tau(\mathrm{x})\big) + O(1), \quad \mathrm{x} \in V(k),$$
$$l\hat{\mathrm{h}}\big(\phi'(\mathrm{x})\big) = \hat{\mathrm{h}}\big(\tau(\mathrm{x})\big) + O(1), \quad \mathrm{x} \in V(k).$$

Therefore if l in (9.2.28) is changed, $\hat{\mathrm{h}}(\mathrm{x}; L)$ differs only by $O(1)$-term.

(ii) This is deduced by arguments similar to those in (i) above based on Lemma 9.2.27. □

For a divisor of the complete linear system $|L|$ of an ample line bundle L we take a regular section $\sigma_D \in H^0(V, L)$ defining D. An approximation function $m(\mathrm{x}; S, D)$ and a counting function $N(\mathrm{x}; S, D)$ for D are defined with notation of (9.2.28) by

$$\left\|\sigma_D(\mathrm{x})\right\|_v = \frac{\|\sigma_D(\mathrm{x})\|_v}{(\max\{\|\phi_j(\mathrm{x})\|_v; 0 \leqq j \leqq n\})^{1/l}},$$

(continued)

$$m(\mathrm{x}; S, D) = \frac{1}{[k:\mathbf{Q}]} \sum_{v \in S} \log \frac{1}{|||\sigma_D(\mathrm{x})|||_v}, \tag{9.2.34}$$

$$N(\mathrm{x}; S, D) = \frac{1}{[k:\mathbf{Q}]} \sum_{v \in M_k \setminus S} \log \frac{1}{|||\sigma_D(\mathrm{x})|||_v}.$$

Remark 9.2.35 Note that $\log |||\sigma_D(\mathrm{x})|||^{-1}$ is always bounded from below on $V(k)$, and for the different choices of σ_D, l, ϕ_j these quantities differ only by an M_k-constant. Hence $m(\mathrm{x}; S, D)$ and $N(\mathrm{x}; S, D)$ are bounded from below on $V(k)$, and uniquely determined by x, S and D up to addition with $O(1)$.

A *truncated counting function* to level λ is defined as in (9.2.24):

(9.2.36)

$$N_\lambda(\mathrm{x}; S, D) = \frac{1}{[k:\mathbf{Q}]} \sum_{v \in M_k \setminus S} \min\left\{\lambda, \left(\operatorname{ord}_v \sigma_D(\mathrm{x}) - \frac{1}{l} \max\{\operatorname{ord}_v \phi_j(\mathrm{x}); 0 \leqq j \leqq n\}\right)^+\right\} \times \log N_{k/\mathbf{Q}}(\mathfrak{p}_v).$$

For a general line bundle $H \to V$ and a divisor $D \in |H|$ we take ample line bundles L_i, $i = 1, 2$, and divisors $D_i \in |L_i|$ such that $H = L_1 \otimes L_2^{-1}$ and $D = D_1 - D_2$. We set

$$\begin{aligned} \hat{\mathrm{h}}(\mathrm{x}; H) &= \hat{\mathrm{h}}(\mathrm{x}; L_1) - \hat{\mathrm{h}}(\mathrm{x}; L_2), \\ m(\mathrm{x}; S, D) &= m(\mathrm{x}; S, D_1) - m(\mathrm{x}; S, D_2), \\ N(\mathrm{x}; S, D) &= N(\mathrm{x}; S, D_1) - N(\mathrm{x}; S, D_2). \end{aligned} \tag{9.2.37}$$

These quantities are well-defined modulo $O(1)$. As in Theorem 9.2.25 one sees

Theorem 9.2.38 (Analogue of the First Main Theorem) *For a line bundle H on V and a divisor $D \in |H|$ we have*

$$\hat{\mathrm{h}}(\mathrm{x}; H) = m(\mathrm{x}; S, D) + N(\mathrm{x}; S, D) + O(1), \quad \mathrm{x} \in V(k) \setminus D.$$

9.3 Theorems of Roth and Schmidt

Let k be a number field. We take a subset S with $M_k^\infty \subset S \subset M_k$, $|S| < \infty$. *Roth's Theorem* (Roth [55]) is often stated in the following form.

Theorem 9.3.1 *Let $\alpha_v \in k$ for $v \in S$. For every $\varepsilon > 0$ there are only finitely many $x \in k$ satisfying*

$$\prod_{v \in S} \min\{1, \|x - \alpha_v\|_v\} \leqq \frac{1}{H_k(x)^{2+\varepsilon}}.$$

By Theorem 9.2.10, Roth's Theorem 9.3.1 is equivalent to the following.

Theorem 9.3.2 *For every $\varepsilon > 0$ there is a constant $C > 0$ such that*

$$\prod_{v\in S} \min\{1, \|x-\alpha_v\|_v\} \geqq \frac{C}{H_k(x)^{2+\varepsilon}}, \quad x \in k \setminus \{\alpha\}.$$

Take distinct elements $\alpha_i \in k$, $1 \leqq i \leqq q$. Then there is a positive constant $c_1 < 1$ satisfying the following property: Let $x \in k \setminus \{\alpha_i\}$ and $v \in S$ be arbitrary elements. If $\|x-\alpha_i\|_v < c_1$ for some α_i, then $\|x-\alpha_j\|_v > c_1$ for other α_j. Fix any $x \in k \setminus \{\alpha_i\}$. For each $v \in S$ we order the indices so that

$$\|x-\alpha_{i_1(v)}\|_v \leqq \cdots \leqq \|x-\alpha_{i_q(v)}\|_v.$$

By the choice of c_1 and Theorem 9.3.2

$$\begin{aligned}
\prod_{i=1}^{q}\prod_{v\in S} \min\{1, \|x-\alpha_i\|_v\} &\geqq \prod_{v\in S}\prod_{i=1}^{q} \min\{c_1, \|x-\alpha_i\|_v\} \\
&= \prod_{v\in S} \min\{c_1, \|x-\alpha_{i_1(v)}\|_v\} \cdots \min\{c_1, \|x-\alpha_{i_q(v)}\|_v\} \\
&\geqq c_1^{(q-1)|S|} \prod_{v\in S} \min\{c_1, \|x-\alpha_{i_1(v)}\|_v\} \\
&= c_1^{(q-1)|S|} \prod_{v\in S} c_1 \min\left\{1, \frac{1}{c_1}\|x-\alpha_{i_1(v)}\|_v\right\} \\
&\geqq c_1^{q|S|} \prod_{v\in S} \min\{1, \|x-\alpha_{i_1(v)}\|_v\} \geqq \frac{C c_1^{q|S|}}{H_k(x)^{2+\varepsilon}}.
\end{aligned}$$

We use the convention $\|x-\infty\|_v = \|1/x\|_v$. Then more generally, for q distinct points $\alpha_i \in k \cup \{\infty\}$, $1 \leqq i \leqq q$, we have with a re-chosen positive constant C

$$\prod_{i=1}^{q}\prod_{v\in S} \min\{1, \|x-\alpha_i\|_v\} \geqq \frac{C}{H_k(x)^{2+\varepsilon}}, \quad x \in k \setminus \{\alpha_i\}_{i=1}^{q}. \tag{9.3.3}$$

Taking the logarithms of the both sides, we obtain

$$\sum_{i=1}^{q} m\left(\frac{1}{x-\alpha_i}; S\right) \leqq (2+\varepsilon)h(x) + \log C, \quad x \in k \setminus \{\alpha_i\}_{i=1}^{q}. \tag{9.3.4}$$

From (9.3.4) and Theorem 9.2.13 we get the following theorem.

Theorem 9.3.5 (Analogue of the Second Main Theorem) *Let $\alpha_i \in k \cup \{\infty\}$, $1 \leqq i \leqq q$, be q distinct points. Then for every $\varepsilon > 0$ there exists a positive constant C*

such that

$$(q-2-\varepsilon)h(x) \leqq \sum_{i=1}^{q} N\left(\frac{1}{x-\alpha_i}; S\right) + C, \quad x \in k \setminus \{\alpha_i\}_{i=1}^{q}. \tag{9.3.6}$$

Deduction of Roth's Theorem 9.3.1 from Theorem 9.3.5 For the proof we write $S = \{v_i\}_{i=1}^{q} (q = |S|)$ and set $\alpha_i = \alpha_{v_i}$. After a change of indices we may assume that α_i $(i = 1, \ldots, q')$ are the distinct elements of $\{\alpha_i\}_{i=1}^{q}$. It is clear that (9.3.6) is equivalent to (9.3.3) except for the choice of C. Thus, assuming Theorem 9.3.5 we have by (9.3.3)

$$\prod_{i=1}^{q'} \prod_{v \in S} \min\{1, \|x-\alpha_i\|_v\} \geqq \frac{C}{H_k(x)^{2+\varepsilon}}, \quad x \in k \setminus \{\alpha_i\}_{i=1}^{q}. \tag{9.3.7}$$

Note that the product in the left-hand side of (9.3.7) contains the product $\prod_{i=1}^{q} \min\{1, \|x-\alpha_i\|_{v_i}\}$; that is, with $\prod'$ denoting the remaining part,

$$\prod_{i=1}^{q} \min\{1, \|x-\alpha_i\|_{v_i}\} \times \prod{}' \min\{1, \|x-\alpha_i\|_v\} \geqq \frac{C}{H_k(x)^{2+\varepsilon}}, \quad x \in k \setminus \{\alpha_i\}_{i=1}^{q}.$$

Since every factor in the above left-hand side is at most 1,

$$\prod_{i=1}^{q} \min\{1, \|x-\alpha_i\|_{v_i}\} \geqq \frac{C}{H_k(x)^{2+\varepsilon}}, \quad x \in k \setminus \{\alpha_i\}_{i=1}^{q}.$$

This is nothing but Roth's Theorem 9.3.2.

It might be very impressive to observe the similarity between Theorem 9.3.5 and Nevanlinna's Second Main Theorem 1.2.5. Nevanlinna theory has developed to the case of projective spaces and hyperplanes as Cartan's theory as well as the Weyl–Ahlfors theory, and then to the Cartan–Nochka Theorem. Roth's Theorem has been extended along the same path. In the course of this, Schmidt's Subspace Theorem 9.3.8 which we describe below will play an essential role.

Linear functions

$$\hat{H}_i(x) = \sum_{j=0}^{n} c_{ij} x_j, \quad c_{ij} \in k, \ 1 \leqq i \leqq q,$$

which are not identically zero are called linear forms in $n+1$ variables $x = (x_0, \ldots, x_n)$ with k-coefficients. The linear forms $\hat{H}_i$ define hyperplanes H_i of $\mathbf{P}_k^n$, and if they are in general position, $\hat{H}_i$ are said to be in general position.

In the sequel a point of $\mathbf{P}^n(k)$ represented by $x = (x_0, \ldots, x_n) \in k^{n+1} \setminus \{0\}$ is denoted by x.

Theorem 9.3.8 (Schmidt's Subspace Theorem; Schmidt [91], p. 178, Theorem 1D'; Schlickewei [77]) *Assume that for each $v \in S$ linear forms $\hat{H}_{vi}$, $1 \leqq i \leqq n+1$,*

in $x = (x_0, \dots, x_n)$ with k-coefficients are given in general position. Then for every $\varepsilon > 0$ there exists a finite union E_ε of proper linear subspaces of $\mathbf{P}^n_k$ such that $E_\varepsilon \supset \bigcup_{i=1}^q H_i$ and

$$\prod_{v\in S}\prod_{i=1}^{n+1}\frac{\|\hat{H}_{vi}(x)\|_v}{\max\{\|x_j\|_v;\, 0\leqq j\leqq n\}} \geqq \mathrm{H}_k(\mathrm{x})^{-(n+1+\varepsilon)} \tag{9.3.9}$$

for $\mathrm{x} \in \mathbf{P}^n(k) \setminus E_\varepsilon$.

Theorem 9.3.10 (Analogue of the Second Main Theorem) *Let H_i, $1 \leqq i \leqq q$, be hyperplanes of $\mathbf{P}^n_k$ in general position. Then for every $\varepsilon > 0$ there exists a finite union E_ε of proper linear subspaces of $\mathbf{P}^n_k$ such that $E_\varepsilon \supset \bigcup_{i=1}^q H_i$ and*

$$\sum_{i=1}^{q} m(\mathrm{x}; S, H_i) \leqq (n+1+\varepsilon)\hat{\mathrm{h}}(\mathrm{x}), \quad \mathrm{x} \in \mathbf{P}^n(k) \setminus E_\varepsilon, \tag{9.3.11}$$

or equivalently

$$(q-n-1-\varepsilon)\hat{\mathrm{h}}(\mathrm{x}) \leqq \sum_{i=1}^{q} N(\mathrm{x}; S, H_i), \quad \mathrm{x} \in \mathbf{P}^n(k) \setminus E_\varepsilon. \tag{9.3.12}$$

Proof Set the index set $Q = \{1, \dots, q\}$. By the assumption that the H_i be in general position, there is a constant $C > 1$ such that for every subset $I \subset Q$ with $|I| = n+1$ and every $v \in M_k$,

$$C^{-1} \leqq \frac{\max\{\|\hat{H}_i(x)\|_v;\, i \in I\}}{\max\{\|x_j\|_v;\, 0 \leqq j \leqq n\}} \leqq C, \quad x \in k^{n+1} \setminus \{0\}. \tag{9.3.13}$$

For any x and $v \in S$ there is a re-ordering of indices in Q, $i_1(v), \dots, i_q(v)$, so that

$$\left\|\hat{H}_{i_1(v)}(x)\right\|_v \leqq \cdots \leqq \left\|\hat{H}_{i_q(v)}(x)\right\|_v. \tag{9.3.14}$$

Put $I(v) = \{i_1(v), \dots, i_{n+1}(v)\}$. Since there are only finitely many such $I(v)$'s, it follows from Theorem 9.3.8 that

$$\prod_{v\in S}\prod_{i\in I(v)}\frac{\|\hat{H}_i(x)\|_v}{\max\{\|x_j\|_v;\, 0\leqq j\leqq n\}} \geqq \mathrm{H}_k(\mathrm{x})^{-(n+1+\varepsilon)}, \quad \mathrm{x} \in \mathbf{P}^n(k) \setminus E_\varepsilon, \tag{9.3.15}$$

where E_ε is a finite union of proper linear subspaces of $\mathbf{P}^n_k$ independent of the sets $I(v)$.

For $i \in Q \setminus I(v)$ (9.3.14) and (9.3.13) imply that

$$\begin{aligned}\left\|\hat{H}_i(x)\right\|_v &\geqq \left\|\hat{H}_{i_{n+1}(v)}(x)\right\|_v = \max\left\{\left\|\hat{H}_i(x)\right\|_v;\, i \in I(v)\right\}\\ &\geqq C^{-1}\max\left\{\|x_j\|_v;\, 0 \leqq j \leqq n\right\}.\end{aligned}$$

Therefore

$$\prod_{v\in S}\prod_{i\in Q\setminus I(v)}\frac{\|\hat{H}_i(x)\|_v}{\max\{\|x_j\|_v;\,0\leqq j\leqq n\}}\geqq C^{-(q-n-1)|S|}. \tag{9.3.16}$$

We see by (9.3.15) and (9.3.16) that

$$\prod_{i\in Q}\prod_{v\in S}\frac{\|\hat{H}_i(x)\|_v}{\max\{\|x_j\|_v;\,0\leqq j\leqq n\}}\geqq C^{-(q-n-1)|S|}\mathrm{H}_k(\mathrm{x})^{-(n+1+\varepsilon)},\quad \mathrm{x}\in\mathbf{P}^n(k)\setminus E_\varepsilon.$$

Note that by Theorem 9.2.18 there are only finitely many $\mathrm{x}\in\mathbf{P}^n(k)$ with $\mathrm{H}_k(\mathrm{x})^\varepsilon\times C^{-(q-n-1)|S|}<1$. Taking a larger E_ε containing them, we have

$$\prod_{i\in Q}\prod_{v\in S}\frac{\|\hat{H}_i(x)\|_v}{\max\{\|x_j\|_v;\,0\leqq j\leqq n\}}\geqq \mathrm{H}_k(\mathrm{x})^{-(n+1+2\varepsilon)},\quad \mathrm{x}\in\mathbf{P}^n(k)\setminus E_\varepsilon.$$

We take the logarithms of the both sides, divide them by $[k:\mathbf{Q}]$, and thus obtain

$$\sum_{i=1}^{q}m(\mathrm{x};S,H_i)\leqq(n+1+2\varepsilon)\hat{\mathrm{h}}(\mathrm{x}),\quad \mathrm{x}\in\mathbf{P}^n(k)\setminus E_\varepsilon.$$

□

The above proof is very similar to that of Lemma 4.2.3 in the case of $N=n$, after which we prove the next result.

Theorem 9.3.17 (Ru–Wong [91]) *Let $N\in\mathbf{N}$, $N\geqq n$. Assume that H_i, $1\leqq i\leqq q$, are hyperplanes of $\mathbf{P}^n_k$ in N-subgeneral position. For every $\varepsilon>0$ there is a finite union $E_\varepsilon\supset\bigcup_{i=1}^q H_i$ of proper linear subspaces of $\mathbf{P}^n_k$ such that*

$$(q-2N+n-1-\varepsilon)\hat{\mathrm{h}}(\mathrm{x})\leqq\sum_{i=1}^{q}N(\mathrm{x};S,H_i),\quad \mathrm{x}\in\mathbf{P}^n(k)\setminus E_\varepsilon. \tag{9.3.18}$$

Proof We may assume that $q-2N+n-1>0$. Note that Nochka's Theorem 4.1.10 and Lemma 4.1.17 hold not only over $\mathbf{C}$ but also over an arbitrary field. Set $Q=\{1,\dots,q\}$. Let $\{\omega(i)\}_{i\in Q}$ be Nochka's weights for $\{H_i\}_{i\in Q}$, and let $\tilde{\omega}$ be Nochka's constant. For every $\mathrm{x}\in\mathbf{P}^n(k)$ and every $v\in S$ there is a re-ordering $i_1(v),\dots,i_q(v)$ of the elements of Q such that

$$\left\|\hat{H}_{i_1(v)}(x)\right\|_v\leqq\cdots\leqq\left\|\hat{H}_{i_{N+1}(v)}(x)\right\|_v\leqq\cdots\leqq\left\|\hat{H}_{i_q(v)}(x)\right\|_v.$$

By the definition of N-subgeneral position and (9.3.13) there is a constant $C>1$ such that

$$\frac{\max\{\|x_j\|_v;\,0\leqq j\leqq n\}}{\|\hat{H}_{i_h(v)}(x)\|_v}\leqq C,\quad N+2\leqq h\leqq q. \tag{9.3.19}$$

Put $d = [k : \mathbf{Q}]$. It may be assumed that all the coefficients of the linear forms $\hat{H}_i$ are elements of $\mathscr{O}_k$. As in (9.2.20) one takes $c_v \geqq 1$, $v \in M_k$, so that

$$\frac{c_v \max\{\|x_j\|_v; 0 \leqq j \leqq n\}}{\|\hat{H}_i(x)\|_v} \geqq 1.$$

We apply Lemma 4.1.17 for $R(v) = \{i_1(v), \ldots, i_{N+1}(v)\}$. Then there is a subset $R^\circ(v) \subset R(v)$ with $|R^\circ(v)| = \mathrm{rk}(R^\circ(v)) = \mathrm{rk}(R(v)) = n+1$ such that by Theorem 9.3.8,

$$\begin{aligned}
&\sum_{i\in Q} \omega(i) m(\mathrm{x}; S, H_i) \\
&\quad \leqq \frac{1}{d} \sum_{v\in S} \log \prod_{i\in Q} \left(\frac{c_v \max\{\|x_j\|_v; 0 \leqq j \leqq n\}}{\|\hat{H}_i(x)\|_v} \right)^{\omega(i)} \\
&\quad \leqq \frac{1}{d} \sum_{v\in S} \log \prod_{i\in R(v)} \left(\frac{c_v \max\{\|x_j\|_v\}}{\|\hat{H}_i(x)\|_v} \right)^{\omega(i)} \prod_{i\notin R(v)} \left(\frac{c_v \max\{\|x_j\|_v\}}{\|\hat{H}_i(x)\|_v} \right)^{\omega(i)} \\
&\quad \leqq \frac{1}{d} \sum_{v\in S} \log \prod_{i\in R^\circ(v)} \frac{c_v \max\{\|x_j\|_v\}}{\|\hat{H}_i(x)\|_v} + \frac{1}{d} \sum_{v\in S} \sum_{i\notin R(v)} \log(c_v C)^{\omega(i)} \\
&\quad \leqq (n+1+\varepsilon)\hat{\mathrm{h}}(\mathrm{x}) + (n+1)\frac{|M_k^\infty|}{d} \log c + \frac{q|S|}{d} \log(cC) \\
&\quad \leqq (n+1+\varepsilon)\hat{\mathrm{h}}(\mathrm{x}) + (n+1)\log c + \frac{q|S|}{d} \log(cC), \quad \mathrm{x} \notin E_\varepsilon.
\end{aligned}$$

Thus there is a constant $C_1 > 0$ such that

$$\sum_{i\in Q} \omega(i) m(\mathrm{x}; S, H_i) \leqq (n+1+\varepsilon)\hat{\mathrm{h}}(\mathrm{x}) + C_1, \quad \mathrm{x} \notin E_\varepsilon.$$

Here we use the First Main Theorem 9.2.2. There is a constant $C_2 > 0$ such that

$$\sum_{i\in Q} \omega(i)\hat{\mathrm{h}}(\mathrm{x}) - \sum_{i\in Q} \omega(i) N(\mathrm{x}; S, H_i) \leqq (n+1+\varepsilon)\hat{\mathrm{h}}(\mathrm{x}) + C_2, \quad \mathrm{x} \notin E_\varepsilon.$$

It follows from Nochka's Theorem 4.1.10 (ii) that

$$\{\tilde{\omega}(q - 2N + n - 1) - \varepsilon\}\hat{\mathrm{h}}(\mathrm{x}) \leqq \sum_{i\in Q} \omega(i) N(\mathrm{x}; S, H_i) + C_2, \quad \mathrm{x} \notin E_\varepsilon.$$

Therefore

$$\begin{aligned}
\left(q - 2N + n - 1 - \frac{\varepsilon}{\tilde{\omega}}\right)\hat{\mathrm{h}}(\mathrm{x}) &\leqq \sum_{i\in Q} \frac{\omega(i)}{\tilde{\omega}} N(\mathrm{x}; S, H_i) + \frac{C_2}{\tilde{\omega}} \\
&\leqq \sum_{i\in Q} N(\mathrm{x}; S, H_i) + \frac{C_2}{\tilde{\omega}}, \quad \mathrm{x} \notin E_\varepsilon.
\end{aligned}$$

By making use of $\tilde{\omega} \geqq \frac{n+1}{2N-n+1}$ (Theorem 4.1.10 (iii)) we get

$$\left(q - 2N + n - 1 - \frac{2N-n+1}{n+1}\varepsilon\right)\hat{\mathrm{h}}(\mathrm{x}) \leqq \sum_{i\in Q} N(\mathrm{x}; S, H_i) + \frac{2N-n+1}{n+1}C_2, \quad \mathrm{x} \notin E_\varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, we obtain (9.3.18), enlarging E_ε if necessary. □

9.4 Unit Equations

In the present section we deduce an analogue of Borel's Theorem (Corollary 4.2.18) from Theorem 9.3.10 in the same way as we derived Borel's Theorem from Cartan's Second Main Theorem (Corollary 4.2.15). As an application we will prove a finiteness theorem for the S-unit point set (cf. Definition 9.4.5) for projective hypersurfaces constructed in Sect. 7.4.

As in the previous section we take an algebraic number field k and a finite subset S of M_k containing M_k^∞.

Theorem 9.4.1 *Let $a_i \in k^*$, $1 \leqq i \leqq s$. Let $\mathscr{F} = \{(x_1, \ldots, x_n)\}$ denote the set of all S-unit solutions $(x_1, \ldots, x_n) \in (\mathscr{O}_{kS}^*)^n$ of the equation*

$$a_1x_1 + \cdots + a_sx_s = 0 \quad (s \geqq 2).$$

Then $\mathscr{F}$ decomposes into finitely many subfamilies, $\mathscr{F} = \bigcup \mathscr{F}_\mu$, and for each $\mathscr{F}_\mu$ there exists a decomposition of indices

$$I = \{1, \ldots, s\} = \bigcup_{l=1}^{m} I_l,$$

which satisfies the following conditions:

(i) $|I_l| \geqq 2$ *for all* l.

(ii) *We write $\mathscr{F}_\mu = \{(x_i(\zeta))\}_{\zeta\in\mathscr{F}_\mu}$ and fix I_l arbitrarily. Then for every $i, j \in I_l$,*

$$\frac{x_i(\zeta)}{x_j(\zeta)} = c_{jk} \in \mathscr{O}_{kS}^*$$

is a constant independent of ζ.

(iii) *For every $\zeta \in \mathscr{F}_\mu$ and $l = 1, 2, \ldots, m$,*

$$\sum_{i\in I_l} a_i x_i(\zeta) = 0.$$

Proof We use the induction on s. The case of $s = 2$ is trivial. We assume that the case of less than or equal to $s - 1$ holds for an arbitrary S.

We prove the theorem in the case of s unknowns. Enlarging S if necessary, the equation is reduced to

$$x_1 + \cdots + x_s = 0. \tag{9.4.2}$$

We regard $\mathrm{x} = [x_1, \ldots, x_{s-1}]$ as a homogeneous coordinate system of $\mathbf{P}_k^{s-2}$. We consider the following s linear forms:

$$\hat{H}_1 = x_1, \ldots, \hat{H}_{s-1} = x_{s-1},$$
$$\hat{H}_s = -(x_1 + \cdots + x_{s-1}).$$

These are in general position. Let $n = s-2, q = s, \varepsilon = \frac{1}{2}$ and apply Theorem 9.3.10. Then there is $E_{1/2}$ such that

$$\frac{1}{2}\hat{\mathrm{h}}(\mathrm{x}) \leqq \sum_{i=1}^{s} N(\mathrm{x}; S, H_i) = 0, \quad \mathrm{x} \in \mathbf{P}^{s-2}(k) \setminus E_{1/2}.$$

Therefore all $\mathrm{x} \in E_{1/2}$ and hence $\{\mathrm{x}\}$ in consideration decomposes into finitely many families, each of which is contained in a hypersurface. It suffices to show our assertion for each subfamily. Thus we may assume that x satisfies the following non-trivial linear equation:

$$c_1 x_1 + \cdots + c_{s-1} x_{s-1} = 0.$$

Changing the indices and picking up only non-zero coefficients, we have that $c_1 = 1, c_2 \cdots c_{s'} \neq 0, s' \leqq s - 1$ and

$$x_1 + c_2 x_2 + \cdots + c_{s'} x_{s'} = 0. \tag{9.4.3}$$

This with (9.4.2) implies

$$(1 - c_2)x_2 + \cdots + (1 - c_{s'})x_{s'} + x_{s'+1} + \cdots + x_s = 0. \tag{9.4.4}$$

Every variable x_i appears in (9.4.3) or (9.4.4) with non-zero coefficient. We apply the induction hypothesis to (9.4.3) and (9.4.4). Thus we obtain two decompositions to subfamilies $\mathscr{F}_\mu, \mathscr{F}'_{\mu'}$ satisfying conditions (i), (ii), and (iii), and the corresponding decompositions of indices $I = \bigcup I_l$, $I = \bigcup I'_{l'}$. We take a decomposition $\mathscr{F} = \bigcup \mathscr{F}''_\nu$ of $\mathscr{F}$ into subfamilies which is induced by $\mathscr{F}_\mu \cap \mathscr{F}'_{\mu'}$. If I_l and $I'_{l'}$ contains a common index, we take the union $I_l \cup I'_{l'}$, and hence obtain a decomposition $I = \bigcup I''_h$ of indices. By the construction these satisfy (i), (ii), (iii). □

Definition 9.4.5 A point $\mathrm{x} = [x_0, \ldots, x_n] \in \mathbf{P}^n(k)$ is called an S-unit point if every x_j is either 0 or an S-unit. We denote by $\mathbf{P}^n(\mathscr{O}_{kS}^*)$ the set of those points.

We take a family $\{M_j(x)\}_{j=1}^s$ of monomials in variables $x = (x_1, \ldots, x_n)$, $c_j \in k^*$, $1 \leqq j \leqq s$, and consider a hypersurface of $\mathbf{P}_k^n$ defined by

$$X: \quad c_1 M_1(x) + \cdots + c_s M_s(x) = 0. \tag{9.4.6}$$

Recall that if $\{M_j\}_{j=1}^s$ is $(n+1)$-admissible (Definition 7.4.4), then the projective hypersurface defined by

$$c_1 M_1^d(x) + \cdots + c_s M_s^d(x) = 0, \quad c_j \in \mathbf{C}^*, \ d > s(s-2),$$

is Kobayashi hyperbolic (Theorem 7.4.19).

Let $X(\mathcal{O}_{kS}^*)$ denote the set of all S-unit points in X; i.e.,

$$X\big(\mathcal{O}_{kS}^*\big) = X(k) \cap \mathbf{P}^n\big(\mathcal{O}_{kS}^*\big).$$

Theorem 9.4.7 (Noguchi [97]) *If $\{M_j(x)\}_{j=1}^s$ is an $(n+1)$-admissible family, then $X(\mathcal{O}_{kS}^*)$ is finite.*

Proof By the definition of an $(n+1)$-admissible family it suffices to consider the case where all $x_j \in \mathcal{O}_{kS}^*$. We apply Theorem 9.4.1 for $x_j = M_j(x)$. Then $\mathcal{F} = \{(M_j(x))\}$ decomposes into finitely many subfamilies $\mathcal{F}_\mu$ and for each $\mathcal{F}_\mu$ there is a decomposition $\{I_l\}$ of the index set $I = \{1, \ldots, s\}$ such that Theorem 9.4.1 (i), (ii), and (iii) hold. By the proof of Theorem 8.4.8 (ii) we see that all the mutual ratios of elements in $x = (x_0, \ldots, x_n)$ given by each $\mathcal{F}_\mu = \{M_j(x)\}$ are constant; that is, as a point of $\mathbf{P}^n(k)$, it is one point x. Therefore $|X(\mathcal{O}_{kS}^*)| < \infty$. □

Remark 9.4.8 To construct a Kobayashi hyperbolic projective hypersurface it was necessary to take higher powers M_j^d of M_j, but here it is unnecessary to do so, for x's are assumed to be S-unit points and hence M_j are S-units.

9.5 The *abc*-Conjecture and the Fundamental Conjecture

The motivation of Theorem 9.4.7 lies in Lang's Conjecture 8.1.1 and Kobayashi's Conjecture 7.1.8. In view of those conjectures there should exist a Kobayashi hyperbolic projective hypersurface defined over $\mathbf{Q}$ that always carries only finitely many rational points over an arbitrary number field.

Let us introduce one example. Following Shirosaki [98], we set out as follows. Let $d, e \in \mathbf{N}$ be co-prime and satisfy

$$d > 2e + 8.$$

We define a homogeneous polynomial $P(w_0, w_1)$ in two variables by

$$P(w_0, w_1) = w_0^d + w_1^d + w_0^e w_1^{d-e}.$$

Then inductively we set

$$P_1(w_0, w_1) = P(w_0, w_1),$$
$$P_n(w_0, w_1, \dots, w_n) = P_{n-1}\big(P(w_0, w_1), \dots, P(w_{n-1}, w_n)\big), \quad n = 2, 3, \dots.$$

Then P_n is a homogeneous polynomial of degree d^n. Assume that $e \geqq 2$. Then

$$X = \big\{P_n(w_0, w_1, \dots, w_n) = 0\big\} \subset \mathbf{P}^n_{\mathbf{Q}} \tag{9.5.1}$$

is Kobayashi hyperbolic (Shirosaki [98]).

Theorem 9.5.2 (Noguchi [03a]) *Let X be defined by* (9.5.1) *with $e \geqq 2$. Then $X(k)$ is finite for an arbitrary number field k.*

It is interesting that there exists such an example. For the proof cf. Noguchi [03a]. In view of Theorem 9.4.1 it is plausible to conjecture that those Kobayashi hyperbolic projective hypersurfaces constructed in Sect. 7.4 should have at most finitely many k-rational points over a number field k if the coefficients are taken from k. However, if one would like to carry out a similar proof, Roth's Theorem 9.3.1 (Theorem 9.3.5) or Schmidt's Subspace Theorem 9.3.8 (Theorem 9.3.10) are insufficient, and the abc-Conjecture 8.3.2 below due to Masser–Österlé and its extension, $abc\cdots$-Conjecture 9.5.6 are necessary.

Let $\mathrm{x} = [x_0, x_1]$ be a homogeneous coordinate system of $\mathbf{P}^1_k$, and take three linear forms in general position:

$$F_1 = x_0, \qquad F_2 = x_1, \qquad F_3 = -x_0 - x_1.$$

In abc-Conjecture 8.3.2 we consider $\mathrm{x} = [a, b] \in \mathbf{P}^1(\mathbf{Q})$. Put $x = (a, b)$. Then (8.3.6) is equivalent to the following estimate:

$$(1 - \varepsilon)\hat{\mathrm{h}}(\mathrm{x}) \leqq \sum_{i=1}^{3} N_1\big(F_i(x); M^{\infty}_{\mathbf{Q}}\big) + C(\varepsilon). \tag{9.5.3}$$

This is a height estimate from above by counting functions truncated to level one. For the reformulation to represent the approximation we introduce the *multiplicity function* defined by

$$N^{\lambda}\big(F(x); S\big) = N\big(F(x); S\big) - N_{\lambda}\big(F(x); S\big).$$

This diverges to infinity as $\mathrm{ord}_p F(x)$ grows larger at every prime point p, but $F(x)$ gets close to zero in p-adic topology. We may write (9.5.3) as

$$\sum_{i=1}^{3}\Big(m\big(F_i(x); M^{\infty}_{\mathbf{Q}}\big) + N^{1}\big(F_i(x); M^{\infty}_{\mathbf{Q}}\big)\Big) \leqq (2 + \varepsilon)\hat{\mathrm{h}}(\mathrm{x}) + C(\varepsilon). \tag{9.5.4}$$

If the multiplicity function in the second term of the above formula is taken out, it is Roth's Theorem 9.3.1. Thus it is seen that abc-Conjecture claims a rather strong assertion.

Conjecture 9.5.5 (*abc*-Conjecture) *Under the same hypothesis as in Theorem 9.3.5 we have*

$$(q-2-\varepsilon)h(x) \leqq \sum_{i=1}^{q} N_1\left(\frac{1}{x-\alpha_i}; S\right) + C(\varepsilon), \quad x \in k \setminus \{\alpha_i\}_{i=1}^{q}.$$

Thus the $abc\cdots$-Conjecture, a version of the *abc*-Conjecture in several variables, is formulated as follows.

Conjecture 9.5.6 ($abc\cdots$-Conjecture; Noguchi [96]; Vojta [98]) *Under the same hypothesis as in Theorem 9.3.10 we have*

$$(q-n-1-\varepsilon)\hat{\mathrm{h}}(\mathrm{x}) \leqq \sum_{i=1}^{q} N_n(\mathrm{x}; S, H_i), \quad \mathrm{x} \in \mathbf{P}^n(k) \setminus E_\varepsilon.$$

Remark If the $abc\cdots$-Conjecture holds, those Kobayashi hyperbolic projective hypersurfaces X obtained in Sect. 7.4 with coefficients in k^* carry an at most finite rational point set $X(k)$; the proof is similar to those of Theorems 7.4.19 and 9.4.7.

We generalize the above problem not only over projective spaces but also over a general projective algebraic variety V defined over k. Assume that V is non-singular, and denote by K_V the canonical bundle over it. For a divisor D on V we define the multiplicity function by

$$N^{\lambda}(\mathrm{x}; S, D) = N(\mathrm{x}; S, D) - N_{\lambda}(\mathrm{x}; S, D).$$

As modeled by the fundamental Conjecture 4.10.2 for entire curves, we may raise the following fundamental conjecture.

Conjecture 9.5.7 *Let V be a non-singular projective algebraic variety defined over k. Let $L \to V$ be an ample line bundle and let D be an s.n.c. divisor on V. Then there exists a number $\lambda \in \mathbf{N}$ such that for every $\varepsilon > 0$ there is a proper algebraic subset $E_\varepsilon \supset D$ of V satisfying*

$$m(\mathrm{x}; S, D) + N^{\lambda}(\mathrm{x}; S, D) + \hat{\mathrm{h}}(\mathrm{x}; K_V) \leqq \varepsilon \hat{\mathrm{h}}(x; L), \quad \mathrm{x} \in V(k) \setminus E_\varepsilon.$$

Vojta posed the following *Vojta Conjecture* introducing the logarithmic discriminant "$d(\mathrm{x})$" for $\mathrm{x} \in V(k)$ (Vojta [87], Chap. 8, Sect. 1).

Conjecture 9.5.8 (Vojta [87], Conjecture 5.2.6) *With the same notation as in Conjecture 9.5.7 we have*

$$m(\mathrm{x}; S, D) + \hat{\mathrm{h}}(\mathrm{x}; K_V) \leqq d(\mathrm{x}) + \varepsilon \hat{\mathrm{h}}(\mathrm{x}; L), \quad \mathrm{x} \in V(k) \setminus E_\varepsilon.$$

9.6 The Faltings–Vojta Theorem

In Sects. 9.2 and 9.3 we described the analogy between the Diophantine approximation theory of Roth and Schmidt and the value distribution theory of Nevanlinna and Cartan, and in Sect. 9.4 an application was given from that viewpoint.

In the present section we advance further, and provide an analogue of the Logarithmic Bloch–Ochiai Theorem 4.8.17 in Diophantine approximation. We will give some applications.

We take a number field k and a finite subset $S \subset M_k$ with $S \supset M_k^\infty$. Let V be a projective algebraic variety over k and let D be a divisor on V defined over k. In what follows, subvarieties are all defined over k, and rational points are those over k.

Definition 9.6.1 A subset $Z \subset V(k) \setminus D$ of rational points is said to be (S, D)-*integral* if there is an M_k-constant $\{C(v)\}_{v \in M_k}$ satisfying (cf. (9.2.34))

$$\log \frac{1}{|||\sigma_D(\mathrm{x})|||_v} \leqq C(v), \quad \mathrm{x} \in Z,\ v \in M_k \setminus S.$$

Remark (i) By Remark 9.2.35 the property of $Z \subset V(k) \subset D$ being (S, D)-integral is independent of the choices of σ_D and $|||\cdot|||$. If Z is a finite set, it is necessarily (S, D)-integral by definition; the converse does not hold.

(ii) We consider the case where D is very ample. By making use of rational regular sections of $L(D)$ we embedded $V \setminus D$ into an affine space $\mathbf{A}_k^N$ over k

$$\Phi = (\Phi_1, \ldots, \Phi_N) : V \setminus D \to \mathbf{A}_k^N.$$

A subset $Z \subset V(k) \setminus D$ is (S, D)-integral if and only if there is a constant $c \in \mathscr{O}_k \setminus \{0\}$ satisfying

$$c\Phi_j(\mathrm{x}) \in \mathscr{O}_k, \quad \mathrm{x} \in Z,\ 1 \leqq j \leqq N.$$

In this sense the notion of being (S, D)-integral is an extension of S-integers in Definition 9.2.15, independent of coordinates.

The next theorem had been conjectured by Mordell [22] in 1922, and was proved by Faltings.

Theorem 9.6.2 (Faltings [83b]; Vojta [91]; Bombieri [90]) *The rational point set $C(k)$ of a non-singular smooth algebraic curve C of genus $\geqq 2$ defined over k is finite.*

Lang's Conjecture 8.1.1 is the one of conjectures that generalize Theorem 9.6.2 to the case of higher dimensions. The criterion of the Kobayashi hyperbolicity is very simple for submanifolds of abelian varieties as seen in Theorem 7.3.8. Moreover, even if X is not Kobayashi hyperbolic and admits a non-constant entire curve $f : \mathbf{C} \to X$, the Zariski closure of the image $f(\mathbf{C})$ is known to be a translation of an abelian subvariety (the Bloch–Ochiai Theorem (Corollary 4.8.18)). Henceforth it is

interesting to specialize Lang's Conjecture 8.1.1 to the case of abelian varieties. In fact, Faltings proved the following.

Theorem 9.6.3 (Faltings [91], [94]) *Let A be an abelian variety over k.*

(i) *For an algebraic subvariety $X \subset A$ there exist finitely many translations $Y_i \subset X$ of abelian subvarieties such that*

$$X(k) = \bigcup Y_i(k).$$

(ii) *Let D be a reduced ample divisor on A. Then every (S, D)-integral subset $Z \subset A(k) \setminus D$ is a finite set.*

Note that (i) is an analogue of the Bloch–Ochiai Theorem (Corollary 4.8.18), and (ii) is that of Corollary 6.3.5 (ii). Let C be an algebraic curve in Theorem 9.6.2. Then C is embedded into its Albanese variety A_C, and hence Theorem 9.6.3 (ii) implies Theorem 9.6.2. Theorem 9.6.3 (ii) in the case of $\dim A = 1$ is a famous result of Siegel [26]. Theorem 9.6.3 (ii) was proved before Corollary 6.3.5 (ii); this is a rare case in the sense that the arithmetic case was ahead. It is natural to think of the logarithmic version of these theorems of Faltings.

Theorem 9.6.4 (Vojta [96]) *Let V be a non-singular projective algebraic variety over k, and let D be a reduced divisor on V. Assume that the logarithmic irregularity $q(V \setminus D) = \dim H^0(V, \Omega^1_V(\log D)) > \dim V$. Then for an (S, D)-integral subset $Z \subset V(k) \setminus D$ there is a proper algebraic subset $W \subsetneqq V$ such that $Z \subset W(k)$.*

9.7 Distribution of Rational Points

In Sect. 4.9 we investigated the image of an entire curve omitting divisors by making use of the Logarithmic Bloch–Ochiai Theorem 4.8.17. Here, instead of the Logarithmic Bloch–Ochiai Theorem 4.8.17 we use Vojta's Theorem 9.6.4 to study an (S, D)-integral subset Z.

Before dealing with general V, we discuss the case of projective space $\mathbf{P}^n_k$ and a finite union D of hyperplanes.

Theorem 9.7.1 (Ru–Wong [91]) *Let H_i, $1 \leqq i \leqq q$, be hyperplanes of $\mathbf{P}^n_k$ in general position, and set $D = \sum_{i=1}^q H_i$. Assume that there is a linear subspace E of $\mathbf{P}^n_k$ and an (S, D)-integral subset $Z \subset (\mathbf{P}^n(k) \setminus D) \cap E(k)$ satisfying the following condition.*

9.7.2 There is no finite subset $Y \subset Z$ such that $Z \setminus Y$ is contained in a proper linear subspace of E.

Then $\dim E \leqq 2n + 1 - q$.

In particular, if $q \geqq 2n+1$, then an arbitrary (S, D)-integral subset Z is a finite set.

Proof Let $\dim E = m \geqq 1$ and $E \cong \mathbf{P}^m_k$. The family $\{H_i \cap E\}$ of hyperplanes of E is in n-subgeneral position. By the definition of (S, D)-integral subset there is a constant C_1 such that

$$\sum_{i=1}^{q} N(\mathrm{x}; S, H_i \cap E) = \sum_{i=1}^{q} N(\mathrm{x}; S, H_i) \leqq C_1, \quad \mathrm{x} \in Z.$$

Therefore by Theorem 9.3.17 with $\varepsilon = \frac{1}{2}$ there is a finite union $E_{1/2}$ of proper linear subspaces such that

$$\left(q - 2n + m - 1 - \frac{1}{2}\right)\hat{\mathrm{h}}(\mathrm{x}) \leqq C_1, \quad \mathrm{x} \in Z \setminus E_{1/2}.$$

The assumption 9.7.2 implies that $q - 2n + m - 1 - 1/2 \leqq 0$. Thus $m \leqq 2n + 1 - q$. □

Remark 9.7.3 Theorem 9.7.1 makes sense when $q \geqq n+1$. In that case an (S, D)-integral subset $Z \subset \mathbf{P}^n(k) \setminus D$ is an S-unit point set. To see this we take linear forms $\hat{H}_i$ defining H_i and a homogeneous coordinate system $[x_0, \ldots, x_n]$ of $\mathbf{P}^n_k$ so that $\hat{H}_j = x_{j-1}$, $1 \leqq j \leqq n+1$. Let $\phi_0, \ldots, \phi_N$ denote the bases of homogeneous polynomials of degree q in $x_0, \ldots, x_n$. Here we set $\phi_0 = x_0 \cdots x_n \hat{H}_{n+2} \cdots \hat{H}_q$. Then

$$\Phi : \mathbf{P}^N_k \setminus D \to \left(\frac{\phi_1}{\phi_0}, \ldots, \frac{\phi_N}{\phi_0}\right) \in \mathbf{A}^N_k$$

gives rise to an embedding. For instance, if

$$\phi_1 = (x_1)^2 x_2 \cdots x_n \hat{H}_{n+2} \cdots \hat{H}_q, \qquad \phi_2 = (x_0)^2 x_2 \cdots x_n \hat{H}_{n+2} \cdots \hat{H}_q,$$

then

$$\frac{\phi_1}{\phi_0} = \frac{x_1}{x_0}, \qquad \frac{\phi_2}{\phi_0} = \frac{x_0}{x_1}.$$

If these are defined by a point of Z, there is an M_k-constant $\{C(v)\}_{v \in M_k}$ such that

$$\left\| \frac{x_1}{x_0} \right\|_v \geqq e^{C(v)}, \quad v \in M_k \setminus S,$$

$$\left\| \frac{x_0}{x_1} \right\|_v \geqq e^{C(v)}, \quad v \in M_k \setminus S.$$

Therefore, enlarging S if necessary, we see that

$$\left\| \frac{x_1}{x_0} \right\|_v = \left\| \frac{x_0}{x_1} \right\|_v = 1, \quad v \in M_k \setminus S.$$

That is, the affine coordinate x_1/x_0 is an S-unit.

Therefore Theorem 9.7.1 is essentially a statement for S-units.

As an analogue of Theorem 4.3.2, we have the following with a dimension estimate better than that of Theorem 9.7.1.

Theorem 9.7.4 *Let H_i, $1 \leqq i \leqq l$, be hyperplanes of $\mathbf{P}^n_k$ in general position, and set $D = \sum H_i$. Let W denote the Zariski closure of an (S, D)-integral subset $Z \subset \mathbf{P}^n_k(k) \setminus D$ in $\mathbf{P}^n_k$. Then*

$$\dim W \leqq \frac{n}{l-n}$$

for $l > n$. In particular, if $l \geqq 2n+1$, every such Z is finite.

For a general projective algebraic variety V over k we have the analogues of Theorems 4.9.7 and 7.3.4. Using the notation same as in Sect. 4.9, we obtain

Theorem 9.7.5[1] *Let $\{D_i\}_{i=1}^l$ be a family of divisors on V which are independent in supports, and set $D = \sum_i D_i$. If $l > \dim V + r(\{D_i\}) - q(V)$ (in particular, if $l > \dim V + \operatorname{rank}_{\mathbf{Z}} \mathrm{NS}(V) - q(V)$), then no (S, D)-integral subset is Zariski dense.*

Theorem 9.7.6 *Let $\{D_i\}_{i=1}^l$ be a finite family of ample divisors on V in general position and set $D = \sum D_i$. Assume that*

$$l \geqq 2\dim V + r\big(\{D_i\}\big),$$

or, more strongly that $l \geqq 2n + \operatorname{rank}_{\mathbf{Z}} \mathrm{NS}(X)$ (cf. (4.9.2)). Then, every (S, D)-integral subset is finite.

Since the proofs are exactly the same as those of Theorems 4.9.7 and 7.3.4 except for the usage of Vojta's Theorem 9.6.4 in place of the Logarithmic Bloch–Ochiai Theorem 4.8.17, we do not repeat it.

[1] Cf. the footnote to Theorem 4.9.7.

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Remark There are two collected volumes of Kiyoshi Oka, [Iw] and [Sp]: In [Iw] and [Sp] the volume number of Jpn. J. Math. of K. Oka [53], the IXth paper, was misprinted as **27**, which should be **23** as here. Moreover, the VIIth paper of Oka in [Iw] is the original version sent to Bull. Soc. Math. France, in which it was rewritten by H. Cartan and published as [50]. There are some differences among them;

e.g., the last sentence of the introduction of [Iw] mentioning Hartogs' inverse problem (the Levi Problem) was deleted. The English translation of the VIIth paper in [Sp] is taken from that in [Iw], and hence it includes that sentence. In [Sp] the volume number of Oka [41c] was not given. Furthermore, the records of the received dates of all papers were deleted from the print in [Sp].

There is thus considerable confusion in the reference records of K. Oka's works. Because of the fundamental importance of K. Oka's papers, we here give a list of the complete data of all his published papers.

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